Algebraic Geometry

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

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Contents

Algebraic Geometry originated as the study of the zero-sets of polynomials, namely it was (and in many ways behind all the abstraction predominantly still is) the solution to a system of polynomials. By explicitly working with the zero sets of polynomials, we can discover a great multitude of geometric results within algebra, however as the field progressed it was discovered zero sets are in many ways insufficient to capture all the geometric data. For some quick examples:

1. it was discovered in the mid 20th century that results in number theory seem eerily similar to the study of 1 dimensional curves.
2. The galois group of a field share many of the same properties of as the fundamental group of a topological space.
3. The fact that a polynomial may have a root of high multiplicity is lost in the classical geometric picture $V(f)$, but this is often key information.
4. Number theorists do not work over algebraically closed fields as they are looking for integer (or integral) solutions to polynomial equations. However, these do not have the same nice geometric correspondence within classical algebraic geometry as the Nullstellensatz works well when the field is algebraically closed.
5. All rings can be represented as the quotient of $\mathbb{Z}[x_i, \mid i \in \mathcal{I}]$ or $k[x_i, \mid i \in \mathcal{I}]$ for some indexing set I, hence all ring (not just reduced finitely generated rings over algebraically closed fields) have some notion of being a function with evaluation
6. When studying a family of varieties (ex. $x^2 + ty^2 = 1$ as t varies) the resulting object will often have "bad" singularities which varieties are poorly equipped to work with.

Such realizations lead to the need to ground algebraic geometry with a new object that encapsulates all of these ideas. this new object is the *scheme*. As compared to varieties that requires an ambient space and working over polynomial rings over fields, schemes are *intrinsic* in that they don't require any ambient space to be defined in, and they generalize classical algebraic geometry to be able to capture the geometry of both Diophantine equations (for number theory) as well as the geometry of (quasi-projective) varieties and other more novel. Perhaps key in their success over other variations is how well they also remember "multiplicity" information of roots of polynomials. This data shall be paramount in many results we shall introduce[1].

This book shall be written with graduate or well-read undergraduates in mind. The reason is to show how inter-connected schemes are to various other fields of mathematics. For example:

1. The geometry of schemes draws many parallels to complex geometry and some results from complex analysis (ex. Hartog's lemma from Complex analysis over several variables shall have a strong appearance when discussing normal schemes).
2. A sheaf draws many similarities to the bundles that are put on manifolds; in fact a manifold can be defined equivalently by using a sheaf of differentiable functions instead of an atlas. Some results such as Poincaré duality shall be proven for schemes (see 6.3)
3. Chapter 6 shall strongly rely on the readers intuitions from algebraic topology and use modern homological algebra.

[1]We shall see for Elliptic Curves that the finite-generation of certain torsion groups shall rely on Schemes in a key way to remember multiplicity

4. Many of the properties in commutative algebra shall become interesting geometric properties in algebraic geometry.

Essentially, anything non-optional material from EYNTKA Undergraduate Algebra ([9]), EYNTKA Differential Geometry ([5]), and EYNTKA Algebraic Topology ([3]) will be considered as assumed prerequisites. Some results from EYNTKA complex analysis ([4]) will be drawn on as well given the strong connection between complex geometry and the geometry of schemes.

All essential category theory will be assumed; it can be reviewed in EYNTKA Undergraduate Algebra.

Throughout this book, we shall use the following terminology:

- Let k be a field, $\overline{k}$ the algebraic closure of k
- Let R an arbitrary ring, $M_n(R)$ the matrix ring over R, $R_\mathfrak{p}$ the localization of R by $R \setminus \mathfrak{p}$, R_f the localization of R by $\{f, f^2, ...\}$, $\mathrm{Frac}(R)$ the field of fractions of R is R is an integral domain
- A is an arbitrary R-algebra.

The material covered in the preliminary chapter is at a break-neck pace and is only there for expository purposes for the reader to see if they have the right intuitions to cover algebraic geometry in its greater generality. The first main Part, Scheme Theory, focuses heavily on building up all the necessary theoretical framework to have a comfortable grasp of the material and is heavily inspired by the many famous texts covering the material (ex. Hartshorne, Shafarevich, Vakil, Serre, Eisenbud, and so forth).

Preliminary Chapter

In this chapter, we shall quickly review elementary concepts linking algebra and geometry without doing any proofs, that is we are going to explore some classical algebraic geometry. For all the important details, see [9].

0.1 Recall Classical Algebraic Geometry

If A is an R-algebra, then by the universal property of free R-algebras we have if S is a set of generators for A,

$$\begin{tikzcd} R[S] \arrow[r, "\varphi"] & A \\ S \arrow[u, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

If S is finite, then $R[S] \cong R[x_1, x_2, \dots, x_n]$ by relabelling S. Since every ring is a $\mathbb{Z}$-algebra, we see that polynomial rings are central to the study of general rings. It is thus fruitful to realize that polynomial rings can naturally be interpreted as functions, that is:

$$f \in R[S],\ R^S \to R[S]^\vee \qquad f \mapsto f(s_i)_{i \in S}$$

In the finite case:

$$f \in R[x_1, x_2, \dots, x_n], \qquad R^n \to R[x_1, x_2, \dots, x_n]^\vee \qquad (a_1, a_2, \dots, a_n) \mapsto (f \mapsto f(a_1, a_2, \dots, a_n))$$

We don't want to consider R^S as having any structure besides being an affine space, hence we shall label it as $\mathbb{A}_R^S$ or in the finite case $\mathbb{A}_R^n$. If the ring is evident in context, we shall write $\mathbb{A}^n$.

0.1.1 Conventions in Number Theory In a number-theoretic context, $\mathbb{A}^n$ often implicitly means $\mathbb{A}_{\mathbb{C}}^n$ (or $\mathbb{A}_{\overline{\mathbb{Q}}}^n$), with $\mathbb{A}^n(R)$ denoting the R-points. This convention arises because polynomials over $\mathbb{C}$ always have solutions, which is not guaranteed over $\mathbb{R}$ or $\mathbb{Q}$.

For now, we shall stick to the finitely generated case. Note that in $\mathbb{A}^n_R$, the point $(0, \ldots, 0)$ has no special property, as we are treating this space as simply a collection of points. To emphasize this geometric interpretation, any $p \in \mathbb{A}^n_R$ will be called a *point*, and for

$$p = (r_1, r_2, \ldots, r_n)$$

each r_i is called a *coordinate* of p. Hence, we may view any $f \in R[x_1, x_2, \ldots, x_n]$ as a function:

$$f : \mathbb{A}^n_R \to R \qquad (r_1, r_2, \ldots, r_n) \mapsto f(r_1, r_2, \ldots, r_n)$$

Algebraic Sets and the Zariski Topology

The zeros of polynomials form many interesting algebraic shapes. A prototypical example is $V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{R}}$, the unit circle.

Definition 0.1.2: Zero Set

For $S \subseteq R[x_1, \ldots, x_n]$, the *zero set* (or *vanishing locus*) of S is

$$\mathcal{Z}(S) = \{p \in \mathbb{A}^n_R \mid f(p) = 0, \ \forall f \in S\}$$

Definition 0.1.3: Algebraic Set

A subset $X \subseteq \mathbb{A}^n_R$ is called an *algebraic set* (or *closed set*) if there exists $S \subseteq R[x_1, \ldots, x_n]$ such that $\mathcal{Z}(S) = X$.

The collection of algebraic sets satisfies the axioms for closed sets of a topology:

1. $\mathbb{A}^n_R = \mathcal{Z}(0)$ and $\emptyset = \mathcal{Z}(1)$ are algebraic.
2. Finite unions: $\mathcal{Z}(S_1) \cup \mathcal{Z}(S_2) = \mathcal{Z}(S_1 \cdot S_2)$ where $S_1 \cdot S_2 = \{fg : f \in S_1, g \in S_2\}$.
3. Arbitrary intersections: $\bigcap_\alpha \mathcal{Z}(S_\alpha) = \mathcal{Z}\big(\bigcup_\alpha S_\alpha\big)$.

Definition 0.1.4: Zariski Topology

The *Zariski topology* on $\mathbb{A}^n_R$ is the topology whose closed sets are precisely the algebraic sets.

0.1.5 Properties of the Zariski Topology The Zariski topology is quite coarse compared to the Euclidean topology on $\mathbb{R}^n$ or $\mathbb{C}^n$:

1. It is generally *not* Hausdorff: in $\mathbb{A}^1_k$, the only proper closed subsets are finite sets of points, so any two nonempty open sets intersect.
2. Every nonempty open set is dense.
3. Despite this, the Zariski topology is well-suited to algebraic geometry because its closed sets are precisely the sets defined by polynomial equations.

The Ideal–Variety Correspondence

Conversely, given a subset $S \subseteq \mathbb{A}^n_R$, we can ask which polynomials vanish on S:

Definition 0.1.6: Ideal of a Set

For $S \subseteq \mathbb{A}^n_R$, the *ideal of* S is

$$\mathcal{I}(S) = \{f \in R[x_1, \ldots, x_n] \mid f(p) = 0, \ \forall p \in S\}$$

One readily checks that $\mathcal{I}(S)$ is always an ideal in $R[x_1, \ldots, x_n]$. Moreover, $\mathcal{I}(S)$ is always a *radical* ideal, i.e., $f^n \in \mathcal{I}(S)$ implies $f \in \mathcal{I}(S)$. This is immediate: if f^n vanishes on S, then f vanishes on S.

The operators $\mathcal{Z}$ and $\mathcal{I}$ form an order-reversing correspondence:

$$S_1 \subseteq S_2 \implies \mathcal{I}(S_1) \supseteq \mathcal{I}(S_2), \qquad I_1 \subseteq I_2 \implies \mathcal{Z}(I_1) \supseteq \mathcal{Z}(I_2)$$

However, this correspondence is not bijective in general. The precise relationship requires the notion of radical ideals and irreducibility.

Definition 0.1.7: Radical Ideal

An ideal $I \subseteq R$ is *radical* if $f^n \in I$ implies $f \in I$ for all $f \in R$ and $n \geq 1$. Equivalently, $I = \sqrt{I}$ where $\sqrt{I} = \{f \in R \mid \exists n \geq 1, \ f^n \in I\}$.

Definition 0.1.8: Irreducible Set

An algebraic set V is *irreducible* if whenever $V = V_1 \cup V_2$ with V_1, V_2 closed, then $V = V_1$ or $V = V_2$. Equivalently, V is irreducible if and only if any two nonempty open subsets of V intersect.

Definition 0.1.9: Variety (Classical)

An algebraic set $V \subseteq \mathbb{A}^n_k$ is an (affine) *variety* if V is irreducible.

The key algebraic characterization of irreducibility is:

Proposition 0.1.10: Irreducibility and Prime Ideals

An algebraic set V is irreducible if and only if $\mathcal{I}(V)$ is a prime ideal.

***Proof*:**
See [9].

Example 0.1.11: Reducible vs. Irreducible
Consider $V(xy) \subseteq \mathbb{A}^2_k$. This is the union of the two coordinate axes:

$$V(xy) = V(x) \cup V(y)$$

The ideal (xy) is radical but not prime (since $xy \in (xy)$ but $x \notin (xy)$ and $y \notin (xy)$). Hence $V(xy)$

is an algebraic set but *not* a variety.

In contrast, $V(y - x^2) \subseteq \mathbb{A}^2_k$ (a parabola) is irreducible: the ideal $(y - x^2)$ is prime.

We can now state the Nullstellensatz precisely. From here on, we assume k is an algebraically closed field.

Theorem 0.1.12: Hilbert's Nullstellensatz

Let k be algebraically closed. Then:

1. **(Weak form)** If $I \subsetneq k[x_1, \dots, x_n]$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$.
2. **(Strong form)** For any ideal $I \subseteq k[x_1, \dots, x_n]$, we have $\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I}$.
3. **(Maximal ideal form)** The maximal ideals of $k[x_1, \dots, x_n]$ are precisely those of the form $\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n)$ for points $p = (a_1, \dots, a_n) \in \mathbb{A}^n_k$.

***Proof*:**
See [9].

The strong form immediately gives:

Corollary 0.1.13: The Ideal–Variety Correspondence

Let k be algebraically closed. The maps $\mathcal{Z}$ and $\mathcal{I}$ induce inclusion-reversing bijections:

$$\{\text{radical ideals in } k[x_1, \dots, x_n]\} \xleftrightarrow{\mathcal{Z}} \{\text{algebraic sets in } \mathbb{A}^n_k\}$$

Under this correspondence:

Ideals	$\longleftrightarrow$	**Algebraic Sets**
radical ideals	$\longleftrightarrow$	algebraic sets
prime ideals	$\longleftrightarrow$	varieties (irreducible)
maximal ideals	$\longleftrightarrow$	points

The Coordinate Ring

The passage from geometry to algebra is achieved through the coordinate ring:

Definition 0.1.14: Coordinate Ring

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set. The *coordinate ring* (or *ring of regular functions*) of V is

$$k[V] := k[x_1, \dots, x_n]/\mathcal{I}(V)$$

The elements of $k[V]$ are equivalence classes of polynomials, where two polynomials are equivalent if they agree on V. Thus $k[V]$ can be interpreted as the ring of *polynomial functions* $V \to k$.

Proposition 0.1.15: Properties of the Coordinate Ring

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic set with coordinate ring $k[V]$. Then:

1. $k[V]$ is a finitely generated k-algebra.
2. $k[V]$ is reduced (i.e., has no nilpotents) if and only if $\mathcal{I}(V)$ is radical.
3. $k[V]$ is an integral domain if and only if V is irreducible.
4. Points $p \in V$ correspond to maximal ideals $\mathfrak{m}_p \subseteq k[V]$.

***Proof*:**
See [9].

Example 0.1.16: Coordinate Rings

1. $k[\mathbb{A}^n] = k[x_1, \ldots, x_n]$, since $\mathcal{I}(\mathbb{A}^n) = (0)$.
2. For the parabola $V = V(y - x^2) \subseteq \mathbb{A}^2$, we have $k[V] = k[x,y]/(y - x^2) \cong k[x]$, reflecting that $V \cong \mathbb{A}^1$.
3. For the circle $V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}$, we have $k[V] = \mathbb{C}[x,y]/(x^2 + y^2 - 1)$. This is an integral domain since $(x^2 + y^2 - 1)$ is prime over $\mathbb{C}$.
4. For the nodal cubic $V = V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2$, the coordinate ring is $k[V] = k[x,y]/(y^2 - x^3 - x^2)$. This is an integral domain (the curve is irreducible), but V is singular at the origin.

Morphisms of Affine Varieties

To form a category of varieties, we need morphisms.

Definition 0.1.17: Regular Map / Morphism

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be algebraic sets. A *regular map* (or *morphism*) $\varphi : V \to W$ is a map of the form

$$\varphi(p) = (f_1(p), f_2(p), \ldots, f_m(p))$$

where each $f_i \in k[V]$. That is, φ is given by polynomial functions in coordinates.

Proposition 0.1.18: Morphisms and Algebra Homomorphisms

A regular map $\varphi : V \to W$ induces a k-algebra homomorphism

$$\varphi^* : k[W] \to k[V], \qquad g \mapsto g \circ \varphi$$

called the *pullback*. Conversely, every k-algebra homomorphism $k[W] \to k[V]$ arises as φ^* for a unique regular map $\varphi : V \to W$.

***Proof*:**
See [9].

Note the contravariance: a map $V \to W$ gives a map $k[W] \to k[V]$ in the opposite direction. This leads to the fundamental equivalence:

Theorem 0.1.19: Equivalence of Categories

Let k be algebraically closed. There is a contravariant equivalence of categories:

$$\text{Aff-Var}_k \underset{\text{mSpec}}{\overset{V\mapsto k[V]}{\rightleftarrows}} \left(\text{Alg}_k^{\text{f.g., red}}\right)^{\text{op}}$$

where Aff-Var_k is the category of affine varieties over k with regular maps, and $\text{Alg}_k^{\text{f.g., red}}$ is the category of finitely generated reduced k-algebras with k-algebra homomorphisms.

***Proof*:**
The functor $V \mapsto k[V]$ sends varieties to their coordinate rings and sends morphisms $\varphi : V \to W$ to pullbacks $\varphi^* : k[W] \to k[V]$. The quasi-inverse sends a finitely generated reduced k-algebra A to its maximal spectrum $\text{mSpec}(A)$, the set of maximal ideals with the Zariski topology. The Nullstellensatz guarantees that for $A = k[V]$, the points of V are in bijection with maximal ideals of A. See [9] for details.

0.1.20 Why Reduced Algebras? The restriction to reduced algebras (no nilpotents) corresponds to the fact that classical varieties are determined by their point-sets. Modern algebraic geometry (via schemes) removes this restriction, allowing nilpotent elements to encode infinitesimal data – this is one of the key generalizations.

Local Rings and Regularity

To study the local behavior of a variety at a point, we first introduce the algebraic machinery of localization. This process allows us to systematically make elements of a ring invertible, essentially allowing us to study rings "locally" by focusing on fractions whose denominators do not vanish at the point of interest.

Definition 0.1.21: Localization of a Ring

Let R be a commutative ring. A subset $S \subseteq R$ is *multiplicatively closed* if $1 \in S$ and for all $a, b \in S$, we have $ab \in S$. The *localization* of R at S, denoted $S^{-1}R$, is the set of equivalence classes of fractions $\frac{r}{s}$ with $r \in R$ and $s \in S$, modulo the equivalence relation:

$$\frac{r}{s} \sim \frac{r'}{s'} \iff \text{there exists } t \in S \text{ such that } t(rs' - r's) = 0$$

This becomes a ring under the standard addition and multiplication of fractions.

Localization is completely characterized by the following universal property, which states that $S^{-1}R$ is the "most efficient" way to force the elements of S to become units.

Proposition 0.1.22: Universal Property of Localization

Let S be a multiplicative set of a ring R. The natural map $\iota : R \to S^{-1}R$ given by $\iota(r) = \frac{r}{1}$ is a ring homomorphism that maps elements of S to units in $S^{-1}R$. Furthermore, if A is any ring and $g : R \to A$ is a ring homomorphism such that $g(S) \subseteq A^\times$ (the group of units of A), then there exists a unique ring homomorphism $h : S^{-1}R \to A$ such that $h \circ \iota = g$.

***Proof*:**
We define $h\left(\frac{r}{s}\right) = g(r)g(s)^{-1}$. To see that h is well-defined, suppose $\frac{r}{s} = \frac{r'}{s'}$. Then there is some $t \in S$ such that $t(rs' - r's) = 0$. Applying g gives $g(t)(g(r)g(s') - g(r')g(s)) = 0$. Since $t \in S$, $g(t)$ is a unit in A, so we can multiply by $g(t)^{-1}$ to obtain $g(r)g(s') = g(r')g(s)$. Multiplying by $g(s)^{-1}g(s')^{-1}$ yields $g(r)g(s)^{-1} = g(r')g(s')^{-1}$, confirming h is well-defined.

It is straightforward to verify that h is a ring homomorphism. For uniqueness, if $h' : S^{-1}R \to A$ satisfies $h' \circ \iota = g$, then $h'\left(\frac{r}{1}\right) = g(r)$ and $h'\left(\frac{1}{s}\right) = h'\left(\frac{s}{1}\right)^{-1} = g(s)^{-1}$. Thus, $h'\left(\frac{r}{s}\right) = g(r)g(s)^{-1} = h\left(\frac{r}{s}\right)$.

An important consequence of this universal property is that localization is a functor.

Proposition 0.1.23: Functoriality of Localization

Let R, R' be rings with multiplicative sets S, S' respectively. If $\varphi : R \to R'$ is a ring homomorphism such that $\varphi(S) \subseteq S'$, then there is a unique induced ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ making the following diagram commute:

$$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$$

where ι and ι' are the natural localization maps.

***Proof*:**
Consider the composition $\iota' \circ \varphi : R \to R' \to (S')^{-1}R'$. For any $s \in S$, we have $\varphi(s) \in S'$. Therefore, $\iota'(\varphi(s))$ is a unit in $(S')^{-1}R'$. By the universal property of $S^{-1}R$ applied to the homomorphism $\iota' \circ \varphi$, there exists a unique ring homomorphism $\widetilde{\varphi} : S^{-1}R \to (S')^{-1}R'$ satisfying

$\widetilde{\varphi} \circ \iota = \iota' \circ \varphi$. Explicitly, this map is given by $\widetilde{\varphi}\left(\frac{r}{s}\right) = \frac{\varphi(r)}{\varphi(s)}$.

We now apply this general algebraic construction to the coordinate ring of a variety to study its local geometry.

Definition 0.1.24: Local Ring at a Point

Let V be an affine variety and $p \in V$ a point with corresponding maximal ideal $\mathfrak{m}_p \subseteq k[V]$. The set $S = k[V] \setminus \mathfrak{m}_p$ is multiplicatively closed. The *local ring of V at p* is the localization of $k[V]$ at S:

$$\mathcal{O}_{V,p} := k[V]_{\mathfrak{m}_p} = \left\{ \frac{f}{g} \;\middle|\; f, g \in k[V],\ g(p) \neq 0 \right\}$$

This is a local ring with unique maximal ideal $\mathfrak{m}_{V,p} = \mathfrak{m}_p \cdot \mathcal{O}_{V,p}$.

The local ring $\mathcal{O}_{V,p}$ captures the "germs" of regular functions near p: two functions are identified if they agree on some neighborhood of p.

Definition 0.1.25: Zariski Tangent Space

The *Zariski tangent space* to V at p is the k-vector space

$$T_pV := \left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2\right)^\vee = \operatorname{Hom}_k\left(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2, k\right)$$

The *Zariski cotangent space* is $T_p^*V := \mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2$.

0.1.26 Tangent Space via Derivations Equivalently, T_pV can be defined as the space of k-derivations $\mathcal{O}_{V,p} \to k$ at p, or as the space of "ε-points": morphisms $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to V$ centered at p. This latter perspective generalizes naturally to schemes.

Definition 0.1.27: Smooth Point, Singular Point

Let V be a variety of dimension d (see below). A point $p \in V$ is:

- *smooth* (or *regular*) if $\dim_k T_pV = d$;
- *singular* if $\dim_k T_pV > d$.

A variety is *smooth* (or *nonsingular*) if every point is smooth.

Example 0.1.28: Singularities

1. The parabola $V(y - x^2)$ is smooth everywhere. At any point, $\dim T_pV = 1 = \dim V$.
2. The cuspidal cubic $V(y^2 - x^3)$ has a singularity (cusp) at the origin. One computes $\dim T_0V = 2 > 1 = \dim V$.
3. The nodal cubic $V(y^2 - x^2(x + 1))$ has a node at the origin where two branches cross.

Proposition 0.1.29: Regularity and Local Rings

A point $p \in V$ is smooth if and only if $\mathcal{O}_{V,p}$ is a *regular local ring*, i.e., $\dim_k(\mathfrak{m}_{V,p}/\mathfrak{m}_{V,p}^2) = \dim \mathcal{O}_{V,p}$ (Krull dimension).

***Proof*:**
See [9].

Rational Functions and Dimension

For an irreducible variety, we can form the field of fractions of its coordinate ring:

Definition 0.1.30: Function Field

Let V be an affine variety (irreducible). The *function field* of V is

$$k(V) := \mathrm{Frac}(k[V])$$

Elements of $k(V)$ are called *rational functions* on V. A rational function f/g is defined on the open set $\{p \in V : g(p) \neq 0\}$.

Definition 0.1.31: Dimension

The *dimension* of an affine variety V is

$$\dim V := \operatorname{tr.deg}_k k(V)$$

the transcendence degree of the function field over k. Equivalently, $\dim V = \dim k[V]$, the Krull dimension of the coordinate ring.

Example 0.1.32: Dimensions

1. $\dim \mathbb{A}^n = n$, since $k(\mathbb{A}^n) = k(x_1, \ldots, x_n)$ has transcendence degree n.
2. Any plane curve $V(f) \subseteq \mathbb{A}^2$ with f irreducible has dimension 1.
3. A hypersurface $V(f) \subseteq \mathbb{A}^n$ with f irreducible has dimension $n - 1$.

Proposition 0.1.33: Dimension and Chains of Primes

For an affine variety V, we have $\dim V = \dim k[V]$, where the Krull dimension of $k[V]$ is the supremum of lengths of chains of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

in $k[V]$.

***Proof*:**
See [9].

0.2 Recall Projective Spaces

Projective space is the natural setting for algebraic geometry. Many phenomena that are awkward or incomplete in affine space become clean and uniform in projective space: intersection theory works properly (Bézout's theorem), curves have well-defined degrees, and there are no "missing points at infinity." This section reviews the essential constructions.

Definition and Homogeneous Coordinates

Definition 0.2.1: Projective Space

Let k be a field. *Projective n-space* over k is

$$\mathbb{P}^n_k := \left(k^{n+1} \setminus \{0\}\right) / \sim$$

where $(a_0, a_1, \ldots, a_n) \sim (\lambda a_0, \lambda a_1, \ldots, \lambda a_n)$ for all $\lambda \in k^\times$. The equivalence class of $(a_0, \ldots, a_n)$ is denoted $[a_0 : a_1 : \cdots : a_n]$ and called a *point* in $\mathbb{P}^n$. The a_i are *homogeneous coordinates*.

0.2.2 Projective Space as Lines Equivalently, $\mathbb{P}^n_k$ is the set of 1-dimensional linear subspaces (lines through the origin) in k^{n+1}. This is sometimes denoted $\mathbb{P}(k^{n+1})$ or $\mathrm{Proj}(k^{n+1})$.

Example 0.2.3: Low-Dimensional Projective Spaces

1. $\mathbb{P}^0$ is a single point.
2. $\mathbb{P}^1$ is the *projective line*. Points are $[a_0 : a_1]$ with $(a_0, a_1) \neq (0, 0)$. If $a_1 \neq 0$, we can normalize to $[t : 1]$ for $t = a_0/a_1 \in k$. The remaining point $[1 : 0]$ is the "point at infinity." Thus $\mathbb{P}^1 \cong \mathbb{A}^1 \sqcup \{\infty\}$.
3. $\mathbb{P}^2$ is the *projective plane*, the natural home for plane curves.

Affine Charts

Projective space is covered by affine open sets:

Definition 0.2.4: Standard Affine Charts

For each $i \in \{0, 1, \dots, n\}$, define the *standard affine open*

$$U_i := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid a_i \neq 0\}$$

There is an isomorphism $\varphi_i : U_i \xrightarrow{\sim} \mathbb{A}^n$ given by

$$\varphi_i([a_0 : \cdots : a_n]) = \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i}\right)$$

where the hat denotes omission.

Proposition 0.2.5: Affine Cover of Projective Space

The standard affine opens cover $\mathbb{P}^n$:

$$\mathbb{P}^n = U_0 \cup U_1 \cup \cdots \cup U_n$$

The complement of U_i is the hyperplane $H_i = \{[a_0 : \cdots : a_n] : a_i = 0\} \cong \mathbb{P}^{n-1}$.

Example 0.2.6: The Projective Line Revisited

For $\mathbb{P}^1$, we have two charts:

- $U_0 = \{[a_0 : a_1] : a_0 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_1/a_0$
- $U_1 = \{[a_0 : a_1] : a_1 \neq 0\} \cong \mathbb{A}^1$ via $[a_0 : a_1] \mapsto a_0/a_1$

On the overlap $U_0 \cap U_1$, if $t = a_1/a_0$ is the coordinate on U_0 and $s = a_0/a_1$ on U_1, then $s = 1/t$. This is the standard gluing of two copies of $\mathbb{A}^1$ to form $\mathbb{P}^1$.

Homogeneous Polynomials and Graded Rings

Polynomials do not define functions on $\mathbb{P}^n$ directly, since $f(\lambda a_0, \dots, \lambda a_n) \neq f(a_0, \dots, a_n)$ in general. However, the *vanishing* of homogeneous polynomials is well-defined.

Definition 0.2.7: Homogeneous Polynomial

A polynomial $f \in k[x_0, x_1, \dots, x_n]$ is *homogeneous of degree d* if every monomial in f has total degree d. Equivalently, $f(\lambda x_0, \dots, \lambda x_n) = \lambda^d f(x_0, \dots, x_n)$ for all $\lambda \in k$.

Definition 0.2.8: Graded Ring

The polynomial ring $S = k[x_0, \dots, x_n]$ has a *grading* $S = \bigoplus_{d \geq 0} S_d$, where S_d is the k-vector space of homogeneous polynomials of degree d (including 0). We have $S_d \cdot S_e \subseteq S_{d+e}$.

Definition 0.2.9: Homogeneous Ideal

An ideal $I \subseteq k[x_0, \ldots, x_n]$ is *homogeneous* if it is generated by homogeneous polynomials. Equivalently, $I = \bigoplus_{d \geq 0}(I \cap S_d)$.

Proposition 0.2.10: Characterization of Homogeneous Ideals

An ideal I is homogeneous if and only if for every $f \in I$, all homogeneous components of f are also in I.

***Proof*:**
See [9].

Projective Varieties

Definition 0.2.11: Projective Algebraic Set

For a set T of homogeneous polynomials in $k[x_0, \ldots, x_n]$, the *projective zero set* is

$$\mathcal{Z}_+(T) := \{[a_0 : \cdots : a_n] \in \mathbb{P}^n \mid f(a_0, \ldots, a_n) = 0, \ \forall f \in T\}$$

This is well-defined since f homogeneous implies $f(\lambda \mathbf{a}) = \lambda^d f(\mathbf{a})$, so $f(\mathbf{a}) = 0 \iff f(\lambda \mathbf{a}) = 0$. A subset $V \subseteq \mathbb{P}^n$ is a *projective algebraic set* if $V = \mathcal{Z}_+(T)$ for some set T of homogeneous polynomials.

Definition 0.2.12: Homogeneous Ideal of a Set

For $V \subseteq \mathbb{P}^n$, define

$$\mathcal{I}_+(V) := \text{ideal generated by } \{f \in k[x_0, \ldots, x_n] \text{ homogeneous} \mid f(p) = 0, \ \forall p \in V\}$$

This is always a homogeneous radical ideal.

The projective algebraic sets form the closed sets of the *Zariski topology on* $\mathbb{P}^n$.

Definition 0.2.13: Projective Variety

A projective algebraic set $V \subseteq \mathbb{P}^n$ is a *projective variety* if it is irreducible in the Zariski topology.

0.2.14 The Irrelevant Ideal The ideal $\mathfrak{m}_+ = (x_0, x_1, \ldots, x_n) \subseteq k[x_0, \ldots, x_n]$ is called the *irrelevant ideal.* It satisfies $\mathcal{Z}_+(\mathfrak{m}_+) = \emptyset$, since the only common zero of all x_i in k^{n+1} is the origin, which does not correspond to a point in $\mathbb{P}^n$. This is an important subtlety in the projective Nullstellensatz.

Theorem 0.2.15: Projective Nullstellensatz

Let k be algebraically closed and let $I \subseteq k[x_0, \dots, x_n]$ be a homogeneous ideal. Then:

1. $\mathcal{Z}_+(I) = \emptyset$ if and only if $\sqrt{I} \supseteq \mathfrak{m}_+^N$ for some $N \geq 1$ (equivalently, $\sqrt{I} = \mathfrak{m}_+$ or $\sqrt{I} = (1)$).
2. If $\mathcal{Z}_+(I) \neq \emptyset$, then $\mathcal{I}_+(\mathcal{Z}_+(I)) = \sqrt{I}$.

Hence $\mathcal{Z}_+$ and $\mathcal{I}_+$ give a bijection between:

$$\left\{\text{homogeneous radical ideals } I \text{ with } \sqrt{I} \neq \mathfrak{m}_+\right\} \longleftrightarrow \{\text{projective algebraic sets in } \mathbb{P}^n\}$$

***Proof*:**
See [9].

Homogeneous Coordinate Ring and the Proj Construction

Definition 0.2.16: Homogeneous Coordinate Ring

For a projective variety $V = \mathcal{Z}_+(I) \subseteq \mathbb{P}^n$, the *homogeneous coordinate ring* is the graded ring

$$S(V) := k[x_0, \dots, x_n]/I$$

with the induced grading $S(V) = \bigoplus_{d \geq 0} S(V)_d$.

0.2.17 Homogeneous Coordinate Ring is Not Intrinsic Unlike the affine case, the homogeneous coordinate ring $S(V)$ depends on the embedding $V \hookrightarrow \mathbb{P}^n$, not just on V as an abstract variety. Different embeddings of the same variety can give non-isomorphic graded rings.

Definition 0.2.18: The Proj Construction

For a graded ring $S = \bigoplus_{d \geq 0} S_d$, we define

$$\operatorname{Proj}(S) := \{\mathfrak{p} \in \operatorname{Spec}(S) \mid \mathfrak{p} \text{ is homogeneous and } \mathfrak{p} \not\supseteq S_+\}$$

where $S_+ = \bigoplus_{d>0} S_d$ is the irrelevant ideal. This set is equipped with a natural topology and structure sheaf making it a scheme. For $S = k[x_0, \dots, x_n]$, we recover $\operatorname{Proj}(S) = \mathbb{P}^n_k$.

Relation Between Affine and Projective

The affine charts give a way to study projective varieties locally:

Definition 0.2.19: Affine Cone

For a projective variety $V \subseteq \mathbb{P}^n$, the *affine cone* over V is

$$C(V) := \mathcal{Z}(\mathcal{I}_+(V)) \subseteq \mathbb{A}^{n+1}$$

This is an affine variety in $\mathbb{A}^{n+1}$, consisting of all lines through the origin that correspond to points of V, together with the origin itself.

Proposition 0.2.20: Affine Pieces of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ be a projective variety and $U_i \cong \mathbb{A}^n$ a standard affine chart. Then $V \cap U_i$ is an affine variety in $\mathbb{A}^n$.
Explicitly, if $V = \mathcal{Z}_+(f_1, \ldots, f_r)$ with each f_j homogeneous of degree d_j, then $V \cap U_i = \mathcal{Z}(f_1^{(i)}, \ldots, f_r^{(i)})$ where

$$f_j^{(i)}(y_0, \ldots, \widehat{y_i}, \ldots, y_n) := f_j(y_0, \ldots, y_{i-1}, 1, y_{i+1}, \ldots, y_n)$$

is the *dehomogenization* of f_j with respect to x_i.

Definition 0.2.21: Homogenization

Conversely, given $g \in k[y_1, \ldots, y_n]$ of degree d, its *homogenization* with respect to x_0 is

$$g^h(x_0, x_1, \ldots, x_n) := x_0^d \cdot g\left(\frac{x_1}{x_0}, \ldots, \frac{x_n}{x_0}\right)$$

This is homogeneous of degree d.

Example 0.2.22: Projective Closure

The parabola $V(y - x^2) \subseteq \mathbb{A}^2$ has homogenization $yz - x^2 \in k[x, y, z]$. Its projective closure is $\overline{V} = \mathcal{Z}_+(yz - x^2) \subseteq \mathbb{P}^2$. The "new" point $[0 : 1 : 0]$ (where $z = 0$ and $x = 0$) is the point at infinity where the parabola "closes up."

Quasi-Projective Varieties

Definition 0.2.23: Quasi-Projective Variety

A *quasi-projective variety* is a locally closed subset of $\mathbb{P}^n$, i.e., an open subset of a projective variety. Equivalently, it is the intersection of an open set and a closed set in $\mathbb{P}^n$.

Proposition 0.2.24: Affine and Projective are Quasi-Projective

Both affine varieties and projective varieties are quasi-projective:

1. A projective variety $V \subseteq \mathbb{P}^n$ is quasi-projective (it is closed, hence locally closed).
2. An affine variety $V \subseteq \mathbb{A}^n \cong U_0 \subseteq \mathbb{P}^n$ is quasi-projective (it is closed in the open set U_0).

The category of quasi-projective varieties with regular maps is the classical category of "algebraic varieties" in much of the literature.

Morphisms of Projective Varieties

Defining morphisms of projective varieties requires care, since homogeneous polynomials do not define functions.

Definition 0.2.25: Morphism of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. A *morphism* $\varphi : V \to W$ is a map that is locally given by homogeneous polynomials of the same degree. That is, for each $p \in V$, there exist homogeneous polynomials $f_0, \ldots, f_m \in k[x_0, \ldots, x_n]$ of the same degree d such that:

1. $f_0, \ldots, f_m$ do not simultaneously vanish on a neighborhood of p in V.
2. $\varphi(q) = [f_0(q) : \cdots : f_m(q)]$ for all q in this neighborhood.

0.2.26 Local Nature of Morphisms Different homogeneous polynomials may be needed on different open subsets of V. On overlaps, they must agree as maps to $\mathbb{P}^m$. This local definition is necessary because no single tuple of homogeneous polynomials may work globally.

Example 0.2.27: The Veronese Embedding

The *Veronese embedding* of degree d is

$$\nu_d : \mathbb{P}^n \to \mathbb{P}^N, \qquad [x_0 : \cdots : x_n] \mapsto [\cdots : x_0^{i_0} \cdots x_n^{i_n} : \cdots]$$

where the coordinates on $\mathbb{P}^N$ are indexed by monomials of degree d, so $N = \binom{n+d}{d} - 1$.

For example, $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$ is given by $[s : t] \mapsto [s^2 : st : t^2]$. The image is the conic $\mathcal{Z}_+(xz - y^2)$, showing that $\mathbb{P}^1$ is isomorphic to this conic.

Example 0.2.28: The Segre Embedding

The *Segre embedding* is

$$\sigma : \mathbb{P}^n \times \mathbb{P}^m \to \mathbb{P}^{(n+1)(m+1)-1}, \qquad ([x_i], [y_j]) \mapsto [x_i y_j]_{0 \le i \le n,\, 0 \le j \le m}$$

For example, $\sigma : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ is given by

$$([s : t], [u : v]) \mapsto [su : sv : tu : tv]$$

The image is the quadric surface $\mathcal{Z}_+(xw - yz) \subseteq \mathbb{P}^3$.

The Segre embedding shows that $\mathbb{P}^n \times \mathbb{P}^m$ can be given the structure of a projective variety.

Products of Projective Varieties

Proposition 0.2.29: Products of Projective Varieties

Let $V \subseteq \mathbb{P}^n$ and $W \subseteq \mathbb{P}^m$ be projective varieties. Then $V \times W$, embedded via the Segre embedding $\mathbb{P}^n \times \mathbb{P}^m \hookrightarrow \mathbb{P}^{(n+1)(m+1)-1}$, is a projective variety.

***Proof*:**
See [9].

0.2.30 Contrast with Affine Products For affine varieties, products are straightforward: $V \times W \subseteq \mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$, and $k[V \times W] \cong k[V] \otimes_k k[W]$. The projective case requires the Segre embedding because $\mathbb{P}^n \times \mathbb{P}^m \ncong \mathbb{P}^{n+m}$.

Completeness and Properness

Projective varieties enjoy a crucial property not shared by affine varieties:

Definition 0.2.31: Complete Variety

A variety V is *complete* if for every variety W, the projection $\pi : V \times W \to W$ is a closed map (sends closed sets to closed sets).

Theorem 0.2.32: Projective Varieties are Complete

Every projective variety is complete.

***Proof*:**
See [9].

0.2.33 Completeness as Compactness Completeness is the algebraic analogue of compactness. Over $\mathbb{C}$, a projective variety is compact in the classical (Euclidean) topology. Affine varieties of positive dimension are never complete: for example, the projection $\mathbb{A}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$ does not send the closed hyperbola $\mathcal{Z}(xy - 1)$ to a closed set.

Proposition 0.2.34: Affine and Complete Implies Finite

If V is both affine and complete, then V is a finite set of points.

***Proof*:**
See [9].

0.3 Noetherian Induction or Well-founded Induction

This is something that will come up in later proofs (for example, see lemma 5.1.14). This is a form of induction that generalizes the usual induction where the well-ordered set is $\mathbb{N}$. Recall that a binary relation R on S is a subset of $S \times S$. For $(a, b) \in R$, we may write $a \preceq b$ or $b \succeq a$. Then the symbol $\preceq$ is sometimes called the binary relation. A binary relation is a strict partial order if it is:

1. Irreflexive: $\forall x \in A, \neg(x < x)$
2. Transitive: $\forall x, y, z \in A, (x < y \wedge y < z) \implies x < z$
3. Asymmetric: $\forall x, y \in A, x < y \implies \neg(y < x)$

Given a set S with a strict partial order $\preceq$, a descending chain is a sequence $a_1, a_2, a_3, \dots$ of elements in A such that $a_1 \succeq a_2 \succeq a_3 \succeq \dots$

Definition 0.3.1: Well-founded Relation

A strict partial order $\preceq$ on a set A is well-founded if there exist no infinite descending chains in A with respect to $\preceq$.

Definition 0.3.2: Minimal Element

Given a subset S of a well-founded set A, an element $m \in S$ is minimal if there is no element $x \in S$ such that $x \preceq m$.

Theorem 0.3.3: Noetherian Induction

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$, and let P be a property defined on A. If for all $x \in A : [\forall y \in A, (y \preceq x \implies P(y))] \implies P(x)$ then $P(x)$ holds for all $x \in A$.

***Proof*:**
see cite:HERE.

The following may be an insightful alternative:

Theorem 0.3.4: Noetherian Induction - Alternative Form

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$. If $S \subseteq A$ is a subset such that for all minimal elements m of $A \backslash S$, we have $m \in S$, then $S = A$.

***Proof*:**
see cite:HERE.

The connection between these forms lies in taking S to be the set of elements where P holds, and showing that if S isn't all of A, then any minimal element of $A \setminus S$ must actually be in S, giving a contradiction.

0.4 Categorical Reminder

Category theory provides the language and organizational framework for modern algebraic geometry. Schemes are defined as certain locally ringed spaces, morphisms are defined via universal properties, and many constructions (fiber products, sheafification, etc.) are instances of general categorical machinery. This section collects the essential concepts.

Categories, Functors, and Natural Transformations

Definition 0.4.1: Category

A *category* $\mathcal{C}$ consists of:

1. A class $\mathrm{Ob}(\mathcal{C})$ of *objects*.
2. For each pair of objects X, Y, a set of *morphisms* from X to Y denoted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ or $\mathrm{Mor}_{\mathbb{C}}(X, Y)$.
3. For each triple X, Y, Z, a *composition* map $\mathrm{Hom}(Y, Z) \times \mathrm{Hom}(X, Y) \to \mathrm{Hom}(X, Z)$.
4. For each object X, an *identity* morphism $\mathrm{id}_X \in \mathrm{Hom}(X, X)$.

These satisfy associativity of composition and the identity laws.

Example 0.4.2: Categories in Algebra and Geometry

1. **Set**: sets and functions.
2. **Grp**, **Ab**, **Ring**, **CRing**: groups, abelian groups, rings, commutative rings.
3. $\mathbf{Mod}_R$: R-modules and R-linear maps.
4. **Top**: topological spaces and continuous maps.
5. $\mathbf{Aff}_k$: affine varieties over k and regular maps.

Definition 0.4.3: Functor

A *(covariant) functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

1. A map $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$.
2. For each $X, Y \in \mathcal{C}$, a map $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$.

These must satisfy $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(g) \circ F(f)$.
A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ reverses arrows: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(Y), F(X))$ with $F(g \circ f) = F(f) \circ F(g)$. Equivalently, it is a covariant functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Example 0.4.4: Functors in Algebraic Geometry

1. The coordinate ring functor $V \mapsto k[V]$ is a contravariant functor $\mathbf{Aff}_k \to \mathbf{CRing}$.
2. Forgetful functors: $\mathbf{Grp} \to \mathbf{Set}$, etc.
3. We shall define the spectrum functor $A \mapsto \mathrm{Spec}\,(A)$ is a contravariant functor $\mathbf{CRing} \to \mathbf{Sch}$.
4. We shall define the global sections functor $X \mapsto \Gamma(X, \mathcal{O}_X)$ is a contravariant functor $\mathbf{Sch} \to \mathbf{CRing}$.

Definition 0.4.5: Natural Transformation

Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\eta : F \Rightarrow G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ for each $X \in \mathcal{C}$, such that for every morphism $f : X \to Y$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} F(X) \arrow[r, "\eta_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\eta_Y"'] & G(Y) \end{tikzcd}$$

A *natural isomorphism* is a natural transformation where each η_X is an isomorphism.

Definition 0.4.6: Equivalence of Categories

An *equivalence of categories* between $\mathcal{C}$ and $\mathcal{D}$ consists of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ together with natural isomorphisms $G \circ F \cong \mathrm{id}_{\mathcal{C}}$ and $F \circ G \cong \mathrm{id}_{\mathcal{D}}$.
A functor F is an equivalence if and only if it is:

- *Fully faithful*: $F : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is a bijection for all X, Y.
- *Essentially surjective*: every object of $\mathcal{D}$ is isomorphic to $F(X)$ for some $X \in \mathcal{C}$.

Example 0.4.7: The Classical Equivalence

The equivalence $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g., red}})^{\mathrm{op}}$ from section 0.1 is an equivalence of categories: the coordinate ring functor is fully faithful and essentially surjective.

Universal Properties: Limits and Colimits

Many constructions in mathematics are characterized by universal properties. Limits and colimits provide a unified framework.

Definition 0.4.8: Diagram and Cone

Let J be a small category (the "index category"). A *diagram* of shape J in $\mathcal{C}$ is a functor $D : J \to \mathcal{C}$.
A *cone* over D with vertex $X \in \mathcal{C}$ is a collection of morphisms $\pi_j : X \to D(j)$ for each $j \in J$, such that for every morphism $\alpha : j \to j'$ in J, we have $D(\alpha) \circ \pi_j = \pi_{j'}$.

Definition 0.4.9: Limit

The *limit* of a diagram $D : J \to \mathcal{C}$, denoted $\varprojlim D$ or $\lim_J D$, is a cone $(\pi_j : L \to D(j))_{j \in J}$ that is universal: for any other cone $(f_j : X \to D(j))$, there exists a unique morphism $f : X \to L$ such that $\pi_j \circ f = f_j$ for all j.

$$\begin{tikzcd}
X \arrow[d, dashed, "\exists! f"'] \arrow[dr, "f_j"] & \\
L \arrow[r, "\pi_j"'] & D(j)
\end{tikzcd}$$

Definition 0.4.10: Colimit

Dually, a *cocone* under D with vertex X is a collection $\iota_j : D(j) \to X$ compatible with the diagram. The *colimit* $\varinjlim D$ is the universal cocone.

Example 0.4.11: Specific Limits and Colimits

Shape J	**Limit**	**Colimit**
Discrete (no arrows)	Product $\prod_i X_i$	Coproduct $\coprod_i X_i$
$\bullet \rightrightarrows \bullet$	Equalizer	Coequalizer
$\bullet \to \bullet \leftarrow \bullet$	Pullback (fiber product)	—
$\bullet \leftarrow \bullet \to \bullet$	—	Pushout
Empty	Terminal object $*$	Initial object $\emptyset$
Directed poset	Directed (inverse) limit	Directed (direct) limit

Products and Coproducts

Definition 0.4.12: Product

The *product* of objects $\{X_i\}_{i \in I}$ is an object $\prod_i X_i$ together with projection morphisms $\pi_j : \prod_i X_i \to X_j$ such that for any object Y with morphisms $f_j : Y \to X_j$, there exists a unique $f : Y \to \prod_i X_i$ with $\pi_j \circ f = f_j$.

Definition 0.4.13: Coproduct

Dually, the *coproduct* $\coprod_i X_i$ has inclusion morphisms $\iota_j : X_j \to \coprod_i X_i$ universal for morphisms out of the X_j.

Example 0.4.14: Products and Coproducts in Various Categories

Category	**Product**	**Coproduct**
Set	Cartesian product	Disjoint union
Grp	Direct product	Free product
Ab, $\mathbf{Mod}_R$	Direct product	Direct sum
CRing	Direct product	Tensor product $A \otimes_{\mathbb{Z}} B$
Top	Product topology	Disjoint union topology
Sch	Fiber product over $\operatorname{Spec} \mathbb{Z}$	Disjoint union

Fiber products are ubiquitous in algebraic geometry: base change, intersections, and fiber constructions all use them.

Definition 0.4.15: Fiber Product / Pullback

Given morphisms $f : X \to S$ and $g : Y \to S$, the *fiber product* (or *pullback*) is an object $X \times_S Y$ with morphisms $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ such that $f \circ p = g \circ q$, universal with this property:

$$\begin{tikzcd} X \times_S Y \arrow[r, "q"] \arrow[d, "p"'] \arrow[dr, phantom, "\square"] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & S \end{tikzcd}$$

For any Z with morphisms to X and Y making the square commute, there is a unique morphism $Z \to X \times_S Y$.

Example 0.4.16: Fiber Products in Sets

In **Set**, the fiber product is

$$X \times_S Y = \{(x, y) \in X \times Y : f(x) = g(y)\}$$

If $S = *$ is a point, this is the ordinary Cartesian product.

Adjoint Functors

Adjunctions capture many important relationships between categories.

Definition 0.4.17: Adjoint Pair

Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ be functors. We say F is *left adjoint* to G (and G is *right adjoint* to F), written $F \dashv G$, if there is a natural bijection

$$\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

for all $X \in \mathcal{C}$ and $Y \in \mathcal{D}$.

Proposition 0.4.18: Characterizations of Adjunctions

The following are equivalent to $F \dashv G$:

1. Natural bijection $\mathrm{Hom}_{\mathcal{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$.
2. Existence of a *unit* $\eta : \mathrm{id}_{\mathcal{C}} \Rightarrow G \circ F$ and *counit* $\varepsilon : F \circ G \Rightarrow \mathrm{id}_{\mathcal{D}}$ satisfying the triangle identities.
3. F preserves colimits and satisfies a solution set condition (Freyd's adjoint functor theorem).

***Proof*:**
See any standard reference on category theory, e.g., Mac Lane [9].

Example 0.4.19: Adjunctions in Algebra and Geometry

1. **Free-Forgetful**: The free group functor $F : \mathbf{Set} \to \mathbf{Grp}$ is left adjoint to the forgetful functor $U : \mathbf{Grp} \to \mathbf{Set}$:

$$\mathrm{Hom}_{\mathbf{Grp}}(F(S), G) \cong \mathrm{Hom}_{\mathbf{Set}}(S, U(G))$$

2. **Tensor-Hom**: For R-modules, $(-) \otimes_R M \dashv \mathrm{Hom}_R(M, -)$:

$$\mathrm{Hom}_R(N \otimes_R M, P) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, P))$$

3. **Inverse and Direct Image**: For a continuous map $f : X \to Y$, the inverse image functor f^{-1} on sheaves is left adjoint to the direct image f_*.

Two adjunctions we shall encounter are:

1. **Spec-Global Sections**: The functor $\mathrm{Spec} : \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$ is left adjoint to $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$:

$$\mathrm{Hom}_{\mathbf{Sch}}(\mathrm{Spec}\,(A), X) \cong \mathrm{Hom}_{\mathbf{CRing}}(\Gamma(X, \mathcal{O}_X), A)$$

2. **Sheafification**: The sheafification functor is left adjoint to the inclusion of sheaves into presheaves.

Proposition 0.4.20: Adjoints and Limits

1. Left adjoints preserve colimits.
2. Right adjoints preserve limits.

In particular, if $F \dashv G$, then F preserves coproducts, coequalizers, and pushouts, while G preserves products, equalizers, and pullbacks.

***Proof*:**
See [9].

Representable Functors and the Yoneda Lemma

The Yoneda perspective is fundamental to modern algebraic geometry: schemes can be understood through their "functor of points."

Definition 0.4.21: Representable Functor

A functor $F : \mathcal{C}^{\text{op}} \to \textbf{Set}$ is *representable* if there exists an object $X \in \mathcal{C}$ and a natural isomorphism

$$F \cong \text{Hom}_{\mathcal{C}}(-, X) =: h_X$$

The object X is called the *representing object*. We say "X represents F" or "F is represented by X."

Theorem 0.4.22: Yoneda Lemma

Let $\mathcal{C}$ be a category, $X \in \mathcal{C}$, and $F : \mathcal{C}^{\text{op}} \to \textbf{Set}$ a functor. There is a natural bijection

$$\text{Nat}(h_X, F) \cong F(X)$$

where $\text{Nat}(h_X, F)$ is the set of natural transformations from h_X to F. The bijection sends $\eta : h_X \Rightarrow F$ to $\eta_X(\text{id}_X) \in F(X)$.

***Proof*:**
See [9].

Corollary 0.4.23: Yoneda Embedding

The functor $h : \mathcal{C} \to \textbf{Fun}(\mathcal{C}^{\text{op}}, \textbf{Set})$ given by $X \mapsto h_X = \text{Hom}(-, X)$ is fully faithful. In particular:

1. If $h_X \cong h_Y$, then $X \cong Y$ (representing objects are unique up to unique isomorphism).
2. $\text{Hom}_{\mathcal{C}}(X, Y) \cong \text{Nat}(h_X, h_Y)$.

Example 0.4.24: Moduli Problems as Functors
Many moduli problems are phrased as functors $F : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$:

- $F(S) = \{\text{families of elliptic curves over } S\}$
- $F(S) = \{\text{closed subschemes of } \mathbb{P}^n_S \text{ flat over } S\}$ (Hilbert functor)

If F is representable by a scheme M, then M is the *moduli space*. Often F is not representable by a scheme but by an algebraic stack.

Filtered Colimits

Filtered colimits arise naturally when taking stalks of sheaves and germs of functions.

Definition 0.4.25: Filtered Category

A category J is *filtered* if:

1. J is nonempty.
2. For any $i, j \in J$, there exists $k \in J$ with morphisms $i \to k$ and $j \to k$.
3. For any parallel morphisms $f, g : i \to j$, there exists $h : j \to k$ with $h \circ f = h \circ g$.

A *directed set* (poset where every pair has an upper bound) is filtered when viewed as a category.

Definition 0.4.26: Filtered Colimit

A *filtered colimit* (or *direct limit*) is a colimit over a filtered index category.

Example 0.4.27: Localization as Filtered Colimit
For a ring A and multiplicative set S, the localization is

$$S^{-1}A = \varinjlim_{s \in S} A_s$$

where $A_s = A[1/s]$ and the colimit is over S ordered by divisibility.

Proposition 0.4.28: Filtered Colimits Commute with Finite Limits

In **Set** (and more generally in any *finitely presentable* category), filtered colimits commute with finite limits. In particular, filtered colimits are exact in module categories.

***Proof*:**
See [9].

Abelian Categories

For homological algebra and cohomology, we need abelian categories.

Definition 0.4.29: Abelian Category

A category $\mathcal{A}$ is *abelian* if:

1. $\mathcal{A}$ has a zero object (both initial and terminal).
2. $\mathcal{A}$ has all binary products and coproducts (which coincide: biproducts).
3. Every morphism has a kernel and cokernel.
4. Every monomorphism is a kernel; every epimorphism is a cokernel.

Example 0.4.30: Abelian Categories

1. $\mathbf{Ab}$, $\mathbf{Mod}_R$ for any ring R.
2. $\mathbf{Sh}(X, \mathbf{Ab})$: sheaves of abelian groups on X.
3. $\mathbf{QCoh}(X)$: quasi-coherent sheaves on a scheme X.
4. $\mathbf{Coh}(X)$: coherent sheaves on a Noetherian scheme X.

0.4.31 Homological Algebra in Abelian Categories In an abelian category, one can define:

- Exact sequences: $A \to B \to C$ is exact at B if $\ker(B \to C) = \operatorname{im}(A \to B)$.
- Chain complexes and homology.
- Derived functors (if there are enough injectives or projectives).

Sheaf cohomology $H^i(X, \mathcal{F})$ is defined as the right derived functors of the global sections functor $\Gamma(X, -)$.

0.4.1 Summary: Categorical Tools for Schemes

Concept	Use in Algebraic Geometry
Equivalence of categories	Classical: $\mathbf{Aff}_k \simeq (\mathbf{Alg}_k^{\text{f.g.,red}})^{\text{op}}$
Fiber products	Base change, fibers, intersections
Adjunctions	Spec $\dashv \Gamma$; $f^{-1} \dashv f_*$; sheafification
Representable functors	Schemes via functor of points
Yoneda lemma	Universal properties; moduli problems
Filtered colimits	Stalks, localizations
Abelian categories	Sheaf cohomology, derived categories

0.5 *Key Results from Differential Geometry

Many constructions in algebraic geometry—tangent bundles, differential forms, cohomology, metrics—have their origins in differential geometry. This section reviews the essential concepts from the smooth category that will inform and motivate their algebraic counterparts.

Throughout this section, "smooth" means C^∞ (infinitely differentiable).

Smooth Manifolds

Definition 0.5.1: Smooth Manifold

A *topological manifold* of dimension n is a Hausdorff, second-countable topological space M that is locally homeomorphic to $\mathbb{R}^n$. A *chart* is a pair (U, φ) where $U \subseteq M$ is open and $\varphi : U \xrightarrow{\sim} \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism.
A *smooth structure* on M is a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are smooth (as maps between open subsets of $\mathbb{R}^n$).
A *smooth manifold* is a topological manifold equipped with a smooth structure.

Example 0.5.2: Examples of Smooth Manifolds

1. $\mathbb{R}^n$ with the standard atlas (single chart $\mathrm{id} : \mathbb{R}^n \to \mathbb{R}^n$).
2. Open subsets of $\mathbb{R}^n$ (or of any smooth manifold).
3. The n-sphere $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$.
4. Real and complex projective spaces $\mathbb{RP}^n$, $\mathbb{CP}^n$.
5. Lie groups: $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}(n)$, $\mathrm{U}(n)$, $\mathrm{SL}_n(\mathbb{R})$, etc.
6. Any smooth algebraic variety over $\mathbb{R}$ or $\mathbb{C}$ (with its classical topology).

Definition 0.5.3: Smooth Map

A map $f : M \to N$ between smooth manifolds is *smooth* if for every $p \in M$ and charts (U, φ) around p and (V, ψ) around $f(p)$, the composition

$$\psi \circ f \circ \varphi^{-1} : \varphi(U \cap f^{-1}(V)) \to \psi(V)$$

is smooth as a map between open subsets of Euclidean spaces.
A *diffeomorphism* is a smooth bijection with smooth inverse.

Definition 0.5.4: Tangent Space (via Derivations)

Let M be a smooth manifold and $p \in M$. A *derivation at* p is an $\mathbb{R}$-linear map $v : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule:

$$v(fg) = f(p) \cdot v(g) + g(p) \cdot v(f)$$

The *tangent space* T_pM is the vector space of all derivations at p.

Proposition 0.5.5: Dimension of Tangent Space

If $\dim M = n$, then $\dim T_pM = n$ for all $p \in M$. In local coordinates $(x^1, \ldots, x^n)$ centered at p, a basis for T_pM is given by the partial derivative operators:

$$\left\{ \left.\frac{\partial}{\partial x^1}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p \right\}$$

Definition 0.5.6: Differential / Pushforward

Let $f : M \to N$ be smooth and $p \in M$. The *differential* (or *pushforward*) of f at p is the linear map

$$df_p = f_{*,p} : T_pM \to T_{f(p)}N, \qquad (df_p(v))(g) := v(g \circ f)$$

for $v \in T_pM$ and $g \in C^\infty(N)$.

Definition 0.5.7: Tangent Bundle

The *tangent bundle* of M is the disjoint union

$$TM := \bigsqcup_{p \in M} T_pM = \{(p, v) : p \in M,\ v \in T_pM\}$$

equipped with the natural projection $\pi : TM \to M$, $(p, v) \mapsto p$. This is a smooth manifold of dimension $2n$ and a vector bundle of rank n over M.

Definition 0.5.8: Vector Field

A *vector field* on M is a smooth section of the tangent bundle, i.e., a smooth map $X : M \to TM$ such that $\pi \circ X = \mathrm{id}_M$. Equivalently, X assigns to each $p \in M$ a tangent vector $X_p \in T_pM$, varying smoothly with p.
The space of vector fields is denoted $\mathfrak{X}(M)$ or $\Gamma(TM)$.

Proposition 0.5.9: Vector Fields as Derivations

A vector field $X \in \mathfrak{X}(M)$ defines a derivation $X : C^\infty(M) \to C^\infty(M)$ by $(Xf)(p) := X_p(f)$. This gives an isomorphism between $\mathfrak{X}(M)$ and the $C^\infty(M)$-module of derivations of $C^\infty(M)$.

Definition 0.5.10: Lie Bracket

The *Lie bracket* of vector fields $X, Y \in \mathfrak{X}(M)$ is the vector field $[X, Y]$ defined by

$$[X, Y](f) := X(Y(f)) - Y(X(f))$$

This makes $\mathfrak{X}(M)$ into a Lie algebra over $\mathbb{R}$.

Cotangent Spaces and Differential Forms

Definition 0.5.11: Cotangent Space

The *cotangent space* at $p \in M$ is the dual vector space

$$T_p^*M := (T_pM)^* = \mathrm{Hom}_{\mathbb{R}}(T_pM, \mathbb{R})$$

Elements of T_p^*M are called *cotangent vectors* or *covectors*.

Definition 0.5.12: Differential of a Function

For $f \in C^\infty(M)$ and $p \in M$, the *differential* $df_p \in T_p^*M$ is defined by

$$df_p(v) := v(f) \quad \text{for } v \in T_pM$$

In local coordinates, $df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$, where $\{dx^1, \ldots, dx^n\}$ is the dual basis to $\{\partial/\partial x^1, \ldots, \partial/\partial x^n\}$.

Definition 0.5.13: Cotangent Bundle

The *cotangent bundle* is $T^*M := \bigsqcup_{p \in M} T_p^*M$, a vector bundle of rank n over M.

Definition 0.5.14: Differential k-Forms

A *differential k-form* (or simply *k-form*) on M is a smooth section of $\bigwedge^k T^*M$. That is, a k-form ω assigns to each $p \in M$ an alternating k-linear map

$$\omega_p : \underbrace{T_pM \times \cdots \times T_pM}_{k \text{ times}} \to \mathbb{R}$$

varying smoothly with p.
The space of k-forms is denoted $\Omega^k(M)$. We set $\Omega^0(M) = C^\infty(M)$.

Example 0.5.15: Differential Forms in Coordinates

In local coordinates $(x^1, \dots, x^n)$, every k-form can be written as

$$\omega = \sum_{i_1 < \cdots < i_k} f_{i_1 \dots i_k} \, dx^{i_1} \wedge \cdots \wedge dx^{i_k}$$

where $f_{i_1 \dots i_k} \in C^\infty(M)$ and $\wedge$ denotes the wedge product.

Definition 0.5.16: Wedge Product

The *wedge product* $\wedge : \Omega^k(M) \times \Omega^\ell(M) \to \Omega^{k+\ell}(M)$ is the unique bilinear, associative operation satisfying:

1. $df \wedge dg = -dg \wedge df$ for $f, g \in C^\infty(M)$.
2. $(f\omega) \wedge \eta = f(\omega \wedge \eta) = \omega \wedge (f\eta)$ for $f \in C^\infty(M)$.
3. $\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega$ for $\omega \in \Omega^k$, $\eta \in \Omega^\ell$.

This makes $\Omega^*(M) := \bigoplus_{k \geq 0} \Omega^k(M)$ into a graded-commutative algebra.

Definition 0.5.17: Pullback of Forms

Let $f : M \to N$ be smooth. The *pullback* $f^* : \Omega^k(N) \to \Omega^k(M)$ is defined by

$$(f^*\omega)_p(v_1, \dots, v_k) := \omega_{f(p)}(df_p(v_1), \dots, df_p(v_k))$$

The pullback satisfies $f^*(\omega \wedge \eta) = f^*\omega \wedge f^*\eta$ and $(g \circ f)^* = f^* \circ g^*$.

Exterior Derivative and de Rham Cohomology

Definition 0.5.18: Exterior Derivative

The *exterior derivative* is the unique family of $\mathbb{R}$-linear maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying:

1. On $\Omega^0(M) = C^\infty(M)$, d is the differential: $(df)(X) = X(f)$.
2. $d \circ d = 0$.
3. $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for $\omega \in \Omega^k$ (graded Leibniz rule).

Example 0.5.19: Exterior Derivative in Coordinates

For $\omega = \sum_I f_I \, dx^I$ (where I is a multi-index), we have

$$d\omega = \sum_I \sum_{j=1}^n \frac{\partial f_I}{\partial x^j} dx^j \wedge dx^I$$

For instance, if $\omega = f\,dx + g\,dy$ on $\mathbb{R}^2$, then

$$d\omega = \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right) dx \wedge dy$$

Proposition 0.5.20: Naturality of d

The exterior derivative commutes with pullback: for $f : M \to N$ smooth,

$$f^* \circ d = d \circ f^*$$

That is, $f^*(d\omega) = d(f^*\omega)$ for all $\omega \in \Omega^*(N)$.

Definition 0.5.21: de Rham Complex

The *de Rham complex* of M is the cochain complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \to 0$$

A form ω is *closed* if $d\omega = 0$, and *exact* if $\omega = d\eta$ for some η. Since $d^2 = 0$, every exact form is closed.

Definition 0.5.22: de Rham Cohomology

The *k-th de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) := \frac{\ker(d : \Omega^k \to \Omega^{k+1})}{\mathrm{im}(d : \Omega^{k-1} \to \Omega^k)} = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}}$$

This is a real vector space measuring the "failure of closed forms to be exact."

Theorem 0.5.23: de Rham's Theorem

For a smooth manifold M, there is a natural isomorphism

$$H^k_{\mathrm{dR}}(M) \cong H^k(M; \mathbb{R})$$

where $H^k(M; \mathbb{R})$ is the singular cohomology of M with real coefficients. The isomorphism is given by integration of forms over cycles.

***Proof*:**
See [5]

Example 0.5.24: de Rham Cohomology Examples

1. $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for $k > 0$, and $H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$ (Poincaré lemma).
2. $H^0_{\mathrm{dR}}(M) = \mathbb{R}^{\pi_0(M)}$, where $\pi_0(M)$ is the number of connected components.

3. $H^1_{\mathrm{dR}}(S^1) = \mathbb{R}$, generated by $[d\theta]$.
4. $H^k_{\mathrm{dR}}(S^n) = \mathbb{R}$ for $k = 0, n$, and 0 otherwise.

0.5.25 Algebraic de Rham Cohomology For a smooth algebraic variety X over a field k of characteristic zero, one can define *algebraic de Rham cohomology* using the complex of algebraic differential forms $\Omega^*_{X/k}$. Over $\mathbb{C}$, this agrees with singular cohomology by Grothendieck's algebraic de Rham theorem.

Vector Bundles

Definition 0.5.26: Vector Bundle

A *(real) vector bundle of rank* r over a smooth manifold M is a smooth manifold E together with:

1. A smooth surjection $\pi : E \to M$.
2. For each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ has the structure of an r-dimensional real vector space.
3. Local triviality: each $p \in M$ has a neighborhood U with a diffeomorphism $\varphi : \pi^{-1}(U) \xrightarrow{\sim} U \times \mathbb{R}^r$ such that φ is linear on each fiber.

Example 0.5.27: Vector Bundle Examples

1. The *trivial bundle* $M \times \mathbb{R}^r \to M$.
2. The tangent bundle TM and cotangent bundle T^*M.
3. The *tautological line bundle* $\mathcal{O}(-1)$ over $\mathbb{RP}^n$ or $\mathbb{CP}^n$: the fiber over a line ℓ is ℓ itself.
4. Exterior powers $\bigwedge^k T^*M$ (whose sections are k-forms).
5. The *canonical bundle* $K_M := \bigwedge^n T^*M$ (top exterior power).

Definition 0.5.28: Section of a Vector Bundle

A *section* of $\pi : E \to M$ is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$. The space of sections is denoted $\Gamma(E)$ or $\Gamma(M, E)$. This is a $C^\infty(M)$-module.

Definition 0.5.29: Operations on Vector Bundles

Given vector bundles E, F over M:

- *Direct sum*: $(E \oplus F)_p = E_p \oplus F_p$.
- *Tensor product*: $(E \otimes F)_p = E_p \otimes_{\mathbb{R}} F_p$.
- *Dual*: $E_p^* = (E_p)^*$.
- *Exterior powers*: $(\bigwedge^k E)_p = \bigwedge^k E_p$.
- *Pullback*: for $f : N \to M$, $(f^*E)_q = E_{f(q)}$.

Definition 0.5.30: Transition Functions

If $\{U_\alpha\}$ is an open cover of M with local trivializations $\varphi_\alpha : E|_{U_\alpha} \xrightarrow{\sim} U_\alpha \times \mathbb{R}^r$, the *transition functions* are

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}_r(\mathbb{R}), \qquad \varphi_\alpha \circ \varphi_\beta^{-1}(p, v) = (p, g_{\alpha\beta}(p) \cdot v)$$

These satisfy the cocycle condition: $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps.

0.5.31 Line Bundles and Divisors A *line bundle* is a vector bundle of rank 1. Line bundles are classified by $H^1(M; C^\infty(M, \mathbb{R}^\times))$ via transition functions. In algebraic geometry, line bundles on a variety X correspond to Cartier divisors and form the Picard group $\mathrm{Pic}(X)$.

Connections and Curvature

Definition 0.5.32: Connection on a Vector Bundle

A *connection* (or *covariant derivative*) on a vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E) = \Omega^1(M, E)$$

satisfying the Leibniz rule: for $f \in C^\infty(M)$ and $s \in \Gamma(E)$,

$$\nabla(fs) = df \otimes s + f \nabla s$$

For a vector field X, we write $\nabla_X s := \iota_X(\nabla s) \in \Gamma(E)$ for the covariant derivative of s in direction X.

0.5.33 Connections Exist Every vector bundle over a paracompact manifold admits a connection. Connections form an affine space over $\Omega^1(M, \mathrm{End}(E))$: if ∇ is a connection and $A \in \Omega^1(M, \mathrm{End}(E))$, then $\nabla + A$ is also a connection.

Definition 0.5.34: Curvature

The *curvature* of a connection ∇ is the $\mathrm{End}(E)$-valued 2-form $F_\nabla \in \Omega^2(M, \mathrm{End}(E))$ defined by

$$F_\nabla(X, Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}$$

for vector fields X, Y. A connection is *flat* if $F_\nabla = 0$.

Proposition 0.5.35: Curvature as Obstruction

A connection ∇ is flat if and only if parallel transport is path-independent (depends only on the homotopy class of the path). Flat connections on a trivial bundle correspond to representations $\pi_1(M) \to \mathrm{GL}_r(\mathbb{R})$.

Riemannian Metrics

Definition 0.5.36: Riemannian Metric

A *Riemannian metric* on M is a smooth assignment of an inner product $g_p : T_pM \times T_pM \to \mathbb{R}$ to each $p \in M$. In local coordinates, $g = \sum_{i,j} g_{ij}\, dx^i \otimes dx^j$ where $(g_{ij}(p))$ is a positive definite symmetric matrix for each p.
A *Riemannian manifold* is a pair (M, g).

Example 0.5.37: Riemannian Metrics

1. The standard metric on $\mathbb{R}^n$: $g = dx^1 \otimes dx^1 + \cdots + dx^n \otimes dx^n$.
2. The round metric on S^n (induced from $\mathbb{R}^{n+1}$).
3. The hyperbolic metric on the upper half-plane: $g = \frac{dx^2+dy^2}{y^2}$.
4. The Fubini–Study metric on $\mathbb{CP}^n$.

Proposition 0.5.38: Existence of Riemannian Metrics

Every smooth manifold admits a Riemannian metric. (This uses partitions of unity.)

Definition 0.5.39: Levi-Civita Connection

On a Riemannian manifold (M, g), there is a unique connection ∇ on TM that is:

1. *Metric-compatible*: $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$.
2. *Torsion-free*: $\nabla_X Y - \nabla_Y X = [X, Y]$.

This is called the *Levi-Civita connection*.

Definition 0.5.40: Geodesics

A *geodesic* is a curve $\gamma : I \to M$ satisfying $\nabla_{\dot\gamma}\dot\gamma = 0$. Geodesics are locally length-minimizing curves.

Integration and Orientation

Definition 0.5.41: Orientation

An *orientation* on an n-manifold M is a choice of smoothly varying orientation on each tangent space T_pM. Equivalently, it is a nowhere-vanishing n-form $\omega \in \Omega^n(M)$, or a trivialization of the canonical bundle $K_M = \bigwedge^n T^*M$.
A manifold is *orientable* if it admits an orientation, and *oriented* if one has been chosen.

Example 0.5.42: Orientability

1. $\mathbb{R}^n$, S^n, and $\mathbb{CP}^n$ are orientable.
2. $\mathbb{RP}^n$ is orientable if and only if n is odd.
3. The Möbius strip is not orientable.

Definition 0.5.43: Integration of Top Forms

Let M be an oriented n-manifold and $\omega \in \Omega^n_c(M)$ a compactly supported n-form. The *integral* $\int_M \omega$ is defined by:

1. In a single oriented chart (U, φ) with $\mathrm{supp}(\omega) \subseteq U$: write $\omega = f\, dx^1 \wedge \cdots \wedge dx^n$ and set
$$\int_M \omega := \int_{\varphi(U)} f(\varphi^{-1}(x))\, dx^1 \cdots dx^n$$
2. For general ω, use a partition of unity to reduce to the local case.

Theorem 0.5.44: Stokes' Theorem

Let M be an oriented n-manifold with boundary ∂M (given the induced orientation), and let $\omega \in \Omega^{n-1}_c(M)$. Then
$$\int_M d\omega = \int_{\partial M} \omega$$
In particular, if M is closed (compact without boundary), then $\int_M d\omega = 0$.

***Proof*:**
See [5]

0.5.45 Classical Theorems as Special Cases Stokes' theorem generalizes:

- The fundamental theorem of calculus ($n = 1$).
- Green's theorem ($n = 2$, planar region).
- The classical Stokes theorem ($n = 3$, surface in $\mathbb{R}^3$).
- The divergence theorem ($n = 3$, solid region in $\mathbb{R}^3$).

Partitions of Unity

Definition 0.5.46: Partition of Unity

Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of M. A *partition of unity subordinate to* $\{U_\alpha\}$ is a collection of smooth functions $\{\rho_\alpha : M \to [0, 1]\}_{\alpha \in A}$ such that:

1. $\text{supp}(\rho_\alpha) \subseteq U_\alpha$ for each α.
2. The collection $\{\text{supp}(\rho_\alpha)\}$ is locally finite.
3. $\sum_\alpha \rho_\alpha(p) = 1$ for all $p \in M$.

Theorem 0.5.47: Existence of Partitions of Unity

Every open cover of a smooth manifold admits a subordinate partition of unity.

***Proof*:**
See [5].

0.5.48 Partitions of Unity in Algebraic Geometry Partitions of unity are a crucial tool in differential geometry, enabling:

- Existence of Riemannian metrics.
- Existence of connections.
- Extension of local constructions to global ones.
- Proof that smooth manifolds are paracompact.

Importantly, partitions of unity do not exist in algebraic geometry. Polynomial or algebraic functions cannot be "bump functions." This is one reason why sheaf cohomology becomes essential in algebraic geometry: it replaces the role of partitions of unity in extending local to global.

0.5.1 Summary: Differential vs. Algebraic Geometry

Concept	**Differential Geometry**	**Algebraic Geometry**
Local model	$\mathbb{R}^n$ or $\mathbb{C}^n$	$\mathrm{Spec}\,(A)$
Functions	$C^\infty(M)$	$\mathcal{O}_X(X)$ (regular functions)
Tangent space	T_pM (derivations)	$(\mathfrak{m}_p/\mathfrak{m}_p^2)^\vee$
Cotangent space	T_p^*M	$\mathfrak{m}_p/\mathfrak{m}_p^2$
Vector bundle	Smooth fiber bundle	Locally free sheaf
Differential forms	$\Omega^k(M)$	$\Omega^k_{X/k}$ (Kähler differentials)
Cohomology	de Rham $H^*_{\mathrm{dR}}(M)$	Sheaf cohomology $H^*(X,\mathcal{F})$
Partitions of unity	Exist	Do not exist
Gluing	Smooth	Via sheaf axioms

The absence of partitions of unity in the algebraic world makes sheaf-theoretic methods indispensable. Many constructions that are "automatic" in differential geometry (e.g., existence of metrics) require careful sheaf-cohomological arguments in algebraic geometry.

0.6 *Key Results from Complex Geometry

Complex geometry sits at the intersection of differential geometry and algebraic geometry. Over $\mathbb{C}$, algebraic varieties carry both a Zariski topology (algebraic) and a classical topology (analytic), and Serre's GAGA theorems establish deep connections between these perspectives. This section reviews the essential concepts from complex analytic geometry that inform and enrich algebraic geometry over $\mathbb{C}$.

Complex Manifolds

Definition 0.6.1: Complex Manifold

A *complex manifold* of dimension n is a smooth manifold M of real dimension $2n$ equipped with an atlas $\{(U_\alpha,\varphi_\alpha)\}$ where each $\varphi_\alpha : U_\alpha \xrightarrow{\sim} \varphi_\alpha(U_\alpha) \subseteq \mathbb{C}^n$ is a homeomorphism onto an open subset of $\mathbb{C}^n$, and all transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

are *holomorphic* (i.e., complex analytic).

Definition 0.6.2: Holomorphic Function

A function $f : U \to \mathbb{C}$ on an open set $U \subseteq \mathbb{C}^n$ is *holomorphic* if it satisfies the Cauchy–Riemann equations in each variable, or equivalently, if it is complex differentiable at every point. In coordinates $z_j = x_j + iy_j$, this means

$$\frac{\partial f}{\partial \bar{z}_j} = 0 \quad \text{for all } j$$

where $\frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.3: Holomorphic Map

A map $f : M \to N$ between complex manifolds is *holomorphic* if it is holomorphic in local coordinates. A *biholomorphism* is a holomorphic bijection with holomorphic inverse.

Example 0.6.4: Complex Manifolds

1. $\mathbb{C}^n$ with the standard complex structure.
2. Open subsets of $\mathbb{C}^n$ (or of any complex manifold).
3. Complex projective space $\mathbb{CP}^n$: covered by $n+1$ charts $U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\} \cong \mathbb{C}^n$.
4. Smooth projective varieties over $\mathbb{C}$ (with the classical topology).
5. Complex tori $\mathbb{C}^n/\Lambda$ where $\Lambda \cong \mathbb{Z}^{2n}$ is a lattice.
6. Riemann surfaces: complex manifolds of dimension 1.

0.6.5 Smooth vs. Complex vs. Algebraic We have strict inclusions of categories:

$$\{\text{smooth projective varieties}/\mathbb{C}\} \subsetneq \{\text{compact Kähler manifolds}\} \subsetneq \{\text{compact complex manifolds}\}$$

Not every complex manifold is algebraic (e.g., generic complex tori of dimension ≥ 2), and not every compact complex manifold is Kähler (e.g., most Hopf surfaces).

Almost Complex and Complex Structures

Definition 0.6.6: Almost Complex Structure

An *almost complex structure* on a smooth manifold M is a smooth bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. This makes each tangent space T_pM into a complex vector space via $(a + bi) \cdot v := av + bJ(v)$.

Proposition 0.6.7: Complex Manifolds Have Almost Complex Structures

Every complex manifold has a natural almost complex structure: in holomorphic coordinates $(z_1, \ldots, z_n)$ with $z_j = x_j + iy_j$,

$$J\left(\frac{\partial}{\partial x_j}\right) = \frac{\partial}{\partial y_j}, \qquad J\left(\frac{\partial}{\partial y_j}\right) = -\frac{\partial}{\partial x_j}$$

The converse is not automatic:

Definition 0.6.8: Integrable Almost Complex Structure

An almost complex structure J is *integrable* if it arises from a complex manifold structure. The *Nijenhuis tensor* N_J measures the obstruction to integrability.

Theorem 0.6.9: Newlander–Nirenberg Theorem

An almost complex structure J is integrable if and only if the Nijenhuis tensor vanishes: $N_J = 0$.

***Proof*:**
see [7]

Complexified Tangent Bundle and (p, q)-Forms

Definition 0.6.10: Complexified Tangent Bundle

For a complex manifold M, the *complexified tangent bundle* is $T_{\mathbb{C}}M := TM \otimes_{\mathbb{R}} \mathbb{C}$. The almost complex structure J extends $\mathbb{C}$-linearly and has eigenvalues $\pm i$. This gives a decomposition

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M$$

where $T^{1,0}M$ is the $+i$ eigenspace and $T^{0,1}M$ is the $-i$ eigenspace.
In local coordinates, $T^{1,0}M$ is spanned by $\frac{\partial}{\partial z_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right)$ and $T^{0,1}M$ by $\frac{\partial}{\partial \bar{z}_j} := \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right)$.

Definition 0.6.11: (p, q)-Forms

The complexified cotangent bundle similarly decomposes: $T^*_{\mathbb{C}}M = (T^{1,0}M)^* \oplus (T^{0,1}M)^*$. A *differential form of type* (p, q) is a section of

$$\bigwedge^{p,q} T^*_{\mathbb{C}}M := \bigwedge^{p}(T^{1,0}M)^* \otimes \bigwedge^{q}(T^{0,1}M)^*$$

The space of (p, q)-forms is denoted $\Omega^{p,q}(M)$ or $\mathcal{A}^{p,q}(M)$.

Proposition 0.6.12: Decomposition of Forms

The space of complex k-forms decomposes as

$$\Omega^k(M, \mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M)$$

In local coordinates, a (p, q)-form can be written as

$$\omega = \sum_{|I|=p,|J|=q} f_{I,J}\, dz^I \wedge d\bar{z}^J$$

where $dz^I = dz_{i_1} \wedge \cdots \wedge dz_{i_p}$ and $d\bar{z}^J = d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$.

The Dolbeault Operators

Definition 0.6.13: Dolbeault Operators

The exterior derivative $d : \Omega^k(M, \mathbb{C}) \to \Omega^{k+1}(M, \mathbb{C})$ decomposes as $d = \partial + \bar{\partial}$ where:

$$\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M)$$
$$\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M)$$

These satisfy $\partial^2 = 0$, $\bar{\partial}^2 = 0$, and $\partial\bar{\partial} + \bar{\partial}\partial = 0$.

Example 0.6.14: Dolbeault Operators in Coordinates

For $f \in C^\infty(M, \mathbb{C}) = \Omega^{0,0}(M)$:

$$\partial f = \sum_{j=1}^{n} \frac{\partial f}{\partial z_j} dz_j, \qquad \bar{\partial} f = \sum_{j=1}^{n} \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j$$

A function is holomorphic if and only if $\bar{\partial} f = 0$.

Definition 0.6.15: Dolbeault Complex

For each p, the *Dolbeault complex* is

$$0 \to \Omega^{p,0}(M) \xrightarrow{\bar{\partial}} \Omega^{p,1}(M) \xrightarrow{\bar{\partial}} \Omega^{p,2}(M) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \Omega^{p,n}(M) \to 0$$

Definition 0.6.16: Dolbeault Cohomology

The *Dolbeault cohomology groups* are

$$H^{p,q}_{\bar{\partial}}(M) := \frac{\ker(\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\bar{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

These are complex vector spaces measuring the failure of the $\bar{\partial}$-Poincaré lemma.

Theorem 0.6.17: Dolbeault's Theorem

For a complex manifold M, there is a natural isomorphism

$$H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p_M)$$

where Ω^p_M is the sheaf of holomorphic p-forms.

***Proof*:**
See [7]

Hermitian and Kähler Metrics

Definition 0.6.18: Hermitian Metric

A *Hermitian metric* on a complex manifold M is a smoothly varying Hermitian inner product h_p on each $T^{1,0}_p M$. Equivalently, it is a Riemannian metric g such that $g(JX, JY) = g(X, Y)$ for all tangent vectors X, Y.
In local coordinates, $h = \sum_{j,k} h_{j\bar{k}}\, dz_j \otimes d\bar{z}_k$ with $(h_{j\bar{k}})$ a positive definite Hermitian matrix.

Definition 0.6.19: Fundamental Form

The *fundamental form* (or *associated* $(1,1)$*-form*) of a Hermitian metric h is

$$\omega := -\operatorname{Im}(h) = \frac{i}{2} \sum_{j,k} h_{j\bar{k}}\, dz_j \wedge d\bar{z}_k$$

This is a real $(1,1)$-form satisfying $\omega(X, JX) > 0$ for nonzero X.

Definition 0.6.20: Kähler Manifold

A Hermitian metric h is *Kähler* if its fundamental form ω is closed: $d\omega = 0$. A *Kähler manifold* is a complex manifold equipped with a Kähler metric. The form ω is then called the *Kähler form*.

Proposition 0.6.21: Equivalent Characterizations of Kähler

The following are equivalent for a Hermitian manifold (M, h):

1. $d\omega = 0$ (the Kähler condition).
2. The Levi-Civita connection ∇ of $g = \mathrm{Re}(h)$ satisfies $\nabla J = 0$.
3. Locally, $\omega = i\partial\bar{\partial}\varphi$ for some real function φ (a *Kähler potential*).
4. The holonomy of g is contained in $\mathrm{U}(n) \subset \mathrm{SO}(2n)$.

Proof:
See [7].

Example 0.6.22: Kähler Manifolds

1. $\mathbb{C}^n$ with the standard metric $\omega = \frac{i}{2}\sum_j dz_j \wedge d\bar{z}_j$.
2. $\mathbb{CP}^n$ with the *Fubini–Study metric*:

$$\omega_{\mathrm{FS}} = \frac{i}{2\pi}\partial\bar{\partial}\log\left(|z_0|^2 + \cdots + |z_n|^2\right)$$

3. Any smooth projective variety over $\mathbb{C}$ (inherits Kähler structure from $\mathbb{CP}^n$).
4. Complex submanifolds of Kähler manifolds are Kähler.
5. Complex tori $\mathbb{C}^n/\Lambda$ with the flat metric.

Example 0.6.23: Non-Kähler Manifolds

1. The *Hopf surface* $(\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z\rangle \cong S^1 \times S^3$ is a compact complex surface with $b_2 = 0$, hence not Kähler (Kähler manifolds have $b_2 \geq 1$ by the existence of $[\omega]$).
2. Most compact complex manifolds are not Kähler.

Hodge Theory

Hodge theory provides a bridge between the topology, complex structure, and Riemannian geometry of a Kähler manifold.

Definition 0.6.24: Hodge Star and Laplacian

On a Riemannian manifold (M, g) of dimension n, the *Hodge star* $* : \Omega^k(M) \to \Omega^{n-k}(M)$ is defined by

$$\alpha \wedge *\beta = g(\alpha, \beta) \operatorname{vol}_g$$

The *Hodge Laplacian* is $\Delta := dd^* + d^*d$ where $d^* := (-1)^{n(k+1)+1} * d*$ is the formal adjoint of d.
A form is *harmonic* if $\Delta\omega = 0$.

Theorem 0.6.25: Hodge Theorem (Riemannian)

On a compact oriented Riemannian manifold M:

1. Every de Rham cohomology class contains a unique harmonic representative.
2. $H^k_{\mathrm{dR}}(M) \cong \mathcal{H}^k(M) := \{\omega \in \Omega^k(M) : \Delta\omega = 0\}$.

***Proof*:**
See [7]

On a Kähler manifold, there are additional Laplacians:

Definition 0.6.26: Dolbeault Laplacians

On a Hermitian manifold, define

$$\Delta_\partial := \partial\partial^* + \partial^*\partial, \qquad \Delta_{\bar\partial} := \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$$

Theorem 0.6.27: Kähler Identities

On a Kähler manifold:

$$\Delta = 2\Delta_\partial = 2\Delta_{\bar\partial}$$

In particular, a form is harmonic if and only if it is ∂-harmonic if and only if it is $\bar\partial$-harmonic.

***Proof*:**
See [7]

Theorem 0.6.28: Hodge Decomposition

On a compact Kähler manifold M:

1. There is a decomposition
$$H^k(M,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(M)$$
where $H^{p,q}(M) \cong H^{p,q}_{\bar\partial}(M) \cong H^q(M,\Omega^p_M)$.
2. Complex conjugation gives $\overline{H^{p,q}(M)} = H^{q,p}(M)$.
3. The decomposition is independent of the choice of Kähler metric.

***Proof*:**
See [7]

Definition 0.6.29: Hodge Numbers

The *Hodge numbers* of a compact Kähler manifold are
$$h^{p,q}(M) := \dim_{\mathbb{C}} H^{p,q}(M)$$
These satisfy:

1. $h^{p,q} = h^{q,p}$ (complex conjugation).
2. $h^{p,q} = h^{n-p,n-q}$ (Serre duality).
3. $b_k = \sum_{p+q=k} h^{p,q}$ (Betti numbers).

The Hodge numbers are often displayed in the *Hodge diamond*.

Example 0.6.30: Hodge Diamond of $\mathbb{CP}^n$
For $\mathbb{CP}^n$, the only nonzero Hodge numbers are $h^{p,p} = 1$ for $0 \le p \le n$:
$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & \cdots & & \cdots & \end{array}$$
This reflects $H^{2k}(\mathbb{CP}^n,\mathbb{C}) = \mathbb{C}$ and $H^{2k+1}(\mathbb{CP}^n,\mathbb{C}) = 0$.

Example 0.6.31: Hodge Diamond of a Curve
For a smooth projective curve C of genus g:

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Here $h^{1,0} = h^{0,1} = g = \dim H^0(C, \Omega^1_C) = \dim H^1(C, \mathcal{O}_C)$.

0.6.1 Summary: Complex Geometry and Algebraic Geometry

This table has more than what was covered for the purpose of reference:

Concept	Complex Analytic	Algebraic
Space	Complex manifold	Smooth variety / scheme
Functions	Holomorphic	Regular (polynomial)
Topology	Classical (Euclidean)	Zariski
Local model	$\mathbb{C}^n$	$\operatorname{Spec}(A)$
Differential forms	(p,q)-forms, $\bar{\partial}$	Kähler differentials $\Omega^p_{X/k}$
Cohomology	Dolbeault $H^{p,q}_{\bar{\partial}}$	Sheaf cohomology $H^q(X, \Omega^p)$
Vector bundles	Holomorphic bundles	Locally free sheaves
Divisors	Analytic hypersurfaces	Weil/Cartier divisors
Line bundles	$\operatorname{Pic}(M)$	$\operatorname{Pic}(X)$
Metrics	Hermitian, Kähler	(No direct analogue)
Key theorem	Hodge decomposition	Serre duality, GAGA

0.6.32 Foreshadowing GAGA For projective varieties over $\mathbb{C}$, one can freely pass between algebraic and analytic viewpoints via GAGA:

- **Algebraic methods**: Work over any field, use scheme theory, étale cohomology.
- **Analytic methods**: Use Hodge theory, L^2-methods, Kähler geometry, transcendental techniques.

Many deep results (e.g., Hodge conjecture, Torelli theorems) involve both perspectives essentially.

Part I

Scheme Theory

I believe there will be more than just basic alg geo. This part will go over the core theory, Hartshorne, Vakil, Eisenbud, and so forth. Other parts will surely be specializations.

1

Sheaves

Roughly speaking, Sheaf Theory is the theory of augmenting a topological space[1] that adds information to each open set (adds "local" information) which can be "glued" to get information in larger open sets, and can be "restricted" to get information on smaller sets. The prototypical example of a sheaf is the set of smooth functions on a manifold. Smooth functions on a manifold can be restricted to open subsets of a manifold which are homeomorphic to $U \subseteq \mathbb{R}^n$ and if a smooth function on each part of the manifold is defined in a compatible way (i.e. they agree on intersections), they can be glued back together to form a function on the entire manifold. As an open covering of M can always be chosen so that each U is simply connected connected, the non-trivial information is in the ability or choice of gluing of functions.

A key motivator behind sheaves is that they can hold the "geometric" information of a space. The reader may recall that a vector-space and subspaces can be characterized via their dual spaces, that is all functions from the vector space to $\mathbb{F}$. A similar result holds for Varieties with the coordinate ring uniquely characterizing a Variety. This duality was found to exist everywhere in mathematics, to name three more examples:

- A classical example that started the duality between algebra and geometry is that of $C(X)$ and X where X is a compact space and $C(X)$ all continuous function on X. By taking the stone-Cech compactification of a space X, the space can be achieved for any space (see [10, p.144] for details)

- Similarly, a radon measure's can be characterized using the duals of $C_c(X)$ where X is a locally compact space (classical result can be found in EYTNKA at [20])

- Spin manifolds can be characterized by spectral triples (see [2])

[1] more generally, a topoi, which concretely translates in our case to a category whose objects are open sets and morphisms are inclusions

- Sets with set-function can be thought of as the most "abstract" geometric object (they only have information about containment). There is an appropriate dual called the *complete atomic Boolean algebras*. The reader may be interested in knowing that Boolean algebra's characterize logical operations, which gives an interesting geometric interpretation to logic[2].

There is a long list of such examples (the reader may be curious to checkout this link). This gave rise to the idea that a geometric space can be understood by studying some subset of function on this space with certain. In algebraic cases, there are relatively few functions that are defined globally on a space, but many that are defined on local components (the prototypical example of this would be $\mathbb{C}$[3]).[4]

Why do functions capture this geometric information? One can think of functions as tools for representing relations. If our function is of the form $f : X \to \mathbb{R}$, then this is relating X to numbers, i.e. quantification, usually X being a topological space and f being continuous for some notion of closeness. If we have a continuous function of the form $f : X \to \mathbb{R}^n$ (or $\mathbb{C}^n$), with X a topological space, then we may pullback a lot of geometric results from $\mathbb{R}^n$ (or $\mathbb{C}^n$) onto X, depending on how f is defined (ex. it can be shown that f is a diffeomorphism if and only if it pulls back the differential structure of $\mathbb{R}^n$). Very often, it is not possible to define a diffeomorphism $f : X \to \mathbb{R}^n$, however we may define it on components. Functions have the property that we may restrict them or construct them by defining what happening at each point (or in the continuous case, on each open set where they agree on intersections[5]). But you may say that morphisms in a category already capture this intuition. This is indeed the case, what makes functions special is that you can construct larger morphisms from smaller morphisms with an appropriate notion of *gluing* and *restricting*. In particular, the closeness given by the notion of topological spaces and the locality properties inspired by functions shall be the two main properties of sheaves. As is the course of mathematics, it may be asked if these properties of functions can be generalized, and we may ask if we can take special categories which have these features; this is called a *site*. For an introduction to algebraic geometry, limiting to the sites defined on topological spaces is plenty sufficient and shall be the focus of this book.

A particularly important realization will be that commutative rings will in many ways behave like functions on a space, that is if R is a ring, there is a sense in which the elements of R will be the "functions" on a special space (namely, the space $\operatorname{Spec}(R)$). This shall be elaborated on in chapter 2 and was commented on in the front-matter. At this stage it is motivating to know that there is a reason why we may care about sheaves other than the sheaf of functions, namely the interpolation between commutative rings and functions.

Motivating Example from Vakil

Before exploring sheaves in generality, it is useful to have the prototypical example of a sheaf in mind in more detail:

[2] This connection is further refined using Topoi

[3] entire holomorphic are necessarily power series, while holomorphic on (not necessarily simply connected) open sets are analytic and may even have non-trivial Laurent expansion around holes

[4] see this link for more details

[5] by the gluing lemma

Example 1.0.1: Sheaf Of Differentiable Functions On Manifold

Let M be a smooth manifold. Then the we shall constuct the sheaf of differentiable function on M. For each open subset $U \subseteq M$, we shall assocaite to it the set of smooth function $C^\infty(U)$. Denote this collection by $\mathcal{O}(U)$ (this will be common notation for sheaves). Then if we have an open subset of U, $V \subseteq U$, we may restrict the function from U to function on V, namely we have a map $\mathrm{res}_{U,V} : \mathcal{O}(U) \to \mathcal{O}(V)$ mapping $f \mapsto f|_V$. Naturally, if we have $W \hookrightarrow V \hookrightarrow U$, then the following commutative diagram.

$$\begin{tikzcd}
 & V \arrow[dr, "\mathrm{res}_{V,W}"] & \\
U \arrow[ur, "\mathrm{res}_{U,V}"] \arrow[rr, "\mathrm{res}_{U,W}"'] & & W
\end{tikzcd}$$

This collection of maps satisfy two important properties. In the following let and $U = \bigcup_i U_i$.

1. *Defining Locally*: Let's say $f_1, f_2 \in \mathcal{O}(U)$, Then if f_1 and f_2 agree on all restriction, that is $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_j} f_2$, then $f_1 = f_2$
2. *Gluing*: If there exists a collection $f_i \in \mathcal{O}(U_i)$ such that f_i and f_j agree on intersections, that is $\mathrm{res}_{U_i, U_i \cap U_j} f_i = \mathrm{res}_{U_j, U_i \cap U_j} f_j$, then there is a $f \in \mathcal{O}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.

Then a sheaf will respect these properties. Note that in fact, we may define a manifold M to be $(M, \mathcal{O}_M)$ where M Is a topological space and $\mathcal{O}_M$ is a sheaf of smooth functions on M. If instead $\mathcal{O}(U)$ was the collection of continuous functions, we would get a topological manifold. Even more generally, we may let the functions be set functions, for set functions work well under restrictions and we may define set functions locally and construct set function via gluing.

Note that Diffeology can be thought of as a generalization of this result or of a special case of sheaf theory, and we shall explore it briefly at the end of this chapter.

So if even set functions can form a sheaf, then why did I write "nice" function earlier? As it turns out, what we shall do is abstract the notion of function! In particular, we shall look at systems where we may have the behavior of localizing and gluing "functions" on a topological space. Hence, in the following abstraction, we will consider $\mathcal{O}_M$ to be the data that stores a collection of objects that *behave like functions*.

A key property that distinguishes functions from general morphisms is the notion of evaluation (taking $f(x)$ for some element x and function f). Since we are abstracting the notion of function, we shall need a notion of *evaluation*. To motivate this generalization, let's come back to $(M, \mathcal{O}_M)$ and consider some $\mathcal{O}(U) = C^\infty(U)$ for some open $U \subseteq M$ and take $p \in U$. Recall that $\mathcal{O}(U)$ is a ring and that the *germ* at p is collection of equivalence relations:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \ \Leftrightarrow W \subseteq U, W \subseteq V, p \in W,\ f|_W = g|_W$$

that is, $[f] = [g]$ if and only if there is an open neighborhood $W \subseteq U, V$ containing p ($p \in W$) such that $f|_W = g|_W$ (in the language of sheaf theory, $\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} g$). Denote this collection $\mathcal{O}(U)_p$. The key is that the germ of $\mathcal{O}(U)$ at p is a *local ring* with maximal ideal $\mathfrak{m}_p$ of germs such that $[f](p) = 0$ (or simply $f(p) = 0$ if unambiguous). Then:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U)_p \xrightarrow{f \mapsto f(p)} \mathbb{R} \to 0$$

and all elements $\mathcal{O}(U) \setminus \mathfrak{m}_p$ are invertible. Hence, we shall see that we may thing of "evalutation" of f at some point $p \in U$ as taking first the germ $\mathcal{O}(U)_p$ and quotient it by the maximal ideal $\mathfrak{m}_p$, $\mathcal{O}(U)_p/\mathfrak{m}_p$, and then taking $\overline{f} \in \mathcal{O}(U)_p/\mathfrak{m}_p$ to be the value of f. Note that the above construction required that $\mathcal{O}(U)$ is a *local ring*. This is evidently the case when $\mathcal{O}(U) = C^\infty(U)$, however in general sheaf theory this ring will not in general be local. In fact, $\mathcal{O}(U)$ will not even need to be a ring! When each $\mathcal{O}(U)$ is a ring, then we shall say that X is a *ringed space*, and when all the germs are local rings, then we shall say that X is a *locally ringed space*.

A final word on generalization: we have been working over a topological space, however we may abstract further and work over a general category; in this way, we would define the notion of a "topology" on a category, known as a *Grothendeick topology* (a category with this structure is called a *site*), and develop sheaf theory accordingly. Though this is fruitful, this generlization shall be ommited for now to concentrate on the geoemtric properties at hand when using a topological space.

1.1 Definitions and Examples

We start by formalizing the above intuitions.

Definition 1.1.1: Presheaf Axiomatic Definition

Let X be a topological space. Then a *presheaf* over a category $\mathbf{C}$ on X, denoted $\mathscr{F}_X$ contains the following data:

1. for each open set $U \subseteq X$, there is an object c of $\mathbf{C}$ and denote it by $\mathscr{F}(U)$. If C is a category whose objects are sets with additioanl structure, then the elements of $\mathscr{F}(U)$ are called the *sections* of $\mathscr{F}$ over U. If we say that s is a section of $\mathscr{F}$, then we mean that s is a section of $\mathscr{F}$ over X.

2. For each inclusion map $U \hookrightarrow V$, there is a morphism $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ called the *restriction* morphism (or *restriction map*) such that

   (a) $\mathrm{res}_{U,U} = \mathrm{id}_U$

   (b) If $W \hookrightarrow V \hookrightarrow U$, then the following diagram commutes

$$\begin{tikzcd} & \mathscr{F}(V) \arrow[dr, "\mathrm{res}_{V,W}"] & \\ \mathscr{F}(U) \arrow[ur, "\mathrm{res}_{U,V}"] \arrow[rr, "\mathrm{res}_{U,W}"'] & & \mathscr{F}(W) \end{tikzcd}$$

The collection of all presheaves on a topological space X is denoted $\mathbf{PSh}_{\mathbf{C}}(X)$ (or $\mathbf{PSh}(X)$ if unambiguous). If U is clear from context, we shall often write $\mathrm{res}_{U,V} f$ as $f|_V$.

The nomenclature "section" will be made clear in proposition 1.1.13. When it is no ambiguous the image of the restriction morphism for $s \in \mathscr{F}(U)$ shall be denoted:

$$\mathrm{res}_{U,V}(s) = s|_V$$

This notation can be ambiguous as it is not explicit which $U \subseteq X$ is taken from which $s \in \mathscr{F}(U)$, however it will be unambiguous in-context.

When constructing sections of non-affine scheme's in chapter 2, it shall be important to have internalized the following consequence of the restriction maps. Let $U = U_1 \cup U_2$ and $W = U_1 \cap U_2$. Then $W \subseteq U_1 \subseteq U$ and $W \subseteq U_2 \subseteq U$. Then by definition of the restriction map, the following diagram commutes:

$$\begin{tikzcd} & \mathscr{F}(U_1) \arrow[dr, "\mathrm{res}_{U_1,W}"] & \\ F(U) \arrow[ur, "\mathrm{res}_{U,U_1}"] \arrow[rr, "\mathrm{res}_{U,W}"'] \arrow[dr, "\mathrm{res}_{U,U_2}"'] & & F(W) \\ & \mathscr{F}(U_2) \arrow[ur, "\mathrm{res}_{U_2,W}"'] & \end{tikzcd}$$

Importantly:

$$f|_W = (f|_{U_1})|_W = (f|_{U_2})|_W$$

The key take-away is that a larger section on U can be restricted to different sections on an open cover of U, but these sections map to the same restriction on intersections. In this way, presheaves preserve elements "going down". We shall see example of how "going up" fails after introducing some more basic terminology.

For practical computational purposes, it shall be important to re-interpret a presheaf in more categorical language, Notice that it is equivalent to define a presheaf as a *contravariant functor*. Namely

Definition 1.1.2: Presheaf Functorial Definition

Let X be a topological space and let $O(X)$ be the category whose objects are the open sets of X and whose morphism are $\subseteq$ (i.e. the category induced by the partial order given by $\subseteq$). Then a *presheaf* over $\mathbf{C}$ is a contravariant functor from $O(X)$ to $\mathbf{C}$, that is a functor $\mathscr{F} : O(X)^{op} \to \mathbf{C}$.

Usually, our presheaves shall be over **Set**, **Ab**, Mod_k or **CRing**[6].

An important notion for geometric objects is their behavior locally. The extreme (or limit) of the notation of locality is the behavior around a point p. This idea originated when studying germs of differential functions in Differential Geometry. These are well-defined on presheaves;

[6]Technically, **Ab** is $\mathbf{Mod}_{\mathbb{Z}}$, however distinguishing these two is useful as we usually mean k to be a field

Definition 1.1.3: Stalk as Equivalence Relation

let $(X, \mathcal{O}_X)$ be a presheaf. Then the *stalks* at $p \in X$ is the collection of equivalence class:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \quad \Leftrightarrow \quad W \subseteq U, W \subseteq V, p \in W,\ \operatorname{res}_{U,W} f = \operatorname{res}_{V,W} g$$

The stalks at p is usually denoted $\mathcal{O}_{X,p}$ or $\mathcal{O}_p$ if unambiguous in context.

If $\mathbf{C}$ is a category whose objects are augmented sets[a], then if $p \in U$ and $f \in \mathscr{F}(U)$, $[f] \in \mathscr{F}_p$ is called the *germ* of f at p.

[a]ex. **Set**, **Ring**, **Grp**, etc.

Note that in general, the notion of "evaluation" of a stalk/germ as it is usually defined on differential geometry won't make sense; though the symbol f is written the open sets need not contain functions or objects that act like function. We shall later define nice presheaves in which we may recover the notion of evaluation and treat the elements like functions. There is again a categorical way of interpreting stalks that is useful to us:

Definition 1.1.4: Stalks As Colimits

Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ be a presheaf where $\mathbf{C}$ is a category with small filtered colimits. Then a *stalk* at $p \in X$ is:

$$\mathscr{F}_p := \varinjlim \mathscr{F}(U)$$

that is $\mathscr{F}_p$ is the colimit over all $\mathscr{F}(U)$ where $p \in U$.

The adjectives "small filtered colimit" are only there to guarantee that the colimit exists. Most base category that will be worked over shall have the stronger property of bieng *cocompelte* (having all colimits) and often also being *complete* (having all limits). Categories that are complete and cocomplete include:

1. **Set** (Category of Sets)
2. **Grp** (Category of Groups)
3. **Ab** (Category of Abelian Groups)
4. **Ring** (Category of Rings)
5. R-**Mod** (Category of modules over a ring R)
6. **Top** (Category of Topological Spaces)

Hence, this shows us that as long as the colimit exist in our category, we may define the notion of stalks for presheaves over that category without needing to reference the notion of elements. Note that it is the colimit and not the limit since we need $\mathscr{F}_p$ to "represent" the behaviour around p rather than "embed" germs.

One of the most important properties of stalks is their ability to be defined locally. First, if $U \subseteq X$ is an open set, we can define the sub-presheaf on U by letting $\mathcal{F}|_U(V) = \mathcal{F}(V \cap U)$. The following proposition will be really useful later when we need to work with stalks between larger and smaller (pre)sheaves, and mirrors the property in differential geometry (recall [5, chapter 2.1])

Proposition 1.1.5: Stalk Isomorphism

Let X be a presheaf of sets and $U \subseteq X$ an open set. Then for any $x \in U$:

$$\mathscr{F}_{X,x} \cong_{\mathbf{C}} \mathscr{F}_{U,x}$$

***Proof*:**
We shall prove this in the case where $\mathbf{C} = \mathbf{Set}$. The general case is left as an exercise.

Let us start by defining the map $\varphi : \mathscr{F}_{X,x} \to \mathscr{F}_{U,x}$. Take $[(f, V)] \in \mathcal{F}_{X,x}$ such that $x \in V$. As $x \in U$, $x \in U \cap V$. Hence, notice that:

$$(f, V) \sim (f|_{U \cap V}, U \cap V) \qquad \text{as} \qquad \mathrm{res}_{U, U\cap V} f = \mathrm{res}_{U\cap V, U\cap V} f|_{U \cap V} = f|_{U\cap V}$$

Then define the map between stalk via:

$$\varphi : (f, V) \sim (f|_{U\cap V}, U \cap V) \mapsto (f|_{U\cap V}, U \cap V)$$

This map is well-defined. Take $(f, V_1) \sim (g, V_2)$ where $V_1, V_2 \subseteq X$ so that we have to show that $(f|_{U\cap V_1}, U \cap V_1) \sim (g|_{U\cap V_2}, U \cap V_2)$. As $(f, V_1) \sim (g, V_2)$ there exists $W \subseteq V_1, V_2$ such that

$$f|_W = g|_W$$

Take $U \cap W \subseteq W \subseteq V_1, V_2$. Then by the restriction axioms we have that:

$$f|_{U\cap W} = g|_{U\cap W}$$

So that $(f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W)$. Then as:

$$\mathrm{res}_{U\cap V_1, U\cap W}(f|_{U\cap V_1}) = \mathrm{res}_{U\cap W, U\cap W}(f|_{U\cap W})$$

(and similarly for V_2), we get:

$$(f|_{U\cap V_1}, U \cap V_1) \sim (f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W) \sim (g|_{U\cap V_2}, U \cap V_2)$$

and hence the map is well-defined. This map has a natural inverse, namely

$$\varphi^{-1} : \mathscr{F}_{U,x} \to \mathscr{F}_{X,x} \qquad (f, V) \mapsto (f, V)$$

which can be shown to be well-defined and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}$, hence these sets are bijective, which are the isomorphisms in set. To generalize to categories that are augmented sets, take advantage of the restriction maps being morphisms.

Now, presheaves allow for the notion of restricting and localizing, that is there is structure preserved "going down". But there are no condition on any structure being preserved "going up". In particular, not all presheaves are amenable to gluing local information to create more global information (think of bounded holomorphic functions on open subsets of $\mathbb{C}$). We must add this condition axiomatically:

Definition 1.1.6: Sheaf using Axioms

Let $\mathscr{F} \in \mathbf{PSh}_C(X)$ be a presheaf over a category $\mathbf{C}$ whose objects are sets possibly with additional structure. Then $\mathscr{F}$ is also a sheaf if it satisfies the following two axioms;

1. *Gluing Axiom (Existence)*: Let $U = \bigcup_i U_i$. If there exists a collection $f_i \in \mathscr{F}(U_i)$ such that $\mathrm{res}_{U_i, U_i \cap U_j} f_i = \mathrm{res}_{U_j, U_i \cap U_j} f_j$, then there is a $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.

2. *Locality Axiom (Uniqueness)*. Let $U = \bigcup_i U_i$. If $f_1, f_2 \in \mathscr{F}(U)$, then if $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_i} f_2$ for all i, then $f_1 = f_2$

3. $\mathscr{F}(\emptyset)$ is associated with the terminal object (if we're working over set, $\mathscr{F}(\emptyset)$ is a singleton).

If only the locality axiom is satisfied, then our presheaf is called a *separated presheaf.* The collection of all sheaves on a topological space X is denoted $\mathbf{Sh}_C(X)$ (or $\mathbf{Sh}(X)$ if unambiguous)

We may think of the locality axiom telling us there is *at most* one way to glue, while the gluing axiom is telling us there is *at least* one way to glue. A separated presheaf is so named as it allows sections to "separate" points (there a cannot be two sections identical on restrictions of an open covering).

1.1.7 Separation and Evaluation Later, the notion of "evaluation" shall be defined for sections where the objects of $\mathbf{C}$ behave like sets of functions. Being separated *does not* imply a function is defined by the evaluation at every point as the intuition of separated algebra for the Stone-Weierstrass Theorem would imply. This is because evaluation on locally ringed spaces will carry more data that needs to be taken into account (ex. see lemma 2.2.6).

The following is sometimes (redundantly) included in the definition of a presheaf:

Lemma 1.1.8: Sheaf and Zero Section

Let $\mathbf{C}$ be a category with the zero object 0 and $\mathscr{F}$ a sheaf (or particularly, a separated presheaf) on X. Let $\bigcup_{i \in I} U_i = X$ be an open cover and $s \in \mathscr{F}(X)$. Then if $s|_{U_i} = 0$ for all i, then $s = 0$.

***Proof*:**
exercise.

Notice that we are taking elements of each $\mathscr{F}(U)$ because $\mathscr{F}(U)$ is an object in a category whose objects are sets with possibly additional structure. This is a deviation from the generalization presented in the definition of presheaves as $\mathbf{C}$ may be more general. There is in fact a way to define a sheaf for general categories:

Proposition 1.1.9: Sheaf Categorical Interpretation

Let $\mathscr{F} \in \mathbf{PSh}(X)$ be a presheaf, and let $\cdot$ be a terminal object (in **Set**, an arbitrary 1-point set). Let $U = \bigcup_i U_i$, and consider:

$$\cdot \to \mathscr{F}(U) \to \prod_{i \in I} \mathscr{F}(U_i) \rightrightarrows \prod_{i,j \in I} \mathscr{F}(U_i \cap U_j)$$

for any indexing set I. Then if the sequence is exact at $\mathscr{F}(U)$, the locality axiom holds, and if it is exact at $\prod \mathscr{F}(U_i)$, the gluing axiom holds.

***Proof*:**
This shall be proven when $\mathbf{C} = \mathbf{Ab}$, the general result can be proven by the reader interested in practicing their category theory.

Assume $\mathscr{F}(U) \xrightarrow{\alpha} \prod \mathscr{F}(U_i)$ is injective so that

$$s \hookrightarrow (s|_{U \cap U_i})_i$$

Then by definition of injectivity any s_a, s_b mapping to this collection of restricted sections is equal, $s_a = s_b$, giving the locality.

Now, let $s_i = \mathscr{F}(U_i)$. Take the equalizer:

$$\prod_{i \in I} \mathscr{F}(U_i) \xrightarrow{\beta,\gamma} \prod_{j \in J} \mathscr{F}(U_i \cap U_j)$$

where:

$$\beta((s_i)_i) = \left(s_i|_{U_i \cap U_j}\right)_{i,j \in I} \qquad \text{and} \qquad \gamma((s_i)_i) = \left(s_j|_{U_i \cap U_j}\right)_{i,j \in I}$$

For example Let $I = \{1, 2\}$ and $U_1 \cup U_2 = U$. Let us not put in the "redundant" elements for the pairs $(1, 1), (2, 2)$, and $(2, 1)$; the last pair is not automatically excluded in the definition to avoid putting an unnecessary ordering on I, and the former two are not excluded for the simplicity of notation. As we are working with abelian groups, we can turn the equalizer into:

$$\cdot \to \mathscr{F}(U) \to \prod_{i \in I} \mathscr{F}(U_i) \xrightarrow{s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j}} \prod_{i,j \in I} \mathscr{F}(U_i \cap U_j)$$

Thus the kernel of the equalize is when these two are equal, namely:

$$\beta(s_1, s_2) = (s_1|_{U_1 \cap U_2}) \qquad \text{and} \qquad \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

are equal:

$$(s_1|_{U_1 \cap U_2}) = \beta(s_1, s_2) = \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

Giving the gluing condition. In general, let us denote the kernel of the equalizer $\mathrm{Eq}(\beta, \gamma)$. By exactness we get that:

$$\mathrm{im}(\alpha) = \mathrm{Eq}(\beta, \gamma)$$

Thus, every element in the equalizer comes from a section in U. If α is also a monomorphism, it is unique, as we sought to show.

1.1.10 Foreshadowing Čech Cohomology

What if we extend this to have further equalizers? Sheaves that satisfy higher-order intersection-exactness conditions are a proper subset of sheaves (over non-trivial categories and topological spaces). Such sheaves can be thought of as never having any gluing obstruction, and hence have trivial cohomology. We shall return to these ideas when exploring Čech cohomology in section 6.1.1.

Example 1.1.11: Presheaves failing exactness (i.e. gluability and locality)

Let X be any topological space and define:

$$\mathcal{F}(X) = \mathbb{Z} \qquad \mathcal{F}(U) = 0 \qquad (\text{for } U \subsetneq X)$$

Then this sheaf is gluable, but not separated. If $U \subsetneq X$ is covered, then it is clear that gluing must hold. If $U = X = \cup_i U_i$, then that restrictions holds as any $z \in \mathbb{Z}$ has $z|_{U_i} = 0$ and so they certainly agree on all intersections. Hence, we see that each $z \in \mathbb{Z}$ is glued by a bunch of 0-elements!

Example 1.1.12: Presheaves And Sheaves

Most of these examples will be on sheaves of sets.

1. Verify that the sheaf of differentiable functions over M, namely $\mathcal{O}_M$, is indeed a sheaf. More generally, if X is any topological space, then the data defined by $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{C}}(U, \mathbb{R})$ where $\mathbf{C}$ is either **Top** or **Set** is a sheaf.

   Note how in a sheaf, we are "forced" to have interesting global behaviour if there are local section on non-intersecting sets. Consider $U_1 = (0, 1)$ and $U_2 = (3, 4)$. Then $U_1 \cap U_2 = 0$, so $\mathscr{F}(U_1 \cap U_2) = \{*\}$ is the terminal object (a singleton). If there is a function $f \in \mathscr{F}(U_1)$ and $g \in \mathscr{F}(U_2)$, then the restriction map

   $$\mathrm{res}_{U_1,\emptyset} f = * = \mathrm{res}_{U_2,\emptyset}$$

   Meaning they *have* to agree on intersection. Thus, by the gluability axiom, there must exist a function F on $U_1 \cup U_2$ that restricts to f, g. For functions, this is obvious as it is the function defined by:

   $$F(x) = \begin{cases} f(x) & x \in (0,1) \\ g(x) & x \in (3,4) \end{cases}$$

   This becomes more interesting when the category in question is not obviously interpret able as functions on a space (ex. some sheaves represent logical proposition, see ref:HERE).

2. (*Restriction of Sheaves*) Let $\mathscr{F} \in \mathbf{Sh}(X)$ and $U \subseteq X$ an open subset of X. Then the *restriction* of $\mathscr{F}$ to U, denoted $\mathscr{F}|_U$ has as data for each open $V \subseteq U$ $\mathscr{F}|_U(V) = \mathscr{F}(V)$ (note hos it is important that U is open for well-definedness). We shall later see how we may define restrictions of sheaves to arbitrary sets[a]

3. (*Skyscraper Sheaf*) Let X be a topological space, $p \in X$, and S be any set. Let $\iota_p : \{p\} \to X$

be the inclusion map. Then define the sheaf of sets $(\iota_p)_* S$ on X (not on S!) to be:

$$(\iota_p)_* S(U) = \begin{cases} S & p \in U \\ \{e\} & p \notin U \end{cases}$$

This sheaf is called the *skyscraper sheaf.* The name comes from the fact that the informal picture for this sheaf looks like a skyscraper at p. For sheaves of Abelian groups, $\{e\}$ is the trivial group. More generally, $(\iota_p)_*(U)$ would be the final object if $p \notin U$.

4. (*Constant Presheaf*) Let X be a topological space and S be any set. Define $\underline{S}_{\text{pre}}(U) = S$ for all open $U \subseteq X$ Then it should be verified that $\underline{S}_{\text{pre}}$ forms a presheaf on X, called the *constant presheaf associated to* S. This set is not in general a sheaf if $|S| > 1$: take $X = U_1 \coprod U_2$ to be the disjoint union of two (open) sets, take the constant presheaf so that all restriction maps are constant maps. Chose $s_1 \in S$ and $s_2 \in S$ which are distinct: $s_1 \neq s_2$. Then by the gluing axiom, there would have to exist a $s \in S$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$ and their intersection is empty and hence tautologically follow the gluing axiom, however as we have the identity map we have $s_1 = s = s_2$ however $s_1 \neq s_2$.

   This is fixed by a process called *sheafification* which will add the appropriate section equivalent to (s_1, s_2) that would restrict to both of these elements.

   What makes the sky-scrapper sheaf special is the "distinguishing" of a point making it act like a generalization of an indicator function. In the above example, one of the two sets will not have the point p.

5. (*Constant Sheaf*) Again let X be a topological space and S be any set. For every open $U \subseteq X$, let $\mathscr{F}(U)$ be the collection of all *locally constant* continuous functions from U to S, that is if $f \in \mathscr{F}(U)$, for each $p \in U$ there is an open neighborhood $V \subseteq U$ containing p, $p \in V \subseteq U$, such that $f|_V : V \to S$ is constant (on euclidean space, this is equivalent to having a constant function on each connected component). Then this sheaf is called the *constant sheaf associated to* S, and is denoted $\underline{S}$.

   Another way of describing this sheaf is if we endow S with the discrete topology and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.

6. A great example of a pre-sheaf on $\mathbb{C}$ that is not a sheaf is the sheaf of bounded holomorphic functions on $\mathbb{C}$. Recall that bounded entire holomorphic functions (defined on all of $\mathbb{C}$) must be constant by Louisville's Theorem, however there do exist bounded functions on open subsets of $\mathbb{C}$. For example , on each disk of radius r, $\mathbb{D}^2_r$, the holomorphic function z^2 is bounded but $\mathbb{C} = \cup_r \mathbb{D}^2_r$ has no bounded holomorphic function for which the restriction to any disk $\mathbb{D}^2_r$ is z^2; hence the gluing axiom fails.

   Another example would be the sheaf composed of holomorphic functions admitting holomorphic square roots, as there is no global square-root function on $\mathbb{C}$ due to non-trivial monodromy.

7. (*Sheaf of continuous maps*) Let X and Y be topological spaces. Define $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{Top}}(U, Y)$. This sheaf is a generalization sheaf of differentiable functions and the constant sheaf, namely

   - The sheaf of differentiable functions has $Y = \mathbb{R}$
   - the constant sheaf has $Y = S$ with the discrete topology.

   If Y is also a topological group, then the continuous maps to Y form a sheaf of groups.

8. (*Sheaf of section of a maps*) This is an important special case of the above sheaf. Let X, Y be topological spaces and let $\mu : Y \to X$ be a continuous map. Then the sections of μ (in the sense of the existence of a map $s : X \to Y$ such that $\mu \circ s = \text{id}$) form a sheaf. In particular, for each open $U \subseteq X$, $\mathscr{F}(U)$ is the set of section maps $s : U \to Y$ (continuous maps $s : U \to Y$ such that $\mu \circ s = \text{id}|_U$). Vector bundles and vector fields are a good example of such a sheaf.

9. The following examples shows how different sheaves can be put on the same object to characterize different type of information. Throughout this example, let $X = \mathbb{A}^1_k$ be the (classical) affine line (see definition 0.1.9). The open subsets in the Zariski topology are all cofinite subsets. If $U \subseteq \mathbb{A}^1_k$ is open let $a_1, ..., a_n$ be the points that are omitted from U.

   (a) (Sheaf of regular function) Then letting $f_i = (x - a_i)$ and $F = \{f_1, ..., f_n\}$, define $\mathscr{F}(U) = k[x]_F$ i.e. $\mathscr{F}(U)$ is the ring generated by $k[x]$ and $1/f_i$ for each i. The reader should verify that this is indeed a pre-sheaf, and even a sheaf.

   (b) (Sheaf of invertible functions) Let X be as above, and let $\mathscr{F}(U) = (k[x]_F)^*$, that is consider only the invertible functions. Then X is again a sheaf.

   (c) (sheaf of roots of unit) Again consider X as above. Then let $\mathscr{F}(U)$ be the subgroup of $k[x]^*_F$ consisting of all roots of unity present in $\mathscr{F}(U)$. This forms a sheaf

   (d) (Sheaf of differential forms) If the reader is familiar with differential geometry, let $k = \mathbb{R}$ and let $\mathscr{F}(U)$ be the collection of all closed forms on U (or more generally, all smooth forms on U). Note that only $\mathscr{F}(\mathbb{A}^1_{\mathbb{R}})$ are all the closed forms exact.

[a] see ref:HERE

If $\mathscr{F} \in \mathbf{PSh}(U)$ and $f \in \mathscr{F}(U)$ for some open $U \subseteq X$, then f is called a section. The following proposition motivates this idea

Proposition 1.1.13: Space of Sections of (Pre)Sheaf: étale spaces

Let $\mathscr{F}$ be a presheaf of sets. Then there exists a topological space F and a local homeomorphism $\pi : F \to X$ so that $\mathscr{F}$ can be represented as the sheaf of sections from F to X. The space F is called the *étalé space*. In particular, F is the *étalé space* of $\mathscr{F}$.

Furthermore, there is an equivalence between the category $\mathbf{Sh}(X)$ and the category of étalé spaces over X (with bundle morphisms as morphisms).

The idea is that (pre)sheaves correspond to a space F that is it is locally homeomorphic to X. Thus, geometrically sheaves could be thought of as local homeomorphisms, though the total space F may be rather ugly as shall be shown in example 1.1.14. Note that local homeomorphisms need not be surjective; this is a common misconceptions as many examples of surjective local homeomorphisms are covering spaces, but examples such as the sky-scrapper sheaf should show why this need not be the case.

***Proof*:**
Let F be the disjoint union of the stalks on F.

$$F = \coprod_{x \in X} \mathscr{F}_x$$

Then $\pi : F \to X$ is naturally a set map. Define a basis for F as follows: for each $s \in \mathscr{F}(U)$, take the collection

$$\{[s]_x \in \mathscr{F}_x \mid x \in U\}$$

These form a basis for F (note that each stalk has the discrete topology). With the setup, it is now easy to see the equivalence of categories.

The étale space may seem like it ought to be pretty similar to X being locally homeomorphic, however this can be far from the case:

Example 1.1.14: Non-Hausdorff Étale Space

Take $X = \mathbb{R}$ be the real numbers and take $\mathcal{F}$ to be the smooth functions on $\mathbb{R}$. Take the constant germ around 0 as well as the germ which can be represented by the function:

$$g(x) = \begin{cases} x\sin(1/x) & x\sin(1/x) > 0 \\ 0 & x\sin(1/x) \leq 0 \\ 0 & x < 0 \end{cases}$$

In other words, g is zero when x is negative and when the function $x\sin(1/x)$ would be negative; we want to keep when $x\sin(1/x)$ is positive. Then these two are different points in the étale space, but *cannot* be separated by open sets.

Note that if we were working with $X = \mathbb{C}$ and took analytic or holomorphic functions, then this cannot happen as we have a non-isolated 0 without all of g being zero. This turns out to be the main impingement for the étale space being Hausdorff. In general sheaves of analytic functions shall form Hausdorff étale spaces.

1.1.1 Stalks and Sheaves

As mentioned earlier, presheaves induce stalks. It was mentioned that sheaves but *not* presheaves can be characterized on the level of stalks, meaning the stalks of a sheaf shall give us all the information about the morphism a sheaf. This in fact almost characterizes sheaves from presheaves, as it in fact characterizes *separated presheaves* (recall they satisfy the locality but not gluing axiom). The following shows the value of working with separated presheaves:

Lemma 1.1.15: Sections Determined By Germs

Let $\mathscr{F}$ be a sheaf (more specifically, a separated presheaf). Then the map:

$$\mathscr{F}(U) \to \prod_{p \in U} (\mathscr{F}_p)$$

is injective

This map being injective justifies the term “separated”, namely the points are separated by the sections.

***Proof*:**
Let f be the map and consider $f(s) = (s_x)_{x \in U}$ Say:

$$(s_x)_{x \in U} = (t_x)_{x \in U} \qquad \text{equiv.} \qquad [(s, V_x)]_{x \in U} = [(t, W_x)]_{x \in U}$$

Then by definition there exists a $C_x \subseteq V_x \cap W_x$ such that

$$s|_{C_x} = t|_{C_x}$$

Now as $x \in C_x$ for each $x \in U$ we have $U = \bigcup_{x \in U} C_x$ and furthermore for for any pair C_x, C_y in the open cover we have:

$$\text{res}_{C_x, C_x \cap C_y} s = \text{res}_{C_y, C_x \cap C_y} t$$

hence, by the gluing axiom $s = t$, as we sought to show.

Note that we can show the locality axiom form this argument:

Corollary 1.1.16: Global Section Over Local Sections

Let $\mathscr{F}$ be a sheaf on X and $U_i \subseteq X$ is an open cover. Then:

$$\mathscr{F}(X) \to \prod_i \mathscr{F}(U_i)$$

is injective

***Proof*:**
Map $s \mapsto \left(s|_{U_i}\right)_i$. Consider

$$\left(s|_{U_i}\right)_i = \left(t|_{U_i}\right)_i$$

then s and t agree on intersections of U_i, hence there exists a unique element in $\mathcal{F}(X)$ for both these elements, namely $s = t$

In example 1.1.11, the sheaf $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$ was not a separated presheaf, and certainly $\mathcal{F}(X) \to \prod_{x \in X} \mathcal{F}_x$ is *not* injective, as each $\mathcal{F}_x$ consist of a single equivalence class representing the zero element (and hence $\prod_{x \in X} \mathcal{F}_x \equiv \{[0]\}$, and $\mathcal{F}(X) = \mathbb{Z}$. This example shows that we can somehow construct a nonzero object from a bunch of zero-objects. This can be thought of as being the key-blocker, especially in the case of abelian categories where most of the time the pre-image of 0 is only 0. This motivate the following:

Definition 1.1.17: Support of Sections

Let $\mathscr{F}$ be a sheaf of an abelian group on X, and s a global section of $\mathscr{F}$. Then the *support* of s, denoted $\text{Supp}(s)$, is the points $p \in X$ where s has a nonzero germ:

$$\text{Supp}(s) := \{p \in X \mid s_p \neq 0 \text{ in } \mathscr{F}_p\}$$

This notion differs a bit from the usual notion of support in analysis, in particular notice that it is the germ itself that is nonzero and not that the evaluation of the germ s_p at the point p that is nonzero.

Lemma 1.1.18: Support is Closed

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $s \in \mathscr{F}(X)$. Then $\operatorname{Supp}(s)$ is closed

***Proof*:**
We'll show $\operatorname{Supp}(s)^c$ is open, that is the set of all points $p \in X$ such that $s_p = 0$ is open. If $s_p = 0$ in the stalk, there exists an open subset $W_p \subseteq X$ such that $s|_{W_p} = 0$. Taking the union of arbitrary open sets is open, and hence taking the union of all W_p cover $\operatorname{Supp}(s)^c$ making $\operatorname{Supp}(s)$ closed, completing the proof.

The equivalent in analysis is the definition of the *closed support* of a function. Due to this, the compliment of the support of an element s is the 0-section. The section can also be defined on presheaves, however there are examples of presheaves where there are nonzero sections that "live nowhere" because it is 0 in every stalk[7]

Let us now take a closer look at $\prod_{p \in U}(\mathscr{F}_p)$. This set is much bigger than sections of a sheaf. We thus take the special sub-collection that associates with $\mathscr{F}(U)$ (essentially characterizing the image of $\mathscr{F}(U)$:

Definition 1.1.19: Compatible Germs

Let $\mathscr{F}$ be a sheaf on X and $U \subseteq X$ an open set. Then we say an elements $(s_p)_{p \in U} \in \prod_{p \in U} \mathscr{F}_p$ consists of *compatible germs* if there exists some representative open sets U_p for s_p:

$$U_p \subseteq U \text{ open}, \;\; \tilde{s}_p \in \mathscr{F}(U_p)$$

such that the germ of $\tilde{s}_p$ for all $q \in U_p$ is s_q. In other words, if $\{U_i\}$ is an open cover of U where $p \in U_i$ for all i, then s_p is the germ of f_i at p.

Hence, a compatible germ is the collection of germs that are "compatible" when taking subsets, that is *restrictions*. Using the gluing axiom, it is a good exercise to show that any choice of compatible germs for a sheaf $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$ over U, that is we have successfully identified the right hand side of the equation in lemma 1.1.15. Note how important it is that we use sheaves (or at least separated presheaves) for the proof. The proof also motivate the agricultural terminology of "sheaf", for in agriculture a sheaf is a collection of "Stalks" (which is the main stem of a plant).

1.1.20 Remark When working with sheaves of sets or some structures on sets, the elements of the section can literally be thought of as functions using the above definition, namely function of the form $s : U \to \coprod_{p \in U} \mathscr{F}_p$. The generalization presented above allows to generalize the notion of "open set U" so that the topological space X is replaced special category called a *site*, see ref:HERE.

[7] Take $\mathscr{F}$ to be the sheaf of holomorphic functions on $\mathbb{C}$, $\mathscr{G}$ the sheaf of bounded holomorphic functions. Then take the sheaf $\mathcal{H} = \mathscr{F}/\mathscr{G}$. Then show that $\mathcal{H}_x = 0$, however $\mathcal{H}(\mathbb{C}) = \{\text{entire functions}\}/\mathbb{C}$

1.1.2 Sheaf from a Topological Base

When defining a differential or topological structure, one of the strategies that is taken is to define pieces of the manifold in a compatible way, or said differently define it on some basis of the manifold. This section endeavours to do this for sheaves.

Let X be a topological space with a sheaf $\mathscr{F}$. Take a basis $\{B_i\}$ for X and take the subset of $\mathscr{F}$ consisting of $\mathscr{F}(B_i)$ along with all the appropriate restriction maps. Then there is enough Stalk information to recover all of $\mathscr{F}$ (show this), and hence we my want to start with such a collection to generate a sheaf. This collection alone will generate a presheaf, we will require the basis to satisfy the locality and gluability axioms:

Definition 1.1.21: Sheaf On A Base

Let X be a topological space and $\mathcal{B}$ be a topological base. Then this collection is caleld a *sheaf on a base* if it satisfies:

1. **Base locality axiom**: If $B = \cup_i B_i$, then if $f, g \in \mathscr{F}(B)$ are such $\text{res}_{B,B_i} f = \text{res}_{B,B_i} g$ for all i, then $f = g$.
2. **Base Gluability**: If $B = \cup_i B_i$, and $f_i \in \mathscr{F}(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$, then there exists $f \in \mathscr{F}(B)$ such that $\text{res}_{B,B_i} f = f_i$ for all i

Theorem 1.1.22: Define Sheaf From Base

Let $\{B_i\}$ be a base for the topological space X and let F be a of sets on the base. Then there is a sheaf $\mathscr{F}$ extending F with isomorphisms $\mathscr{F}(B_i) \cong F(B_i)$ agreeing with restrictions. Furthermore, this sheaf is unique up to unique isomorhpism

***Proof*:**
The sheaf $\mathscr{F}$ will be defined as the sheaf of compatible germs of F. To this end, define the stalk of a presheaf F on a base at $p \in X$ by taking:

$$F_p = \varinjlim F(B_i)$$

Define:

$$\mathscr{F}(U) = \\ \{(f_p \in F_p)_{p \in U} \mid \forall p \in U, \exists B \text{ with } p \in B \subseteq U,\ s \in F(B) \text{ that is compatible: } s_q = f_q\ \forall q \in B\}$$

Then the map $F(B) \to \mathscr{F}(B)$ is an isomorphism.

Lots more to cover here that is worth going over

1.1.3 Gluing Sheaves on an Open Cover

Just as we can construct a sheaf from data on a topological base, we can construct a sheaf from sheaves defined on the pieces of an open cover, provided the local sheaves are compatible on overlaps.

This result is foundational: it will be used in chapter 2 to construct the structure sheaf when gluing schemes.

Theorem 1.1.23: Gluing Sheaves On An Open Cover

Let X be a topological space and $\{U_i\}_{i\in I}$ an open cover. Suppose we are given:

1. a sheaf $\mathscr{F}_i$ on each U_i,
2. isomorphisms of sheaves $\varphi_{ij} : \mathscr{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathscr{F}_j|_{U_i\cap U_j}$ for all i, j

satisfying the cocycle condition on triple overlaps:

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik} \quad \text{on } U_i \cap U_j \cap U_k$$

Then there exists a sheaf $\mathscr{F}$ on X, unique up to unique isomorphism, together with isomorphisms $\psi_i : \mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$ such that $\varphi_{ij} \circ \psi_i = \psi_j$ on $U_i \cap U_j$.

***Proof*:**
For each open set $V \subseteq X$, define:

$$\mathscr{F}(V) = \left\{(s_i)_{i\in I} \;\middle|\; s_i \in \mathscr{F}_i(V\cap U_i), \quad \varphi_{ij}\big(s_i|_{V\cap U_i\cap U_j}\big) = s_j|_{V\cap U_i\cap U_j}\ \forall i,j\right\}$$

with restriction maps defined componentwise: if $W \subseteq V$ then $\mathrm{res}_{V,W}((s_i)_i) = (s_i|_{W\cap U_i})_i$.

$\mathscr{F}$ is a presheaf. The restriction of a compatible tuple is compatible (as φ_{ij} commutes with restriction, being a sheaf morphism), and the identity and composition axioms are inherited componentwise.

Locality axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i), (t_i) \in \mathscr{F}(V)$ satisfy $(s_i)|_{V_\alpha} = (t_i)|_{V_\alpha}$ for all α. Then for each i, we have $s_i|_{V_\alpha\cap U_i} = t_i|_{V_\alpha\cap U_i}$ for all α. Since $\{V_\alpha \cap U_i\}_\alpha$ is an open cover of $V \cap U_i$ and $\mathscr{F}_i$ is a sheaf (satisfying locality), $s_i = t_i$ for all i.

Gluing axiom. Let $V = \bigcup_\alpha V_\alpha$ and suppose $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ agree on overlaps, i.e. $s_i^{(\alpha)}|_{V_\alpha\cap V_\beta\cap U_i} = s_i^{(\beta)}|_{V_\alpha\cap V_\beta\cap U_i}$ for all α, β, i. For each fixed i, the sections $s_i^{(\alpha)} \in \mathscr{F}_i(V_\alpha \cap U_i)$ agree on overlaps $V_\alpha \cap V_\beta \cap U_i$. As $\mathscr{F}_i$ is a sheaf, there exists a unique $s_i \in \mathscr{F}_i(V \cap U_i)$ with $s_i|_{V_\alpha\cap U_i} = s_i^{(\alpha)}$ for all α.

It remains to check the compatibility condition for the glued tuple $(s_i)_i$: we need $\varphi_{ij}(s_i|_{V\cap U_i\cap U_j}) = s_j|_{V\cap U_i\cap U_j}$. Both sides are sections of $\mathscr{F}_j(V \cap U_i \cap U_j)$ that agree on each $V_\alpha \cap U_i \cap U_j$ (since $(s_i^{(\alpha)})_i \in \mathscr{F}(V_\alpha)$ satisfies the compatibility condition). By the locality axiom for $\mathscr{F}_j$, they are equal.

Hence $\mathscr{F}$ is a sheaf. The isomorphisms $\psi_i : \mathscr{F}|_{U_i} \to \mathscr{F}_i$ are given by projection onto the i-th component: $(s_j)_j \mapsto s_i$, which is an isomorphism because the compatibility conditions determine all other components on U_i via the φ_{ij}. Uniqueness of $\mathscr{F}$ follows from corollary 1.3.4 applied to the open cover.

1.2 Morphisms and Exactness

Now that we have defined (pre)sheaves, it is natural to define morphism between sheaves to show that $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ form a category. Sheaf morphisms shall be over the same object, and hence can be thought of as the equivalent of looking at local homeomorphisms between two étale spaces:

Definition 1.2.1: Morphism Of Sheaves

Let $\mathscr{F}, \mathscr{G} \in \mathbf{PSh}(X)$ (note that $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)$). Then a *morphism of (pre)sheaves* (over the chosen category) $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation between the two functors. Concretely, it contains the data $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ such that if $V \subseteq U$ then the following diagram commutes:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\mathrm{res}_{U,V}"'] & \mathscr{G}(U) \arrow[d, "\mathrm{res}_{U,V}"] \\ \mathscr{F}(V) \arrow[r, "\varphi(V)"'] & \mathscr{G}(V) \end{tikzcd}$$

that is

$$\mathrm{res}_{U,V}(\varphi(U)(-)) = \varphi(V)(\mathrm{res}_{U,V}(-))$$

The collection of morphisms between two sheaves is denoted $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$ or $\mathrm{Mor}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$

To emphasize when the domain and codomain of a presheaf morphism is a sheaf, the morphism shall be called a *sheaf morphism*. A useful geometric intuition of sheaf morphisms is that they apply a transformation to each map (ex. $\exp(f)(z) = e^{f(z)}$). With the morphisms and objects defined, it is easy to show that they form a category. We shall denote the category of sheaves over a category $\mathbf{C}$ on a space X as $\mathbf{C}_X$. Thus:

$$\mathbf{Set}_X \qquad \mathbf{Ab}_X \qquad \mathbf{Ring}_X$$

are the categories of sheaves over sets, abelian groups, and rings respectively. Since the morphisms are the same, the category $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$. For presheaves, we shall denote the category with an extra superscript:

$$\mathbf{Set}_X^{\mathrm{pre}} \qquad \mathbf{Ab}_X^{\mathrm{pre}} \qquad \mathbf{Ring}_X^{\mathrm{pre}}$$

Example 1.2.2: Morphism of Presheaves and Sheaves

(If the reader is familiar with Riemann surface) Let X be a Riemann surface and $\mathcal{O}_X(-P)$ be the collection of meromorphic function $f \in K^*(X)$ such that $\mathrm{div}(f) + (-p) \geq 0$ ($P \in X$ a point). Then if $\mathcal{O}_X$ is the sheaf of holomorphic functions on X there are sheaf morphisms:

$$0 \to \mathcal{O}_K(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

where κ_P is the sky-scrapper sheaf at P, namely:

$$\kappa_P = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

This is even an exact sequence of sheaves, but we need to build-up to show how to properly define exactness in $\mathbf{C}_X$.

From now on, we shall be working with *abelian categories* unless stated otherwise: they all have 0 objects, they are closed under finite products and coproducts, and their kernel/cokernels behave well with epimorphisms and monomorphisms; see section 0.4 for a refresher or [9] for more details. We shall build-up to showing that $\mathbf{C}_X$ and $\mathbf{C}_X^{\text{pre}}$ are abelian categories.

As a presheaf morphism is a natural transformation, it important to emphasize that it does not make sense to say that a sheaf morphism is injective or surjective. Instead, it can be a *monomorphism* or an *epimorphism* and have corresponding kernel, cokernels, and images. Let us find objects in $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ that can represent these kernel, cokernel, and image. Starting with the kernel of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ which is defined to be be object $\mathcal{K}$ with the morphism $K : \mathcal{K} \to \mathcal{F}$ such that $\varphi \circ K = 0$ universally.

Proposition 1.2.3: Kernels in Presheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\ker \varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\mathcal{K}(U) = \ker(\varphi(U))$$

The resulting presheaf is denoted $\ker_{\text{pre}}(\varphi)$. Furthermore, if $\mathscr{F}$ is a sheaf and $\mathscr{G}$ satisfies at least the locality axiom (in particular if it is also a sheaf), then $\mathcal{K}$ is a sheaf.

***Proof*:**
chase the diagram, and use a similar separated/locality condition in the case of a sheaf as was done before.

Note that as the representation of the kernel for sheaves is the same as the characterization of presheaves, we can just write $\ker(\varphi)$ without any ambiguity. As the monomorphism condition can be checked on open sets, the following usual definition of injectivity is recovered on the level of open sets:

Definition 1.2.4: Injective Presheaf and Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between shaves or presheaves. Then φ is *injective* if $\ker \varphi(U) = 0$ for all open $U \subseteq X$.

The image of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is defined to be be object $\mathscr{I}$ with the monomorphism $I : \mathscr{G} \to \mathscr{I}$ and a morphism $E : \mathscr{F} \to \mathscr{I}$ such that $\varphi = E \circ I$ universally, namely:

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, "\varphi"] \arrow[dr, "E"] \arrow[ddr, bend right, "E'"'] & & \mathscr{G} \\
& \mathscr{I} \arrow[ur, "I"] \arrow[d, dashed, "\exists! v"] & \\
& \mathscr{I}' \arrow[uur, bend right, "I'"'] &
\end{tikzcd}$$

The reader familiar with category may be more familiar with images in abelian categories defined as:

$$\text{im}(\varphi) = \ker(\text{coker}(\varphi))$$

As we are working with abelian categories, $\ker(\operatorname{coker}(\varphi)) \cong \operatorname{coker}(\ker(\varphi))$, and hence there is no need of defining the coimage.

Proposition 1.2.5: Images in Presheaf and Sheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\operatorname{im}_{\text{pre}}\varphi$ can be defined to be the image of each $\varphi(U)$:

$$\operatorname{im}_{\text{pre}}\varphi(U) = \operatorname{im}(\varphi(U))$$

and similarly for cokernels:

$$\operatorname{coker}_{\text{pre}}(\varphi(U)) = \frac{\mathscr{G}(U)}{\operatorname{im}(\varphi(U))}$$

The resulting $\operatorname{im}_{\text{pre}}(\varphi)$ and $\operatorname{coker}_{\text{pre}}(\varphi)$ is called the *image presheaf* and *cokernel presheaf* respectively.

This is *not* necessarily the case if $\mathscr{F}$ is a sheaf. Images and cokernel in $\mathbf{C}_X$ will require a bit more work to be characterized.

***Proof*:**
chase the diagram.

Definition 1.2.6: Surjective Presheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then φ is *surjective* if $\operatorname{im}(\varphi(U)) = \mathscr{G}(U)$ for all open $U \subseteq X$.

As noted earlier, images of sheaves need not be sheaves:

Example 1.2.7: Sheaves: difficulty With Cokernels and Images

1. Let $X = \mathbb{C}$ with the usual topology, $\underline{\mathbb{Z}}$ be the constant sheaf on X, $\mathcal{O}_X$ the sheaf of holomorphic functions, and $\mathscr{F}$ the *presheaf* of functions admitting a holomorphic logarithm. Then there is an exact sequence of presheaves on X:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathscr{F} \to 0$$

where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion on each open set, and $\mathcal{O}_X \to \mathscr{F}$ is given by $f \mapsto \exp(2\pi i f)$ (again, over each open set). This is indeed a short exact sequence, for if $U \subseteq V \subseteq \mathbb{C}$, then:

$$\begin{tikzcd}
0 \arrow[r] & \underline{\mathbb{Z}}(V) \arrow[r] \arrow[d, "\text{res}"] & \mathcal{O}_X(V) \arrow[r] \arrow[d, "\text{res}"] & \mathscr{F}(V) \arrow[r] \arrow[d, "\text{res}"] & 0 \\
0 \arrow[r] & \underline{\mathbb{Z}}(U) \arrow[r] & \mathcal{O}_X(U) \arrow[r] & \mathscr{F}(U) \arrow[r] & 0
\end{tikzcd}$$

commutes. Certainly each $\underline{\mathbb{Z}}(V) \to \mathcal{O}_X(V)$ is an inclusion, and $\mathcal{O}_X(V) \to \mathscr{F}(V)$ is surjective (by definition) with kernel being the constant functions $\mathbb{Z}$ since if f admits a logarithmic inverse, then $f = \log(g)$, so $g \mapsto \exp(2\pi i \log g) = f$.
Then it should be clear that $\mathscr{F}$ is *not* a sheaf: take $U = \mathbb{C}^*$, cover it with $U_1 = \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and $U_2 = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Then $\mathscr{F}(U)$, $\mathscr{F}(U_1)$, and $\mathscr{F}(U_2)$ each have multiple non-zero functions

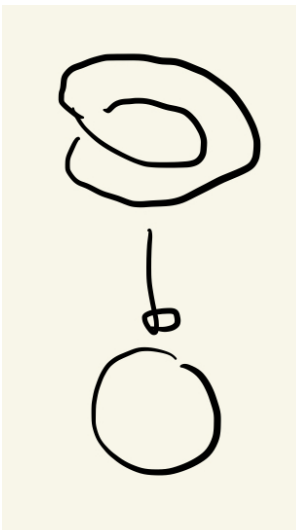
admitting logarithms (see [4]). If $\mathscr{F}$ were a sheaf, then as z admits logarithmic inverse on U_1 and U_2 and tautologically $z|_{U_1 \cap U_2} = z|_{U_1 \cap U_2}$ there should be a function defined on U restricting to these two, that restricts to both of these, but no such function exists (consider that such a function would have to be continuous, but going around the origin introduces non-trivial monodromy).

2. Consider the following similar but subtly different example. Consider $\mathcal{O}_{\mathbb{C}}^*$ the sheaf of all non-vanishing holomorphic functions (this sheaf is strictly "bigger" than the above $\mathscr{F}$, ex. $\mathscr{F}(\mathbb{C}^*) \subsetneq \mathcal{O}_{\mathbb{C}}^*(\mathbb{C}^*)$). Consider the following short exact sequence of *sheaves*:

$$0 \to \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \xrightarrow{!} 0$$

Notice that on $U = \mathbb{C}^*$, z does not vanish and so $z \in \mathcal{O}_{\mathbb{C}}^*(U)$. However, z does not admit a logarithmic inverse (again see [4]). This would imply that $\exp(\mathbb{C}^*)$ is *not* surjective and so exp is not surjective on all open sets. And yet it is indeed the case that there exists a morphism $\xrightarrow{!}$, namely exp is an *epimorphism* on sheaves. This will be an example of an epimorphism that is *not* characterize by surjectivity on open sets, as shall be explained by defining sheafification after these examples.

3. For the reader less comfortable with complex analysis, here is a more rudimentary example: take the sheaf of continuous sections of the circle S^1. For our first sheaf of sections, let $\pi : S^1 \to S^1$ be the identity (the trivial bundle), and let $\mathcal{F}_1$ be the sheaf of sections of this bundle. The second sheaf of section will with the projection from the twisted circle, which can be visualized as:



Label this shape S_2^1. The second sheaf $\mathcal{F}_2$ is all the continuous sections of all the open subsets of the codomain for the map $\pi : S_2^1 \to S^1$ with a reasonable definition of projection map mirroring the above image (over the complex numbers, we can take $z \mapsto z^2$). Note that $\mathcal{F}_2(S^2) = 0$, i.e. it has no global sections. Then there is certainly a surjective continuous map φ satisfying the commutative diagram:

$$\begin{tikzcd} S^1_2 \arrow[rr, "\varphi"] \arrow[dr] & & S^1 \arrow[dl] \\ & S^1 & \end{tikzcd}$$

However, there is no surjective maps:

$$\mathcal{F}_2(X) \to \mathcal{F}_1(X)$$

namely since there no continuous section from S^1 to S^1_2 that would satisfy the projection condition.

In the above, the first example shows that exactness has no issue with pre-sheaves, and in the second example we claimed the short exact sequence of sheaves was still exact in the category of sheaves. Notice that the image of exp in $\mathcal{O}^*_{\mathbb{C}}$ was the presheaf $\mathscr{F}$ in the first example. What we shall show is that this image uniquely determines $\mathcal{O}^*_{\mathbb{C}}$. In the same vein in which we can from an abelian group via a abelian semi-group using groupification, or a normal subgroup by normalizing a subgroup, we can from a sheaf from a presheaf via sheafification. As is usual, this is done in a universal way that is adjoint to the forgetful functor. This sheafification shall be the solution to defining right notion of surjection for sheaves:

Theorem 1.2.8: Sheafification

Let $\mathscr{F}$ be a presheaf on X. Then there exists a sheaf $\mathscr{F}^{\text{sh}}$ and a map $\text{sh} : \mathscr{F} \to \mathscr{F}^{\text{sh}}$ such that for any sheaf $\mathscr{G}$ and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique morphism of sheaves $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathscr{F}^{\text{sh}} \arrow[d, "g\text{, sheaf mor.}"] \\ \mathscr{F} \arrow[ur, "\text{sh}"] \arrow[r, "f", "\text{presheaf mor.}"'] & \mathscr{G} \end{tikzcd}$$

The sheafification of the presheaf $\mathscr{F}$ is often denoted $\mathscr{F}^+$.

The idea will be to remove all section that violate locality, and add sections to respect gluability. Another good take is the following post here.

***Proof*:**

Let $\mathscr{F}$ be the presheaf in the question. Define $\mathscr{F}^{\text{sh}}$ as the set of compatible germs of the presheaf $\mathcal{F}$ over U, namely:

$$\mathscr{F}^{\text{sh}}(U) := \{(f_p \in \mathscr{F}_p)_{p\in U} \mid \forall p \in U, \exists p \in V \subseteq U \text{ open and } s \in \mathscr{F}(V) \text{ such that } s_q = f_q, \forall q \in V\}$$

Note this resemble to the étale space. To see that $\mathscr{F}^{\text{sh}}$ is a sheaf, first note that restriction is defined by restricting the domain of the tuple of germs. For the gluing axiom, if we have sections $s^{(i)} \in \mathscr{F}^{\text{sh}}(U_i)$ agreeing on overlaps, we can define a global tuple s on $U = \bigcup U_i$ by setting $s_p = s^{(i)}_p$ for $p \in U_i$. This tuple belongs to $\mathscr{F}^{\text{sh}}(U)$ because the condition of being "locally represented by a section of $\mathscr{F}$" is satisfied locally by the witnesses for each $s^{(i)}$.

We define the map $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ by sending a section $\sigma \in \mathscr{F}(U)$ to the tuple of its germs $((\sigma)_p)_{p \in U}$, which is clearly a presheaf morphism.

To prove the universal property, let $\mathscr{G}$ be a sheaf and $g : \mathscr{F} \to \mathscr{G}$ a presheaf morphism. We wish to construct a unique $f : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$. Let $s \in \mathscr{F}^{\mathrm{sh}}(U)$. By definition, there exists an open cover $\{V_i\}$ of U and sections $\sigma_i \in \mathscr{F}(V_i)$ such that $s|_{V_i} = \mathrm{sh}(\sigma_i)$ (i.e., the germs of s match the germs of σ_i on V_i). We propose to define $f(s)$ by gluing the sections $g_{V_i}(\sigma_i) \in \mathscr{G}(V_i)$.

To do this, we must verify that $g(\sigma_i)$ and $g(\sigma_j)$ agree on $V_i \cap V_j$. For any point $p \in V_i \cap V_j$, we have $(\sigma_i)_p = s_p = (\sigma_j)_p$. By the definition of germs, there exists a small neighborhood $W_p \subseteq V_i \cap V_j$ such that $\sigma_i|_{W_p} = \sigma_j|_{W_p}$. Since g is a presheaf morphism, it respects restriction, so $g(\sigma_i)|_{W_p} = g(\sigma_j)|_{W_p}$. Because $\mathscr{G}$ is a sheaf (specifically, separated), and the sections $g(\sigma_i)$ and $g(\sigma_j)$ agree on a neighborhood of every point in the intersection, they must be equal on the entire intersection $V_i \cap V_j$. Thus, by the gluing axiom of $\mathscr{G}$, there exists a unique section $T \in \mathscr{G}(U)$ such that $T|_{V_i} = g(\sigma_i)$. We define $f_U(s) = T$. This map is unique because any morphism f must respect restriction, forcing it to agree with this local construction.

The "failure" of a presheaf being a sheaf can be measured as the lack of exactness of at the $\prod_i F(U_i)$ in proposition 1.1.9.

Example 1.2.9: Sheafification

To gain intuition, show the following sheafifications:

1. Show that $(\underline{S}_{\mathrm{pre}})^{\mathrm{sh}} = \underline{S}$, that is the presheaf of constant function sheafifies to the sheaf of constant functions.
2. Let $\mathscr{F}$ on $\mathbb{R}$ be all bounded continuous functions on each open set. Show that $\mathscr{F}^{\mathrm{sh}}$ is *all* continuous functions on each open subset.
3. Show that $\mathcal{H}$ on $\mathbb{C}$ of non-vanishing holomorphic functions that admit a logarithm sheafify to all non-vanishing holomorphic functions.
4. Let $\mathscr{F}$ be a non-trivial presheaf where all stalks are trivial (see exercise ref:HERE). Show that $\mathscr{F}^{\mathrm{sh}}$ is trivial.

Corollary 1.2.10: Sheafification and Stalks

Let $\mathcal{F} \in \mathbf{PSh}(X)$ be a presheaf. Then for all $p \in X$

$$\mathcal{F}_p \cong \mathcal{F}^{\mathrm{sh}}_p$$

***Proof*:**
Use something similar given in the proof of proposition 1.1.5

Let us now put together the universal properties we've shown so far to get that the category of sheaves also have cokernels:

Proposition 1.2.11: Sheaf Morphisms Have Cokernels

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a sheaf morphism. Then it has a cokernel and it is the sheafification of the cokernel-presheaf.

***Proof*:**
Recall that the universal property of the cokernel can be seen as the colimit of the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \\ 0 & \end{tikzcd}$$

As we know, the cokernel forms a presheaf $\mathcal{H}_{\text{pre}}$. By sheafification we have a unique map $\text{sh} : \mathcal{H}_{\text{pre}} \to \mathcal{H}$. Now let $\mathcal{A}$ be any sheaf and consider the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \arrow[d] \arrow[ddrr, bend left] & & \\ 0 \arrow[r] \arrow[drrr] & \mathcal{H}_{\text{pre}} \arrow[r, "\text{sh}"] & \mathcal{H} \arrow[dr, dashed, "\exists!"] & \\ & & & \mathcal{A} \end{tikzcd}$$

By the universal property of cokernels for presheaves, we have a unique map $\mathcal{H}_{\text{pre}} \to \mathcal{A}$. By the universal property of sheafification, we will get a unique map composed from the two unique universal maps giving us $\mathcal{H} \to \mathcal{A}$, as we sought to show.

Corollary 1.2.12: Image and Sheafification

Let $\varphi \in \mathbf{Sh}(X)$ be a morphism of sheaves over an abelian category. Then the image of φ is the sheafification of the image presheaf.

***Proof*:**
Recall that in an abelian category, the image of a morphism is defined as the kernel of its cokernel:

$$\text{Im}(\varphi) \cong \text{Ker}(\mathcal{G} \to \text{coker}(\varphi))$$

Let $\mathcal{K}_{\text{pre}}$ be the presheaf cokernel of φ. We have, by definition, a short exact sequence of presheaves:

$$0 \to \text{Im}_{\text{pre}}(\varphi) \to \mathcal{G} \to \mathcal{K}_{\text{pre}} \to 0$$

Since the sheafification functor is exact (it preserves exact sequences and finite limits), applying it to the sequence above yields an exact sequence of sheaves:

$$0 \to (\text{Im}_{\text{pre}}(\varphi))^{\text{sh}} \to \mathcal{G}^{\text{sh}} \to (\mathcal{K}_{\text{pre}})^{\text{sh}} \to 0$$

Since $\mathcal{G}$ is already a sheaf, $\mathcal{G}^{\text{sh}} \cong \mathcal{G}$. By proposition 1.2.11, $(\mathcal{K}_{\text{pre}})^{\text{sh}}$ is the sheaf cokernel of φ. Consequently, the object $(\text{Im}_{\text{pre}}(\varphi))^{\text{sh}}$ is the kernel of the morphism $\mathcal{G} \to \text{coker}(\varphi)$, which is precisely the definition of the image sheaf.

Definition 1.2.13: Surjective Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves and $\mathrm{im}\varphi$ the image presheaf. Then the cokernel and image of φ is the sheafification of the cokernel-presheaf and image-presheaf respectively. The morphism φ is *surjective* if
$$(\mathrm{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} = \mathscr{G}$$

Using this, we see that the map $\mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}^*_{\mathbb{C}}$ is *surjective* in the category of sheaves, even though it is not surjective in the category of presheaves. Hence, exactness in presheaves and sheaves is *different*. The following example using simpler categories should demonstrate why this is not so strange:

Example 1.2.14: Ab Vs TF-Ab

Let **Ab** be the category of abelian groups and TF-Ab be the category of torsion-free abelian groups. Consider the morphism $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(n) = 2n$. This morphism exists in both categories. The cokernel of φ in **Ab** is $\mathbb{Z}/2\mathbb{Z}$, which is torsion. To find the cokernel in TF-AB, the “best” torsion-free group must be found so that its composition with φ is universal. It isn't hard to see this is 0, so:
$$\mathrm{coker}_{\mathbf{Ab}}(\varphi) = \mathbb{Z}/2\mathbb{Z} \neq 0 = \mathrm{coker}_{\text{TF-Ab}}(\varphi)$$
Thus, though the morphism in each category are the same, the epimorphism differ.

Thus, a theoretical understanding of epimorphisms in $\mathbf{C}_X$ is done. However, it is computationally intractable to always sheafify the image sheaf to check for surjectivity. As sheaves are created to be “local”, we shall be able to shall be able to check their condition on *stalks*. Proving this shall fit in better into the broader framework on when the functor taking $\mathscr{F} \mapsto \mathscr{F}(U)$ and $\mathscr{F} \mapsto \mathscr{F}_p$ is exact.

Functors from (pre)sheaves: simplifying checking exactness

As presheaves and sheaves store all local information, we may want that there are some natural functors that preserve some exactness to extract that information. The two natural candidates for functors to would be $-(U)$ taking a (pre)sheaf to its sections over an open set U, and $(-)_p$ the functor taking a (pre)sheaf to it's stalk at p.

It is clear that $-(U) : \mathbf{C}_X \to \mathbf{C}$ is a functor for:
$$\mathscr{F} \mapsto \mathscr{F}(U) \qquad (\varphi : \mathscr{F} \to \mathscr{G}) \mapsto (\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U))$$
preserves composition's an identities. Let us show $(-)_p$ is also a functor:

Proposition 1.2.15: Morphism Of (Pre)Sheaves Induces Morphism On Stalks

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of (pre)sheaves over any category **C** having colimits (so that stalks are well-defined). Then there is an induced morphism on stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$, that is, it induced a functor $\mathbf{C}_X \to \mathbf{C}$.

***Proof*:**
This is definition pushing: For each open $U \subseteq X$ we have the morphism $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$.

Define the map:

$$[(f, U)]_p \mapsto [(\varphi(U)(f), U]_p$$

For this map to be well-defined if $(f, U) \sim (f', V)$ then we need:

$$[(f', V)]_p \mapsto [(\varphi(V)(f'), V]_p = [(\varphi(U)(f), U]_p$$

Hence we need that $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$. As $(f, U) \sim (f', V)$ we have there exists $W \subseteq U, V$ such that

$$\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} f' = g$$

Then as φ is a pre-sheaf morphism we have:

$$\begin{tikzcd} \varphi(U) : f \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{U,W}"] & \varphi(U)(f) \arrow[d, "\mathrm{res}_{U,W}"] \\ \varphi(W) : \mathrm{res}_{U,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd} \qquad \begin{tikzcd} \varphi(V) : f' \arrow[r, mapsto] \arrow[d, "\mathrm{res}_{V,W}"] & \varphi(U)(f') \arrow[d, "\mathrm{res}_{V,W}"] \\ \varphi(W) : \mathrm{res}_{V,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd}$$

which implies

$$\mathrm{res}_{U,W} \varphi(U)(f) = \mathrm{res}_{V,W} \varphi(V)(f')$$

But that means they are similar, $((\varphi(V)(f'), V) \sim ((\varphi(U)(f), U)$, as we sought to show

Let us now see how these functors preserve monomorphisms/epimorphisms (i.e. injectivity and surjectivity on opens/stalks). Starting with open subsets:

Proposition 1.2.16: On Presheaf: Open Set Functor is Exact

The functor $-(U) : \mathbf{PSh_C}(X) \to \mathbf{C}$ (where $\mathbf{C}$ is abelian) is an exact functor

On $\mathbf{Sh}$, the functor $-(U)$ is left-exact, and hence preserves injective morphisms.

***Proof*:**
Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a monomorphism of presheaves between presheaves. By definition, for any presheaf $\mathcal{H}$ and any morphisms $f, g : \mathcal{H} \to \mathcal{F}$, the relation $\varphi \circ f = \varphi \circ g$ implies $f = g$. Since morphisms of presheaves are natural transformations defined component-wise, the equality of morphisms holds if and only if it holds for every open set U. Thus, the condition implies:

$$\varphi(U) \circ f(U) = \varphi(U) \circ g(U) \implies f(U) = g(U)$$

This is precisely the definition of a monomorphism in the target abelian category $\mathbf{C}$, giving left-exactness and preservation of injectivity. A symmetric argument holds for epimorphisms: if $\psi : \mathcal{G} \to \mathcal{Q}$ is an epimorphism, then $\alpha \circ \psi = \beta \circ \psi \implies \alpha = \beta$, which translates component-wise to $\psi(U)$ being an epimorphism and preserving surjectivity in $\mathbf{C}$. Since the functor $-(U)$ preserves both monomorphisms (kernels) and epimorphisms (cokernels), it is exact on presheaves.

Turning to the category of sheaves $\mathbf{Sh}(X)$, the functor $-(U)$ remains left-exact. This is because the inclusion $\mathbf{Sh}(X) \hookrightarrow \mathbf{PSh}(X)$ preserves limits: the sheafification function $(-)^{\mathrm{sh}} : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$ is *left adjoint* to the inclusion $\iota : \mathbf{PSh}(X) \to \mathbf{Sh}(X)$, hence the inclusion is a *right adjoint* so by ref:HERE (put it in section 0.4) it preserves limits. As kernels are limits they are preserved. Consequently, if φ is a monomorphism of sheaves, it is also a monomorphism of presheaves, and as shown above, $\varphi(U)$ is a monomorphism (and thus injective) in $\mathbf{C}$. However, $-(U)$ is not right-exact on sheaves. While epimorphisms in presheaves imply component-wise surjectivity, there are not enough sheaves for component-wise surjectivity of sheaves and sheafification is required. As seen in example 1.2.7, a map can be an epimorphism of sheaves (locally surjective) without the component map $\varphi(U)$ being surjective, failing exactness on the right (ex. take the global section functor Γ which is will not perserve exactness of example 2).

1.2.17 Remark In Vakil's proof of this result, [15], he uses “”indicator sheaves” to verify these cancellations. In categorical terms, to test the value of a section $s \in \mathcal{F}(U)$, one uses the Yoneda Lemma. The indicator sheaf mentioned corresponds to the representable presheaf h_U (which is the sheafification of the constant section on subsets of U and empty elsewhere). Mapping out of h_U allows one to probe the specific algebraic properties of $\mathcal{F}(U)$ essentially element-by-element.

Proposition 1.2.18: Exactness of Presheaves On The Level of Open Sets

An sequence of presheaves

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \cdots \to \mathscr{F}_n \to 0$$

is exact if and only if the sequence

$$0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$$

is exact for all U.

***Proof*:**
The only if direction is induction on proposition 1.2.16. Conversely, if $0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$ is exact for each open $U \subseteq X$, then as we already have morphism of presheaves we have these commute with restrictions and $\mathrm{im}\mathscr{F}_i = \ker \mathscr{F}_i$, completing the proof.

Corollary 1.2.19: Exactness of Stalk For Presheaves

Let $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ be an exact sequence of presheaves. Then if filtered colimits exists (so that stalks exist), then

$$0 \to \mathscr{F}_p \to \mathscr{G}_g \to \mathscr{H}_p \to 0$$

is exact.

***Proof*:**
Use that each functor $-(U)$ is exact for each U; so we can take their colimit.

Note that for presheaves, exactness on all stalks does not imply exactness on presheaves as example 1.2.7(2) shows.

Theorem 1.2.20: Presheaves Over Ab Category Form Ab Category

The category $\mathbf{PSh_C}(X)$ over an abelian category C is an abelian category

***Proof*:**
The main thing that wasn't checked was how to define addition of morphism. By the above work, this can be done on each open set: if $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ then $\varphi + \psi$ defined via open sets as:

$$\varphi + \psi(U) := \varphi(U) + \psi(U)$$

is a map of presheaf (it respects restriction by the commutativity of both natural transformation), and so the hom-sets form an abelian group. Then as we also have a 0-object, and we have finite products (again, something the reader could verify), we have an additive category.

To show kernels and cokernels exist, by proposition 1.2.18 it suffices to check on the level of open sets. Then we define:

$$(\ker_{\text{pre}} \varphi)(U) = \ker \varphi(U)$$

The key is to consider the following diagram where $U \hookrightarrow V$:

$$\begin{tikzcd} 0 \arrow[r] & \ker_{\text{pre}}(\varphi(V)) \arrow[r] \arrow[d, dashed, "\exists!"] & \mathcal{F}(V) \arrow[r] \arrow[d, "\text{res}_{V,U}"] & \mathcal{G}(V) \arrow[d, "\text{res}_{V,U}"] \\ 0 \arrow[r] & (\ker_{\text{pre}} \varphi)(U) \arrow[r] & \mathcal{F}(U) \arrow[r] & \mathcal{G}(U) \end{tikzcd}$$

which shows that this is indeed well-defined. The reader can further check that this does indeed satisfy the universal property of the kernel. By dualizing the argument, we can do the same for the cokernel, and hence the additive category of presheaves has kernels and cokernels, making it abelian.

To keep track of all morphisms between open sets, we box the following definition:

Definition 1.2.21: Sheaf Hom

Let $\mathscr{F}, \mathscr{G} \in \mathbf{Sh_C}(X)$. Then the *sheaf hom* is defined to be:

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) := \text{Mor}(\mathscr{F}_U, \mathscr{G}|_U)$$

where $\text{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$ are all morphism of (pre)sheaves. The dual of $\mathscr{F}$ is:

$$\mathscr{F}^\vee = \text{Hom}_{\text{Mod}_{\mathcal{O}_X}}(\mathscr{F}, \mathcal{O}_X)$$

Just like Hom, *Hom* is a (co/contra)variant functor, depending on which position you choose.

The situation is not so clean for $(-)_p$, namely the functor on $\mathbf{Sh_C}(X)$ is not always exact for *any* abelian category; it need not be the case that filtered colimits preserve exactness, and hence that $(-)_p$ preserve exactness (think of the category of finite abelian groups which doesn't have $\varinjlim_p \mathbb{Z}/p^n\mathbb{Z}$). Grothendieck axiomatically added this condition to abelian categories (such categories are called *Grothendieck categories*). Fortunately for us, this will never be an issue as all the categories of interest to us shall have this property, namely:

1. abelian groups, more generally R-Mod
2. Sheaves of abelian groups (more generally $\mathcal{O}_X$-modules which are defined later)
3. the product of Grothendieck categories
4. In ref:HERE, it shall be important to have internal Hom's: given a small category $\mathbf{C}$ and a Grothendieck category $\mathbf{G}$, all functors from $\mathbf{C}$ to $\mathbf{G}$ (with morphism being natural transformation) form Grothendieck categories.

Proposition 1.2.22: Sheaves and exactness For Stalk

Let $\mathbf{Sh}(X)$ be the category of sheaves over any R-Mod. Then $(-)_p : \mathbf{Sh}(X) \to R\text{-Mod}$ is exact.

***Proof*:**
Recall that the stalk of a sheaf $\mathcal{F}$ at a point p is defined as the colimit over the filtered system of open neighborhoods of p:

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U)$$

By the general properties of abelian categories, the colimit functor is always right-exact (it preserves cokernels). The non-trivial property required is left-exactness. In the category of R-modules (unlike the category of finite abelian groups mentioned above), filtered colimits commute with finite limits. Consequently, the filtered colimit commutes with kernels.

Therefore, if $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is an exact sequence of sheaves, applying the stalk functor yields a sequence of filtered colimits. Since exactness is preserved by filtered colimits in R-Mod, the resulting sequence of stalks

$$0 \to \mathcal{F}_p \to \mathcal{G}_p \to \mathcal{H}_p \to 0$$

is exact.

With the right notion of compatible stalks defined, we may show how to determine morphisms of sheaves at the level of stalks, namely a scheme morphisms is determined by stalks, and a scheme morphism that is an isomorphism on stalks induces an inverse scheme morphism:

Proposition 1.2.23: Morphisms Determined By Stalks

Let φ_1, φ_2 be morphisms from a presheaf of sets $\mathscr{F}$ to a *sheaf* of sets $\mathscr{G}$ that induce the same maps on each stalk. Then $\varphi_1 = \varphi_2$

***Proof*:**
For every open U, $\varphi_1(U) = \varphi_2(U)$. Let $s \in \mathscr{F}(U)$. Since $\mathscr{G}$ is a sheaf, the section $t = \varphi_1(U)(s)$ is uniquely determined by its germs $(t_p)_{p\in U}$.

For any $p \in U$, the germ of the image is the image of the germ:

$$(\varphi_1(s))_p = (\varphi_1)_p(s_p)$$

By assumption, $(\varphi_1)_p = (\varphi_2)_p$, so:

$$(\varphi_1)_p(s_p) = (\varphi_2)_p(s_p) = (\varphi_2(s))_p$$

Since $\varphi_1(s)$ and $\varphi_2(s)$ have identical germs at every point in U, and $\mathscr{G}$ is a sheaf (satisfying the identity axiom), the sections must be identical. Thus $\varphi_1(U)(s) = \varphi_2(U)(s)$.

The above proof can be summarized by the following diagram:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r] \arrow[d, hook] & \mathscr{G}(U) \arrow[d, hook] \\ \prod_{p\in U} \mathscr{F}_p \arrow[r] & \prod_{p\in U} \mathscr{G}_p \end{tikzcd}$$

Proposition 1.2.24: Stalks-Local Isomorphism induces Isomorphism

Let φ be a morphism of sheaves. Then φ is an isomorphism if and only if it induces an isomorphism for all stalks.

***Proof*:**
Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a sheaf morphism. If φ is an isomorphism, then certainly each φ_p is an isomorphism, hence say each φ_p is an isomorphism. Then it must be shown that each $\varphi(U)$ is an isomorphism. For injectivity, say for $s \in \mathscr{F}(U)$ $\varphi(s) = 0 \in \mathscr{G}(U)$. Then for each $p \in U$, $\varphi(s)_p = 0 \in \mathscr{G}_p$. As φ_p is injective at each p, we have $s_p = 0 \in \mathscr{F}_p$. Using the definition of stalk, we have that s is zero on some open set, namely $s|_{W_p} = 0$ for $W_p \subseteq U$. As U is covered by such neighborhoods, by the gluing axiom we have $s = 0$

For surjectivity, let $t \in \mathscr{G}(U)$. Consider $t_p \in \mathscr{G}_p$. As φ_p is surjective, we have $s_p \in \mathscr{F}_F$ such that $\varphi_p(s_p) = t_p$. Now, let $(s(p), V_p)$ be a representative of s_p on the set V_p containing the point p. Then $\varphi(s(p))$ and $t|_{V_p}$ are two elements of $\mathscr{G}(V_p)$ whose germs at p are the same. Thus, replacing V_p by a smaller neighborhood if necessary, we have $\varphi(s(p)) = t|_{V_p} \in \mathscr{G}(V_p)$.

Now, U is covered by open sets V_p where on each V_p we have $s(p) \in \mathscr{F}(V_p)$ and $\varphi(s(p)) = t|_{V_p}$. If p, q are two points, then $s(p)|_{V_p\cap V_q}$ and $s(q)|_{V_p\cap V_q}$ are two sections of $\mathscr{F}(V_p \cap V_q)$ which are both sent to $t|_{V_p\cap V_q}$ via φ. As φ was shown to be injective, they are equal. then by the gluing axiom, there exists an $s \in \mathscr{F}(U)$ such that

$$s|_{V_p} = s(p) \qquad \forall p \in U$$

Finally, to show that $\varphi(s) = t$, notice that $\varphi(s)$ and t are two section of $\mathscr{F}(U)$ where for each p $\varphi(s)|_{V_p} = t|_{V_p}$, hence they glue to create $\varphi(s) = t$, as we sought to show.

This shows that a *morphism* $\varphi : \mathscr{F} \to \mathscr{G}$ can be determined on the level of stalks. This does not imply that if two *sheaves* have isomorphic stalks, then they are isomorphic! You need to already have a sheaf homomorphism which in turn induces the isomorphic stalks. Conceptually, this is saying that being locally isomorphic does not imply it is globally isomorphic; there are plenty concrete example form differential geometry to choose from.

Example 1.2.25: Isomorphic Stalks, Not Isomorphic Sheaves

1. Let X be a Hausdorff space. Consider the constant sheaf $\underline{\mathbb{Z}}$ and the sky-scrapper sheaf $\iota_*(\mathbb{Z})$. Show that the stalks at each point is $\mathbb{Z}$, however these two sheaves are not isomorphic.
2. The following will rely on the reader's intuitions of varieties (see [9]). Take the variety associated to the coordinate ring:
$$A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$$
which is the 2-sphere in $\mathbb{R}^3$. Take the tangent bundle, i.e. A-module, of all derivations $E = \mathrm{Der}_{\mathbb{R}}(A)$. Then E is a rank 2 locally trivial module since for any open subset or distinguished open subset, $E_x \cong \mathbb{R}^2$, showing each stalk is isomorphic. However, $E \not\cong A^2$.
3. Another famous example would be the trivial line bundle on S^1 vs. the möbius bundle.

As stalks define maps on some open neighbourhood, we can use this condition to find the closest condition that allows to check if a map between topological spaces is sheaf map on the level of open sets:

Proposition 1.2.26: Characterizing Stalks Inducing Sheaf Morphism

Let X be a topological space with two sheaves $\mathscr{F}, \mathscr{G}$. Let $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ be a collection of stalk morphisms for all $p \in X$. Then this gives a map of sheaves $\Phi : \mathscr{F} \to \mathscr{G}$ if and only if for all open subsets $U \subseteq X$, for all sections $s \in \mathscr{F}(U)$, there is an open cover $U = \cup_i U_i$ and sections $t_i \in \mathscr{G}(U_i)$ such that
$$\varphi_p(s_p) = (t_i)_p$$
for all $p \in U_i$ and all i.

***Proof*:**
($\Rightarrow$) Assume $\Phi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves. Then for any open U and section $s \in \mathscr{F}(U)$, we can simply take the trivial cover $U_1 = U$. Define $t_1 = \Phi(U)(s) \in \mathscr{G}(U)$. By the definition of the induced map on stalks (proposition 1.2.15), for any $p \in U$, the map φ_p sends the germ of s to the germ of $\Phi(U)(s)$. Thus, $\varphi_p(s_p) = (t_1)_p$, satisfying the condition.

($\Leftarrow$) Conversely, assume the local condition holds. We must construct a morphism $\Phi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for every open set U. Let $s \in \mathscr{F}(U)$. By hypothesis, there exists a cover $\{U_i\}$ and sections $t_i \in \mathscr{G}(U_i)$ such that for every $p \in U_i$, the germ of t_i at p is $\varphi_p(s_p)$.

We first show that these local sections t_i glue together. Consider the intersection $U_{ij} = U_i \cap U_j$. For any point $p \in U_{ij}$, we have:
$$(t_i)_p = \varphi_p(s_p) = (t_j)_p$$
Since $\mathscr{G}$ is a sheaf (specifically, a separated presheaf), sections are determined by their germs (lemma 1.1.15). Because t_i and t_j have identical germs at every point in U_{ij}, they must be equal as sections:
$$\mathrm{res}_{U_i, U_{ij}}(t_i) = \mathrm{res}_{U_j, U_{ij}}(t_j)$$
Since the sections agree on overlaps, the Gluing Axiom for $\mathscr{G}$ implies there exists a unique section $t \in \mathscr{G}(U)$ such that $t|_{U_i} = t_i$. We define $\Phi(U)(s) = t$.

To ensure Φ is well-defined, suppose there was another cover $\{V_j\}$ and sections $\{r_j\}$ satisfying the condition, yielding a glued section t'. For any $p \in U$, $t_p = \varphi_p(s_p)$ (by the construction on the cover U_i) and $t'_p = \varphi_p(s_p)$ (by the construction on V_j). Thus t and t' have the same germs everywhere, and since $\mathscr{G}$ is a sheaf, $t = t'$.

Finally, Φ commutes with restrictions naturally because the construction is defined pointwise via germs: for $V \subseteq U$, the germs of $\Phi(V)(s|_V)$ are $\varphi_p((s|_V)_p) = \varphi_p(s_p)$, which matches the germs of $(\Phi(U)(s))|_V$.

Summarizing the results for morphisms of sheaves:

Theorem 1.2.27: Equivalence of Mono- and Epimorphisms

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X (over **Set** or an abelian category).

The following conditions are equivalent:

1. φ is a monomorphism in the category of sheaves.
2. φ is injective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is injective for all $p \in X$.
3. φ is injective on the level of open sets: $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open $U \subseteq X$.

The following conditions are equivalent:

1. φ is an epimorphism in the category of sheaves.
2. φ is surjective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ is surjective for all $p \in X$.

If $\varphi(U)$ is surjective for all open sets U, then φ is an epimorphism (surjectivity on sections implies surjectivity on stalks). The converse is *false*: an epimorphism of sheaves need not be surjective on global sections (as seen in the exponential sequence).

***Proof*:**
This is combining the results from earlier. In particular, apply the results just proven about stalks to the proof of proposition 1.2.16.

Theorem 1.2.28: Sheaves Over Ab Category Form Ab Category

The category $\mathbf{Sh_C}(X)$ over an abelian category C is an abelian category

***Proof*:**
exercise.

Morphisms and Stalks

Since the stalk functor $(-)_p$ is exact by proposition 1.2.22, it preserves all finite limits and colimits. This yields the following immediate corollaries.

Corollary 1.2.29: Commuting Kernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the kernel is the kernel of the stalk:

$$(\ker(\varphi))_p \cong \ker(\varphi_p)$$

Proof:
Since $(-)_p$ is an exact functor, it is left-exact. Left-exact functors preserve kernels.

Corollary 1.2.30: Commuting Cokernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the cokernel is the cokernel of the stalk:

$$(\operatorname{coker}(\varphi))_p \cong \operatorname{coker}(\varphi_p)$$

Proof:
Since $(-)_p$ is an exact functor, it is right-exact. Right-exact functors preserve cokernels.

Due to sheafification, the surjectivity of a sheaf morphism is no longer conditional on the surjectivity of sections on all open sets. This leads to the following counter-intuitive examples.

Example 1.2.31: Surjective on Stalks $\neq$ Surjective on Global Sections

1. (if you know Complex Analysis): Let $X = \mathbb{C} \setminus \{0\}$. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and Ω^1_X be the sheaf of holomorphic differential 1-forms. Define $\varphi : \mathcal{O}_X \to \Omega^1_X$ by $f \mapsto df$.
   - *Locally:* On any simply connected disk U, every closed form is exact. Since holomorphic forms are closed, $\omega = df$ has a solution. Thus φ_p is surjective.
   - *Globally:* The form $\omega = \frac{dz}{z}$ is a global section of $\Omega^1_X(X)$. However, its primitive is $\log(z)$, which is not a function on $\mathbb{C} \setminus \{0\}$. Thus $\frac{dz}{z} \notin \operatorname{im}(\varphi(X))$.
2. Let $X = \mathbb{R}$, $\mathscr{F} = \underline{\mathbb{Z}}$ (constant sheaf), and $\mathscr{G} = i_{p*}(\mathbb{Z}) \oplus i_{q*}(\mathbb{Z})$ (skyscrapers at distinct points p, q). Define $\varphi : \mathscr{F} \to \mathscr{G}$ by restriction.
   - *Stalks:* At p, $\mathscr{F}_p = \mathbb{Z}$ and $\mathscr{G}_p = \mathbb{Z} \oplus 0 \cong \mathbb{Z}$. The map is identity (surjective). Similarly at q. Elsewhere, both are 0.
   - *Globally:* $\mathscr{F}(X) = \mathbb{Z}$ (constant functions). $\mathscr{G}(X) = \mathbb{Z} \oplus \mathbb{Z}$. The map sends $n \mapsto (n, n)$. The image is the diagonal, which is not the whole plane $\mathbb{Z} \oplus \mathbb{Z}$.

Corollary 1.2.32: Commuting Image and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the image is the image of the stalk:

$$(\mathrm{im}(\varphi))_p \cong \mathrm{im}(\varphi_p)$$

***Proof*:**
In an abelian category, $\mathrm{im}(\varphi) = \ker(\mathrm{coker}(\varphi))$. Since $(-)_p$ commutes with both kernels and cokernels, it commutes with images.

1.2.33 Summary

- **Presheaves** form an abelian category because computation is done on the level of open sets.
- **Sheaves** form an abelian category because computation is done on the level of stalks.

Exercise 1.2.1

1. Show that $\varphi : \mathscr{F} \to \mathscr{G}$ is surjective if and only if for every $s \in \mathscr{G}(U)$, there exists an open covering $\bigcup_i U_i = U$ and elements $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ (hint: recall φ is surjective if and only if it is surjective on stalks).

   This exercise will come back in proposition 5.1.15

2. Let $(X, \mathcal{O}_X)$ be a ringed space and $U \subseteq X$ an open subset. Show that we can define non-vanishing distinguished open subset by:

$$\begin{tikzcd} U_f \arrow[r] \arrow[d] & \mathcal{O}_X^* \arrow[d] \\ U \arrow[r, "f"] & \mathcal{O}_X \end{tikzcd}$$

   that is, $U_f = f^{-1}(\mathcal{O}_X^*) = \left\{ x \in U \;\middle|\; f(x) \in \mathcal{O}_{X,x}^* \right\}$. Show these form a sheaf.

3. Define a *local ringed space* to be a ringed space $(X, \mathcal{O}_X)$ such that for all open $U \subseteq X$ and all $f \in \mathcal{O}_X(U)$, $U = U_f \cup U_{1-f}$, namely for all $x \in U$, f or $1 - f$ is invertible.
   a) Show that if X, $\mathrm{mSpec}\,\mathcal{O}_X(X)$ is a singleton.
   b) Let $C(X)$ be all continuous real function on X. Show that $(X, \mathbb{C}(X))$ is local.

4. Let $(X, \mathcal{O}_X)$ is a local ringed space, $U \subseteq X$ open, and $f \in \mathcal{O}_X(U)$. Then if $f_1 + \cdots + f_r = 1$, then $\bigcup U_i = U$.

5. Let $\mathscr{F}$ be a sheaf of sets, show that $\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$

6. Let $\mathscr{F}$ be a sheaf of groups, show that $\mathcal{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$

7. It is tempting to then think that $\mathcal{H}om(\mathscr{F}, \mathscr{G})_p \cong \mathrm{Hom}(\mathscr{F}_p, \mathscr{G}_p)$. However this is *not* the case: think of a counter-example. There is a map from one of these sets to another, what is it?

1.3 Grothendieck's Functors: Pushforward, Pullback, and more

We next endeavor to move sheaves around from one topological space to another given a continuous map $f : X \to Y$. Grothendieck identified 4 key ways of doing so that come in adjoint pairs:

1. The pushforward and pullback: f_* and f^*

2. the proper direct image and proper inverse image: $f_!$ and $f^!$

These are some of the key functors in Grothendieck *6 functor formalism* where the adjoint pair Hom and $\otimes$ is missing. These functor's are more naturally introduced in a cohomological setting and will be present in chapter 6.

1.3.1 Pushforward and Pullback

The easiest of these is the pushforward. If $U \subseteq V \subseteq Y$ and $f : X \to Y$, then $f^{-1}(U) \subseteq f^{-1}(V)$. Hence, f^{-1} can be thought of as a contravariant functor preserving $\subseteq$. In fact, more is true: f^{-1} also preserves the properties in proposition 1.1.9, and hence works on sheaves as well:

Definition 1.3.1: Pushforward Of (Pre)sheaf

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{PSh}(X)$. Then the *pushforward* or *direct image* of $\mathscr{F}$ through f is defined on every open set $U \subseteq Y$ to be

$$f_*\mathscr{F}(U) = \mathscr{F}(f^{-1}(U))$$

and induces a (per)sheaf on Y. The pushforward defines a [covariant] functor $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ sending:

$$f_* : \mathscr{F} \to f_*(\mathscr{F})$$

If working with a particular category, for example **Set**, we can write $f_* : \mathbf{Set}_X \to \mathbf{Set}_Y$.

The term "pushforward" comes from the fact that we are taking a sheaf on X and defining a sheaf on Y through the map $f : X \to Y$. The pullback shall do the same thing in reverse.

It is perhaps unsurprising that f^{-1} preserves pre-sheaves, however it is a little less obvious that it preserves sheaves as well; it is worth thinking this through. To me, this shows that sheaves are a very natural concept, as the usually topological function f^{-1} naturally preserves them. This also motivates the Grothendieck topology that will be introduced much later. We have already seen an example of a pushforward: the skyscraper sheaf is induced that pushing forward along the inclusion map $\iota_p : \{p\} \to X$ the constant sheaf $\underline{S}$.

Proposition 1.3.2: Pushforward Induces Map On Stalks

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{Sh}(X)$. Then for each $\pi(p) = q$, there exists a map

$$(f_*\mathscr{F})_q \to \mathscr{F}_p$$

between the stalks of $f_*\mathscr{F}$ and $\mathscr{F}$.

***Proof*:**
exercise. This can be verified using the equivalence class definition or the universal property. Doing it using equivalence classes, define the map

$$[s]_q \mapsto [f^{-1}(s)]_p$$

Proposition 1.3.3: Sheaf Morphisms Determined by a Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have a morphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf morphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
Let $U \subseteq X$ be an arbitrary open set. Since $\mathcal{B}$ is a basis, there exists a cover $U = \bigcup_{i \in I} B_i$ where $B_i \in \mathcal{B}$. We wish to construct a map $\Phi_U : \mathcal{F}(U) \to \mathcal{G}(U)$.

Take a section $s \in \mathcal{F}(U)$. We restrict this section to the basis elements to get $s_i = s|_{B_i} \in \mathcal{F}(B_i)$. Using our hypothesis, we can map these to the target sheaf:

$$t_i := \varphi_{B_i}(s_i) \in \mathcal{G}(B_i)$$

We now claim that the collection $\{t_i\}$ glues together to form a global section $t \in \mathcal{G}(U)$. By the sheaf axioms for $\mathcal{G}$, it suffices to show that t_i and t_j agree on the overlap $B_i \cap B_j$.

Unlike the specific case of affine schemes, the intersection $B_i \cap B_j$ might not be a single basis element. However, by the definition of a topology basis, we can cover the intersection by smaller basis elements:

$$B_i \cap B_j = \bigcup_k V_k \quad \text{where } V_k \in \mathcal{B}$$

Since $V_k \subseteq B_i$, our compatibility hypothesis tells us that restricting the mapped section is the same as mapping the restricted section:

$$t_i|_{V_k} = (\varphi_{B_i}(s|_{B_i}))|_{V_k} = \varphi_{V_k}(s|_{V_k})$$

By the exact same logic for j, we have $t_j|_{V_k} = \varphi_{V_k}(s|_{V_k})$. Thus, t_i and t_j agree on every V_k covering the intersection. Since $\mathcal{G}$ is a sheaf (specifically, using the identity axiom locally), this implies t_i and t_j agree on the entire intersection $B_i \cap B_j$.

Since the sections are compatible, there exists a unique $t \in \mathcal{G}(U)$ such that $t|_{B_i} = t_i$. We define $\Phi_U(s) = t$.

To ensure this is well-defined, one must check that choosing a different basis cover for U yields the same result. This follows from standard refinement arguments (taking the union of two covers and applying the uniqueness of gluing on the common refinement). Uniqueness of Φ is guaranteed because two sheaf morphisms agreeing on a basis must agree everywhere.

Corollary 1.3.4: Sheaf Isomorphism Determined By A Basis

Let X be a topological space and $\mathcal{B}$ a basis for the topology. Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. Suppose that for every basis set $B \in \mathcal{B}$, we have an isomorphism $\varphi_B : \mathcal{F}(B) \to \mathcal{G}(B)$ such that for any inclusion of basis sets $V \subseteq U$ (where $U, V \in \mathcal{B}$), the maps commute with restriction:

$$\varphi_V(s|_V) = (\varphi_U(s))|_V$$

Then there exists a unique sheaf isomorphism $\Phi : \mathcal{F} \to \mathcal{G}$ extending this collection.

***Proof*:**
As each φ_B is an isomorphism, they each have an inverse. It is easy to show that they satisfy the same restriction compatibility condition, hence by proposition 1.3.3 there exists a global inverse morphism, and hence φ is an isomorphism.

Pullback

So far, most of the discussion of sheaves where with the assumption of a topological X. In example 1.1.12, there were examples of sheaves being pushed forward (Definition 1.3.1) via a continuous map $f : X \to Y$. Namely the sheaf $\mathscr{F}$ maps to the sheaf $f_*\mathscr{F}$ on Y defined via:

$$\pi_*\mathscr{F}(U) = \mathscr{F}\left(f^{-1}(U)\right)$$

There is an adjoint, the *pullback*, called the inverse image sheaf. It is called a pullback as it takes a sheaf $\mathscr{F}$ and Y and defines a sheaf $f^*(\mathscr{F})$ on X. It is not as clear as the case for the push-forward as the image of an open set need not be open, hence the existence is more subtle (similar to the existence of cokernels for groups). The adjoint will be important not only for having more than one perspective on sheaves, but for definition morphisms of schemes.

Definition 1.3.5: Inverse Image Sheaf

Let $f : X \to Y$ be a function function. and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectively. Then the *inverse image sheaf* or *pullback* $f^*\mathscr{G}$ on X is the sheaf associated to the presheaf:

$$f^*_{\text{pre}}\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V)$$

or more concisely

$$f^*\mathscr{G} = (f^*_{\text{pre}}\mathscr{G})^{\text{sh}}$$

Giving us a map $f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$.

1.3.6 On Vakil's Notation Vakil uses the f^{-1} notation intend of f^* as he wants to reserve the f^* notation for pulling back $\mathcal{O}_X$-modules:

$$f^*\mathcal{G} := f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$$

This is because we still want pullback as defined in this section to apply to $\mathcal{O}_X$-modules. We shall instead keep track of which category we are performing the operations.

The construction of the sheaf is done in the usual quotient way: the elements of $f^*_{\text{pre}}\mathscr{F}(U)$ are equivalent classes $(s,U),(t,V)$ where $(s,U) \sim (t,V)$ if there exists a $W \subseteq U \cap V$ where $f(U) \subseteq W$ and the restriction of s,t are equal on W.

The sheafification is crucial:

Example 1.3.7: Presheaf Inverse Image

Let $f : Y \to X$ where $Y = X \coprod X$ and f is the natural projection map and a sheaf $\mathscr{G}$ defined on X. Then by definition, we would get $f^*_{\text{pre}}\mathscr{G}(U) = \mathscr{G}(U)$, however the sheafification will get:

$$f^*\mathscr{G}(U) = \mathscr{G}(U) \times \mathscr{G}(U)$$

showing the two are in general not equal.

Proposition 1.3.8: push-forward Sheaf Adjoint

Let $f : X \to Y$ be a continuous functions and $\mathscr{F},\mathscr{G}$ be sheaves on X and Y respectfully. Then there is a bijection:

$$\text{Mor}_X(f^*\mathscr{G}, \mathscr{G}) \leftrightarrow \text{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$$

That is, they are adjoints

***Proof*:**
exercise.

One advantage of the inverse image sheaf is the computation of stalk

Proposition 1.3.9: Stalks Via Inverse Image Sheaf

Let $f : X \to Y$ be a continuous function with $\mathscr{G}$ a sheaf on Y. Then:

$$(f^*\mathscr{G})_p = \mathscr{F}_{f(p)}$$

***Proof*:**
exercise, follows more naturally via the definition.

In fact, the adjoint of this map is the *skyscraper sheaf*, namely given $f : \{p\} \to X$ where $p \in X$, then $f^*\mathscr{G} = \mathscr{G}_p$. Conversely, if we define the sheaf $\mathscr{F}(\{p\}) = S$, then $f_*\mathscr{F}(U)$ is the skyscraper sheaf.

will get back to this when it is more needed

1.3.2 Proper Direct and Inverse Image

While the standard pushforward f_* and pullback f^* are sufficient for many geometric operations, they lack certain "clean" properties regarding the supports of sections:

Example 1.3.10: Smeared Section

Consider $X = \mathbb{R} - \{0\}$ with the euclidean topology and take the inclusion $j : X \hookrightarrow \mathbb{R}$. Then let $\mathscr{F}$ on X be the $\mathbb{Z}$-constant sheaf. Then what is $j_*\mathscr{F}$ at 0? Consider that on any open neighbourhood $(-\epsilon, \epsilon)$, the pushforward is $(-\epsilon, 0) \cup (0, \epsilon)$. We can choose two arbitrary choices of constant functions. Hence:

$$j_*(\mathscr{F})_0 \cong \mathbb{Z} \oplus \mathbb{Z}$$

it is *not* 0!

For those who already encountered the Zariski topology on $\mathbb{A}^1_k$ (k infinite), consider $X = \mathbb{A}^1_k \setminus \{0\}$ and the inclusion $j : X \hookrightarrow \mathbb{A}^1_k$. Then as the open sets of $\mathbb{A}^1_k$ are cofinite, hence any open subset U containing 0 is cofinite; it is easy to show that $X \cap U$ is always *connected*. Hence, if we put the $\mathbb{Z}$-constant sheaf on X, $j_*(\mathscr{F})_0 = \mathbb{Z}$. The functor $j_!$ that will be defined soon will be characterized by $j_!(\mathscr{F})_x = 0$ if $x \notin X$.

To address this, Grothendieck introduced the "shriek" functors (denoted with an exclamation mark in the subscript, ex. $j_!$, or in Hartshorne with a tick in the subscript, ex. $j_{\prime}$), which handle supports more strictly. These functors behave differently depending on whether the map is an open immersion or a closed immersion.

Extension by Zero (Open Immersion)

Let $j : U \hookrightarrow X$ be an inclusion of an open set. The standard pushforward j_* "smears" the information from U up to the boundary in X. Often, we wish to extend a sheaf from U to X such that it is exactly zero outside of U.

Definition 1.3.11: Extension by Zero

Let $j : U \to X$ be an open immersion and $\mathscr{F}$ a sheaf on U. The *extension by zero*, denoted $j_!\mathscr{F}$, is the sheaf on X associated to the presheaf:

$$j_!(V) = \begin{cases} \mathscr{F}(V) & \text{if } V \subseteq U \\ 0 & \text{otherwise} \end{cases}$$

This functor $j_! : \mathbf{Sh}(U) \to \mathbf{Sh}(X)$ is also called the *proper direct image* or *lower shriek*.

The behavior of this functor is best understood at the level of stalks. Unlike j_*, which may have non-zero stalks on the boundary ∂U, $j_!$ is "clean":

$$(j_!\mathscr{F})_p = \begin{cases} \mathscr{F}_p & \text{if } p \in U \\ 0 & \text{if } p \notin U \end{cases}$$

Just as f^* is the left adjoint to f_*, the extension by zero provides a *left* adjoint to the restriction.

Proposition 1.3.12: Adjunction of Extension by Zero

Let $j : U \to X$ be an open immersion. Then $j_!$ is the left adjoint to the pullback (restriction) j^*. That is, for $\mathscr{F} \in \mathbf{Sh}(U)$ and $\mathscr{G} \in \mathbf{Sh}(X)$:

$$\mathrm{Mor}_X(j_!\mathscr{F}, \mathscr{G}) \cong \mathrm{Mor}_U(\mathscr{F}, j^*\mathscr{G})$$

***Proof*:**
exercise. (Hint: A map from zero is unique, so the morphisms on X are entirely determined by their behavior on the open set U).

Sections with Support (Closed Immersion)

Conversely, let $i : Z \hookrightarrow X$ be a closed immersion. The standard pullback i^* restricts sections to Z, effectively "forgetting" everything outside. The right adjoint to the pushforward i_* is the functor that finds sections supported *only* on Z.

Definition 1.3.13: Exceptional Inverse Image

Let $i : Z \to X$ be a closed immersion. The *exceptional inverse image*, denoted $i^!\mathscr{F}$ (or sometimes $\mathcal{H}^0_Z(\mathscr{F})$), is the functor $i^! : \mathbf{Sh}(X) \to \mathbf{Sh}(Z)$ defined by the subgroup of sections:

$$\Gamma(V, i^!\mathscr{F}) = \{s \in \Gamma(V, i^*\mathscr{F}) \big| \mathrm{supp}(s) \subseteq Z\}$$

It serves as the right adjoint to the pushforward i_*.

1.3.3 Hom and Tensor

here (I think this is a good place for it, to round off the 6-functor formalism.)

Summary of the Six Operations

It is common to group these functors into a "Six Operation" formalism. While defined generally for derived categories, for sheaves on topological spaces, they exhibit a specific symmetry between open and closed maps.

1. **For an Open Immersion j:** The exceptional inverse image collapses to the standard pullback ($j^! \cong j^*$), leaving us with the triple $(j_!, j^*, j_*)$.
2. **For a Closed Immersion i:** The proper direct image collapses to the standard pushforward ($i_! \cong i_*$), leaving us with the triple $(i^*, i_*, i^!)$.

These functors allow for the decomposition of a sheaf $\mathscr{F}$ on X into its parts on an open set U and the closed complement Z via the fundamental exact sequence:

$$0 \to j_!j^*\mathscr{F} \to \mathscr{F} \to i_*i^*\mathscr{F} \to 0$$

Exercise 1.3.1

1. Let $U \subseteq Y$ open and $\iota : U \hookrightarrow Y$ be the inclusion. Show that $\iota^*\mathscr{G} \cong \mathscr{G}|_U$.
2. Show that f^* is an exact functor (naturally on abelian categories, however is just as good to show it for any ${}_R\text{Mod}$)
3. Let $U \subseteq X$ be an open subset of a topological space so that $V = X \setminus U$ is closed. Let $\mathscr{F}$ be a sheaf on V, and $\iota : V \hookrightarrow X$ the inclusion.
   a) Show that $(\iota_*\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in V$ and 0 otherwise.
   b) Let $j_!(\mathscr{F})$ be a sheaf on X given by
   $$W \mapsto \mathscr{F}(W) \qquad W \mapsto 0$$
   where the first happens when $W \subseteq V$, and the second in all other cases. Show that $(j_!\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in U$ and 0 otherwise. .
   c) conclude there is a short exact sequence $0 \to j_!(\mathscr{F}|_U) \to \mathscr{F} \to \iota_*(\mathscr{F}|_V) \to 0$.

1.4 Ringed Space and Locally Ringed Space

Having defined the morphisms in $\mathbf{Sh}(X)$, an important class of sheaves to distinguish are sheaves over rings, as one of the ultimate goals in this Part of the textbook is the generalize the Nullstellensatz to arbitrary (commutative) rings:

Theorem 1.4.1: (Commutative) Ringed Objects in Sheaf

let $\mathbf{Sh}_{\mathbf{C}}(X)$ be the category of sheaves over a category $\mathbf{C}$ that has finite products and a terminal object. Then commutative ringed object exists, and the subcategory $\mathcal{R} \subseteq \mathbf{Sh}_{\mathbf{C}}(X)$ of commutative ringed object is equivalent to the category of sheaves $\mathbf{Sh}_{\mathscr{R}(C)}(X)$ on the *commutative ringed objects* of $\mathbf{C}$.

***Proof*:**
This is a lot of diagram chasing. With some of the proofs later in this section, this result can be proved more easily on the level of stalks.

We shall almost always work in the case of $\mathbf{C} = \mathbf{Set}$ so that $\mathscr{R}(\mathbf{C}) = \mathbf{Ring}$:

Definition 1.4.2: Ringed Space And Structure Sheaf

Let $\mathbf{Sh}_{\mathbf{Ring}}(X)$ be the category of sheaves over $\mathbf{Ring}$ on X. Then given a sheaf of [commutative] rings over X, denote it $\mathcal{O}_X$, the tuple $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*, and the sections of the structure sheaf $\mathcal{O}_X$ over an open subset U are called *[regular] functions on U*.

If it is unambiguous, the $\mathcal{O}_X$ is dropped and X is simply called the ringed space.

1.4.3 Warning Though these are called functions, they are elements of an arbitrary [commutative] ring. The reader should have regular function from [9, chapter 22.3] in mind.

Before defining morphisms formally, it is useful to understand why the sheaf morphism appears to go in the "opposite" direction. Geometrically, if we have a continuous map $f : X \to Y$ and a "function" φ on an open set $V \subseteq Y$ (i.e., $\varphi \in \mathcal{O}_Y(V)$), we naturally obtain a function on X by *pulling back* φ via composition: $\varphi \circ f$. The domain of this new function is $f^{-1}(V)$.

In the language of sheaves, this operation assigns to a section in $\mathcal{O}_Y(V)$ a section in $\mathcal{O}_X(f^{-1}(V))$. Recall that by definition of the pushforward sheaf, $(f_*\mathcal{O}_X)(V) := \mathcal{O}_X(f^{-1}(V))$. Therefore, the geometric notion of pulling back functions induces an algebraic map $\mathcal{O}_Y(V) \to (f_*\mathcal{O}_X)(V)$ for every open set V, or globally, a sheaf morphism $\mathcal{O}_Y \to f_*\mathcal{O}_X$.

Definition 1.4.4: Ringed Space Morphism

A map $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_X)$ is a *ringed space morphism* if $f : X \to Y$ is a continuous map and $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a sheaf morphism.

It is easy to see that these form a category and shall often be denoted **RingedSp**.

Example 1.4.5: Ringed Space

1. The simplest and prototypical ringed space is again the sheaf of [real/complex] continuous/smooth/holomorphic functions on a manifold/variety with component-wise addition and multiplication.
2. Schemes in the next part shall form a large class of ringed spaces
3. vector fields on a manifold do not form a ringed space as they are not closed under multiplication (they are closed under the lie bracket operation). However, they shall form a $\mathcal{O}_X$-module over manifold, as shall be explored in chapter 5.

An important sheaf that shall often be used is the sheaf of units:

Definition 1.4.6: Unit Sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathcal{O}_X^*$ is the sheaf given by the fibered product:

$$\begin{tikzcd}
\mathcal{O}_X^* \arrow[r] \arrow[d] & \{1\} \arrow[d] \\
\mathcal{O}_X \times \mathcal{O}_X \arrow[r, "\cdot"] & \mathcal{O}_X
\end{tikzcd}$$

We shall almost never refer to the categories interpretations of the sheaf of rings in this Part of the book (it shall be important in ref:HERE), however ringed spaces are the central object of study in algebraic geometry.

As elements the sections in $\mathcal{O}_X(U)$ are called functions, there should be a notion of evaluation. When working over coordinate rings, evaluation of f at a point a is equivalent to modding f by the (maximal) ideal $(x-a)$. This works over coordinate rings since by the Nullstellensatz the maximal ideals correspond to points. For a general (commutative) ring, there can be many more pathological behaviours, namely it is not clear how to associate to each point $x \in X$ a maximal ideal $\mathfrak{m} \subseteq R$.

The first natural place to consider is the behaviour at the stalk, namely each point x is associated to the ring $\mathcal{O}_{X,x}$. Is there a natural choice of ideal in this ring? In this case, a natural choice would mean there is only *one* choice of ideal, in which case any $f \in \mathcal{O}_X(U)$ where $x \in U$ maps into $\mathcal{O}_{X,x}$ and can be modded by this maximal ideal. In general the answer is no:

Example 1.4.7: Stalk that is not a Local Ring

Take $X = \{p, q\}$ with the topology given by $\mathcal{T}_X = \{\emptyset, \{p\}, X\}$. Let:

$$\mathcal{O}_X(\emptyset) = \mathbf{0} \qquad \mathcal{O}_X(\{p\}) = k \qquad \mathcal{O}_X(X) = k \times k$$

where k is a field with the only non-trivial restriction map being

$$\mathcal{O}_X(X) \to \mathcal{O}_X(\{p\}) \qquad (a, b) \mapsto a$$

Then the stalk at p and q are:

$$\mathcal{O}_{X,p} = k \qquad \text{and} \qquad \mathcal{O}_{X,q} = k \times k$$

showing that the stalk at q has more than one maximal ideal (namely $k \times \{0\}$ and $\{0\} \times k$).

When working over coordinate rings of varieties, all stalks will be local rings (exercise 1.4.1(1)). In fact, the same holds for (smooth, complex, topological) manifolds [5, section 2.1]. This motivate focusing on ringed spaces whose stalks are local rings:

Definition 1.4.8: Locally Ringed Space

Let X be a ringed space. Then X is a *locally ringed space* if its stalks are local rings.

Definition 1.4.9: Evaluation on Locally Ringed Space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Then as every stalk is a local ring, it has a unique maximal ideal, hence a unique field

$$\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$$

called the *residue field*. Define *evaluation* of an element $f \in \mathcal{O}_X(U)$ where $x \in U$ by defining:

$$f(x) := \overline{f} \bmod \mathfrak{m}_x$$

where $\overline{f}$ is the image of f in $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$.

In general, ringed space morphisms *do not* preserve the local ring structure of the stalks (an example is worked through at the beginning of chapter 3, see example 3.1.1). This condition must be imposed:

Definition 1.4.10: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $(\pi, \pi^{\#}) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^\sharp : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms (the pre-image of the maximal ideal is contained in the maximal ideal).

It can be checked that locally ringed spaces with locally ringed morphisms form a category which shall often be denoted **LocRingedSp** or **LRS**.

Example 1.4.11: Manifolds and Locally ringed sheaves

The two basic examples are varieties and manifolds. Abstracting, topological spaces with continuous function form locally ringed spaces. Complex analytic space (which generalize varieties by taking the vanishing set of holomorphic functions) also form locally ringed spaces. p-adic spaces (central to number theory and modern representation theory) form locally ringed spaces. More generally, Algebraic Spaces and Stacks (defined in Part ref:HERE) form locally ringed spaces.

Overall, locally ringed spaces are the "right" over-arching category when considering the duality between algebra and geometry.

Definition 1.4.12: Isomorphism Of Ringed Spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two ringed spaces. Then an isomorphism or ringed space is a map $\pi : X \to Y$ where:

1. π is a homeomorphism
2. There is an isomorphism of sheaves Between $\mathcal{O}_X$ and $\mathcal{O}_Y$ where two two sheaves are given via the pushforward π_* (i.e. $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ on Y)[a]

[a]By proposition 1.2.24, it is enough to check this isomorphism on stalks

Exercise 1.4.1

1. Show that the stalks of coordinate rings at maximal ideals are all local rings

1.5 *Diffeology

(would love to add a word here)

2

Schemes I: Basics

Recall the following correspondence:

$$\left\{\text{Affine Varieties}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{matrix}\text{Finitely Generated,} \\ \text{Reduced rings} \\ \text{over algebraically closed fields}\end{matrix}\right\} \tag{2.1}$$

This setups an algebraic-geometric correspondence, where the algebraic object can be thought of as functions on the geometric object, or conversely that the algebraic objects can induce a new geometric space with a notion of "points" (or more generally local information).

finish this above!!

The question naturally arises then if this duality can be extended to general commutative rings. In particular, does there exist some topological space such that:

$$\left\{ ??? \right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\text{Commutative Ring}\right\}$$

The necessity of commutativity stems from the need of prime ideals being well-defined[1]. For a more geometric motivation, smooth/holomorphic functions over a real/complex manifold $\mathbb{C}^\infty(M)/\mathcal{H}(M)$ form a commutative ring since they map into $\mathbb{R}$ or $\mathbb{C}$, and hence we copy this behaviour.

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

[1]The theory of "non-commutative geometry" using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see [6])

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some arbitrary (commutative) ring R (the coordinate ring being $R[x]$) currently does not have a good representation since the Nullstellensatz requires k to be an algebraically closed fields. Another example that may at first seem "un-geometric" but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (see [9, chapter 22]). Mathematicians have found that these corresponds naturally to looking for integer solutions (or integral solutions when working over an algebraic ring) of a curves (see [9, chapter 22]). In particular, the element of $\mathbb{Z}[i] = \mathbb{Z}[x]/(x^2 + 1)$ should be functions that "evaluate" at points that would find information mod $\mathfrak{p}$ to solutions of $x^2 + 1$.

2. As we saw in front-matter, localization gives us powerful tools to explore the behavior of a variety at a point. In the theoretical framework given by the Nullstellensatz, we cannot consider such rings, for example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we loose finite generation. However, as we saw these certainly still have interpretations as functions on a geometric space (the former being the rational functions on a variety, the latter being the "germs"[2] around $0 \in \mathbb{A}^1_k$).

3. Varieties loose multiplicity information. Take for example two varieties over $\mathbb{C}$ defined by the equations
$$y = x^2 \qquad \text{and} \qquad y = 0$$
then if we take their intersection, we get a point $(0, 0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:
$$\frac{k[x, y]}{(y, x^2 - y)} = \frac{k[x]}{(x^2)}$$
Usually, we would omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ "doubly intersects" the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a "double-point", while $k[x]/(x)$ does not. This is a achieved by the fact that the ring has nilpotent elements.

4. (If you know differential geometry) A geometrical object that has many algebraic parallels is the vector bundle. Most vector bundles are very algebraic, however many are not projective (cannot be embedded into $\mathbb{P}^n_k$, or even $\mathbb{P}^n_R$ for some ring R). We thus need need to generalize better our notion of projective space. This shall lead to different sheaves on spectra and will be studied in chapter 5.

This build-up will introduce to us the *affine scheme*. Many familiar geometric objects shall look like affine schemes, but not all familiar geometric objects shall look like them (notably, projective space). We will need to generalize a bit more and define a *scheme* which is locally an affine scheme. This shall complete the liaison that algebraic geometry promises between algebra and geometry.

We shall then focus on how what properties of commutative rings translate to geometric properties on a scheme, and conversely when imposing desirable geometric properties how does this translate as algebraic properties. Properties such as being an integral domain, PID, UFD, Noetherian, reduced, and so forth will have geometric interpretations, and conversely notions like dimension, smoothness, open/closed immersions, and so forth shall have algebraic counter-parts.

[2]Somewhat surprisingly, germs are also a "purely algebraic" notion given by nilpotent elements and localization, and hence appear as schemes

Pontificating about Points

(I think it is worth writing here my thoughts on what really constitutes a point. Points are fundamental in geometry, you can say there are the necessary indivisible object, and Treating points as secial collections of functions *I think* makes sense because of the result on locally compact spaces withe the upsisde down pi isomorphism, and the more elementary result on compact groups)

(Also, This may veer into "functor of points" as well as "Yoneda's lemma" as well as Distributions as well as Mitchell's theorem were that any abelian category can be seen as an R-module, and as R-modules have elements, we get something reminiscent to recovering points)

2.1 Defining Spec

Recall that the maximal ideals of $k[x_1, x_2, \cdots, x_n]/V$ correspond to points in a variety V (theorem 0.1.12). It may then be natural to say that the collection of maximal ideals should be considered the "points" of an arbitrary ring R. These collection were called the spectrum (plural spectra). For such an association to work, there would need to be a functor translating maps between rings to maps between the collection of maximal ideals. In particular, we would want that the functor translating a ring homomorphism $\varphi : R \to S$ to the associated continuous function mapping the maximal ideal $\mathfrak{m} \subseteq S$ to a maximal ideal $\varphi^{-1}(\mathfrak{m})$ (recall that regular function corresponds contra-variantly to k-algebra homomorphisms), However, the pre-image of a maximal ideal need not be maximal breaking functoriality (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$).

When the theory was originally being developed within spectral theory from functional analysis (see [6]), the spaces were compact Hausdorff spaces, in which case it would have sufficed to work with the maximal ideals as within this setting the pre-image of maximal ideals (of the ring of continuous functions) is usually maximal. However, as demonstrated in the example above this doesn't hold for general rings.

Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(P)$ is prime, making it the more natural choice of "points" for functorial purposes, even though it shall produce some more "complicated" points (see example 2.1.3). Working with primes also brings the added benefit of incurring a lot more structure which shall turn out to be very insightful and make the addition of primes not only necessary for the definition to work, but to explain a lot of phenomena[3].

Definition 2.1.1: Spectrum

Let R be an [arbitrary] commutative ring. Then $\operatorname{Spec} R$ is the collection of prime ideals of R. Let $\operatorname{mSpec}(R)$ represent the collection of maximal ideals of R.

We call an element of $\operatorname{Spec} R$ a *point*, and we call an element of R a *regular functions*

In the special case where R is a coordinate ring $k[x_1, ..., x_n]/I(V)$, $\operatorname{Spec} R$ corresponds to points of an affine variety as well as "extra" points associated to primes that are not maximal (ex. $(y-x^2) \subseteq k[x, y]$)

[3]For example, recall that if k was not algebraically closed, there exists regular functions that are not polynomials. Another is the distinct characterization of divisors in number theory which unifies describing the same cohomological properties of schemes

along with the zero ideal which is prime in integral coordinate rings; the intuition behind having these extra points shall be elucidated throughout this section.

To avoid confusion between points of $\operatorname{Spec} R$ and ideals of R, we shall write points in square brackets, for example $[\mathfrak{p}] \in \operatorname{Spec} R$ for $\mathfrak{p} \subseteq R$. There is some merit in writing it in the same notation as the equivalence relation notation, as points on in the topology are defined up to the radical of an ideal, for example for $(x^2) \subseteq \mathbb{C}[x]$, $V(\{(x^2)\}) = [(x)]$ (the $V(-)$ function will be defined in section 2.1.1, or see [9, chapter 21]). As we shall see, once a structure sheaf is introduced, the sets $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ and $\operatorname{Spec} \mathbb{C}[x]/(x)$ which each have one element shall not be have the same structure sheaf added to them. We shall see that in most cases, we can interpret the maximal ideals as points and can write an ideal of the form $[(x_1 - a_1, ..., x_n - a_n)]$ as $(a_1, ..., a_n)$ without ambiguity, however this is not always the case (we shall define non-trivial schemes that shall have no such point[4]). In example 2.1.3, we shall see that we can write any point associated to a maximal ideal from a coordinate ring in this way.

Another advantage of working with spectra is that there is no need to reference an ambient space for the algebraic subsets, the information is "within" the space! In particular, every quasi-projective variety in [9] was a subset of $\mathbb{P}^n_k$, while a spectra need not be a subset of $\mathbb{P}^n_k$! In fact, there shall be schemes that are not embed-able in projective space, though they are few and far between[5]. We shall later see under which condition we can embed a scheme into $\mathbb{P}^n_R$ for a ring R, and naturally this shall always be be the case for schemes associated to quasi-projective varieties (covered in section 5.2.1).

When working with coordinate rings R of a variety V, we may consider R to be the set of functions on V, since we may have a notion of "evaluation" for each element of R. We can do the same for a general commutative ring R, motivating the notation f for a general element of a commutative ring R ($f \in R$) and for calling such elements *regular functions*. Notice that if $p(x) \in k[x]$, then we may find the value of the evaluation $p(a)$ for some $a \in k$ by doing

$$\frac{p(x)}{(x-a)} = p(a) \bmod (x-a)$$

To represent the above more formally, every element $f \in k[x]$ (for some algebraically closed field k) is associated to an element of $\operatorname{Hom}_{\mathbf{Set}}\left(\operatorname{Spec}(k[x]), \coprod_{(x-a)\in \operatorname{Spec}(k[x])} k[x]/(x-a)\right)$, in particular:

$$k[x] \to \operatorname{Hom}_{\mathbf{Set}}\left(\operatorname{Spec}(k[x]), \coprod_{(x-a)\in \operatorname{Spec}(k[x])} \frac{k[x]}{(x-a)}\right)$$

$$f \mapsto ((x-a) \mapsto f \bmod (x-a))$$

Stare at the above until it makes sense; this is essentially a result of the Nullstellensatz ([9]). Notably, we can think of a function as holding local information (indeed, the "most local" as it gives point-wise information rather than open-set local, component local, germ local, and so forth) at each point of the spectrum[6]. We extract this point-wise information by evaluating, or in the case of spectra by modding.

Recall that if we take subsets of the classical affine space, we can often define more regular functions[7], for example by taking the (classical) affine line minus the origin, $\mathbb{A}^{\setminus}_k\{0\}$, we have $f(x) = 1/x$ is

[4]In particular, there shall be no closed points

[5]We shall see in ref:HERE that all proper scheme are covered by a projective scheme, hence we shall require some "awkward blow-ups" to forcefully construct a scheme that cannot be embedded in projective space

[6]or more generally, at each point of a geometric space, like a manifold

[7]though not always, for example the punctured plane has the same algebraic functions as the plane, see example 2.3.14

a well-defined regular function in the classical sense. We shall thus want to focus on all possible functions from $\operatorname{Frac}(R)$[8] that are regular at a point $\mathfrak{p} \in \operatorname{Spec} R$; we saw in [9, chapter 21.6] that this collection is given by localizing at the prime: $R_{\mathfrak{p}}$ (for $R = k[x]$ with $\mathfrak{p} = (x-a)$ we had $k[x]_{(x-a)}$). This collection is called the *stalk* at the point $\mathfrak{p}$. Notice that the codomain of the hom-set can be thought of as the disjoint union of field of fractions of the germs $k[x]_{(x-a)}$ as

$$\frac{k[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}} \cong \operatorname{Frac}\left(\frac{k[x]}{(x-a)}\right) \cong \operatorname{Frac}(k) = k$$

which show that we recover the same notion of evaluation when working over the localization. This greater generality shall be important when we define the sheaf on $\operatorname{Spec} R$ namely since the ring $R_{\mathfrak{p}}$ shall be a *local ring* that has only a single maximal ideal $\mathfrak{m}_{\mathfrak{p}}$ making it unambiguous which ideal to mod by (as explored in section 1.4), and so we shall use it for our definition:

Definition 2.1.2: Value Of Regular Function

Let $f \in R$. Then the *value* of f at the point $\mathfrak{p} \in \operatorname{Spec}(R)$ is:

$$f(\mathfrak{p}) := \overline{f} \bmod \mathfrak{m}_{\mathfrak{p}} \in R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$$

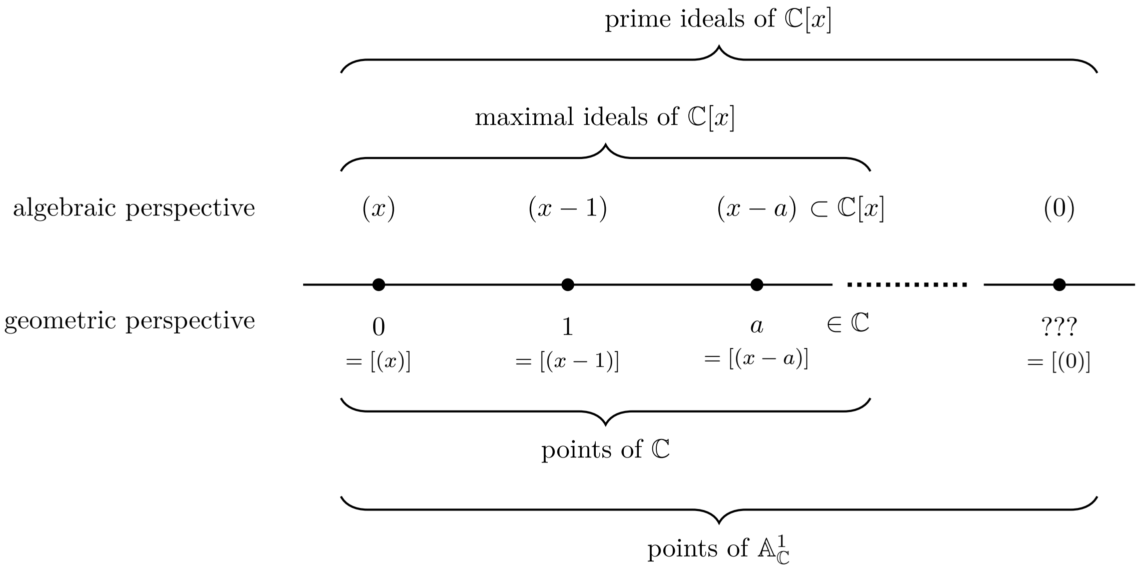
Let us now give multiple examples to get some grasp on spec and the functions on Spec

Example 2.1.3: Spectra and [regular] functions

This is a long set of examples, the reader should cover as many as they need to feel comfortable with the notion of spectrum; they may return here later for further elucidation on spectra.

1. Take $\operatorname{Spec}(\mathbb{C}[x])$ which consists of all prime (in fact maximal) ideals $(x-a)$ for $a \in \mathbb{C}$ and the prime ideal (0). This spectrum is denoted $\mathbb{A}^1_{\mathbb{C}}$; more generally the spectrum of $R[x_1, \ldots, x_n]$ is denoted $\mathbb{A}^n_R$. As $\mathbb{C}[x]$ is a finitely generated reduced ring over an algebraically closed field, the maximal ideals correspond exactly to the classical points of $\mathbb{A}^1_{\mathbb{C}}$ by the Nullstellensatz, with the exception of the special point (0) called the *generic point* which we shall return to soon. To visualize this set, imagine a real line (to remember that it is a 1-dimensional object over $\mathbb{C}$):

[8] Technically, it is required that R be an integral domain as this corresponds to irreducible subsets. Each irreducible subsets will have its own $\operatorname{Frac}(R_i)$ of possible regular functions



The generic point was located "away" from the line, but that is not in fact accurate. The problem is that it is a single point that is near every other point, while still being a single point. What it means for points to be near shall be defined shortly when the Zariski topology on spectra is introduced, but for now you can imagine that this causes issues when trying to draw such a point on a Euclidean piece of paper. There are many ways to visualize it: some write it as a cloud of points in the corner, others at some edge of the visual. Personally, I find it fruitful to think of it as a "higher-dimensional" point that lives over all other points of $\mathbb{A}^1_{\mathbb{C}}$. This perspective shall be corroborated in section 2.6.

Let us evaluate some regular functions $f \in \mathbb{C}[x]$. Let's say $f = x^2 + 2x + 1$ and input $[(x-3)]$. Then the answer is

$$16 \bmod (x-3) = f(3) \bmod (x-3)$$

which can be thought of as our usual answer $f(3)$! Notice that $\mathbb{C}[x]/(x-a) \cong \mathbb{C}$, and hence the codomain when evaluating at each point is the same. We shall see this is not the case soon. We can think of the fact that all the codomains of evaluation coincide as being a special symmetry for varieties over algebraically closed fields (that the evaluation information locally can all be embedded into a single global function-space).

Let us now plug in the generic point (0) into our function f (i.e. the point that isn't in our usual visualization of the complex line). We get:

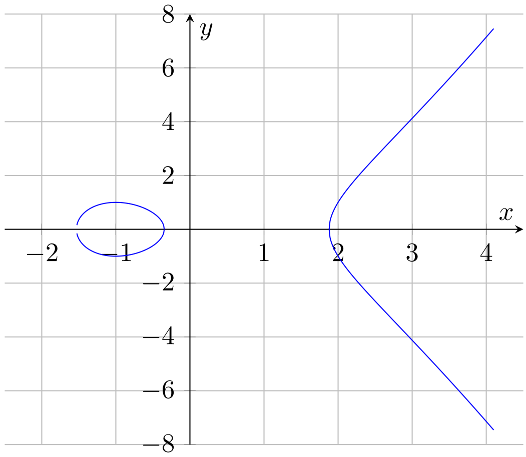
$$f \bmod 0 \in \mathbb{C}[x]$$

that is, it gives back the function itself, and is the only point for which the codomain of f is not isomorphic to $\mathbb{C}$. You can think of (0) as a "1"-[complex]-dimensional point. When we have spectra of higher dimensional (irreducible) schemes, say dimension n, then the generic point will be "n"-dimensional. We can't yet define dimension as we are still working generically with sets; this shall be rectified in section 2.7.

More generally, this entire discussion works just as well over any algebraically closed field $k = \overline{k}$, as the completeness of $\mathbb{C}$ did not factor into the above results.

Often when working with $\overline{k}$-polynomials over several variables, we shall write the maximal ideals as tuples of points, e.g. $[(x-2, y-3)] \in \mathbb{A}^2_{\overline{k}}$ will be represented as $(2,3) \in \mathbb{A}^2_{\overline{k}}$. We shall often then denote the generic point $[(0)]$ as η to distinguish it from the point 0 which is associated to the ideal (x).

2. Take $y^2 - x^3 + 3x - 1 \in \mathbb{C}[x,y]$ and consider Spec $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$. This spectrum corresponds to the elliptic curve $y^2 = x^3 - 3x + 1$. The spectrum will contain all the maximal ideals of the ring $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$ as well as a generic point η corresponding to $[(0)]$ representing the entire space, and hence can be visualized as:



More generally, all the points of an affine algebraic set correspond to the maximal ideals of a spectrum, and so a subset of the geometry of spectra includes the classical geometry of curves, surfaces, and so forth!

3. Here is a Spectra that has no corresponding algebraic set in the classical sense: $\text{Spec}(\mathbb{Z}) = \{(2),(5),(7),\cdots,(p),...,(0)\}$. In other words, $\text{Spec}(\mathbb{Z}) = \{(p) \big| p \text{ is prime}\} \cup \{0\}$ and $\text{mSpec}(\mathbb{Z}) = \text{Spec}(\mathbb{Z}) \backslash \{(0)\}$.

   Let us evaluate some of the regular function $n \in \mathbb{Z}$ on the points $\text{Spec}\,(\mathbb{Z})$. The regular function 15 at the point $[(7)] \in \text{Spec}\,(\mathbb{Z})$ is $15 \bmod (7) = 1 \bmod (7)$. The regular function 9 evaluated at $[(3)]$ is $0 \bmod (3)$. We may even say it has a double-zero at 3.

   Hence, functions on $\text{Spec}\,(\mathbb{Z})$ return some information about divisibility by primes.

4. For any field k, $\text{Spec}\, k = \text{mSpec}\, k = \{(0)\}$ (or $\text{Spec}\, k = \{\eta\}$). For example, $\text{Spec}\, \mathbb{Z}/3\mathbb{Z} = \{(0)\}$. Thus, fields are like "points" within algebraic geometry. An advantage of adding a structure sheaf will be to remember the original field k: we shall see that as schemes $\text{Spec}\, K \ncong \text{Spec}\, L$ if $K \ncong L$, as their sheaves will differ. Though the geometric picture as spectra is trivial, they shall still hold valuable non-trivial information.

   The evaluation of each function at the single point returns the function modulo (0), namely the function itself as an element of k:

$$3 \bmod 0 = 3 \in k$$

5. Take Spec $\mathbb{R}[x]$. This consists of $[(0)]$, all prime (even maximal) ideals of the form $[(x-a)]$, and all prime (even maximal) ideals that can be represented by an irreducible 2nd degree polynomial $\left[\left(x^2+bx+c\right)\right]$, for example $\left[\left(x^2+1\right)\right]$. Evaluating at any of these points (besides $[(0)]$) will give a field: if evaluating points of the form $(x-a)$ the field will be isomorphic to $\mathbb{R}$, and it evaluating points of the form $\left(x^2+ax+b\right)$, the field will be isomorphic to $\mathbb{C}$. As every irreducible quadratic splits into conjugate pairs of linear terms over $\mathbb{C}$, we can think of Spec($\mathbb{R}[x]$) as being a copy of $\mathbb{C}$ folded over itself attaching the conjugate pairs (don't continue reading until this is clear)
Let us take the function x^3-1 and evaluate it as some point. At the point $(x-2)$ we get 7 mod $(x-2)$, or in our classical interpretation, $f(2)$. At the point $\left(x^2+1\right)$, we get:

$$x^3-1 \equiv -x-1 \bmod \left(x^2+1\right)$$

which we can think of as being $-i-1$. Hence, the Spectra of polynomials over fields that are not algebraically closed will behave more like algebraically closed fields as the non-linear maximal primes shall play the role of the required "additional points"! Hence, one piece of the puzzle for the "additional points" in a spectra is that they allow us to think of the spectrum as being "algebraically closed"!

6. Generalizing the above result, show that the points of Spec $\mathbb{Q}[x]$ (minus the generic point 0) are in bijective correspondence with the points of $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, where the points of $\overline{\mathbb{Q}}$ are glue to their galois-conjugates: $\mathrm{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$.

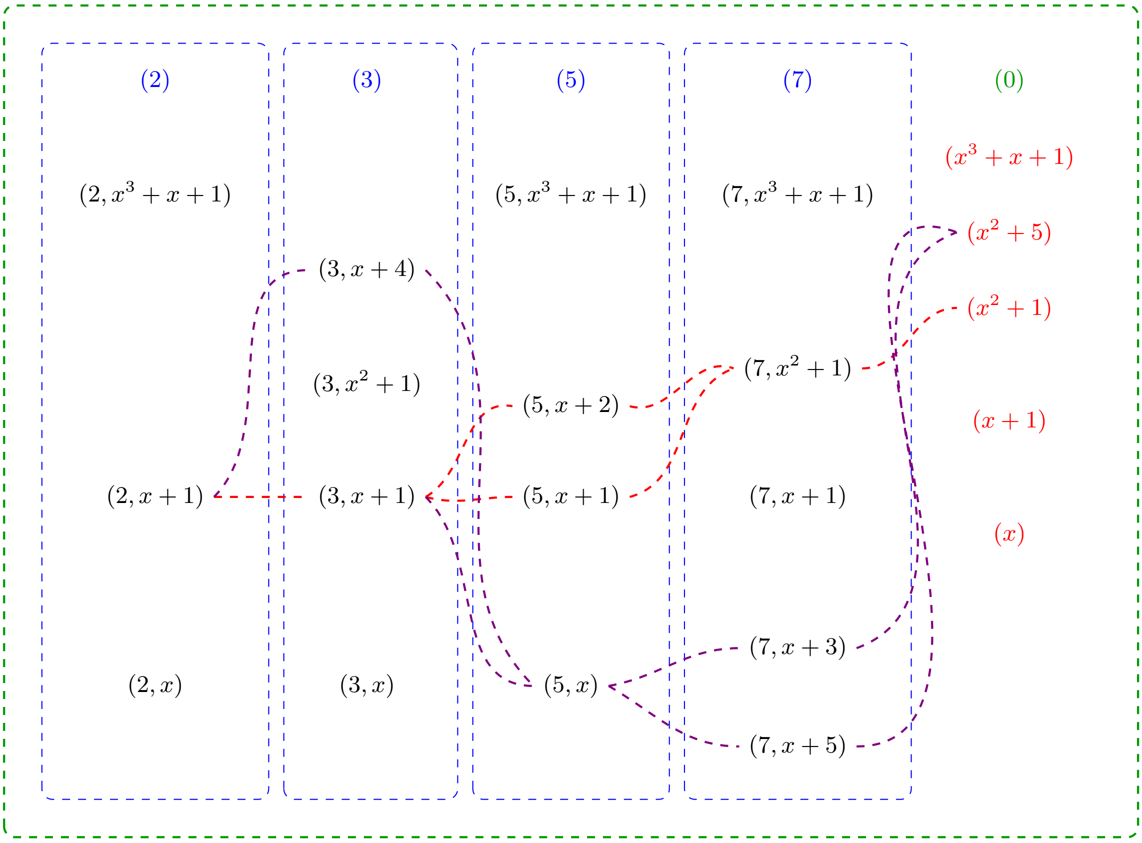
7. Consider Spec $\mathbb{Z}[x]$. Unlike $\mathbb{Q}[x]$, the ring $\mathbb{Z}[x]$ is not a PID and has Krull dimension 2, As it is not a PID, it has prime ideals of the form $(p, f(x))$ where $f(x)$ and $f(x)$ mod p is irreducible. This is the first example where the irreducible functions of a polynomial ring are not in one-to-one correspondence with the points, there are more points than there are irreducible functions[a]! The maximal ideals are of the form

$$(p, f(x))$$

while the prime ideals are of the form:

$$(p) \quad (f(x)) \quad (p, f(x))$$

There are now non-maximal prime ideals, namely the principal ideals (p) for prime p and principal ideals $(f(x))$ where $f(x)$ is irreducible over $\mathbb{Z}$ (or equivalently by Gauss's lemma over $\mathbb{Q}$). To understand this spectrum better, the following is a part of a visual of Spec $\mathbb{Z}[x]$:



Let us parse this image: the maximal points are all of the form $(p,f(x))$ and they are the black points. The prime (p) are contained in each maximal ideal $(p,f(x))$. They are like a more specialized generic point, where they are not near all the points, but near every point for which the ideal representing the point contains (p). Due to this, each column is surround by dotted blue lines that indicating the generic location of each prime $(\ p\ )$. The entire image is surround by a green dotted line which represents the generic point $[(0)]$ and how it is close to every point. Notice that there are gaps in the columns exactly where the irreducible polynomial $f(x)$ is reducible mod$p$. For example x^2+1 mod 2 is reducible with solution $x=1$, and so $\left(2,x^2+1\right)$ is not prime, namely given $\mathbb{Z}[x]/\left(2,x^2+1\right)=(\mathbb{Z}/2\mathbb{Z})[\omega]$ we have

$$(\omega+1)(\omega+1)=\omega^2+2\omega+1=1+1=0$$

showing that $\left(2,x^2+1\right)$ is not prime, and hence not a point.
Next, each point $(f(x))$ is also a generic point, however their "shape" is not a simple as the points (p). The simplest case is when a point $(p,g(x))$ contains $f(x)$, namely $f(x)=g(x)$. In the image, we see that $\left[\left(3,x^2+1\right)\right]$ is a point and hence $\left[\left(x^2+1\right)\right]$ is near this point. On the other hand, both $(5,x+2),(5,x+1)$ are contained in $\left(x^2+1\right)$, precisely because x^2+1 factors into a product of those two linear terms mod 5 . Hence $\left[\left(x^2+1\right)\right]$ is near $[(5,x+2)]$ and $[(5,x+1)]$. This closeness is shown via a red-line passing near these points and represents the fact that $\left[\left(x^2+1\right)\right]$ is near each of these points. Essentially, this represents the quadratic residues of x^2+1 !

The same thing has been done for the point $\left[\left(x^2+5\right)\right]$. Some extra points have been added to the image to demonstrate more clearly the branching behavior. Notice that $\left[\left(x^2+5\right)\right]$ and $\left[\left(x^2+1\right)\right]$ "intersect", for example at $[(2, x+1)]$ and at $[(13, x+5)]$. These give us some number-theoretic information about these equations!
The way I like to interpret these "fuzzy" points corresponding to the irreducible polynomials from some intuition given by complex analysis. This is not unreasonable as the spectrum in many ways mirror the behavior of algebraically closed fields and complex analysis is the analysis of functions that are locally infinite polynomials, and indeed these two fields are very closed linked, something we shall cover in ref:HERE.
For now, consider the following: if we have a "germ" around a single point of a holomorphic function (i.e. the information given by its power-series representation around an arbitrarily small neighborhood around the point), we in fact have information about the entire holomorphic function. From this perspective, it doesn't matter at which point we choose to take the power-series representation, it will return equivalent information. Thus, the information does not live at any point point in particular but "generically" at all these points. If we choose to "specialize" and specify which data we want, we shall evaluate further to a single maximal ideal.
Notice that $\mathbb{Z}[x]$ is seen on a two-dimensional grid. We shall later see that $\operatorname{Spec} \mathbb{Z}[x]$ is naturally seen as a 2 -dimensional scheme, while $\operatorname{Spec} \mathbb{Q}[x]$ shall be 1-dimensional. Furthermore, the red-lines and blue dotted lines will be seen as 1-dimensional points, the generic point will be seen as a 2 -dimensional point, while all the maximal ideals will be 0 -dimensional points.

Finally, let us consider evaluation of functions $f \in \mathbb{Z}[x]$ at points. At maximal points, we shall get values in the field $\mathbb{F}_p(\omega)$[b] where ω is a root of the irreducible polynomial. For example, take $[(7, x^3+x+1)] \in \operatorname{Spec} \mathbb{Z}[x]$ and $x^2-4 \in \mathbb{Z}[x]$. Then the value will be in $(\mathbb{Z}/7\mathbb{Z})(\omega)$ where ω is an element such that $\omega^3+\omega+1=0$. Then we would get that the value of x^2-4 is $\omega^2+3 \in \mathbb{F}_7(\omega)$.

Another example would be evaluating at the point $[(3, x^2+1)]$ the function x^4-1. Working in $\mathbb{Z}[x]/(3, x^2+1) \cong \mathbb{F}_3[x]/(x^2+1) \cong \mathbb{F}_9$, we use $x^2 \equiv -1$ to compute:

$$x^4-1 \equiv (x^2)^2-1 \equiv (-1)^2-1=1-1=0 \pmod{3,\, x^2+1}$$

In other words, $f([(3, x^2+1)])=0 \in \mathbb{F}_3(i)$. As we are evaluating at maximal ideals, we shall always get a "number" as the output; in particular, since we can always embed finite fields into an algebraic closure, the output of a function at a maximal point always lies within an algebraically closed field.

Let us now evaluate at point that corresponds to a non-maximal prime ideal, say at 3. The output of each function $f \in \mathbb{Z}[x]$ will then be:

$$\bar{f} \in (\mathbb{Z}/3\mathbb{Z})[x]=\mathbb{F}_3[x]$$

The way I like to think about these output is that we are asking for some of the information stored in f, namely it shall return enough information about f to determine the generic behavior of that function around $[(3)]$. If we wanted the generic behavior around a different generic point, for example $\left[\left(x^2+1\right)\right]$, then we would get:

$$f(i) \in \mathbb{Z}[i]$$

Namely, if we choose any irreducible polynomial, we are choosing which new element of an integral extension we are adding that we then evaluate the function f at. We may then extract further information by modding by different primes. It shall often be the case that evaluating at a generic point corresponds to reducing the information of a function down to a smaller geometric object or to modding information.

8. If you know some number theory, $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is the quintessential example of a ring that has primes that are *not* principle, for example $(2, 1+\sqrt{-5})$ is prime but is provably not principal (see [9, chapter 22]). Evaluating at (2) and $(1+\sqrt{-5})$ will return 0, however the zero set of both of these function's shall be larger than $(2, 1+\sqrt{-5})$; we shall require the vanishing set to contain both of these.

9. The following is another example of a spectrum with a single element. Take $\operatorname{Spec}(k[x]/(x^2))$, the spectrum of a ring with nilpotent elements. This ring can be thought of as an extension of k by an element $\epsilon = x \bmod x^2$ that is thought to be "infinitely small". It has the interesting property that $\epsilon^2 = 0$, even though ϵ itself is nonzero. As the spectrum only contains one point[c], namely $[(\epsilon)]$, this can be thought of as a "thicker" point (as shall be explored in greater detail in section 2.2.2). The ring $k[x]/(x^2)$ is called the *dual numbers* and shall be a recurring ring to study the behaviour of nilpotent elements. This ring has the interesting property of having the element $\epsilon \in k[\epsilon]$ which always evaluates to 0 even though it is not the zero element! This gives us our first example of a function[d] that is not defined by what it does at every point, namely because the functions 0 and ϵ have the same outputs.

   More generally, $k[x]/(x^n)$ has a single element in its spectrum. As k and $k[x]/(x^n)$ have the same underlying topological space, what shall distinguish them will be the sheaf structure associated to these two spectra (they shall not be isomorphic as schemes).

10. The above ring only had one point in its spectrum. There are many spectrum's with only two points: a generic point and a non-generic point. A large class of such rings shall be *local rings*. For example $k[x]_{(x-a)}$ will have (0) and $(x-a)$ as the two points of the spectrum. Another example example of a spectrum with only two points is $k[[x]]$ which has (x) and (0) as its two points. As spectra, $k[x]_{(x)}$ and $k[[x]]$ are indistinguishable. What shall distinguish these two spectra shall be the sheaf-structure associated to these two spectra.

11. Let us take a look at the number theoretic equivalent of localization: take $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at $\mathbb{Z}\backslash(p)$. This spectrum contains two points, $[(0)]$ and $[(p)]$. In particular, $\operatorname{Spec}\mathbb{Z}_{(p)}$ is not a single point like $\operatorname{Spec}\mathbb{F}_p$, it somehow has some more "fuzz" around it's single point $[(p)]$. This can be thought of as $\operatorname{Spec}\mathbb{Z}_{(p)}$ "remembering" that there is some arbitrarily small neighborhood around the point $[(p)]$ that it must keep track of. Once we introduce a topology on spectra's, we'll see that the closure of $[(p)]$ is itself (it is a closed point), while the closure of $[(0)]$ is $\{[(0)], [(p)]\}$, motivating the idea that $[(0)]$ "remembers" that there is some more information outside of just the single point! This shall be even more clear once we introduce the notion of dimension: the point $[(p)]$ will be zero-dimensional, while $[(0)]$ will be one-dimensional!

12. Let $R = k[x] \times k[x]$. What is $\operatorname{Spec} R$? Notice that $\operatorname{Spec} R$ does not contain just one generic point, but two! Importantly, $[(0)]$ is not prime as $(1,0)\cdot(0,1) \in (0)$ but $(1,0),(0,1) \notin (0)$.

    In general, the primes of a product ring $A \times B$ are all of the form $\mathfrak{p} \times B$ or $A \times \mathfrak{q}$ where $\mathfrak{p} \in \operatorname{Spec}(A)$ and $\mathfrak{q} \in \operatorname{Spec}(B)$ (exercise). Hence $\operatorname{Spec}(A \times B) \cong \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B)$: the spectrum of a product is the disjoint union of the spectra. In our case, $\operatorname{Spec}(k[x] \times k[x]) \cong \mathbb{A}^1_k \sqcup \mathbb{A}^1_k$, two disjoint copies of the affine line, each with its own generic point.

13. Take $\operatorname{Spec} \mathbb{F}_p[x] = \mathbb{A}^1_{\mathbb{F}_p}$. As it is a Euclidean domain, the points are (0) and $(f(x))$ for all irreducible polynomials over $\mathbb{F}_p$. One of the interesting facets of $\mathbb{A}^1_{\mathbb{F}_p}$ is that its closed points are far richer than the p elements of $\mathbb{F}_p$: for each $n \geq 1$, each irreducible polynomial of degree n over $\mathbb{F}_p$ gives a closed point whose residue field is $\mathbb{F}_{p^n}$. Since there are irreducible polynomials of every degree, $\mathbb{A}^1_{\mathbb{F}_p}$ has infinitely many closed points despite $\mathbb{F}_p$ being finite. This richness in points will allow us to distinguish functions on $\mathbb{F}_p$ by their evaluation: recall that a polynomial is not distinguished by evaluation at all points in $\mathbb{F}_p$ (e.g. $x^p - x$ vanishes everywhere on $\mathbb{F}_p$ but is nonzero), however a polynomial *will* be distinguished by its evaluation at all points in $\operatorname{Spec}(\mathbb{F}_p[x])$, as we shall show using proposition 2.1.4.

14. Take $\operatorname{Spec}(\overline{k}[x, y]) = \mathbb{A}^2_{\overline{k}}$ (e.g. $\overline{k} = \mathbb{C}$). Then the points are:

$$(0) \qquad (x - a, y - b) \qquad (f(x, y))$$

where $f(x, y)$ is an irreducible polynomial in $\overline{k}[x, y]$. Visually, all the primes of the form $(x - a, y - b)$ correspond to the usual points in $\mathbb{C}^2$. The points $(f(x, y))$ can be thought of as tracing out their curve in $\mathbb{C}^2$; for example $(y - x^2)$ is the point that is the curve $y = x^2$. Notice how the maximal ideals correspond to "0"-dimensional points, while the ideal $(y - x^2)$ corresponds to a "1"-dimensional point. Similarly, (0) is a "2"-dimensional point. We shall come back to this through Krull dimensions in section 2.7.

More generally, by the Nullstellensatz the maximal points of $\overline{k}[x_1, \ldots, x_n]$ are the ideals of the form:

$$(x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$$

The prime ideals of this ring don't lend themselves to a nice classification. Each prime ideal corresponds to an irreducible variety of a certain dimension, and hence the classification of the prime ideals is equivalent to the classification of irreducible varieties, which is a rather difficult task that is carried out in special cases, and more generally leads to the theory of moduli spaces. Some context on this classification in a classical setting was covered in [9].

15. Find the points of $\operatorname{Spec}(k[x_1, \ldots, x_n])$ where k is not algebraically closed, ex. $\mathbb{Q}$; generalize example 5[e].

[a] This sort of exploration on the correspondence between objects that "principal" (or locally principal) and how much they are not will be the focus of section 5.3

[b] Recall that $k[\omega] = k(\omega)$, or re-prove this

[c] Prove that $[(0)]$ is not a prime ideal: $\epsilon \cdot \epsilon = 0 \in (0)$ but $\epsilon \notin (0)$

[d] granted, a generalization of the notion of function

[e] If you solved it, note that $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but *not* to $(\sqrt{2}, -\sqrt{2})$

2.1.1 Zariski Topology

Having defined the set that represents the space, the next thing to do is give a topology to this set. This topology would have to be a generalization of the Zariski topology defined on $k[x_1, ..., x_n]$ using the definition of regular function that was just introduced:

Definition 2.1.4: Zariski Topology

Let R be a commutative ring and $\operatorname{Spec}(R)$ the collection of prime ideals of R. Then define the *closed sets* for the Zariski topology on $\operatorname{Spec}(R)$ to be:

$$V(S) = \{x \in \operatorname{Spec}(R) \mid f(x) = 0,\ \forall f \in S\} = \{[\mathfrak{p}] \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}$$

Take a moment to see how this parallels the condition that all the functions in the coordinate ring must equal zero when evaluated at $\mathfrak{p}$. the notation $f(x) = 0$ will eventually be correctly re-interpreted when we show the sheaf naturally induced by R is a *locally ringed space*, and hence evaluation is well-defined for any ring element. It is an exercise in commutative algebra to check that this is indeed a topology, namely:

Lemma 2.1.5: Zariski-closed Sets Form Topology

Let $\operatorname{Spec}(R)$ be the collection of primes and let:

$$\mathcal{T} = \{V(S) \mid S \subseteq R\}$$

Then:

1. $\mathcal{T}$ contain the empty set and $\operatorname{Spec}(R)$ (these are both the vanishing set of single elements)
2. $\mathcal{T}$ is closed under arbitrary unions and finite intersections, namely:

$$\cap_i V(I_i) = V\left(\sum_i I_i\right)$$

for arbitrary indexing set, and that

$$V(I_1) \cup V(I_2) = V(I_1 I_2)$$

Furthermore, for any radical $I \subseteq R$, $V(I) \cong \operatorname{Spec} R/I$ as as topological spaces[a].

[a] and as sheaves, which we'll see later

***Proof*:**
see [9, chapter 21.1].

Note that the vanishing set $V(-)$ defined for the spectra of rings usually have more points than their counterpart for coordinate rings in classical algebraic geometry as they will contains primes that are not maximal ideals (particularly for rings of "dimension" ≥ 2 as we have briefly discussed), and these primes will be within $V(S)$ for appropriate S.

The reader should check how this topology compares to the topology for classical algebraic sets. For example, as there exist generic points in most spectra we have that this space is no longer T_1 (it shall soon be shown that generic points are not closed). If R is Noetherian, then $\operatorname{Spec} R$ is a Noetherian topological space; see exercise 2.1.1 and section 2.1.3 for more comparisons.

Just like for varieties, we may define the inverse function $I(-)$, that is for [nonempty] $S \subseteq \text{Spec}(R)$,

$$I(S) \subseteq R \qquad I(S) = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p}$$

Hence $I(S)$ is a radical ideal (recall the intersection of prime ideals is a radical ideal, think $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$). Just like in classical algebraic geometry:

Proposition 2.1.6: Properties of $I(-)$ and $V(-)$

1. $I(S)$ is an ideal of R, and furthermore it is always *radical*.
2. $I(-)$ is inclusion-reversing
3. $I(A \cup B) = I(A) \cap I(B)$, and $I(A \cap B) = \sqrt{I(A) + I(B)}$.
4. $I(\overline{S}) = I(S)$
5. $V(I(S)) = \overline{S}$ and $I(V(J)) = \sqrt{J}$.
6. If S is closed, then $V(I(S)) = S$. If I is radical, then $I(V(J)) = J$.
7. $I(-)$ and $V(-)$ are an inclusion-reversing bijection between closed subsets of $\text{Spec}(R)$ and radical ideals of R. This is called a *galois correspondence*.

Hence:

$$\text{closed subset} \subseteq \text{Spec}(R) \qquad \Leftrightarrow \qquad \text{radical ideal of } R$$

***Proof*:**
exercise.

Thus, we get the (surjective) correspondence:

$$\{\text{Radical ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed sets of Spec}(R)\} \tag{2.2}$$

Notice that prime ideals are radical and hence are part of this association. This correspondence does not extend to all ideals until a sheaf, the structure sheaf, is introduced (see section 2.2.2).

As we see, the Zariski topology on $\text{Spec}(R)$ is a generalization of the Zariski topology on $k[x_1, ..., x_n]$. The closed sets of $\text{Spec}(R)$ will contain the "traditional" points as well as the new points, for example $V(xy, yz)$ will contain all the "traditional" points where $y = 0$ or $x = z = 0$, as well as the generic point that vanish on $V(xy)$ and $V(yz)$ (i.e. $[(y)]$ vanishes on this set). Just like for affine varieties, $\text{Spec}(R)$ is almost never Hausdorff. This is even clearer in $\text{Spec}(R)$, for if there are primes that are not maximal, these are *not* closed points (in a Hausdorff or T_1 space, every point is closed)

Example 2.1.7: Non-closed Points
Take $\mathbb{A}^2_{\mathbb{C}}$ with $[(y^2 - x)]$ and $[(x - 2, y - 4)]$. The point $[(y^2 - x)]$ is not closed. Notice that:

$$[(y - 4, x - 2)] \in V(I([(y^2 - x)])) = V(y^2 - x) \overset{\text{pr. 2.1.6}}{=} \overline{\{[(y^2 - x)]\}}$$

Showing that the point is *not* closed, namely points associated to non-maximal prime ideals are not closed. However, notice the set $V(I([(y^2-x)]))$ is closed by definition. In a moment, the point $[(y^2-x)]$ will be seen as the generic point of the above closed set.

In fact, due to Hilbert's Nullstellensatz, we know that in $\mathbb{A}_k^n$ the maximal ideals correspond to the "traditional points", that is the closed points. From here on, we shall call the "traditional points" (more generally points associated to maximal ideals) the *closed points*, and the "extra" or "bonus" points that were being referred to shall be called the *non-closed points* or *generic points*.

With a topology, we can now link points like $[(x-2, y-4)]$ to points like $[(y-x^2)]$. As these are point, it doesn't make sense to say one point contains another. However we we be able to link these points using closure:

Definition 2.1.8: Specialization And Generization

Let $x, y \in X$ be two points in a topological space. Then we say that x is a *specialization* of y and y a *generization* of x if

$$x \in \overline{\{y\}}$$

Algebraically, $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$, hence $V(\mathfrak{p}) = \overline{\{[\mathfrak{p}]\}}$.

We can now define generic point more precisely:

Definition 2.1.9: Generic Point

Let X be a topological space. Then $p \in X$ is a *generic point* for a closed subset K iff

$$\overline{\{p\}} = K$$

By this definition, closed points are technically generic points associated the closet set of a maximal idea; we shall only use the term generic point for closed sets that are not given by a maximal ideal. With generic points and the Zariski topology defined, the idea that generic points of a closed set (including all of Spec (R)) are close to all the points in the closed subset (it is contained in all the open subsets (or all the open subsets of $V(S) \cap U$).

Let us end with upgrading proposition proposition 2.2.7

Proposition 2.1.10: Nilpotent Spectrum Homeomorphism

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism.

***Proof*:**
The map is a bijective continuous map. Show the inverse is continuous.

Non-closed points shall have their use in maps such as $k[x,y] \to k[x](y)$ as well as in intersection theory where the intersection or "creases" of schemes shall often be generic points. We shall see that there is a "local-ring" associated to both closed and non-closed points alike and their reduction behaviour is the same (though naturally being in different dimensions means that the behaviour may differ in exactitude's).

2.1.2 (Topological) Basis: Distinguished Open Sets

As the topology of $|\mathrm{Spec}\,(R)|$ is generated by the closed sets $V(S)$, and these sets are the intersection of:

$$V(S) = \bigcap_{f \in S} V(f)$$

the sets $V(f)$ generate the topology of $\mathrm{Spec}\,(R)$. We may want to call these sets "distinguished closed sets", however we shall favor their open counterpart due to their simpler behaviour when introducing sheaf theory to spectra[9]:

Definition 2.1.11: Distinguished Open Sets

The sets $D(f)$ are called the *distinguished open sets*, *basic open sets*, or *principal open sets*, and are defined to be

$$D(f) = X_f = |\mathrm{Spec}\,(R)| \setminus V(f) = \{\mathfrak{p} \in \mathrm{Spec}\,(R) \mid f \notin \mathfrak{p}\}$$

that is, it is the collection of prime ideals of R that do not contain f. Any open set U can be written as:

$$U = \mathrm{Spec}\,(R) \setminus V(S) = \mathrm{Spec}\,(R) \setminus \bigcap_{f \in S} V(f) = \bigcup_{f \in S} D(f)$$

Writing it as $D(f)$ may be helpful; Prof. Ravi Vakil calls these "doesn't-vanish sets" which may serve as a helpful mnemonic. These sets are usually not closed (especially if $\mathrm{Spec}\,(R)$ is connected, in which case unless $D(f) = \emptyset$ it cannot be), and hence they are not the vanishing set for any functions in R. However, they are still isomorphic to a spectrum:

Proposition 2.1.12: Distinguished Open Sets and Localization

Let $D(f) \subseteq \mathrm{Spec}\,(R)$ be a distinguished open set. Then:

$$D(f) \cong_{\mathbf{Top}} \mathrm{Spec}\,(R_f)$$

justifying the X_f notation. Furthermore, if $g = f_1 f_2 \cdots f_n$, then

$$\bigcap_{i=1,\cdots,n} (\mathrm{Spec}\,R)_{f_i} = (\mathrm{Spec}\,R)_g$$

or as distinguished open sets if $D(f_1), \ldots, D(f_n) \subseteq \mathrm{Spec}\,(R)$, then their intersection is the distinguished open

$$D(f_1 \cdots f_n)$$

[9]There shall be a natural unique open subscheme structure up to isomorphism, but infinitely many natural closed subscheme structures up to isomorphism due to the possibility of nilpotents.

***Proof*:**
Recall that $R_f = R[f^{-1}]$, and that $\mathfrak{p} \subseteq R[f^{-1}]$ iff "$f \notin \mathfrak{p}$ and $\mathfrak{p} \in R$". This is exactly the points of $D(f)$; it is left to check that the map is bi-continuous. The second part is left as an exercise.

Corollary 2.1.13: Distinguished Open Sets Form Topological Basis

The collection of distinguished open sets $\{D(f)\}_{f\in R}$ from a topological basis for $\operatorname{Spec} R$ with the Zariski topology

***Proof*:**
Distinguished opens cover $\operatorname{Spec} R$ by definition, and proposition 2.1.12 show they are closed under intersections.

A mistake the reader may make is to believe that each $D(f)$ can be thought of as the spectrum minus some points, but this need not be the case:

Example 2.1.14: Spectrum and Distinguished Open Sets
Take $X = \operatorname{Spec}(k[x, y])$ and take $U = X \setminus \{[(x, y)]\}$, i.e. X minus the origin (which is a closed set as it is a maximal ideal). Show that $U \ncong D(f)$ for any f (hint: notice that (x, y) is not principal). This shows an important distinction between points and functions.

Show further that U can be represented as the union of two distinguished open sets.

Another result we may want to conclude is that each open set $U \subseteq \operatorname{Spec}(R)$ corresponds to $\operatorname{Spec}(R_S)$ for an appropriate S, i.e. $U \cong_{\mathbf{Top}} \operatorname{Spec}(R_S)$. This shall be shown to be *false*; the curious reader may jump to example 2.3.14 for a counter-example.

The following result, while elementary, is used repeatedly in gluing arguments and in the proof of the Affine Communication Lemma. It says that a distinguished open inside a distinguished open is itself distinguished in the ambient spectrum.

Proposition 2.1.15: Distinguished Open of a Distinguished Open

Let $D(f) \subseteq \operatorname{Spec}(R)$ be a distinguished open, so that $D(f) \cong \operatorname{Spec}(R_f)$. Let $g' \in R_f$ and consider the distinguished open $D(g') \subseteq \operatorname{Spec}(R_f)$. Then $D(g')$, viewed as a subset of $\operatorname{Spec}(R)$, is itself a distinguished open of $\operatorname{Spec}(R)$.
More precisely, if $g' = h/f^n$ with $h \in R$ and $n \geq 0$, then:

$$\operatorname{Spec}((R_f)_{g'}) = \operatorname{Spec}(R_{fh}).$$

In particular, $D(g') = D(fh)$ as subsets of $\operatorname{Spec}(R)$.

***Proof*:**
Since f is already a unit in R_f, inverting $g' = h/f^n$ in R_f is the same as inverting h in R_f:

$$(R_f)_{h/f^n} = (R_f)_h.$$

Now $(R_f)_h$ is the localization of R at the multiplicative set generated by f and h, which is the same as localizing at fh:

$$(R_f)_h = R_{fh}.$$

(Formally: $(R_f)_h = R[f^{-1}][h^{-1}] = R[(fh)^{-1}] = R_{fh}$, since inverting fh inverts both f and h, and conversely.) The corresponding open set is $D(fh) = D(f) \cap D(h) \subseteq \operatorname{Spec}(R)$.

2.1.16 Remark The proposition says that the collection of distinguished open sets is closed under "refinement": passing to a smaller distinguished open inside a distinguished open stays within the class. This is what makes distinguished opens a good basis for defining sheaves—one never needs to leave the class when restricting.

Let us record two further useful consequences.

Corollary 2.1.17: Unit Ideal and Distinguished Cover

Elements $f_1, \dots, f_n \in R$ generate the unit ideal $(f_1, \dots, f_n) = R$ if and only if $\operatorname{Spec}(R) = \bigcup_{i=1}^n D(f_i)$.

***Proof*:**
A prime $\mathfrak{p}$ is not in $\bigcup D(f_i)$ if and only if $f_i \in \mathfrak{p}$ for all i, if and only if $(f_1, \dots, f_n) \subseteq \mathfrak{p}$. Thus $\bigcup D(f_i) = \operatorname{Spec}(R)$ if and only if no prime contains all the f_i, if and only if $(f_1, \dots, f_n) = R$.

Corollary 2.1.18: Powers Generate the Unit Ideal

If $f_1, \dots, f_n$ generate the unit ideal in R, then for any positive integers $k_1, \dots, k_n$, the powers $f_1^{k_1}, \dots, f_n^{k_n}$ also generate the unit ideal.

***Proof*:**
We have $D(f_i) = D(f_i^{k_i})$ (a prime contains f_i if and only if it contains $f_i^{k_i}$), so $\bigcup D(f_i^{k_i}) = \bigcup D(f_i) = \operatorname{Spec}(R)$, which by the previous corollary means $(f_1^{k_1}, \dots, f_n^{k_n}) = R$.

2.1.3 Topological Properties

Let us see what classical topological properties applies to $\operatorname{Spec} R$. An important consequence is that for any open cover of $\operatorname{Spec}(R)$ given by distinguished open sets, there exists a finite open sub-cover. To see this, notice that $\cup_{i \in I} D(f_i) = \operatorname{Spec}(R)$ For some indexing set I if and only if $(\{f_i\}_{i \in I}) = R$ that is the ideal generated by the set $\{f_i\}_{i \in I}$. Then 1 must be inside that ideal, and must be a finite linear combination, and the result follows by the algebraic/geometric correspondence. This shows that $\operatorname{Spec}(R)$ is always compact in the topological sense. We may wish that this implies that $\operatorname{Spec}(R)$ has the usual nice properties of compactness, but this is very much not the case (notice this implies $\mathbb{A}^n_{\mathbb{C}}$ is compact). This turns out to be a consequence of lacking Hausdroffness. For this reason, a different name is given to compactness:

Definition 2.1.19: Quasicompact

Let X be a topological space. Then if X is compact and non-Hausdorff, X is *quasicompact*

As an exercise, show that if R is Noetherian, every subset of $\mathrm{Spec}\,(R)$ is quasicompact. In section 3.5.4, we shall show how the notion of proper maps from analysis is the right way of defining the notion of compactness which recovers many of the nice results expected from compactness.

The companion property of [quasi]compactness is connectedness. Recall a set is connected if it cannot be written as a disjoint of two nonempty open sets. Disconnectedness of Spec is equivalent to a ring representable as the product of two rings, $R = R_1 \times R_2$, or if the reader remembers the brief discussion in Representation theory(see [9, chapter 23.2]), if R has idempotent elements

Proposition 2.1.20: Spectrum Connectedness And Idempotent Elements

IF $R = R_1 \times \cdots \times R_n = \prod_i^n R_i$ is a ring, then show that

$$\mathrm{Spec}\left(\prod_i^n R_i\right) = \coprod_i \mathrm{Spec}\,(R_i)$$

More generally, show that $\mathrm{Spec}\,(R)$ is not connected if R is isomorphic to the product of two zero-rings, that is there exists $a_1, a_2 \in R$ st $a_1^2 = a_2, a_2^2 = a_2, a_1 + a_2 = 1$.

***Proof*:**
good exercise.

Note that for the homeomorphism, the product must be finite, if the product is infinite then $\mathrm{Spec}\,(\prod_i R_i)$ is larger than the right hand side. In fact, an infinite disjoint union of $\mathrm{Spec}\,(R_i)$ cannot be represented as $\mathrm{Spec}\,(R)$ for an appropriate R; it can be seen as the "simplest" example of a scheme that is not an affine scheme (see example 2.2.12(6)).

A notion similar to connectedness is *irreducibility*. Recall that a topological space is said to be *irreducible* if it is nonempty and it is not the union of two proper closed subsets. Written differently, X is irreducible if whenever $X = Y \cup Z$ with Y, Z closed in X, then $Y = X$ or $Z = X$[10]. Note that Y, Z may have nonempty intersection. In the euclidean topology, this notion is much less important (think $\mathbb{C}$), however in the Zariski topology we shall see that many interesting sets are irreducible. The intuition stems from traditional varieties where irreducibility corresponds to irreducible polynomials, and indeed this intuition when properly generalized translates over to irreducible [closed] sets in the Zariski topology of Spectra.

Let $X = Y \cup Z$ for closed sets Y, Z. Then there are radical ideals J_X, J_Y, J_Z such that $V(J_X) = V(J_Y) \cup V(J_Z) = V(J_Y J_Z) = V(J_Y \cap J_Z)$. As these are all radical ideals, $J_X = J_Y \cap J_Z$. Then if J_X is prime, it is an exercise in commutative algebra to show that either $J_Y \subseteq J_Z$ or $J_Z \subseteq J_Y$. Hence, irreducibility is in correspondence with prime ideals:

[10]The finiteness is important, for there may be irreducible sets that contain within them reducible sets of lower dimension

$$\{\text{prime ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed irreducible sets of Spec}\,(R)\} \tag{2.3}$$

If we look at irreducible closed sets, then $V(-)$ and $I(-)$ gives a bijection between the irreducible closed subsets of Spec (R) and the prime ideals of R, in other words there is a bijection between the *points* of Spec (R) and the irreducible closed subsets of Spec (R). Since it is a bijection, every irreducible closed subset of Spec (R) has *exactly* one generic point associated to it.

As usual, we define the Noetherian property on Spec (R) to mean that the descending chain condition for closed sets is satisfied (to mirror the ascending chain of ideals in R). In the Noetherian case, all closed sets can be decomposed into irreducibles:

Proposition 2.1.21: Noetherian Decomposition

Let X be a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union

$$Z = Z_1 \cup \cdots \cup Z_n$$

of irreducible closed subsets, none contained in another.That is , any closed subset Z has a finite number of pieces.

***Proof*:**
see [9].

The reader should check the equivalent result for prime ideals

2.1.4 Ring to Top Functor: Induced Spectra Maps

As spectra are given by rings, we shall want to translate over ring homomorphisms to maps between spectra:

Definition 2.1.22: Induced Spectra Map

let $Y = \text{Spec}\,(S)$ and $X = \text{Spec}\,(R)$ be two spectrum. Then given the ring homomorphism: $\varphi : R \to S$, the induced *spectrum map* $\varphi^a : \text{Spec}\,(S) \to \text{Spec}\,(R)$ is given by:

$$\varphi^a(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$$

The reader ought to verify that if $\varphi : R \to S$ is a ring homomorphism, then $\varphi^a : \text{Spec}\,(S) \to \text{Spec}\,(R)$ sending $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is functorial (recall or reprove that the pre-image of a prime ideal is prime). We now have two functions happening: φ^a given by ring homomorphisms, and regular functions given by ring elements. Often, φ^a is called a map, while the elements of R are called functions (following a similar convention from differential geometry).

Example 2.1.23: Maps between Spectra: φ^a

1. Show that the surjective map $R \to R/I$ induce an injective map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Algebraically, the surjective map adds relations. Geometrically, we are taking the points with the extra algebraic relations given by I and associating them to the points in R.

2. Show that the injective map $R \to S^{-1}R$ induce a injective map $\operatorname{Spec}(S^{-1}R) \to \operatorname{Spec}(R)$. Algebraically, the map $R \to S^{-1}R$ takes (integral) elements[a] and maps it to units.

3. Here is a concrete example generalizing the usual notion of a regular morphism as introduced in[9]: Take $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x]$ which by Nullstellensatz we can think of as the traditional points $\mathbb{C}$ along with the generic point (0). Consider the ring homomorphism between these rings defined by:
$$\varphi : \mathbb{C}[y] \to \mathbb{C}[x] \qquad y \mapsto x^2$$
Note this is not the squaring map, as that is not a ring morphism, but rather the map that maps the variable y to x^2 and then extends linearly. Then the map on the sets Spec's is given by:
$$\varphi^a : \operatorname{Spec}(\mathbb{C}[x]) \to \operatorname{Spec}(\mathbb{C}[y])$$
$$\begin{aligned}\varphi^a(\mathfrak{p}) &= \varphi^{-1}((x-a)) \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) \in (x-a) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; x-a \middle| f(x^2) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) = (x-a)q(x) \right\} \\ &= (y-a^2)\end{aligned}$$
In one line:
$$\varphi^a([x-a]) = [(y-a^2)]$$
verify that
$$(\varphi^a)^{-1}((y-a)) = \{(x-\sqrt{a}), (x+\sqrt{a})\}$$
Perhaps more intuitively, this is the Spec-equivalent of the more usual map $\mathbb{C} \to \mathbb{C}$ given by $z \to z^2$ which can be thought of as the double-covering of $\mathbb{C}$ with branching point 0. Note the importance of visualizing schemes using complex numbers, or more generally an algebraically closed field, even if we replaced $\mathbb{C}$ with $\mathbb{R}$; for example the image of (x^2+1) is $(y+1)$ (i.e. $-i^2 = -1$) and $(\varphi^a)^{-1}\left((y^2+1)\right) = \{(x^2+1)\}$ where the information about the 2 points is captured in the degree of the (irreducible) polynomial in the pre-image (recall they are glued together in $\operatorname{Spec}\mathbb{R}[x]$).

4. More generally, any collection of regular functions can produce a spectra map, a "regular map" (in particular, this mirror's regular maps between affine varieties). Let k be any field and take polynomial $f_1, ..., f_n \in k[x_1, ..., x_m]$. Then define:
$$\varphi : k[y_1, ..., y_n] \to k[x_1, ..., x_m] \qquad y_i \to f_i$$
Let A represent the domain and B the codomain. Then show that for any ideal $I \subseteq A$, $J \subseteq B$ such that $\varphi(I) \subseteq J$, φ induces a map of sets:
$$\operatorname{Spec}(k[x_1, ..., x_m]/J) \to \operatorname{Spec}(k[y_1, ..., y_n]/I)$$

More particularly, show that this maps sends the point $[(x_1 - a_1, ..., x_m - a_m)]$ (traditionally, $(a_1, ..., a_m) \in k^m$ to the point traditionally represented as:

$$(f_1(a_1, ..., a_m), ..., f_n(a_1, ..., a_m)) \in k^n$$

From now on, we shall sometimes take liberties in spectra maps between spectra resembling varieties by defining polynomial maps on the maximal ideals (this shall pose no problems with extending the definition to the prime ideals or to the later sheaf structure that shall be added by theorem 3.8.1).

5. The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $a \mapsto p(a)$ can always be represented as a map to the graph:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad a \mapsto (a, p(a))$$

Furthermore, by the projection map $\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k$ given by $(a, b) \mapsto a$, we shall see that this maps $\mathbb{A}^1_k$ isomorphic to all of its graphs (this works even for higher dimensions). Note that this does not extend if we have more variable where more than one term is not the identity, for example:

$$\mathbb{A}^1_k \to \mathbb{A}^3_k \qquad t \mapsto (t, t^2, t^3)$$

maps to the point of $\mathbb{A}^1_k$ to the associated to the variety $\mathcal{Z}(x - y^2, x - z^3)$ which in [9] was shown to be rational but not isomorphic. Another example that isn't even rational is a map from $\mathbb{P}^1_k$ to an elliptic curve E.

6. This example gives a way of thinking of $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[x_1, ..., x_n])$ as an "$\mathbb{A}^n$-bundle" on $\operatorname{Spec}(\mathbb{Z})$ and is an important example for number theory. Define the map $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$ given by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[x_1, ..., x_n]$. Then show there is a bijection between $\pi^{-1}([(p)])$ where $(p) \neq (0)$and $\mathbb{A}^n_{\mathbb{F}_p}$! If $(p) = (0)$ then the pre-image is $\mathbb{A}^n_k$. The picture of this bundle can be seen as follows:

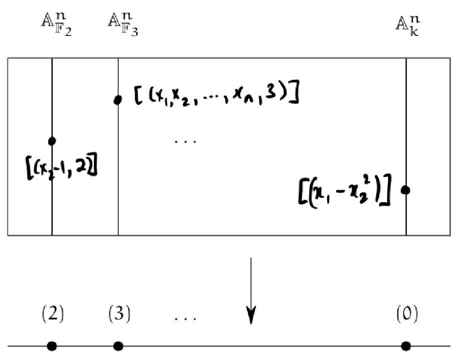


Figure 2.1: Bundle on $\operatorname{Spec}\mathbb{Z}$

7. Consider $\varphi : \mathbb{Z} \hookrightarrow k[x]$. Then $\varphi^a : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{Z}$ is the zero map, namely $\varphi^a(\operatorname{Spec} k[x]) = \{[(0)]\}$. From this perspective, we can see that classical field theory has trivial number theory as it has no interesting fibers over $\operatorname{Spec}\mathbb{Z}$.

8. Let $\varphi : R \to S$ be an integral extension. Show that $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective (hint: Lying Over). This results also motivates the terminology "lying over" geometrically. Recall that is R or S is a field, the other must be too. Show that S being a field and the map being surjective can be used to conclude R is a field. What is the geometric argument if R was a field?

9. Let us see why it is important to work with primes instead of maximal ideals. Consider $\mathrm{Spec}\ k(x)[y] \to \mathrm{Spec}\ k[x]$ induced by $k[x] \to k(x)[y]$ where $k[x]$ maps to itself (notably, all elements become units). Then as $k(x)[y]$ is a PID, all of its primes are maximal and are of the form $(y - f(x))$ for some $f(x) \in k(x)$. Then the pre-image of this ideal under the inclusion map must be the ideal (0), which is certainly not maximal (in fact it is the generic point). Hence the map $\mathrm{Spec}\ k(x)[y] \to \mathrm{Spec}\ k[x]$ will map $[(y - f(x))] \mapsto \eta$, notably a 0 -dimensional point mapped to a 1-dimensional point!

10. Let R be a ring with nilpotence and $I = \sqrt{(0)}$ the nilradical. Take $\varphi : R \to R/I$ to be the usual projection. Then φ^a is a bijection; the key is to recall that the nilradical intersection of all prime ideals, and hence any prime $\overline{\mathfrak{p}} \subseteq R/I$ which is of the for $\mathfrak{p} + I$ must in fact equal itself: $\mathfrak{p} + I = \mathfrak{p}$ (as $I \subseteq \mathfrak{p}$). The example to keep in mind should be $R = \mathbb{C}[x]/(x^2)$ with $I = (x)$.

[a]Note that if an element is nilpotent, localizing it will map it to zero

Lemma 2.1.24: Induced Sepctra Map Is Continuous

Let $\varphi : R \to S$ be a ring homomorphism. Then φ^a is a continuous map

***Proof*:**
It suffices to show the pre-image of a closed set is closed, so take $V(S) \subseteq \mathrm{Spec}\,(R)$ and consider $(\varphi^a)^{-1}(V(I))$. Show that this is equal to $V(\varphi(I))$, showing it is closed.

The map φ^a given by the natural inverting function is a naturally continuous map, that is φ^a is a continuous function between the Spec of two rings with the Zariski topology. Hence Spec is a [contravariant] functor from **CRing** to **Top**. Furthermore, continuity can be shown on distinguished open sets, namely as they form a basis we can show that:

$$(\varphi^a)^{-1}\left(D([\mathfrak{p}]) = D(\varphi([\mathfrak{p}])\right)$$

Note that there will be continuous maps between the spectrum of rings that are not φ^a,

Example 2.1.25: Continuous Not Induced By Ring Homomorphism
The map from $\mathbb{A}^1_{\mathbb{C}}$ to itself that is the identity on all points except $[(x)]$ and $[(x-1)]$ (which will be represented with 0 and 1) where it swaps these two values is *not* a polynomial and hence cannot have been induced by a ring homomorphism, however it *is* continuous. If $0, 1 \notin V(S)$, then $\varphi^{-1}(V(S)) = V(S)$. Similarly if both values are inside. If 0 or 1 is in $V(S)$, say $0 \in V(S)$, then $\varphi^{-1}(V(S)) = V(S) \setminus \{0\} \cup \{1\}$ which is closed as all finite subsets of $\mathbb{A}^1_{\mathbb{C}}$ are closed.

Can you now find two ring homomorphisms that induce the same Spectra map? See exercise 2.1.1(13)

> Exercise 2.1.1(14) goes over some of the properties preserved by continuous maps. Most of the properties that we shall be want for schemes to have (ex. separatedness, properness, smoothness, multiplicity information, finiteness conditions) shall require more than continuity.

The correspondence is terse as Spec does not carry enough information to fully and faithfully correspond to ring homomorphism. In chapter 3 it shall be shown that scheme morphisms are the right object to carry this information (namely, they shall insure that around every point, that is each stalk, the spectra map behaves like a polynomial).

2.1.5 *Subsets of Spectra via quotient and Localization

Two common constructions that came up in a few example is the process of taking the quotient and the localization of a ring R. These form important subsets of $\operatorname{Spec}(R)$, namely by the 4th isomorphism theorem and the prime-correspondence theorem for prime ideals in localization (see [9, chapter 9.3] and [9, chapter 20.5]). By these theorems, $\operatorname{Spec}(R/I), \operatorname{Spec}(S^{-1}R) \subseteq \operatorname{Spec}(R)$. If $I = \mathfrak{p}$ and $S = R \setminus \mathfrak{p}$,the first corresponds to all prime ideals containing $\mathfrak{p}$, while the second corresponds to all prime ideals that are contained in the prime ideal $\mathfrak{p}$[11] These two sets need not union to the entire $\operatorname{Spec}(R)$ (take $\mathbb{Z}$ and 5), however they do correspond to two "opposite" ways of carving out a part of $\operatorname{Spec}(R)$. In the following visual is a very common illustration:

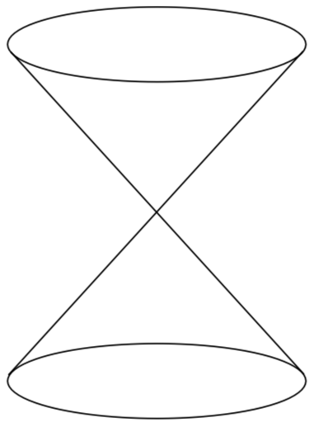


Figure 2.2: partial decomposition of $\operatorname{Spec}(R)$

Geometrically, $\operatorname{Spec}(R/I)$ can be thought of as "carving out" a part of $\operatorname{Spec}(R)$. For example, take $R = \mathbb{C}[x, y, z]$ and $\mathfrak{p} = (x^2 + y^2 - z)$. If we were working with affine complex varieties, then we would say this is the 3D-parabola in $\mathbb{C}^3$, however as we are on $\operatorname{Spec}(R/\mathfrak{p})$ this would come with some extra points (which ones?). For $\operatorname{Spec}(S^{-1}R)$, we may think it as the opposite (we shall cover the special case of $R_\mathfrak{p}$ afterwards). For example let's say $S = \{1, f, f^2, ...\}$ and take $S^{-1}R$. Then the primes of $\operatorname{Spec}(S^{-1}R)$ would be all primes that do not containing f, that is all points where f doesn't vanish!

[11] More generally it is all prime ideals not having trivial intersection with the multiplicative set.

As an illustration, the set $\operatorname{Spec}\left(\mathbb{C}[x,y]_{y-x^2}\right)$ are the points in the affine plane, minus all the points on the parabola $y-x^2$, namely all the singletons (a,a^2) for $a\in\mathbb{C}$ (so $[(x-a,y-a^2)]$), as well as the point $[(y-x^2)]$.

Proposition 2.1.26: Closed And Open Subsets

Let R be a commutative ring, $I\subseteq R$ an ideal, $f\in R$, $\mathfrak{p}\subseteq R$ prime, and S a multiplicative set. Then:

1. $\operatorname{Spec}(R/I)\subseteq\operatorname{Spec}(R)$ is a closed subset
2. $\operatorname{Spec}(R_f)$ is an open subset
3. $\operatorname{Spec}\left(S^{-1}R\right)\subseteq\operatorname{Spec}(R)$ need not be open or closed in general. In particular $\operatorname{Spec}(R_\mathfrak{p})\subseteq\operatorname{Spec}(R)$ need not be open or closed. However, it is an embedding, namely $\iota:\operatorname{Spec}S^{-1}R\hookrightarrow\operatorname{Spec}R$ is bijective, continuous, and closed.

***Proof*:**

1. Think of $V(I)$ (induce the topology on $\operatorname{Spec}(R/I)$ via the inclusion map)
2. Think $\operatorname{Spec}(R)\setminus V(f)$
3. think of $\operatorname{Spec}(\mathbb{Q})\subseteq\operatorname{Spec}(\mathbb{Z})$ and $\operatorname{Spec}(\mathbb{Z}_\mathfrak{p})\to\operatorname{Spec}(\mathbb{Z})$ induced by $\mathbb{Z}\to\mathbb{Z}_\mathfrak{p}$ (this shows that $\operatorname{Spec}(R_\mathfrak{p})$ need not be open.

For the case of $S=R-\mathfrak{p}$, we can think of $\operatorname{Spec}(R_\mathfrak{p})$ as keeping near the $\mathfrak{p}$, namely since the prime ideals away from $\mathfrak{p}$ will be turned into units, the primes intersecting $R\setminus\mathfrak{p}$ will be eliminated. For example, if $R=\mathbb{C}[x,y]$ and $\mathfrak{p}=(x,y)$, then we keep all the points corresponding to being "near" the origin, that is the point $[(x,y)]$, i.e. $(0,0)$, and the 1-dimensional points $[(f(x,y))]$ where $f(0,0)=0$ (i.e. where $f\equiv 0 \bmod (x,y)$) that is irreducible curves that pass through the origin. In the special case where $R=k[x]$, then the localization will always only have two points, usually called the classical point and the generic point. These shall eventually be seen as geometric interpretations of DVR.

Using these geometric results, they can lend intuition towards algebraic manipulation:

Example 2.1.27: Illustration Of Quotient And Localization

Using the above intuitions, we can quickly come up with some intuition isomorphisms. Take $R=\mathbb{C}[x,y]/(xy)$. Visually, this is the union of the y and x-axis. If we localize this at x, then by our above intuition this should eliminated all points where the "function" x vanishes. The function x vanishes at the y-axis, and so we must be left with the x-axis minus the origin. This happens to be $\operatorname{Spec}\mathbb{C}[x]_x$. In fact, these two rings are isomorphic!

$$\left(\frac{\mathbb{C}[x,y]}{(xy)}\right)_x\cong\mathbb{C}[x]_x$$

which the reader should check.

Exercise 2.1.1

1. Show that $D(f) \cap D(g) = D(fg)$.
2. $D(f) \subseteq D(g)$ iff $f^n \in (g)$ iff g is invertible in R_f. How would you prove this via "geometric explanation"? For an algebraic explanation, change to $V(f) \supseteq V(g)$ and take $I(-)$ to inverse and conclude.
3. $D(f) = \emptyset$ iff f is in the Nilradical.
4. Show that any connected component is the union of irreducible components (think of the + shape, i.e. $\mathrm{Spec}\,(\mathbb{C}[x,y]/(xy))$).
5. Show that $\mathbb{A}^1_{\mathbb{C}}$ is irreducible.
6. Show that in an irreducible topological space, any nonempty open set is dense.
7. If X is a topological and Z an irreducible subspace, then $\overline{Z}$ is irreducible as well
8. Show that an irreducible space is connected.
9. If A is a integral domain, $\mathrm{Spec}\,(A)$ is irreducible.
10. Show that $\mathrm{Spec}\,(\mathbb{C}[x,y]/(xy))$ is connected but reducible.
11. Show that there is a natural bijection irreducible components (maximal irreducible subsets) of $\mathrm{Spec}\,(R)$ and the *minimal* prime ideals of R.
12. these next two exercises show us how the theory of classical varieties persists with Schemes
    a) Let A be a finitely generated k-algebra. Then show that the closed points of $\mathrm{Spec}\,(A)$ are dense; show that if $f \in A$ and $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\mathrm{Spec}\,(A)$[12]
    b) Let $A = \overline{k}[x_1, ..., x_n]/I$ where the Nilradical is trivial, and let $X = \mathrm{Spec}\,(A) \subseteq \mathbb{A}^n_{\overline{k}}$. Show that [regular] functions on X are determined by their valued on the closed points. Hint: if f, g are different [regular] functions on X, then $f - g$ is nowhere zero on an open subset of X. Use the above result.
13. Find two ring homomorphism that induce the same spectra map.
14. Let $f : X \to Y$ be a continuous map.
    a) Show that irreducible sets map to irreducible sets to irreducible sets.
    b) Show that if X is Noetherian, then the image of f is Noetherian
    c) Show that the dimension of X is greater than or equal to the dimension of the image of Y (also see 2.7).

2.2 Schemes

The spectrum of a ring with the Zariski Topology does not hold all data provided by a commutative ring, for example nilpotence is forgotten. The *structure sheaf* will be added to $\mathrm{Spec}\,R$ in order to add this extra information (namely, we shall have the collection of regular function defined on each

[12]Hint: A_f is also a finitely generated k-algebra use generalized Nullstellensatz. Show the closed points of the Spec of a finitely generated k-algebra are those for which the residue field is a finite extension of k

open set as part of the data we are tracking). The resulting object is called the *affine scheme*.

After defining the affine scheme, we shall require one more crucial abstraction that links it to differential geometry: we shall want our object to the glued-together open sets. In differential geometry, these would all be open subsets of euclidean space, usually simply connected and all isomorphic. A key distinguishing feature in algebraic geometry is that the sets that can be glued can be very different, namely $\operatorname{Spec} R$ for all commutative rings R will be the sets from which we can select, and as the reader has already seen these sets can have very different topologies.

2.2.1 The Structure Sheaf for Affine Schemes

When working with geometric objects (varieties, real and complex manifolds, etc), we study the local behaviour of the space, namely by studying some given information on open sets or germs/stalks. In the theory of differential manifold, the information at each local was the "linear" equation (recall that one way of defining the tangent bundle is by taking all the derivations of smooth functions, which can be shown to "preserve" the linear part of each smooth function). For an , we are interested in the functions functions are *regular* on open sets. By section 2.1.2, it is enough to define the regular functions on distinguished open sets. For those motivated by analytic parallels, Complex manifold have a sheaf of meromorphic functions which are defined on the open subsets of the complex manifold, and we usually have holomorphic functions on the entire manifold.

To motivate the following sheaf, recall that on $\mathbb{C}^n$, if we take open subsets there are many more holomorphic functions that could be defined on that open subset, namely that rational functions are well-defined when the roots of the denominator (which are not cancel by the numerator) are outside the open set. For example, on $\mathbb{C}^2 \setminus \{x\text{-axis} \cup y\text{-axis}\}$, the function

$$\frac{x^2+4x+y+1}{x^5y^4}$$

is well-defined. Moving this to Spec, in $\mathbb{A}^2_{\mathbb{C}}$, we ought to have the above function to be well-defined on $D(xy)$. Furthermore, we want the collection of global sections, $\mathcal{O}_X$, to be R, mirroring the "global section" on $\mathbb{C}^2$ being the (non-rational) holomorphic functions in two variables.

As we saw in theorem 1.1.22, it suffice to define a sheaf on a base of $\operatorname{Spec}(R)$, namely on the distinguished open sets. Thus, we may define on $D(f) \subseteq \operatorname{Spec}(R)$ the ring that is the localization of R at the multiplicative subset of all function that do not vanish on $V(f)$ (note how this is subtly different than functions that do not vanish on f, what if f is nilpotent?). This can be represented as:

$$\mathcal{O}_X(D(f)) = \mathcal{O}_X(X_f) \cong R_f$$

First off, this is a presheaf: if $D(f) \supseteq D(g)$, by exercise 2.1.1 there is a minimal $n \in \mathbb{N}_{>0}$ such that $g^n \in (f)$ so $rf = g^n$. Then as $D(g^n) = D(g)$, we may define the restriction map:

$$\operatorname{res}_{X_f, X_g} : R_f \to R_g \qquad \frac{s}{f^m} \mapsto \frac{r^m s}{g^{nm}}$$

which shows we have defined a presheaf on a base. We next must show that it satisfies the sheaf on base axioms.

Proposition 2.2.1: Distinguished Opens form Sheaf-Basis for $\operatorname{Spec} R$

Let $X = \operatorname{Spec} R$, and suppose that $D(f)$ is covered by distinguished open sets $D(f_a) \subseteq D(f)$. Then:

1. If $g, h \in R_f$ are equal in R_{f_a} when passing through the restriction map for every a, then $g = h$.
2. If for each a, there exists a $g_a \in R_{f_a}$ such that for each pair a, b, the images of g_a and g_b in $R_{f_a f_b}$ are equal, then there is an element $g \in R_f$ whose image in R_{f_a} is g_a.

In other words, $\mathcal{O}_X$ defines a $\mathcal{B}$-sheaf on the distinguished base, and by theorem 1.1.22, $\mathcal{O}_X$ extends to a sheaf on X.

***Proof*:**
Let $\operatorname{Spec}(R) = \bigcup_{i \in I} D(f_i)$, which is equivalent to saying the f_i generate the unit ideal: $(f_i)_{i \in I} = R$. We show the locality and gluability axioms, or in categorical terms the exactness of:

$$0 \longrightarrow R \longrightarrow \prod_{i \in I} R_{f_i} \rightrightarrows \prod_{i \neq j \in I} R_{f_i f_j}$$

Starting with locality. As $s_1 = s_2$ is equivalent to $s_1 - s_2 = 0$, it suffices to show the result in the case where $s = 0$. By quasicompactness we may pass to a finite subcover:

$$\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i) \qquad (f_1, \dots, f_n) = R$$

Suppose that for some $s \in R$, $\operatorname{res}_{\operatorname{Spec}(R), D(f_i)} s = 0$ for all i. We must show $s = 0$. By definition of R_{f_i}:

$$s|_{D(f_i)} = 0 \qquad \text{if and only if} \qquad f_i^{m_i} s = 0 \quad \text{for some } m_i \geq 0$$

Taking $m = \max_i(m_i)$, we have $f_i^m s = 0$ for all i. Now, as $D(f_i) = D(f_i^m)$, the cover $\bigcup D(f_i^m) = \operatorname{Spec}(R)$ still holds, so $(f_1^m, \dots, f_n^m) = R$. Thus, for appropriate choices $r_i \in R$ with $\sum_{i=1}^n r_i f_i^m = 1$, we get:

$$s = 1 \cdot s = \left(\sum_i r_i f_i^m \right) s = \sum_i r_i (f_i^m s) = 0$$

For general $D(f)$, repeat the above argument replacing R with R_f (quasi-compactness of $D(f)$ ensures the cover can still be taken to be finite, and the same algebraic manipulations apply in R_f).

We next show gluability[a]. Starting with the case $\operatorname{Spec}(R) = \bigcup_i D(f_i)$, suppose there are elements in each R_{f_i} that agree on overlaps $R_{f_i f_j}$ (note that intersections of distinguished open sets are distinguished open sets). As there may be infinitely many sets on which there are overlaps, it cannot be assumed that I is finite. However, the finite case is a good start, say $I = \{1, \dots, n\}$. Then we have elements $\frac{a_i}{f_i^{l_i}} \in R_{f_i}$ agreeing on the overlaps $R_{f_i f_j}$, meaning:

$$(f_i^{l_i} f_j^{l_j})^{m_{ij}} (f_j^{l_j} a_i - f_i^{l_i} a_j) = 0$$

Simplifying notation by writing $g_i = f_i^{l_i}$:

$$(g_i g_j)^{m_{ij}}(g_j a_i - g_i a_j) = 0$$

As there are finitely many m_{ij}, take $m = \max_{i,j}(m_{ij})$:

$$(g_i g_j)^m (g_j a_i - g_i a_j) = 0$$

Next, convert to a different set of distinguished opens: take $b_i = a_i g_i^m$ and $h_i = g_i^{m+1}$ so that $D(h_i) = D(g_i)$. Then on each $D(h_i)$, we have the section b_i/h_i, and the overlap condition simplifies to:

$$h_j b_i = h_i b_j$$

Noting that $\bigcup_i D(h_i) = \operatorname{Spec}(R)$ gives $1 = \sum_{i=1}^n r_i h_i$ for appropriate $r_i \in R$, define:

$$r = \sum_i r_i b_i$$

Then r is the element of R that restricts to b_j/h_j on each $D(h_j)$:

$$rh_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = \left(\sum_i r_i h_i\right) b_j = b_j$$

showing gluability in the finite case. For the infinite case, we use the previously shown base-locality axiom. By quasi-compactness, we can choose a finite subcover $D(f_1), \ldots, D(f_n)$ and construct r as above. For any $z \in I \setminus \{1, \ldots, n\}$, repeat the argument on $\{1, \ldots, n, z\}$ to get $r' \in R$ which restricts to a_i for $i \in \{1, \ldots, n, z\}$. By base locality, $r' = r$. Hence r restricts to $a_z/f_z^{l_z}$ for all z.

To finish off the proof, we must handle the case of an arbitrary $D(f)$ rather than all of $\operatorname{Spec}(R)$. This follows by applying the same argument with R replaced by R_f throughout.

[a] Vakil pointed out that Serre called this a "partition of unity argument," which is the local-to-global principle physically manifested in differential geometry – a rather good perspective on what is about to happen.

It may be tempting to say that $\mathcal{O}_{\operatorname{Spec}(R)}(U)$ for any open subset U is a localization of R at some multiplicatively closed subset S, however this is not true:

Example 2.2.2: Subtle Local Section Of Affine Scheme

Let $R = k[x, y, z, w]/(xw - yz)$; $\operatorname{Spec} R$ is called the *quadric cone*. Note that R is an integral domain as $(xw - yz)$ is prime[a]. Let $U = D(y) \cup D(w)$. Then a section $s \in \mathcal{O}(U)$ is determined by a choice of $f \in R_y$ and $g \in R_w$ that agree on R_{yw}, namely $\mathcal{O}(U)$ is determined by the left exact sequence:

$$0 \to \mathcal{O}(U) \hookrightarrow R_y \oplus R_w \xrightarrow{(f,g) \mapsto f-g} R_{yw}$$

The kernel of the difference map is $\mathcal{O}(U)$. Take $(x/y,\, z/w)$ and consider:

$$\frac{x}{y} = \frac{z}{w} \qquad \text{if and only if} \qquad xw - zy = 0$$

where there is no need for a factor of $(yw)^n$ in front, as there are no zero-divisors. In R, this is exactly the defining relation of the quotient, hence this equality holds. Therefore there is an element $t \in \mathcal{O}(U)$ such that:

$$t \mapsto \left(\frac{x}{y}, \frac{z}{w}\right)$$

Next, for any $r \in R$, certainly $r \in R_y, R_w$ and $r - r = 0$, so $r \in \mathcal{O}(U)$. Is it possible that $t = r$ for some $r \in R$? As the map $\mathcal{O}(U) \hookrightarrow R_y \oplus R_w$ is injective, if $t = r$ then since $r \mapsto (r, r)$ we would need $r = x/y = z/w$, but y and w are not invertible in R, and hence $t \notin R$. Thus the ring $R[t] \neq R$. Certainly $R[t] \subseteq \mathcal{O}(U)$ and any R-polynomial $f(t) \in R[t]$ maps to:

$$f(t) \mapsto \left(f\left(\frac{x}{y}\right), f\left(\frac{z}{w}\right)\right)$$

Let us show that $yt - x = 0$ (equivalently $wt - z = 0$). By definition of a sheaf, this can be checked on the open cover $\{D(y), D(w)\}$. By construction (recall that restriction maps are ring homomorphisms and that $r \mapsto r$ under these maps):

$$(yt - x)|_{D(y)} = y \cdot \frac{x}{y} - x = x - x = 0$$

and

$$(yt - x)|_{D(w)} = y \cdot \frac{z}{w} - x = \frac{yz - xw}{w} = \frac{0}{w} = 0$$

giving $yt - x = 0$. Thus:

$$R[t] \cong \frac{R[T]}{(yT - x)} \cong R\left[\frac{x}{y}\right]$$

Note that y is *not* a unit in $R[x/y]$. Let us show $\mathcal{O}(U) \cong R[t]$. If $s \in \mathcal{O}(U)$,

$$s|_{D(y)} = \frac{a}{y^n} \qquad \text{and} \qquad s|_{D(w)} = \frac{b}{w^m}$$

for some $a, b \in R$ and $n, m \in \mathbb{N}$. They must agree on $D(yw)$:

$$\left.\frac{a}{y^n}\right|_{D(yw)} = \frac{aw^n}{(yw)^n} = \frac{by^m}{(yw)^m} = \left.\frac{b}{w^m}\right|_{D(yw)}$$

Since R is an integral domain, this simplifies to $aw^m = by^n$ in R.

To show that $s \in R[t]$, we use the key relations $x = ty$ and $z = tw$ (which hold in $R[t]$ since $yt = x$ and $wt = z$). In R_y, the element a/y^n can be rewritten by substituting $x = ty$ and $z = tw$: every monomial in a involving x or z can be expressed using t, y, and w. More precisely, any element of R_y can be written as a polynomial in t with coefficients in $k[y, y^{-1}, w]$ (after eliminating x and z using the relations). Similarly, any element of R_w is a polynomial in t with coefficients in $k[y, w, w^{-1}]$. The overlap condition $aw^m = by^n$ ensures that the two expressions, when restricted to R_{yw}, give the same polynomial in t. By the locality axiom (already proved), this polynomial is unique, and it lies in $R[t]$ (no negative powers of y or w are needed, as one can verify by comparing degrees). Hence $s \in R[t]$, and so $\mathcal{O}_X(U) \cong R[x/y]$.

Now, $R[x/y]$ is *not* a localization of R. Indeed, $R[x/y] \ncong R$ (since $x/y \notin R$), yet $R[x/y]$ has the same units as R (namely k^*): no element becomes invertible. Since any nontrivial localization R_S contains units not present in R, and isomorphisms must preserve units, no localization of R is isomorphic to $R[x/y]$.

More generally, given an arbitrary open subset U, we can cover it by affine opens $\{U_i\}_{i\in I}$ and cover the intersections $U_i \cap U_j$ by affine opens $\{U_{ijk}\}_{k\in K_{ij}}$. Then the sheaf condition gives that $\mathcal{O}_X(U)$ is the equalizer of:

$$\prod_{i\in I} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j\in I,\, k\in K_{ij}} \mathcal{O}_X(U_{ijk})$$

The general idea is that $\mathcal{O}_X(U)$ may contain rational functions f/g where g is nonzero on all of U. In practice, we shall often only require the local sections on distinguished open sets or on affine open sets, and many computations reduce to these two cases.

Note that even the following weaker hope fails: given $U \subseteq \operatorname{Spec} R$, is $\mathcal{O}_X(U)$ isomorphic to the localization of R at the multiplicative set S of all elements that do not vanish at any point of U? Consider $U = D(x) \cup D(y) \subseteq \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$. We will show in example 2.3.14 that U is not affine. However, $\mathcal{O}_{\mathbb{A}^2_k}(D(x) \cup D(y)) \cong k[x, y]$ (a Hartogs-type phenomenon: regular functions extend across a codimension-2 locus). This is just $k[x, y]$ itself, yet $D(x) \cup D(y) \ncong \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$.

This shows that the relationship between local geometry and algebra is more nuanced than the affine case suggests.

[a] As $k[x, y, z, w]$ is a UFD, a principal ideal (p) where p is irreducible is prime. It is tedious but rudimentary to show that $xw - yz$ is irreducible.

Keeping track of the structure sheaf $\mathcal{O}_{\operatorname{Spec} R}$ along with $\operatorname{Spec}(R)$ gives us our first definition we need:

Definition 2.2.3: Affine Scheme

Let $X = \operatorname{Spec}(R)$ be the spectrum of the ring R with the Zariski topology, and let $\mathcal{O}_{\operatorname{Spec}(R)}$ be the structure sheaf. Then $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ is called an *affine scheme*.

In some textbooks, an affine scheme is defined as a ringed space (recall definition 1.4.2) $(X, \mathcal{O}_X)$ is an *affine scheme* if it is ringed-space isomorphic to $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for appropriate ring R.

With the sheaf defined, we have now "justified" thinking of elements on an arbitrary ring R as functions, as

$$\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec} R) = \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(1)) = R_1 = R$$

Even more is true, due to the section's being constructed through localization of R, they may be ratio's of functions, and we give elements of section a particular name:

Definition 2.2.4: Regular Functions

Let $X = \operatorname{Spec} R$ be an and $U \subseteq X$ an open subset. Then the set $\mathcal{O}_X(U)$ is called the set of *regular functions* on U.

Note that like complex manifolds (and unlike general smooth manifolds), there are in general relatively few functions in the global section of an as compared to the global section (in the global section of an affine scheme, all rational functions are regular, that is they are R). On the other hand, $\mathcal{O}_X(U)$ may have many interesting rational functions. In fact, in many cases all open subsets of X are dense meaning the local behavior can give a lot of fruitful information on the entire scheme (as shall be the case for abstract varieties, see ref:HERE)

Example 2.2.5: Affine Schemes

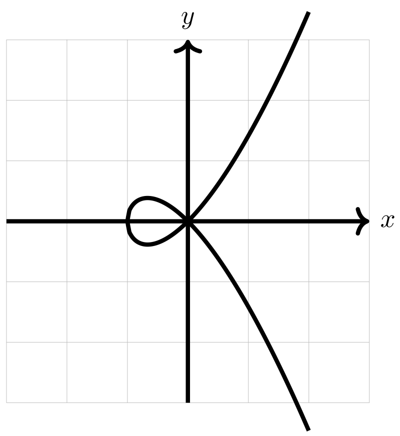
All spectra can be equipped with a structure sheaf by definition and so they are all affine schemes. For less obvious examples:

1. If $k \ncong k'$ are distinct fields fields, then as schemes $\operatorname{Spec}(k) \ncong \operatorname{Spec}(k')$ as their sheaves are different. Since there is only one point $[(0)]$ an all such Spectra, the functions (elements of the sheaf) will all return themselves, which the reader can think of as there only being constant functions, where each sheaf have different constant functions.

2. Let $R = k[x]_{(x)}$. Then $\operatorname{Spec} R$ contains two points, $[(x)]$ and $[(0)]$ and has 3 open sets, namely
$$\emptyset \quad \{[(0)]\} \quad \{[(0)], [(x)]\} = \operatorname{Spec} R$$
Note that U and $\emptyset$ are distinguished open sets (since $\{(0)\} = X_x$). The corresponding structure sheaf is:
$$\begin{aligned} \mathcal{O}_X(X) &= R = k[x]_{(x)} \\ \mathcal{O}_X(U) &= k(x) \end{aligned}$$
The restriction map is the natural inclusion. This schemes is an example of a *local scheme*.

3. Let $R = k[[x]]$, the completion of $k[x]$ at x. The only points of Spec R are again $[(0)]$ and $[(x)]$, and so it has the same open sets. The corresponding structure sheaf is:
$$\begin{aligned} \mathcal{O}_X(X) &= R = k[[x]] \\ \mathcal{O}_X(U) &= \operatorname{Frac}(k[[x]]) \end{aligned}$$
Thus, we see that there can be schemes with very similar sheaves and identical spectra, but are distinct.

   This scheme is considered to be *even more local* than localization. Consider $\mathbb{A}^2_k$. Then the sequence of maps:
$$k[x, y] \hookrightarrow k[x, y]_{(x,y)} \hookrightarrow k[[x, y]]$$
induces a sequence of inclusion spectra maps $Y \to X \to \mathbb{A}^2_k$, showing how the completion at the point (x, y) is "smaller" than the localization at that point. To demonstrate this, assume k does not have characteristic 2 or 3 and consider the affine scheme given by $y^2 = x^2 + x^3$:



Note that $y^2 - x^2 - x^3$ is irreducible, and localizing at (x, y) keeps the corresponding $(\frac{k[x,y]}{(y^2-x^2-x^3)})_{(x,y)}$ with only two points. However, in

$$\frac{k[[x, y]]}{(y^2 - x^2 - x^3)}$$

we have the access to power series expansion around $(0, 0)$, and we have that the power series:

$$x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \cdots$$

belongs to this ring, and notably this power series is the square-root of $x^2 + x^3$. Thus, we may write:

$$y^2 - x^3 - x^2 = (y - \sqrt{x^3 + x^2})(y + \sqrt{x^3 + x^2})$$

and so we get that $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ is *not* an integral domain, and in particular we shall see that $\operatorname{Spec} \frac{k[[x,y]]}{(y^2-x^3-x^2)}$ has *two* irreducible components corresponding to the two branches at $(0, 0)$.

4. The reader may recall that most power series do not satisfy an algebraic equation. For this reason, the above ring is often considered too big, and instead we take an intermediary ring H which contains all power series that satisfy algebraic equations over $\operatorname{Frac}(R)$. This is called the *Henselization* of $\operatorname{Spec} R$ at a point $\mathfrak{p}$.

5. If X, Y are then the product of X and Y are:

$$\operatorname{Spec}(R) \times \operatorname{Spec}(S) = \operatorname{Spec}(R \otimes_{\mathbb{Z}} S) = \operatorname{Spec}(R \otimes S)$$

where the tensor product with no base ring by default assume it is over the initial ring. The generalization of this concept shall bee shown in section 3.2.

2.2.2 Ideal Correspondence in Affine Schemes

So far, the dictionary between certain types of ideals of R and the geometric equivalent in $\operatorname{Spec}(R)$ has been establish, namely through $V(-)$ and $I(-)$:

$$\begin{aligned} \text{maximal ideal of } R &\quad\Leftrightarrow\quad \text{closed points of } \operatorname{Spec}(R) \\ \text{minimal prime ideals of } R &\quad\Leftrightarrow\quad \text{irreducible components of } \operatorname{Spec}(R) \\ \text{prime ideal of } R &\quad\Leftrightarrow\quad \text{irreducible closed subset of } \operatorname{Spec}(R) \\ \text{radical ideals of } R &\quad\Leftrightarrow\quad \text{closed subsets of } \operatorname{Spec}(R) \end{aligned}$$

What of a general ideal $J \subseteq R$? In section 2.1.1, this information was "lost" as $I(V(J)) = \sqrt{J}$. However, a sheaf would capture this information!

Let's start with the simplest case of $R = \mathbb{C}[x]/(x^2)$ with $X = \operatorname{Spec}(R)$ that was mentioned in example 2.1.3(6). If we consider the set $\operatorname{Spec}(R)$, then this is just the origin, i.e. $[(x)]$. Namely, by proposition 2.1.10[13],

$$\operatorname{Spec}(\mathbb{C}[x]/(x^2)) \cong_{\mathbf{Top}} \operatorname{Spec}(\mathbb{C}[x]/(x)) \cong_{\mathbf{Top}} \{\overline{(x)}\} \equiv \{0\}$$

[13] Note that this is a topological isomorphism, these are not scheme isomorphic as we shall see in section 3.1.1!

Note that (0) is no longer prime since $x^2 \in (0)$ but $x \notin (0)$, and thus $[(0)] \notin \operatorname{Spec}(\mathbb{C}[x]/(x^2))$. To show that the $\left(\operatorname{Spec}\left(\mathbb{C}[x]/\left(x^2\right)\right), \mathcal{O}_{\operatorname{Spec}(\mathbb{C}[x]/(x^2))}\right)$ is different from $\left(\operatorname{Spec}(\mathbb{C}[x]/(x)), \mathcal{O}_{\operatorname{Spec}\,(\mathbb{C}[x]/(x))}\right)$, let us first label:

$$(X, \mathcal{O}_X) = \left(\operatorname{Spec}\frac{\mathbb{C}[x]}{(x^2)}, \mathcal{O}_{\operatorname{Spec}\mathbb{C}[x]/(x^2)}\right)$$

Consider the image of a polynomial $f(x)$ (i.e. an element in the global section) in $\mathbb{C}[x]/\left(x^2\right)$. For example if $\overline{f(x)} = \overline{(x-a)^2} \in \mathbb{C}[x]/\left(x^2\right)$, then:

$$\bar{f} = \overline{(x-a)^2} = \overline{x^2 - 2ax + a^2} = \overline{-2ax + a^2} \in \mathbb{C}[x]/\left(x^2\right) = \mathcal{O}_X(X)$$

Evaluating at the single point $[(x)] = \epsilon$ we get

$$\bar{f}(\epsilon) = a^2$$

showing that evaluating at ϵ acts like plugging in zero. as x is a function, we get:

$$x(\epsilon) = 0$$

even though x is *not* the zero-function. In fact, one very important algebraic fact that is "obvious" for traditional functions but is not necessarily the case for ring with nilpotents is that a function that is zero everywhere will have to be zero:

Lemma 2.2.6: Nilpotence and Nonzero "Zero" Functions

Let R be a ring with nilpotence (i.e. a nonreduced ring). Then $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R))$ contains a function that evaluate to 0 everywhere but is not zero.

***Proof*:**
Let $f \in R$ be a nilpotence element so that $f \neq 0$ but $f^n = 0$. Then f is contained in each prime ideal, for $ff^{n-1} = f^n = 0 \in \mathfrak{p}$, and so either $f, f^{n-1} \in \mathfrak{p}$, which we can then reduce to having $f \in \mathfrak{p}$. Hence it is in the Nilradical, and so as a function

$$f(\mathfrak{p}) = 0$$

However $f \neq 0$, as we sought to show.

For completeness, we register this result whose proof was outlined in example 2.1.23(8). As it turns out, the failure of a regular function to be determined by its evaluation at points can completely be put at the feet of nilpotents:

Proposition 2.2.7: Spectrum And Nilpotent Elements

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism. As a consequence, two regular functions (elements of R) have the same value if they differ by a nilpotent element.

Thus, as topological spaces, nilpotents have no effect. Their presence is felt within the sheaf we put on spectra (see section 2.2.1)

> ***Proof*:**
> Hint: recall that the nilradical is equal to the intersection of all prime elements (see [9, chapter 22]).

Hence if the nilradical is zero, $\sqrt{(0)} = (0)$, functions (in global sections of affine schemes) are determined by their points[14]! In the new language we developed, this say that a function vanishes at every point of an affine scheme if and only if it is nilpotent! Note how this contrasts to the case of (classical) varieties where we required k to be algebraically closed for the same result, showing the greater generality achieved by having the extra prime points!

How to then visualize the $\operatorname{Spec}(R)$ along with its structure sheaf? It is usually done by picturing a "cloud" or "fuzz" around the point:

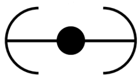


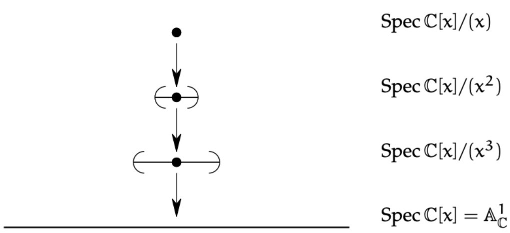
Figure 2.3: Picture of $(X, \mathcal{O}_X)$, not just $X = \operatorname{Spec}(R)$

This can be thought of as the "first degree germ" information around the point (recall that in $\mathbb{C}[\epsilon]$, $f(a+\epsilon) = f(a) + f'(a)\epsilon$, showing that the output of f at a plus the "fudge" of ϵ gives $f(a)$ plus an infinitesimal amount in the direction $f'(a)$. The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be thought of as inching closer to $f(0)$:

$$\begin{tikzcd}
\mathbb{C}[x] \arrow[r, two heads] & \mathbb{C}[x]/(x^2) \arrow[r, two heads] & \mathbb{C}[x]/(x) \\
f(x) \arrow[rr, mapsto] & & f(0),
\end{tikzcd}$$

The reason the point is visualized to have the fuzz is that it is the point that creates this "effect" when a function evaluates at this point, in other words remembering the functions behaviour changes the geometric interpretation of the point. If we take, $\mathbb{C}[x]/(x^3)$, then then the spectrum will again be a single point, and when finding an element from the global section, the value, 1st derivative, and 2nd derivative will be remembered. This can either be thought of a larger fuzz or a fuzz that is "bending" with more degree's of precision. We can have the following chain of inclusions:

[14]Proposition 2.5.5 shall push this result to its maximal generalization on quasicompact schemes



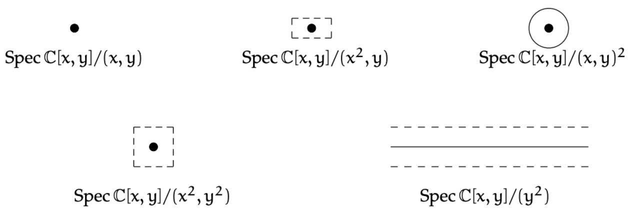
Going to higher dimensions and considering:

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(x,y)^2}\right)$$

this can be thought of as remembering the derivative in all directions, and can be represented by a circle.

Example 2.2.8: Geometric Interpretation of Nilpotence

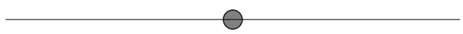
1. The reader should stare at this image to make sure it makes sense:



2. consider,

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2,xy)}\right)$$

Then this can be thought of as the intersection of a thickened x-axis and the "+" created by xy, that is this can be thought of as the x-axis as well as the first-order differential information around the origin.



Once we introduce stalks in section 2.2.5, these points will be more detectable. Notice that $[(x)]$ is *no longer* a generic point since $y^2 = 0 \in \overline{(x)}$ but $y \notin \overline{(x)}$, and hence the ideal is not prime, thus no longer a point. On the other hand, the function y when evaluating at the points $[\overline{(x-a, y-b)}] = [\overline{(x-a, y)}]$ is 0 at each point

3. Consider
$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2)}\right)$$
By proposition 2.1.10, this is homeomorphic to $\mathbb{A}^1_{\mathbb{C}}$. As this is a rather simple example, this can be seen concretely: if $\mathfrak{p} \subseteq \mathbb{C}[x,y]/(y^2)$ is prime and contains y^2, then by primality $y \in \mathfrak{p}$, and hence the prime ideals correspond to those of $\mathbb{C}[x,y]/(y^2,y) \cong \mathbb{C}[x]$, i.e. those of $\mathbb{A}^1_{\mathbb{C}}$.

Now, in $\mathbb{C}[x,y]/(y^2)$, there cannot be y-terms of degree higher than 1: if $F(x,y) \in \mathbb{C}[x,y]/(y^2)$, then:
$$F(x,y) = f(x) + g(x)\, y$$
This can be thought of as $F(x,y)$ "remembering" some first-order information in the y-direction, as though it were the restriction of a function on $\mathbb{A}^2_{\mathbb{C}}$ to an infinitesimal neighborhood of the x-axis. In this decomposition, $f(x)$ is the value on the x-axis and $g(x)$ plays the role of $\partial F/\partial y|_{y=0}$ – a tangent direction at each point. Algebraic operations respect this: for instance,
$$(f_1 + g_1 y)(f_2 + g_2 y) = f_1 f_2 + (f_1 g_2 + f_2 g_1)\, y$$
which mirrors the product rule for derivatives.

Another behaviour that may seem strange is the addition of new units! Consider $1 + xy$. Since y is nilpotent ($y^2 = 0$), the element xy is also nilpotent: $(xy)^2 = x^2 y^2 = 0$. We can verify directly that $1 + xy$ is a unit:
$$(1 + xy)(1 - xy) = 1 - (xy)^2 = 1 - 0 = 1$$
More generally, an element $f(x) + g(x)\, y$ is a unit in $\mathbb{C}[x,y]/(y^2)$ if and only if $f(x)$ is a unit in $\mathbb{C}[x]$, i.e. $f(x) \in \mathbb{C}^*$ is a nonzero constant. This is because the nilpotent part $g(x)\, y$ can always be "absorbed" by the geometric series trick $(1+n)^{-1} = 1 - n$ when $n^2 = 0$, but the non-nilpotent part $f(x)$ must already be invertible.

Geometrically, this means the unit group of $\mathbb{C}[x,y]/(y^2)$ is *much larger* than that of $\mathbb{C}[x]$. For $\mathbb{C}[x]$, the only units are the nonzero constants $\mathbb{C}^*$. For the "fuzzy line" $\mathbb{C}[x,y]/(y^2)$, every element $a + g(x)\, y$ with $a \in \mathbb{C}^*$ is a unit – regardless of the choice of $g(x)$. The unit group is thus $\mathbb{C}^* \times \mathbb{C}[x]$, an infinite-dimensional group. All of these units evaluate to the constant a at every point (since y maps to 0 in every residue field), yet they are genuinely distinct elements of the ring, distinguished by their "tangent data" $g(x)$. In this sense, $1 + g(x)\, y$ is an invertible function that is infinitesimally close to 1: it equals 1 on the underlying topological space but carries a nontrivial first-order perturbation. This is the multiplicative counterpart of the earlier observation that y is a nonzero function that evaluates to 0 everywhere – here, $1 + g(x)\, y$ is a non-constant unit that evaluates to the *same* constant everywhere.

Observe that $D(x)$ is the "punctured fuzzy line": it removes the point $[(x)] \in \operatorname{Spec}(\mathbb{C}[x,y]/(y^2))$ but retains the nilpotent thickening at every remaining point. On the other hand, $D(y) = \emptyset$, since every prime ideal contains y (as $y^2 = 0 \in \mathfrak{p}$ implies $y \in \mathfrak{p}$ by primality). Hence the underlying topological space $|\operatorname{Spec} R|$ does not detect the nilpotent "germ" information carried by y – but the structure sheaf does, as y is a nonzero section of $\mathcal{O}_X$ that vanishes at every point.

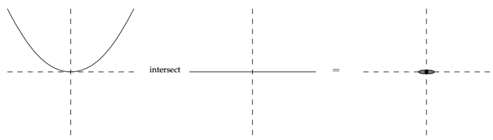
Intersection Theory and Family of Schemes

We shall cover intersection theory extensively in Part II. The intersection theory of curves as covered in [9, chapter 21.6] gives insightful ways in which the discussion above has practical applications. Recall that the multiplicity information of a curve is defined as an equivalence class of elements of $k[x, y]$ up to scaling. This was to distinguish f and f^2 in $k[x, y]$, something varieties cannot do in classical algebraic geometry. Schemes naturally capture multiplicity information.

For example, consider the following:

$$\begin{aligned}\operatorname{Spec}\left(\mathbb{C}[x,y]/(y-x^2)\right) \cap \operatorname{Spec}\left(\mathbb{C}[x,y]/(y)\right) &= \operatorname{Spec}\left(\mathbb{C}[x,y]/(y-x^2,y)\right)\\ &= \operatorname{Spec}\left(\mathbb{C}[x,y]/(y,x^2)\right)\end{aligned}$$

That is, intersecting the parabola $y = x^2$ with the x-axis produces a fuzzy point that remembers the multiplicity:



Let us push this idea further and identify the local subscheme associated to a curve passing through the origin with a given tangent direction. Let $R = k[\epsilon]/(\epsilon^2)$ be the dual numbers, and consider a surjection:

$$\varphi : k[x, y] \to R$$

that sends the origin to the unique point of $\operatorname{Spec} R$, i.e. $\varphi(x), \varphi(y) \in (\epsilon)$. The most general such map sends $x \mapsto b\epsilon$ and $y \mapsto -a\epsilon$ for some $a, b \in k$ not both zero (different choices of (a, b) up to scaling give different tangent directions). As $\epsilon^2 = 0$, the map φ vanishes on $(x, y)^2 = (x^2, xy, y^2)$, and hence φ factors through:

$$\overline{\varphi} : \frac{k[x, y]}{(x^2, xy, y^2)} \to R$$

As R is 2-dimensional while $k[x, y]/(x^2, xy, y^2)$ is 3-dimensional (with basis $\{1, x, y\}$), the map $\overline{\varphi}$ has a 1-dimensional kernel. One can check that this kernel is generated by $ax + by$ (the linear form vanishing in the chosen tangent direction), and hence we get the isomorphism:

$$X_{a,b} = \operatorname{Spec} \frac{k[x, y]}{(x^2, xy, y^2, ax + by)} \cong \operatorname{Spec} R$$

How can we interpret this scheme?

- Algebraically, we can interpret $X_{a,b} \subseteq \mathbb{A}^2_k$ as the subscheme associated to the ideal containing all functions $f \in k[x, y]$ that vanish at the origin *and* have partial derivatives satisfying:

  $$b \left.\frac{\partial f}{\partial x}\right|_0 - a \left.\frac{\partial f}{\partial y}\right|_0 = 0$$

  This says that the gradient of f at the origin is proportional to (a, b), or equivalently, $f = c(ax + by) + g(x, y)$ where $g(x, y) \in (x, y)^2$ consists of higher-order terms.

- Geometrically, $X_{a,b}$ is the first-order neighborhood of the origin along the line $ax + by = 0$. It arises as the composite
$$\operatorname{Spec}\frac{k[t]}{(t^2)} \hookrightarrow \mathbb{A}^1_k \xrightarrow{t\mapsto(bt,\,-at)} \mathbb{A}^2_k$$
which embeds a "fuzzy point" into $\mathbb{A}^2$ along the tangent direction $(b, -a)$.

We interpret $X_{a,b}$ as containing the point $(0, 0)$ together with the infinitesimally near information in the direction specified by the line $ax + by = 0$ (the "tangent-line" information!). Such a scheme arises naturally when intersecting two curves non-transversally (the example with the parabola and the line is the simplest such case).

Another place where such schemes naturally arise is when looking at the *limit of families of schemes*. To understand this, take the case of two points $(0, 0), (a, b) \in \mathbb{A}^2_k$. These two points together form a scheme:

$$X_1 = \operatorname{Spec} R = \operatorname{Spec}\frac{k[x,y]}{(x,y)\cap(x-a,\,y-b)} = \operatorname{Spec}\frac{k[x,y]}{(x^2-ax,\,xy-bx,\,xy-ay,\,y^2-by)}$$

It should not be hard to see (e.g. via the Chinese Remainder Theorem) that $R \cong k \times k$.

Now, suppose (a, b) moves towards $(0, 0)$ along a polynomial curve. In particular, say there are two polynomial functions $a(t), b(t)$ such that $(a(0), b(0)) = (0, 0)$. Let $X_t = \{(0, 0), (a(t), b(t))\}$. Then what scheme would we expect to get when $t \to 0$? The advantage of using schemes for intersection theory, as outlined above, suggests that this limit should remember what direction the points came from – and indeed it does. Even more than this, the limit shall still be a length-2 scheme: two points that have "collided" but retained a memory of their approach direction.

Let us elucidate the details. Take

$$I_t = (x,y)\cap(x-a(t),\,y-b(t)) = (x^2-a(t)x,\ xy-b(t)x,\ xy-a(t)y,\ y^2-b(t)y)$$

As $t \to 0$, the ideal I_0 must contain x^2, xy, and y^2. Furthermore, I_t always contains a linear form:

$$a(t)y - b(t)x = (xy - b(t)x) - (xy - a(t)y)$$

Writing $a(t) = a_1t + a_2t^2 + \cdots$ and $b(t) = b_1t + b_2t^2 + \cdots$ (since $a(0) = b(0) = 0$), we can divide by t to obtain an element of I_t for $t \neq 0$:

$$\frac{a(t)y - b(t)x}{t} = a_1y - b_1x + t\cdot(a_2y - b_2x + \cdots)$$

Taking $t \to 0$, the limit ideal contains:

$$(x^2,\,xy,\,y^2,\,a_1y - b_1x) \subseteq I_0$$

These are in fact equal. To see this, note that the quotient $k[x, y]/I_t$ is a k-vector space of dimension 2 for each $t \neq 0$ (since I_t cuts out two distinct points). By a continuity argument (in the sense of flat limits), the limiting quotient $k[x, y]/I_0$ also has dimension 2[15]. Since the quotient by the left-hand side $(x^2, xy, y^2, a_1y - b_1x)$ already has dimension 2 (with basis $\{1, b_1x\}$ if $b_1 \neq 0$, or $\{1, a_1y\}$ if $a_1 \neq 0$), the inclusion must be an equality. Labelling $\alpha = a_1$, $\beta = b_1$ and $X_0 = X_{\alpha,\beta}$, we may write:

$$\lim_{t\to0} X_t = X_{\alpha,\beta}$$

[15]This argument is more subtle without the finite-dimensionality hypothesis, which shall be addressed when we cover this in fuller generality in section 6.4.

In higher dimensions, things get more complicated, and when our schemes are no longer local even more so. We shall see in section 6.4 that what we have been examining is a *flat family* of schemes. Flatness provides the correct conditions for a parameterized family of schemes to have well-behaved limits, generalizing the "continuity" argument used above.

Exercise 2.2.1

1. Show that for any polynomial $f \in k[x]$, if $a + b\epsilon \in \mathbb{C}(\epsilon)$, then
$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$
2. let k be algebraically closed, R be a local k-algebra of dimension 2 with maximal ideal $\mathfrak{m}$. Show that $R \cong k[x]/(x^2)$ (hint: use some dimension argument and Nakayama's lemma to conclude $\mathfrak{m}^2 = 0$).
3. Let k be algebraically closed. Find two non-isomorphic local k-algebras of dimension 3. Geometrically, we can think of this as representing the extra flexibility in directions or local schemes can capture.
4. Let k be algebraically closed and $\operatorname{Spec} R = \operatorname{Spec} k[x_1, ..., x_n]/I \subseteq \mathbb{A}^n_k$ be a local subscheme of dimension 0 and degree 3 (R has dimension 3 over k) supported at the origin. Then $\operatorname{Spec} R$ is either isomorphic to $\operatorname{Spec} k[x]/(x^3)$ or $\operatorname{Spec} k[x, y]/(x^2, xy, y^2)$, and these two schemes are not isomorphic to each other.
5. Let R be a ring that is not a field. Show that the restriction map on the structure sheaf is almost never surjective. Hence, there are more local functions than global function.
6. Let R be a reduced ring (contains no nilpotent elements). Show that the restriction maps are injective, that is each function on a larger open set can only restrict to one function. Give a counter-example for a ring with nilpotent elements.

2.2.3 Schemes

The last step in the definition of schemes is to observe that many geometric objects are not the spectrum of a ring, but are *locally* the spectrum of a ring. The prototypical examples are projective spaces, their closed subsets, and sequences of blow ups—none of which are in general affine.

Definition 2.2.9: Scheme

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ that is locally affine: there exists an open cover $X = \bigcup_i U_i$ such that for each i, there is a ring R_i and an isomorphism of locally ringed spaces
$$(U_i,\ \mathcal{O}_X|_{U_i}) \cong_{\mathbf{LRS}} (\operatorname{Spec}(R_i),\ \mathcal{O}_{\operatorname{Spec}(R_i)}).$$
Such a cover is called an *affine open cover* of X.

The requirement that $(X, \mathcal{O}_X)$ be a locally ringed space—that is, that each stalk $\mathcal{O}_{X,x}$ be a local ring—is essential. Without it, the "evaluation" of a section at a point (via the residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$) would not be well-defined, and the connection to geometry would be lost. We saw in section 1.4 that locally ringed spaces are the correct generalization of the spaces that arise in both differential and algebraic geometry. Since the stalk of $\mathcal{O}_X$ at a point $x \in X$ is a local ring (by the locally ringed space condition), we denote it $\mathcal{O}_{X,x}$, its maximal ideal $\mathfrak{m}_x$, and its residue field $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

The topology of affine schemes is generated by distinguished open sets. Lemma 2.2.29 shall show that given $\operatorname{Spec} R, \operatorname{Spec} S \subseteq X$ where their intersection is non-empty, the intersection can be covered by distinguished open sets that are distinguished on both $\operatorname{Spec} R$ and $\operatorname{Spec} S$. Hence the topology of schemes is also generated by distinguished open sets. See section 2.9 for a deeper discussion about the purely topological properties of schemes.

The way to interpret the agreement on intersections is that locally a scheme respects the algebraic relations imposed by a ring R_i, and the compatibility on $U_i \cap U_j$ (when nonempty) corresponds to the algebraic relations between R_i and R_j being consistent. A detailed example is presented in example 2.3.17.

2.2.10 Lack of $V(-)$ and $I(-)$? The topology of an affine scheme is characterized by the ideals of its global ring. General schemes are no longer tethered to the global sections (as shall be momentarily demonstrated). However, there is a generalization of $V(-)$ and $I(-)$ that works with the (quasi-coherent) *ideal sheaves* of a scheme X; the details are covered in section 5.1.3. From this perspective, the structure sheaf $\mathcal{O}_X$ can be thought of as generalizing the notion of a ring to a "local" object, remembering information that no single ring can capture.

Definition 2.2.11: Morphism of Schemes

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. Then a *morphism* of schemes is a morphism of locally ringed spaces $(\varphi, \varphi^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$.

A *isomorphism of schemes* is an isomorphism of locally ringed spaces: a homeomorphism $f : X \to Y$ together with a sheaf isomorphism $f^\# : \mathcal{O}_Y \xrightarrow{\sim} f_*\mathcal{O}_X$.

Not every morphism of ringed spaces preserves local rings (see example 3.1.1). The full exploration of the nuances shall be covered in section section 3.1.1.

Example 2.2.12: Schemes

1. All affine schemes are schemes, covered by a single open set.
2. The key non-affine scheme is projective space $\mathbb{P}^n_k$. We shall construct the general projective scheme $\mathbb{P}^n_R$ in section 2.4; for now, the reader can verify that the usual projective space as defined in [9] is a scheme by exhibiting the standard affine charts $U_i = \{x_i \neq 0\} \cong \mathbb{A}^n_k$. Any closed subvariety of $\mathbb{P}^n_k$ (with the appropriate structure sheaf) is also a scheme, typically not affine.
3. The following is the smallest non-affine scheme. Take $X = \{p, q_1, q_2\}$ with the topology generated by $X_1 = \{p, q_1\}$ and $X_2 = \{p, q_2\}$ as a sub-base. Define the sheaf of rings by
$$\mathcal{O}(X) = \mathcal{O}(X_1) = \mathcal{O}(X_2) = K[x]_{(x)}, \qquad \mathcal{O}(\{p\}) = K(x),$$
with restriction maps $\mathcal{O}(X) \to \mathcal{O}(X_i)$ the identity and $\mathcal{O}(X_i) \to \mathcal{O}(\{p\})$ the natural localization. This presheaf is a sheaf, and $(X, \mathcal{O})$ is a scheme but *not* an affine scheme: the

ring $K[x]_{(x)}$ has two points in its spectrum, but X has three. Geometrically, this is the "germ of a doubled point"—the generic point of $\mathbb{A}^1$ has been split into two.

4. By proposition 2.1.20, a finite disjoint union of schemes is a scheme. An arbitrary (infinite) disjoint union of affine schemes is a scheme but not an affine scheme, as it fails to be quasi-compact.

5. All affine schemes are "separated" (a property to be defined in section 3.5.1). There exist non-separated schemes; the simplest is the "line with doubled origin," constructed by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$.

6. If the reader has already encountered blow-ups, the schemes associated to a blow-up are typically not affine.

7. The open subset $D(x) \cup D(y) \subseteq \mathbb{A}^2_k$ is a scheme that is quasi-compact, separated, and a subset of an affine scheme, yet is not itself affine. The details are in Example 2.3.14.

8. Let $X = \{*\}$ be a singleton and $\mathcal{O}_X(X) = R$ for a commutative ring R. Is $(X, \mathcal{O}_X)$ a scheme? First, it must be affine so $X = \operatorname{Spec}(R)$. Next, $\operatorname{Spec} R$ must have a single prime ideal. This must be a 0-dimensional is a local ring. For example:

$$R \in \{k, \mathbb{Z}/p^n\mathbb{Z}, k[x]/(x^m)\}$$

Note there are local rings with more than one point, for example $k[[x]]$.

It is not always immediate whether a scheme is affine. For example, $\mathbb{A}^1_{\mathbb{C}} \setminus \{[(x)]\}$ may seem non-affine since it is not an affine variety in the classical sense, but it is isomorphic to $\operatorname{Spec}(\mathbb{C}[x, x^{-1}])$. On the other hand, $\mathbb{A}^2_{\mathbb{C}} \setminus \{[(x, y)]\}$ is genuinely not affine (Example 2.3.14). We shall establish tools for detecting affineness shortly.

Given a scheme X, we write $|X|$ for the underlying topological space. The elements of $\mathcal{O}_X(U)$ are called *sections* over U, and sections of $\mathcal{O}_X(X)$ are called *global sections*. These are variously denoted

$$\mathcal{O}_X(U), \qquad \Gamma(U, \mathcal{O}_X), \qquad H^0(U, \mathcal{O}_X),$$

depending on context: the first is convenient when only the structure sheaf is present; the second will be useful when multiple sheaves on X are in play; and the third hints at the fact that the global section functor $\Gamma(X, -)$ is left-exact but generally not exact, with its right-derived functors $H^i(X, -)$ forming the cohomology theory developed in chapter 6.

Recovering the ring from an affine scheme

A natural question arises: if a scheme X happens to be affine, can we recover the ring R from the locally ringed space $(X, \mathcal{O}_X)$? The answer is yes, and the ring is simply the global sections.

Proposition 2.2.13: Distinguished Open Sets and Schemes

Let R be a ring and $f \in R$. Then:

$$(D(f),\ \mathcal{O}_{\operatorname{Spec}(R)}|_{D(f)}) \ \cong\ (\operatorname{Spec}(R_f),\ \mathcal{O}_{\operatorname{Spec}(R_f)}).$$

In particular, taking $f = 1$: if $(X, \mathcal{O}_X)$ is an affine scheme, meaning $(X, \mathcal{O}_X) \cong (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ for some ring R, then R is recovered as the global sections:

$$\mathcal{O}_X(X) \ \cong\ R.$$

***Proof*:**
The first statement is the scheme-theoretic upgrade of Proposition 2.1.12: the homeomorphism $D(f) \cong \operatorname{Spec}(R_f)$ extends to an isomorphism of locally ringed spaces because the structure sheaf on $D(f)$ is defined by the same localization data as $\mathcal{O}_{\operatorname{Spec}(R_f)}$.

For the recovery of R: since X is affine, $X = D(1) = \operatorname{Spec}(R)$, and

$$\mathcal{O}_X(X) = \mathcal{O}_X(D(1)) = R_1 \cong R,$$

where the middle equality is the definition of the structure sheaf on $\operatorname{Spec}(R)$.

The motivation behind this proposition is that for affine schemes, the global sections completely determine the scheme. This is emphatically *not* the case for general schemes: the global sections of $\mathbb{P}^n_k$ are just k (for $n \geq 1$), which carries no information about the geometry of projective space. The structure sheaf, not the global sections, is the fundamental invariant.

Proposition 2.2.14: Characterization of Affine Schemes

Let $(X, \mathcal{O}_X)$ be a locally ringed space with $R = \mathcal{O}_X(X)$. Then $(X, \mathcal{O}_X) \cong (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ if and only if:

1. For every $f \in R$, $\mathcal{O}_X(U_f) \cong R_f$, where $U_f = \{x \in X : f \notin \mathfrak{m}_x\}$.
2. The natural map $X \to |\operatorname{Spec}(R)|$ (sending x to the prime $\mathfrak{m}_x \cap R$) is a homeomorphism.

***Proof*:**
Exercise. (The content is that conditions (1) and (2) together give an isomorphism of locally ringed spaces, not merely of topological spaces or of ringed spaces.)

Theorem 2.2.15: Ring Homomorphisms and Affine Scheme Isomorphisms

Let R and S be rings. An isomorphism of locally ringed spaces $\pi : \operatorname{Spec}(R) \to \operatorname{Spec}(S)$ induces an isomorphism $\pi^{\#} : S \xrightarrow{\sim} R$ on global sections. Conversely, an isomorphism $\varphi : S \xrightarrow{\sim} R$ induces an isomorphism $\operatorname{Spec}(\varphi) : \operatorname{Spec}(R) \xrightarrow{\sim} \operatorname{Spec}(S)$, and these two constructions are inverse to each other.

***Proof*:**
Corollary of theorem 3.1.4.

Given a scheme X, the collection of distinguished open sets from any affine cover forms a basis for the topology on X. This is important because the intersection of two affine opens need not itself be affine (see section 3.5.1), but distinguished opens within each affine chart suffice.

Proposition 2.2.16: Distinguished Open Sets Form a Basis

Let X be a scheme and $\{U_i = \operatorname{Spec}(R_i)\}$ an affine open cover. Then the collection of all distinguished open sets $D(f) \subseteq U_i$, as i ranges over the index set and f ranges over R_i, forms a basis for the topology on X.

***Proof*:**
Every open subset of X restricts to an open subset of each U_i, which is a union of distinguished opens in U_i (since distinguished opens form a basis for the Zariski topology on $\operatorname{Spec}(R_i)$). The union over all i covers the original open set.

Finiteness conditions and base ring

In algebra, every ring is a $\mathbb{Z}$-algebra, and additional structure arises by specifying a base ring. The scheme-theoretic counterpart is the following.

Definition 2.2.17: R-scheme and Finite Type Schemes

An *R-scheme* (or a *scheme over R*) is a scheme X together with a morphism $X \to \operatorname{Spec}(R)$, called the *structure morphism*. Equivalently, the sections of $\mathcal{O}_X$ are R-algebras and all restriction maps are R-algebra homomorphisms.
An R-scheme X is *locally of finite type over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-algebra. If X is also quasi-compact, then X is *of finite type over R*.

In number theory, integral extensions S/R (where S is a finitely generated R-module) play a central role. The scheme-theoretic analogue is:

Definition 2.2.18: Finite R-Schemes

An R-scheme X is *locally finite over R* if it can be covered by affine opens $\operatorname{Spec}(S_i)$ where each S_i is a finitely generated R-module. If X is quasi-compact, then X is *finite over R*.

If $R = k$ is a field (or more generally an Artinian ring), then a finite R-scheme is a finite collection of "fat points"—spectra of Artinian local k-algebras.

2.2.19 Terminology Warning Do not confuse *finite type* with *finite*. A scheme is locally of finite type if its coordinate rings are finitely generated R-*algebras* (finitely many generators as a ring); it is locally finite if its coordinate rings are finitely generated R-*modules* (finitely many generators as a module). The latter is a much stronger condition: $k[x]$ is a finitely generated k-algebra but not a finitely generated k-module.

Example 2.2.20: Finite Type R-schemes

1. Affine n-space $\mathbb{A}_k^n = \operatorname{Spec}(k[x_1, \ldots, x_n])$ is of finite type over k.
2. For polynomials $f_1, \ldots, f_m \in R[x_1, \ldots, x_n]$, the scheme $\operatorname{Spec}(R[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ is of finite type over R. Most classical intuitions from algebraic sets concern $\overline{k}$-schemes of finite type over $\overline{k}$ (abstract varieties shall require further properties; see section 3.8).
3. If R is the localization of a polynomial ring over k at a prime, then $\operatorname{Spec}(R)$ is generally not of finite type over k. Intuitively, local rings live in a "different dimension" from the global picture.
4. If the fibers of $X \to \operatorname{Spec}(R)$ are infinite-dimensional, then X is not of finite type over R.

2.2.4 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to catalogue how schemes and real smooth manifolds have some fundamentally different geometric properties:

1. The connected scheme given by $V(x) \cup V(yz) \subseteq \mathbb{A}_k^3$ has two irreducible components of different dimensions (as we shall more rigorously show in section 2.7). A connected manifold (smooth or even topological) is of a single, well-defined dimension at every point.
2. An abstract manifold cannot self-intersect[16], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}_k^2$, which has a node at the origin.
3. The patches of a manifold are very regular, being diffeomorphic to $\mathbb{R}^n$; in this sense manifold patches all carry "linear smooth" information, and there is essentially no local diversity from one patch to another. The affine patches of a scheme, by contrast, may be "pathological"—for example, the local rings can be non-reduced or non-Noetherian (in the latter case there may even be uncountably many connected components). This reflects how affine schemes represent "local algebraic information," which is far richer than the "local smooth linear information" that real manifolds encode. Some of the conditions and properties we shall introduce later (e.g. Noetherianity, regularity) will restore some order.
4. In Manifold Theory, the notion of irreducible component is trivial, since in any Hausdorff space the only irreducible subsets are singletons. From an analytic perspective, this existence of non-trivial irreducible components in schemes can be thought of as a consequence of the failure of the Zariski topology to separate points by functions; part of this phenomenon is the

[16] An *immersion* of a manifold may of course have self-intersections in the target, but the source manifold itself is not singular at such points.

existence of "large" (i.e. non-closed) points for schemes. More generally, the correspondence between irreducible closed sets and non-closed points (in at least Noetherian schemes) shows that schemes encode much more "global" information about codimension subsets than real smooth manifolds do.

5. The topology of $\mathbb{R}^n$ (and hence of $\mathbb{C}^n$ as a real manifold) is the product topology: the sets
$$\prod_{i=1}^{n}(a_i, b_i), \qquad a_i, b_i \in \mathbb{R},\ a_i < b_i,$$
form a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}^n_k$: the Zariski topology on $\mathbb{A}^n_k$ is *not* the product of the Zariski topologies on $\mathbb{A}^1_k$ (indeed, the product would be the cofinite topology, whose closed sets are always finite unions of points, whereas $\mathbb{A}^2_k$ has closed curves). However, there will still be a meaningful way to construct the scheme $\mathbb{A}^n_k$ from $\mathbb{A}^1_k$ via the *fibered product* (see section 3.2).

6. If a scheme X is irreducible, all its nonempty open sets are dense. Many of the schemes we shall work with will be unions of irreducible subschemes, where for each irreducible subscheme every nonempty affine open subset is dense in that subscheme. Dense open subsets are far from typical for manifolds (e.g. a disjoint union of two manifolds has open subsets contained in a single component, which are not dense).

7. More generally, the behaviour of schemes matches more closely with the behaviour of complex analytic geometry than with smooth real geometry. We shall draw such analogies often: for example, *Hartogs's Extension Lemma* ([4, chapter 8]) will have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions will tie into divisor theory, compact complex Riemann surfaces will correspond to projective curves, and so forth.

8. (embedding properties) Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N (Whitney Embedding Theorem). For classical affine (resp. projective) varieties of dimension n, they essentially by definition embed into $\mathbb{A}^N_k$ (resp. $\mathbb{P}^N_k$) for some N. For complex manifolds, it is *not* the case that they all embed in $\mathbb{C}^N$[17], and in particular no compact complex manifold of positive dimension embeds into $\mathbb{C}^N$. This is because complex manifolds possess much more rigidity than real manifolds [7][18]. Schemes exhibit this same type of rigidity.

   By the Kodaira Embedding Theorem [7], a compact Kähler manifold with a positive line bundle can be embedded into projective space. Proper schemes (the algebraic analogue of compact complex manifolds) with an *ample line bundle* (the algebraic analogue of a positive line bundle) similarly embed into projective space. In fact, this means that compact Kähler manifolds with positive line bundle are (as complex analytic spaces) isomorphic to a smooth proper scheme with an ample bundle. This is the closest to a general embedding theorem that is widely used, and will be proven in section 5.2.1.

   For a closed immersion $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong V \subseteq \mathbb{A}^n_R$ where V is a closed subscheme, X is topologically a sub-variety, though it may have a more complicated sheaf structure (for more details see section 3.8).

[17] If a compact complex manifold does embed holomorphically, it must reduce to a point, since any global holomorphic function on a compact complex manifold is constant. Non-compact complex manifolds that embed holomorphically in $\mathbb{C}^N$ are called *Stein manifolds* and share many properties of quasi-projective varieties.

[18] For instance, the complex torus $\mathbb{C}/\Lambda$ depends on the lattice Λ up to biholomorphism, giving a one-complex-parameter family of non-isomorphic structures.

For an open embedding $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong U \subseteq \mathbb{A}^n_R$ where U is open, X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more subtle: the map $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding and, as will be discussed in section 5.1.1, all quasi-coherent sheaves of $\mathcal{O}_X$-modules must be generated by global sections. Verifying that every $\mathcal{O}_X$-module is generated by global sections is difficult; there is a simpler characterization given by the Goodman–Hartshorne Theorem [11], but it is still a nontrivial condition, illustrating the greater complexities that arise when working with non-proper schemes.

More generally, much later in ref:HERE, there shall be other interesting ambient spaces into which certain types of schemes may be embedded (e.g. toric varieties, Grassmannians, flag varieties, etc.), each serving different purposes depending on the moduli problem under study.

9. Schemes can have "doubled points": recall that the line with two origins is a standard counterexample for manifolds, as it fails to be Hausdorff at the origin. In scheme theory the analogous phenomenon is permitted; indeed, many moduli spaces that are schemes are naturally non-separated schemes (see section 3.3.6).

10. A scheme remembers multiplicity. For example, on a classical point there is little room for interesting functions, but scheme-theoretic functions (i.e. regular functions on Spec $k[x]/(x^2)$) can encode multiplicity and infinitesimal information. A smooth manifold $(M, \mathcal{A})$ with its maximal atlas does not retain such data.

11. There can be multiple structure sheaves on the same underlying topological space, defining genuinely different schemes (think of the different fields that can serve as the structure sheaf over a single point). In contrast, manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism (by work of Radó in dimension 2 and Moise in dimension 3); in fact, the categories of piecewise-linear, topological, and smooth manifolds coincide in these dimensions, showing how "linear" manifold theory is in low dimensions. In section 3.2 we shall introduce the process of *base change*, by which one can change the structure sheaf without changing the underlying topological space.

12. Though we have not defined an atlas for schemes, Theorem 2.3.16 will show that schemes can be constructed by gluing affine patches, just as a manifold is built from its atlas, see section 2.3. (want to say why not as useful here).

13. The natural additional structure on a smooth manifold is a *tangent bundle*, whose transition functions take the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$; that is, they are not just smooth maps between open subsets but must respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the Jacobian. We shall define the algebraic analogue for schemes (the sheaf of Kähler differentials and the tangent sheaf) in chapter 5, section 5.4, though more care is required since schemes are not "locally linear" everywhere in the way smooth manifolds are. Much of the intuition from Poincaré duality (see [3, section 4.3]) will inform the generalization to Serre duality.

14. Schemes can better capture different levels of *local* behaviour. For example, a germ of a smooth manifold at a point is not itself a smooth manifold, but the analogous object for a scheme—the spectrum of a local ring—is again a scheme, called a *local scheme*. This is because commutative rings faithfully encode this local information, and (affine) schemes are the geometric realization of commutative rings.

15. Scheme functions are more "rigid": on a smooth manifold, partitions of unity exist thanks to the flexibility of the Euclidean topology, but no exact analogue holds for schemes. This rigidity,

however, also adds structure and sometimes simplifies definitions (e.g. closed subschemes, introduced in section 3.1.3, are a rather "natural" concept compared to the work that goes into verifying that closed submanifolds are locally diffeomorphic to $\mathbb{R}^k \subseteq \mathbb{R}^n$).

As smooth functions on each $U \subseteq M$ form a sheaf, sheaves that admit a "partition of unity" are called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section 6.1; this is the closest scheme-theoretic analogue of the partition-of-unity property.

16. Schemes unify "arithmetic" and "geometry" in a way that manifold theory is not equipped for. As examples, there are no manifold analogues of $\operatorname{Spec} \mathbb{Z}$, $\operatorname{Spec} \mathcal{O}_K$ (where $\mathcal{O}_K$ is a ring of integers of a number field), or $\operatorname{Spec} \mathbb{Z}[t]$. In this way, scheme theory gives geometric meaning to objects that are not tractable in differential geometry. A striking consequence is that different points of a single scheme can have residue fields of different characteristics (e.g. on $\operatorname{Spec} \mathbb{Z}$, the residue field at the prime (p) is $\mathbb{F}_p$ while the residue field at the generic point is $\mathbb{Q}$)—a phenomenon with no manifold analogue. To work in a more "purely geometric" setting, one often restricts to $\overline{k}$-schemes where $\overline{k}$ denotes the algebraic closure of a field of characteristic 0. In fact, when defining Z-valued points in section 3.3.2 for a scheme Z, if $Z = \operatorname{Spec} \overline{k}$ then such points are called *geometric points*. Working over a non-algebraically closed field often signals that the scheme carries arithmetic information.

17. Schemes have non-closed ("generic") points. Such points are essential: for instance, the ring map $k[x, y] \to k(x)[y]$ (localizing away from the irreducible polynomials in x) induces a morphism $\operatorname{Spec} k(x)[y] \to \operatorname{Spec} k[x, y]$ whose image consists of points lying over the generic point of $\operatorname{Spec} k[x]$. We shall later see that the image of such a morphism is *dense*.

18. Though the above list highlights many differences, many of the usual geometric tools remain applicable in the study of schemes. For example, $\mathbb{C}[x, y]/(x^2 - y^3)$ is not a normal ring. This can be hard to verify algebraically, but viewing $\operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y^3)$ as a curve in $\mathbb{C}^2$ (in the Euclidean topology), the fact that it has a cusp at the origin—and hence is singular there, at a codimension-0 point of the curve—is precisely why it fails to be normal. (Normality for varieties is equivalent to regularity in codimension 1, also known as Serre's condition R_1 together with S_2.) This type of geometric reasoning can be powerful: one can study the smoothness (or more specifically the normality) of a variety to determine the integral closure of its coordinate ring.

Though this is a long list consisting mostly of differences, very often in practice (especially in the setting of k-schemes, or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that are proper (the correct generalization of compact manifolds) and have many methods to "resolve" singularities (e.g. eliminate nodes, cusps, etc.) and make the resulting scheme regular[19]. In fact, as we shall see in section 6.6, a result known as *GAGA* (Géométrie Algébrique et Géométrie Analytique) shows that proper schemes over $\mathbb{C}$ with appropriate coherent sheaves are equivalent, as categories, to compact complex analytic spaces with their coherent analytic sheaves. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

[19] Though as we shall see, there is no universal resolution of singularities in all characteristics and dimensions. Resolution has been completely proved in characteristic 0 (Hironaka) and in low dimensions, but in positive characteristic the general case remains open. Moreover, the singularity information itself is sometimes important and should not always be discarded.

2.2.5 Evaluation and Stalks on Schemes

In differential geometry, the stalks of a manifold can be used define the evaluation of a function. This same trick is endeavored for scheme:

Proposition 2.2.21: Stalks of Schemes

Let $\operatorname{Spec}(R)$ be a scheme. Then the stalk at $\mathcal{O}_{\operatorname{Spec}(R)}$ at the point $[\mathfrak{p}]$ is isomorphic to the local ring $R_{\mathfrak{p}}$.

$$\mathcal{O}_{\operatorname{Spec}(R),\mathfrak{p}} \cong R_{\mathfrak{p}}$$

Furthermore, for any $[\mathfrak{p}] \in X$, as $[\mathfrak{p}] \in \operatorname{Spec}(R_i)$ for appropriate R_i, we have

$$\mathcal{O}_{X,[\mathfrak{p}]} \cong \mathcal{O}_{\operatorname{Spec}(R_i),[\mathfrak{p}]}$$

***Proof*:**
For the first part, prove that $R_{\mathfrak{p}}$ is the colimit of R_f where $f \notin \mathfrak{p}$. The second part comes from proposition 1.1.5.

Definition 2.2.22: Local Scheme

Let X be a scheme. Then X is a *local scheme* if it has a unique closed points.

The scheme $(\operatorname{Spec}\mathcal{O}_{\operatorname{Spec}R,\mathfrak{p}}, \mathcal{O}_{\operatorname{Spec}R,\mathfrak{p}})$ is called a *local scheme* at the point $\mathfrak{p}$.

The reader should show that the local scheme at the point $\mathfrak{p}$ is the intersection of all open subsets containing $\mathfrak{p}$. Hence, local schemes are the germs of schemes. Note that as there are "points within points", namely as there are non-closed points in schemes, then there is also stalks that are subsets of stalks, in particular if $\mathfrak{p}$ and $\mathfrak{m}$ is a prime and maximal ideal respectively where $\mathfrak{p} \subsetneq \mathfrak{m} \in \operatorname{Spec}R$ so that $[\mathfrak{m}] \in \overline{[\mathfrak{p}]}$, then $\mathcal{O}_{X,\mathfrak{p}} \cong R_{\mathfrak{p}}$ is a localization of $\mathcal{O}_{X,\mathfrak{m}} \cong R_{\mathfrak{m}}$. Due to this, notice that the local schemes in higher dimensions (section 2.7) can have local schemes contained within them. For example, the scheme $\operatorname{Spec}k[x,y]_{(x,y)}$ is the local scheme at the origin, it contain a single closed and a non-closed point for every irreducible curve in $\mathbb{A}^2_k$. Notice too how the local schemes are still schemes, as compared to the collection of germs at a point along with the point is not considered to be a manifold.[20]

The main idea is that the single maximal ideal represents that there is one point at which we can evaluate for each germ. As all local rings have a maximal ideal, this motivates remembering the following notion form commutative algebra:

Definition 2.2.23: Residue Field

Let R be a commutative ring and $\mathfrak{p} \subseteq R$ a prime ideal. Then the ring of fractions of the integral domain domain $R/\mathfrak{p}$ is called the *residue field*, i.e. $\operatorname{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ (with respect to R). This field is often denoted $\kappa(\mathfrak{p})$.

[20] In fact, germs not being manifolds can be seen as a lacking property in the category of manifolds as this shows manifolds are not closed under filtered colimits. Synthetic differential geometry is an attempt to rectify this.

The use of kappa in $\kappa(\mathfrak{p})$ is due to the fact that there can be many different residue fields at different points. For example in $\operatorname{Spec}\mathbb{R}[x]$, all residue fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ (depending if the localization is being taken at a non-maximal or maximal prime ideal).

Using residue fields, we can generalize the evaluation of a regular function (i.e. the evaluation of any element in the sections of $\mathcal{O}_X$). Recall that:

$$R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \operatorname{Frac}(R/\mathfrak{p})$$

as localization and taking the quotient commutes (see [9, section 20.5]). This makes it much easier to evaluate a function if we can determine R. Then as the valuation of a function is usually in $R/\mathfrak{p}$, as rational functions are now permitted on (proper) open subsets of X, the evaluation will be in $\operatorname{Frac}(R/\mathfrak{p})$. For a general point $x \in X$ in a scheme, we may diminish to the to define evaluation.

Definition 2.2.24: Evaluation of Function [on a Scheme]

Let X be a scheme, $U \subseteq X$ an open subset, and $f \in \mathcal{O}_X(U)$ a section. Then the value of f at the point $[\mathfrak{p}] \in U$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \operatorname{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$$

where $\operatorname{Spec}(R)$ is some affine local neighborhood of $\mathfrak{p}$.

As the quotient of any ring by a prime ideal is an integral domain, by taking its field of fraction we are forcing the codomain of each function to be a disjoint union of fields:

$$f : U \to \coprod_{\mathfrak{p}} \mathcal{O}_{U,\mathfrak{p}} \quad \text{(and in affine case)} \quad f : U \to \coprod_{\mathfrak{p}\in U} \operatorname{Frac}(R/\mathfrak{p})$$

If $\mathfrak{p}$ is maximal, then $\operatorname{Frac}(R/\mathfrak{p}) = R/\mathfrak{m}$. For certain rings, it suffices to evaluate sections only over maximal points (namely Jacobson Schemes, see corollary 3.8.6). In the case of affine Jacobson schemes, we can consider:

$$f : U \to \coprod_{\mathfrak{p}\in U} R/\mathfrak{m}$$

Varieties will be an example of Jacobson schemes. If $R/\mathfrak{m}_i \cong R/\mathfrak{m}_j$ for all maximal ideals (which shall be shown to be the case for varieties), then we can simply write:

$$f : U \to R/\mathfrak{m} \cong k$$

which recovers the notion of evaluation from classical algebraic geometry!

2.2.25 Foreshadowing: What Type of Function? As f is defined above, it is a set function between a relatively nice space U and a bad codomain. Can this function be characterized, or represented, by a known nicer function? In fact, it shall be representable with relatively simple scheme morphisms, recovering how in classical algebraic geometry sections are regular functions; details in proposition 3.3.7.

Example 2.2.26: Evaluation On Affine Schemes

The key idea is that the stalk $R_\mathfrak{p}$ gives all function's that will not vanish at $\mathfrak{p}$, and hence can divide by the function at that point, and then taking the quotient by the maximal ideal evaluate the function at that point, equivalently considering the value of the rational function in $\mathrm{Frac}(R/\mathfrak{p})$:

1. Take $X = \mathbb{C}[x]$. Then each $f/g \in \mathcal{O}_X(U)$ is a rational polynomial where $g(\mathfrak{p}) \neq 0$ for all $\mathfrak{p} \in U$. For example if $U = D(x)$, then $h = \frac{x+1}{x} \in \mathcal{O}_X(D(x))$ and:

$$h(3) = \frac{3+1}{3} = \frac{4}{3} \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x-3)} \cong \mathcal{O}_{X,3}/(x-3)$$
$$h(-1) = \frac{-1+1}{-1} = 0 \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x+1)} \cong \mathcal{O}_{X,-1}(x+1)$$

   As the points of $\mathbb{C}[x]$ correspond to the variety, we see that the evaluation will always be in $\mathbb{C}$, and there is no need to take the field of fractions as $\mathrm{Frac}(\mathbb{C}) = \mathbb{C}$.
   Another important example, take (want the vanishing at 0 , but something feels off)

2. Take $\mathbb{A}^2_k$ for a field k where char $\neq 2$. Then in the open set U away from the y-axis and the curve $y^2 - x^5$ has in it's local section $\mathcal{O}_{\mathbb{A}^2_k}(U)$ the rational function $\left(x^2+y^2\right)/x\left(y^2-x^5\right) \in \mathcal{O}_{\mathbb{A}^2_k}(U)$. It's value at the point $(2,4)$ (i.e. $[(x-2,y-4)]$) give a values in $\mathrm{Frac}(R/(x-2,y-4)) = \mathrm{Frac}(k[x,y]/(x-2,y-4)) \cong R_{(x-2,y-4)}/\overline{(x-2,y-4)}$. Then:

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (x-2,y-4) = \frac{2^2+4^2}{2\left(4^2-2^5\right)} \in \mathbb{C} = \frac{\mathbb{C}[x,y]}{(x-2,y-4)} \cong \mathcal{O}_{\mathbb{A}^2_k,(2,4)}$$

   The value at the point $[(y)]$, which is the generic point of the x axis, is

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (y) = \frac{x^2}{-x^6} = -1\frac{1}{x^4} \in \mathbb{C}(x) \cong \mathrm{Frac}(\mathbb{C}[x,y]/(y)) \cong \mathcal{O}_{\mathbb{A}^2_k,[(y)]}$$

3. Take $X = \mathrm{Spec}(\mathbb{Z})$ and take the open set $U = \mathrm{Spec}(\mathbb{Z})\backslash\{(2),(7)\}$. This open set has many rational function, but let's take $27/4$. Then evaluating this function at different points of U we get:

$$\begin{aligned} &(5) : 2/(-1) \equiv -2 \equiv 3 \bmod 5 \\ &(0) : 27/4 \\ &(3) : 0/1 = 0 \bmod 3 \end{aligned}$$

Notice that at 3, the function became 0. A (rational) function whose output is 0 is said to *vanish* at that point. If $\mathfrak{m}_\mathfrak{p}$ represents the maximal ideal of $\mathcal{O}_{X,\mathfrak{p}}$, then functions on an open subset U (i.e. section in $\mathscr{F}(U)$) have values at each point of U whose value lies in $\kappa(\mathfrak{p})$. A function *vanishes* at $\mathfrak{p}$ if its value at $\mathfrak{p}$ is 0. This idea is not well-defined on ringed spaces in general! There is furthermore the following properties mirroring manifold theory:

Proposition 2.2.27: Properties of Functions on Locally Ringed Spaces

Let f be function on a locally ringed space X. Then:

1. The subset of X where f vanishes is closed
2. if f vanishes nowhere, then f is invertible

Note that the property "if f vanishes everywhere, then $f = 0$" is not one of the properties. This will be addressed in proposition 2.4 .10 .

***Proof*:**

1. the subset of X where the germ of f is invertible is open
2. If $f \not\equiv 0 \bmod \mathfrak{p}$, then in particular $f \not\equiv 0 \bmod \mathfrak{m}$ for all maximal ideal, meaning f is contained in no maximal ideal. But then f must be a unit. If f where not a unit, then there would exist an $\mathfrak{m}$ that f belongs to, namely the maximal ideal containing f. But only units are not contained in any maximal ideal (as the ring is non-trivial)

Note how this is is *not* true on a variety defined in the classical sense:

Example 2.2.28: On Variety: Vanishes Nowhere, Not Invertible

Take

$$\overline{x - 100} \in \frac{\mathbb{R}[x, y]}{(x^2 + y^2 - 1)}$$

The reader should check that this function vanishes nowhere on $V(x^2 + y^2 - 1)$ as a variety, but it is *not* in invertible (i.e it is not a unit). However, $\overline{x - 100}$ *does* vanish on $\operatorname{Spec} \mathbb{R}[x, y]/(x^2 + y^2 - 1)$; can the reader find what prime ideal it vanishes on?

This ties in with the intuition that the spectrum of $\mathbb{R}[x, y]/(x^2 + y^2 - 1)$ would be some "gluing" of $\mathbb{C}[x, y]/(x^2 + y^2 - 1)$ via galois conjugate.

Finally, a moment should be taken to interpret evaluating at generic points. Let us consider $X = \mathbb{A}^1_k$ and let η be the generic point. Then looking at the fiber at η, we see that:

$$\mathcal{O}_{\mathbb{A}^1_k, \eta} = (k[x])_{(0)} = k(x)$$

that is, everything is inverted. The earlier intuition about $R_\mathfrak{p}$ is that we only evaluate functions *near* the point $\mathfrak{p}$ and every other $\mathfrak{q} \in \operatorname{Spec} R$ can be inverted, so this means that η is near *no* points as we can invert every point! This contrasts with the intuition that $\overline{\eta} = \mathbb{A}^1_k$ which seems to suggest that it is *close* to every point. This is where the nomenclature *generic* can be of use: it doesn't "see" what global sections capture, and can be thought of as "ignoring" any constraints given by points. This also explains the evaluation at the generic point, it returns the function itself motivating how it is not "particularly" extracting any information.

2.2.6 Independence of Affine Covering

A scheme X can be covered by affine open subsets in many different ways. Unlike the situation for smooth manifolds in low dimensions—where Moise's theorem [17] guarantees a unique smooth structure on topological manifolds of dimension ≤ 3—schemes admit no such rigidity. Even in dimension zero, the topological space underlying a field is always a single closed point, yet different fields give non-isomorphic affine schemes. It is therefore essential to know when a property of a scheme can be checked on *any* affine covering, rather than depending on a particular choice.

The tool for this is the *Affine Communication Lemma*. It identifies a class of properties—those that are stable under localization and can be recovered from a generating cover—for which checking on a single affine cover suffices. Such properties are called *affine-local*, and they include most of the "geometric" properties of schemes (reducedness, Noetherianness, integrality, normality, etc.). Properties that depend on "arithmetic" or global information—such as the number of connected components—are typically not affine-local.

The key preliminary observation is that the intersection of two affine opens, while generally not affine, can always be covered by open sets that are simultaneously distinguished in both.

Lemma 2.2.29: Build-up Lemma

Let $\operatorname{Spec}(R)$ and $\operatorname{Spec}(S)$ be affine open subschemes of a scheme X. Then $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ can be covered by open sets that are simultaneously distinguished open in $\operatorname{Spec}(R)$ and distinguished open in $\operatorname{Spec}(S)$.

***Proof*:**
Let $[\mathfrak{p}] \in \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Since $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$ is open in $\operatorname{Spec}(R)$, and the distinguished open sets form a basis for the Zariski topology (section 2.1.2), there exists $f \in R$ with $[\mathfrak{p}] \in \operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S)$. Now, as

$$\operatorname{Spec}(R_f) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(S)$$

$\operatorname{Spec}(R_f)$ is an open subset of $\operatorname{Spec}(S)$. Thus, by construction of the topology on $\operatorname{Spec}(S)$ using distinguished opens (given by S), there exists $g \in S$ with

$$[\mathfrak{p}] \in \operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f) \cap \operatorname{Spec}(S) \subseteq \operatorname{Spec}(R) \cap \operatorname{Spec}(S).$$

In particular, $\operatorname{Spec}(S_g) \subseteq \operatorname{Spec}(R_f)$.

Now, $g \in \mathcal{O}_X(\operatorname{Spec}(S))$ restricts to an element $g' \in \mathcal{O}_X(\operatorname{Spec}(R_f)) = R_f$, and the locus in $\operatorname{Spec}(R_f)$ where g is nonvanishing is exactly $\operatorname{Spec}(S_g)$, so:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}((R_f)_{g'}).$$

By definition of R_f, write $g' = h/f^n$ for some $n \geq 0$ and $h \in R$. By Proposition 2.1.15, a distinguished open of a distinguished open is distinguished:

$$(R_f)_{g'} = (R_f)_{h/f^n} = R_{fh}$$

where the last equality holds since f is already invertible in R_f. Hence:

$$\operatorname{Spec}(S_g) = \operatorname{Spec}(R_{fh}) = D(fh) \subseteq \operatorname{Spec}(R).$$

This is a distinguished open in $\operatorname{Spec}(R)$, and it is simultaneously equal to $\operatorname{Spec}(S_g) = D(g) \subseteq \operatorname{Spec}(S)$, a distinguished open in $\operatorname{Spec}(S)$. Since $[\mathfrak{p}]$ was arbitrary, these sets cover $\operatorname{Spec}(R) \cap \operatorname{Spec}(S)$.

With this in hand, we can state and prove the main tool.

Theorem 2.2.30: Affine Communication Lemma

Let X be a scheme and let Q be a property enjoyed by some affine open subschemes of X. Suppose Q satisfies:

1. *(Localization)* If $\operatorname{Spec}(A) \subseteq X$ satisfies Q and $f \in A$, then $\operatorname{Spec}(A_f)$ satisfies Q.
2. *(Gluing)* If $f_1, \ldots, f_n \in A$ generate the unit ideal $(f_1, \ldots, f_n) = A$ and each $\operatorname{Spec}(A_{f_i})$ satisfies Q, then $\operatorname{Spec}(A)$ satisfies Q.

If there exists an affine open cover $X = \bigcup_i \operatorname{Spec}(R^{(i)})$ such that each $\operatorname{Spec}(R^{(i)})$ satisfies Q, then *every* affine open subscheme of X satisfies Q.

***Proof*:**
Let $\operatorname{Spec}(A) \subseteq X$ be an arbitrary affine open subscheme. We must show $\operatorname{Spec}(A)$ satisfies Q. For each i, the intersection $\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})$ is an open subset of both $\operatorname{Spec}(A)$ and $\operatorname{Spec}(R^{(i)})$. By the Build-up Lemma (lemma 2.2.29), this intersection can be covered by open sets that are simultaneously distinguished in $\operatorname{Spec}(A)$ and distinguished in $\operatorname{Spec}(R^{(i)})$.

Since $\operatorname{Spec}(R^{(i)})$ satisfies Q, condition (1) implies that each of these distinguished opens (viewed as localizations of $R^{(i)}$) also satisfies Q. But these same opens are also distinguished opens of $\operatorname{Spec}(A)$, say of the form $\operatorname{Spec}(A_{g_j})$ for various $g_j \in A$.

Now, $\operatorname{Spec}(A)$ is quasi-compact, and the union $\bigcup_i (\operatorname{Spec}(A) \cap \operatorname{Spec}(R^{(i)})) = \operatorname{Spec}(A)$ covers it. Extracting a finite subcover from among the $\operatorname{Spec}(A_{g_j})$, we obtain elements $g_1, \ldots, g_m \in A$ such that:

(a) each $\operatorname{Spec}(A_{g_j})$ satisfies Q (inherited from the $R^{(i)}$ via localization), and

(b) $\operatorname{Spec}(A) = \bigcup_{j=1}^m \operatorname{Spec}(A_{g_j})$, which means $g_1, \ldots, g_m$ generate the unit ideal in A.

By condition (2), $\operatorname{Spec}(A)$ satisfies Q.

The power of the lemma is that once the two conditions (localization and gluing) are verified for a property Q—which is typically a purely algebraic statement about rings—the conclusion is that Q holds for *every* affine open, not just those in the chosen cover. This means the property can be attributed to the scheme X itself, independent of the presentation.

The Affine Communication Lemma gives a uniform method for proving that properties of schemes are well-defined. We collect some important instances;

Example 2.2.31: Affine-local Properties

1. *Reducedness*: Let Q be the property "A is a reduced ring" (i.e., A has no nonzero nilpotents). We verify the two conditions:

   (a) *Localization:* If A is reduced, then A_f is reduced. Indeed, if $(a/f^n)^m = 0$ in A_f, then $f^k a^m = 0$ in A for some k, hence $(f^j a)^m = 0$ for j large enough, so $f^j a = 0$, hence $a/f^n = 0$ in A_f.

(b) *Gluing:* If $(f_1, \ldots, f_n) = A$ and each A_{f_i} is reduced, then A is reduced. Indeed, if $a^m = 0$ in A, then $a^m = 0$ in each A_{f_i}, so $a = 0$ in each A_{f_i} (as A_{f_i} is reduced), meaning $f_i^{k_i} a = 0$ in A for some k_i. Since the f_i generate the unit ideal, so do the $f_i^{k_i}$ (see section 2.1.2), hence $a = 0$.

By the Affine Communication Lemma, a scheme X is reduced if and only if some (equivalently, every) affine open cover consists of reduced rings. This shall be the foundation for definition given in section 2.5.1.

2. *Noetherian*: Let Q be the property "A is a Noetherian ring."

   (a) *Localization:* If A is Noetherian, then A_f is Noetherian (a standard result in commutative algebra: every ideal of A_f is an extension of an ideal of A, and extensions of finitely generated ideals are finitely generated).

   (b) *Gluing:* Suppose $(f_1, \ldots, f_n) = A$ and each A_{f_i} is Noetherian. Let $I \subseteq A$ be an ideal. Each extension $I \cdot A_{f_i}$ is finitely generated, say by elements $a_{i,1}/f_i^{k_1}, \ldots, a_{i,r_i}/f_i^{k_{r_i}}$. Clearing denominators, we obtain finitely many elements of I whose images generate $I \cdot A_{f_i}$ for each i. Since the f_i generate the unit ideal, these elements generate I itself.

   A scheme is *locally Noetherian* if it admits an affine cover by spectra of Noetherian rings. By the Affine Communication Lemma, this is equivalent to every affine open being the spectrum of a Noetherian ring.

3. *Normality*: Let Q be the property "A is a normal ring" (integrally closed in its total ring of fractions). This can be checked via the Affine Communication Lemma: if A is normal then A_f is normal, and normality can be recovered from a generating cover. Thus, a scheme is *normal* if and only if every affine open is the spectrum of a normal ring.

Local Properties and the Affine Communication Lemma

As more properties of schemes are defined, we must be precise about what it means for a scheme to "have" a property and how to check for it. There at least two natural levels at which a property can be detected, and they form a hierarchy.

Definition 2.2.32: Property of Schemes

A *property* P of schemes is a class of schemes closed under isomorphism: if X satisfies P and $Y \cong X$, then Y satisfies P.

Definition 2.2.33: Affine-local and Stalk-local Properties

Let P be a property of schemes.

1. P is *affine-local* if for any scheme X and any affine open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
2. P is *Zariski-local* if for any scheme X and any (Zariski) open cover $X = \bigcup U_i$, the scheme X satisfies P if and only if every U_i satisfies P.
3. P is *stalk-local* if for any scheme X, X satisfies P if and only if for every point $x \in X$, the affine scheme $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies P.

The Affine Communication Lemma provides a practical test for affine-locality: it suffices to verify the localization and gluing conditions for the corresponding ring property. As sheaf maps can always be verified stalk-locally, stalk-locality is another natural condition to consider. We shall see that morphism properties are not always verified affine-locally; those that are give rise to *affine morphisms* (see definition 3.4.5).

Note that $\operatorname{Spec} \mathcal{O}_{X,x}$ *need not* be open in X, hence $\{\operatorname{Spec} \mathcal{O}_{X,x}\}_{x \in X}$ need not be an affine open cover. For example $X = \mathbb{A}^1_{\mathbb{C}}$ and $\mathcal{O}_{X,2} \cong \operatorname{Spec} k[x]_{(x-2)}$ so that $\operatorname{Spec} \mathcal{O}_{X,2}$ has only 2 points but any open subscheme of X has cofinitely many points plus the generic point. Thus, a stalk-local is not the condition that it suffices to check the specific "stalk-cover".

Zariski-local properties are another natural choice, but as any scheme has a topological basis of distinguished open sets, these are identical to affine-locals:

Proposition 2.2.34: Affine-local Iff Zariski-local

Let P be a property of schemes. Then P is affine-local if and only if it is Zariski-local.

***Proof*:**
exercise.

It is natural to ask whether stalk-locality is a stronger or weaker condition than affine-locality. The answer is that it is strictly stronger.

Proposition 2.2.35: Stalk-local implies Affine-local

If a property of a scheme is stalk-local, then it is affine-local.

***Proof*:**
Let P be stalk-local and let $X = \bigcup U_i$ be an affine open cover.

($\Rightarrow$) Suppose X satisfies P. For any $x \in U_i$, the stalk $\mathcal{O}_{U_i,x} \cong \mathcal{O}_{X,x}$ by proposition 1.1.5, and $\operatorname{Spec} \mathcal{O}_{X,x}$ satisfies property P since X satisfies P. As this holds for all $x \in U_i$, the scheme U_i satisfies P by stalk-locality.

($\Leftarrow$) Suppose every U_i satisfies P. For any $x \in X$, choose some U_i containing x. Then $\mathcal{O}_{X,x} \cong \mathcal{O}_{U_i,x}$ satisfies the local-ring property since U_i satisfies P. As x was arbitrary, X satisfies P.

Common examples are:

1. "X is reduced" is stalk-local with respect to "$\mathcal{O}_{X,x}$ has no nonzero nilpotents"

2. "X is regular" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is a regular local ring"

3. "X is normal" is stalk-local with respect to "$\mathcal{O}_{X,x}$ is an integrally closed domain."

The converse is false: there exist affine-local properties that cannot be detected at the level of stalks. The issue is that stalks capture "infinitely close" behavior but lose information about finiteness conditions that involve the global structure of a neighborhood.

Example 2.2.36: Affine-local Does Not Imply Stalk-local

Consider the property "locally Noetherian." As shown in example 2.2.31, this is affine-local. We show it is not stalk-local.

Let $R = \prod_{i=1}^{\infty} \mathbb{F}_2$, the countably infinite product of copies of $\mathbb{F}_2$. The ring R is not Noetherian: the ideals $I_n = \{(a_1, a_2, \ldots) : a_1 = \cdots = a_n = 0\}$ form a strictly ascending chain. Thus $\operatorname{Spec}(R)$ is not locally Noetherian.

However, every stalk of $\mathcal{O}_{\operatorname{Spec}(R)}$ is Noetherian. For any prime ideal $\mathfrak{p} \subseteq R$, the localization $R_{\mathfrak{p}}$ is isomorphic to $\mathbb{F}_2$. To see this, recall that R is a Boolean ring ($a^2 = a$ for all a), so every prime ideal is maximal with residue field $\mathbb{F}_2$. Since R is zero-dimensional and Boolean, the localization at any prime is a field.

If "locally Noetherian" were stalk-local, the fact that all stalks are Noetherian fields would force R to be Noetherian—a contradiction. Thus stalk-locality is strictly stronger than affine-locality.

The upshot is a hierarchy of "locality" conditions for scheme properties:

$$\begin{tikzcd}
 & & \text{closed under open restrictions} \\
\text{stalk-local} \arrow[r, Rightarrow] & \text{affine-local} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & \\
 & & \text{closed under gluing over open covers}
\end{tikzcd}$$

The first implication is strict, as the example above shows. In practice, most "algebraic" properties (reduced, normal, regular, Cohen–Macaulay) are stalk-local, while "finiteness" properties (Noetherian, finite type) are affine-local but not stalk-local.

2.2.37 Advanced Foreshadowing: Other Covers Notice how the collection of maps $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ for all $x \in X$ can still be seen as a cover which is neither open nor closed. Such a cover still has important properties; in this case it fpqc (fidèlement plate et quasi-compacte). We shall not address such covers in the textbook, however the idea of replacing open sets with covering maps with "desirable" properties will be a key intuition in defining the *étale topology* using *étale coverings* in section ref:HERE.
Readers interested in Descent Theory [1] may be interested to know that the above hierarchy is often extended to the following:

$$\text{fpqc-local} \implies \text{fppf-local} \implies \text{étale-local} \implies \text{Zariski-local}$$

where stalk-local becomes a part of fpqc-local.

2.3 *Affine Atlases

By definition, a scheme is a locally ringed space that admits some affine open cover. The structure sheaf $\mathcal{O}_X$ is part of the data: it is given, and the affine cover witnesses that it is "locally affine." But there is an equivalent formulation—closer in spirit to the definition of a smooth manifold—in which the structure sheaf is not given a priori, but is *recovered* from an atlas. This perspective clarifies the sense in which a scheme is "independent of coordinates."

Recall that a smooth manifold of dimension n can be defined in two equivalent ways:

1. A topological space M equipped with a sheaf C^∞_M of smooth functions such that (M, C^∞_M) is locally isomorphic to $(\mathbb{R}^n, C^\infty_{\mathbb{R}^n})$.
2. A topological space M equipped with a *maximal smooth atlas*: a maximal collection of charts (U_i, φ_i) where $\varphi_i : U_i \xrightarrow{\sim} V_i \subseteq \mathbb{R}^n$ is a homeomorphism, and the transition maps $\varphi_j \circ \varphi_i^{-1}$ are smooth (see section 0.5)

The two definitions are equivalent because the sheaf C^∞_M can be recovered from the atlas: a function $f : U \to \mathbb{R}$ is smooth if and only if $f \circ \varphi_i^{-1}$ is smooth for every chart (U_i, φ_i) with $U_i \subseteq U$. Conversely, the atlas is recoverable from the sheaf: a chart is precisely a local isomorphism of the sheaf with the standard model.

In the manifold setting, most textbooks adopt definition (2) because the transition maps are concrete and checkable. In scheme theory, definition (1) has historically been preferred because the structure sheaf interacts well with the algebraic machinery of localization, sheaf cohomology, and base change. Nevertheless, the atlas formulation works equally well and is illuminating.

Affine atlases for schemes

Definition 2.3.1: Affine Chart

Let X be a topological space. An *affine chart* on X is a pair (U, φ) where $U \subseteq X$ is open and $\varphi : U \xrightarrow{\sim} |\operatorname{Spec}(R)|$ is a homeomorphism for some commutative ring R. The ring R is called the *coordinate ring* of the chart.

Two charts (U_i, φ_i) with coordinate ring R_i and (U_j, φ_j) with coordinate ring R_j must be compatible on their overlap. In the manifold case, compatibility means the transition map is smooth. For schemes, it means the transition map is "algebraic" in the following sense.

Definition 2.3.2: Compatibility of Charts

Two affine charts (U_i, φ_i) and (U_j, φ_j) on a topological space X are *compatible* if for every point $x \in U_i \cap U_j$, there exist elements $f \in R_i$ and $g \in R_j$ such that:

1. $x \in \varphi_i^{-1}(D(f)) \cap \varphi_j^{-1}(D(g))$,
2. $\varphi_i^{-1}(D(f)) = \varphi_j^{-1}(D(g))$ as subsets of X, and
3. The composite homeomorphism

$$D(f) \xrightarrow{\varphi_i^{-1}} \varphi_i^{-1}(D(f)) \;=\; \varphi_j^{-1}(D(g)) \xrightarrow{\varphi_j} D(g)$$

is induced by a ring isomorphism $(R_j)_g \xrightarrow{\sim} (R_i)_f$.

Condition (3) is the key algebraic content: the transition "map" between two charts must not merely be a homeomorphism of topological spaces, but must arise from an isomorphism of the corresponding localized rings. This is the condition that fails for arbitrary homeomorphisms between spectra—and it is exactly what the structure sheaf encodes in the locally ringed space definition.

Definition 2.3.3: Affine Atlas

An *affine atlas* on a topological space X is a collection $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ of pairwise compatible affine charts such that $X = \bigcup_{i \in I} U_i$.

Definition 2.3.4: Maximal Affine Atlas

An affine atlas $\mathcal{A}$ is *maximal* if it contains every affine chart on X that is compatible with all charts already in $\mathcal{A}$. That is, if (V, ψ) is an affine chart compatible with every $(U_i, \varphi_i) \in \mathcal{A}$, then $(V, \psi) \in \mathcal{A}$.

Definition 2.3.5: Scheme via Atlas

A *scheme* is a pair $(X, \mathcal{A})$ where X is a topological space and $\mathcal{A}$ is a maximal affine atlas on X.

This definition makes no reference to sheaves, locally ringed spaces, or stalks. The algebraic data is encoded entirely in the transition isomorphisms between charts: the atlas definition is equivalent to definition 2.2.9. The equivalence rests on two constructions: recovering the structure sheaf from an atlas, and recovering the maximal atlas from a locally ringed space.

Proposition 2.3.6: From Atlas to Structure Sheaf

Let $(X, \mathcal{A})$ be a scheme in the sense of definition 2.3.5. Then there exists a unique sheaf of rings $\mathcal{O}_X$ on X such that $(X, \mathcal{O}_X)$ is a scheme in the sense of definition 2.2.9, and for every chart $(U, \varphi) \in \mathcal{A}$ with coordinate ring R:

$$(U,\ \mathcal{O}_X|_U)\ \cong\ (\mathrm{Spec}\,(R),\ \mathcal{O}_{\mathrm{Spec}\,(R)}).$$

***Proof*:**
Define $\mathcal{O}_X$ on the basis of opens $\{U : (U, \varphi) \in \mathcal{A} \text{ for some } \varphi\}$ by setting $\mathcal{O}_X(U) = R$ whenever $\varphi : U \xrightarrow{\sim} \mathrm{Spec}\,(R)$. For a distinguished open $D(f) \subseteq \mathrm{Spec}\,(R)$, the chart $(\varphi^{-1}(D(f)),\ \varphi|_{D(f)})$ is in $\mathcal{A}$ (by maximality) with coordinate ring R_f, so we set $\mathcal{O}_X(\varphi^{-1}(D(f))) = R_f$. The restriction maps are the natural localization maps $R \to R_f$.

The compatibility condition on charts ensures that this is well-defined on overlaps: if two charts (U_i, φ_i) and (U_j, φ_j) both contain a point x, the transition isomorphism $(R_j)_g \cong (R_i)_f$ identifies the sections. By the sheaf axiom for the structure sheaf of $\mathrm{Spec}\,(R)$, this basis assignment extends uniquely to a sheaf on all of X.

The stalks of $\mathcal{O}_X$ are local rings: at a point $x \in U_i$, the stalk $\mathcal{O}_{X,x}$ is the localization $(R_i)_{\mathfrak{p}}$ where $\mathfrak{p} = \varphi_i(x)$, which is local. Hence $(X, \mathcal{O}_X)$ is a locally ringed space, and each chart gives an affine open, so it is a scheme.

Proposition 2.3.7: From Locally Ringed Space to Maximal Atlas

Let $(X, \mathcal{O}_X)$ be a scheme in the sense of definition 2.2.9. Then the collection

$$\mathcal{A}_{\max} = \left\{ (U, \varphi) : \begin{array}{l} U \subseteq X \text{ is open, } \varphi : (U, \mathcal{O}_X|_U) \xrightarrow{\sim} (\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)}) \\ \text{is an isomorphism of locally ringed spaces, for some } R \end{array} \right\}$$

is a maximal affine atlas on X. The atlas $\mathcal{A}_{\max}$ depends only on $(X, \mathcal{O}_X)$, not on any particular affine open cover.

***Proof*:**
The charts in $\mathcal{A}_{\max}$ are pairwise compatible: if (U_i, φ_i) and (U_j, φ_j) are in $\mathcal{A}_{\max}$, then on $U_i \cap U_j$ the isomorphisms φ_i and φ_j are both isomorphisms of locally ringed spaces, and the transition map is an isomorphism of the appropriate localizations (this is the content of the Build-up Lemma, lemma 2.2.29). The atlas covers X because the definition of a scheme guarantees the existence of at least one affine cover. It is maximal by construction: every affine open is included.

These two constructions are inverse to each other: starting from an atlas, building the sheaf, and taking the maximal atlas of the resulting scheme recovers the original atlas; starting from a scheme, taking $\mathcal{A}_{\max}$, and building the sheaf from it recovers the original structure sheaf.

Just as for manifolds, the practical consequence is that one need not specify a maximal atlas. Any atlas suffices.

Proposition 2.3.8: Uniqueness of Maximal Atlas

Let $\mathcal{A}$ be any affine atlas on a topological space X. Then there exists a unique maximal affine atlas $\mathcal{A}_{\max}$ containing $\mathcal{A}$. Explicitly,

$$\mathcal{A}_{\max} = \{(V, \psi) : (V, \psi) \text{ is compatible with every } (U, \varphi) \in \mathcal{A}\}.$$

***Proof*:**
One must verify that any two charts in $\mathcal{A}_{\max}$ are compatible with each other (not just with the charts in $\mathcal{A}$). Let (V_1, ψ_1) and (V_2, ψ_2) be in $\mathcal{A}_{\max}$, and let $x \in V_1 \cap V_2$. Choose a chart $(U, \varphi) \in \mathcal{A}$ with $x \in U$. By compatibility of (V_1, ψ_1) with (U, φ), there exist distinguished opens around x where the transition is an isomorphism of localizations. By compatibility of (V_2, ψ_2) with (U, φ), the same holds. Composing (and using that distinguished opens of distinguished opens are distinguished, by proposition 2.1.15), we obtain a transition isomorphism between localizations of V_1 and V_2.

Uniqueness is immediate: any maximal atlas containing $\mathcal{A}$ must contain every compatible chart, hence must equal $\mathcal{A}_{\max}$.

The upshot is that defining a scheme requires only specifying a topological space together with *any* affine atlas—the maximal atlas is then uniquely determined. The Affine Communication Lemma (theorem 2.2.30) is the tool that ensures properties of the scheme can be checked on the given atlas rather than on the full maximal atlas: its localization and gluing conditions propagate a property from any covering atlas to all affine opens.

The atlas viewpoint also gives an explicit construction of schemes from purely algebraic data, analogous to constructing a manifold by specifying charts and transition maps. Suppose we are given:

1. Rings $R_1, \ldots, R_n$.

2. For each pair (i, j), elements $f_{ij} \in R_i$ specifying open subsets $U_{ij} = D(f_{ij}) \subseteq \operatorname{Spec}(R_i)$.

3. Ring isomorphisms $\varphi_{ij} : (R_i)_{f_{ij}} \xrightarrow{\sim} (R_j)_{f_{ji}}$.

Subject to the *cocycle condition*: for each triple (i, j, k), the composition $\varphi_{jk} \circ \varphi_{ij}$ agrees with φ_{ik} on the appropriate localizations.

These data determine a unique scheme X (up to unique isomorphism) by the gluing construction of section 2.3.2: the underlying set is the disjoint union $\coprod_i |\operatorname{Spec}(R_i)|$ modulo the identifications given by the φ_{ij}, the topology is the quotient topology, and the affine atlas consists of the images of the $\operatorname{Spec}(R_i)$. Conversely, every scheme with a given finite affine cover arises in this way: the rings are the coordinate rings of the charts, and the transition isomorphisms are read off from the structure sheaf on overlaps.

2.3.1 Subschemes: Open

We seek the natural notions of subschemes induced by the parent scheme. As a scheme has a topology, a natural starting point is to look for open or closed subsets with a sheaf naturally induced by the parent scheme[21]. To define the topology of open and closed subschemes is easy. The difficult part is to insure the subscheme has a sheaf structure that is in a natural way compatible with the parent scheme. This shall require a deeper discussion on monomorphic scheme morphisms as the sheaf structure of subschemes is "non-trivial", and hence this discussion is mostly delayed till section 3.1.3. There is one type of subscheme that shall be used in this chapter and requires no use of scheme morphism since it is directly compatible with sheaves: *open subschemes*.

Proposition 2.3.9: Open Subscheme

Let X be a scheme and $U \subseteq X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

***Proof*:**
U is certainly a subspace and a subsheaf with local rings as stalks. To show it is locally affine, for any $x \in U$ choose affine open $\operatorname{Spec} R \cong V \subseteq X$ where $x \in U \cap V$. As $D(f)$ forms a (topological) basis for $\operatorname{Spec} R$, there exists an f_0 such that $x \in D(f_0) \subseteq U \cap V \subseteq U$, showing that U is covered in affine, completing the proof.

Definition 2.3.10: Open Subscheme

Let X be a scheme. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme*. If U is also an affine scheme, then U is often said to be an *affine open subscheme* or *affine open subset*.

If a scheme is isomorphic to an open subscheme, we shall say that the isomorphism map (onto the image) is an open embedding:

Definition 2.3.11: Open Embedding

A map of schemes $\pi : X \to Y$ is an open embedding if:

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

2.3.12 Remark about Hartshorne's Definition of Open Subscheme Note that in Hartshorne, an open subscheme was technically not defined as $(U, \mathcal{O}_X|_U)$ but as $(U, [\mathcal{O}_X|_U])$ where $[\mathcal{O}_X|_U]$ is the equivalence class of all isomorphic sheaves. This is in fact not correct: isomorphic sheaves shall give technically different open subschemes (which are of course isomorphic). Hence, Hartshorne claims that an open subscheme has a unique subscheme structure, not unique up to isomorphism.

Corollary 2.3.13: Subspace and Zariski Topology On Open Subscheme

Let $U = \operatorname{Spec}(R) \subseteq X$ be an affine open subset and take $(U, \mathcal{O}_X|_U)$. Then the Zariski topology on U is the subspace topology on U with respect to X

[21] Categorically, a natural starting point is to look for subobjects, but surprisingly these are not so useful in **Sch**; see section 3.1.3

***Proof*:**
exercise.

The prototypical example of an open subscheme would be the open sets $D_+(x_i) \subseteq \mathbb{P}_k^n$ of projective space or $D(f) \subseteq \operatorname{Spec} R$ of affine space. Note that an open sub-scheme of an affine sub-scheme need not be affine:

Example 2.3.14: Open Sub-Scheme need not be Affine

Let $R = k[x, y]$, $X = \mathbb{A}_k^2 = \operatorname{Spec}(k[x, y])$, and consider $U = \mathbb{A}_k^2 \setminus \{(x, y)\}$ which may be considered the plane without the origin. Before seeing why this is a scheme, try to show that U is indeed an open subset (it is the compliment of intersection of certain closed subsets, equivalently the union of certain open sets, there is more than one way of creating it).

Now, there does not exist an R such that $\operatorname{Spec}(R) \cong U$. It may be tempting to say that there ought to be a distinguished open set of $\mathbb{A}_k^2$ that produces U, but the reader should think through why this is not the case (hint: (x, y) is not principle). However, U can certainly be described as the union of two distinguished open sets, namely:

$$U = D(x) \cup D(y)$$

What is the global section on the open sub-scheme U? This would be all functions in $D(x) \cup D(y)$ that agree on the intersection, $D(x) \cap D(y) = D(xy) = R_{xy} = k[x, y]_{xy}$. Computing the local sections, as $D(x)$ is all points in $\mathbb{A}_k^2$ that do not vanish in the function x, and $D(y)$ is all the points that do not vanish in the function y, we get:

$$\mathcal{O}_U(D(x)) = R_x = k\left[x, y, \frac{1}{x}\right] \qquad \mathcal{O}_U(D(y)) = R_y = k\left[x, y, \frac{1}{y}\right]$$

Now consider the functions that agree on the intersection: $R_x \cap R_y \subseteq R_{xy}$. These are all rationals functions with only powers of x in the denominator, and only powers of y in the denominator (as it must be in the intersection). But this leaves only the elements of R: $R_x \cap R_y = R$ (where R is identified with it's canonical embedding in R_{xy}). Thus:

$$\mathcal{O}_{\mathbb{A}_k^2}(U) = \mathcal{O}_U(U) = k[x, y]$$

in other words, the global section of U is identical to the global section of $\mathbb{A}_k^2$. Now, if U was affine, then by proposition 2.2.13, we would have that the is determined by the global ring, that is:

$$U \cong \mathbb{A}_k^2$$

that is, this is a scheme isomorphism, and hence also a homeomorphic map between prime ideals that must respect $V(-)$ and $I(-)$. But in U we have:

$$V(x) \cap V(y) = \emptyset$$

while this is certainly not the case in $\mathbb{A}_k^2$ (as it gives $[(x - 0, y - 0)]$). Hence U is *not* an affine scheme!

The main idea is that $D(f)$ misses a codimension 1 subset if R is Noetherian integral domain (see exercise 1), where Noetherian is to avoid infinite pathologies and integral domain is to insure there is only one irreducible component (see section 2.5). More generally, given R is Noetherian and an integral domain, $\operatorname{Spec} R_S$ for some $S \subseteq R$ not containing units misses a (pure) codimension 1 subset (as it is open). The reason that higher codimensional sets "don't matter" when considering global sections very reminiscent of complex geometry, namely this is an algebraic version of *Hartog's Lemma* (see [4, chapter 8]).

It is possible to have an affine open subscheme of an affine scheme, and hence by dualizing we may look for algebraic conditions on the ring homomorphisms. We state the result for those familiar with flatness (see [9]) but leave the proof till flatness is introduced in ref:HERE:

Theorem 2.3.15: Characterizing Affine Open Immersions

$\text{Spec}\,(A) \to \text{Spec}\,(R)$ is an open immersion if and only $R \to A$ is flat, of finite presentation, and an epimorphism.

***Proof*:**
see https://mathoverflow.net/questions/20782/ring-theoretic-characterization-of-open-affines.

2.3.2 Gluing (Coproduct) of Schemes

In Differential Geometry, one of the most common way of constructing a manifold, and indeed what is at the heart of geometry, is the process of gluing pieces together with some amount of consistency (see [5, chapter 1]). More generally, gluing manifolds, and gluing schemes, is a special case of a pushout (or fibered coproduct) $X \cup_Z Y$:

$$\begin{tikzcd}
& Z \arrow[dl] \arrow[dr] & \\
X \arrow[dr] \arrow[ddr, dashed, bend right] & & Y \arrow[dl] \arrow[ddl, dashed, bend left] \\
& X \cup_Z Y \arrow[d, dashed, "\exists!"] & \\
& Q &
\end{tikzcd}$$

The fibered product $X \times_Z Y$ can be thought of as taking a product of spaces $X \times Y$ and restricting to where the maps agree (i.e., carving out a subspace), while the fibered coproduct can be thought of as taking the disjoint union of spaces and doing some gluing. The fibered product shall be more nuanced as defining $X \times Y$ does not sense and hence is left for section 3.2.

Just like for manifolds, we can't glue schemes any way we want and get another scheme, we must be a bit more careful. Example 3.2.7 shall give an example of "bad" gluing.

The key idea for coproducts of schemes is to ensure that the gluing preserves the local affine structure (note that for a manifold it also has to preserve Hausdorffness). Hence, it is marginally easier to construct schemes via gluing:

Theorem 2.3.16: Coproduct (Pushout) of Schemes

Let $\{X_i\}_{i \in I}$ be a collection of schemes. Then if there exists:

1. A collection of open subschemes, not necessarily an open cover, $X_{ij} \subseteq X_i$ (where $X_{ii} = X_i$)
2. a collection of isomorphisms $f_{ij} : X_{ij} \to X_{ji}$ (where f_{ii} would be the identity) where

$$f_{ij}(X_{ij} \cap X_{ik}) \subseteq X_{ji} \cap X_{jk}$$

such that the isomorphisms agree on triple intersection[a]:

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}$$

then there is a unique scheme X (up to unique isomorphism) with open subsets isomorphic to X_i respecting the above gluing conditions.

[a] this is called the *cocycle condition*

This is a translation of the techniques from differential geometry (this condition may look more familiar if written as $\mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U \cap V) \to \mathbf{y}(U \cap V)$). The cocycle condition can be visualized like so:

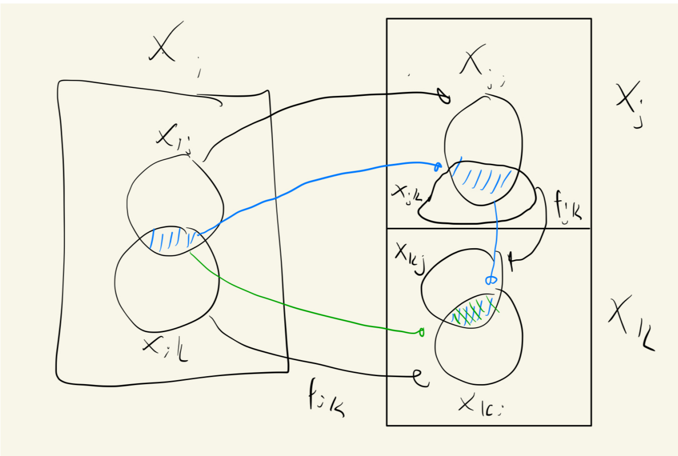


Figure 2.4: Green map must match blue map

***Proof*:**
Define X as the quotient of the disjoint union of the schemes X_i:

$$X = \frac{\coprod_{i \in I} X_i}{(X_{ij} \ni x \sim f_{ij}(x) \in X_{ji})}$$

where the equivalence relation is defined by identifying points via the gluing maps: $x \sim y$ if $x \in X_{ij}$, $y \in X_{ji}$, and $f_{ij}(x) = y$. The cocycle condition ensures that this relation is transitive (and symmetry follows from $f_{ji} = f_{ij}^{-1}$, which is a consequence of the cocycle condition with $k = i$), so this is a valid equivalence relation. We endow X with the quotient topology relative to the natural projection map $\pi : \coprod X_i \to X$.

Let $\iota_i : X_i \to X$ denote the restriction of π to the component X_i, and let $U_i = \iota_i(X_i)$ be its image in X. Because the X_{ij} are open sets, one can verify that ι_i is a homeomorphism onto the open set U_i. The sets U_i form an open cover of X, where each U_i is a topological copy of X_i.

With the topological space established, we define the structure sheaf $\mathcal{O}_X$. For any open set $V \subseteq X$, we define a section $s \in \mathcal{O}_X(V)$ to be a collection of sections $\{s_i\}_{i\in I}$ where $s_i \in \mathcal{O}_{X_i}(\iota_i^{-1}(V))$, satisfying the compatibility condition on overlaps:

$$s_i\big|_{\iota_i^{-1}(V)\cap X_{ij}} = f_{ij}^{\#}\left(s_j\big|_{\iota_j^{-1}(V)\cap X_{ji}}\right)$$

This defines a valid sheaf of rings by theorem 1.1.23, applied to the open cover $\{U_i\}$ with sheaves $(\iota_i)_*\mathcal{O}_{X_i}$ and transition isomorphisms induced by $f_{ij}^{\#}$.

Finally, we must verify that the ringed space $(X, \mathcal{O}_X)$ is indeed a scheme. It suffices to check this locally. Consider the open set U_i equipped with the restricted sheaf $\mathcal{O}_X|_{U_i}$. By our construction of the global sections, the sheaf $\mathcal{O}_X|_{U_i}$ is canonically isomorphic to the sheaf $\mathcal{O}_{X_i}$ via the map ι_i. Thus, we have an isomorphism of ringed spaces $(U_i, \mathcal{O}_X|_{U_i}) \cong (X_i, \mathcal{O}_{X_i})$. Since each X_i is a scheme by hypothesis, each open set U_i is a scheme. As X is covered by open subsets that are schemes (and hence covered by affine schemes), X is a scheme as we sought to show.

What would the resulting global sections of a glued scheme look like? Consider the case of two schemes X_1, X_2 glued along $X_{12} \subseteq X_1$ and $X_{21} \subseteq X_2$ via an isomorphism $f_{12} : X_{12} \xrightarrow{\sim} X_{21}$, with inclusion maps $\iota_i : X_i \to X$. Then:

$$\mathcal{O}_X(U) =$$
$$\left\{(s_1, s_2) \;\middle|\; s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)),\ s_2 \in \mathcal{O}_{X_2}(\iota_2^{-1}(U)) \text{ such that } s_1\big|_{\iota_1^{-1}(U)\cap X_{12}} = f_{12}^{\#}\left(s_2\big|_{\iota_2^{-1}(U)\cap X_{21}}\right)\right\}$$

This formula is common enough to be given a name: we call it the *gluing formula*.

For three schemes X_1, X_2, X_3, the formula extends naturally: a global section is a triple $(s_1, s_2, s_3) \in \mathcal{O}_{X_1}(X_1) \times \mathcal{O}_{X_2}(X_2) \times \mathcal{O}_{X_3}(X_3)$ satisfying the pairwise compatibility conditions on all overlaps simultaneously:

$$s_i\big|_{X_{ij}} = f_{ij}^{\#}\left(s_j\big|_{X_{ji}}\right) \qquad \text{for all } i, j \in \{1, 2, 3\}$$

Note that one does not glue two schemes first and then attach the third; the cocycle condition ensures that all pairwise constraints are mutually consistent, so all three (or finitely many) pieces are handled at once. The pattern for n schemes is identical: global sections are n-tuples satisfying $\binom{n}{2}$ compatibility conditions.

When thinking about global sections of glued schemes, it is helpful to view the gluing maps as imposing additional *relations*: the global sections of the glued scheme are the elements of the product $\prod_i \mathcal{O}_{X_i}(X_i)$ that are compatible under all the transition isomorphisms. The more gluing constraints there are, the smaller the ring of global sections becomes.

Example 2.3.17: Gluing [affine] Schemes

Let $X = \operatorname{Spec}(k[x])$ and $Y = \operatorname{Spec}(k[y])$. Take $D(x) \subseteq X$ and $D(y) \subseteq Y$, which is the collection of all prime ideals that do not contain x and y respectively. As shown in proposition ref:HERE, these subsets are isomorphic to:

$$D(x) \cong \operatorname{Spec}\left(k\left[x, \frac{1}{x}\right]\right) \qquad D(y) \cong \operatorname{Spec}\left(k\left[y, \frac{1}{y}\right]\right)$$

As a diagram, we have:

$$\begin{tikzcd} \operatorname{Spec}(k[x]) & & \operatorname{Spec}(k[y]) \\ & \operatorname{Spec}(k[x,1/x]) \arrow[ul] \arrow[ur] & \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] & \end{tikzcd}$$

or it may be more helpful to see this as:

$$\begin{tikzcd} \operatorname{Spec} k[x] & & \operatorname{Spec} k[y] \\ & \operatorname{Spec} k[x,1/x] \cong \operatorname{Spec} k[y,1/y] \arrow[ul] \arrow[ur] & \end{tikzcd}$$

and

$$\begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] \cong k[y,1/y] & \end{tikzcd}$$

where the cocycle condition must be satisfied by the isomorphisms. Notice how these are similar to restriction conditions: we are "manually" figuring out how the restrictions should work so that by uniqueness of gluing local sections on a sheaf we can get the global sections. These two open subsets can be glued in at least two different ways to produce two non-affine schemes that will be familiar to the reader:

1. (Line with double origin). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^{\#} : k[y, 1/y] \to k[x, 1/x]$ given by $y \mapsto x$. This is a scheme by theorem 2.3.16. Label this scheme S_1. We claim S_1 is *not* affine, even though its global sections coincide with those of $\mathbb{A}^1_k$.

   By the gluing formula, a global section of $\mathcal{O}_{S_1}$ is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying the compatibility condition:

   $$f(x) = \varphi^{\#}(g(y)) = g(x) \qquad \text{in } k[x, 1/x]$$

   Since $k[x] \hookrightarrow k[x, 1/x]$ is injective, this simply says $f(x) = g(x)$ as polynomials (after identifying the variable names). In particular, any polynomial satisfies this condition, and so:

   $$\mathcal{O}_{S_1}(S_1) \cong k[x]$$

By proposition 2.2.13, an affine scheme is determined by its global sections, so if S_1 were affine it would be isomorphic to $\operatorname{Spec}(k[x]) = \mathbb{A}^1_k$. But this is impossible: S_1 has two distinct closed points (the two origins) that are not separated by any open set, while $\mathbb{A}^1_k$ does not. Hence S_1 is not affine. The same construction works in higher dimensions (e.g. $\mathbb{A}^n_k$ with double origin).

In differential geometry, such spaces are examples of non-Hausdorff manifolds, however all non-trivial schemes already fail to be Hausdorff. The relevant condition for schemes is *separatedness*: spaces with double-origins fail to be separated (as a hint: show that the intersection of the two copies of $\mathbb{A}^1_k$ inside S_1 is $D(x)$, which is not affine), while $\mathbb{A}^n_k$ is separated.

2. (Projective space). Define the isomorphism $D(x) \cong D(y)$ via the ring isomorphism $\varphi^\# : k[y, 1/y] \to k[x, 1/x]$ defined by $y \mapsto 1/x$ (equivalently, $1/y \mapsto x$). The reader will recognize the resulting scheme as $\mathbb{P}^1_k$. When $k \in \{\mathbb{R}, \mathbb{C}\}$, this is isomorphic (as a manifold) to a circle and a sphere respectively, while in dimension n, $\mathbb{P}^n_k$ cannot be embedded in $\mathbb{A}^n_k$. For schemes, one must account for the generic point (0) and the non-closed points given by irreducible polynomials (recall that linear factors that are Galois conjugates are identified).

Let us show that $\mathbb{P}^1_k$ is not affine by computing $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. A global section is a pair $(f(x), g(y)) \in k[x] \times k[y]$ satisfying:

$$f(x) = \varphi^\#(g(y)) = g(1/x) \qquad \text{in } k[x, 1/x]$$

Write $f(x) = \sum_{n=0}^N a_n x^n$ and $g(y) = \sum_{m=0}^M b_m y^m$, so that:

$$g(1/x) = \sum_{m=0}^M b_m \, x^{-m}$$

In $k[x, 1/x]$, the Laurent monomials $\{x^n\}_{n\in\mathbb{Z}}$ are linearly independent over k. The left-hand side $f(x)$ involves only non-negative powers of x, while the right-hand side $g(1/x)$ involves only non-positive powers. For these to be equal, both must be constant:

$$a_n = 0 \text{ for } n \geq 1, \qquad b_m = 0 \text{ for } m \geq 1, \qquad a_0 = b_0$$

Hence f and g are both the same constant, giving:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$$

By proposition 2.2.13, if $\mathbb{P}^1$ were affine then $\mathbb{P}^1 \cong \operatorname{Spec}(\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)) = \operatorname{Spec}(k)$, which is a single point. But $\mathbb{P}^1_k$ has more than one point, so $\mathbb{P}^1_k$ is not affine[a].

[a]Those familiar with complex analysis may recall that the only holomorphic functions on $\mathbb{P}^1_{\mathbb{C}}$ (the Riemann sphere) are constants, which is a manifestation of the same phenomenon.

2.3.3 *Gluing Affine Schemes and Fiber Products of Rings

Let us now specialize to the case where the pieces being glued are affine. Recall that the functors $\operatorname{Spec}(-)$ and $\Gamma(-)$ are contravariant and form an adjunction that exchanges limits and colimits. Hence, the cofibered product (pushout) of schemes dualizes to the fiber product (pullback) of rings.

If $X_1 = \operatorname{Spec}(A)$, $X_2 = \operatorname{Spec}(B)$, and the overlap maps factor through a ring C, then the gluing formula for global sections becomes:

$$\mathcal{O}_X(X) = A \times_C B := \{(a,b) \in A \times B \mid \text{images of } a \text{ and } b \text{ agree in } C\}$$

This is dual to the familiar fact that the fibered product of affine schemes corresponds to the tensor product of rings:

$$\operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B \cong \operatorname{Spec}(A \otimes_C B)$$

In both cases, the contravariance of Spec exchanges limits and colimits, and hence exchanges tensor products (colimits in **CRing**) with fiber products (limits in **CRing**). More generally, for an arbitrary number of affine pieces, the global sections are the limit of the diagram of rings formed by all the overlap maps.

It is natural to ask: since **CRing** is both complete and cocomplete, does $\operatorname{Spec}(A \times_C B)$ give the pushout of $\operatorname{Spec}(A)$ and $\operatorname{Spec}(B)$ over $\operatorname{Spec}(C)$ in the category of schemes? The answer is *no* in general. The subtlety is that the universal property must be tested against different classes of objects depending on the ambient category:

- *In **Aff***: the object $\operatorname{Spec}(A \times_C B)$ satisfies the universal property of the pushout with respect to maps into affine schemes, and this pushout always exists since **CRing** is complete (so fiber products of rings always exist).
- *In **Sch***: the pushout must satisfy the universal property for maps into *all* schemes. A morphism from an affine scheme to a non-affine target Q is not determined by a ring homomorphism out of $\Gamma(Q, \mathcal{O}_Q)$ (since Q is not affine, the adjunction $\operatorname{Spec} \dashv \Gamma$ does not apply), so the affine pushout may fail the universal property.

In categorical language, the inclusion **Aff** $\hookrightarrow$ **Sch** does not preserve pushouts. The pushout computed in **Sch** can be genuinely non-affine: the line with doubled origin and the projective line from example 2.3.17 are both pushouts of affine schemes in **Sch** that are not affine. In each case, $\operatorname{Spec}(A \times_C B)$ exists as an affine scheme, but it is *not* isomorphic to the glued scheme X – it is only the "best affine approximation" to the pushout.

The preceding discussion raises a natural question: if $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, what *is* it? The answer is given by the adjunction between global sections and Spec.

Proposition 2.3.18: $\Gamma \dashv \operatorname{Spec}$ Adjunction

The global sections functor $\Gamma : \mathbf{Sch} \to \mathbf{CRing}^{op}$ is left adjoint to $\operatorname{Spec} : \mathbf{CRing}^{op} \to \mathbf{Sch}$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \Gamma(X, \mathcal{O}_X))$$

for any scheme X and any ring R.

***Proof*:**
See proposition 3.1.12

Since left adjoints preserve colimits and right adjoints preserve limits, this single adjunction encodes both dualities between schemes and rings:

- Γ (left adjoint) sends pushouts in **Sch** to colimits in $\mathbf{CRing}^{op}$, i.e. limits (fiber products) in **CRing**:
$$\Gamma(X_1 \cup_Z X_2) \cong \Gamma(X_1) \times_{\Gamma(Z)} \Gamma(X_2)$$
This is why the gluing formula computes global sections as a fiber product of rings – it is a formal consequence of the adjunction.

- Spec (right adjoint) sends limits in $\mathbf{CRing}^{op}$ (i.e. colimits in **CRing**, such as tensor products) to limits (fiber products) in **Sch**:
$$\operatorname{Spec}(A \otimes_C B) \cong \operatorname{Spec}(A) \times_{\operatorname{Spec}(C)} \operatorname{Spec}(B)$$

- Spec, being a right adjoint, does *not* preserve colimits in general. The fiber product $A \times_C B$ is a limit in **CRing**, hence a colimit in $\mathbf{CRing}^{op}$, and there is no reason for Spec to preserve it.

Definition 2.3.19: Affinization

Let X be a scheme. The *affinization* of X is the affine scheme $X^{\mathbf{Aff}} := \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$, together with the canonical morphism
$$\eta_X : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$$
given by the unit of the $\Gamma \dashv \operatorname{Spec}$ adjunction.

By the universal property of adjunctions, η_X is the universal map from X to an affine scheme: any morphism $X \to \operatorname{Spec}(R)$ factors uniquely through η_X. For the pushout $X = X_1 \cup_Z X_2$ of affine schemes, since $\Gamma(X) \cong A \times_C B$, this gives:
$$X_1 \cup_Z X_2 \xrightarrow{\eta} \operatorname{Spec}(A \times_C B)$$
Hence $\operatorname{Spec}(A \times_C B)$ is not the pushout in **Sch**, but rather the *best affine approximation* to the pushout – the closest affine scheme that the glued scheme maps to.

Example 2.3.20: Affinization of Glued Schemes

Returning to the two gluings from example 2.3.17:

1. *Line with doubled origin.* Here $A \times_C B \cong k[x]$, so the affinization is $\mathbb{A}^1_k$. The map $\eta : S_1 \to \mathbb{A}^1_k$ identifies the two origins to a single point, collapsing exactly the non-affine pathology.
2. *Projective line.* Here $A \times_C B \cong k$, so the affinization is $\operatorname{Spec}(k)$, a single point. The map $\eta : \mathbb{P}^1_k \to \operatorname{Spec}(k)$ collapses the entire projective line to a point, reflecting the fact that $\mathbb{P}^1_k$ has only constant global functions.

In both cases, the affinization map η "kills" precisely the geometric information that cannot be captured by global sections.

Exercise 2.3.1

1. Let X be the line with two origins, $Y = \mathbb{A}^1_k$ the affine line, and $\pi : X \to Y$ the natural projection. Compute $f_*(\mathcal{O}_X)_0$, the talk of the pushforward at 0.

2. Let X be a scheme. Show that the affine open subsets form a base.
3. A subset Z of a scheme X is closed if and only if there exists an open cover of X by affine subschemes, $\{U_i = \mathrm{Spec}(A_i)\}_{i \in I}$, such that for every $i \in I$, the intersection $Z \cap U_i$ is a closed subset of the affine scheme U_i. In other words, a closed subset of X is characterized as being locally closed on an affine open covering.

2.4 Projective Schemes

Recall projective space from [5]. In [9] a few key properties of projective space $\mathbb{P}^n_{\mathbb{C}}$ were proved, namely:

1. Projective space was constructed by gluing together affine spaces. The cocycle condition is straightforward to verify over any ring (the transition maps are rational functions that do not require algebraic closure), though understanding the closed points is simpler when $k = \overline{k}$. We shall construct Proj in a more "natural" way without any choice of coordinates.
2. There shall be a homogeneous coordinate ring S from which we can construct the scheme $\mathrm{Proj}\, S$. Just like in the classical setting, S shall not be the ring of regular functions on $\mathrm{Proj}\, S$: the global sections of $\mathrm{Proj}\, S$ shall be constant, and $\mathrm{Proj}\, S$ shall behave like the "compact" sets in algebraic geometry (namely the proper sets, as mentioned in [9]).

   We shall show that the algebraic object S will be associated to a different sheaf over $\mathrm{Proj}\, S$ that is not the structure sheaf, see section 5.2.
3. The properties of Bézout's theorem shall turn out to be the special case of the theory of *divisors* on projective space. See section 7.2 for more details.

Classically, n-dimensional projective space $\mathbb{R}\mathrm{P}^n$ or $\mathbb{C}\mathrm{P}^n$ is defined on $\mathbb{R}^{n+1} \setminus \{0\}$ or $\mathbb{C}^{n+1} \setminus \{0\}$ by taking an equivalence class:

$$(x_0, \ldots, x_n) \sim (y_0, \ldots, y_n) \qquad \Leftrightarrow \qquad (x_0, \ldots, x_n) = (\lambda y_0, \ldots, \lambda y_n) \qquad \text{for some } \lambda \neq 0$$

This produces an n-dimensional smooth manifold with the usual patches defined as:

$$U_i = D(x_i) = \{[x_0 : \cdots : x_n] \in \mathbb{P}^n \mid x_i \neq 0\}$$

where, using the notation $x_{k,j} = x_k / x_j$ for the affine coordinates on U_j, the transition maps were defined as (see [5, chapter 1]):

$$\varphi_i \circ \varphi_j^{-1}(x_{0,j}, \ldots, \widehat{x_{j,j}}, \ldots, x_{n,j}) = \left(\frac{x_{0,j}}{x_{i,j}}, \ldots, \frac{1}{x_{i,j}}, \ldots, \frac{\widehat{x_{i,j}}}{x_{i,j}}, \ldots, \frac{x_{n,j}}{x_{i,j}} \right)$$

In [9], we saw we can do the same for any field k, namely since inverses exist. Let us do the same but with a general ring R and the graded polynomial ring $S = R[x_0, x_1, \ldots, x_n]$ with the standard grading ($\deg x_i = 1$, R in degree 0), and $S_+ = \bigoplus_{d>0} S_d$ being the irrelevant ideal. Though S is naturally $\mathbb{N}$-graded, we shall work in the setting of $\mathbb{Z}$-graded rings since localizations introduce negative degrees[22].

[22] More general gradings are sometimes useful, however we shall not need this either in this book or for a while

As a set, define:

Definition 2.4.1: Projective Space – as a Set

$$\operatorname{Proj} S = \{[\mathfrak{p}] \mid \mathfrak{p} \subseteq S \text{ is a homogeneous prime with } S_+ \not\subseteq \mathfrak{p}\}$$

The condition $S_+ \not\subseteq \mathfrak{p}$ excludes the "irrelevant" prime consisting of all positive-degree elements, which would correspond to the non-existent point $[0 : \cdots : 0]$ in projective space. With the set defined, $\operatorname{Proj} S$ has the Zariski topology induced by homogeneous elements of positive degree:

$$V_+(T) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid T \subseteq \mathfrak{p}\}$$

The basis for the topology can be given by the positive-degree homogeneous elements:

$$D_+(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S \mid f \notin \mathfrak{p}\}$$

If $f = x_i$, we would like this to correspond to the usual affine space $\mathbb{A}^n_R$. In the classical case, we got this isomorphism by doing:

$$\begin{aligned}
\{[x_0 : \cdots : x_n] \mid x_i \neq 0\} &= \left\{\left[\frac{x_0}{x_i} : \cdots : 1 : \cdots : \frac{x_n}{x_i}\right] \,\middle|\, x_i \neq 0\right\} \\
&\cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\
&= k[x_{0,i}, \ldots, x_{n,i}] && \text{relabeling} \\
&= \mathbb{A}^n_k
\end{aligned}$$

Notice how these are all degree 0 in the grading of S. These sets are not quite the same as $\operatorname{Spec} S_f$; however it turns out that taking $D_+(f)$ and working with the degree-0 part of the localization gives us exactly the desired result:

Lemma 2.4.2: Characterizing Basic Open Sets For Projective Space

Let $f \in S$ be a homogeneous element of positive degree, and define $S_{(f)} := (S_f)_0$ to be the degree-0 part of the localization. Then:

$$D_+(f) \cong \operatorname{Spec}\left(S_{(f)}\right)$$

***Proof*:**
We construct explicit inverse maps. Define the forward map $\Phi : D_+(f) \to \operatorname{Spec}\left(S_{(f)}\right)$ by:

$$\Phi(\mathfrak{p}) = \mathfrak{p} \cdot S_f \cap S_{(f)}$$

Since $f \notin \mathfrak{p}$, the extension $\mathfrak{p} \cdot S_f$ is a proper homogeneous prime of S_f, and its intersection with the subring $S_{(f)}$ is a prime ideal of $S_{(f)}$.

For the inverse, let $e = \deg f$ and define $\Psi : \operatorname{Spec}\left(S_{(f)}\right) \to D_+(f)$ by:

$$\Psi(\mathfrak{q}) = \bigoplus_{d \geq 0} \left\{a \in S_d \,\middle|\, \frac{a^e}{f^d} \in \mathfrak{q}\right\}$$

Note that a^e has degree de and f^d has degree de, so $a^e/f^d \in S_{(f)}$ and the condition makes sense.

$f \notin \Psi(\mathfrak{q})$: If $f \in \Psi(\mathfrak{q})$, then $f^e/f^e = 1 \in \mathfrak{q}$, contradicting $\mathfrak{q}$ being proper.

Closure under addition: Let $a, b \in S_d$ with $a^e/f^d, b^e/f^d \in \mathfrak{q}$. Consider a binomial term $a^j b^{e-j}$ with $0 \leq j \leq e$. Observe:

$$\left(\frac{a^j b^{e-j}}{f^d}\right)^e = \left(\frac{a^e}{f^d}\right)^j \cdot \left(\frac{b^e}{f^d}\right)^{e-j} \in \mathfrak{q}$$

Since $\mathfrak{q}$ is prime (hence radical: $x^n \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$), we get $a^j b^{e-j}/f^d \in \mathfrak{q}$ for each j. Summing over all binomial terms gives $(a+b)^e/f^d \in \mathfrak{q}$.

Closure under S-multiplication: If $a \in \Psi(\mathfrak{q})_d$ and $c \in S_k$, then $(ca)^e/f^{d+k} = (c^e/f^k)(a^e/f^d)$. The second factor lies in $\mathfrak{q}$ and the first in $S_{(f)}$, so the product lies in $\mathfrak{q}$.

Primality: If $a \in S_d, b \in S_k$ with $ab \in \Psi(\mathfrak{q})$, then $(ab)^e/f^{d+k} = (a^e/f^d)(b^e/f^k) \in \mathfrak{q}$. Since $\mathfrak{q}$ is prime, either $a^e/f^d \in \mathfrak{q}$ or $b^e/f^k \in \mathfrak{q}$, giving $a \in \Psi(\mathfrak{q})$ or $b \in \Psi(\mathfrak{q})$.

One can verify that Φ and Ψ are inverse to each other: the key step is that $a \in \mathfrak{p}_d$ implies $a^e/f^d \in \mathfrak{p}S_f \cap S_{(f)}$, and conversely if $a^e/f^d \in \mathfrak{p}S_f$ then $a \in \mathfrak{p}$ by primality (since $f \notin \mathfrak{p}$). For the homeomorphism, the preimage of a basic open $D(g^e/f^{\deg g}) \subseteq \operatorname{Spec}(S_{(f)})$ is $D_+(fg) \cap D_+(f) = D_+(fg)$, so both sides have matching bases.

Let us apply this to $f = x_i$:

$$\begin{aligned} D_+(x_i) &= \{[\mathfrak{p}] \in \operatorname{Proj} S \mid x_i \notin \mathfrak{p}\} \\ &\cong \operatorname{Spec}\left(S_{(x_i)}\right) \\ &= \operatorname{Spec}\left(\text{degree-0 elements of } R[x_0, \ldots, x_n, 1/x_i]\right) \\ &\overset{!}{=} \operatorname{Spec} R\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong \operatorname{Spec} \frac{R[x_{0,i}, \ldots, x_{n,i}]}{(x_{i,i} - 1)} \\ &\cong \mathbb{A}^n_R \end{aligned}$$

where $\overset{!}{=}$ comes from the fact that $1/x_i$ is now a unit of degree -1, hence each x_j/x_i is of degree 0, and these elements generate $S_{(x_i)}$ over R. Relabeling x_j/x_i as $x_{j,i}$ and "omitting" $x_{i,i}$ (which equals one, equivalently we quotient by $(x_{i,i} - 1)$, a form of dehomogenizing), we see that it is isomorphic to the usual $\mathbb{A}^n_R$, and hence we recover the usual affine cover! Note the importance of taking the degree-0 component, for

$$\operatorname{Spec} R[x_0, \ldots, x_n, 1/x_i] \cong \mathbb{A}^{n+1}_R \setminus \{x_i = 0\}$$

Let us now construct the structure sheaf on $\operatorname{Proj} S$. We shall see that $\operatorname{Proj} S$ is not affine (we hope not, as the classical projective variety is not affine!), and so it is not determined by its global sections in the way that affine schemes are. Nonetheless, the construction below endows $\operatorname{Proj} S$ with a natural scheme structure. As we just saw, each $D_+(f)$ is affine and hence we certainly have an affine cover. Let us ensure that the gluing satisfies the cocycle condition from theorem 2.3.16, namely that all intersections glue appropriately to form a scheme.

Theorem 2.4.3: Structure Sheaf On Proj

Take the Zariski topology on $\operatorname{Proj} S$ given by the basis $D_+(f) \cong \operatorname{Spec}(S_{(f)})$. Then $\operatorname{Proj} S$ is a scheme.

***Proof*:**
Let $f, g \in S$ be homogeneous elements of positive degree and take $D_+(f), D_+(g)$ with non-trivial overlap. It is not difficult to show that $D_+(f) \cap D_+(g) = D_+(fg)$. We verify the cocycle condition from theorem 2.3.16. Consider:

$$D(g^{\deg f}/f^{\deg g}) \subseteq D_+(f) \qquad \text{and} \qquad D(f^{\deg g}/g^{\deg f}) \subseteq D_+(g)$$

We shall show that:

$$S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \cong S_{(fg)} \cong S_{(g)}\left[\frac{1}{f^{\deg g}/g^{\deg f}}\right]$$

This will induce the desired isomorphism on spectra. There is a natural map:

$$\varphi : S_{(f)} \to S_{(fg)}$$

mapping $h/f^k \mapsto hg^k/(fg)^k$. Taking $U = \left\{(g^{\deg f}/f^{\deg g})^k \;\middle|\; k \in \mathbb{N}\right\}$, notice that $\varphi(U)$ consists of units in $S_{(fg)}$, thus there is a natural map:

$$\widetilde{\varphi} : S_{(f)}\left[\frac{1}{g^{\deg f}/f^{\deg g}}\right] \to S_{(fg)}$$

It is straightforward to verify that this map is in fact an isomorphism (injectivity is easy, and matching degrees gives surjectivity). The same argument works in the other direction, giving the desired result. Thus, we have established the desired ring isomorphism, which induces the required isomorphism of schemes, completing the proof.

Definition 2.4.4: Projective Scheme

Let $S_\bullet$ be a $\mathbb{Z}$-graded ring with $S_0 = R$. Then $\operatorname{Proj} S_\bullet$ is called the projective scheme associated to $S_\bullet$. If $S = R[x_0, \ldots, x_n]$ with the standard grading, then the above construction defines the classical projective n-scheme or prototypical projective scheme over R, and is denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n$ if unambiguous.

We can naturally recover the notion of quasiprojective as open subschemes of projective schemes:

Definition 2.4.5: Quasiprojective Scheme

A *quasiprojective R-scheme* is an open subscheme of a projective R-scheme.

Notice that the scalars can be defined for any ring, which generalizes the result from classical algebraic geometry. If $R = k$ is an algebraically closed field, the closed points of $\mathbb{P}^n_k$ will behave like the traditional points of projective space defined in classical algebraic geometry.

Proposition 2.4.6: Global Sections of $\operatorname{Proj} S_\bullet$

Let $S_\bullet$ be an $\mathbb{N}$-graded ring generated by S_1 as an S_0-algebra (e.g. $S = R[x_0, \ldots, x_n]$ with the standard grading), and let $X = \operatorname{Proj} S_\bullet$. Then:

$$\mathcal{O}_X(X) = S_0$$

Hence $\mathbb{P}^n_R$ is not an affine scheme (the case $n = 1$ was given in example 2.3.17).

***Proof*:**
exercise

Recall from [9] that homogeneous polynomials of $k[x_0, \ldots, x_n]$ cut out closed sets from $\mathbb{P}^n_k$. This is because if $a \in k^{n+1}$ is a root of a homogeneous polynomial, so are all scalar multiples of a. In modern algebraic geometry, the global sections $\mathcal{O}_X(X)$ are regarded as the "functions" on a scheme X. However, $\mathcal{O}_{\mathbb{P}^n_R}(\mathbb{P}^n_R) = R$ by proposition 2.4.6, so the graded ring S that *generates* $\operatorname{Proj} S$ does not manifest as the structure sheaf.

How, then, should we interpret S? The answer is that each graded piece S_d gives rise to a sheaf $\mathcal{O}(d)$ on $\mathbb{P}^n_R$ called a *twisting sheaf* (or *Serre twist*). These are examples of *line bundles* (invertible sheaves), and the structure sheaf $\mathcal{O}_{\mathbb{P}^n_R}$ is the special case $\mathcal{O}(0)$. We shall show in proposition 5.2.7 that the graded ring S can be recovered from these sheaves:

$$S_d \cong \Gamma(\mathbb{P}^n_R, \mathcal{O}(d))$$

This is significant because line bundles and their global sections control the geometry of morphisms *from* projective space: a morphism $\mathbb{P}^n_R \to \mathbb{P}^m_R$ is determined by a choice of $m+1$ global sections of a line bundle on $\mathbb{P}^n_R$ (with no common vanishing), a correspondence that will be made precise in section 5.2.1. In this way, the graded ring S – despite not being the ring of global functions – encodes the "embedding data" of projective space.

The reader ought to remember that if $f(x_{0,i}, \ldots, x_{n,i})$ is a homogeneous polynomial of degree d on $D(x_i)$, then when converting coordinates to $D(x_j)$ we get:

$$f(x_{0,i}, \ldots, x_{n,i}) = x_{j,i}^{\deg f} f(x_{0,j}, \ldots, x_{n,j})$$

Example 2.4.7: Projective Scheme Homogeneous Polynomial Cut

There are many examples in [9], and hence we shall just put a few as a refresher.

1. Take $C = V(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$ for an algebraically closed field k. Then if $z = 0$, we would get $[(0,0,0)] \in C$, however this point doesn't exist in $\mathbb{P}^n_k$. Hence consider $z \neq 0$. Then for each solution $[(a,b,c)] = [(\lambda a, \lambda b, \lambda c)] \in C$. In k^3, the solutions form a cone of points. Loosely visualizing $\mathbb{P}^2_k$ as a sphere (we should glue the antipodal points), we see that this resembles the image:

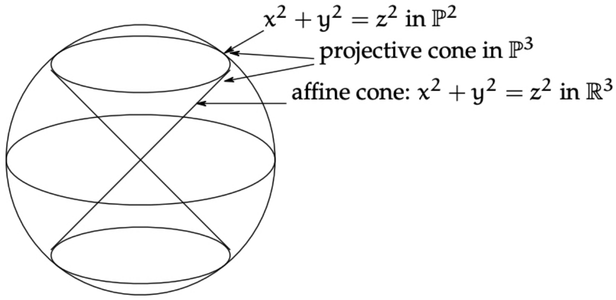


Figure 2.5: Visualization of $V(x^2 + y^2 - z^2)$

Such vanishing sets are called *vanishing cones.*

2. $\mathbb{P}^0_R = \operatorname{Proj} R[x_0] \cong \operatorname{Spec} R$, hence all 0-dimensional projective schemes are affine. Note then that $\mathbb{P}^0_k = \{\overline{0}\}$ since $\operatorname{Spec} k$ is a singleton; or this can be verified directly by noticing that the only non-irrelevant homogeneous prime ideal of $k[x]$ is (0), for all other homogeneous prime ideals are of the form (x^n) and hence contain S_+.

3. For an example that is not a hypersurface, take $V(wz - xy, x^2 - wy, y^2 - xz)$. This is called the *twisted cubic curve.* This ideal is prime, and the variety is isomorphic to $\mathbb{P}^1_k$ (hint: consider $k[w, x, y, z] \to k[s, t]$, $(w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$).

4. When $R = k$ is a field, the definition of $\mathbb{P}^n_k$ as a scheme matches the classical definition of projective space as a quotient of $k^{n+1} \setminus \{0\}$ by $k^\times$. The points in the scheme correspond exactly to lines through the origin (plus the generic point). However, when R is not a field, $\mathbb{P}^n_R$ behaves like a "compact" object in a very powerful way. Consider the case where R is a Discrete Valuation Ring (DVR), for example $R = k[[t]]$ or $\mathbb{Z}_{(p)}$, with fraction field K. Consider a point $P \in \mathbb{P}^n_K$ (i.e., a point in projective space over the fraction field). We can write this point in coordinates $[f_0 : \cdots : f_n]$ where $f_i \in K$. We might ask: does this point "extend" to a point in $\mathbb{P}^n_R$? In other words, if we have a curve defined for $t \neq 0$, does it have a unique limit as $t \to 0$?

   The answer is yes. Since we are in projective space, we can scale coordinates by any non-zero element.

   (a) Let $v(f_i)$ be the valuation of each coordinate.
   (b) Multiply the entire tuple by t^{-m}, where $m = \min_i(v(f_i))$.
   (c) Now all coordinates are in R, and at least one coordinate is a unit (valuation 0).
   (d) We can now safely set $t = 0$ to find the unique limit point in the special fiber $\mathbb{P}^n_k \subset \mathbb{P}^n_R$.

   For example, if $R = k[[t]]$ and $P = [1/t^2 : 1/t] \in \mathbb{P}^1_K$, we scale by t^2 to get $[1 : t]$. Setting $t = 0$ gives the limit point $[1 : 0]$. This property – that any map from the punctured curve $\operatorname{Spec} K$ extends uniquely to the whole curve $\operatorname{Spec} R$ – is called the *Valuative Criterion for Properness.* It essentially states that projective schemes are "compact" (curves don't run off to infinity without hitting a limit). This implies that $\mathbb{P}^n_R$ is connected in a way affine schemes are not (an affine line with a missing origin cannot be patched!). See section 3.5.4 for more details.

5. In an affine scheme, two polynomials over a field cut out the same vanishing set if they differ by a scalar. This is not the case for homogeneous polynomials cutting out parts of $\mathbb{P}^n$: consider $x^2y = 0$ and $xy^2 = 0$. The problem is that these two are homogenizations of different polynomials that produce the same set. However, as we know from section 2.2.2, schemes remember multiplicity. This behaviour shall later be properly taken into account by saying that two polynomials shall cut out different *closed subschemes* (rather than the same *closed subset*). With this addendum, it shall once again be the case that two polynomials cut out the same closed subscheme if and only if they differ by a scalar multiple, as closed subschemes shall essentially correspond to polynomial cut-outs (as sets) that have sheaves remembering multiplicity information.

Proposition 2.4.8: Irreducibility of classical projective scheme

Let R be a domain. Then the scheme $\mathbb{P}^n_R$ is irreducible and is of finite type over R.

As irreducible spaces are connected, $\mathbb{P}^n_k$ is connected.

***Proof*:**
Since R is a domain, $S = R[x_0, \ldots, x_n]$ is a domain, so the zero ideal (0) is a homogeneous prime not containing S_+. Hence $(0) \in \operatorname{Proj} S$ is a point. For each i, $x_i \notin (0)$, so $(0) \in D_+(x_i)$ for every i. Since $\{D_+(x_i)\}_{i=0}^n$ covers $\operatorname{Proj} S$, the generic point (0) lies in every basic open of the standard cover. As any nonempty open set contains some $D_+(x_i)$, it contains (0), so $\{(0)\}$ is dense and $\mathbb{P}^n_R$ is irreducible. Finite type over R follows from each chart $D_+(x_i) \cong \mathbb{A}^n_R = \operatorname{Spec} R[x_1, \ldots, x_n]$ being of finite type over R, with finitely many charts.

To conclude this section, let us formally define the affine and projective cone:

Definition 2.4.9: Affine Cone

Let $S_\bullet$ be a graded ring. Then the affine cone of $\operatorname{Proj} S_\bullet$ is $\operatorname{Spec}(S_\bullet)$.

Note that $\operatorname{Spec}(S_\bullet)$ depends on $S_\bullet$, not just $\operatorname{Proj}(S_\bullet)$. Think of the affine cone as eliminating the points at infinity and adding the origin. In fact, given a field k, there is a natural map $\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet$ (more generally, if $S_\bullet$ is a finitely generated graded ring over a base ring R, there is a natural morphism).

Definition 2.4.10: Projective Cone

Take $\operatorname{Spec} S_\bullet$. Then the projective cone or projective completion is $\operatorname{Proj}(S_\bullet[t])$ for some indeterminate t of degree 1.

Take for example $\operatorname{Proj}(k[x,y,z]/(z^2 - x^2 - y^2))$. Then the projective cone is $\operatorname{Proj}(k[x,y,z,t]/(z^2 - x^2 - y^2))$. The vanishing locus $V_+(t)$ returns the original projective scheme, and taking the complement (the distinguished open set $D_+(t)$) gives $\operatorname{Spec}(S_\bullet)$. This construction can be thought of as the opposite of the affine cone, where we attach the points at infinity of a conic curve.

The Non-Uniqueness of Graded Rings

Before moving on, we must highlight a fundamental difference between the Spec and Proj constructions. The spectrum functor provides a perfect anti-equivalence between commutative rings and affine schemes; if $\operatorname{Spec} A \cong \operatorname{Spec} B$ as schemes, then $A \cong B$ as rings.

The Proj construction completely fails this property. Two wildly different graded rings can produce the exact same projective scheme. For example, let $S_\bullet = k[x, y]$ and let $T_\bullet = k[x^2, xy, y^2] \cong k[u, v, w]/(uw - v^2)$. As standard rings, these are not isomorphic: $S_\bullet$ is a Unique Factorization Domain, while $T_\bullet$ is not (since $uw = v^2$ represents two distinct factorizations into irreducibles). Geometrically, their affine cones are drastically different: $\operatorname{Spec} S_\bullet$ is the smooth plane $\mathbb{A}^2_k$, whereas $\operatorname{Spec} T_\bullet$ is a quadric cone which is singular at the origin.

However, because Proj effectively removes the origin (by restricting to homogeneous prime ideals not containing S_+) and takes a quotient by the grading, these differences vanish: $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} T_\bullet \cong \mathbb{P}^1_k$ (proven in theorem 3.1.32).

In general, the Proj construction only determines a graded ring "up to truncation at finite degrees". If two graded rings $S_\bullet$ and $T_\bullet$ share the exact same graded pieces for all degrees $d \geq N$ for some large integer N, they will yield isomorphic projective schemes. We will formalize exactly how to recover the "canonical" coordinate ring of a projective scheme using twisted sheaves in section 5.2.

Proj-like Functors for other schemes?

here

2.5 Properties of Schemes

In this section, we shall cover various different properties a scheme may have. In the process, we shall see if these properties can be checked affine-locally or stalk-locally Many of the topological properties inquired about affine schemes can be asked about for general schemes. For example, general schemes need not be quasicompact, while all affine schemes are quasicompact. Properties that have been explored already are:

1. (dis)connected: In the affine case, this corresponds to the existence of non-trivial idempotent elements, for if $e \in R$ is such an element then $R \cong eR \times (1 - e)R$, and so the spectrum is a disjoint union. For the general case of schemes, it is usually given by the disjoint union of known schemes.

2. (ir)reducible: In the affine case, this corresponds to the ring having *no zero divisors*, i.e. an integral domain. Proposition 2.5.19 shall characterize irreducible schemes[23].

   Recall that an equivalent conditions to irreducibility is that every non-empty open subset is dense (as the reader should verify).

3. quasicompact: Essentially all schemes we shall work with will be quasicompact (with the "trivial" exception being the infinite disjoint union of schemes). Most of the time, non-quasicompact schemes will come from some infinite gluing.

[23] with the reducedness condition insuring no "fuzz" in the sheaf

These are global topological properties, hence they cannot be checked affine- or stalk-locally (as the reader should verify with some counter-examples).

Finiteness Conditions

The most common types of rings in commutative algebra are Noetherian rings. The geometric equivalent is the following:

Definition 2.5.1: Noetherian Scheme

Let X be a scheme. If X can be covered by affine open sets where each ring R_i of $\operatorname{Spec}(R_i)$ is Noetherian, then X is said to be a *locally Noetherian Scheme*. If X is also quasicompact, then X is said to be a *Noetherian Scheme*.

All coordinate rings are Noetherian, and hence we may think of varieties as examples of Noetherian schemes[24]. We can think of the Noetherian condition as insuring that the open affine don't have "difficult" infinite behavior. Most examples which involve non-Noetherian rings involve infinitely many variables (ex. $k[x_1, x_2, ...]$, $k[x, y, y/x, y/x^2, ...]$ and so forth). In part I; examples of non-Noetherian topological spaces we shall encounter will be the fibered product of Noetherian rings (see ref:HERE), and the special product topology put on the ring of adèles. In general, we shall see that there are many results that do not require finite generation, and hence it is worth keeping track of it when this condition is necessary.

One important consequence of being Noetherian is the existence of finitely many irreducible components:

Lemma 2.5.2: Finitely Many Components

Let X be a Noetherian scheme. Then X has finitely many irreducible components.

***Proof*:**
As X is Noetherian, it is locally Noetherian, so let it be covered be Noetherian affine open schemes. As it is quasicompact, we may take a finite subcover so $X = \bigcup_i^n \operatorname{Spec} R_i$. As each R_i is Noetherian, it can have only finitely many irreducible components (check this), and hence their union can have only finitely many irreducible components, completing the proof.

This is not true if the ring is not Noetherian: the spectrum of the ring

$$\varinjlim_{n\to\infty} \frac{k[x_i \,|\, 1 \leq i \leq n]}{\prod_i^n x_i}$$

has infinitely many components, but it is a quasicompact scheme (as it is affine).

Let us next turn to an important property of quasicompact schemes. When working with affine schemes, we always have that there exists closed points corresponding to the maximal ideals of the ring. When working over general schemes, there may be no closed points[25]. Fortunately, all quasicompact scheme have closed points, and even better:

[24] Once we introduce theorem 3.8.1 this will be made precise
[25] see https://math.stanford.edu/~vakil/files/schwede03.pdf

Proposition 2.5.3: Quasicompact Schemes and Closed Points

Let X be a quasicompact scheme. Then X contains closed points. Furthermore, the closure of any non-closed point contains closed points.

***Proof*:**
First, if X is affine, it certainly contains closed points as $\mathcal{O}_X(X) = R$ contain at least one maximal ideal. Furthermore, the closure of any non-closed point would contain maximal ideal as any prime ideal is contained in a maximal ideal.

Now, let X be some quasicompact scheme. Let $X = \bigcup_i^n U_i$ be a finite covering of affine open subsets. Take any closed point $\mathfrak{p}_1 \in U_1$. If it is closed in X we're done. If not, take its closure $\overline{\mathfrak{p}_1}$ in X. As $\mathfrak{p}_1$ is not closed in X, its closure contains more than $\mathfrak{p}_1$, call this point $\mathfrak{p}_2$, and without loss of generality say it is in U_2. If $\mathfrak{p}_2$ is closed in X, we are done. If not, take $\overline{\mathfrak{p}_2}$ in X. Its closure must contain a point $\mathfrak{p}_3$ that is not in U_2 or U_1 (as $\mathfrak{p}_3$ would be in the closure of $\mathfrak{p}_1$ or $\mathfrak{p}_2$ in U_1 and U_2 respectively). Continue this process until a point is closed, or if we get to $\mathfrak{p}_n \in U_n$, then the closure of this point can't contain any other point in $U_1, \ldots, U_n$, making $\mathfrak{p}_n$ closed and completing the proof.
Note that $\mathfrak{p}_n$ is in the closure of $\mathfrak{p}_1$ in X : by a topological argument it is easy to show that:

$$\overline{\mathfrak{p}_1} \supseteq \overline{\mathfrak{p}_2} \supseteq \cdots \supseteq \overline{\mathfrak{p}_n} = \mathfrak{p}_n$$

and hence the closure of any non-closed point contains a closed point.

Note that this does not mean that the closed points are dense, that is it does not imply that the closure of the collection of closed point is the entire space. Such schemes are called Jacobson schemes and are related to Jacobson rings and Jacobson radicals (see [9]). Once we introduce (abstract) varieties, we shall see that the closed points are dense for varieties. The following gives a good set of (affine) schemes whose closed points are dense:

Lemma 2.5.4: Dense Closed Points

Let A be a finitely generated k-algebra. Then the closed points of $\operatorname{Spec} A$ are dense

***Proof*:**
It suffices show that for each $f \in A$ where $D(f) \neq \emptyset$ (i.e. f is not nilpotent, see exercise ref:HERE), $D(f)$ $\operatorname{Spec}(A_f) \cong D(f)$, and A_f certainly contains maximal ideals, there is a maximal ideal $\mathfrak{p}_f \subseteq A_f$ which corresponds to a (at least) prime ideal $\mathfrak{p} \subseteq A$ not containing f. If we show that $A/\mathfrak{p}$ is a field, we're done. By Zariski's lemma, F is a finitely generated k-algebra if it is a finitely generated k-module, hence it suffice to show that $A/\mathfrak{p}$ is a finitely generated k-vector space. As $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, if we show that $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space, we can lift to a finite basis for $A/\mathfrak{p}$.

Now, as $A_f = A[1/f]$, it is a finitely generated k-algebra and $\mathfrak{p}_f$ is maximal, $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space by Zariski. But then we get the desired conclusion

We can also show it in the following more "straightforward" with a more subtle application of Nullstellensatz. For the sake of contradiction, suppose $D(f)$ has no maximal ideals. Then for each maximal ideal $\mathfrak{m} \subseteq A$, $f \in \mathfrak{m}$. Thus:

$$f \in \bigcap_{\mathfrak{m} \subseteq A} \mathfrak{m} \stackrel{!}{=} \sqrt{(0)}$$

where $\stackrel{!}{=}$ comes as a consequence of the Nullstellensatz (key here is the finitely generation as a k-algebra). But then f is nilpotent, and so $D(f) = \emptyset$. Hence, any nonempty set contains a closed point, completing the proof.

Recall in proposition 2.3.18 it was mentioned that invertibility can be determined if a function is non-zero at every point, however a function being zero at all points requires more care to study. Part of this is because this is true for quasicompact schemes

Proposition 2.5.5: Quasicompact Schemes and Vanishing Function

Let X be a quasicompact scheme. Then if $f \in \mathcal{O}_X(X)$ vanishes at all points in X, then there exists an n such that $f^n = 0$.

***Proof*:**
Say $f([\mathfrak{p}]) = 0$ so that $f \in \mathfrak{p}$ for all $\mathfrak{p}$. Then on $U_i \cong \operatorname{Spec} R_i \subseteq X, f \in \sqrt{R_i}$, implying $f^{n_i} = 0$ for some n. Cover X with these open sets, and by Quasicompactness we only require finitely many such sets, where in each set we have an f_i and n_i such that $f_i^{n_i} = 0$. As X is a scheme, on intersection $U_i \cap U_j$ we have $g_{i,j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and as ring homomorphisms must map 0 to 0, $g_{i,j}^{\max n_i, n_j} = 0$. Take $n = \max_i n_i$ so that for each intersection $U_i \cap U_j$:

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} = g_{i,j} \quad g_{i,j}^n = 0$$

Now, as these agree on intersections and cover X, by definition there must exist a function $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Then as $\operatorname{res}_{X,U_i}(f^n) = \operatorname{res}_{X,U_i}(f)^n = f_i^n f_i^{n_i n'} = 0 f^{n'} = 0$, we have:

$$(f^n)|_{U_i} = \left(f|_{U_i}\right)^n = 0$$

which implies $(f^n)|_{U_i} = 0 \in R_i$. Then by guability, it must be that $f^n = 0$, completing the proof.

This is not necessarily true for a general scheme, namely since we can have nilpotence of every degree:

Example 2.5.6: Non-vanishing Function On Scheme
Take

$$X = \coprod_{n \geq 1} \operatorname{Spec}\left(k[x]/\left(x^n\right)\right)$$

Then the global function ϵ is zero every-where, namely there is only one point to evaluate to, but essentially by definition there cannot exist an n such that $\epsilon^n = 0$.

Lemma 2.5.7: Projective Scheme is Noetherian

Let $X = \mathbb{P}^n_{\mathbb{Z}}$ or $\mathbb{P}^n_k$. Then X is Noetherian.

***Proof*:**
exercise.

To conclude this section, we shall define one more type of scheme here due to the fact that all affine and projective schemes have this property, however it shall not be of great use to us until chapter ref:HERE. The following establishes some finiteness conditions on the intersection of important open subsets:

Definition 2.5.8: Quasi-separated

Let X be a topological space. Then X is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 2.5.9: Characterizing Quasi-separated Schemes

Let X be scheme. Then X is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets

***Proof*:**
This is very similar to a compactness argument and is left as an exercise

Proposition 2.5.10: Affine Scheme and Quasiseparatedned

Let $X = \operatorname{Spec} R$ be an affine scheme. Then X is quasiseparated.

***Proof*:**
The intersection of affine open subsets is affine (example ref:HERE shows that $U \cap V \cong \operatorname{Spec}(S \otimes_R S')$ for $U, V \subseteq \operatorname{Spec} R$ and $U \cong \operatorname{Spec} S$, $V \cong \operatorname{Spec} S'$. Then by quasi-compactness this can be covered by finitely many distinguished open subsets.

From this, we can reduce to the case of affine opens and the rest follows quickly.

A large family of schemes that are quasi-separated and quasicompact are projective R-schemes, and as we shall see many schemes we shall be working with can be embedded into a projective R-scheme, and hence these two finiteness properties will be very common. There do exist non quasi-separated schemes:

Example 2.5.11: Non-quasiseparated Scheme
Let $X = \operatorname{Spec}(k[x_1, x_2, \ldots])$ and let $U = X \backslash [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \ldots)$. Take two copies of X and glue along U to create a double-origin scheme, namely the infinite version of the double-origin scheme seen in example 2.15. Show this is not quasiseparated.

Lemma 2.5.12: Projective Scheme Quasicompact And Quasiseperated

Let X be a projective R-scheme. Then X is quasicompact and quasiseparated

Note that R need not be Noetherian, hence any projective $\mathbb{P}^n_R$ scheme is quasicompact.

***Proof*:**
hint: affine communication lemma.

Exercise 2.5.1

1. Let X be a scheme. Show that X is a locally Noetherian scheme if and only if every open affine subset corresponds to a Noetherian ring.
2. Show that $\prod_i^\infty \operatorname{Spec} R_i$ properly embeds in $\operatorname{Spec}\left(\prod_i^\infty R_i\right)$. More precisely, the right hand side is (significantly) larger than the left hand side.

2.5.1 Reducedness and Integrality

The next important property that general rings have that coordinate rings for varieties don't have is the presence of zero-divisors and nilpotence. A ring with zero-divisors can be decomposed into a product to two rings and corresponds to a function vanishing on separate irreducible components (there must be more than 1), while a ring with nilpotence have effect on the sheaf of a ring as discussed in section 2.2.2. A ring with no nilpotent elements is said to be *reduced* (note that it may have zero-divisors[26]). A scheme that has this property of being reduced for every ring of sections is given a name:

Definition 2.5.13: Reduced Scheme

Let X be a scheme. Then X is a *reduced scheme* if $\mathcal{O}_X(U)$ is reduced for every open set U of X

Naturally, on a reduced scheme by proposition 2.5.5 a global section that evaluates to zero at every point will be the zero-section.

Proposition 2.5.14: Reducedness Is Stalk-local

Let X be a scheme. Then X is reduced if and only if non of the stalks have nonzero nilpotents. As a consequence, if $f, g \in \mathcal{O}_X(U)$ are two functions and agree on all points, then $f = g$

***Proof*:**
if X is a reduced scheme, then the localization will preserve reducedness, and hence each stalk is reduced. For the converse, we can show the contrapositive and assume there exists a $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Note that f cannot be a unit (as the reader could verify). Then some affine open subset $\operatorname{Spec} R \subseteq U$ we have $f|_U^n = f^n|_U = 0|_U$, showing that f is still nilpotent in $\mathcal{O}_X(\operatorname{Spec} R)$. Then it is a non-unit element in $\operatorname{Spec} R$, and hence belongs in some maximal ideal

[26] if D is the set of all zero-divisors of a reduced ring, then it is the union of all minimal ideals

$f \in \mathfrak{m}$. Then localizing at $[\mathfrak{m}]$, it is easy to see that f remains to be a nilpotent element, showing that not all stalks are reduced.

For the next assertion, this will be a simpler version of proposition 2.5.5 as we do not need to keep track of any exponents, and hence the Quasicompactness condition is not needed! Assume $f, g \in \mathcal{O}_X(U)$ such that $f(x) = g(x)$ for all $x \in U$. We want to show that $f = g$. Take $h = f - g$. Now, by definition $h(x) = f(x) - g(x) = 0$, so h must be contained in every prime ideal on every affine open subset of U. As $\mathcal{O}_X(U)$ is reduced, $\sqrt{(0)} = (0)$ on each affine open, and so $h|_{\operatorname{Spec} R} = 0$ for each element in the affine open covering, which by the gluging gives us $h = 0$, completing the proof.

Lemma 2.5.15: Reducedness For Affine And Projective Schemes

Let R be a reduced ring. Then $\operatorname{Spec}(R)$ is reduced. Hence, $\mathbb{A}^n$ and $\mathbb{P}^n$ are reduced.

***Proof*:**
exercise.

Proposition 2.5.16: Reducedness On Quasicompact Schemes

Let X be a quasicompact scheme. Then X is reduced if and only if it is reduced at closed points

***Proof*:**
Assume it is reduced at all closed points. As X is quasi-compact, the closure of each non-closed point contains a closed point; say $Y \in X$ is a non-closed and $x \in X$ is a closed point contained in its closure. Then $\mathcal{O}_{X,Y}$ is the localization of $\mathcal{O}_{X,x}$. As localization preserves reducedness, $\mathcal{O}_{X,Y}$ must be reduced, completing the proof.

The above showed that reducedness for quasicompact schemes is a "closed-point" local condition. It is not an open set local condition. Furthermore, it is not enough to check the global sections:

Example 2.5.17: Reducedness Edge Case

1. The scheme given by:

$$X = \operatorname{Spec} \frac{\mathbb{C}[x, y_1, y_2, \ldots]}{(y_1^2, y_2^2, \ldots, (x-1)y - 1, (x-2)y_2, \ldots)}$$

   is nonreduced at the closed points corresponding to the positive integers. The complement of this set is not Zariski open.
   The key is that if X is also locally Noetherian, then reducedness is a locally open condition.

2. Another mistake that is easy to make: let X be scheme where the global sections is reduced. This doesn't imply X is reduced (take the scheme cut out by $x^2 = 0$ in $\mathbb{P}^2_k$; note that the global section will be k, but X will be non-reduced).

One of the most important type of reduced ring is an integral domain. If a scheme is an integral domain for each section, it is given a name:

Definition 2.5.18: Integral Scheme

Let X be a scheme. Then X is *integral* if it is nonempty and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset $U \subseteq X$.

Recall that the product of integral domain is not an integral domain, or geometrically, the disjoint union of integral scheme need not be integral. This global behavior cannot be captured through stalks, as the reader should verify, and hence integrality is not stalk-local. Integral schemes can be thought of as schemes that are in "one piece" and have "no fuzz":

Proposition 2.5.19: Integral, then Irreducible and Reduced

Let X be a scheme. Then if X is an integral scheme, it is irreducible and reduced

It is in fact an if and only if, which we shall show in proposition 2.5.25.

***Proof*:**
If X is integral, it is certainly reduced as each ring of local sections is integral (and hence reduced). To show it is irreducible, assume $X = X_1 \cup X_2$, X_1, X_2 closed. For the sake of contradiction, let's say these sets don't contain each other. Then there exists open subsets $U \subseteq X_1, V \subseteq X_2$ such that $U \cap V = \emptyset$. Indeed, given $X_1 \not\subseteq X_2$, then here exists a point $p \in X_1 \setminus X_2$. Let

$$U = (X \setminus X_2) \cap X_1$$

Then U is open in X_1, and hence by construction in X. Similarly, define $V = (X \setminus X_1) \cap X_2$. Then $U \cap V = \emptyset$.

Then, by exercise ref:HERE, we have by an almost direct consequence of the sheaf axioms that:

$$\mathcal{O}_X(U \coprod V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

But then $\mathcal{O}_X(U \coprod V)$ certainly contains zero-divisors, and hence cannot be integral.

An important property that integral domains is the ability to localize to obtain a fractional field. Localizing at an integral schemes generic point shall give us a "global" function field in which all local sections can embed, similarity to varieties! We first prove that an integral scheme has a unique generic point. Note the proof can be done if we replace integrality with irreducibility, that is we don't need every ring of local sections to be integral, see exercise ref:HERE.

Lemma 2.5.20: Irreducible Schemes and Generic Points

Let X be an integral scheme. Then for every irreducible closed subset $Z \subseteq X$, there exists a unique generic point η as $\overline{\eta} = Z$. In particular, as X is irreducible, then there exists a point η such that $\overline{\eta} = X$.

***Proof*:**
Let $Z \subseteq X$ be an irreducible subscheme. Then $Z = \bigcup_i U_i$ is an affine open cover. As $U_i = \operatorname{Spec} R_i$ is open and Z is irreducible, $\overline{U_i} = Z$ (exercise in topology). Now, as X is integral, $[(0_i)]$ is the generic point of U_i as (0_i) is prime. Let $\eta_i = [(0_i)]$. Then:

$$\overline{\eta_i} = Z$$

This shows a generic point exists, let's shows the generic point is unique. Say η_1, η_2 are generic points of Z so that $\overline{\eta_1} = \overline{\eta_2} = Z$. Take any affine open $U_i \subseteq Z$. Then we must have $\eta_1, \eta_2 \in U_i$. If they weren't, then $\eta_i \in Z \setminus U_i$ would be a closed and hence there exists a smaller closed set containing η_i which would imply $\overline{\eta_i} = Z \setminus U_i$, a contradiction. Thus, we get $\eta_1, \eta_2 \in U_i$. Hence, in the subspace topology

$$\overline{\eta_1} = \overline{\eta_2} = U_i$$

Showing that these are also the generic points of U_i. But as U_i is affine and integral, there can only exist a single generic point, hence $\eta_1 = \eta_2$.

In the special case where X is itself irreducible, we can conclude it has a unique generic point η, as we sought to show.

This is a purely topological fact, and hence can also be proven using only irreducibility, see exercise ref:HERE.

The existence was given by reducing to an affine open and "using" its generic point. Using this, we can quickly calculate the stalk at the generic point of an integral scheme. We give this stalk a name:

Definition 2.5.21: Rational Field

Let X be an integral scheme. Then the function field is the collection

$$\mathcal{O}_{X,\eta} = R_{(0)} = \operatorname{Frac}(R)$$

for any R given by $\operatorname{Spec}(R) \subseteq X$, and it is often denoted $\kappa(X)$. If X is a k-scheme, then it is denoted $k(X)$.

Thus, integral schemes have a similar property to that of classical affine and projective varieties, namely that their sections over different open sets can be considered as subsets of a single ring! Gluing is easy as we can see that two elements are glued if and only if they are the same element of $\kappa(X)$.

Proposition 2.5.22: Integral Scheme: Restriction Maps and Function Field

Let X be a scheme that is irreducible and reduced. If $U \subseteq V$ where $V \neq \emptyset$, then

$$\operatorname{res}_{V,U} : \mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U)$$

is an inclusion.

***Proof*:**
As X is reduced, its sections are all determined by their value on the points, and hence can be treated as function (see example 2.5.23 on why this proof fails otherwise). Take $U \subseteq V \subseteq X$, $f, g \in \mathcal{O}_X(U)$ and assume $f|_U = g|_U$. We want to show that $f|_V = g|_V$. Define:

$$Z = \{x \in V \mid f(x) = g(x)\}$$

Z is closed in V as it is the zero set of $f - g$. By definition $U \subseteq Z$, $Z \subseteq V$, and U is open in V. As X is integral, the open subset V is irreducible. But then U is *dense* in V. As Z is closed in V and contains U, it must be that $Z = V$, but then

$$f|_V = g|_V$$

as we sought to show

Note that we require *both* the irreducibility and the reducedness, having *only* irreducibility is not enough

Example 2.5.23: Non-injective Non-reduced Scheme

Let $R = \frac{k[x,y]}{(xy, y^2)}$ so that $\operatorname{Spec} R$ is the x axis there is a bit of fluff in direction of the y-axis at the point 0. Then given $D(x) \subseteq \operatorname{Spec} R$, the map $\text{res} : \mathcal{O}_{\operatorname{Spec} R}(\operatorname{Spec} R) \to \mathcal{O}_{\operatorname{Spec} R}(D(x))$ maps $y \mapsto 0$ as

$$y \mapsto \frac{y}{1} = \frac{xy}{x} = \frac{0}{x} = 0$$

Corollary 2.5.24: Integral Scheme: Generic Point

Let X be an integral scheme. Then there exists a generic point η such that given any choice of $\operatorname{Spec}(R) \subseteq X$, there is always an inclusion:

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,\eta} \cong \kappa(R)$$

***Proof*:**
By definition:

$$\mathcal{O}_{X,\eta} = \varinjlim_{\eta \in U} \mathcal{O}_X(U)$$

and as we saw lemma 2.5.20 the generic point is in each open set. By proposition 2.5.22, each map in the commutative diagram are injective, hence the map $\mathcal{O}_X(U)$ is injective into $\mathcal{O}_{X,\eta}$.

Proposition 2.5.25: Irreducible and Reduced, then Integral

Let X be a scheme. If X is irreducible and reduced, it is integral

***Proof*:**
Let X be irreducible and reduced. Then any open subscheme is irreducible and reduced (the irreducibility is proved topologically, and the reducedness can be deduced by the stalks). In

particular, any affine open subscheme $\operatorname{Spec} R$ must be irreducible and reduced. Certainly R must then be reduced, hence to show that $\operatorname{Spec} R$ is integral consider $xy \in R$ such that $xy = 0$. Then

$$\operatorname{Spec} R = V(0) = V(xy) = V(x) \cup V(y)$$

But that contradicts irreducedness. Hence, all ring of sections of affine open subsets are integral. For general open $U \subseteq X$, we must have an affine open $\operatorname{Spec} R \subseteq U$, and hence a restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(\operatorname{Spec} R)$. By proposition 2.5.22, this map is injective, and hence $\mathcal{O}_X(U)$ is a subring of an integral domain, making it an integral domain. Hence all rings of local sections are integral domains, completing the proof.

If we have the added condition of being Noetherian, then we only need to check that it is connected and that all its stalks are integral domains:

Proposition 2.5.26: Characterizing Integrality in Noetherian Schemes

Let X be a Noetherian Scheme. Then X is integral if and only if X is nonempty, connected, and all stalks $\mathcal{O}_{X,\mathfrak{p}}$ are integral domains.

Intuitively, the stalk at the intersection of irreducible components will "remember" the generic points of the respective irreducible component.

***Proof*:**
By proposition 2.5.19 and proposition 2.5.25, X is integral if and only if it is reduced and irreducible. Then X is certainly non-empty, connected, and the stalks are integral (the localization of integral domains is integral).

Conversely, assume X is nonempty, connected, with integral stalks. First of all, as the stalks are integral, they are reduced. We must show that X is irreducible. As X is Noetherian, by lemma 2.5.2 X can be decomposed into finitely many irreducible components:

$$X = X_1 \cup X_2 \cup \cdots \cup X_n$$

As X is connected, for each i, j we must have $X_i \cap X_j \neq \emptyset$. Take an affine open subset $\operatorname{Spec} R \subseteq X$ and:

$$\operatorname{Spec} R = \bigcup_{i}^{n} (X_i \cap \operatorname{Spec} R)$$

Then as each X_i is irreducible, it is a topological argument to see that the sets $(X_i \cap \operatorname{Spec} R)$ form an irreducible decomposition of $\operatorname{Spec} R$. Hence, they must each have a generic point that corresponds to a minimal prime ideal. Hence, if the X_i are distinct, R_i must contains at least two distinct minimal prime ideals. Then if $[\mathfrak{p}] \in X_i \cap X_j$, $\mathcal{O}_{X,[\mathfrak{p}]} = (R_i)_{\mathfrak{p}}$ contains two distinct minimal prime ideals. But integral domains could only contain a single minimal prime ideal (namely the one associated to the generic point $[(0)]$), and so this stalk cannot be integral. Thus, it must be that $X_i = X_j$ for all i, j, showing that X is irreducible and completing the proof.

Exercise 2.5.2

1. Show that every point $x \in X$ in a topological space must be contains in an irreducible component (you will use Zorn's Lemma).

2. Show that connected components (maximal connected subset) are closed. Show that the connected components of Spec $(\prod_i^\infty \mathbb{F}_2)$ are not open.

3. Show that $D(f)$ is empty if and only if f is nilpotent. Show that if X is an integral scheme, then $D(f)$ is always nonempty and contains the generic point.

4. If $f \in R$ is nilpotent, then R_f is the zero-ring.

5. Let X be an irreducible scheme. Show that there is a unique generic point η such that $\overline{\eta} = X$ (note that lack of reducedness). More generally, show that if $Z \subseteq X$ is a (nonempty) irreducible subset, there is a unique generic point η such that $\overline{\eta} = Z$

6. Let X be a reduced scheme and $Y \hookrightarrow X$ a closed immersion where $\iota(Y) = X$. Then ι is an isomorphism.

2.5.2 Normality and Factoriality

Recall that a normal ring R has the property that it an integral domain that is is integrally closed in it's field of fractions, $\overline{R}^{\mathrm{Frac}(R)} = R$. Due to their importance in number theory, we shall take the time to develop the theory of Schemes for which all the rings of sections are normal. These shall correspond to schemes that have some nicer behaviour for their singularities. It can be thought of as a condition weaker than smoothness (smoothness shall be called regular in section REF:HERE) as normality shall still allow for singularities, however these singularities shall be more tame. In particular, the reader can intuitively have the following geometric picture in mind for subschemes of Normal Schemes:

1. In codimension 1, all singularities are resolved, meaning it is "smooth" in codim 1

2. In codimension ≥ 2, being normal means being nice enough so that rational functions defined on the compliment of the singularities can be extended to the singularity (i.e. Algebraic Hartogs's lemma applies)

Definition 2.5.27: Normal Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is normal domains.

Proposition 2.5.28: Normal Scheme Stalk Local

Let X be a scheme. Then X is a normal scheme if and only if $\mathcal{O}_{X,\mathfrak{p}}$ is normal for each $\mathfrak{p} \in X$

***Proof*:**
Recall (or reprove) that normality is preserved when localizing (hint: recall the proof that a UFD is normal)

Note that reducedness was stalk-local, and so it is immediate that normal schemes are reduced. As being integrally closed is well-behaved under localization (recall the proof that a UFD is normal), for an if R is normal then Spec (R) is a normal scheme. If each stalk is a UFD, then it is immediately a Normal scheme. Let's give a name to such a scheme

Definition 2.5.29: Factorial Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is a UFD.

If R is a UFD, then $\operatorname{Spec} R$ is factorial (by the above reasoning), however the converse need be true! Namely, we may have factorial schemes where the stalks are all UFD's but the global sections do not form a UFD. A quintessential example will be the spectra of elliptic curves.

Example 2.5.30: Normal Schemes

1. The schemes $\mathbb{A}^n_k$, $\mathbb{P}^n_k$ and $\operatorname{Spec} \mathbb{Z}$ are normal. If R is normal, by definition $\operatorname{Spec} R$ is normal.
2. Starting on p.164 section 5.4.8, lots and lots of good examples

2.6 Associated Points

Associated points can be thought of as the points that capture the components structure of a scheme and how it comes together at intersection points. These points include points that come from schemes $k[x,y]/(y^2, xy)$ that "remember" that $V(xy)$ would be two components if it were not for the "fuzz factor" of y^2.

Example 2.6.1: Support Of Non-reduced Ring

Let $X = \operatorname{Spec}(k[x,y]/(y^2, xy))$. Show that the support of any function X is either empty, the origin, or all of X. Note that the function y evaluated to 0 everywhere on X, but the support is nonzero at the origin!

This will be linked to the irreducible components of the above X, namely the x-axis and the origin. This gives the following definition tha shall be expanded upon:

Definition 2.6.2: Associated Points

Let X be a scheme. Then the associated points are precisely the generic points of irreducible components of the support of some element of X

Proposition 2.6.3: Affine Integral Schemes And Associated Points

Let R be an integral domain. Then the generic point is the only associated point of $\operatorname{Spec} R$.

***Proof*:**
Certainly the generic point is an associated point as it is associate to the support of the function 1 on the irreducible component $\operatorname{Spec} R$ (i.e. the entire space itself). We must show the nonzero support of all functions (besides 0) is all of $\operatorname{Spec} R$. Let $f \in R$, and let $f_{\mathfrak{p}} \in R_{\mathfrak{p}}$ be the associated localization at the point $[\mathfrak{p}]$. If $f_{\mathfrak{p}} = 0_{\mathfrak{p}}$, then there must be a $g \in R \setminus \mathfrak{p}$ such that $f/1 = fg/g = 0/g = 0$. But R is an integral domain, hence this is not possible.

Finish this here, p.167

Exercise 2.6.1

1. Find the structure for the following rings:
   a) $\operatorname{Spec} \mathbb{C}[x]/(x^3 - x^2)$
   b) $\operatorname{Spec} \mathbb{C}[x]/(x^2 - d)$
   c) $\operatorname{Spec} \mathbb{C}[x]/(x^2 + 1)$
2. Let X be a quasicompact scheme. Show that the closure of each point contains a closed point (this is not true for non quasicompact schemes in general)
3. Let X be a quasicompact scheme. Suppose f vanishes at all point of X. Then there exists some N st $f^n = 0$. Show that this fails when X is not quasicompact (hint: take infinite disjoint unions of $\operatorname{Spec}(R_n)$ where $R_n := k[x]/(x^n)$)
4. Let X be a quasicompact scheme. Then it suffices to cheek reducedness at closed points (show that any nonreduced point has a nonreduced closed point in its closure)
5. Show that reducedness is not in general an open condition by considering:
$$X = \operatorname{Spec}\left(\frac{\mathbb{C}[x, y_1, y_2, ...,}{(y_1^2, y_2^2, y_3^3, ..., (x-1)y_1, (x-2)y_2, (x-3)y_3), ...}\right)$$
and $Y = \operatorname{Spec}[x]$. Show the nonreduced points of X are the closed points corresponding to the positive integers. The compliment of this set is *not* Zariski open.
6. Let X be an integral scheme. Show that if $\emptyset \neq V \subseteq U$, then the map $\operatorname{res}_{U,V} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is an inclusion. very important, see p.156 under exercise
7. Let X be an integral scheme, γ the generic point, and $\operatorname{Spec}(R) \cong U \subseteq X$ is any nonempty affine open subset of X. Then $\mathcal{O}_X(U) \to \mathcal{O}_{X,\gamma} = K(A)$ is an inclusion. Conclude that gluing is easy: every f_i in U_i for an open cover for U glue together if they are same element in $\operatorname{Frac}(X)$ (recall that $\operatorname{Frac}(X)$ is defined to be $\operatorname{Frac}(R)$ where R is given by $\mathcal{O}_X(\operatorname{Spec}(R))$
8. Show that integrality is not stalk local via the example $\operatorname{Spec}(A) \coprod \operatorname{Spec}(B) = \operatorname{Spec}(A \times B)$.
9. Show that reducedness is an affine local property.
10. Let X be a locally Noetherian Scheme. Then X is quasiseparated.
11. Let X be a Noetherian Scheme. Then it has a finite number of connected components, each a finite union of irreducible components.
12. Show that the ring of sections of a Noetherian Scheme need not be Noetherian.
13. Let X be a scheme of locally finite type over $\mathbb{Z}$ or k. Then X is locally Noetherian.
14. A scheme that is of finite type over any Noetherian ring is Neotherian.
15. Let X be a quasiprojective R-scheme. Then if R is Noetherian, X is a Noetherian scheme, and hence has a finite number of irreducible components.
16. Let X be a locally Noetherian scheme. Then the reduced locus is open. (Vakil p.156)
17. Let $U \subseteq X$ be an open subscheme of a projective R-scheme X. Show that U is locally of finite type R

18. Find a counter-example showing that normal schemes are not in general integral schemes (hint: think of the disjoint union of two normal scheme. For more hint, Vakil p. 162).

19. Let X be a Noetherian scheme. Then X is normal if and only if if is a finite disjoint union of integral normal schemes.

2.7 Dimension

The reader may want to review [9, section 20] for an in-depth look at the algebraic counterpart. Recall that the *Krull dimension* of a commutative ring R is the supremum of the lengths of proper chains of prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

with indexing starting at 0. Geometrically, for $\operatorname{Spec} R$ and topological spaces more generally (including schemes), the *dimension* is the supremum of lengths of proper chains of closed irreducible subsets with indexing starting at 0. Since the closed irreducible subsets of $\operatorname{Spec} R$ are exactly the sets $V(\mathfrak{p})$ for $\mathfrak{p}$ prime, we have $\dim \operatorname{Spec} R = \operatorname{kdim} R$. Furthermore:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} (R/\sqrt{0})$$

since $\operatorname{Spec} R$ and $\operatorname{Spec} (R/\sqrt{0})$ are homeomorphic (quotienting by the nilradical does not change the prime ideals).

If the space is not irreducible, then the dimension per component can vary (for example $\mathbb{A}^1 \cup \mathbb{A}^2 \subseteq \mathbb{A}^3$, interpreted as the x-axis union the yz-plane). Hence we usually work with irreducible schemes. If necessary, we say a scheme has *pure dimension* n (or is *equidimensional*) if all of its irreducible components have the same dimension n. Dimension respects inclusions: if $X \subseteq Y$ then $\dim X \leq \dim Y$.

Dimension	**Name**
0	point (or collection of points)
1	curve
2	surface
$n \geq 3$	n-fold

Lemma 2.7.1: Dimension of Open Subset

Let $U \subseteq X$ be an open subset of a topological space. Then there is a bijection between irreducible closed subsets of U and the irreducible closed subsets of X that meet U, given by $Z \mapsto \overline{Z}$ (closure in X) with inverse $W \mapsto W \cap U$.

***Proof*:**
If $Z \subseteq U$ is irreducible and closed in U, then $\overline{Z}$ is irreducible and closed in X with $\overline{Z} \cap U = Z$ (since Z is closed in U, it equals the intersection of its closure with U). Conversely, if $W \subseteq X$ is irreducible closed with $W \cap U \neq \emptyset$, then $W \cap U$ is irreducible (an open subset of an irreducible set is irreducible) and closed in U, with $\overline{W \cap U} = W$ (since $W \cap U$ is dense in W). These operations are mutually inverse.

Proposition 2.7.2: Dimension of Scheme

Let X be a scheme. Then $\dim X = n$ if and only if X admits an open cover by affine open subsets of dimension at most n, where equality is achieved by at least one affine open.

***Proof*:**
If $\dim X = n$, then each open subset in an affine cover has dimension $\leq n$ (since irreducible closed subsets of an open set correspond to those of X meeting it, by lemma 2.7.1).

Conversely, suppose every affine open in some cover has dimension $\leq n$ and at least one has dimension exactly n. We must show $\dim X = n$. The dimension is at least n since some affine open has dimension n, and a chain in U_i lifts to a chain in X via lemma 2.7.1. For the reverse inequality, suppose $\dim X > n$. Then there exists a chain of irreducible closed subsets in X:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_{n+1}$$

Since Z_0 is nonempty, it meets some affine open U_i in the cover. Then each Z_j also meets U_i (since $Z_0 \subseteq Z_j$ and $Z_0 \cap U_i \neq \emptyset$). By lemma 2.7.1, the chain:

$$Z_0 \cap U_i \subsetneq Z_1 \cap U_i \subsetneq \cdots \subsetneq Z_{n+1} \cap U_i$$

consists of distinct irreducible closed subsets of U_i (since the closure map is a bijection). This gives $\dim U_i \geq n + 1 > n$, contradicting our assumption.

Lemma 2.7.3: Integral Extensions and Dimension

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ correspond to an integral extension $S \hookrightarrow R$. Then:

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} S$$

***Proof*:**
The going-up theorem gives $\dim \operatorname{Spec} R \geq \dim \operatorname{Spec} S$ (any chain in S lifts to a chain in R), and the incomparability theorem gives $\dim \operatorname{Spec} R \leq \dim \operatorname{Spec} S$ (primes lying over the same prime cannot be comparable). See [9, chapter 17] for the details.

2.7.4 A word on Noetherian Schemes It is *not* the case that a Noetherian scheme must be finite-dimensional: Nagata constructed a Noetherian ring of infinite Krull dimension (see Vakil, chapter 11).
However, Noetherian *local* rings are always finite-dimensional (by [9, corollary 20.42]), and so points of locally Noetherian schemes always have finite codimension.

2.7.1 Codimension

We are often concerned with subspaces that are a few dimensions lower than the ambient scheme X. Since schemes can be unions of irreducible components of different dimension, we define codimension only for irreducible subschemes:

Definition 2.7.5: Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\begin{aligned}&\operatorname{codim}_X Y\\ &= \sup\{n : \text{there exists a chain } Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \text{ of irreducible closed subsets of } X\}\end{aligned}$$

For a scheme X and a point $x \in X$, the codimension of $\overline{\{x\}}$ in X equals the Krull dimension of the local ring $\mathcal{O}_{X,x}$.

For the affine case: if $\mathfrak{p} \in \operatorname{Spec} R$, then $\operatorname{codim}_{\operatorname{Spec} R} V(\mathfrak{p})$ equals the *height* of $\mathfrak{p}$, i.e. the supremum of lengths of chains:

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_h = \mathfrak{p}$$

and this equals $\mathrm{kdim} R_{\mathfrak{p}}$. For example, in $\operatorname{Spec} \mathbb{Z}$, the generic point $\eta = [(0)]$ has codimension 0, while any closed point $[(p)]$ has codimension 1.

If Y has codimension 0 in X, then Y must be an irreducible component of X.

Definition 2.7.6: Hypersurface

Let $Y \subseteq X$ be an irreducible closed subscheme. Then Y is called a *hypersurface* if $\operatorname{codim}_X Y = 1$.

Lemma 2.7.7: Sum of Dimension and Codimension

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. Then:

$$\operatorname{codim}_X Y + \dim Y \leq \dim X$$

***Proof*:**
A chain of length c descending from Y and a chain of length d ascending from Y can be concatenated to give a chain of length $c + d$ in X. Taking suprema gives $\operatorname{codim}_X Y + \dim Y \leq \dim X$.

Equality holds for irreducible varieties of finite type over a field, but *not* in general:

Example 2.7.8: Codimension Pathology

Consider the scheme $X = \operatorname{Spec}(k[x]_{(x)}[t])$, which can be thought of as a surface with a line attached at one point. Then $\dim X = 2$, but the closed point $p = (x, t)$ has dimension 0 and codimension 1 (not 2!), since the maximal chain from p is:

$$(0) \subsetneq (x) \subsetneq (x, t)$$

but (x) already has codimension 1 and dimension 1 (the "line" component), while the "surface" component (0) has dimension 2. Note too that this scheme has open subsets of *different dimension* – a phenomenon that does *not* occur for open subsets of n-manifolds.

The pathology above is excluded by the following class of rings:

Definition 2.7.9: Catenary Ring

A commutative ring R is *catenary* if for every pair of nested prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$, all maximal chains of prime ideals between $\mathfrak{q}$ and $\mathfrak{p}$ have the same length.

Many rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains, Cohen–Macaulay rings, and in particular all localizations of finitely generated k-algebras (see exercise 2.7.1).

2.7.2 Transcendence Dimension

Working with Krull dimension directly can be difficult. For finitely generated algebras over a field, we have a more computable invariant:

Definition 2.7.10: Transcendence Dimension

Let A be an integral domain that is a k-algebra. Then the *transcendence dimension* of A over k is:

$$\operatorname{trdeg}_k A := \operatorname{trdeg}_k \operatorname{Frac}(A)$$

i.e. the transcendence degree of the fraction field of A over k.

Theorem 2.7.11: Transcendence and Krull Dimension

Let A be a finitely generated integral domain over a field k. Then:

$$\dim \operatorname{Spec} A = \operatorname{trdeg}_k \operatorname{Frac}(A)$$

Thus, if X is an irreducible k-variety with function field $k(X)$, then $\dim X = \operatorname{trdeg}_k k(X)$.

***Proof*:**
See [9, chapter 20] for the algebraic proof. The key steps are: Noether normalization gives a finite (hence integral) extension $k[y_1, \ldots, y_d] \hookrightarrow A$ with $d = \operatorname{trdeg}_k \operatorname{Frac}(A)$; by lemma 2.7.3, $\dim \operatorname{Spec} A = \dim \mathbb{A}^d_k = d$.

For catenary rings, dimension and codimension add up:

Theorem 2.7.12: Dimension Formula for Finite Type k-Schemes

Let X be an irreducible scheme locally of finite type over a field k, $Y \subseteq X$ an irreducible closed subset, and η the generic point of Y. Then:

$$\operatorname{codim}_X Y + \dim Y = \dim X$$

equivalently:

$$\dim \mathcal{O}_{X,\eta} + \dim Y = \dim X$$

***Proof*:**
The question is local, so we may reduce to $X = \operatorname{Spec} A$ for a finitely generated k-domain A. Let $\mathfrak{p}$ be the prime corresponding to η. Then $\operatorname{codim}_X Y = \operatorname{kdim} A_\mathfrak{p}$ and $\dim Y = \operatorname{kdim} A/\mathfrak{p}$, so we must show $\operatorname{kdim} A_\mathfrak{p} + \operatorname{kdim} A/\mathfrak{p} = \operatorname{kdim} A$. This follows from the fact that finitely generated k-algebras are catenary and equidimensional (see [9, chapter 20] or Vakil, chapter 11 for a guided exercise).

2.7.3 Krull's Height Theorem and Algebraic Hartogs' Lemma

We now turn to two fundamental results about codimension-1 phenomena. Krull's theorem gives an upper bound on codimension in terms of the number of defining equations, while the algebraic Hartogs' lemma characterizes regular functions on normal schemes in terms of codimension-1 behavior. Both will be used extensively in the theory of divisors (section 5.3).

Krull's Height Theorem

The following theorem, proved in [9, corollary 20.47], is one of the most important tools for computing codimension:

Theorem 2.7.13: Krull's Height Theorem (Hauptidealsatz)

Let R be a Noetherian ring. If $\mathfrak{a} = (f_1, \ldots, f_n)$ is a proper ideal generated by n elements, then every minimal prime $\mathfrak{p}$ over $\mathfrak{a}$ has:

$$\operatorname{height}(\mathfrak{p}) \leq n$$

In particular, if $f \in R$ is neither a zero-divisor nor a unit, then every minimal prime over (f) has height exactly 1.

***Proof*:**
See [9, section 20.5]. The proof uses the dimension theorem for Noetherian local rings: localizing at $\mathfrak{p}$, the ideal $\mathfrak{a}_\mathfrak{p}$ is $\mathfrak{p}_\mathfrak{p}$-primary and generated by n elements, so $\operatorname{kdim} R_\mathfrak{p} \leq n$ by the characterization of dimension as the minimal number of generators of a parameter ideal.

Geometrically, Krull's theorem says: in a Noetherian scheme, the vanishing locus of n functions has all irreducible components of codimension $\leq n$. In particular:

Corollary 2.7.14: Principal Ideal Theorem – Geometric Form

Let X be a locally Noetherian scheme and $f \in \mathcal{O}_X(X)$ a regular function that is not a zero-divisor on any irreducible component. Then every irreducible component of $V(f)$ has codimension exactly 1 in X.

Corollary 2.7.15: Codimension Bound for Complete Intersections

Let X be a locally Noetherian scheme and $Z = V(f_1, \ldots, f_n) \subseteq X$. Then every irreducible component of Z has codimension $\leq n$. If equality holds for every component, Z is called a *(set-theoretic) complete intersection.*

Example 2.7.16: Krull's Theorem in Practice

1. In $\mathbb{A}^3_k = \operatorname{Spec} k[x, y, z]$, the ideal (xy, xz) defines the vanishing locus $V(x) \cup V(y, z)$, which is the yz-plane $\cup$ the x-axis. The yz-plane has codimension 1, while the x-axis has codimension 2. Krull's theorem guarantees codimension ≤ 2, and both bounds are achieved.
2. The ideal $(x^2 + y^2 + z^2, x + y + z) \subseteq k[x, y, z]$ defines a curve in $\mathbb{A}^3_k$. By Krull's theorem, each component has codimension ≤ 2. In this case, the curve has codimension exactly 2, so it is a complete intersection.
3. The twisted cubic curve $C \subseteq \mathbb{A}^3_k$ (the image of $t \mapsto (t, t^2, t^3)$) has codimension 2 but ideal $I(C) = (y - x^2, z - x^3)$, which requires 2 generators. However, in $\mathbb{P}^3_k$ the twisted cubic requires 3 generators of its homogeneous ideal (it is *not* a complete intersection).

The converse direction of Krull's theorem also holds and is equally useful:

Proposition 2.7.17: Converse to Krull's Theorem

Let R be a Noetherian ring and $\mathfrak{p}$ a prime of height n. Then there exist $f_1, \ldots, f_n \in \mathfrak{p}$ such that $\mathfrak{p}$ is a minimal prime over $(f_1, \ldots, f_n)$.

***Proof*:**
See [9, proposition 20.48]. One constructs the f_i inductively using prime avoidance: at each step, choose f_i in $\mathfrak{p}$ but outside all minimal primes of $(f_1, \ldots, f_{i-1})$ of height $< i$.

Algebraic Hartogs' Lemma

In complex analysis, Hartogs' extension theorem states that a holomorphic function defined on the complement of a codimension-≥ 2 set extends uniquely to the ambient space. There is a beautiful algebraic analogue for normal schemes:

Theorem 2.7.18: Algebraic Hartogs' Lemma

Let A be a Noetherian normal domain. Then:

$$A = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ \operatorname{height}(\mathfrak{p}) = 1}} A_{\mathfrak{p}}$$

where the intersection takes place inside $\operatorname{Frac}(A)$.

In geometric language: a rational function on a Noetherian normal scheme with no poles along any codimension-1 subvariety is in fact a regular function. Equivalently, regular functions on a normal scheme are determined by their behavior at codimension-1 points – you cannot have a function that is "only irregular in codimension ≥ 2."

***Proof*:**
The inclusion $A \subseteq \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_\mathfrak{p}$ is clear (any element of A lies in every localization). For the reverse, suppose $x \in \mathrm{Frac}(A)$ lies in $A_\mathfrak{p}$ for every height-1 prime $\mathfrak{p}$. Let $I = \{a \in A \mid ax \in A\}$ be the ideal of "denominators" of x. If $x \notin A$, then $I \neq A$. Let $\mathfrak{q}$ be a minimal prime over I.

Claim: height$(\mathfrak{q}) \leq 1$.

If height$(\mathfrak{q}) = 0$, this is fine. If height$(\mathfrak{q}) \geq 2$, then $x \in A_\mathfrak{p}$ for all height-1 primes $\mathfrak{p} \subseteq \mathfrak{q}$, and since $A_\mathfrak{q}$ is a normal local domain, $A_\mathfrak{q} = \bigcap_{\mathrm{ht}(\mathfrak{p})=1,\, \mathfrak{p} \subseteq \mathfrak{q}} (A_\mathfrak{q})_{\mathfrak{p}A_\mathfrak{q}}$ (applying the result in the local case, where it holds by the characterization of normal domains as S_2 and R_1). This gives $x \in A_\mathfrak{q}$, contradicting $I \subseteq \mathfrak{q}$. So height$(\mathfrak{q}) = 1$.

But then $x \in A_\mathfrak{q}$ by hypothesis, so there exist $a \in A$, $s \notin \mathfrak{q}$ with $x = a/s$, giving $sx = a \in A$ and hence $s \in I$. But $I \subseteq \mathfrak{q}$ and $s \notin \mathfrak{q}$, a contradiction. Therefore $I = A$ and $x \in A$.

Example 2.7.19: Algebraic Hartogs in Practice

1. Consider $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$ and the open set $U = D(x) \cup D(y) = \mathbb{A}^2 \setminus \{(0,0)\}$. We showed in example 2.2.2 that $\mathcal{O}_{\mathbb{A}^2}(U) = k[x,y]$. This is now immediate from algebraic Hartogs: the complement of U is $\{(0,0)\}$, which has codimension 2, and $k[x,y]$ is a normal domain. Hence any regular function on U extends to all of $\mathbb{A}^2$.
2. The same argument fails for $\mathbb{A}^1$: removing a point from $\mathbb{A}^1$ removes a codimension-1 subset, and indeed $\mathcal{O}_{\mathbb{A}^1}(D(x)) = k[x, 1/x] \neq k[x]$.
3. Let $A = k[x,y,z]/(x^2 + y^2 - z^2)$, the quadric cone, which is a normal domain (exercise). The singular point at the origin has codimension 2, so algebraic Hartogs implies that regular functions on the smooth locus extend to all of $\operatorname{Spec} A$.
4. The algebraic Hartogs' lemma *fails* without normality. Consider the nodal cubic $A = k[x,y]/(y^2 - x^2(x+1))$, which is not normal at the origin. The element y/x in the fraction field is regular at all codimension-1 points away from the node, yet $y/x \notin A$.

The connection between these two results is deep: Krull's theorem says that the zero locus of a single function has codimension ≤ 1, while Hartogs' lemma says that on normal schemes, a rational function is determined by its poles, which also occur in codimension 1. Both point toward the centrality of codimension-1 phenomena, which will be formalized in the theory of *Weil and Cartier divisors* in section 5.3.

Exercise 2.7.1

1. Show that a Noetherian scheme of dimension 0 has finitely many points.
2. Let $f : X \to Y$ be an integral morphism. Show that $\dim f^{-1}(y) = 0$ for all $y \in Y$ (i.e., the fibers are zero-dimensional).

3. If $\nu : \widetilde{X} \to X$ is the normalization of a scheme, show that $\dim \widetilde{X} = \dim X$.
4. If K/k is an algebraic field extension and X is a k-scheme of finite type, show that $\dim X = \dim X_K$ where $X_K = X \times_k K$ is the base change.
5. Generalize the above result: if K/k is an arbitrary field extension and X is an irreducible k-scheme of finite type, show that $\dim X_K = \dim X$.
6. Show the following rings are catenary: localizations of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains.
7. Let X be an integral scheme locally of finite type over a field k, and let $U \subseteq X$ be a nonempty open subset. Show that $\dim U = \dim X$.
8. (*Hartogs' lemma for schemes*) Let X be a normal Noetherian integral scheme, and let $U \subseteq X$ be an open subset whose complement has codimension ≥ 2. Show that the restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(U)$ is an isomorphism. (Hint: reduce to the affine case and apply theorem 2.7.18.)
9. Show that the quadric cone $\operatorname{Spec} k[x, y, z]/(x^2 + y^2 - z^2)$ is a normal domain ($\operatorname{char} k \neq 2$). (Hint: show it is R_1 and S_2, or use the Jacobian criterion to show regularity in codimension 1 and then invoke Serre's criterion.)

2.8 Regular and Smooth Schemes

The notion of smoothness is central to both differential and algebraic geometry. In differential geometry, a manifold is smooth if it locally looks like $\mathbb{R}^n$. In algebraic geometry, the analogous condition is that the local ring at each point behaves like a polynomial ring – this is the notion of *regularity*. Over perfect fields (in particular, algebraically closed fields of any characteristic), regularity and smoothness coincide; over imperfect fields, smoothness is the stronger and more geometrically natural condition.

The algebraic foundations for this section – Hilbert functions, the dimension theorem, regular local rings, and their characterization via homological methods – were developed in [9, sections 20.5–20.7]. We recall the key definitions and build on them.

Definition 2.8.1: Tangent Space

Let X be a scheme and $x \in X$ a point. Then the *(Zariski) cotangent space* at x is the $\kappa(x)$-vector space:

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the stalk at x, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is the residue field. The *(Zariski) tangent space* is its dual:

$$T_xX := (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \operatorname{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2,\, \kappa(x))$$

Definition 2.8.2: Regular Local Ring

Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then R is *regular* if:

$$\operatorname{kdim} R = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

By [9, corollary 20.44], we always have $\operatorname{kdim} R \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$, so regularity asks for equality.

Definition 2.8.3: Regular Scheme

A scheme X is *regular* if $\mathcal{O}_{X,x}$ is a regular local ring for every point $x \in X$. A scheme is *singular* at a point x if $\mathcal{O}_{X,x}$ is not regular.

Example 2.8.4: Regular and Singular Schemes

1. *$\mathbb{A}^2_k$ is regular.* For a closed point $\mathfrak{m} = (x-a, y-b)$, first observe that $\kappa(\mathfrak{m}) = k$ (since k is algebraically closed). The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is spanned by the images of $x-a$ and $y-b$, so $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$. The local ring $k[x,y]_{\mathfrak{m}}$ has a maximal chain $0 \subsetneq (x-a) \subsetneq (x-a, y-b)$, giving $\operatorname{kdim} = 2$.

   Since we are working over schemes, we also have higher-dimensional points. Consider $[\mathfrak{p}] = [(x^2 - y)] \in \mathbb{A}^2_k$. The local ring $k[x,y]_{\mathfrak{p}}$ has dimension 1 (one can show the chain $0 \subsetneq (x^2 - y)$ is maximal), and $\mathfrak{m}_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}^2$ is a 1-dimensional $\kappa(\mathfrak{p})$-vector space. Hence this point is regular.

   An important observation: $[(x^3 - y^2)] \in \mathbb{A}^2_k$ is *also* a regular point of $\mathbb{A}^2_k$, even though $x^3 - y^2$ defines a singular curve! This is because regularity at a generic point of $\mathbb{A}^2_k$ concerns the local ring of the *ambient* scheme, not the subscheme $V(x^3 - y^2)$. The point $[(x^3 - y^2)]$ is "generically regular" in $\mathbb{A}^2_k$ – further motivating the name *generic* point.

2. *The cuspidal cubic is singular.* Let $X = \operatorname{Spec} k[x,y]/(x^3 - y^2)$. Then X is not regular at the origin: the local ring $\mathcal{O}_{X,(x,y)}$ has $\operatorname{kdim} = 1$ (since X is an irreducible curve), but the cotangent space $\mathfrak{m}/\mathfrak{m}^2$ at the origin has $\dim_k = 2$, since neither $\overline{x}$ nor $\overline{y}$ is redundant modulo $\mathfrak{m}^2$ (the relation $x^3 = y^2$ lies in $\mathfrak{m}^3$, not $\mathfrak{m}^2$). On the other hand, X is regular at every other closed point (exercise).

3. *Normal does not imply regular.* If X is a normal scheme, then its stalks are integrally closed, and all codimension-1 points are regular (by [9, proposition 20.53]). However, normality does *not* imply regularity in dimension ≥ 2. For example, $\operatorname{Spec} k[x,y,z]/(x^2 + y^2 - z^2)$ is normal but not regular at the origin (exercise: show $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 3$ while $\operatorname{kdim} = 2$).

   However, for a 1-dimensional connected Noetherian scheme, normality *does* imply regularity: the codimension-1 points are exactly the closed points, and the generic point is always regular. In fact, a 1-dimensional normal local domain is a DVR by [9, corollary 20.55], which is a regular local ring of dimension 1.

The singular locus of a scheme is typically "small." For varieties over algebraically closed fields, this was shown in [9, proposition 20.57]: the set of singular points forms a proper closed subset.

For schemes of finite type over a field, regularity can be checked via partial derivatives, generalizing the classical criterion from [9, proposition 20.56]:

Proposition 2.8.5: Jacobian Criterion for Schemes

Let k be a field and $X = \operatorname{Spec}(k[x_1, \ldots, x_n]/(f_1, \ldots, f_m))$ a closed subscheme of $\mathbb{A}^n_k$. Let $x \in X$ be a closed point with residue field $\kappa(x)$. Then X is regular at x if and only if:

$$\operatorname{rank}\left(\frac{\partial f_i}{\partial x_j}(x)\right) = n - \dim \mathcal{O}_{X,x}$$

where the partial derivatives are evaluated in $\kappa(x)$.

***Proof*:**
This is the scheme-theoretic translation of [9, proposition 20.56]. The key computation is that the cotangent space satisfies:

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \cong \frac{\mathfrak{m}_{\mathbb{A}^n,x}}{(\mathfrak{m}_{\mathbb{A}^n,x}^2 + (f_1, \ldots, f_m))}$$

and so $\dim_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = n - \operatorname{rank}(J_x)$. Regularity is the condition that this equals $\dim \mathcal{O}_{X,x}$, which is equivalent to $\operatorname{rank}(J_x) = n - \dim \mathcal{O}_{X,x}$.

Example 2.8.6: Jacobian Criterion in Practice

1. The cuspidal cubic $X = V(y^2 - x^3) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2, 2y)$. At the origin, this is $(0, 0)$, which has rank $0 < 2 - 1 = 1$. At any other closed point (a, b) with $b \neq 0$, the rank is $1 = n - \dim X$. Hence X is singular only at the origin.

2. The nodal cubic $X = V(y^2 - x^2(x + 1)) \subseteq \mathbb{A}^2_k$. The Jacobian is $(-3x^2 - 2x, 2y)$. At the origin, this is $(0, 0)$, so the origin is again singular. At any other point on X where $y \neq 0$, the rank is 1.

3. The quadric surface $X = V(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3_k$ ($\operatorname{char} k \neq 2$). The Jacobian is $(2x, 2y, -2z)$, which has rank 0 only at the origin. Hence X is regular away from the origin.

Smooth Morphisms and Smooth Schemes

Regularity is a property of a scheme in isolation, while *smoothness* is a relative notion: it concerns a morphism $X \to S$ and asks whether the fibers are "non-singular" in a way that varies well over the base. The two notions coincide over perfect fields.

Definition 2.8.7: Smooth Morphism (Preliminary)

A morphism $f : X \to S$ of schemes locally of finite type is *smooth of relative dimension* n at a point $x \in X$ if there exists an open neighborhood U of x and a factorization:

$$\begin{tikzcd} U \arrow[r, hook, "\text{open}"] \arrow[dr, "f|_U"'] & \operatorname{Spec} \frac{\mathcal{O}_S(V)[x_1,\ldots,x_{n+r}]}{(f_1,\ldots,f_r)} \arrow[d] \\ & V \end{tikzcd}$$

where V is an open neighborhood of $f(x)$ in S, and the Jacobian matrix $(\partial f_i/\partial x_j)$ has rank r at x.
The morphism f is *smooth* if it is smooth at every point. It is *étale* if it is smooth of relative dimension 0.

Definition 2.8.8: Smooth Scheme Over a Field

Let k be a field. A k-scheme X is *smooth over* k (or k*-smooth*) if the structure morphism $X \to \operatorname{Spec} k$ is smooth.

The relationship between smoothness and regularity is as follows:

Proposition 2.8.9: Smooth Implies Regular

Let k be a field and X a k-scheme locally of finite type. If X is smooth over k, then X is regular.

***Proof*:**
Smoothness over k implies that the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ has dimension equal to the local dimension of X at x for every point x, which is exactly regularity. The key point is that smoothness gives the Jacobian rank condition over every residue field, not just over k itself.

The converse holds over perfect fields:

Theorem 2.8.10: Regularity and Smoothness Over Perfect Fields

Let k be a perfect field (e.g., any field of characteristic 0, or any finite field) and X a k-scheme locally of finite type. Then X is smooth over k if and only if X is regular.

***Proof*:**
The forward direction is proposition 2.8.9. For the converse, regularity gives the Jacobian rank condition at all k-rational points. The perfection of k ensures that this extends to all points: if k is perfect, then every finitely generated field extension $\kappa(x)/k$ is separably generated, and the cotangent space computation over $\kappa(x)$ agrees with the one over k. The details require the theory of Kähler differentials, which will be developed in section 5.4.

Example 2.8.11: Regularity $\neq$ Smoothness Over Imperfect Fields

Let $k = \mathbb{F}_p(t)$ (an imperfect field) and consider $X = \operatorname{Spec} k[x]/(x^p - t)$. Then X is a single closed point with residue field $k(\alpha)$ where $\alpha^p = t$. The local ring is $k(\alpha)$, which is a field and hence a regular local ring of dimension 0.

However, X is *not* smooth over k: the polynomial $f(x) = x^p - t$ has $f'(x) = px^{p-1} = 0$, so the Jacobian criterion fails. Geometrically, the extension $k(\alpha)/k$ is purely inseparable, and the fiber over $\operatorname{Spec} k$ is “infinitesimally thickened” from the perspective of k. This is the prototypical example of a regular but non-smooth scheme.

The Singular Locus

Definition 2.8.12: Singular Locus

Let X be a locally Noetherian scheme. The *singular locus* of X is the set:

$$\operatorname{Sing}(X) = \{x \in X \mid \mathcal{O}_{X,x} \text{ is not a regular local ring}\}$$

Proposition 2.8.13: Singular Locus is Closed

Let X be a scheme locally of finite type over a field k. Then $\operatorname{Sing}(X)$ is a closed subset of X.

***Proof*:**
The question is local, so we may assume $X = \operatorname{Spec}(k[x_1, \dots, x_n]/(f_1, \dots, f_m))$. By the Jacobian criterion, the singular locus is cut out by the vanishing of all $(n - \dim X) \times (n - \dim X)$ minors of the Jacobian matrix, which defines a closed set.

For irreducible varieties, the singular locus is not just closed but *proper*:

Theorem 2.8.14: Generic Smoothness

Let X be an irreducible variety over an algebraically closed field k. Then $\operatorname{Sing}(X)$ is a proper closed subset of X. In particular, the smooth locus is a dense open subset.

***Proof*:**
See [9, proposition 20.57]. The key idea is that if X is a hypersurface $V(f)$ and $\operatorname{Sing}(X) = X$, then all partial derivatives of f vanish identically on X. In characteristic 0, this forces f to be constant. In characteristic $p > 0$, it forces $f = g^p$ for some polynomial g (using algebraic closure to take p-th roots), contradicting the irreducibility of f. The general case reduces to the hypersurface case via birational equivalence.

Over non-algebraically-closed fields, the correct generalization replaces “regular” with “smooth”:

Corollary 2.8.15: Generic Smoothness Over Perfect Fields

Let X be a geometrically integral[a] k-scheme of finite type over a perfect field k. Then the smooth locus of $X \to \operatorname{Spec} k$ is a dense open subset.

[a]geometrically integral means change the base ring k to $\overline{k}$; see ref:HERE

2.8.1 Properties of Regular Schemes

We collect some important structural consequences of regularity:

Proposition 2.8.16: Regular Local Rings are Domains

Every regular local ring is an integral domain.

***Proof*:**
By [9, lemma 20.54], if $G_{\mathfrak{m}}(R) \cong k[t_1, \ldots, t_d]$ is a domain and $\bigcap_n \mathfrak{m}^n = 0$ (which holds by the Krull intersection theorem for Noetherian local rings), then R is a domain.

Proposition 2.8.17: Regular Local Rings are Normal

Every regular local ring is integrally closed (normal).

***Proof*:**
See [9, corollary 20.56]. This follows from the fact that the associated graded ring of a regular local ring is a polynomial ring (hence a UFD), and one can lift the integral closure property from the associated graded ring.

Proposition 2.8.18: Regular Local Rings are UFDs

Every regular local ring is a unique factorization domain.

***Proof*:**
This is a deep theorem of Auslander–Buchsbaum, using the characterization of regular local rings via finite global dimension (Serre's theorem, [9, theorem 20.52]). The key step is showing that every height-1 prime in a regular local ring is principal, and then invoking the Nagata criterion for UFDs.

The chain of implications for Noetherian local domains is:

$$\text{regular} \implies \text{UFD} \implies \text{normal} \implies \text{regular in codimension 1}$$

None of these implications reverse in general (except in dimension 1, where regular, normal, and DVR are equivalent by [9, corollary 20.55]).

Proposition 2.8.19: Regular Scheme and Irreducible Components

A regular scheme is a disjoint union of its irreducible components (each of which is integral).

***Proof*:**
Since each stalk is a domain, every point lies on a unique irreducible component. Hence distinct components cannot meet, and the scheme decomposes as a disjoint union. Each component is integral (irreducible and reduced) since regularity implies the stalks are domains.

2.8.2 Dimension and Regularity for Schemes Over $\mathbb{Z}$

An important feature of the scheme-theoretic framework is that the notion of regularity applies uniformly to schemes over any base, including $\operatorname{Spec}\mathbb{Z}$. This allows number-theoretic objects to be treated geometrically:

Example 2.8.20: Regularity of Arithmetic Schemes

1. $\operatorname{Spec}\mathbb{Z}$ is a regular scheme of dimension 1: for each prime (p), the local ring $\mathbb{Z}_{(p)}$ is a DVR (hence a regular local ring of dimension 1), and the generic point has local ring $\mathbb{Q}$, a field.
2. $\operatorname{Spec}\mathbb{Z}[i]$ is a regular scheme. At each maximal ideal $\mathfrak{p}$, the local ring $\mathbb{Z}[i]_{\mathfrak{p}}$ is a DVR (since $\mathbb{Z}[i]$ is a Dedekind domain).
3. $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is *not* regular at the prime $\mathfrak{p} = (2, 1+\sqrt{-5})$: the local ring $\mathbb{Z}[\sqrt{-5}]_{\mathfrak{p}}$ is a 1-dimensional local domain that is not a DVR (since $\mathfrak{p}$ is not principal). See [9, chapter 22] for the failure of unique factorization in this ring. The normalization (integral closure in the fraction field) produces the ring of integers $\mathbb{Z}\left[\frac{1+\sqrt{-5}}{2}\right]$, which *is* a Dedekind domain and hence regular.
4. More generally, if K is a number field and $\mathcal{O}_K$ its ring of integers, then $\operatorname{Spec}\mathcal{O}_K$ is a regular scheme if and only if $\mathcal{O}_K$ is a principal ideal domain (equivalently, the class number of K is 1). This follows from the equivalence of regularity and DVR in dimension 1 (by [9, corollary 20.55]).

Exercise 2.8.1

1. Show that $\mathbb{A}^n_k$ is a regular scheme for any field k.
2. Let $X = \operatorname{Spec} k[x,y]/(y^2 - x^3 + x)$. Show that X is a regular scheme (assuming $\operatorname{char} k \neq 2$).
3. Show that the generic point of any integral scheme is a regular point.
4. Let $X = \operatorname{Spec} k[x,y,z,w]/(xw - yz)$ be the quadric cone (from example 2.2.2). Find $\operatorname{Sing}(X)$ and show it consists of a single point.
5. Let k be a perfect field and $f \in k[x_1, \ldots, x_n]$ irreducible. Show that $V(f)$ is smooth over k if and only if f and its partial derivatives have no common zero.

6. Show that the product of regular schemes is regular (hint: use that regularity is detected at stalks, and compute the stalks of a product).
7. Let R be a regular local ring and $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Show that $R/(x)$ is again a regular local ring (of dimension one less).

2.9 *How Much Does the Topology Know?

Throughout this chapter, we have constructed schemes as locally ringed spaces: a topological space X together with a structure sheaf $\mathcal{O}_X$. The topological space carries the Zariski topology, which for affine schemes $\operatorname{Spec}(R)$ is entirely determined by the lattice of prime ideals of R. A natural question arises: how much of the scheme-theoretic structure is already visible at the level of topology?

One might initially expect a clean dichotomy: that the topological spaces arising from affine schemes form a special class, and that the topology of a non-affine scheme would look fundamentally "different"—perhaps failing some property that all affine topologies share. This turns out to be false, and understanding *why* it is false sharpens our appreciation for the role of the structure sheaf.

Spectral spaces

To state the result precisely, we need to isolate the topological properties that characterize the spaces $|\operatorname{Spec}(R)|$.

Definition 2.9.1: Spectral Space

A topological space X is a *spectral space* (or *coherent space*) if it satisfies the following four conditions:

1. X is quasi-compact.
2. X is sober: every irreducible closed subset has a unique generic point.
3. The quasi-compact open subsets of X are closed under finite intersection (i.e. X is quasi-separated, see definition 3.5.6) and form a basis for the topology.
4. Every non-empty irreducible closed subset of X has a generic point.

Condition (2) already implies (4), so the definition can be stated with just (1)–(3), but we include (4) for emphasis as it is the condition most often violated by "pathological" topological spaces. Note that condition (3) asks for two things at once: the quasi-compact opens form a basis (so there are enough of them), and a finite intersection of quasi-compact opens is again quasi-compact (a condition that can fail even for well-behaved spaces—it fails, for instance, for the Zariski topology on an infinite product of affine lines, though not for any Noetherian space).

The reader should recognize that every affine scheme satisfies these conditions. In $\operatorname{Spec}(R)$: the space is quasi-compact (section 2.1.3); it is sober by construction (generic points of irreducible closed sets correspond to prime ideals); and the distinguished open sets $D(f)$, which are quasi-compact, form a basis that is closed under finite intersection ($D(f) \cap D(g) = D(fg)$).

The remarkable fact, proved by Hochster in 1969, is that these conditions are not just necessary but *sufficient*.

Theorem 2.9.2: Hochster's Theorem

A topological space X is homeomorphic to $\operatorname{Spec}(R)$ for some commutative ring R if and only if X is a spectral space.

***Proof*:**
see [16]

The "only if" direction is the easy half: we have just verified that every $\operatorname{Spec}(R)$ is spectral. The "if" direction is the deep content: given any spectral space X, Hochster explicitly constructs a commutative ring R such that $|\operatorname{Spec}(R)| \cong X$. The construction uses the lattice of quasi-compact open subsets of X to build R as a quotient of a polynomial ring, with generators corresponding to the quasi-compact opens and relations encoding their intersection and covering behavior.

We do not reproduce Hochster's construction here, but the theorem has an immediate and perhaps surprising consequence.

Non-affine schemes with affine topology

Consider projective space $\mathbb{P}^n_k$ over a field k, with $n \geq 1$. This is emphatically not an affine scheme: its ring of global sections is just k, and $\operatorname{Spec}(k)$ is a single point, while $\mathbb{P}^n_k$ has many points. Yet the underlying topological space $|\mathbb{P}^n_k|$ is:

1. quasi-compact (it is covered by finitely many affine charts $U_i \cong \mathbb{A}^n$),
2. sober (it is a scheme, and every scheme is sober),
3. equipped with a basis of quasi-compact opens closed under finite intersection (the distinguished opens in each affine chart, together with their finite intersections, which are again quasi-compact by proposition 2.1.15).

Hence $|\mathbb{P}^n_k|$ is a spectral space, and by Hochster's theorem, there exists a commutative ring R with $|\mathbb{P}^n_k| \cong |\operatorname{Spec}(R)|$ as topological spaces. The topology of projective space is indistinguishable from that of an affine scheme.

This is not an isolated phenomenon. Any quasi-compact and quasi-separated scheme has an underlying spectral space, and hence is homeomorphic to $\operatorname{Spec}(R)$ for some ring R. This includes virtually all schemes encountered in practice: all Noetherian schemes, all schemes of finite type over a field or over $\mathbb{Z}$, all separated schemes.

Proposition 2.9.3: Topology of Quasi-Compact Quasi-Separated Schemes

Let X be a quasi-compact and quasi-separated scheme. Then $|X|$ is a spectral space, and hence there exists a commutative ring R such that $|X| \cong |\operatorname{Spec}(R)|$ as topological spaces.

***Proof*:**
Quasi-compactness of X gives condition (1). Every scheme is sober (a point $x \in X$ has closure $\overline{\{x\}}$ which is irreducible, and if Z is irreducible closed then its generic point is the unique point η with $\overline{\{\eta\}} = Z$; this follows from the corresponding fact for affine schemes, which glues). For condition (3): the quasi-compact opens of X include all affine opens and their finite unions, which form a basis. The quasi-separated hypothesis says exactly that the intersection of two quasi-compact opens is quasi-compact, giving closure under finite intersection.

To make this concrete, here are some examples.

Example 2.9.4: Non-Affine Schemes with Affine Topology

1. *Projective space.* As discussed above, $|\mathbb{P}^n_k|$ is spectral. Over $k = \overline{k}$ algebraically closed, the closed points of $\mathbb{P}^1_k$ are in bijection with $\overline{k} \cup \{\infty\}$, and the topology is the cofinite topology on this set together with a generic point whose closure is the whole space. This is homeomorphic to $|\mathrm{Spec}\,(k[x])|$ (which also has the cofinite topology on its closed points, plus a generic point). In particular, $|\mathbb{P}^1_k| \cong |\mathbb{A}^1_k|$ as topological spaces, even though $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$ are not isomorphic as schemes.

2. *The line with doubled origin.* The non-separated scheme obtained by gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ is quasi-compact and quasi-separated, so its topology is spectral. This scheme is not affine (it is not even separated), yet topologically it is homeomorphic to $\mathrm{Spec}\,(R)$ for some ring R.

3. *The punctured plane.* The scheme $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ from Example 2.3.14 is quasi-compact and quasi-separated (it is separated, being an open subscheme of an affine scheme), hence spectral. It is not affine, yet its topology is that of some $\mathrm{Spec}\,(R)$.

The constructible topology and Stone spaces

Hochster's theorem tells us that a topological space is the spectrum of a ring if and only if it is spectral. But the definition of a spectral space involves a somewhat asymmetric collection of conditions. There is a cleaner perspective that explains *what structure a spectral space actually carries*, and it comes from a classical duality in topology and logic.

Given a spectral space X, define the *constructible topology* (also called the *patch topology*) on the underlying set of X by declaring both the quasi-compact open subsets *and their complements* to be open. Equivalently, the constructible topology is the coarsest topology in which every quasi-compact open of X is clopen (both open and closed).

Proposition 2.9.5: The Constructible Topology is a Stone Space

Let X be a spectral space. Then X equipped with the constructible topology is a *Stone space*: a topological space that is compact, Hausdorff, and totally disconnected.

***Proof*:**
(Sketch) The constructible topology is Hausdorff because any two distinct points of a spectral space can be separated by a quasi-compact open (one point lies in it, the other does not), which is clopen in the constructible topology. It is compact because the constructible topology is finer than the original spectral topology (which is already quasi-compact), and one can verify that any cover by constructible opens has a finite subcover using the quasi-compactness of the spectral topology together with the finiteness of intersections. Total disconnectedness follows because the clopen sets (which include all quasi-compact opens and their complements) separate points.

We denote the set X equipped with the constructible topology by X^{cons}. The identity map $X^{\mathrm{cons}} \to X$ is continuous (the constructible topology is finer), but not a homeomorphism: the constructible topology is strictly finer than the spectral topology whenever X has a non-closed point (i.e., whenever X is not discrete).

The name "constructible topology" is not a coincidence. A subset $C \subseteq X$ of a topological space is *constructible* if it belongs to the Boolean algebra generated by the open sets—equivalently, if C is a finite union of locally closed subsets ($U \cap Z$ where U is open and Z is closed). In a spectral space, the constructible subsets have a clean characterization:

Proposition 2.9.6: Constructible Sets are Clopens

Let X be a spectral space. A subset $C \subseteq X$ is constructible (in the classical sense: a finite Boolean combination of quasi-compact opens) if and only if C is clopen in the constructible topology.

This connects directly to Chevalley's theorem (section 3.7): if $f : X \to Y$ is a morphism of finite type between Noetherian schemes, then the image of a constructible set is constructible. In the language of the constructible topology, this says that f induces a continuous map $X^{\mathrm{cons}} \to Y^{\mathrm{cons}}$ between the associated Stone spaces, and continuous maps between Stone spaces preserve clopens. From this perspective, Chevalley's theorem is a statement about the *continuity of scheme morphisms with respect to the constructible topology*—a viewpoint that is both conceptually cleaner and technically powerful.

Stone duality and the lattice of opens

The appearance of Stone spaces is not accidental; it reflects a duality from mathematical logic and lattice theory.

Stone duality (1936) establishes an anti-equivalence of categories between Boolean algebras and Stone spaces: every Boolean algebra B is the algebra of clopen subsets of a unique Stone space $S(B)$, and conversely, every Stone space arises in this way. Restricting to distributive lattices (which need not be Boolean), one obtains *spectral spaces*: a spectral space is precisely a Stone space together with a distinguished sublattice of its Boolean algebra of clopens.

More precisely, let X^{cons} be a Stone space. A sublattice $L \subseteq \mathrm{Clopen}(X^{\mathrm{cons}})$ that forms a basis for some topology on the underlying set, and is closed under finite intersection and finite union, determines a spectral topology on X: the topology generated by L. Different choices of L give different spectral topologies on the same underlying Stone space.

This gives a clean factorization of the data in a spectral space:

$$\text{Spectral space} \;=\; \text{Stone space} \;+\; \text{Choice of lattice.}$$

The Stone space is the "combinatorial skeleton"—it knows which subsets are constructible (i.e., definable by first-order formulas in model-theoretic language). The lattice records which constructible sets are "open," i.e., which direction the specialization order runs.

Example 2.9.7: Different Spectral Structures on the Same Stone Space

Consider the Cantor set $C \subseteq [0,1]$, which is a Stone space (compact, Hausdorff, totally disconnected). Every countable Boolean algebra has C as its Stone space. In particular, the spaces $|\mathrm{Spec}\,(\mathbb{Z})|$ and $|\mathrm{Spec}\,(k[x])|$ (for k algebraically closed) both have the same constructible topology—they are both countable spectral spaces whose Stone space is the Cantor set—but they carry different spectral topologies (different lattices of opens) corresponding to the different lattice structures of their prime ideals.

More concretely, both spaces consist of a generic point η together with countably many closed points, and the topology in both cases has $\{\eta\}$ dense and the closed points discrete in the complement. But the "arithmetic" of which opens contain which closed points differs: in $\mathrm{Spec}\,(\mathbb{Z})$, the closed point (p) lies in $D(q)$ for $q \neq p$, while in $\mathrm{Spec}\,(k[x])$, the closed point $(x-a)$ lies in $D(x-b)$ for $b \neq a$. These are different lattices on the same Stone space.

What does the topology actually store?

The data of a scheme $(X, \mathcal{O}_X)$ decomposes into three layers:

1. *The Stone space* X^{cons}: the combinatorial skeleton. It knows the set of points, the constructible subsets, and (via Stone duality) the Boolean algebra they generate. This is the coarsest invariant—it is shared by many different schemes.
2. *The spectral topology*: a choice of sublattice L of the clopens of X^{cons}, determining which constructible sets are "open." This encodes the specialization order (which points are limits of which others) and hence the notion of "generic" and "closed" points. Different spectral topologies on the same Stone space give genuinely different spaces (e.g., $\mathrm{Spec}\,(\mathbb{Z})$ vs. $\mathrm{Spec}\,(k[x])$), but by Hochster's theorem, both are realizable as $\mathrm{Spec}\,(R)$ for some ring.
3. *The structure sheaf* $\mathcal{O}_X$: the algebraic data. It determines which continuous maps are "algebraic," what the global sections and cohomology are, and whether the scheme is affine, projective, smooth, etc. This is the finest invariant, and it is where the geometry lives.

The passage from (3) to (2) forgets the algebra; the passage from (2) to (1) forgets the specialization order. Neither passage is reversible: many non-isomorphic schemes share the same topology (as we saw with $\mathbb{P}^1_k$ and $\mathbb{A}^1_k$), and many non-homeomorphic spectral spaces share the same Stone space (as with $\mathrm{Spec}\,(\mathbb{Z})$ and $\mathrm{Spec}\,(k[x])$).

The conclusion is that the Zariski topology, while important for defining the notion of "local" and for the combinatorics of specialization and generization, carries far less information than one might expect. The structure sheaf is the essential datum: it determines which continuous maps are "algebraic," which open subsets are "distinguished," and what "functions" look like. Two schemes can share the same topological space—or even the same constructible topology—while having completely different algebraic geometry.

3

Schemes II: Morphisms and Constructions

Scheme morphisms are inherently more *algebraic* than *smooth* in nature: they are in some ways more stringent than holomorphic maps (only "polynomial-like" functions are allowed), but they also permit singularities. In this chapter, we shall properly explore morphisms of schemes, classify many important types, and interpret them geometrically. In the process, we will be able to "recover" the correct analogues of Hausdorffness and compactness for schemes in the form of *separatedness* and *properness*. The condition of being proper will turn out to be closely related to being projective: by Chow's lemma, every proper scheme over a Noetherian base admits a surjective birational morphism from a projective scheme, showing that projective schemes are the natural "compact schemes" to consider. The chapter concludes by defining an abstract variety and showing that $\mathbf{Var}_k$ is a fully faithful subcategory of $\mathbf{Sch}_k$.

Algebraically, smooth and holomorphic morphisms share an important feature with scheme morphisms: a continuous map $f : M \to N$ between smooth manifolds is a *smooth map* if and only if $f^*(C^\infty(N)) \subseteq C^\infty(M)$ where

$$f^* : C^\infty(N) \to C^\infty(M) \qquad f^*(\varphi) = \varphi \circ f$$

and in particular if f is a homeomorphism, it is a diffeomorphism if and only if f^* is a ring isomorphism. In essence, f "preserves" and is "compatible with" all the relations on N given by smooth functions. As schemes are defined by their algebraic relations, this pullback approach generalizes naturally[1].

With this in mind, we shall require the following properties from scheme morphisms:

1. A scheme morphism $f : X \to Y$ must preserve the structure sheaf (which encodes all algebraic

[1]In the smooth case, the "if and only if" makes this an equivalent definition of smooth map; since there is only one type of "affine patch" (open subsets of $\mathbb{R}^n$), one can then recover the usual definition. Proposition 3.1.10 establishes the analogous result for schemes.

information). Concretely, for every open set $V \subseteq Y$, there should be a ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ that pulls back local sections on Y to local sections on X. The intuition is that we treat sections as genuine (scheme) functions, and we require their composition with f to remain a scheme function. Thus, a scheme morphism must be a *morphism of ringed spaces* (a continuous map together with a compatible sheaf map).

This additional structure is genuinely necessary: as we saw in example 2.1.25, not all continuous maps between spectra are induced by ring homomorphisms, and there is no natural way to recover a ring homomorphism from a mere continuous map. A simple instance is that $\operatorname{Spec} k$ and $\operatorname{Spec} k'$ for two non-isomorphic fields k, k' have homeomorphic underlying spaces (both are single points) yet are not isomorphic as schemes.

2. The pullback of functions must be compatible with evaluation: if $s \in \mathcal{O}_Y(V)$ vanishes at $f(x) \in Y$, then $f^\#(s)$ must vanish at $x \in X$. This means the pullback must respect the local ring structure at each point, making f a morphism of *locally* ringed spaces.

3. Ring homomorphisms between global sections and scheme morphisms between affine schemes must be dual to each other (generalizing the fact that a map between k-varieties is regular if and only if the contravariant map between coordinate rings is a k-algebra homomorphism).

This final condition means we want an equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\operatorname{Spec}(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

In section ref:HERE, this equivalence is generalized from affine schemes to all schemes by replacing $\mathbf{CRing}^{\mathrm{op}}$ with the appropriate category of sheaves of rings on a site (the functor-of-points perspective).

For those familiar with differential geometry [5], let us take a moment to compare and contrast smooth maps with scheme morphisms.

In differential geometry, continuous maps between manifolds can be arbitrarily well approximated by smooth maps (as a consequence of *Whitney's Approximation Theorem*). Hence, smooth maps can be thought of as "smoothing out" continuous maps – the smooth and continuous worlds are not far apart. This is *not at all* the case in the Zariski topology. Consider, for example, any permutation $\sigma : \overline{k} \to \overline{k}$ that preserves finite subsets (where $\overline{k}$ is an algebraically closed field). Such a σ induces a homeomorphism of $\mathbb{A}^1_{\overline{k}}$ in the Zariski topology – since the closed sets are exactly the finite sets and the whole space, any bijection preserving finiteness is continuous. Yet a "wild" permutation (one not given by a polynomial) will not be a scheme morphism[2]. Thus, there is no hope of approximating arbitrary continuous maps by regular maps: scheme morphisms are fundamentally more rigid than smooth maps.

In fact, regular maps behave much more like *holomorphic* maps. Both are determined by their local algebraic/analytic expressions, both satisfy analogues of the identity theorem (a regular map vanishing on a dense open must vanish everywhere), and both have fibers of controlled dimension. This similarity becomes an exact correspondence for the right categories: the GAGA theorems of Serre establish an equivalence between algebraic and analytic coherent sheaves on projective varieties over $\mathbb{C}$ (see section 6.6). Hence, when building geometric intuition for scheme morphisms via [5]

[2] Over $\mathbb{C}$, any discontinuous field automorphism $\sigma \in \operatorname{Aut}(\mathbb{C}/\mathbb{Q})$ gives such an example; these exist in abundance (assuming the axiom of choice) but none are $\mathbb{C}$-algebra maps.

and [4], it is holomorphic maps, not smooth maps, that provide the better mental model. A key difference is that scheme morphisms freely allow singularities and have unique algebraic-geometric features absent from the complex-analytic or smooth settings (for example, the Frobenius morphism from example 3.1.11, or the behavior of morphisms between schemes over $\operatorname{Spec}\mathbb{Z}$).

3.1 Elementary Concepts

Some of the discussion here shall overlap with the content in section 1.4; this was done as the self-completeness of the material in this section is a priority.

Recall (definition 1.4.4) that a morphism of ringed spaces $\pi : X \to Y$ is a continuous map of topological spaces together with a map $\pi^\# : \mathcal{O}_Y \to \pi_*\mathcal{O}_X$ that replicates composing any function in $\mathcal{O}_Y$ with π[3]. However, this notion is not restrictive enough for scheme theory for two reasons:

(i) Not all *continuous* maps between spectra are induced by ring homomorphisms (see Example 2.1.25), so specifying a sheaf map is genuinely additional data beyond the continuous map. Again, to abuse the same example, $\operatorname{Spec}(k)$ and $\operatorname{Spec}(k')$ for non-isomorphic fields k, k' are homeomorphic (both are single points) but carry different structure sheaves.

(ii) Even among ringed space morphisms, the global sections functor $\Gamma(-)$ is *not injective* on morphism sets: distinct ringed space morphisms can induce the same ring homomorphism on global sections. The following example demonstrates this.

Example 3.1.1: Two Ringed Space Morphisms for One Ring Homomorphism
Take $R = \mathbb{Z}_{(p)}$ and $S = \mathbb{Q}$ so that $\operatorname{Spec}(R) = \{[(p)], [(0)]\}$ and $\operatorname{Spec}(S) = \{[(0)]\}$. Let us drop the parentheses and square brackets to declutter notation. The open sets of $\operatorname{Spec}(R)$ are:

$$\emptyset, \quad \{0\}, \quad \{0, p\} = \operatorname{Spec}(R)$$

The stalks of $\operatorname{Spec}(R)$ are $\mathbb{Z}_{(p)}$ at p and $\mathbb{Q}$ at 0. The stalk of $\operatorname{Spec}(S)$ is $\mathbb{Q}$ at 0.

There is only one ring homomorphism $\psi : R \to S$, namely the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

In the category **Sch**, this should correspond to a unique map $\iota^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $\psi^{-1}((0)) = (0)$. However, in the category **RingedSp** we can define two morphisms between these affine schemes:

1. *Map 1 (The "correct" map):* $\pi_1 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto 0$. The sheaf map is determined by the stalk map at the unique point $0 \in \operatorname{Spec}(S)$:

$$(\pi_1)_0^\# : \mathcal{O}_{R,0} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Q} \xrightarrow{\text{id}} \mathbb{Q}$$

This is a local ring homomorphism (both sides are fields, with maximal ideal (0) mapping to (0)). On global sections, it induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

2. *Map 2 (The "phantom" map):* $\pi_2 : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ defined by $0 \mapsto p$. Since $\{p\}$ is closed, this is a continuous map. The sheaf map is determined by the stalk map at $0 \in \operatorname{Spec}(S)$:

$$(\pi_2)_0^\# : \mathcal{O}_{R,p} \to \mathcal{O}_{S,0} \qquad \Longrightarrow \qquad \mathbb{Z}_{(p)} \xrightarrow{\text{inclusion}} \mathbb{Q}$$

[3] Or equivalently by adjointness, the map $\pi^{-1}(\mathcal{O}_Y) \to \mathcal{O}_X$.

This is a valid ring homomorphism, so π_2 is a valid morphism of ringed spaces. On global sections, this *also* induces the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$.

Thus, we have found two distinct ringed space morphisms $\pi_1 \neq \pi_2$ that induce the *same* ring homomorphism on global sections. We reject π_2 because the stalk map $\mathbb{Z}_{(p)} \to \mathbb{Q}$ is *not* a local ring homomorphism: the maximal ideal $(p) \subseteq \mathbb{Z}_{(p)}$ maps to the unit ideal in $\mathbb{Q}$ (since p is invertible in $\mathbb{Q}$), violating the condition $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. The definition of locally ringed space morphism correctly filters out this non-geometric map.

Thus, if $\mathbf{RingedSp}^{\mathbf{Aff}}$ denotes the category whose objects are $(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ and whose morphisms are ringed space morphisms, the functor:

$$\Gamma(-) : \mathbf{RingedSp}^{\mathbf{Aff}} \to \mathbf{CRing}^{\mathrm{op}}$$

is surjective on hom-sets (every ring homomorphism $R \to S$ induces at least one ringed space morphism via the standard contraction-of-primes construction), but not injective (as the example above demonstrates).

The key realization is that ring homomorphisms induce local structure that must be preserved. By proposition 2.2.21, the stalks of schemes are all *local rings*. A map between local rings need not preserve maximal ideals (for instance, the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$ sends the maximal ideal (p) to the unit ideal). We thus single out maps that do preserve this structure:

Definition 3.1.2: Local Ring Homomorphism

Let $\varphi : (R, \mathfrak{m}_R) \to (S, \mathfrak{m}_S)$ be a ring homomorphism between local rings with maximal ideals $\mathfrak{m}_R$ and $\mathfrak{m}_S$ respectively. Then φ is a *local ring homomorphism* if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$, or equivalently $\varphi^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$.

A ringed space morphism that is a local ring homomorphism at each stalk is given a name:

Definition 3.1.3: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. A *morphism of locally ringed spaces* $(\pi, \pi^{\#}) : X \to Y$ is a morphism of ringed spaces such that the induced stalk map $\pi_p^{\#} : \mathcal{O}_{Y,\pi(p)} \to \mathcal{O}_{X,p}$ is a local ring homomorphism for every $p \in X$.

Observe that ring homomorphisms *automatically* induce local maps on localizations: if $\varphi : R \to S$ is a ring homomorphism and $\mathfrak{p} \subseteq S$ is prime, then the induced map $\varphi_{\mathfrak{p}} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ given by $r/t \mapsto \varphi(r)/\varphi(t)$ satisfies $\varphi_{\mathfrak{p}}(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. Indeed, if $r/t \in \mathfrak{m}_R$ then $r \in \varphi^{-1}(\mathfrak{p})$, so $\varphi(r) \in \mathfrak{p}$, giving $\varphi(r)/\varphi(t) \in \mathfrak{p}S_{\mathfrak{p}} = \mathfrak{m}_S$. Equivalently, $\mathfrak{m}_R = \varphi_{\mathfrak{p}}^{-1}(\mathfrak{m}_S)$[4]. Example 3.1.1 shows that this locality condition need *not* hold for arbitrary ringed space morphisms, even between locally ringed spaces. The local ring condition forces the morphism to "respect" not just the lattice structure of opens but also the evaluation of functions at each point.

With this added structure, we recover a bijection:

[4] Note this does not imply $\varphi(\mathfrak{m}_R) = \mathfrak{m}_S$: the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Z}_{(p)}[1/q]$ for primes $p \neq q$ is a local ring homomorphism where $\varphi(\mathfrak{m}_R) \subsetneq \mathfrak{m}_S$.

Theorem 3.1.4: Ring Homomorphisms and Locally Ringed Space Morphisms

Let R, S be commutative rings. Then:

1. Every ring homomorphism $\varphi : R \to S$ induces a natural morphism of locally ringed spaces:
$$(f, f^{\#}) : (\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$$
2. Every morphism of locally ringed spaces $(\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}) \to (\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)})$ induces a ring homomorphism $\varphi : R \to S$ (via global sections).
3. These operations are inverse to each other. In particular, the locally ringed space morphism is uniquely determined by the ring homomorphism on global sections:
$$\operatorname{Hom}_{\mathbf{LRS}}(\operatorname{Spec}(S), \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, S)$$

***Proof*:**
(1) Ring homomorphism $\Rightarrow$ locally ringed space morphism. Let $\varphi : R \to S$ be a ring homomorphism. Define the continuous map $f : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ by the standard contraction $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$.

We construct the sheaf morphism $f^{\#} : \mathcal{O}_{\operatorname{Spec}(R)} \to f_*\mathcal{O}_{\operatorname{Spec}(S)}$ on the basis of distinguished open sets, using proposition 1.3.3. For each $g \in R$, we must define a map
$$f^{\#}(D(g)) : \mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \longrightarrow (f_*\mathcal{O}_{\operatorname{Spec}(S)})(D(g)) = \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(D(g))).$$

The preimage of a distinguished open under f is again distinguished:
$$\begin{aligned} f^{-1}(D(g)) &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; \varphi^{-1}(\mathfrak{p}) \in D(g)\right\} \\ &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \;\middle|\; g \notin \varphi^{-1}(\mathfrak{p})\right\} \\ &= \{[\mathfrak{p}] \in \operatorname{Spec}(S) \mid \varphi(g) \notin \mathfrak{p}\} \\ &= D(\varphi(g)) \end{aligned}$$

This is a special property of the affine setting: the pullback of a “basic” geometric object is again basic. Using the identifications $\mathcal{O}_{\operatorname{Spec}(R)}(D(g)) \cong R_g$ and $\mathcal{O}_{\operatorname{Spec}(S)}(D(\varphi(g))) \cong S_{\varphi(g)}$, the global map φ induces a map on localizations. Since $\varphi(g)$ is a unit in $S_{\varphi(g)}$, the universal property of localization provides a unique ring homomorphism:
$$\Phi_g : R_g \longrightarrow S_{\varphi(g)}, \qquad \frac{a}{g^n} \longmapsto \frac{\varphi(a)}{\varphi(g)^n}.$$

These maps are compatible with restriction: if $D(h) \subseteq D(g)$ (so $h \in \sqrt{(g)}$), the diagram
$$\begin{tikzcd} R_g \arrow[r, "\Phi_g"] \arrow[d, "\mathrm{loc}"] & S_{\varphi(g)} \arrow[d, "\mathrm{loc}"] \\ R_h \arrow[r, "\Phi_h"] & S_{\varphi(h)} \end{tikzcd}$$

commutes because both paths send a/g^n to $\varphi(a)/\varphi(g)^n$ viewed in $S_{\varphi(h)}$. By proposition 1.3.3, the maps Φ_g glue to a unique sheaf morphism $f^{\#}$.

At each stalk, the induced map $f^\#_\mathfrak{p} : R_{\varphi^{-1}(\mathfrak{p})} \to S_\mathfrak{p}$ is the localization of φ at $\mathfrak{p}$, which is a local ring homomorphism as shown above. Hence $(f, f^\#)$ is a morphism of locally ringed spaces.

(2) Locally ringed space morphism ⇒ ring homomorphism. Given a morphism of locally ringed spaces $(f, f^\#) : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$, evaluate the sheaf map on global sections:

$$f^\#(\operatorname{Spec}(R)) : \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_{\operatorname{Spec}(S)}(f^{-1}(\operatorname{Spec}(R)))$$

Since $\mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \cong R$ and $\mathcal{O}_{\operatorname{Spec}(S)}(\operatorname{Spec}(S)) \cong S$, this yields a ring homomorphism $\varphi : R \to S$.

(3) These operations are inverse. Starting with $\varphi : R \to S$, constructing $(f, f^\#)$, and taking global sections recovers φ by construction.

Conversely, starting with a locally ringed space morphism $(f, f^\#)$ and letting $\varphi = f^\#(\operatorname{Spec}(R)) : R \to S$, and g be the morphism defined in step (1) given by $g(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$, we must show that $(f, f^\#)$ equals the morphism $(g, g^\#)$ constructed in step (1).

The continuous maps agree: Let $\mathfrak{p} \in \operatorname{Spec}(S)$. Consider the commutative diagram relating global sections to stalks:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "\text{loc}"] & S \arrow[d, "\text{loc}"] \\ R_{f(\mathfrak{p})} \arrow[r, "f^\#_\mathfrak{p}"] & S_\mathfrak{p} \end{tikzcd}$$

We compute $\varphi^{-1}(\mathfrak{p}) \subseteq R$ by tracing the preimage of $\mathfrak{m}_{S_\mathfrak{p}} \subseteq S_\mathfrak{p}$ along two paths:

- *Algebraic path (right then down):* The localization $S \to S_\mathfrak{p}$ pulls $\mathfrak{m}_{S_\mathfrak{p}}$ back to $\mathfrak{p}$. Then φ pulls $\mathfrak{p}$ back to $\varphi^{-1}(\mathfrak{p})$.
- *Geometric path (down then right):* Since $(f, f^\#)$ is a morphism of *locally* ringed spaces, $f^\#_\mathfrak{p}$ is a local ring homomorphism, so $(f^\#_\mathfrak{p})^{-1}(\mathfrak{m}_{S_\mathfrak{p}}) = \mathfrak{m}_{R_{f(\mathfrak{p})}}$. The localization $R \to R_{f(\mathfrak{p})}$ pulls this maximal ideal back to $f(\mathfrak{p})$.

By commutativity, both paths yield the same ideal: $f(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p}) = g(\mathfrak{p})$. Hence $f = g$ as continuous maps.

The sheaf maps agree: Since $f = g$, both $(f, f^\#)$ and $(g, g^\#)$ are sheaf morphisms over the same continuous map. To show $f^\# = g^\#$, it suffices to check equality on stalks ($\mathcal{O}_{\operatorname{Spec}(R)}$ is a sheaf, so morphisms out of it are determined by stalks). At each $\mathfrak{p} \in \operatorname{Spec}(S)$, both stalk maps are local ring homomorphisms $R_{\varphi^{-1}(\mathfrak{p})} \to S_\mathfrak{p}$ making the above diagram commute. Since the localization map $R \to R_{\varphi^{-1}(\mathfrak{p})}$ is an epimorphism in the category of rings (every element of $R_{\varphi^{-1}(\mathfrak{p})}$ is a fraction r/s with $r, s \in R$), any ring homomorphism out of $R_{\varphi^{-1}(\mathfrak{p})}$ that agrees with φ when precomposed with the localization map must be the unique localization $\varphi_\mathfrak{p}$. Hence $f^\#_\mathfrak{p} = g^\#_\mathfrak{p}$ for all $\mathfrak{p}$, giving $f^\# = g^\#$.

Affine schemes form the correct category to be the opposite category of commutative rings. We can rephrase what we just proved as:

Corollary 3.1.5: Equivalence of Rings and Affine Schemes

The category of affine schemes with locally ringed space morphisms is equivalent to the opposite category of commutative rings:

$$\mathbf{Sch}^{\mathbf{Aff}} \underset{\mathrm{Spec}\,(-)}{\overset{\Gamma(-)}{\rightleftarrows}} \mathbf{CRing}^{\mathrm{op}}$$

where Spec and Γ are inverse equivalences. More generally, for any locally ringed space X and any ring R, this equivalence extends to an adjunction on all of **LRS**:

$$\mathrm{Hom}_{\mathbf{CRing}}(R, \Gamma(X)) \cong \mathrm{Hom}_{\mathbf{LRS}}(X, \mathrm{Spec}\,(R))$$

where $\Gamma(-)$ is the left adjoint and $\mathrm{Spec}\,(-)$ is the right adjoint. On the subcategory of affine schemes, this adjunction restricts to an equivalence because both the unit $(X \to \mathrm{Spec}\,\Gamma(X))$ and the counit $(\Gamma(\mathrm{Spec}\,(R)) \to R)$ are isomorphisms.

Having shown that locally ringed morphisms are the right choice, let us take a moment to understand better how evaluation at a point is "preserved" by locally ringed space morphisms. Given a morphism of locally ringed spaces $\pi : X \to Y$, a point $q \in Y$ and a point $p \in X$ with $\pi(p) = q$, how do we interpret the local ring homomorphism between stalks? Say there is a section $g \in \mathcal{O}_Y(V)$ on an open subset $V \subseteq Y$ that vanishes at q, i.e., $g(q) = 0$ (equivalently, $g \in \mathfrak{m}_q$ in the stalk $\mathcal{O}_{Y,q}$). We want the pullback $\pi^{\#}(g)$ to vanish at p:

$$g(\pi(p)) = 0 \quad \Rightarrow \quad (\pi^{\#}g)(p) = 0.$$

The local ring condition ensures exactly this: since $\pi_p^{\#} : \mathcal{O}_{Y,q} \to \mathcal{O}_{X,p}$ satisfies $\pi_p^{\#}(\mathfrak{m}_q) \subseteq \mathfrak{m}_p$, the vanishing $g \in \mathfrak{m}_q$ implies $\pi^{\#}(g) \in \mathfrak{m}_p$, which means $\pi^{\#}(g)$ vanishes at p.

$$\begin{tikzcd}
\mathcal{O}_{Y,q} \arrow[rr, "\pi_q^{\#}"] & & \mathcal{O}_{X,p} \\
g \in \mathfrak{m}_q & \longmapsto & \pi^{\#}(g) \in \mathfrak{m}_p \\
Y \supseteq V & & X \supseteq \pi^{-1}(V) \\
\in & & \in \\
q = \pi(p) & & p \arrow[ll, "\pi"]
\end{tikzcd}$$

The top row encodes the algebraic condition (the stalk map sends the maximal ideal into the maximal ideal); the bottom row encodes the geometric picture (the point p maps to q under π). The locally ringed space condition is the requirement that these two rows are compatible: vanishing is preserved under pullback.

This is "obvious" when g and π are our usual functions since $g(\pi(p)) = (g \circ \pi)(p)$. However, the phantom map from Example 3.1.1 shows that this can fail for general ringed space morphisms.

Consider again:

$$\pi_2 : \mathrm{Spec}\,(\mathbb{Q}) \to \mathrm{Spec}\,(\mathbb{Z}_{(p)}), \qquad \pi_2([(0)]) = [(p)].$$

The function $p \in \mathcal{O}_Y(Y) = \mathbb{Z}_{(p)}$ vanishes at the closed point $[(p)]$: its value in the residue field $\kappa([(p)]) = \mathbb{Z}_{(p)}/(p) \cong \mathbb{F}_p$ is $p \bmod p = 0$. However, the pullback $\pi_2^{\#}(p)$ is $p \in \mathbb{Q}$ (via the inclusion $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$), and the value of p at the unique point $[(0)] \in \mathrm{Spec}\,(\mathbb{Q})$ is $p \bmod (0) = p \neq 0$ in $\kappa([(0)]) = \mathbb{Q}$. The vanishing was not preserved.

3.1.6 Remark: the generic point of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ The reader may find it surprising that $\{[(0)]\}$ is an open subset of $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$: it is the distinguished open $D(p)$, the complement of the closed point $V(p) = \{[(p)]\}$. This is *not* typical behaviour for generic points. In $\mathrm{Spec}\,(\mathbb{Z})$ or $\mathrm{Spec}\,(k[x, y])$, the generic point is *not* open—its closure is the entire space. The peculiarity here is that $\mathbb{Z}_{(p)}$ is a local domain with exactly two prime ideals, so the space has only two points, and removing a closed point from a two-point space leaves an open point. This makes $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$ the simplest non-trivial example for testing morphism conditions, precisely because the topology is so small that pathological maps are easy to construct explicitly.

3.1.7 Sections as Functions: Three Perspectives The reader may be troubled by a philosophical point: in differential geometry, a section $f \in C^\infty(U)$ *is* a function $f : U \to \mathbb{R}$, and evaluation at a point x is simply $f(x)$. For schemes, a section $f \in \mathcal{O}_X(U)$ is an element of a ring, and evaluation at x is the composite

$$\mathcal{O}_X(U) \longrightarrow \mathcal{O}_{X,x} \longrightarrow \kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x,$$

which is an additional construction. Is the datum of evaluation "missing" from the definition? The answer is no—it is encoded in the locally ringed space condition, but implicitly. There are three equivalent ways to see this.

(a) Hartshorne's approach. Define a section over U to be a function $f : U \to \coprod_{x\in U} \mathcal{O}_{X,x}$ such that $f(x) \in \mathcal{O}_{X,x}$ for each x, and such that f is "locally a quotient of ring elements." This makes sections into genuine functions and evaluation into literal function evaluation. The cost is that the codomain $\coprod \mathcal{O}_{X,x}$ varies from point to point and the definition requires a local coherence condition.

(b) The Vakil–Eisenbud approach. Keep sections as abstract ring elements. The locally ringed space axiom guarantees that each stalk $\mathcal{O}_{X,x}$ is local with a unique maximal ideal $\mathfrak{m}_x$, hence a unique residue field $\kappa(x)$, hence a unique evaluation map. The datum of evaluation is not "added"—it is a *consequence* of the locally ringed space structure. If the stalks were not local (i.e., if there were multiple maximal ideals), there would be multiple residue fields and no canonical notion of "the value of f at x." The LRS axiom is precisely the condition that evaluation is well-defined.

(c) The functor-of-points approach. By the representability of the global section functor (Proposition 3.3.7), a section $f \in \mathcal{O}_X(U)$ corresponds to a morphism $U \to \mathbb{A}^1_k$. In this picture, sections *are* morphisms, and evaluation at a point x is the composite $\mathrm{Spec}\,(\kappa(x)) \to U \to \mathbb{A}^1_k$, which is genuinely a function value.

All three perspectives carry exactly the same information. Approach (a) is most concrete; (b) is most algebraically efficient; (c) is most categorically natural. This text primarily uses (b), with (c) developed in section 3.3.

3.1.1 Scheme Morphisms

Inspired by the algebraic-geometric properties of locally ringed space morphisms, we define morphisms between schemes as locally ringed space morphisms:

Definition 3.1.8: Morphism of Schemes

Let X, Y be schemes. A morphism of locally ringed spaces $\pi : X \to Y$ is called a *morphism of schemes*. Schemes together with scheme morphisms form the category **Sch**.

One of the most important ways to control scheme morphisms is to reduce them to an affine cover. The following result is slightly more general but implies the desired result:

Lemma 3.1.9: Gluing Morphisms of Ringed Spaces

Let X, Y be ringed spaces and $X = \bigcup_i U_i$ an open cover. Suppose there exist ringed space morphisms $f_i : U_i \to Y$ that agree on overlaps:

$$f_i\big|_{U_i \cap U_j} = f_j\big|_{U_i \cap U_j}$$

Then there is a unique morphism of ringed spaces $f : X \to Y$ such that $f\big|_{U_i} = f_i$ for all i. Furthermore, if each f_i is a morphism of locally ringed spaces, then so is f.

***Proof*:**
Continuity. By the topological gluing lemma, the maps f_i glue to a unique continuous map $f : X \to Y$ with $f|_{U_i} = f_i$.

Sheaf morphism. We must construct $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$. For any open $V \subseteq Y$, we need a ring homomorphism $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$.

The open set $f^{-1}(V) \subseteq X$ is covered by $\{f^{-1}(V) \cap U_i\}_i$. Since $f|_{U_i} = f_i$, we have $f_i^{-1}(V) = f^{-1}(V) \cap U_i$. The sheaf morphism $f_i^\# : \mathcal{O}_Y \to (f_i)_*\mathcal{O}_{U_i}$ gives, for each i, a ring homomorphism:

$$f_i^\#(V) : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_{U_i}(f^{-1}(V) \cap U_i)$$

On overlaps $f^{-1}(V) \cap U_i \cap U_j$, the agreement condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ implies $f_i^\#|_{U_i \cap U_j} = f_j^\#|_{U_i \cap U_j}$, so:

$$f_i^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} = f_j^\#(V)(s)\big|_{f^{-1}(V) \cap U_i \cap U_j} \qquad \forall s \in \mathcal{O}_Y(V)$$

The sets $\{f^{-1}(V) \cap U_i\}_i$ cover $f^{-1}(V)$, and the local images $f_i^\#(V)(s)$ are sections of $\mathcal{O}_X$ that agree on overlaps. By the gluability axiom of $\mathcal{O}_X$, there exists a unique section $t \in \mathcal{O}_X(f^{-1}(V))$ restricting to $f_i^\#(V)(s)$ on each $f^{-1}(V) \cap U_i$. Define $f^\#(V)(s) := t$. This is a ring homomorphism since each $f_i^\#(V)$ is, and it commutes with restriction by construction. Uniqueness of $f^\#$ follows from the locality axiom of $\mathcal{O}_X$.

Locally ringed space condition. If each f_i is an LRS morphism, then for any $p \in X$, p lies in some U_i and the stalk map $(f^\#)_p = (f_i^\#)_p$ is a local ring homomorphism by assumption. Hence f is an LRS morphism.

By lemma 3.1.9, we can control morphisms of schemes by looking at ring homomorphisms. We can even ensure the codomain is an affine scheme:

Proposition 3.1.10: Scheme Morphisms are Locally Affine

A morphism $\pi : X \to Y$ between schemes is a morphism of schemes if and only if it is *locally affine*: every point $x \in X$ has affine open neighborhoods $U = \operatorname{Spec} R \ni x$ and $V = \operatorname{Spec} S \ni \pi(x)$ with $\pi(U) \subseteq V$, such that $\pi|_U : U \to V$ is induced by a ring homomorphism $\varphi : S \to R$.

***Proof*:**
(⇒) Scheme morphisms are locally affine. Let $\pi : X \to Y$ be a morphism of schemes and $x \in X$. Since Y is a scheme, there exists an affine open $V = \operatorname{Spec} S$ containing $\pi(x)$. By continuity, $\pi^{-1}(V)$ is an open neighborhood of x in X; since X is a scheme, it can be covered by affine opens, so there exists an affine open $U = \operatorname{Spec} R \ni x$ with $U \subseteq \pi^{-1}(V)$, hence $\pi(U) \subseteq V$.

The restricted map $\pi|_U : \operatorname{Spec} R \to \operatorname{Spec} S$ is a morphism of locally ringed spaces between affine schemes. By theorem 3.1.4, it is uniquely induced by the ring homomorphism on global sections:

$$\varphi := \Gamma(\pi|_U) : S \cong \mathcal{O}_Y(V) \xrightarrow{\pi|_U^\#(V)} \mathcal{O}_X(U) \cong R$$

(⇐) Locally affine morphisms are scheme morphisms. Suppose $\pi : X \to Y$ is an LRS morphism and every $x \in X$ admits affine neighborhoods as above. Let $\{U_i = \operatorname{Spec} R_i\}$ be an open cover of X with corresponding affine opens $\{V_i = \operatorname{Spec} S_i\}$ in Y such that $\pi(U_i) \subseteq V_i$ and $\pi|_{U_i}$ is induced by a ring homomorphism $\varphi_i : S_i \to R_i$.

Each $\pi|_{U_i}$ is an LRS morphism (by theorem 3.1.4), so the stalk map at every $p \in U_i$ is a local ring homomorphism. Since $\{U_i\}$ covers X, the stalk map at every $p \in X$ is local. Hence π is an LRS morphism between schemes, i.e., a morphism of schemes.

Example 3.1.11: Morphisms Of Schemes

1. Recall the ring map $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$. Geometrically, this corresponds to the map $f : \mathbb{A}^1_x \to \mathbb{A}^1_y$ sending $a \mapsto a^2$. Let us verify the sheaf map on a specific open set. Consider the section $s = 3/(y-4) \in \mathcal{O}_{\mathbb{A}^1_y}(D(y-4))$, defined on the open set $V = D(y-4) = \mathbb{A}^1 \setminus \{4\}$. The preimage of this set is:
$$f^{-1}(V) = \left\{a \in \mathbb{C} \mid a^2 \neq 4\right\} = \mathbb{A}^1 \setminus \{-2, 2\}$$
The induced map on sections $f_V^\# : \mathcal{O}_{\mathbb{A}^1_y}(V) \to \mathcal{O}_{\mathbb{A}^1_x}(f^{-1}(V))$ sends s to:
$$f_V^\#(s) = \frac{3}{x^2 - 4}$$
which is indeed a regular function on $f^{-1}(V) = D(x-2) \cap D(x+2)$.
2. Let $U \subseteq Y$ be an open subset of a scheme Y. There is a natural morphism of ringed spaces:
$$\iota : (U, \mathcal{O}_Y|_U) \to (Y, \mathcal{O}_Y)$$

This is the open embedding as seen in definition 2.3.11. It is important to note the behavior of the stalks. For any point $p \in U$, the stalk map $\iota_p^\# : \mathcal{O}_{Y,\iota(p)} \to \mathcal{O}_{U,p}$ is an *isomorphism* (specifically, the identity map). This distinguishes open embeddings from other types of immersions.

3. *Tautological Projection (Affine Cone):* Consider the projection from the punctured affine space to projective space:

$$\pi : \mathbb{A}_k^{n+1} \setminus \{0\} \to \mathbb{P}_k^n \qquad (x_0, \ldots, x_n) \mapsto [x_0 : \cdots : x_n]$$

This is a morphism of schemes. To see the sheaf map, we check it on the standard affine cover of $\mathbb{P}^n$. Let $U_i = D_+(x_i) \cong \mathbb{A}_k^n$ be the standard open charts of projective space. The preimage is the distinguished open set $\pi^{-1}(U_i) = D(x_i) \subseteq \mathbb{A}^{n+1} \setminus \{0\}$. The rings of sections are:

$$\mathcal{O}_{\mathbb{P}^n}(U_i) \cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \qquad \text{and} \qquad \mathcal{O}_{\mathbb{A}^{n+1}\setminus\{0\}}(D(x_i)) \cong k[x_0, \ldots, x_n]_{x_i}$$

The sheaf map $\pi^\#(U_i)$ is the natural inclusion of the degree-0 subring into the full localization:

$$k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \hookrightarrow k[x_0, \ldots, x_n]_{x_i}$$

Since these inclusions commute with further localization (restriction), they glue to form a valid sheaf morphism. Furthermore, the topological map π restricted to $D(x_i) \to U_i$ is exactly the map of spectra induced by this ring homomorphism (given by pulling back prime ideals $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathcal{O}_{\mathbb{P}^n}(U_i)$), confirming this matched pair forms a valid morphism of schemes.

4. *Structure Map of Proj:* Let $S_\bullet$ be a finitely generated graded R-algebra (where $S_0 = R$). There is a natural structure morphism:

$$\pi : \operatorname{Proj} S_\bullet \to \operatorname{Spec}(R)$$

This is induced locally. On the basic open set $D_+(f) \subseteq \operatorname{Proj} S_\bullet$ (for homogeneous f), the ring of sections is $S_{(f)}$. The map $R \to S_\bullet$ induces a map $R \to S_{(f)}$ via $r \mapsto r/1$. These local maps glue to give the global morphism π.

5. Let $\varphi : S_\bullet \to R_\bullet$ be a *surjective* homomorphism of graded rings. This induces a *closed embedding* of schemes in the opposite direction:

$$f : \operatorname{Proj} R_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$$

For example, the projection $k[x,y,z] \to k[x,y,z]/(x^2+y^2-z^2)$ induces the inclusion of the conic into $\mathbb{P}^2$.

6. Consider the projection $\pi : \operatorname{Spec}(\mathbb{C}[x,y]) \to \operatorname{Spec}(\mathbb{C}[x])$ induced by the inclusion $\mathbb{C}[x] \hookrightarrow \mathbb{C}[x,y]$.

   - *Closed Points:* A point $(x-3, y-a)$ maps to the contraction $(x-3)$. Geometrically, $(3,a) \mapsto 3$.
   - *Generic Points:* Consider the generic point of the parabola, $\eta = [(y-x^2)]$. The image is the contraction $(y-x^2) \cap \mathbb{C}[x]$. Since no nonzero polynomial in $\mathbb{C}[x]$ alone lies in $(y-x^2)$, the intersection is (0). Thus, the parabola maps to the generic point of the affine line!

7. Let $p \in X$ be a point of a scheme. We can form a scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$ induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,p}$ for open neighborhoods U of p. For $X = \mathbb{A}^1_k$ and $p = (x)$, this is the map $\operatorname{Spec}(k[x]_{(x)}) \to \operatorname{Spec}(k[x])$.

   The scheme $\operatorname{Spec}(\mathcal{O}_{X,p})$ can be thought of as the "intersection of all open neighborhoods of p," but it is *not* an open embedding: $\operatorname{Spec}(\mathcal{O}_{X,p})$ has only the points of X that specialize to p (i.e., the prime ideals contained in $\mathfrak{m}_p$), which typically includes the generic point but not other closed points. The key property is that the stalk map at p is an isomorphism $\mathcal{O}_{X,p} \xrightarrow{\sim} \mathcal{O}_{\operatorname{Spec}(\mathcal{O}_{X,p}),\,\mathfrak{m}_p}$, since localizing a local ring at its maximal ideal has no effect.

8. *The Frobenius Morphism (Non-isomorphism bijection):* Let $X = \operatorname{Spec}(\mathbb{F}_p[t])$. Consider the ring homomorphism $\varphi : \mathbb{F}_p[t] \to \mathbb{F}_p[t]$ given by $t \mapsto t^p$. This induces a scheme morphism $F : X \to X$.

   - *Topologically:* The map on points is $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. For a closed point $(t-a)$ where $a \in \overline{\mathbb{F}_p}$, we compute:

     $$\varphi^{-1}((t-a)) = \{f \in \mathbb{F}_p[t] \mid f(t^p) \in (t-a)\} = \{f \mid f(a^p) = 0\} = (t - a^p)$$

     Since $a^p = a$ in $\mathbb{F}_p$ (by Fermat's little theorem), we get $\varphi^{-1}((t-a)) = (t-a)$. Similarly, $\varphi^{-1}((0)) = (0)$. Thus, F is the *identity* on the underlying topological space.
   - *Sheaf-theoretically:* The map on sections is $t \mapsto t^p$. This is *not* an isomorphism: it is not surjective, as t is not in the image.

   Thus, F is a homeomorphism that is not an isomorphism of schemes. This distinction is invisible to the Zariski topology but detected by the structure sheaf.

   To give some insight on how we can extract information from this geometric perspective: once we define differentials in section 5.4, we shall see that $d(t^p) = pt^{p-1}\,dt = 0$, and hence the tangent map between tangent spaces will be the *zero map* at each point. We can think of this as "infinitesimal shrinking" at each point, while globally it is the identity!

If the codomain of a scheme morphism is an affine scheme, then we can completely determine the behaviour of the scheme morphisms by looking at the ring morphism between the global sections:

Proposition 3.1.12: Adjunction of Affine Schemes

Let $\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R))$ be the collection of morphism of scheme from a scheme X to an $\operatorname{Spec}(R)$. Then it is in natural bijection to the ring homomorphisms $\operatorname{Hom}_{\text{Ring}}(R, \mathcal{O}_X(X))$:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}(R)) \cong \operatorname{Hom}_{\mathbf{CRing}}(R, \mathcal{O}_X(X))$$

In particular, The map $\theta : X \to \operatorname{Spec}(\Gamma(X, \mathcal{O}_X))$ is an injection. Furthermore, this is an isomorphism if and only if X is an affine scheme.

Note that the ring $\mathcal{O}_X(X)$ can be very complex, even in nice cases where X is a finite-type k-scheme, for example it need not be finitely generated.

***Proof*:**
If $X = \operatorname{Spec}(S)$, then theorem 3.1.4 gives the bijection. Next, map $\pi : X \to \operatorname{Spec}(R)$ certainly induces a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$, we shall show that a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$ induces a unique scheme map $\pi : X \to \operatorname{Spec}(R)$ and that this map is the inverse of the above map. By definition of a scheme:

$$\pi : \bigcup_i \operatorname{Spec}(S_i) \to \operatorname{Spec}(R)$$

where we get a restriction map $\pi_i : \operatorname{Spec}(S_i) \to \operatorname{Spec}(R)$ along with sheaf morphisms, for which the global section map is $\varphi_i : R \to \mathcal{O}_X(\operatorname{Spec}(S_i))$. As $\operatorname{Spec} S_i \subseteq X$ is an open subset, there is a natural restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X(\operatorname{Spec} S_i)$. We thus get a composition:

$$R \to \mathcal{O}_X(X) \to \mathcal{O}_X(\operatorname{Spec} S_i)$$

The map $R \to \mathcal{O}_X(\operatorname{Spec} S_i)$ is uniquely determined by the ring homomorphism $S_i \to R$, and as X is covered by the $\operatorname{Spec} S_i$, by lemma 3.1 .6 there is a unique way to glue the maps of $\mathcal{O}_X(\operatorname{Spec} S_i)$ to $\mathcal{O}_X(X)$, completing the proof.

The converse is not true:

Example 3.1.13: Maps From Affine To Projective

Take $\operatorname{Hom}_{\mathrm{Sch}}\left(\operatorname{Spec}\mathbb{Q}[x], \mathbb{P}^1_{\mathbb{Q}}\right)$ and $\operatorname{Hom}_{\mathbf{Ring}}(\mathbb{Q}, \mathbb{Q}[x])$ (recall that $\mathcal{O}_X\left(\mathbb{P}^1_{\mathbb{Q}}\right) \cong \mathbb{Q})$). Then

$$\operatorname{Hom}_{\mathrm{Ring}}(\mathbb{Q}, \mathbb{Q}[x]) = \{\mathrm{id}\}$$

namely since since the units of $\mathbb{Q}$ must map to the units of $\mathbb{Q}[x]$, which are all contains in $\mathbb{Q}$, and as it must be that $1 \mapsto 1$, we are forced to have the identity. However, there are multiple maps from Spec $\mathbb{Q}[x]$ to $\mathbb{P}^1_k$; simply map Spec $\mathbb{Q}[x]$ into the open subset $U_0 \cong k[y] \subseteq \mathbb{P}^1_k$ by taking $x \mapsto p(y)$ for some polynomial $p(y) \in k[y]$. More generally, we can let choose a field k[a]

The intuition is that $\mathcal{O}_X(X)$ in general contains less information than X as the global sections on a general scheme need to restrict to local sections on affine open subsets, and hence there may be in general less of them.

Show that the canonical morphism $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ is an isomorphism if and only if X is a affine.

[a]Note that an [injective] ring homomorphism $k \to k$ need not be surjective, consider $F(x) \to F(x)$ given by $x \mapsto x^2$, hence the argument requires a bit more work

Using proposition 3.1.12, we can get a few more interesting results:

Example 3.1.14: Morphisms to Affine and Projective Space

1. *Maps to Affine Space are Tuples of Functions:* Let X be *any* scheme (not necessarily affine). What is a morphism $X \to \mathbb{A}^n_k$? Using the Adjunction property (proposition 3.1.12), we have:

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^n_k) \cong \operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} k[t_1, \ldots, t_n])$$

$$\cong \operatorname{Hom}_{\mathbf{Ring}}(k[t_1, \ldots, t_n], \mathcal{O}_X(X))$$

A ring homomorphism from a polynomial ring is uniquely determined by where it sends the variables t_i. Let f_i be the image of t_i in $\Gamma(X, \mathcal{O}_X)$. Thus, a morphism to affine space is exactly equivalent to choosing n global functions $(f_1, \ldots, f_n)$ on X. This generalizes the classical fact that a map to $\mathbb{R}^n$ is given by n coordinate functions.

Notice how this also shows that there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$ (exercise 3.1.2(4)).

s

2. *Maps to Projective Space (The need for Line Bundles):* One might hope to define a map $X \to \mathbb{P}^n_k$ similarly by choosing $n+1$ global functions $f_0, \ldots, f_n$ to serve as homogeneous coordinates $[f_0, \ldots, f_n]$. This naturally fails when using the structure sheaf since global functions on projective space are just constants, which would make such maps trivial (as illustrated in example 3.1.13). Instead, the coordinate functions x_i on $\mathbb{P}^n$ are sections of a twisted sheaf, not the structure sheaf. Consequently, to define a scheme morphism to projective space, one requires a *Line Bundle* (an invertible sheaf $\mathcal{L}$) on X and $n+1$ global sections $s_i \in \Gamma(X, \mathcal{L})$ that do not vanish simultaneously. This distinction – that maps to affine space use functions, while maps to projective space use bundles – is the primary motivation for the study of *Invertible Sheaves* in section 5.2.1.

Our last result in this section is a special case where gluing affine schemes *does* produce an affine scheme:

Corollary 3.1.15: Characterizing Affine Schemes Locally

Let X be a scheme. Then X is an affine scheme if and only if there is a finite subset of elements $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that X_f is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$

***Proof*:**
If X is affine, take $f_1 = 1$. Conversely, assume there exist $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that each X_{f_i} is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$. This ideal condition implies there exist $a_i \in \mathcal{O}_X(X)$ such that $\sum_{i=1}^n a_i f_i = 1$. Let $R = \mathcal{O}_X(X)$. We claim that $X \cong \operatorname{Spec} R$.

By proposition 3.1.12, the identity homomorphism on R induces a canonical morphism of schemes:

$$\theta : X \longrightarrow \operatorname{Spec} R$$

To show θ is an isomorphism, it suffices to verify it locally on an open cover of the target $\operatorname{Spec} R$. The condition $\sum_{i=1}^n a_i f_i = 1$ ensures that the distinguished open sets $D(f_1), \ldots, D(f_n)$ form an open cover of $\operatorname{Spec} R$.

For each i, the preimage of $D(f_i)$ under θ is precisely the locus of points in X where the global section f_i does not vanish, which is by definition X_{f_i}. Since the $D(f_i)$ cover $\operatorname{Spec} R$, their preimages X_{f_i} necessarily cover X.

Consider the restriction of θ to one of these patches:

$$\theta|_{X_{f_i}} : X_{f_i} \longrightarrow D(f_i) \cong \operatorname{Spec}(R_{f_i})$$

By our hypothesis, X_{f_i} is an affine scheme. Because the global sections of a scheme on a distinguished open set defined by a global function f_i are given by the localization of the global sections ring, we have $\mathcal{O}_X(X_{f_i}) \cong R_{f_i}$.

Thus, by corollary 3.1.5, the morphism $\theta|_{X_{f_i}}$ corresponds exactly to the identity ring isomorphism $R_{f_i} \xrightarrow{\sim} R_{f_i}$. Hence, each restriction $\theta|_{X_{f_i}}$ is an isomorphism of schemes.

Since θ is a morphism of schemes that restricts to an isomorphism on the sets of an open cover $\{X_{f_i}\}$ mapping to an open cover $\{D(f_i)\}$, we can apply the gluing lemma for morphisms (lemma 3.1.9). The local inverses glue uniquely to form a global inverse, proving that θ is an isomorphism and $X \cong \operatorname{Spec} R$.

***Proof*:**
If X is affine, say $X = \operatorname{Spec} R$, taking $f_1 = 1$ trivially satisfies the conditions since $X_1 = X$ and $(1) = R$.

Conversely, assume such $f_1, \dots, f_n$ exist. Let $R = \mathcal{O}_X(X)$. By the adjunction property (proposition 3.1.12), the identity map on R induces a canonical scheme morphism:

$$\theta : X \longrightarrow \operatorname{Spec}(R)$$

Because $(f_1, \dots, f_n) = R$, the basic open sets $D(f_i) \cong \operatorname{Spec}(R_{f_i})$ form an affine open cover of $\operatorname{Spec}(R)$.

By the definition of the canonical map θ, the preimage of $D(f_i)$ is exactly the non-vanishing locus of f_i on X, so $\theta^{-1}(D(f_i)) = X_{f_i}$.

Since X_{f_i} is affine by assumption, it is isomorphic to the spectrum of its global sections. A standard property of schemes tells us that the global sections on the non-vanishing locus of a global function are given by localization: $\mathcal{O}_X(X_{f_i}) \cong R_{f_i}$. Thus, $X_{f_i} \cong \operatorname{Spec}(R_{f_i})$.

Under the equivalence of categories $\mathbf{Sch}^{\mathrm{Aff}} \simeq \mathbf{CRing}^{\mathrm{op}}$, the restricted morphism

$$\theta|_{X_{f_i}} : X_{f_i} \longrightarrow D(f_i)$$

corresponds exactly to the identity ring homomorphism $R_{f_i} \xrightarrow{\mathrm{id}} R_{f_i}$. Therefore, $\theta|_{X_{f_i}}$ is an isomorphism of affine schemes for every i.

Since θ is an isomorphism over an open cover of its target, and the preimages form an open cover of its domain, θ is a global isomorphism. Hence, $X \cong \operatorname{Spec}(R)$, meaning X is affine.

3.1.2 S-Schemes

Recall that all abelian rings can be seen as $\mathbb{Z}$-algebras, and that $\mathbb{Z}$ is initial in the category of commutative rings. From this perspective, all ring homomorphism $\varphi : R \to S$ are in bijective correspondence with morphisms that make the following diagram commute:

$$\begin{tikzcd}
R \arrow[rr] & & S \\
& \mathbb{Z} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Then we may generalize the notion of abelian groups to R-algebras by replacing $\mathbb{Z}$ with a ring R. We may further augment the restriction morphisms to R-algebra homomorphisms as we are working with commutative rings; this was done in definition definition 2.2.17. Now, this structure was given by a scheme morphism $X \to \operatorname{Spec} R$, and an R-scheme morphism is a scheme morphism between R-schemes $X \to Y$ where the following diagram commutes:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & \operatorname{Spec}(R) & \end{tikzcd}$$

This is the commutative diagram of fiber-preserving morphism, the same requirement that is imposed of fiber-bundle morphisms. And indeed, R-schemes behave very much like bundles over $\operatorname{Spec} R$ (as the examples bellow will demonstrates), and when the morphism $X \to \operatorname{Spec} R$ is nice enough (ex. it is étale, see ref:HERE), then it shall even be locally trivial.

Thus, it is natural to consider the base space to be an arbitrary scheme:

Definition 3.1.16: S-Scheme

Let $\mathbf{Sch}_S$ be the category who objects are scheme morphisms:

$$X \to S$$

and whose morphisms are commutative diagrams:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

As compositions, $f \circ \pi_X = \pi_Y$. The objects of this category or called *S-scheme*. The morphism of schemes $X \to S$ is called the *structure morphism*.

To see what extra structure is added, let us start the case of $S = \operatorname{Spec}(R)$. Then each open subset of X has the structure of an R-algebra, and these generalize definition 2.2.17. If $S = \operatorname{Spec}(R)$, then we would often say that the S-scheme (i.e. $\operatorname{Spec}(R)$-scheme) is an R-scheme

Proposition 3.1.17: Final Object Of Scheme

In **Sch**, the scheme $\operatorname{Spec} \mathbb{Z}$ is the final object. This makes the category of schemes isomorphic to the category of $\mathbb{Z}$-schemes.

If k is a field, $\operatorname{Spec} k$ is the final object in the category of k-schemes.

***Proof*:**
Recall that $\mathbb{Z}$ is initial in Ring, and thus it is easy to show it is initial in Comm-Ring, then as Schemes (resp. k-schemes) are the co-categories to rings (resp. k-algebras), the result is immediate.

Working over S-schemes shall restrict to much "simpler" cases than over general schemes. For example, if studying complex analysis and complex varieties, we would expect Spec($\mathbb{C}$) to have trivial automorphisms, and we may want to interpret $\operatorname{Spec}\left(\mathbb{R}[x]/\left(x^2+1\right)\right)$ without too much "gluing" done by the automorphism ring, namely we want $\mathbb{C}$ be be trivial.

Example 3.1.18: The Category of S-schemes

An *S-scheme* is a scheme X equipped with a morphism $\pi : X \to S$, called the structure morphism. A morphism of S-schemes $f : X \to Y$ is a morphism of schemes such that the triangle with the base S commutes .

1. *Schemes over k ("Geometry" Schemes)*: If $S = \operatorname{Spec}(k)$ for a field k, an S-scheme closely resembles an algebraic set (over k). By proposition 3.1.12, the structure morphism $X \to \operatorname{Spec}(k)$ is equivalent to the data of a k-algebra homomorphism $k \to \Gamma(X, \mathcal{O}_X)$. This fixes the field of scalars, ensuring that the functions on X are k-algebras, not just rings.

   When working with polynomials, we are often interested in k-linear homomorphisms $k[x] \to k[x]$ where k is fixed (e.g., coordinate changes). If we only looked at ring homomorphisms, include field automorphisms (like complex conjugation or the Frobenius). We enforce linearity by requiring the algebra diagram to commute:

$$\begin{tikzcd} k[x] \arrow[rr, "\varphi"] & & k[x] \\ & k \arrow[ul, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

   In the language of schemes, this corresponds to a morphism of S-schemes (where $S = \operatorname{Spec}(k)$):

$$\begin{tikzcd} \operatorname{Spec}(k[x]) \arrow[rr, "f"] \arrow[dr, "\pi_X"'] & & \operatorname{Spec}(k[x]) \arrow[dl, "\pi_Y"] \\ & \operatorname{Spec}(k) & \end{tikzcd}$$

   The condition $f \circ \pi_X = \pi_Y$ ensures that the morphism preserves the underlying field structure.

2. *Projective Schemes:* Projective schemes over a field, $\mathbb{P}^n_k = \operatorname{Proj}(k[x_0, \ldots, x_n])$, are naturally k-schemes. The structure map $\mathbb{P}^n_k \to \operatorname{Spec}(k)$ is induced by the inclusion of the degree 0 scalars $k \hookrightarrow k[x_0, \ldots, x_n]$.

3. *Field Extensions as Coverings:* Let L/K be a finite field extension. Then $\operatorname{Spec}(L)$ and $\operatorname{Spec}(K)$ are both single points topologically. However, the inclusion $K \hookrightarrow L$ induces a morphism of schemes $\pi : \operatorname{Spec}(L) \to \operatorname{Spec}(K)$. We don't want to include morphisms permuting K, hence we limit to k-scheme morphisms While topologically trivial, this morphism encodes the arithmetic complexity of the extension.

3.1.19 Foreshadowing Étale Fundamental Group Later, we shall see that $\mathrm{Spec}(L)$ acts as a "covering space" for $\mathrm{Spec}(K)$. This analogy can be made more precise K, L be perfect fields and let L/K be a finite extension. Then $L = k[\alpha]$ for a root of the appropriate irreducible polynomial $m_{\alpha,L/K}$. Then taking advantage of the fact that the category of K-field extensions is *isomorphic* (not just equivalent) to the category of rings of integers with base ring $\mathcal{O}_K$ (exercise if the reader forgot) The morphism $\mathrm{Spec}\,(L) \to \mathrm{Spec}\,(K)$ is associated to a (dense) morphism $\mathrm{Spec}\,(\mathcal{O}_L) \to \mathrm{Spec}\,(\mathcal{O}_K)$ (as $\mathcal{O}_K \to \mathcal{O}_L$ is injective). In section 3.5.1, we shall see morphism between affine schemes are uniquely determined by their value on dense sets. As $\mathcal{O}_L$ and $\mathcal{O}_K$ are both dimension 1 rings, these are *covering of curves*. This covering hints at the generalization of covering spaces from algebraic topology, leading to the *Étale Fundamental Group*, which generalizes the Galois group $\mathrm{Gal}(L/K)$ to schemes, allowing us to study loops and homotopy in an arithmetic setting.

Lemma 3.1.20: Stalks of k-schemes

Let X be a k-scheme and $x \in X$ so that $x = [\mathfrak{p}]$ for some prime $\mathfrak{p} \subseteq R_i = k\left[u_i\right]_i$. Then

$$\mathcal{O}_{X,\mathfrak{p}} \cong \left(k\left[u_i\right]_i\right)_{\mathfrak{p}}$$

In particular, we have that the quotient by $\mathfrak{m}_x$ is a k-algebra:

$$\kappa(x) = k(x)$$

***Proof*:**
As X is a k-scheme, each $\mathcal{O}_X(U)$ is a k-algebra, and the colimit of k-algebra's is a k-algebra (they are closed under products and equalizers, see [9]). The rest comes from proposition 1.1.5.

One of the most important first impositions for many algebraic manipulations are finiteness conditions:

Definition 3.1.21: Locally Finite Type And Finite Type

Let X, Y be S-schemes. Then a morphism $f : X \to Y$ is locally of finite type if there exists an affine open cover $V_j = \mathrm{Spec}\,(S_j)$ of Y where $f^{-1}\,(V_i)$ can be covered by affine open subsets $U_{ij} = \mathrm{Spec}\,(A_{ij})$ where A_{ij} is a finitely generated S_i-algebra, that is

$$f^{\#}\,(V_i) : S_j = \mathcal{O}_Y\,(V_j) \to \mathcal{O}_X\left(f^{-1}\,(V_i)\right) \xrightarrow{\text{res}} \mathcal{O}_Y\,(U_{ij}) = R_{ij}$$

induces a finitely generated S_j-algebra on each R_{ij}. If furthermore each $f^{-1}\,(V_j)$ can be covered by finite number of affine subsets, f is said to be of finite type.
An S-scheme X is said to be of locally finite type (resp. finite type) over S if $f : X \to S$ is of locally finite type (resp. finite type).

3.1.3 Subschemes: Closed, Locally Closed, and Neither

(I also want to include that in general the generic point with the localization at 0 is not a open or closed or locally closed subscheme, giving an example of a "believable" sub-object that doesn't fall under any notion of subscheme we looked at)

I think I want to move all immersion discussion into the next chapter. Create a new section in "Elementary Concepts" to put it in.

An overarching driver with the definition of affine schemes is that it ought to be the dual category to the category of rings. Thus, we would like it if the sub-objects in the category of schemes correspond to the quotient objects in the category of rings, and indeed given $R \to R/I$, by proposition 2.1.26 the subset $\operatorname{Spec} R/I \subseteq \operatorname{Spec} R$ is a closed subset and there exists a continuous inclusion map that is also a map of locally-ringed spaces on the sheaves. Intuitively, $R \to R/I$ adds some relations, and hence shrinks the given affine scheme. For a general scheme, this condition shall be generalized to requiring that the map on stalks be surjective (as we will see in definition 3.1.22).

Though categorically correct, it must be pointed out that there can be more interesting classes of subschemes. We may generally ask the question "if $Y \subseteq X$ is a subset and $(X, \mathcal{O}_X)$ is a scheme, is there a sheaf-structure we can put on Y that is naturally induced by X via the pullback". For one, open subsets shall have a natural structure sheaf induced by the original scheme. For another, we shall have inclusions $X \hookrightarrow Y$ where X is neither open nor closed, but it is homeomorphic onto its image and the induced sheaf map is an isomorphism (in particular, this shall be true for *local schemes*). We thus see that monomorphisms are too restrictive to fully capture all the behaviour we may wish out of a subscheme.

I would like to add here somehow that since we have a topology we care about sets that map to closed sets, you're usual monomorphisms, sets that map to open sets which is a good deal more complicated, and then locally closed sets (and unions of them) which are what you encounter most often when working with images of varieties or when as subschemes of projective space. Hence, there are a lot of non-trivial characterizations

Nevertheless, the monomorphisms which are given as open subschemes or locally closed subschemes will create a great class of schemes that we shall encounter very often and deserve to be singled out. Other notions of subschemes shall appear when we need them.

Closed Subschemes

When working with closed subschemes, we have to be a little more careful as there can be multiple scheme structure, namely since we can have varying multiplicity of the zero sets, for example Spec $k[x]/(x)$, Spec $k[x]/\left(x^2\right)$, Spec $k[x]/\left(x^3\right)$, and so forth. This comes down to the very elementary fact that $0^n = 0$ and $n \cdot 0 = 0$. For any non-zero number a, it is not the case that both $a^n \neq a$ and $n \cdot a = a$ (for all $n \in \mathbb{Z}$) be true unless $a = 0$. Hence, up to isomorphism, open subschemes have a unique compatible sheaf-structure with the parent scheme, while closed subscheme can have multiple:

Definition 3.1.22: Closed Immersion and Closed Subschemes

Let $f : Y \to X$ be a continuous map along with the induced pushforward $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$. Then:

1. if f induces a homeomorphism from Y onto it's image which is closed in X, and $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective, then f is said to be a *closed immersion*.
2. A *closed subscheme* $Y \subseteq X$ is a closed subset where the structure sheaf on Y makes $\mathrm{id} : Y \hookrightarrow X$ is a closed immersion.

The main difference between a closed subscheme and closed immersion is that a closed immersion comes with an isomorphism, while a closed subscheme Y is always given with the natural inclusion as the map. Often, the inclusion $\iota : Y \hookrightarrow X$ is called the closed subscheme as it is key to induced the sheaf structure on Y. Note that the compliment of a closed subscheme is a unique open subscheme, but the compliment of an open subscheme has multiple closed subscheme, usually infinitely many!

Example 3.1.23: Closed Subscheme

1. Take $\{0\} \subseteq \mathbb{A}^1_k$. This point corresponds to the Spec $k[x]/(x) \cong \operatorname{Spec} k$. Then f : Spec $k \to$ Spec $k[x]$ mapping $\eta \mapsto 0$ is a closed immersion: its image being closed, the map $f^\#$: $\mathcal{O}_{\operatorname{Spec} k[x]}(\operatorname{Spec} k[x]) \to \mathcal{O}_{\operatorname{Spec} k(x)}(\operatorname{Spec} k(x))$ is the map
$$k[x] \to k$$
is naturally surjective, hence all its localizations are surjective, hence we have a closed subscheme.

2. Now consider Spec $k[x]/\left(x^2\right) \to \operatorname{Spec} k[x]$. The image of f is the same as the above and so is still closed, and the sheaf morphism:
$$k[x] \to k[x]/\left(x^2\right)$$
is surjective, hence this is also a closed subscheme. More generally, $\operatorname{Spec} k[x]/\left(x^n\right) \to \operatorname{Spec} k[x]$ is a closed subscheme. This example shows that the sheaf of a subscheme is *not* a sub-sheaf of the original scheme.

   Furthermore, shows that the single point $\{0\}$ has countably many closed subscheme associated to it, depending how much "infinitesimal data" is preserved!

3. Let $I \subseteq R$ be an ideal. Then $\operatorname{Spec} R/I \to$ Spec R is a closed immersion, it's image being closed as it is equal to $V(I)$, and the maps $R \to R/I$ are all surjective.

4. Let $\mathbb{P}^1_k$ and $\mathbb{P}^2_k$ be usual projective spaces. Then the inclusion $[S : T] \to [S : T : 0]$ is a closed immersion.

Having the sheaf map be surjective is not enough. For example $D(x) \cup D(y) \to \mathbb{A}^2_k$ is open, not closed, but the sheaf map is surjective (even an isomorphism). Another example will come when studying non-separated schemes. Another example is the projection $f : \mathbb{P}^1_k \times \mathbb{A}^1_k \to \mathbb{A}^1_k$, showing just how necessary the condition is.

Furthermore, being a topological closed immersion does not imply the sheaf map is surjective. The normalization of the cuspidal cubic $y^2 = x^3$ given by $\mathbb{A}^1_k \to \operatorname{Spec} k[x, y]/(y^2 - x^3) = Y$ given by $t \mapsto (t^2, t^3)$ is an example (prove this now if you haven't seen this before). The singularity is to blame (or algebraically the lack of integral closure). In particular, notice that y/x is rational on Y, but $t^3/t^2 = t$ is regular.

Lemma 3.1.24: Properties of Closed Immersion

1. f is a closed immersion if and only if for every affine fine open subset $\text{Spec}(S) \subseteq Y$ where $f^{-1}(\text{Spec}(S)) = \text{Spec}(R), S \to R$ is surjective.
2. Every surjective ring homomorphism $S \to R$ induces a closed immersion $\text{Spec}(R) \to \text{Spec}\ (\ S\ )$
3. The composition two closed immersions is a closed immersion.

***Proof*:**

1. see exercise ref:HERE. A proof using ideal sheaves is presented in section 5.1.3
2. exercise
3. Use the two above parts to conclude the result by using the fact that the composition of surjective maps is surjective.

As closed subschemes are given by ideals for each open subsets, it would be good to describe all possible closed subschemes of a scheme by looking at an "ideal sub-sheaf" of the structure sheaf. This requires some more elaboration on the different types of sheaves that can be put on a scheme, which will be covered in section 4.1.2.

Locally Closed Subschemes: Immersion

When working with classical projective space, closed subsets of an open subset need not be open in the projective variety. These are common subscheme and are given a name:

Definition 3.1.25: Locally Closed Subscheme

Let X be a scheme. Then $Y \subseteq X$ is *locally closed* if it is a closed subscheme of an open subscheme that is the intersection of a closed subset $C \subseteq X$ with an open subset $U \subseteq X$ $(Y = U \cap C)$ with the natural subscheme structure.
Often, a map $f : X \to Y$ is simply called an *immersion* if the image is an locally closed immersion.

3.1.26 Closed vs. Locally Closed Note that closedness can be verified on an open cover, namely if $C \subseteq X$ is a subscheme and $\{U_i\}$ is an open cover of X, then C is a closed subscheme if it is closed *on each* $U_i \cap C$. Locally closed requires C to be closed be intersection of a closed set with a single open set.

(worth defining since schemes are inherently local defined, and we would computationally often want to define a subset that is locally closed in some choice of affine open subsets)

(also important because the diagonal map $X \to X \times_S X$ is always a locally closed immersion, and closedness is what makes X a separated S-scheme).

Proposition 3.1.27: Immersions, and Open/Closed Immersions

let f, g be open and closed immersions (respectively) with the appropriate domain and codomain. Then they are both immersions (i.e. both locally closed immersions).

***Proof*:**
exercise.

(perhaps finish somewhere here before moving on that another meaningful and natural inclusion is $\operatorname{Spec} R_\mathfrak{p} \to \operatorname{Spec} R$ (the "point-morphism") which is neither open nor closed. But it will be shown to be (trivially) affine, but more interestingly it is a flat morphism).

Injective Ring Homomorphisms: Dominant Morphism

This feels like the clearly wrong spot for it. This should be in the rational map section, and there should instead be a word at the beginning of this section saying that we shall deal with the case where the ring homomorphism is injective in section REF:HERE.

In section 3.1.3, we showed that surjective ring homomorphisms $f : R \to S$ correspond to closed immersions $f^a : \operatorname{Spec} S \to \operatorname{Spec} R$. If f is injective, we need not get a surective map of affine schemes, but we get the next closest thing:

Definition 3.1.28: Dominant Morphism

A scheme morphism $f : X \to Y$ is called *dominant* if the image of f is dense in Y.

Proposition 3.1.29: Dominant Morphism and Affine Schemes

Let $f : R \to S$ be an injective ring homomorphism. Then the corresponding affine scheme map is a dominant map.

***Proof*:**
exercise.

3.1.4 More on Projective Schemes

In section 2.4, we defined the prototypical projective scheme $\mathbb{P}^n_R = \operatorname{Proj} R[x_0, \ldots, x_n]$. To do geometry, we need to understand its closed subschemes.

As we will prove rigorously in the next chapter using ideal sheaves (specifically corollary 5.2.10), closed subschemes of a projective scheme are governed by homogeneous ideals. For now, we take this as our primary geometric construction: given a graded ring $S_\bullet$ and a homogeneous ideal $I \subseteq S_\bullet$, the canonical surjective graded ring homomorphism $S_\bullet \twoheadrightarrow S_\bullet/I$ induces a closed immersion:

$$\operatorname{Proj}(S_\bullet/I) \hookrightarrow \operatorname{Proj} S_\bullet$$

Conceptually, this is the exact generalization of the affine case. If $S_\bullet$ is a finitely generated graded R-algebra generated in degree 1, we can choose $n+1$ degree 1 generators to form a surjective graded map $R[x_0, \ldots, x_n] \twoheadrightarrow S_\bullet$. This implies $S_\bullet \cong R[x_0, \ldots, x_n]/I$, meaning $\operatorname{Proj} S_\bullet$ naturally embeds as a closed subscheme of $\mathbb{P}^n_R$. Choosing these generators is geometrically equivalent to choosing projective coordinates.

We can classify the simplest closed subschemes of $\mathbb{P}^n_k$ by the generators of their ideals:

Definition 3.1.30: Projective Hypersurface and Linear Spaces

A *hypersurface* of $\mathbb{P}^n_k$ is a closed subscheme $\operatorname{Proj} k[x_0, \ldots, x_n]/(f)$ given by a single homogeneous polynomial f. The *degree* of the hypersurface is the degree of f.

A hypersurface of degree 1 is called a *hyperplane*. A hypersurface of degree $2, 3, 4, 5 \ldots$ is called a *quadric*, *cubic*, *quartic*, *quintic*, and so forth.

If $n = 2$ (so we are in $\mathbb{P}^2_k$), a hypersurface is called a *curve*. A hypersurface of degree $1, 2, 3$ is called a *line*, *conic curve* (or *conic*), and *cubic curve* respectively.

A *linear space* is a closed subscheme cut out by a collection of degree 1 homogeneous polynomials (an intersection of hyperplanes). Any injective linear transformation $V \to W$ of vector spaces induces a closed immersion $\mathbb{P}(V) \hookrightarrow \mathbb{P}(W)$ whose image is a linear space.

Recall that $\mathbb{P}^n_k$ has only constant global functions, so these subschemes are cut out by regular functions defined only on open affine patches. Let us look at some classic examples.

Example 3.1.31: Closed Projective Subschemes

1. *Twisted Cubic*: Take the homogeneous ideal $I = (xy - zw, x^2 - wy, y^2 - xz)$ and look at the closed subscheme $\operatorname{Proj} k[w, x, y, z]/I \subseteq \mathbb{P}^3_k$. This shape is called the *twisted cubic*. It is a *curve*, and it is isomorphic to $\mathbb{P}^1_k$ via the map induced by the graded ring homomorphism:

$$k[w, x, y, z] \to k[s, t] \qquad (w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$$

   To see this is an isomorphism of schemes, one can take the standard affine open sets and show it gives closed immersions on the charts. For example, on the chart $D_+(w)$ where $w \neq 0$ and $s \neq 0$, the map of degree 0 localizations is:

$$k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right] \to k\left[\frac{t}{s}\right] \qquad \frac{x}{w} \mapsto \frac{t}{s},\ \frac{y}{w} \mapsto \frac{t^2}{s^2},\ \frac{z}{w} \mapsto \frac{t^3}{s^3}$$

2. *Lines on a Quadric Surface*: The following shows a way to parameterize certain projective varieties. Take the quadric surface $V_+(wz - xy) \subseteq \mathbb{P}^3_k$ (with projective coordinates w, x, y, z). For any two elements $\alpha, \beta \in k$ that are not both zero, notice that as $[c, d] \in \mathbb{P}^1_k$ varies, $[\alpha c, \beta c, \alpha d, \beta d]$ forms a line completely contained on the surface. Show there is one other parameterization of this surface via lines, and show it must be the case that these are the only two families (hint: one parameterization contains $V_+(w, x)$ and the other $V_+(w, y)$).

Another important construction arises from taking a different grading of the polynomial ring, denoted

$S_{d\bullet}$. This allows us to "linearize" high-degree equations. First, we must establish that altering the grading in this specific way does not change the underlying abstract scheme.

Theorem 3.1.32: Isomorphism of Veronese Subrings

Let $S_\bullet = R[x_0, \ldots, x_n]$ be the standard graded polynomial ring. For any integer $d > 0$, let $S_{d\bullet} = \bigoplus_{m=0}^{\infty} S_{md}$ be the subring consisting of degrees that are multiples of d. Then there is a canonical isomorphism of schemes:

$$\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$$

***Proof*:**
We construct this isomorphism by comparing the standard affine open covers. By lemma 2.4.2, the scheme $\operatorname{Proj} S_\bullet$ is covered by the distinguished open sets $D_+(x_i) \cong \operatorname{Spec}((S_{x_i})_0)$ for $i = 0, \ldots, n$.

For the subring $S_{d\bullet}$, the elements x_i^d have degree d and therefore reside in $S_{d\bullet}$. The corresponding distinguished open sets $D_+(x_i^d)$ cover $\operatorname{Proj} S_{d\bullet}$, and applying lemma 2.4.2 again, each is isomorphic to $\operatorname{Spec}(((S_{d\bullet})_{x_i^d})_0)$.

We claim the rings $(S_{x_i})_0$ and $((S_{d\bullet})_{x_i^d})_0$ are identically the same. An arbitrary element in $(S_{x_i})_0$ is of the form a/x_i^k, where $a \in S_k$ is a homogeneous polynomial of degree k. We can rewrite this fraction by multiplying the numerator and denominator by x_i^{kd-k}:

$$\frac{a}{x_i^k} = \frac{a x_i^{kd-k}}{x_i^{kd}} = \frac{a x_i^{kd-k}}{(x_i^d)^k}$$

The new numerator $a x_i^{kd-k}$ has degree $k + (kd - k) = kd$, meaning it is an element of S_{kd}, which is a valid component of $S_{d\bullet}$. The denominator is a power of x_i^d. Therefore, this fraction is naturally an element of $((S_{d\bullet})_{x_i^d})_0$. Since this containment goes both ways, the rings are equal. These local isomorphisms canonically glue together to yield the global isomorphism $\operatorname{Proj} S_{d\bullet} \cong \operatorname{Proj} S_\bullet$.

Definition 3.1.33: Veronese Embedding

Let $S_\bullet = k[x_0, \ldots, x_n]$ and let $S_{d\bullet}$ be the d-tuple Veronese subring. By theorem 3.1.32, $\operatorname{Proj} S_{d\bullet} \cong \mathbb{P}_k^n$. We can view $S_{d\bullet}$ as being generated in "degree 1" by the monomials of degree d from the original ring. There are $N + 1 = \binom{n+d}{d}$ such monomials. Mapping a new polynomial ring $k[y_0, \ldots, y_N] \twoheadrightarrow S_{d\bullet}$ by sending the variables y_i to these monomials induces a closed immersion:

$$\mathbb{P}_k^n \cong \operatorname{Proj} S_{d\bullet} \hookrightarrow \mathbb{P}_k^N$$

This map is called the *d-tuple Veronese Embedding.*

The geometric utility of the Veronese embedding is that a hypersurface of degree d in $\mathbb{P}_k^n$ gets mapped exactly to a hyperplane intersection (degree 1) in $\mathbb{P}_k^N$.

3.1.34 Foreshadowing: Sheaves and Morphisms In the next chapter, we will show that morphisms to projective space are governed by invertible sheaves. The Veronese embedding is precisely the morphism $X \to \mathbb{P}^N_k$ generated by the global sections of the twisting sheaf $\mathcal{O}_X(d)$.

As noted earlier, choosing a set of degree 1 generators is equivalent to choosing an embedding into a specific ambient projective space $\mathbb{P}^N_R$. Therefore, it is necessary to distinguish the intrinsic geometry of the abstract scheme from the extrinsic geometry of its embedding.

Consider the scheme $X = \mathbb{P}^1_k$. We can embed it in several ways:

1. *Degree 1*: Use $S_\bullet = k[x, y]$. It is generated in degree 1 by 2 elements. This gives the identity embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^1_k$ as a straight line.
2. *Degree 2*: Consider the 2-tuple Veronese subring $S_{2\bullet} = k[x^2, xy, y^2]$. As an abstract scheme, $\operatorname{Proj} S_{2\bullet} \cong \mathbb{P}^1_k$. To write $S_{2\bullet}$ as a ring generated in degree 1, we require 3 generators: let $u = x^2$, $v = xy$, $w = y^2$. They satisfy $uw = v^2$, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^2_k$. The image is a conic curve.
3. *Degree 3*: The 3-tuple Veronese subring $S_{3\bullet} = k[x^3, x^2y, xy^2, y^3]$ requires 4 generators, yielding an embedding $\mathbb{P}^1_k \hookrightarrow \mathbb{P}^3_k$. The image is the twisted cubic.

The line in $\mathbb{P}^1_k$, the conic in $\mathbb{P}^2_k$, and the twisted cubic in $\mathbb{P}^3_k$ are isomorphic as abstract schemes. Extrinsically, however, they are embedded differently, have different geometric degrees, and are cut out by different homogeneous ideals.

One might ask why we do not always restrict to degree 1 generators. Certain equations become unnecessarily complicated if forced into a standard degree 1 embedding, but simplify if we assign different *weights* (degrees) to the variables. For example, embedding certain curves or moduli spaces into standard $\mathbb{P}^N$ via Veronese embeddings increases the ambient dimension N significantly and requires many equations to cut out. This motivates the following construction.

Definition 3.1.35: Weighted Projective Space

Let $S_\bullet = k[a_1, \ldots, a_n]$ be a graded ring where each variable a_i is assigned a specific degree $d_i \in \mathbb{Z}_{>0}$. Then $\operatorname{Proj} k[a_1, \ldots, a_n]$ is called a *weighted projective space* and is denoted $\mathbb{P}(d_1, \ldots, d_n)$.

Geometrically, over a field, assigning different weights corresponds to taking the quotient of $\mathbb{A}^n_k \setminus \{0\}$ by the action of the multiplicative group $\mathbb{G}_m$ (i.e. $(\mathbb{A}^*_k, \cdot)$) where the scaling acts as $\lambda \cdot (x_1, \ldots, x_n) = (\lambda^{d_1} x_1, \ldots, \lambda^{d_n} x_n)$.

Example 3.1.36: Weighted Projective Space Examples

1. *Show that $\mathbb{P}(m, n) \cong \mathbb{P}^1_k$ for any positive integers m, n.*

   Let $S_\bullet = k[x, y]$ with $\deg(x) = m$ and $\deg(y) = n$. The scheme is covered by the two open sets $D_+(x)$ and $D_+(y)$. For $D_+(x) \cong \operatorname{Spec}((S_x)_0)$, the degree 0 elements are generated by $u = y^m/x^n$. Thus, $(S_x)_0 \cong k[u]$, which is isomorphic to $\mathbb{A}^1_k$. Similarly, for $D_+(y) \cong \operatorname{Spec}((S_y)_0)$, the degree 0 elements are generated by $v = x^n/y^m$. Thus, $(S_y)_0 \cong k[v]$, which is also isomorphic to $\mathbb{A}^1_k$. On the intersection $D_+(xy)$, the variable v is exactly u^{-1}. This translates to gluing two copies of $\mathbb{A}^1_k$ along $\mathbb{A}^1_k \setminus \{0\}$ via the map $u \mapsto u^{-1}$, which is the precise definition of $\mathbb{P}^1_k$.

2. *Show that* $\mathbb{P}(1,1,2) \cong \operatorname{Proj} k[u,v,w,z]/(uw - v^2)$.

   Let $S_\bullet = k[x,y,t]$ with $\deg(x) = 1, \deg(y) = 1$, and $\deg(t) = 2$. By theorem 3.1.32, $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{2\bullet}$. The subring $S_{2\bullet}$ is generated by all degree 2 monomials from $S_\bullet$. These monomials are exactly x^2, xy, y^2, and t. Let us map a standard polynomial ring into $S_{2\bullet}$ by setting $u = x^2$, $v = xy$, $w = y^2$, and $z = t$. The only algebraic relation between these generators is $uw = x^2y^2 = (xy)^2 = v^2$. The generator z is algebraically independent of the others. Therefore, $S_{2\bullet} \cong k[u,v,w,z]/(uw - v^2)$. Since the spaces are isomorphic, it follows that $\mathbb{P}(1,1,2)$ can be realized as the quadric cone $\operatorname{Proj} k[u,v,w,z]/(uw - v^2)$ in standard $\mathbb{P}^3_k$.

3. *Smooth Compactifications of Hyperelliptic Curves.*

   Weighted projective spaces are highly practical for compactifying curves without introducing artificial singularities. Consider the affine hyperelliptic curve $y^2 = x^5 + 1$. If we attempt to compactify this by homogenizing it into standard $\mathbb{P}^2_k$ (where all variables have degree 1), we obtain the curve $y^2z^3 = x^5 + z^5$. Calculating the Jacobian reveals that this curve has a singularity at the point at infinity $[0 : 1 : 0]$.

   However, if we embed the curve into the weighted projective space $\mathbb{P}(1,1,3)$ with coordinates (x,z,y) having degrees $1, 1$, and 3 respectively, we can form a homogeneous equation of total degree 6:
   $$y^2 = x^5z + z^6$$
   In this weighted space, the curve remains smooth at infinity.

Exercise 3.1.1

1. Put Exercise 8.1.M from Vakil
2. Show that the composition of two locally closed immersions is locally closed.

3.1.5 The Category of Schemes

Having defined schemes, morphisms, subschemes, and projective schemes, we have formally constructed the category of schemes, denoted **Sch** (or $\mathbf{Sch}_S$ for S-schemes). Before moving on, it is valuable to look at the categorical properties of **Sch** as a whole.

How well-behaved is this category? Does it have limits and colimits? The short answer is that the local affine structure is well-behaved, but the global structure is much more complex.

The Affine Subcategory

Let $\mathbf{Sch}^{\mathbf{Aff}}$ be the full subcategory of affine schemes. By the definition of the spectrum functor, we have an anti-equivalence of categories:

$$\mathbf{Sch}^{\mathbf{Aff}} \simeq \mathbf{CRing}^{op}$$

Because the category of commutative rings is both complete (has all limits) and cocomplete (has all colimits), the category $\mathbf{Sch}^{\mathbf{Aff}}$ is also completely well-behaved: it has all limits and colimits.

As discussed in section 2.3.3, operations here simply mirror the algebraic operations in rings (for example, the categorical coproduct of affine schemes corresponds to the direct product of their rings).

For instance, the initial object in **CRing** is $\mathbb{Z}$, which means the terminal object in $\mathbf{Sch}^{\mathbf{Aff}}$ is $\operatorname{Spec}\mathbb{Z}$. In fact, $\operatorname{Spec}\mathbb{Z}$ is the terminal object in the entirety of **Sch**. To prove this quickly, let X be any scheme, and let X be covered by affine open sets $\operatorname{Spec} A_i$. A morphism $X \to \operatorname{Spec}\mathbb{Z}$ restricts to local morphisms $\operatorname{Spec} A_i \to \operatorname{Spec}\mathbb{Z}$, which correspond exactly to ring homomorphisms $\mathbb{Z} \to A_i$. Since $\mathbb{Z}$ is the initial object in the category of rings, there exists exactly one such ring homomorphism for each A_i. Because these local morphisms are completely unique, they trivially agree on all overlaps and glue together to form a unique global morphism $X \to \operatorname{Spec}\mathbb{Z}$.

The Projective Subcategory

On the opposite end of the spectrum is the subcategory of projective schemes over a fixed base ring. While affine schemes are determined entirely by their global functions, connected projective schemes have only constant global functions.

Categorically, the projective subcategory is highly rigid and lacks the completeness of $\mathbf{Sch}^{\mathbf{Aff}}$. It does not admit infinite limits, and finite colimits (such as general pushouts) often fail to produce a scheme that remains projective. However, what the projective subcategory lacks in categorical completeness, it makes up for in geometric properness. Because projective schemes satisfy the valuative criterion for properness (as we will see later), curves do not escape to infinity without a limit. This makes the projective subcategory the ideal setting for intersection theory and global counting problems, whereas the affine category remains the setting for local algebraic calculations.

Subobjects and Immersions

In standard category theory, a "subobject" is defined as an equivalence class of monomorphisms. If we apply this strict categorical definition to **Sch**, we encounter non-geometric behavior. A morphism of schemes $X \to Y$ being a categorical monomorphism simply means it is left-cancellative.

To see why this is geometrically problematic, consider the affine subcategory $\mathbf{Sch}^{\mathbf{Aff}}$. Because of the anti-equivalence to **CRing**, a monomorphism of affine schemes corresponds exactly to an *epimorphism* of commutative rings. Ring epimorphisms certainly include surjective maps $A \to A/I$ (which yield closed immersions) and principal localizations $A \to A_f$ (which yield basic open immersions). However, ring epimorphisms also include total localizations like $\mathbb{Z} \to \mathbb{Q}$. Therefore, the map $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ is a strict categorical subobject in $\mathbf{Sch}^{\mathbf{Aff}}$, even though geometrically $\operatorname{Spec}\mathbb{Q}$ does not resemble a "piece" of $\operatorname{Spec}\mathbb{Z}$ (it is neither an open nor a closed subset).

For this reason, algebraic geometry rarely focuses on bare categorical monomorphisms. Instead, we restrict our attention to specific geometric embeddings: *open immersions* and *closed immersions* (and their composition, locally closed immersions). These immersions ensure that the subscheme maps homeomorphically onto a topological subspace of the target and that the structure sheaves behave compatibly.

In $\mathbf{Sch}^{\mathbf{Aff}}$, these geometric subobjects are strictly generated by ideal quotients and localizations. In the projective subcategory, the geometric subobjects are almost exclusively closed immersions; because projective schemes are proper (compact), any open immersion that is also a projective scheme must actually be a closed set (a union of connected components). Thus, geometric "subschemes" form a much narrower and better-behaved class of morphisms than general categorical subobjects.

Limits in Sch

When we glue affine schemes together to form general schemes, we lose some of the categorical completeness found in $\mathbf{Sch}^{\mathbf{Aff}}$.

1. *Finite limits exist:* As proved above, the category **Sch** contains a terminal object $\operatorname{Spec}\mathbb{Z}$. Furthermore, as we will see in section 3.2, the category of schemes also admits pullbacks (fibered products). A standard result in category theory dictates that any category with a terminal object and pullbacks automatically has all finite limits (including equalizers and finite products).

2. *Infinite limits generally do not exist:* Unlike affine schemes, general schemes are not closed under infinite products. For example, the infinite product $\prod_{i=1}^{\infty} \mathbb{P}^1_k$ does not exist in the category of schemes. The intuitive reason for this failure is that the Zariski topology behaves poorly under infinite Cartesian products, making it impossible to find a valid affine open cover for the resulting space (for a formal proof involving the failure of the representability of its functor of points, see ref:HERE).

3. *Cofiltered limits exist:* The category of schemes does have cofiltered limits (often called inverse limits), provided the transition maps are affine. This is easily justified: because the spectrum functor is contravariant, taking the inverse limit of these affine schemes translates algebraically to taking the direct limit (filtered colimit) of their corresponding commutative rings, which is a well-behaved algebraic operation. A classic example is the scheme of p-adic integers, which is the limit of affine schemes: $\operatorname{Spec}\mathbb{Z}_p = \varprojlim \operatorname{Spec}(\mathbb{Z}/p^n\mathbb{Z})$.

Colimits in Sch

1. *Arbitrary coproducts exist:* One of the simplest and most useful colimits in **Sch** is the coproduct. Any arbitrary collection of schemes $\{X_i\}$ has a coproduct given by their disjoint union $\coprod X_i$. As we established in section 2.3.2, this holds true for both finite and infinite collections.

2. *Finite colimits generally do not exist:* Outside of disjoint unions, the category of schemes does not have general pushouts or coequalizers. A standard counter-example involves trying to glue a scheme to itself along a generic point. For instance, the pushout of $\operatorname{Spec}(K) \rightrightarrows \mathbb{A}^1_K$ (where we attempt to glue the generic point to something else) fails to be a scheme. We will see this explicitly in **??** 3.2.7.

3. *Pushouts of open immersions exist:* There is one important exception to the failure of pushouts. If $f : X \to Z$ and $g : Y \to Z$ are both open immersions, their pushout always exists. This is exactly the categorical formulation of gluing schemes together, which we covered in section 2.3.2.

4. *Filtered colimits exist:* The category does admit filtered colimits (often called direct limits). A geometric example is infinite-dimensional projective space, $\mathbb{P}^\infty_k = \varinjlim \mathbb{P}^n_k$, which exists as a scheme (specifically $\operatorname{Proj} k[x_0, x_1, \dots]$), though it loses the property of being locally Noetherian.

Schemes as Colimits of Affine Schemes

We have seen that the category **Sch** contains $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$ as a full subcategory, and that general schemes are obtained by "gluing" affine pieces. We can now make the gluing process categorically precise: every scheme is a colimit of affine schemes, and the colimit has a particularly clean form.

The coequalizer picture. Let X be a scheme admitting a finite affine open cover $X = U_1 \cup \cdots \cup U_n$. Form the disjoint union

$$U = U_1 \sqcup \cdots \sqcup U_n,$$

which is itself an affine scheme: if $U_i = \operatorname{Spec}(A_i)$, then $U = \operatorname{Spec}(A_1 \times \cdots \times A_n)$. The natural map $\pi : U \to X$ (the inclusion of each U_i into X) is surjective but not injective: a point lying in $U_i \cap U_j$ appears twice in U, once in the U_i-copy and once in the U_j-copy.

To recover X from U, we must identify these redundant copies. Define the "overlap scheme"

$$R = U \times_X U = \coprod_{i,j} (U_i \cap U_j).$$

By the Build-up Lemma (lemma 2.2.29), each intersection $U_i \cap U_j$ can be covered by open sets that are simultaneously distinguished in U_i and U_j. If X is separated, the intersections $U_i \cap U_j$ are themselves affine (section 3.5.1), and R is affine as well; in general, R can be covered by affines, and the construction still works by refining the cover.

The scheme R comes equipped with two natural morphisms to U: the projections

$$s, t : R \rightrightarrows U$$

where s includes $U_i \cap U_j$ into its U_i-copy and t includes it into its U_j-copy. The scheme X is then recovered as the *coequalizer* of s and t in the category of locally ringed spaces:

$$R \underset{t}{\overset{s}{\rightrightarrows}} U \xrightarrow{\pi} X.$$

That is, X is the universal locally ringed space receiving a map from U that identifies $s(r)$ with $t(r)$ for every point $r \in R$.

Let us form an equivalence relation between the glued points categorically. We shall look at the *groupoid object* in the category of schemes (or, when everything is affine, in **Aff**). The additional structure maps are:

1. *Identity:* $e : U \to R$, the diagonal, sending a point of U_i to the "trivial overlap" of U_i with itself.
2. *Composition:* $c : R \times_{s,U,t} R \to R$, expressing transitivity: if a point of $U_i \cap U_j$ and a point of $U_j \cap U_k$ agree on U_j, they compose to a point of $U_i \cap U_k$.
3. *Inverse:* $\iota : R \to R$, the swap: a point of $U_i \cap U_j$ is also a point of $U_j \cap U_i$.

These satisfy the usual groupoid axioms (associativity, unit, inverse), which encode the fact that the relation "x and y represent the same point of X" is an equivalence relation. On the ring side, dualizing: if $U = \operatorname{Spec}(A)$ and $R = \operatorname{Spec}(B)$, the two ring homomorphisms $s^\#, t^\# : A \rightrightarrows B$ together with the identity, composition, and inverse maps make (A, B) into a *cogroupoid object* in **CRing**.

Conversely, one can *start* with affine groupoid data and produce a scheme. Given rings A and B with homomorphisms $s^\#, t^\# : A \rightrightarrows B$ and maps $e^\# : B \to A$, $c^\# : B \to B \otimes_A B$, $\iota^\# : B \to B$ satisfying the cogroupoid axioms, together with the condition that s and t are open immersions (so that the "gluing maps" identify open subsets), the coequalizer of $\operatorname{Spec}(B) \rightrightarrows \operatorname{Spec}(A)$ in the category of locally ringed spaces is a scheme.

This gives a precise sense in which the category of schemes is built from the category of affine schemes: **Sch** is obtained from **Aff** by freely adjoining coequalizers of affine groupoid presentations whose structure maps are open immersions. Every scheme admits such a presentation (by choosing an affine cover), and the result is independent of the choice (by the Affine Communication Lemma and the uniqueness of colimits).

The reader should keep this in mind when going through the functorial perspective developed in section 3.3.7, where schemes are characterized as Zariski sheaves with affine open covers. The two viewpoints are dual: the functorial approach defines schemes as certain *limits* (sheaves are defined by equalizer conditions), while the groupoid approach constructs schemes as certain *colimits* (coequalizers of equivalence relations). The equivalence of these two perspectives is a manifestation of a general principle: in algebraic geometry, limit-based and colimit-based descriptions of the same object coexist, and switching between them is sometimes the key to a proof.

Exercise 3.1.2

1. Show that the map $\operatorname{Spec} k(x) \to \operatorname{Spec} k[y]_{(y)}$ sending the unique (generic) point η to the closed point of the codomain is a valid morphism of ringed spaces, but *not* a morphism of locally ringed spaces. (This highlights why the local ring condition is necessary for scheme morphisms).
2. Let $p \in X$ be a point in a scheme. Show there is a canonical morphism of schemes $\operatorname{Spec}(\mathcal{O}_{X,p}) \to X$.
3. Using the above, define a canonical morphism $\operatorname{Spec}(\kappa(p)) \to X$. This is the formal scheme-theoretic morphism corresponding to the inclusion of a point.
4. Show there is no non-trivial map $\mathbb{P}^n_k \to \mathbb{A}^m_k$.

3.2 (Fiber) Product and Fibers

https://stacks.math.columbia.edu/tag/01JW

One of the most common tools in constructing new algebraic and geometric objects are limits and colimits. Given a category that is complete and cocomplete (having all limits and colimits), then:

1. limits are a combination of products and equalizers (i.e. cutting)

2. colimits are a combination of coproducts and coequalizers (i.e. gluing)

The category $\mathbf{Sch}_S$ is closed under coproducts (disjoint union), but not necessarily colimits as we require non-trivial gluing conditions (recall theorem 2.3.16). However, it *is* closed under products and equalizes (equalizer of an affine scheme R usually becomes a closed subscheme). To do this, we must show it is closed under products. Starting with the affine case, by corollary 3.1.5 it is equivalence for R-algebra to be closed under coproducts, and indeed the coproduct of A, B is $A \otimes_R B$ (see [9]).

Theorem 3.2.1: Fibered Product Of Schemes

Let X, Y be schemes over S. Then then the fibered product of X and Y over S exists, that is the limit of the diagram exists:

$$\begin{tikzcd} X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

The fibered product is denoted $X \times_S Y$

More commonly, we would say that the fibered product satisfies the universal property that if $Z \to X$ and $Z \to Y$ commutes in the diagram, then there exists a unique morphism $Z \to X \times_S Y$ such that

$$\begin{tikzcd} & Z \arrow[d] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\ & X \times_S Y \arrow[dl] \arrow[dr] & \\ X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

commutes. If S is the final object, or more concisely it is empty or "doesn't matter", then we have the usual product. If S is not specified, the product shall be defined as $X \times Y := X \times_{\mathrm{Spec}\,(\mathbb{Z})} Y$

***Proof*:**
We shall construct the products of s, and then glue them together.

Let $X = \mathrm{Spec}\,(A)$, $Y = \mathrm{Spec}\,(B)$, and $S = \mathrm{Spec}\,(R)$ be affine. Then A, B are R-algebras, and the product of X and Y over S is:

$$\mathrm{Spec}\,(A \otimes_R B)$$

This comes form the universal property of tensor products along with the property of adjoints. Indeed, if $Z \to \mathrm{Spec}\,(A \otimes_R B)$ is a scheme morphism, then this gives a ring homomorphism $A \otimes_R B$ into $\mathcal{O}_Z(Z)$. By the universal property of tensor products, a morphism of $A \otimes_R B$ into is the same as a morphism from A and B to that ring. Then by proposition 3.1.12, this gives a morphism of Z into $\mathrm{Spec}\,(A \otimes_R B)$, and this morphism will satisfy the universal property of fibered products given by the universal property of tensor products.

Let us now construct the more general case via gluing. Let X, Y be schemes. To do this, we shall reduce to a covering of $X \times_S Y$. For any open affine $U \subseteq X$, if the product existed then $\pi_1^{-1}(U) \cong U \times_S Y$ (check this[a]). Now if $\{X_i\}$ is an open covering for x, then I claim that if $X_i \times_S X$ exists, so does $X \times_S Y$. Indeed, for each $U_{ij} = \pi_1^{-1}(X_i) \subseteq X_i \times_S Y$, by the uniqueness of the product there are unique isomorphisms: $\varphi_{ij} : U_{ij} \to U_{ji}$ for each i, j for compatible projections. Furthermore, these isomorphisms are compatible with the gluing condition (as you should check if you are not convinced!). As a remind, the condition was:

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}$$

Thus, we get that we may glue them together by lemma 3.1.9 we have that this glued scheme is isomorphic to $X \times_S Y$, and through this isomorphism we can check that the universal property is satisfied.

By interchanging X and Y, we can now conclude from the proof on affine schemes that if X and Y are any schemes, and S is an affine, then $X \times_S Y$ exists. To show that S can be an arbitrary scheme, let $\{S_i\}$ be an open affine cover of S. Let $\alpha : X \to S$ and $\beta : Y \to S$ be two scheme morphisms, and let $X_i = \alpha^{-1}(S_i)$, $Y_i = \beta^{-1}(S_i)$. Then $X_i \times_{S_i} Y_i$ is well-defined.

Now, what's key is that this scheme is the same as the scheme $X_i \times_S Y$. To see this, we must invoke the uniqueness of the universal property, and indeed if $f : Z \to X_i$ and $g : Z \to Y$ are scheme morphisms over S, by the properties of the commutative diagram we must have that the image of v must land inside Y_i. But by universality they must be equal. Thus $X_i \times_S Y$ is well-defined for each i, and so by the above argument it glues to create $X \times_S Y$, as we sought to show.

[a] If $f : Z \to U$, compose with the inclusion map and invoke the appropriate universal properties

Example 3.2.2: Fibered Products

1. Show that $\mathbb{A}^n_k \times_k \mathbb{A}^m_k \cong \mathbb{A}^{n+m}_k$, which is exactly what we would hope from a product. Notably, prove $k[x] \otimes_k k[y] \cong k[x, y]$.

2. Similarly, if k is a characteristic 0 field show that $\mathbb{Z}[x] \otimes_{\mathbb{Z}} k[y] \cong k[x, y]$. Hence $\mathbb{A}^1_{\mathbb{Z}} \times \mathbb{A}^1_k \cong \mathbb{A}^2_k$.

3. Let $X = \operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y)$ and $Y = \mathbb{C}[z, w]/(z^3 - w^4)$. Then:
$$X \times_{\mathbb{C}} Y \cong \operatorname{Spec} \frac{\mathbb{C}[x, y, z, w]}{(x^2 - y, z^3 - w^3)}$$
The reader should stare at this until they understand why it is called a fibered product.

4. There is new behaviour when considering schemes being tensored over non-algebraically closed fields. Take $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$. As these are affine, this is $\operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$. As $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ (exercise),
$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C}) \amalg \operatorname{Spec}(\mathbb{C})$$
Notice here how the choice of sheaf completely changed the answer, notably $\operatorname{Spec} \mathbb{C}$ and $\operatorname{Spec} \mathbb{R}$ are schemes over a single point, hence the fibered product diagram when only considering points is an identity map on points:
$$\begin{tikzcd} \bullet \arrow[dr] & & \bullet \arrow[dl] \\ & \bullet & \end{tikzcd}$$
and it is easy to see that in **Top** this results is a single point. However, the resulting scheme is *two* points. Thus, the fibered product here somehow "unravelled" the fact that $\operatorname{Spec} \mathbb{R}$ should be thought of as "gluing" conjugates together (think of the intuition for $\operatorname{Spec} \mathbb{R}[x]$ as a $\operatorname{Spec} \mathbb{C}[x]/\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$, the real line then don't glue to another points, but their residue field is $\mathbb{R}$ instead of $\mathbb{C}$, and hence it codimension 2 with respect to $\mathbb{C}$).

More generally show that if L/K is a finite galois extension, then

$$L \otimes_K L \cong \prod_{|\mathrm{Aut}(L/K)|} K$$

is isomorphic to $L^{|G|}$($|G|$ copies of L), hint: this is the linear independence of characters proof in geometric form. Hence, over fields fibered products capture some idea properties of galois groups.

5. Recall that the intersection of rings is a ring. This is a special case of the fibered product in the category of commutative rings, namely that if $S \to R$ and $S' \to R$ exists, then:

$$\begin{tikzcd} & S \cap S' \arrow[dl] \arrow[dr] & \\ S \arrow[dr] & & S' \arrow[dl] \\ & R & \end{tikzcd}$$

Let us see how this translates to schemes and affine schemes. Recall that the product and coproduct of affine schemes is an affine scheme:

$$\operatorname{Spec} S \times \operatorname{Spec} S' = \operatorname{Spec} S \otimes_{\mathbb{Z}} S'$$
$$\operatorname{Spec} S \amalg \operatorname{Spec} S' = \operatorname{Spec} S \times S'$$

If $U, U' \subseteq \operatorname{Spec} R$ are open affine, $U = \operatorname{Spec} S$ and $U' = \operatorname{Spec} S'$, Then as $\operatorname{Spec}(-)$ is contravariant we need that the intersection $(\operatorname{Spec}(S)) \cap (\operatorname{Spec}(S'))$ becomes the "union" of rings. Naturally, the union of rings need not be a ring, and as both S, S' contain units then $\langle S, S' \rangle = R$. Instead, the natural dual is:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \otimes_R S'$$

As we saw, the union of affine need not be affine (see example 2.3.17).

One of the most important uses of the fibered is to change the base scheme:

Definition 3.2.3: Base Change (Scheme-theoretic Pullback)

Let X be an S-scheme, and $T \to S$ a scheme morphism. Then the *base change* of X to T is the T-scheme:

$$X_T = T \times_S X$$

If $f : X \to Y$ is an S-scheme morphism, and $T \to S$ a scheme morphism, then the *base change* of f is the map $f' : X_T \to Y_T$ where $f' = \mathrm{id}_T \times_{\mathrm{id}_S} f$.

Example 3.2.4: Base Change

1. For any R-scheme X and morphism $\operatorname{Spec} S \to \operatorname{Spec} R$ (i.e. ring homomorphism $R \to S$), we

get the S-scheme:

$$X' = S \times_R X$$

The category of R-schemes maps to the category of S schemes via base change.

2. If $X = R[x_1, ..., x_m]/I$, is an R-scheme and $S \to R$ is a ring homomorphism, then:

$$X_S = \frac{S[x_1, ..., x_m]}{IS}$$

that is, the set of constant change.

3. As the map $S \to R$ need not be surjective, we can get something like $X = \operatorname{Spec} \mathbb{C}$ as a $\mathbb{C}$-scheme and base change $\mathbb{R} \hookrightarrow \mathbb{C}$.

4.

As the tensor product is a very versatile operation, the fibered product is as well. Let us demonstrate by translating some important results:

Proposition 3.2.5: Base Change Properties

The following properties hold for fibered products:

1. Let $\iota : U \to Z$ be an open immersion and $\varphi : Y \to Z$ be any scheme morphism. Then $U \times_Z Y$ is isomorphic to the open subscheme of Y given by the preimage of U:

$$U \times_Z Y \cong \left(\varphi^{-1}(U), \mathcal{O}_Y|_{\varphi^{-1}(U)}\right)$$

2. *Stability of Closed Immersions:* If $X \to Z$ is a closed immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is a closed immersion.

3. *Stability of Open Immersions:* If $X \to Z$ is an open immersion, then for any map $Y \to Z$, the base change $X \times_Z Y \to Y$ is an open immersion.

***Proof*:**

Reduction to the Affine Case (for 2 and 3): The properties of being a closed immersion or an open immersion are *local on the target*. That is, a morphism $f : W \to Y$ is a closed (resp. open) immersion if and only if there exists an open cover $\{V_i\}$ of Y such that each restriction $f^{-1}(V_i) \to V_i$ is a closed (resp. open) immersion.

Let $X \to Z$ be the immersion we are base changing. Let $\{Z_\alpha = \operatorname{Spec}(R_\alpha)\}$ be an affine open cover of Z. Since $X \to Z$ is an immersion, the restriction $X_\alpha = X \times_Z Z_\alpha \to Z_\alpha$ corresponds to a map of affine schemes (either a quotient or a localization). Now, let $\{Y_{ij} = \operatorname{Spec}(S_{ij})\}$ be an affine open cover of Y such that each Y_{ij} maps into some Z_α. Since fibered products commute with open restriction, the base change restricted to Y_{ij} is:

$$(X \times_Z Y) \times_Y Y_{ij} \cong X \times_Z Y_{ij} \cong (X \times_Z Z_\alpha) \times_{Z_\alpha} Y_{ij}$$

This reduces the problem to showing that the base change of an affine immersion by an affine morphism is an immersion. If we prove this for rings, the global property follows by gluing.

1. The fibered product is the limit in the category of schemes. Maps $W \to U \times_Z Y$ correspond to pairs of maps $W \to U$ and $W \to Y$ that agree in Z. Since $U \hookrightarrow Z$ is an inclusion, a map to U is just a map to Z whose image lies in U. Thus, a map to the product is a map $W \to Y$ such that the composition $W \to Y \xrightarrow{\varphi} Z$ factors through U. This is exactly the definition of a map into the open subscheme $\varphi^{-1}(U) \subseteq Y$.

2. *Affine Case:* Let $Z = \operatorname{Spec}(R)$, $X = \operatorname{Spec}(R/I)$, and $Y = \operatorname{Spec}(S)$. The closed immersion corresponds to the surjection $\pi : R \twoheadrightarrow R/I$. We compute the base change by tensoring with S over R. Since the tensor product is *right-exact*, applying $S \otimes_R -$ to the surjection preserves surjectivity:
$$S \cong S \otimes_R R \xrightarrow{1\otimes\pi} S \otimes_R (R/I) \longrightarrow 0$$
Thus, the ring map corresponding to the base change $X \times_Z Y \to Y$ is surjective. Since surjective ring maps correspond to closed immersions of affine schemes, and we have established the property is local, the general base change is a closed immersion.

3. *Affine Case:* Let $Z = \operatorname{Spec}(R)$ and let $X \to Z$ correspond to a localization $R \to R_f$ for some $f \in R$ (a basic open immersion). Let $Y = \operatorname{Spec}(S)$ with map $\varphi : R \to S$. We compute the base change by tensoring:
$$S \otimes_R R_f = S \otimes_R R[f^{-1}] \cong S[\varphi(f)^{-1}] = S_{\varphi(f)}$$
The resulting map $S \to S_{\varphi(f)}$ is a localization. Since any open immersion is locally isomorphic to a basic open immersion (localization), and we have established the property is local, the general base change is an open immersion.

This motivates an important definition generalizing affine and projective schemes (idk if this should be here)

Definition 3.2.6: Affine Schemes over a Scheme

Let X be a scheme. Then:
$$\mathbb{A}^n_X = X \times_{\mathbb{Z}} \mathbb{Z}[x_1, \dots, x_n]$$

Example 3.2.7: Pushout Need Not Exist

Consider two copies of $\mathbb{A}^1_{\mathbb{C}}$. Consider:

$$\begin{tikzcd} & \operatorname{Spec}\mathbb{C} \arrow[dl] \arrow[dr] & \\ \mathbb{A}^1_{\mathbb{C}} & & \mathbb{A}^1_{\mathbb{C}} \end{tikzcd}$$

where $[(0)] \mapsto [(0)]$ in both cases. This is like the double-origin example, except every point is a double origin. This shall be shown to not be a scheme as it is "too non-separated".
We cannot fix this by limiting to the category of affine schemes. Consider for example:

$$\begin{tikzcd} & \operatorname{Spec} k[t]_t \arrow[dl] \arrow[dr] & \\ \operatorname{Spec} k[t] & & \operatorname{Spec} k[t] \end{tikzcd}$$

where $\operatorname{Spec} k[t]_t = \mathbb{A}^1_k \backslash \{0\}$, and the maps are the maps given by $t \mapsto t$ and $t \mapsto 1/t$. Then we saw in example 2.15 that the colimit of this diagram is projective space, $\mathbb{P}^1_k$. Hence affine schemes are not in general closed under pushout. Note that the corresponding fibered product does exist in CRing, namely it will have to be:

$$\begin{tikzcd} & k \arrow[dl, dashed] \arrow[dr, dashed] & \\ k[t] \arrow[dr] & & k[t] \arrow[dl] \\ & k[t]_t & \end{tikzcd}$$

However, applying Spec to this will not return the pushout, namely the pair of adjoint functors (Spec, Γ) does not preserve pushouts.

Note however that colimits do exist in the category of locally ringed spaces. See this link (which link??) for more information.

(words here)

Definition 3.2.8: Geometrically Integral

Let S be a k-scheme. Then S is *geometrically integral* if it is integral over the algebraic closure$\overline{k}$.

The term "geometrically PROPERTY" is a common term to specify that a PROPERTY occurs for the k-scheme when taking the algebraic closure of k.

(want to include this somewhere: **Tensor Product Interpretation (Base Change):** The map constructed above is canonically isomorphic to the base change of the ring R_r to S. Using the tensor product:

$$S \otimes_R R_r \cong S \otimes_R (R[x]/\langle xr - 1\rangle) \cong S[x]/\langle x\varphi(r) - 1\rangle \cong S_{\varphi(r)}$$

This interpretation is powerful because it links the *geometric pullback* of sheaves to the *algebraic tensor product.*)

3.2.1 Fibers and family of Schemes

This gives a huge class of new object's that we may define. Let us start with defining fibers:

Definition 3.2.9: Fibers

Let $f : X \to Y$ be a morphism of scheme. Let $y \in Y$, and $k(y)$ be the residue field of y. Let $\operatorname{Spec}(k(y)) \to Y$ be the natural morphism[a]. Then the *fiber* of the morphism f over the point y is the scheme

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y))$$

In the case where $X = \operatorname{Spec}(R), Y = \operatorname{Spec}(S)$ are affine, then for a point $\mathfrak{p} \in Y$:

$$f^{-1}(\mathfrak{p}) = \operatorname{Spec} R \times_S \operatorname{Spec} \kappa(\mathfrak{p}) = \operatorname{Spec}\left(R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}}\right)$$

[a]exercise ref:HERE

Proposition 3.2.10: Topological Space of Fibers

Let $f : X \to Y$ be a morphism of schemes, and let $y \in Y$. The natural map $|X_y| \to f^{-1}(y)$ is a homeomorphism.

Furthermore, $X_y \to X$ is a *closed immersion* identified with the ideal $f^{\#}(\mathfrak{m}_y)$.

***Proof*:**
The question is affine local on source and target (proving bijectivity can be reduced to compatible open covers), so let's assume $X = \operatorname{Spec}(R)$ and $Y = \operatorname{Spec}(S)$ are affine schemes. Then $f : \operatorname{Spec} R \to \operatorname{Spec} S$ corresponds to a ring homomorphism $f^{\#} : S \to R$. Let y correspond to the prime ideal $\mathfrak{p}$ in S. Then:

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y)) = \operatorname{Spec}(R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}})$$

We shall take advantage of example **??**(1) where points are associated to maps to $\kappa(\mathfrak{q})$. Suppose $(f^{\#})^{-1}(\mathfrak{q}) = \mathfrak{p}$ where $\mathfrak{q} \in \operatorname{Spec}(R)$ (i.e. $f(\mathfrak{q}) = \mathfrak{p}$). Then Let g represent the composition $S \to R \to R_{\mathfrak{q}}$. Then $g : S \to R_{\mathfrak{q}}$ factors uniquely through $\overline{g} : S_{\mathfrak{p}} \to R_{\mathfrak{q}}$. We can extend this map by tensoring with $R \otimes_S (-)$ so that:

$$\Phi : R \otimes_S S_{\mathfrak{p}} \to R_{\mathfrak{q}} \qquad (r, s/t) = r\overline{g}(s/t)$$

Note that any element in $\mathfrak{p}S_{\mathfrak{p}}$ maps to $\mathfrak{q}R_{\mathfrak{q}}$, hence have have a unique map:

$$R \otimes_S S_{\mathfrak{p}}/\mathfrak{p}S_{\mathfrak{p}} \twoheadrightarrow R_{\mathfrak{q}}/\mathfrak{q}R_{\mathfrak{q}} = \kappa(\mathfrak{q})$$

As this map is unique, the correspondence between $k(\mathfrak{q})$ and the points of the fiber is unique, and certainly it surjective. Thus, for each point $\operatorname{Spec}(R_{\mathfrak{q}}/\mathfrak{q}R_{\mathfrak{q}})$, it is associated to a unique point in $f^{-1}(\mathfrak{p})$.

Finally, it is clear that $f^{-1}(\{y\})$ satisfies the (topological) pullback

$$\begin{tikzcd}
f^{-1}(y) \arrow[r] \arrow[d] & X \arrow[d, "f"] \\
\operatorname{Spec}(k(y)) \arrow[r] & Y
\end{tikzcd}$$

Taking:

$$\begin{tikzcd} X_y \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec}(k(y)) \arrow[r] & Y \end{tikzcd}$$

The maps are all continuous and follow from properties of fiber product, and the algebra above shows it is a bijection. It is easy to see it is a homeomorphism. For the closed

In some ways, fibers of schemes are more "regular" than fibers of sets, as the extra information of schemes will provide some more possible pre-images:

Example 3.2.11: Fibers

1. Take the familiar projection of the parabola onto the x axis induced by the ring map $\mathbb{Q}[x] \to \mathbb{Q}[y]$ mapping $x \mapsto y^2$. We can think of this as the map $\mathbb{Q}[x] \to \mathbb{Q}[y^2]$ where the codomain is isomorphic to:
$$\mathbb{Q}[y^2] \cong \frac{\mathbb{Q}[x,y]}{x-y^2}$$
This is sometimes called the parameterization of the map. Then the pre-image of 1 is:
$$\begin{aligned} \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x\right) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2-x, x-1\right) \\ &\cong \operatorname{Spec}\left(\mathbb{Q}[y]/\left(y^2-1\right)\right) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)(y+1)) \\ &\overset{\text{C.R.T}}{\cong} \operatorname{Spec}(\mathbb{Q}[y]/(y-1) \times \mathbb{Q}[y]/(y+1)) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)) \amalg \operatorname{Spec}(\mathbb{Q}[y]/(y+1)) \end{aligned}$$
That is, it is the points ± 1. The pre-image of 0 is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x,x) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2)$$
Notice that the pre-image remembers multiplicity information!
As the spectrum of a polynomial ring over rationals is the quotient by the galois action on the polynomial ring over $\bar{k}$, the pre-image of -1 exists! It is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x,x+1) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2+1) \cong \operatorname{Spec} \mathbb{Q}[i] = \operatorname{Spec} \mathbb{Q}(i)$$
This can be thought of as "size 2" point over the base field. The pre-image of the generic point is a reduced point, and can be thought of as "size 2" over the residue field:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2-x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x) \cong \operatorname{Spec} \mathbb{Q}[y] \otimes_{\mathbb{Q}[y]} \mathbb{Q}(y^2)$$
Try working through the pre-image of other points that are not commonly associated to $\mathbb{Q}$. Notice that the pre-image is the two points, or you get nonreduced behaviour that is "of degree 2" or you get a field extension of degree 2.

2. Let $X = \operatorname{Spec}(\mathbb{Z}[i])$ (the Gaussian integers) and $Y = \operatorname{Spec}(\mathbb{Z})$. The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[i]$ induces a morphism $\pi : X \to Y$. Let us analyze the fibers of this map (the pre-images of points).

- *Ramification:* Consider the point $(2) \in \operatorname{Spec}(\mathbb{Z})$. The fiber is $\operatorname{Spec}(\mathbb{Z}[i] \otimes_{\mathbb{Z}} \mathbb{Z}/(2)) \cong \operatorname{Spec}(\mathbb{F}_2[x]/(x^2+1)) = \operatorname{Spec}(\mathbb{F}_2[x]/(x+1)^2)$. Topologically, there is one point, but the ring has a nilpotent element. The prime 2 "ramifies."
- *Splitting:* Consider $(5) \in \operatorname{Spec}(\mathbb{Z})$. In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$. The fiber corresponds to $\mathbb{F}_5[x]/(x^2+1)$. Since x^2+1 has roots $2, 3$ in $\mathbb{F}_5$, this ring is $\mathbb{F}_5 \times \mathbb{F}_5$. Geometrically, the point (5) pulls back to *two* distinct points.
- *Inert:* Consider $(3) \in \operatorname{Spec}(\mathbb{Z})$. x^2+1 is irreducible in $\mathbb{F}_3$. The fiber is $\operatorname{Spec}(\mathbb{F}_9)$, a single point with a larger residue field.

This example demonstrates that scheme morphisms provide the correct geometric framework for Algebraic Number Theory.

3. As $X \to \operatorname{Spec}(\mathbb{Z})$ is always well-defined and unique, there is always fibers over $\operatorname{Spec}(\mathbb{Z})$ and hence they are usually given a name: The fiber over (0) is often labeled $X_{\mathbb{Q}}$ (over $\mathbb{Q}$), and the fibers for all the primes are X_p (over $\mathbb{F}_p$). A common terminology is that X_p *arises by reduction* mod p of the scheme X.

Proposition 3.2.12: Pre-image of Generic (Spreading out)

Let $f : X \to Y$ be a morphism of finite type and Y Noetherian. Then if $f^{-1}(\eta) = \emptyset$ for a generic point $\eta \in Y$, there exists an open subset $U \subseteq Y$ containing η such that $f^{-1}(U) = \emptyset$.

***Proof*:**
Since the problem is local on Y around η, we may assume $Y = \operatorname{Spec}(A)$ is affine. Since Y has a generic point η, Y is irreducible. As Y is Noetherian, we can treat A as an integral domain (modulo nilpotents, which do not affect the topology or emptiness of schemes). Let $K = \operatorname{Frac}(A)$ be the residue field of η.

Since f is of finite type, X is quasi-compact. We can cover X by a finite number of affine open subsets $X = \bigcup_{i=1}^n \operatorname{Spec}(B_i)$, where each B_i is a finitely generated A-algebra.

The hypothesis $f^{-1}(\eta) = \emptyset$ means the fiber product $X \times_Y \operatorname{Spec}(K)$ is the empty scheme. For each affine patch $\operatorname{Spec}(B_i)$, this corresponds to the ring:

$$B_i \otimes_A K = 0$$

Note that $B_i \otimes_A K$ is isomorphic to the localization of B_i by the multiplicative set $S = A \setminus \{0\}$. If a localization is the zero ring, then $1 = 0$ in that ring. By the definition of localization, this implies there exists an element $s_i \in S$ (i.e., a non-zero element $s_i \in A$) such that:

$$s_i \cdot 1_{B_i} = 0 \quad \text{in } B_i$$

Now, consider the basic open set $U_i = D(s_i) \subseteq Y$. Since $s_i \neq 0$, $\eta \in U_i$. The preimage of this open set in the chart $\operatorname{Spec}(B_i)$ is $\operatorname{Spec}((B_i)_{s_i})$. However, since s_i annihilates the unity in B_i, the localized ring is zero:

$$1 = \frac{s_i}{s_i} = \frac{1}{s_i} \cdot (s_i \cdot 1_{B_i}) = 0$$

Thus, $\operatorname{Spec}((B_i)_{s_i}) = \emptyset$.

To handle the entire scheme X, let $s = s_1 s_2 \cdots s_n$. Since A is a domain, $s \neq 0$, so $U = D(s)$ is a non-empty open set containing η. Furthermore, $U \subseteq U_i$ for all i. Consequently, the preimage

$f^{-1}(U)$ is the union of the preimages restricted to each chart. Since s is a multiple of s_i, s acts as zero on every B_i, making the localization $(B_i)_s = 0$ for all i. Therefore, $f^{-1}(U) = \emptyset$.

On p. 260 of Vakil he gives more on general fibers, generic fibers, and generically finite morphisms don't know where to put these examples yet

Example 3.2.13: Parameterization of Curves using Schemes
Take

$$X = \operatorname{Spec} \frac{k[x,y,t]}{(ty - x^2)}$$
$$Y = \operatorname{Spec} k[t]$$

Let $f : X \to Y$ be given by the natural ring homomorphism $k[t] \to \frac{k[x,y,t]}{(ty-x^2)}$. Note both these schemes are integral schemes of finite type over k.

Now, identify the closed points of Y with k. For $a \neq 0 \in k$, we get the fibers correspond to the parabola's $ay = x^2$ or $y = x^2/a$. All of these are irreducible, reduced curves. At $a = 0$, we get $x^2 = 0$, which is a reduced scheme.

More generally, S-schemes can be geometrically interpreted as schemes paramaterized by the points of S whose maps are fiber-bundle morphisms. Hence, the scheme $\operatorname{Spec} k[t]$ *paramaterized* the curve.

A quick note must be done about general products and why it may be preferable to take the fibered product over a field:

Example 3.2.14: Strange Fibered Products

1. Show that $\operatorname{Spec}(\mathbb{Z}/m\mathbb{Z}) \times_{\operatorname{Spec}(\mathbb{Z})} \operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) = \emptyset$.
2. Another product of taking the "absolute" fibered product is that if $X = \operatorname{Spec} \mathbb{Z}[x]$ and $Y = \operatorname{Spec} \mathbb{Z}[y]$ then:

$$\dim X \times Y = \dim(X) + \dim(Y) - \dim(\operatorname{Spec} Z) = \dim X + \dim Y - 1$$

If we restrict to k-schemes for a field k, these two pathologies do not occur.

As these examples show, the fibered product of two schemes need not be the schemes of these two fibered products. In a concrete sense, this is because we must be more specific with what types of points we are dealing with. If we specify the points, we in fact recover the usual "sets"[5].

[5] The topological product can also be thought of as being over the right valued points

Proposition 3.2.15: Fibered Product With Z-valued Points

Let X, Y be schemes and take Z-valued points on each scheme (i.e. morphisms in $\mathrm{Hom}(Z, X)$ and $\mathrm{Hom}(Z, Y)$. Then show that the fibered product will be:

$$\mathrm{Hom}(W, X \otimes_W Y) = \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y)$$

***Proof*:**
exercise.

Put an appropriate definition here for the following

Hence, given a surjective map $Y \to X$, Y can be regarded as a collection of schemes paramaterized by Y. Conversely, we may want to study a scheme X_0, and see if there are some deformations we may want to do. Then a common way of defining a *family of algebraic deformation* is by a scheme morphisms $f : X \to Y$ where one of the fibers of f is isomorphic to X_0.

Some more here on representable Functors

Exercise 3.2.1

1. Let $\varphi : X \to Y$ be a morphism of S-scheme of finite type. Show that if $S \to S'$ is any base extension, $\varphi' : X' \to Y'$ is an S'-scheme morphism preserves finite type.
2. Let $\varphi : X \to Y$ be a scheme morphism of integral schemes. Show the fibers needn't be irreducible or reduced.
3. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map
$$\mathrm{Spec}\,(\mathcal{O}_{X,\mathfrak{p}}) \to X$$
4. Using the above, define a canonical morphism $\mathrm{Spec}\,(\kappa(\mathfrak{p})) \to X$.
5. Perhaps put here some base-change exercises relating the tensor product exercises you had in [9] and now give them their geometric counter-part.
6. Show if $U, V \subseteq X$ are open subschemes then $U \times_X V \cong U \cap V$.
7. Let $\varphi : B \to A$ be a ring morphism, $S \subseteq B$ a multiplicative subset, so $\varphi(S)$ is a multiplicative subset of A. Show that $\varphi(S)^{-1}A \cong A \otimes_B (S^{-1}B)$. What does this imply for the relation between localization and the fibered product?
8. Let X be a k-scheme. Show that if L/k is a field extension, $X \times_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(L)$ is an L-scheme. Show that this in fact gives a functor (define the appropriate maps, and verify).

3.3 The Functor of Points

Throughout the development of scheme theory, we have built our geometric objects from locally ringed spaces: a topological space equipped with a structure sheaf, glued from affine pieces. This is a powerful framework, but it carries a certain asymmetry: the points of a scheme and the morphisms between schemes are defined by quite different-looking data. The goal of this section is to develop a perspective—the *functor of points*—in which schemes are understood entirely through their morphisms. The result is not merely a rephrasing; it reveals structure that the locally ringed space definition obscures, and it provides the language in which most modern moduli problems are formulated.

The idea has a simple origin. Recall from section 2.2.5 that a global section $f \in \Gamma(X, \mathcal{O}_X)$ of a scheme X assigns to each point $x \in X$ an element $f(x)$ in the residue field $\kappa(x)$, but that these residue fields vary from point to point. A global section is therefore *not* a function in the classical sense. Let us trace how this observation leads, step by step, to a functorial description of schemes.

3.3.1 From Evaluation to Morphisms

When the base field is algebraically closed, the situation is familiar. Over an algebraically closed field $k = \overline{k}$, a polynomial $f \in \overline{k}[x_1, \dots, x_n]$ determines a genuine set-theoretic function:

$$\overline{k}[x_1, \dots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Set}}(\overline{k}^n, \overline{k}), \qquad f \longmapsto \big((a_1, \dots, a_n) \mapsto f(a_1, \dots, a_n)\big). \tag{3.1}$$

Here $\overline{k}^n$ can be identified with $\mathrm{mSpec}\,(\overline{k}[x_1, \dots, x_n])$ via the Nullstellensatz, and each evaluation $f(a_1, \dots, a_n)$ lands in $\overline{k} \cong \kappa(\mathfrak{m}_{(a_1,\dots,a_n)})$. The map in equation (3.1) is injective precisely because an algebraically closed field has "enough points" to separate polynomials.

Over a general field k, this breaks down: the maximal ideals of $k[x_1, \dots, x_n]$ no longer correspond to k-rational tuples. As discussed in example 2.1.3, $\mathrm{mSpec}\,(k[x_1, \dots, x_n])$ contains orbits of geometric points under the absolute Galois group. One can partially recover the situation by mapping not into k, but into the affine line $\mathbb{A}^1_k$, which itself carries these additional closed points. Concretely, a polynomial $f \in k[x_1, \dots, x_n]$ induces a morphism of k-algebras

$$k[t] \longrightarrow k[x_1, \dots, x_n], \qquad t \longmapsto f,$$

which, by the equivalence of affine schemes with rings, corresponds to a scheme morphism $\mathbb{A}^n_k \to \mathbb{A}^1_k$. Evaluating at a closed point $\mathfrak{m}$ of $\mathbb{A}^n_k$ yields the image of f in $\kappa(\mathfrak{m})$, and this image determines a closed point of $\mathbb{A}^1_k$. Thus, we arrive at a map

$$k[x_1, \dots, x_n] \longrightarrow \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{mSpec}\, k[x_1, ..., x_n], \mathrm{mSpec}\, k[t]), \qquad f \longmapsto \text{the morphism induced by } t \mapsto f. \tag{3.2}$$

This is already an improvement: it works over any field, and the target is a set of *scheme morphisms* rather than set-theoretic functions. However, it still only detects the behavior of f at closed points. Recall that such schemes are called *Jacobson schemes* (see definition 3.8.4). For general schemes this is insufficient. For instance, on $\mathrm{Spec}\,(k[[x]])$ or $\mathrm{Spec}\,(\mathbb{Z}_{(p)})$, a function is not determined by its values at closed points alone.

The natural fix is to allow *all* prime ideals as test points. For any k-algebra A, a scheme morphism $\mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ is the same data as a k-algebra map $k[t] \to A$, which since this is a k-algebra homomorphism is the same as choosing an element of A for t to map to. This gives:

$$\Gamma(\mathrm{Spec}\,(A), \mathcal{O}_{\mathrm{Spec}\,(A)}) = A \;\cong\; \mathrm{Hom}_{\mathbf{Sch}_k}(\mathrm{Spec}\,(A), \mathbb{A}^1_k). \tag{3.3}$$

The left-hand side is the ring of global sections; the right-hand side is the set of scheme morphisms to the affine line. Each element $f \in A$ corresponds to the unique morphism $\pi_f : \mathrm{Spec}\,(A) \to \mathbb{A}^1_k$ for which $\pi_f^\#(t) = f$. The "value" of f at a prime $\mathfrak{p}$ is recovered by following $t \mapsto f \mapsto f \bmod \mathfrak{p} \in \kappa(\mathfrak{p})$, which is precisely the scheme-theoretic notion of evaluation from section 2.2.5.

Two observations are now in order. First, the construction (3.3) is not specific to affine schemes: it will extend to arbitrary k-schemes X once we establish the representability of the global section functor (Proposition 3.3.7 below). Second, and more importantly, there is nothing special about $\mathbb{A}_k^1$ as a target apart from the fact that it *represents* the functor of global sections. One can ask: what happens if we replace $\mathbb{A}_k^1$ by another scheme T?

The set $\mathrm{Hom}_{\mathbf{Sch}_k}(X, T)$ makes perfect sense for any T, and different choices of T probe different aspects of X. Taking $T = \mathbb{A}_k^1$ recovers global sections; taking $T = \mathbb{G}_m$ recovers invertible global sections; taking $T = \mathbb{P}_k^n$ recovers line bundles generated by $n+1$ global sections (cf. section 5.2.1). The test object T acts as a "measuring device" that detects a particular piece of the geometry of X. The remarkable fact—a consequence of the Yoneda Lemma applied to schemes—is that if we keep track of $\mathrm{Hom}(-, X)$ for *all* possible test objects simultaneously, we recover X entirely.

3.3.2 Z-valued Points

Let us make the above discussion concrete. Consider the scheme

$$X = \operatorname{Spec} \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)},$$

whose underlying topological space consists of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$-conjugate pairs of maximal ideals together with a generic point. The closed points do not individually look like points on a circle—many have residue field $\mathbb{C}$, not $\mathbb{R}$—and so the "usual" picture of the unit circle is not immediately visible.

If we want to recover that picture, we should look for $\mathbb{R}$-valued points: ring homomorphisms

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}$$

sending $x \mapsto a$ and $y \mapsto b$. The quotient relation forces $a^2 + b^2 = 1$, so the collection of all such homomorphisms is in bijection with

$$X(\mathbb{R}) \;\simeq\; \left\{(a,b) \in \mathbb{R}^2 \;\middle|\; a^2+b^2 = 1\right\},$$

which is exactly the unit circle. Equivalently, each such homomorphism factors through the evaluation map $\mathbb{R}[x,y] \to \mathbb{R}$ as a map that kills the ideal (x^2+y^2-1). Via the contravariant equivalence of affine schemes and rings, these ring homomorphisms correspond to scheme morphisms $\operatorname{Spec}(\mathbb{R}) \to X$. The sheaf morphism on global sections is:

$$\mathrm{ev}_{a,b}^\#(X) : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \longrightarrow \mathbb{R}.$$

Now suppose we change the target ring. Over $\mathbb{Z}$, the equation $a^2+b^2 = 1$ has only the four solutions $(\pm 1, 0)$ and $(0, \pm 1)$. Over $\mathbb{Q}$, a Pythagorean-triple parametrization yields infinitely many solutions such as $\left(\frac{3}{5}, \frac{4}{5}\right)$. Over $\mathbb{C}$, the circle becomes a rational curve, and the solution set is much larger still. Each choice of "coefficient ring" yields a different collection of evaluation maps, probing different arithmetic and geometric features of the same scheme.

There is one subtlety to address regarding the base. For the evaluation map $\mathrm{ev}_{a,b}$ to exist as a map of $\mathbb{R}$-algebras, the target ring must receive a structure map from $\mathbb{R}$. There is no nontrivial ring homomorphism $\mathbb{R} \to \mathbb{Q}$, so asking for $\mathbb{Q}$-valued points of the $\mathbb{R}$-scheme X is not well-posed. To allow all these evaluations simultaneously, it is natural to work over the universal base $\mathbb{Z}$. Set $X_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x,y]/(x^2+y^2-1)$. Then for any commutative ring R, the R-valued points of $X_{\mathbb{Z}}$ are the ring homomorphisms $\mathbb{Z}[x,y]/(x^2+y^2-1) \to R$, or equivalently the solutions of $a^2+b^2=1$ in R. If $A \to B$ is a ring homomorphism and X_A, X_B denote the base changes, then $X_A(B) = X_B(B)$: the B-valued points see only the base-changed scheme.

It is worth pausing to understand the $\mathbb{Z}$-points of $X_{\mathbb{Z}}$ scheme-theoretically.

A $\mathbb{Z}$-point of $X_{\mathbb{Z}}$ is a morphism $\operatorname{Spec}(\mathbb{Z}) \to X_{\mathbb{Z}}$, which is a section of the structure map $X_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$. Geometrically, one can visualize $X_{\mathbb{Z}}$ as an arithmetic surface fibered over $\operatorname{Spec}(\mathbb{Z})$, with each fiber X_p being the circle $x^2+y^2=1$ reduced modulo the prime p. A $\mathbb{Z}$-point is then a "horizontal" section of this fibration: it passes through every fiber simultaneously.

These $\mathbb{Z}$-points are *not* closed points of $X_{\mathbb{Z}}$. Consider the point $P=(1,0)$, corresponding to the homomorphism $\varphi : \mathbb{Z}[x,y]/(x^2+y^2-1) \to \mathbb{Z}$ defined by $x \mapsto 1$, $y \mapsto 0$. The kernel is the prime ideal $\mathfrak{p} = (x-1, y)$, and the quotient $\mathbb{Z}[x,y]/\mathfrak{p} \cong \mathbb{Z}$. Since $\mathbb{Z}$ is not a field, $\mathfrak{p}$ is prime but not maximal, and the image of P is a closed subscheme isomorphic to $\operatorname{Spec}(\mathbb{Z})$—it has dimension 1 and residue field $\mathbb{Q}$ at its generic point. In contrast, a closed point of $X_{\mathbb{Z}}$ has dimension 0 and a finite residue field $\mathbb{F}_q$.

The four $\mathbb{Z}$-points interact with the fibers in instructive ways. Modulo 2, since $1 \equiv -1$ in $\mathbb{F}_2$, the points $(1,0)$ and $(-1,0)$ collapse to the same point of X_2, as do $(0,1)$ and $(0,-1)$. Modulo an odd prime $p \geq 3$, all four points remain distinct in $X_p(\mathbb{F}_p)$. These four "horizontal sections" are the only integer solutions on the circle; any other rational point, such as $\left(\frac{3}{5}, \frac{4}{5}\right)$, gives a $\mathbb{Q}$-point but not a $\mathbb{Z}$-point.

More generally, there is no reason to restrict the "value scheme" to be affine. One might wish to evaluate points on a torus with values in a curve, or to study morphisms between projective schemes. The evaluation interpretation as a global ring homomorphism is then lost, but this is precisely what the sheaf was built to handle: evaluation can be performed locally. Geometrically, a morphism $Z \to X$ between arbitrary schemes is the datum of compatible local evaluations. This motivates the following definition.

Definition 3.3.1: Z-valued Points

Let X be a scheme. The *Z-valued points* of X are the morphisms $Z \to X$ in the category of schemes. The set of all such morphisms is denoted

$$X(Z) := \operatorname{Hom}_{\mathbf{Sch}}(Z, X).$$

When $Z = \operatorname{Spec}(R)$ is affine, this is written $X(R)$.

The reader should note a potential source of confusion: a "Z-valued point" is the *morphism* $Z \to X$ itself, not any particular point in Z or in the image of Z. This terminology is justified by the fact that a morphism of schemes is not determined by the underlying continuous map of topological spaces—the sheaf map $\varphi^\#$ carries additional information. The full morphism is the "point."

3.3.3 The Yoneda Embedding for Schemes

Having defined Z-valued points, we can now ask: how much of a scheme X is captured by the totality of its valued points? The answer, given by the Yoneda Lemma, is: *everything.*

Proposition 3.3.2: Scheme Morphisms from Valued Points

Let X and Y be schemes. A morphism of schemes $\varphi : X \to Y$ is equivalent to a natural transformation $\eta : h_X \to h_Y$ between the associated functors of points. That is, giving a collection of maps of sets

$$\eta_Z : X(Z) \longrightarrow Y(Z)$$

for every scheme Z, natural in Z, uniquely determines a scheme morphism $\varphi : X \to Y$, and conversely.

***Proof*:**
Every scheme X defines a contravariant functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ by

$$h_X(Z) = \mathrm{Hom}_{\mathbf{Sch}}(Z, X) = X(Z).$$

A morphism $\varphi : X \to Y$ induces a natural transformation $\varphi_* : h_X \to h_Y$ by post-composition: for each Z and each $f \in X(Z)$, set $(\varphi_*)_Z(f) = \varphi \circ f$. Naturality follows from the associativity of composition.

Conversely, given a natural transformation $\eta : h_X \to h_Y$, we evaluate at $Z = X$. The set $X(X) = \mathrm{Hom}(X, X)$ contains the identity id_X, and we define

$$\varphi := \eta_X(\mathrm{id}_X) \in Y(X) = \mathrm{Hom}(X, Y).$$

By the Yoneda Lemma (section 0.4),

$$\mathrm{Nat}(h_X, h_Y) \;\cong\; h_Y(X) = \mathrm{Hom}(X, Y),$$

and this correspondence is bijective. Thus, the assignment $X \mapsto h_X$ defines a fully faithful embedding $\mathbf{Sch} \hookrightarrow \mathrm{Fun}(\mathbf{Sch}^{\mathrm{op}}, \mathbf{Set})$: the category of schemes embeds into the category of presheaves on $\mathbf{Sch}$.

The proof is short, but the content is not trivial. It says that a scheme is *completely determined* by how all other schemes map into it. No information is lost by passing from the locally ringed space X to the functor h_X.

A useful analogy organizes this perspective. In linear algebra, a vector $v \in V$ is determined by the collection of values $\ell(v)$ for all linear functionals $\ell \in V^*$: the dual space "separates points." Choosing a basis for V^* gives coordinates on V. The Yoneda embedding plays the same role for schemes. The functor h_X is the "dual description" of X: the scheme is determined by the collection of sets $X(Z)$ for all test schemes Z. Each choice of test scheme is analogous to choosing a basis vector in the dual—it extracts one piece of information about X.

Different test objects give genuinely different perspectives on the same scheme:

Test object Z	$X(Z)$	*What it detects*
$\operatorname{Spec}(\bar{k})$	geometric points	the classical variety
$\operatorname{Spec}(k)$	rational points	arithmetic information
$\operatorname{Spec}(k[\epsilon]/(\epsilon^2))$	tangent vectors	infinitesimal geometry (order 1)
$\operatorname{Spec}(k[t]/(t^n))$	$(n-1)$-jets	higher infinitesimal geometry
$\operatorname{Spec}(k[[t]])$	formal arcs	formal local behavior
$\operatorname{Spec}(K),\ K = k(C)$	rational curves	birational geometry
$\operatorname{Spec}(\mathcal{O}_{X,x})^\dagger$	the germ at x	local scheme structure
$\operatorname{Spec}(\hat{\mathcal{O}}_{X,x})^\dagger$	formal neighborhood	formal geometry at x

† The last two entries differ in character from the others: they depend on the scheme X itself, rather than being universal test objects. Their role is not to "probe" X from outside, but to isolate the local structure at a point $x \in X$.

The local ring $\mathcal{O}_{X,x}$ is the stalk of the structure sheaf, and $\operatorname{Spec}(\mathcal{O}_{X,x})$ is the "germ" of X at x: the limit of all open neighborhoods of x, retaining only the information visible in an arbitrarily small neighborhood. There is a canonical morphism $\operatorname{Spec}(\mathcal{O}_{X,x}) \to X$ (induced by the localization maps $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$), and any morphism $f : Y \to X$ sending a point $y \in Y$ to x factors through $\operatorname{Spec}(\mathcal{O}_{X,x})$ at the level of germs:

$$\operatorname{Spec}(\mathcal{O}_{Y,y}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X,x}) \longrightarrow X.$$

In this sense, the local ring "sees" every morphism that passes through x. This is why properties of morphisms (flatness, smoothness, étaleness) can often be checked at the level of local rings.

The completion $\hat{\mathcal{O}}_{X,x}$ captures even finer information: the formal power series behavior at x, analogous to the Taylor expansion of a smooth function. The formal arc entry $\operatorname{Spec}(k[[t]])$ in the table is in fact the special case $\operatorname{Spec}(\hat{\mathcal{O}}_{\mathbb{A}^1,0})$—the formal neighborhood of the origin in the affine line. For a general point x on a smooth n-dimensional variety, the completion is $\hat{\mathcal{O}}_{X,x} \cong k[[t_1, \ldots, t_n]]$, and the formal neighborhood is a "formal polydisk" of dimension n.

The passage from one test object to another is analogous to a change of basis. A morphism $Z' \to Z$ induces, by precomposition, a map $X(Z) \to X(Z')$ that relates the two perspectives. For instance, the closed immersion $\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[\epsilon]/(\epsilon^2))$, dual to the quotient map $k[\epsilon]/(\epsilon^2) \twoheadrightarrow k$, sends a tangent vector to its basepoint. This is the "forgetful" map from infinitesimal data to point data. The inclusions among the infinitesimal test objects form a tower:

$$\operatorname{Spec}(k) \hookrightarrow \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \hookrightarrow \operatorname{Spec}(k[t]/(t^3)) \hookrightarrow \cdots \hookrightarrow \operatorname{Spec}(k[[t]])$$

where each step remembers one more order of infinitesimal information, and the formal arc $\operatorname{Spec}(k[[t]])$ is the limit, retaining all orders simultaneously.

3.3.4 Key Examples

The definition of Z-valued points subsumes many classical constructions. We now develop the most important instances in detail.

Example 3.3.3: Geometric Points

Let X be a scheme locally of finite type over a field k, and let $\overline{k}$ be an algebraic closure. A $\overline{k}$-valued point is a k-morphism

$$\varphi : \operatorname{Spec}(\overline{k}) \longrightarrow X.$$

Since $\operatorname{Spec}(\overline{k})$ has a unique point η, the morphism φ is determined by the image $y = \varphi(\eta) \in X$ together with the stalk map $\varphi_y^\# : \mathcal{O}_{X,y} \to \overline{k}$. Passing to the residue field, we obtain a k-embedding $\kappa(y) \hookrightarrow \overline{k}$.

We claim that y must be a closed point. Indeed, if y were not closed, then $\kappa(y)$ would contain an element transcendental over k; say t. In particular, as $\varphi_y(t) \in \overline{k}$ must be algebraic, there exists a k-polynomial p such that $\varphi_y(p(t)) = p(\varphi_y(t)) = 0$. As $\varphi_y(0) = 0$ and t is transcendental, $p(t) \neq 0$, demonstrating the failure of injectivity—a contradiction, since field homomorphisms are injective. Conversely, if y is closed, then $\kappa(y)/k$ is a finite extension (by Zariski's Lemma, since X is locally of finite type), and any finite extension of k embeds into $\overline{k}$. Thus, y must be closed, and every closed point arises in this way.

However, there may be *more* $\overline{k}$-valued points than closed points. For example, the $\mathbb{R}$-scheme $\operatorname{Spec}(\mathbb{R}[x]/(x^2+1)) \cong \operatorname{Spec}(\mathbb{C})$ has a single closed point, but two $\mathbb{C}$-valued points: these correspond to the two $\mathbb{R}$-algebra homomorphisms $\mathbb{C} \to \mathbb{C}$ (namely the identity and complex conjugation). In general, the number of $\overline{k}$-valued points lying over a closed point y equals the number of k-embeddings $\kappa(y) \hookrightarrow \overline{k}$, which is $[\kappa(y) : k]$ when $\kappa(y)/k$ is separable.

To obtain a bijection, one can either base-change to $\overline{k}$ (replacing X by $X_{\overline{k}} = X \times_k \operatorname{Spec}(\overline{k})$, which identifies all the Galois conjugates; see section 3.2), or work directly over an algebraically closed field. In the latter case, $\kappa(y) = \overline{k}$ for every closed point, and the correspondence

$$\left\{\operatorname{Spec}(\overline{k}) \to X\right\} \xrightarrow{\sim} \{y \in X : y \text{ is closed}\}$$

is bijective. This recovers the classical variety associated to X: the "geometric points" are exactly the closed points over an algebraically closed field.

Example 3.3.4: Generic Points and Rational Maps

Let C be a curve over $\mathbb{C}$ with function field $K = \mathbb{C}(C)$. A K-valued point of a $\mathbb{C}$-scheme X is a $\mathbb{C}$-morphism $\operatorname{Spec}(K) \to X$. Since $\operatorname{Spec}(K)$ is the generic point of C, such a morphism amounts to a rational map $C \dashrightarrow X$: it is defined on some dense open subset of C, but may not extend to all of C.

More concretely, taking $K = \mathbb{C}(t)$ (the function field of $\mathbb{A}^1_{\mathbb{C}}$), any $\mathbb{C}(t)$-valued point fits into the commutative diagram:

$$\begin{tikzcd}
\operatorname{Spec} \mathbb{C}(t) \arrow[rr] \arrow[dr] & & X \arrow[dl] \\
& \operatorname{Spec} \mathbb{C} &
\end{tikzcd}$$

This is the algebro-geometric analogue of a "generic curve" in X: rather than picking out a single closed point, it describes a one-parameter family of points parametrized by the generic point of the affine line. Whether such a rational map extends to a regular morphism $\mathbb{A}^1_{\mathbb{C}} \to X$ (or $C \to X$ in general) is precisely the content of the valuative criteria discussed in section 3.5.1 and section 3.5.4.

Example 3.3.5: Tangent Vectors and the Dual Numbers

Let $D = k[\epsilon]/(\epsilon^2)$ be the ring of dual numbers. The scheme $\mathrm{Spec}\,(D)$ is a "fat point": it has a unique closed point $\overline{o}$ (corresponding to the maximal ideal (ϵ)), but its structure sheaf remembers one infinitesimal direction. The canonical surjection $D \twoheadrightarrow k$ corresponds to a closed immersion $\iota : \mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(D)$, and we write $o = \iota^{-1}(\overline{o})$.

Let X be a k-scheme of finite type and let $x \in X$ be a closed point (so $\kappa(x) = k$). We classify all D-valued points of X based at x, namely all morphisms $\varphi : \mathrm{Spec}\,(D) \to X$ such that $\varphi \circ \iota$ sends o to x. Choose an affine open neighborhood $x \in \mathrm{Spec}\,(R) \subseteq X$; then $\mathrm{Hom}_x(\mathrm{Spec}\,(D), X) \cong \mathrm{Hom}_x(\mathrm{Spec}\,(D), \mathrm{Spec}\,(R))$, where the subscript x indicates that the closed point maps to x. By the equivalence of affine schemes and rings, these are in bijection with k-algebra homomorphisms $f : R \to D$ satisfying $f(\mathfrak{m}_x) \subseteq (\epsilon)$.

Now R decomposes as a k-vector space as $R = k \oplus \mathfrak{m}_x$. Indeed, as x is closed,

$$0 \to \mathfrak{m}_x \hookrightarrow R \twoheadrightarrow R/\mathfrak{m}_x \cong k \to 0$$

and as k is free this splits. Then f is the identity on k, so f is determined by its restriction to $\mathfrak{m}_x$. Since $\epsilon^2 = 0$ in D, we have $f(\mathfrak{m}_x^2) = 0$, and so f descends to a k-linear functional on $\mathfrak{m}_x/\mathfrak{m}_x^2$. Conversely, any linear functional $\ell : \mathfrak{m}_x/\mathfrak{m}_x^2 \to k$, extended by zero on $\mathfrak{m}_x^2$ and by the identity on k, determines a k-algebra map $R \to D$ sending $\mathfrak{m}_x$ into (ϵ). We conclude:

$$\mathrm{Hom}_x(\mathrm{Spec}\,(D),\, X) \;\cong\; (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = \Theta_{X,x},$$

the Zariski tangent space at x. This is a satisfying realization: the tangent space, originally defined via the maximal ideal and its square (cf. section 2.8), is recovered purely by probing X with the fat point $\mathrm{Spec}\,(D)$.

Moreover, if $f : X \to Y$ is a morphism of k-schemes, then post-composition with f sends a D-valued point based at x to one based at $f(x)$:

$$\mathrm{d}_x f :\; \Theta_{X,x} \longrightarrow \Theta_{Y,f(x)}, \qquad v \longmapsto f \circ v.$$

This is the differential of f, now defined without any reference to local rings or Kähler differentials—it is simply the functoriality of D-valued points.

Example 3.3.6: Group Schemes via Valued Points

By Proposition 3.3.2, a group scheme G over k can equivalently be described as a scheme such that for every k-scheme Z, the set $G(Z)$ carries a group structure, and all maps $G(Z) \to G(Z')$ induced by morphisms $Z' \to Z$ are group homomorphisms. In other words, G is a "group object" in the functor-of-points sense: the functor $h_G : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ factors through the forgetful functor $\mathbf{Grp} \to \mathbf{Set}$.

For a concrete instance, consider the multiplicative group scheme $\mathbb{G}_m = \mathrm{Spec}\,(k[t, t^{-1}])$. For any k-algebra A:

$$\mathbb{G}_m(A) = \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(k[t, t^{-1}],\, A) \;\cong\; A^\times.$$

A k-algebra homomorphism out of $k[t, t^{-1}]$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit in A. Since the units of any ring form a group under multiplication, and ring homomorphisms preserve units, $\mathbb{G}_m(A)$ is a group for every A, functorially in A.

Similarly, the additive group scheme $\mathbb{G}_a = \operatorname{Spec}(k[t]) = \mathbb{A}^1_k$ satisfies $\mathbb{G}_a(A) = A$ with its additive group structure; and the general linear group $\mathrm{GL}_{n,k} = \operatorname{Spec}(k[x_{ij}, \det^{-1}])$ satisfies $\mathrm{GL}_{n,k}(A) \cong \mathrm{GL}_n(A)$ for any k-algebra A. These examples demonstrate a recurring theme: when working over a non-algebraically-closed base, or with non-reduced schemes, the functor of points often gives a cleaner description than the underlying topological space. One *defines* the algebraic group by specifying what it does on all test rings, and the scheme structure is a consequence.

3.3.5 Representable Functors in Algebraic Geometry

The examples of the previous subsection probe a fixed scheme X by varying the test object Z. We now reverse the perspective: fix the type of geometric data we wish to extract, and ask whether there exists a "universal" scheme that classifies it. Formally, given a contravariant functor $\mathcal{F} : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$, we say $\mathcal{F}$ is *representable* if there exists a scheme M and a natural isomorphism $\mathcal{F} \cong h_M$. The scheme M is then called a *fine moduli space* for the problem encoded by $\mathcal{F}$, and by the Yoneda Lemma it is unique up to unique isomorphism.

We have already encountered the simplest instance of this idea.

Proposition 3.3.7: Global Sections and the Affine Line

Let X be a k-scheme. There is a natural bijection

$$\Gamma(X, \mathcal{O}_X) \;\cong\; \operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{A}^1_k).$$

Concretely, a global section $f \in \Gamma(X, \mathcal{O}_X)$ corresponds to the unique morphism $\pi_f : X \to \mathbb{A}^1_k$ such that $\pi_f^\#(t) = f$, where t is the coordinate on $\mathbb{A}^1_k = \operatorname{Spec}(k[t])$. In categorical language, the global section functor $\Gamma(-, \mathcal{O})$ is representable by $\mathbb{A}^1_k$.

***Proof*:**
By the adjunction between Spec and Γ (proposition 3.1.12):

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \operatorname{Spec} k[t]) \;\cong\; \operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t], \Gamma(X, \mathcal{O}_X)).$$

The polynomial ring $k[t]$ is the free k-algebra on one generator: a k-algebra homomorphism $k[t] \to R$ is uniquely determined by the image of t. Thus:

$$\operatorname{Hom}_{k\text{-}\mathbf{Alg}}(k[t], \Gamma(X, \mathcal{O}_X)) \;\cong\; \Gamma(X, \mathcal{O}_X).$$

Let us verify that the bijection matches our intuition of evaluation. Let $x \in X$ and let $\pi_f : X \to \mathbb{A}^1_k$ be the morphism corresponding to $f \in \Gamma(X, \mathcal{O}_X)$. The definition of a scheme morphism gives a commutative diagram of local rings:

$$\begin{tikzcd}
k[t] \arrow[r, "\psi"] \arrow[d] & \Gamma(X, \mathcal{O}_X) \arrow[r] & \mathcal{O}_{X,x} \arrow[d, "\text{residue}"] \\
\mathcal{O}_{\mathbb{A}^1, \pi_f(x)} \arrow[rr] & & \kappa(x)
\end{tikzcd}$$

The composition across the top sends $t \mapsto f \mapsto f(x) \in \kappa(x)$, and the image point $\pi_f(x)$ is the prime ideal $\mathfrak{q} = \{P(t) \in k[t] : P(f(x)) = 0 \text{ in } \kappa(x)\}$. This is exactly the scheme-theoretic "value" of f at x.

3.3.8 Remark: no field hypothesis is needed The above proof uses only that $k[t]$ is free on one generator; it did not use that k is a field. For any base scheme S, the relative affine line $\mathbb{A}^1_S$ represents the global section functor on $\mathbf{Sch}_S$. The same remark applies to several of the examples that follow.

The representability of $\Gamma(-, \mathcal{O})$ by $\mathbb{A}^1_k$ is the first instance of a broad pattern. Let us now survey the most important representable functors in algebraic geometry. In each case, a piece of geometric data attached to a scheme X is captured by morphisms from X into a single "classifying" scheme.

The multiplicative group. Replacing the additive structure of $\mathbb{A}^1_k$ with the multiplicative structure of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ gives:

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{G}_m) \;\cong\; \Gamma(X, \mathcal{O}_X)^\times.$$

The proof follows the same pattern: a k-algebra map $k[t, t^{-1}] \to R$ is determined by the image of t, and invertibility of t in the source forces the image to be a unit. The $\mathbb{G}_m$-valued points of X are precisely the invertible global sections.

Projective space and line bundles. The representability of projective space is considerably deeper. Define the functor

$$\mathcal{F}(X) = \{(\mathcal{L},\, s_0, \ldots, s_n) \mid \mathcal{L} \in \operatorname{Pic}(X),\; s_i \in \Gamma(X, \mathcal{L}),\; \text{the } s_i \text{ generate } \mathcal{L}\} \,/\!\cong$$

where two tuples $(\mathcal{L}, s_0, \ldots, s_n)$ and $(\mathcal{L}', s_0', \ldots, s_n')$ are identified if there is an isomorphism $\mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ carrying s_i to s_i' for each i. Then there is a natural bijection

$$\operatorname{Hom}_{\mathbf{Sch}_k}(X, \mathbb{P}^n_k) \;\cong\; \mathcal{F}(X).$$

A morphism to $\mathbb{P}^n$ is the same data as a line bundle equipped with $n+1$ globally generating sections; these sections serve as "homogeneous coordinates" for the map. This is the content of the theory developed in section 5.2.1. When $n = 1$, a map $X \to \mathbb{P}^1$ corresponds to a line bundle with two generating sections—the algebro-geometric analogue of a meromorphic function.

Grassmannians. The Grassmannian $\operatorname{Gr}(r, n)$ generalizes projective space from lines to r-dimensional subspaces. It represents the functor

$$\mathcal{F}(X) = \left\{\mathcal{O}_X^{\oplus n} \twoheadrightarrow \mathcal{E} \;\middle|\; \mathcal{E} \text{ is a locally free sheaf of rank } r\right\} /\!\cong.$$

A morphism $X \to \operatorname{Gr}(r, n)$ classifies a rank-r quotient bundle of the trivial bundle $\mathcal{O}_X^{\oplus n}$, or equivalently, a sub-bundle of rank $n - r$. When $r = 1$, this recovers the description of projective space above. The Grassmannian is itself a smooth projective variety; its functor-of-points description gives a uniform construction that works over any base ring, without the need to choose coordinates or Plücker embeddings.

The general linear group. The group scheme $\operatorname{GL}_{n,k} = \operatorname{Spec}(k[x_{11}, \ldots, x_{nn}, \det^{-1}])$ represents the functor

$$\operatorname{Hom}_{\mathbf{Sch}_k}(\operatorname{Spec}(A), \operatorname{GL}_{n,k}) \;\cong\; \operatorname{GL}_n(A)$$

for any k-algebra A. A map into $\mathrm{GL}_{n,k}$ is the same as giving an invertible $n \times n$ matrix with entries in A. Combined with the group scheme structure from Example 3.3.6, this gives a definition of the general linear group that works uniformly over all base rings, including rings with nilpotents or mixed characteristic.

Hilbert schemes. Perhaps the most striking application of representability is given by the Hilbert scheme, which we develop in the next subsection.

The common thread in each of the above examples is the Yoneda philosophy: a geometric invariant of X—global sections, line bundles, quotient bundles, invertible matrices—is captured by morphisms from X into a single classifying scheme. The Yoneda Lemma ensures that the classifying scheme, when it exists, is unique up to unique isomorphism.

3.3.6 Hilbert Schemes

The representable functors considered so far classify data that is "linear" in nature: sections of a sheaf, quotients of a free module, invertible matrices. The Hilbert scheme is of a fundamentally different character: it parametrizes *subschemes* of a given projective scheme. That such a moduli problem admits a scheme as its solution is one of the deepest results in the functorial approach to algebraic geometry, and it provides a rich source of examples and applications.

Flatness as the continuity condition

Before stating the result, we must address a foundational question: what does it mean for a collection of subschemes to "vary continuously" over a parameter space? Consider a family of closed subschemes $\{Z_s\}_{s \in S}$ of a fixed projective scheme $Y \subseteq \mathbb{P}^n_k$, parametrized by a scheme S. Algebraically, this family is encoded by a closed subscheme $Z \subseteq Y \times_k S$ such that each fiber $Z_s = Z \times_S \{s\}$ is the corresponding member of the family.

Not every such Z should be considered a "nice" family. For instance, one could construct a family where a curve degenerates to a union of a curve and an isolated point, causing the Hilbert polynomial to jump. The condition that prevents such pathologies is *flatness*: we require that Z be flat over S. Flatness ensures that the Hilbert polynomial $P_{Z_s}(m) = \chi(Z_s, \mathcal{O}_{Z_s}(m))$ is constant as s varies over connected components of S. This is the correct algebraic analogue of "continuous variation": the members of the family all share the same discrete numerical invariants.

The reader who has not yet encountered flatness in detail should think of it as a homological condition: a morphism $\pi : Z \to S$ is flat if the fibers Z_s do not "jump" in any cohomological sense. A thorough treatment will be given in section 6.4.

The Hilbert functor and Grothendieck's theorem

Fix a projective scheme $Y \subseteq \mathbb{P}^n_k$ and a numerical polynomial $p \in \mathbb{Q}[t]$ (recall that the Hilbert polynomial of any coherent sheaf on projective space is a polynomial with rational coefficients). The *Hilbert functor* is the contravariant functor $\mathcal{H}ilb^p_Y : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ defined by

$$\mathcal{H}ilb^p_Y(S) = \left\{ Z \subseteq Y \times_k S \;\middle|\; \begin{array}{l} Z \text{ is a closed subscheme, flat over } S, \\ \text{with Hilbert polynomial } p \text{ on each fiber} \end{array} \right\}.$$

For a morphism $f : S' \to S$, the induced map $\mathcal{H}ilb_Y^p(S) \to \mathcal{H}ilb_Y^p(S')$ sends a family $Z \subseteq Y \times S$ to its pullback $Z \times_S S' \subseteq Y \times S'$.

Theorem 3.3.9: Grothendieck's Existence Theorem

Let $Y \subseteq \mathbb{P}_k^n$ be a projective scheme over a Noetherian ring k, and let $p \in \mathbb{Q}[t]$ be a numerical polynomial. Then the Hilbert functor $\mathcal{H}ilb_Y^p$ is representable by a projective scheme Hilb_Y^p over k:

$$\mathrm{Hom}_{\mathbf{Sch}_k}(S, \mathrm{Hilb}_Y^p) \;\cong\; \mathcal{H}ilb_Y^p(S)$$

naturally in S.

The proof is beyond our scope (it relies on a careful analysis of Castelnuovo–Mumford regularity and the Grassmannian embedding), but the statement is worth absorbing. It says that the totality of all subschemes of Y with a given Hilbert polynomial is itself a geometric object—a projective scheme—on which one can do geometry. There is a *universal family*: a closed subscheme $\mathcal{Z} \subseteq Y \times \mathrm{Hilb}_Y^p$, flat over Hilb_Y^p, such that every flat family $Z \subseteq Y \times S$ with Hilbert polynomial p is the pullback of $\mathcal{Z}$ along a unique morphism $S \to \mathrm{Hilb}_Y^p$.

Example: the Hilbert scheme of points

The simplest interesting case arises when p is the constant polynomial $p(m) = n$ for a positive integer n. Then a closed subscheme $Z \subseteq Y$ with Hilbert polynomial $p(m) = n$ is a zero-dimensional subscheme of length n: it is supported on at most n points of Y, possibly with multiplicity (i.e., with nilpotent structure). The corresponding Hilbert scheme is denoted $\mathrm{Hilb}^n(Y)$ and is called the *Hilbert scheme of n points* on Y.

When Y is a smooth surface, $\mathrm{Hilb}^n(Y)$ turns out to be a smooth variety of dimension $2n$. This is a nontrivial theorem (due to Fogarty) and stands in contrast to the naive expectation: the "symmetric product" $\mathrm{Sym}^n(Y) = Y^n/S_n$, which parametrizes unordered n-tuples of points, is singular along the diagonals where points collide. The Hilbert scheme resolves these singularities by remembering the infinitesimal directions in which colliding points approach each other.

To see this concretely, consider $Y = \mathbb{A}_k^2$ and $n = 2$. A length-2 subscheme $Z \subseteq \mathbb{A}^2$ is one of two types:

1. Two distinct points $\{p, q\}$, corresponding to the ideal $I_p \cap I_q \subseteq k[x, y]$ where I_p and I_q are the maximal ideals of p and q. The quotient $k[x, y]/I$ has dimension 2 as a k-vector space.

2. A single point p with a tangent direction $v \in T_p\mathbb{A}^2$. If $p = (0, 0)$ and v points in the x-direction, the corresponding ideal is (y, x^2), and $k[x, y]/(y, x^2) \cong k[x]/(x^2)$ has dimension 2. The extra nilpotent direction records "how the two points came together."

Thus $\mathrm{Hilb}^2(\mathbb{A}^2)$ parametrizes pairs of distinct points together with a "blow-up" of the diagonal: at each point of the diagonal, the fiber is a $\mathbb{P}^1$ recording the tangent direction of approach. This is precisely the blow-up of $\mathrm{Sym}^2(\mathbb{A}^2)$ along its singular locus.

Tangent spaces to the Hilbert scheme

The functor-of-points perspective makes the computation of tangent spaces to moduli spaces almost mechanical. By Example 3.3.5, a tangent vector to Hilb_Y^p at a point $[Z]$ (corresponding to a subscheme

$Z \subseteq Y$) is a morphism $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2)) \to \mathrm{Hilb}^p_Y$ sending the closed point to $[Z]$. By the universal property, this is the same as a flat family of subschemes of Y over $\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))$ whose special fiber is Z. One can show that such a family corresponds to a first-order deformation of Z inside Y, and that:

$$\Theta_{\mathrm{Hilb}^p_Y,\,[Z]} \;\cong\; \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z,\,\mathcal{O}_Z)$$

where $\mathcal{I}_Z$ is the ideal sheaf of Z in Y. This identification—tangent vectors to the Hilbert scheme are the same as homomorphisms from the ideal sheaf to the structure sheaf of the subscheme—is a prototype for the general principle that tangent spaces to moduli spaces are computed by Hom or Ext groups.

3.3.7 Schemes as Functors

The reader may have noticed an apparent circularity in the functor-of-points perspective. We define the functor $h_X : \mathbf{Sch}^{\mathrm{op}} \to \mathbf{Set}$ using the category $\mathbf{Sch}$ of schemes, but if the functor of points is to serve as a *definition* of schemes, we seem to need the category $\mathbf{Sch}$ already in hand.

The resolution, due to Grothendieck and developed systematically by Demazure and Gabriel, proceeds in stages and is worth spelling out.

Step 1: Start with rings. The category $\mathbf{CRing}$ of commutative rings is purely algebraic. We define the category of *affine schemes* as $\mathbf{Aff} = \mathbf{CRing}^{\mathrm{op}}$, the formal opposite category. No geometry has been invoked.

Step 2: Embed into functors. Any functor $F : \mathbf{CRing} \to \mathbf{Set}$ (equivalently, a presheaf on $\mathbf{Aff}$) can be thought of as a "generalized space." An affine scheme $\mathrm{Spec}\,(A)$ defines such a functor by $h_{\mathrm{Spec}\,(A)}(R) = \mathrm{Hom}_{\mathbf{CRing}}(A, R)$. By the Yoneda Lemma, the assignment $\mathrm{Spec}\,(A) \mapsto h_{\mathrm{Spec}\,(A)}$ is a fully faithful embedding $\mathbf{Aff} \hookrightarrow \mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$.

Step 3: Impose a sheaf condition. Not every functor $\mathbf{CRing} \to \mathbf{Set}$ deserves to be called geometric. We require that F be a *sheaf* for the Zariski topology. Concretely, if $f_1, \ldots, f_n$ generate the unit ideal in a ring A (so that $\mathrm{Spec}\,(A) = \bigcup_i D(f_i)$ is a cover by distinguished open sets), then the diagram

$$F(A) \longrightarrow \prod_i F(A_{f_i}) \rightrightarrows \prod_{i,j} F(A_{f_i f_j})$$

should be an equalizer. This is the local-to-global property that geometric objects possess: compatible data on an open cover should glue uniquely. A sheaf satisfying this property is called a *Zariski sheaf.*

Step 4: Require an open cover by affines. A *scheme* is a Zariski sheaf F that admits an open cover by representable subfunctors. That is, there exist rings A_i and natural transformations $h_{\mathrm{Spec}\,(A_i)} \hookrightarrow F$ that are "open immersions" (in the sense defined below) and that jointly cover F.

This definition recovers the usual category of schemes without circularity: we begin with rings, pass to representable functors (affine schemes), impose a sheaf condition (gluing), and characterize schemes as those sheaves admitting an affine open cover. The result is a full subcategory of $\mathrm{Fun}(\mathbf{CRing}, \mathbf{Set})$ defined by intrinsic conditions.

The practical payoff is significant. To define a scheme, it suffices to specify a functor $\mathcal{F} : \mathbf{CRing} \to \mathbf{Set}$ and verify the Zariski sheaf condition and the existence of an affine open cover. This is often considerably easier than constructing a locally ringed space by hand, especially for moduli problems: the Hilbert scheme was originally constructed by Grothendieck precisely by first defining the functor $\mathcal{H}ilb^p_Y$ and then proving it satisfies the above conditions. The same strategy applies to Picard schemes, moduli of curves, and many other classifying spaces that arise in algebraic geometry.

Open subfunctors

Step 4 above relies on a notion of "open immersion" for functors, which we now make precise. Let $F : \mathbf{CRing} \to \mathbf{Set}$ be a Zariski sheaf. A subfunctor $G \subseteq F$ (meaning $G(A) \subseteq F(A)$ for every ring A, compatibly with restriction maps) is called an *open subfunctor* if the following condition holds: for every ring A and every element $\xi \in F(A)$, there exist elements $f_1, \ldots, f_n \in A$ generating the unit ideal such that, for each i, the image of ξ in $F(A_{f_i})$ lies in $G(A_{f_i})$.

Informally, G is open in F if membership in G can be "checked locally." To see why this is the right definition, consider the representable case. If $F = h_{\mathrm{Spec}\,(B)}$ and G corresponds to an open subscheme $U \subseteq \mathrm{Spec}\,(B)$, then a morphism $\xi : \mathrm{Spec}\,(A) \to \mathrm{Spec}\,(B)$ factors through U if and only if its image lands in U. The preimage $\xi^{-1}(U)$ is an open subset of $\mathrm{Spec}\,(A)$, and the condition that $f_1, \ldots, f_n$ generate the unit ideal with $\xi|_{D(f_i)}$ landing in U says exactly that $\xi^{-1}(U)$ covers $\mathrm{Spec}\,(A)$—i.e., ξ factors through U.

A collection of open subfunctors $\{G_i \subseteq F\}$ is said to *cover* F if for every field K, the sets $G_i(K)$ jointly cover $F(K)$. The condition in Step 4 can now be restated: a Zariski sheaf F is a scheme if it admits a cover by open subfunctors, each of which is representable by an affine scheme.

Tangent spaces to functors

One of the practical advantages of the functorial viewpoint is that many geometric constructions can be performed directly on functors, without first knowing that the functor is representable. We have already seen in Example 3.3.5 that the tangent space to a scheme X at a k-rational point x is recovered by probing with the dual numbers $D = k[\epsilon]/(\epsilon^2)$. This construction generalizes immediately to arbitrary functors.

Definition 3.3.10: Tangent Space of a Functor

Let $F : \mathbf{Sch}_k^{\mathrm{op}} \to \mathbf{Set}$ be a functor and let $\xi_0 \in F(\mathrm{Spec}\,(k))$ be a k-rational point of F. The *tangent space* of F at ξ_0 is

$$T_{\xi_0} F := \left\{ \xi \in F(\mathrm{Spec}\,(k[\epsilon]/(\epsilon^2))) \;\middle|\; \xi \text{ maps to } \xi_0 \text{ under } F(\mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(D)) \right\}.$$

When $F = h_X$ for a scheme X and ξ_0 corresponds to a closed point $x \in X$ with $\kappa(x) = k$, this recovers the Zariski tangent space $\Theta_{X,x}$ by Example 3.3.5. The key point is that the definition makes sense even when F is not representable.

For the tangent space to carry a k-vector space structure, one needs the functor F to satisfy a mild condition: $T_{\xi_0} F$ should be compatible with the k-module structure on square-zero extensions of k. This is automatic for representable functors, and holds for most functors arising in practice (including the Hilbert functor and the Picard functor).

As an application, we computed the tangent space of the Hilbert functor at the end of section 3.3.6: for a subscheme $Z \subseteq Y$ with ideal sheaf $\mathcal{I}_Z$, the tangent space $T_{[Z]} \operatorname{Hilb}^p_Y \cong \operatorname{Hom}_{\mathcal{O}_Y}(\mathcal{I}_Z, \mathcal{O}_Z)$. This computation was carried out purely at the level of the functor $\mathcal{H}ilb^p_Y$; the fact that Hilb^p_Y is actually a scheme (Theorem 3.3.9) was not needed for the computation itself, only for the conclusion that this tangent space has geometric meaning.

3.3.8 Moduli Problems and the Limits of Representability

The Hilbert scheme demonstrates the power of the functorial approach: define a moduli functor, prove it is representable, and obtain a geometric parameter space for free. A natural question is how far this method extends. Can every reasonable classification problem in algebraic geometry be solved by a representing scheme?

The answer is no, and understanding *why* it fails is as instructive as the successes.

The moduli problem for elliptic curves

An *elliptic curve* over a field k is a smooth projective curve of genus 1 equipped with a distinguished rational point (the reader unfamiliar with this notion may consult section 7.1.3 and return here afterward). Two elliptic curves are isomorphic if and only if they have the same j-invariant, an element of k.

We would like a "moduli space of elliptic curves": a scheme M whose points correspond to isomorphism classes of elliptic curves, and such that families of elliptic curves over a base S correspond to morphisms $S \to M$. The functor to be represented is:

$$\mathcal{M}_{1,1}(S) = \left\{ (E \to S,\, \sigma : S \to E) \;\middle|\; \begin{array}{l} E \to S \text{ is a smooth proper morphism} \\ \text{with genus-1 geometric fibers,} \\ \sigma \text{ is a section} \end{array} \right\} /\!\cong_S$$

where $\cong_S$ denotes isomorphism of families over S. If this functor were representable by a scheme M, there would exist a *universal family*: an elliptic curve $\mathcal{E} \to M$ such that every family $E \to S$ is the pullback of $\mathcal{E}$ along a unique morphism $S \to M$.

Over an algebraically closed field, the isomorphism classes of elliptic curves are classified by the j-invariant, so the underlying set of M should be $\mathbb{A}^1$. Indeed, there is a scheme—the j-line $\mathbb{A}^1_k$—that serves as a *coarse moduli space*: its closed points are in bijection with isomorphism classes, and for any family $E \to S$ there exists a unique morphism $S \to \mathbb{A}^1$ whose fiber over a point s has j-invariant equal to the j-invariant of E_s. But the j-line is *not* a fine moduli space. The functor $\mathcal{M}_{1,1}$ is not representable.

The failure is easy to spot: every elliptic curve E over a field of characteristic $\neq 2$ has a nontrivial automorphism, namely it has at least the inversion map $\iota : E \to E$ sending $P \mapsto -P$ (with respect to the group law). For the curves with $j = 0$ and $j = 1728$, the automorphism group is even larger ($\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$, respectively). These automorphisms are the obstruction to representability.

To see why concretely, suppose a fine moduli space M existed with universal family $\mathcal{E} \to M$. Consider the trivial family $E \times S \to S$ for a fixed elliptic curve E with nontrivial automorphism α. The identity map $\mathrm{id} : E \times S \to E \times S$ and the automorphism $\alpha \times \mathrm{id}_S : E \times S \to E \times S$ are *different* isomorphisms of the family, but they give rise to the *same* classifying morphism $S \to M$ (since they classify the same family up to isomorphism). This violates the requirement that the classifying morphism be unique.

More precisely, the issue is that the functor $\mathcal{M}_{1,1}$ does not satisfy the sheaf condition in sufficient generality: it fails descent for étale covers. A family that is locally trivial in the étale topology need not be globally trivial, and the automorphisms of the fibers obstruct the gluing.

Coarse moduli spaces

The j-line is an example of a *coarse moduli space*: a scheme M equipped with a natural transformation $\mathcal{M}_{1,1} \to h_M$ that is universal among maps to representable functors, and that induces a bijection $\mathcal{M}_{1,1}(\mathrm{Spec}\,(\bar{k})) \xrightarrow{\sim} M(\bar{k})$ on geometric points. A coarse moduli space classifies objects up to isomorphism at the level of points, but does not carry a universal family and does not faithfully track families over general bases.

Beyond schemes: stacks

The correct solution is to enlarge the category of geometric objects. Rather than forcing $\mathcal{M}_{1,1}(S)$ to be a *set* (of isomorphism classes), one retains it as a *groupoid*: the category whose objects are families $E \to S$ and whose morphisms are isomorphisms of families. The resulting structure is an *algebraic stack*, often denoted $\mathcal{M}_{1,1}$ as well. The stack remembers the automorphisms of each object, and this additional data resolves the descent obstruction.

An intermediate notion is that of an *algebraic space*. These arise when the moduli problem has no nontrivial automorphisms but the functor still fails to be representable by a scheme—for instance, because the "correct" topology for gluing is the étale topology rather than the Zariski topology. Algebraic spaces are Zariski-locally schemes but may not be globally so; they are to schemes what manifolds are to open subsets of $\mathbb{R}^n$.

The hierarchy, in increasing generality, is:

$$\text{Schemes} \subset \text{Algebraic Spaces} \subset \text{Algebraic Stacks} \subset \text{Higher Stacks},$$

where each step relaxes a condition on the moduli functor. Schemes require representability by a Zariski sheaf with an affine cover. Algebraic spaces replace the Zariski topology with the étale topology. Algebraic stacks allow the functor to take values in groupoids rather than sets. The development of this hierarchy is beyond our scope, but the reader should be aware that the functor-of-points perspective developed in this section is the gateway to this broader landscape: every step in the hierarchy is formulated in terms of functors and their properties.

3.3.9 *The Fourier–Mukai Transform

The functor-of-points philosophy associates to a scheme X the collection of all morphisms into X. One can push this idea further by asking: what information is carried not by morphisms, but by *sheaves* on products? This line of thought leads to the Fourier–Mukai transform, which can be understood as a "categorified" version of the change-of-basis perspective developed above. This subsection is intended as a brief orientation for the interested reader; the full theory belongs to the realm of derived categories and will not be developed in detail here.

From graphs to correspondences

Given a morphism $f : X \to Y$, its *graph* $\Gamma_f = \{(x, f(x)) \mid x \in X\}$ is a closed subscheme of $X \times Y$. The morphism f is entirely recoverable from Γ_f: project to X (an isomorphism) and then to Y.

A natural generalization is to drop the requirement that the projection to X be an isomorphism. Any closed subscheme $Z \subseteq X \times Y$—or, more generally, any sheaf $\mathcal{P}$ on $X \times Y$—defines a "correspondence" between X and Y. For a point $x \in X$, the fiber $Z_x = Z \cap (\{x\} \times Y)$ is a subscheme of Y, and as x varies, one obtains a family of subschemes of Y parametrized by X. This is precisely the kind of data that Hilbert schemes classify.

A word on derived categories

To formulate the Fourier–Mukai transform precisely, one works not with the abelian category $\mathbf{Coh}(X)$ of coherent sheaves, but with its *bounded derived category* $D^b(X)$. The motivation is that many natural operations on sheaves—tensor products, pushforwards, global sections—are not exact, and their "corrections" (the higher derived functors Tor, $R^i f_*$, H^i) carry essential geometric information. The derived category is a framework in which a sheaf is replaced by a *complex* of sheaves, and these higher derived functors become part of a single, coherent operation.

An object of $D^b(X)$ is a bounded complex

$$\cdots \to \mathcal{F}^{i-1} \to \mathcal{F}^i \to \mathcal{F}^{i+1} \to \cdots$$

of coherent sheaves, considered up to *quasi-isomorphism*: two complexes are identified if they have isomorphic cohomology sheaves. A single sheaf $\mathcal{F}$ is regarded as a complex concentrated in degree zero. The derived category retains the information of the abelian category (via the cohomology functor H^i) but also remembers the "process" of resolution, which is essential for the functoriality that follows.

The reader familiar with the construction of Ext groups via projective or injective resolutions will recognize the underlying idea: in the derived category, one works with the resolutions themselves rather than discarding them after taking cohomology.

The transform

Let X and Y be smooth projective varieties over a field k, and let $\pi_X : X \times Y \to X$ and $\pi_Y : X \times Y \to Y$ denote the projection morphisms. Given a "kernel" $\mathcal{P} \in D^b(X \times Y)$, the *Fourier–Mukai transform* with kernel $\mathcal{P}$ is the functor

$$\Phi_{\mathcal{P}} : D^b(X) \longrightarrow D^b(Y), \qquad \mathcal{F} \longmapsto R\pi_{Y,*}(\mathcal{P} \otimes^L \pi_X^* \mathcal{F}),$$

where π_X^* is the (exact) pullback, $\otimes^L$ is the derived tensor product, and $R\pi_{Y,*}$ is the derived pushforward. In words: pull the sheaf $\mathcal{F}$ back from X to the product, "multiply" by the kernel $\mathcal{P}$, and "integrate" (push forward) along Y.

The analogy with the classical Fourier transform on $\mathbb{R}^n$ is worth making explicit. Recall that the classical transform sends a function f on $\mathbb{R}^n$ to a function $\hat{f}$ on the dual space $(\mathbb{R}^n)^*$ via

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)\, e^{2\pi i \langle x, \xi \rangle}\, dx.$$

The exponential $e^{2\pi i \langle x, \xi \rangle}$ is a "kernel" on the product $\mathbb{R}^n \times (\mathbb{R}^n)^*$; the multiplication by f is the analogue of tensoring with $\pi_X^* \mathcal{F}$; and the integration over $\mathbb{R}^n$ is the analogue of the derived pushforward $R\pi_{Y,*}$. The Fourier–Mukai transform is, in this precise sense, the algebraic geometer's Fourier transform, with coherent sheaves playing the role of functions and the derived pushforward replacing integration.

From delta-points to character-points

The classical Fourier transform admits a particularly illuminating description in terms of two distinguished "bases" for the space of functions on a locally compact abelian group G. The first basis consists of the *delta functions* δ_x for $x \in G$: any function can be written as $f = \int_G f(x)\,\delta_x\,dx$, expressing f in terms of where it is "supported." The second basis consists of the *characters* χ_ξ for $\xi \in \hat{G}$ (the Pontryagin dual group): the Fourier transform rewrites f as $\hat{f} = \int_G f(x)\,\chi_\xi(x)\,dx$, expressing f in terms of its "frequency content." The Fourier transform is, at its core, a change of basis from delta-points to character-points:

$$\widehat{\delta_x} = \chi_x, \qquad \widehat{\chi_\xi} = \delta_\xi.$$

The Fourier–Mukai transform makes this change of basis precise in algebraic geometry, and the natural setting is that of *abelian varieties.* Let A be an abelian variety of dimension g over k (a smooth projective group variety—the higher-dimensional generalization of an elliptic curve). Its *dual abelian variety* $\hat{A} = \operatorname{Pic}^0(A)$ parametrizes the degree-zero line bundles on A: a point $\xi \in \hat{A}$ corresponds to a line bundle L_ξ on A that is algebraically equivalent to zero. The dual $\hat{A}$ is the algebro-geometric analogue of the Pontryagin dual $\hat{G}$.

On the product $A \times \hat{A}$ there is a canonical line bundle $\mathcal{P}$, the *Poincaré bundle*, characterized by the universal property: for each $\xi \in \hat{A}$, the restriction $\mathcal{P}|_{A\times\{\xi\}} \cong L_\xi$. The Poincaré bundle is the algebro-geometric analogue of the exponential kernel $e^{2\pi i\langle x,\xi\rangle}$: it "pairs" a point of A with a character from $\hat{A}$.

The Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ with kernel $\mathcal{P}$ now enacts the change of basis between delta-points and character-points. The two fundamental computations are:

Deltas map to characters. Let $a \in A$ be a closed point and let $k(a)$ denote the skyscraper sheaf at a (the sheaf supported at the single point a with stalk k). The skyscraper $k(a)$ is the algebro-geometric δ-function: it is "concentrated at a" and detects nothing elsewhere. Applying the transform:

$$\Phi_{\mathcal{P}}(k(a)) = R\pi_{\hat{A},*}(\mathcal{P} \otimes^L \pi_A^* k(a)) = R\pi_{\hat{A},*}(\mathcal{P}|_{\{a\}\times\hat{A}}) = \mathcal{P}|_{\{a\}\times\hat{A}}.$$

The restriction $\mathcal{P}|_{\{a\}\times\hat{A}}$ is a line bundle on $\hat{A}$. Under the double-duality identification $A \cong \hat{\hat{A}}$, this is precisely the "character" on $\hat{A}$ associated to the point a—just as the classical Fourier transform sends δ_x to the character χ_x.

Characters map to deltas. Let $\xi \in \hat{A}$ and let $L_\xi = \mathcal{P}|_{A\times\{\xi\}}$ be the corresponding degree-zero line bundle on A. The line bundle L_ξ is the algebro-geometric "character": tensoring with L_ξ defines a "twist" of sheaves on A, analogous to multiplication by χ_ξ. One computes:

$$\Phi_{\mathcal{P}}(L_\xi) \cong k(\xi)[-g]$$

where $k(\xi)$ is the skyscraper sheaf at the point $\xi \in \hat{A}$ and $[-g]$ denotes a cohomological shift by $g = \dim A$. Up to this shift (which accounts for the difference between underived and derived pushforward on a g-dimensional variety), the character L_ξ is sent to the delta-function $k(\xi)$ on the dual—exactly as $\widehat{\chi_\xi} = \delta_\xi$ in the classical setting.

The cohomological shift $[-g]$ is the price of working in the derived category. It appears because the higher direct images $R^i\pi_{\hat{A},*}$ of L_ξ vanish for $i \neq g$ (a consequence of the fact that L_ξ has no global sections for $\xi \neq 0$, and its cohomology is concentrated in degree g). The shift is harmless: it is an automorphism of $D^b(\hat{A})$, and Mukai's theorem accounts for it by asserting that the *composition*

$$D^b(A) \xrightarrow{\Phi_{\mathcal{P}}} D^b(\hat{A}) \xrightarrow{\Phi_{\mathcal{P}^\vee}} D^b(A)$$

is isomorphic to the shift functor $(-1)_A^*[-g]$, where $(-1)_A$ is the inversion on A. This is the precise analogue of the Fourier inversion formula $\hat{\hat{f}}(x) = f(-x)$.

Mukai's theorem and Orlov representability

The computations above are instances of a fundamental result due to Mukai: the Fourier–Mukai transform $\Phi_{\mathcal{P}} : D^b(A) \to D^b(\hat{A})$ is an *equivalence of categories*. This is a strong statement: it says that the derived category of coherent sheaves on an abelian variety is equivalent to that of its dual. Two geometrically distinct varieties—A and $\hat{A}$ need not be isomorphic—are "the same" from the perspective of their sheaf theory.

More broadly, Orlov's representability theorem asserts that for smooth projective varieties, every equivalence $D^b(X) \xrightarrow{\sim} D^b(Y)$ is isomorphic to a Fourier–Mukai transform $\Phi_{\mathcal{P}}$ for some kernel $\mathcal{P} \in D^b(X \times Y)$. This is the categorified Yoneda Lemma: just as a natural transformation $h_X \to h_Y$ is determined by a single element of $\mathrm{Hom}(X, Y)$, a sufficiently nice functor $D^b(X) \to D^b(Y)$ is determined by a single object of $D^b(X \times Y)$.

The analogy with the change-of-basis perspective from section 3.3.3 is clean. The Yoneda embedding probes X by mapping test objects *into* X; the Fourier–Mukai transform probes X by its category of sheaves. The kernel $\mathcal{P}$ on $X \times Y$ mediates between these two descriptions, playing the role of a "change-of-basis matrix" between two categorical coordinate systems. On abelian varieties, this change of basis is literally the passage from delta-points to character-points—the oldest and most fundamental duality in harmonic analysis, now transplanted into algebraic geometry.

3.4 Finiteness and Affine Conditions

As schemes are rather general notions, so to scheme morphisms can be very general. The following notions are guiding principles presented by Vakil on what a good type of morphism should have. This terminology will be very useful in labeling many of the following lemmas an propositions. Let's say that C is a collection of morphisms between schemes. Then:

1. it is *local on target* if
   (a) $\pi : X \to Y$ is in the class, then for any open subset $V \subseteq Y$, the restricted homomorphism $\pi^{-1}(V) \to V$ is in the class
   (b) $\pi : X \to Y$ is a morphism, $\cup_i V_i = Y$ is an open cover where each $\pi^{-1}(V_i) \to V_i$ is in this class, then π is in this class

   Usually we consider affine opens, but it need not be the case.
2. is *local on source* if there is
   (a) $\pi : X \to Y$ is in the class, then for any open subset $U \subseteq X$, the restricted homomorphism $U \to \pi(U)$ is in the class

   (b) $\pi : X \to Y$ is a morphism, $\cup_i U_i = X$ is an open cover where each $U_i \to \pi(U_i)$ is in this class, then π is in this class

3. it is closed under composition
4. it is closed under fibered products. As a special case, is closed under change of basis from an S-scheme to a T-scheme

Finiteness Conditions

Definition 3.4.1: Quasicompact Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasicompact* if every open affine subset $U \subseteq Y$ has quasicompact pre-image, that is $\pi^{-1}(U)$ is quasicompact.

Naturally, Quasicompactness is is closed under composition, and is local on target.

Example 3.4.2: Quasicompact Morphism

We shall in this section show that other types of morphism shall be quasicompact (finite, embeddings, proper, etc). At the moment, we shall give concrete examples:

1. Any scheme morphism of affine schemes is quasicompact, this relies on the use of the distinguished open base: Let $f : \operatorname{Spec} S \to \operatorname{Spec} R$ and $U \subseteq \operatorname{Spec} R$ be quasi-compact. Then $U = \bigcup_i^n D(g_i)$ for appropriate distinguished open sets. Then since $f^{-1}(D(g_i)) = D(f^{\#}(g_i))$, we have that
$$f^{-1}(U) = \bigcup_i^n D(f^{\#}(g_i))$$
showing the pre-image is quasi-compact.
2. Let $X = \coprod_i^\infty \operatorname{Spec} k$ and $Y = \operatorname{Spec} k$. Then the natural projection $\pi : X \to Y$ is certainly not quasicompact, the pre-image of $\operatorname{Spec} k$ not being quasicompact.
3. Let X be Noetherian schemes and $f : X \to Y$ a scheme morphism. Then f is quasicompact. This follows from the following topological property: If X is a Noetherian topological space, every subset is Noetherian.

   Hence, any non-quasicompact map *must* have X as a non-Noetherian scheme, and so in practice we shall not often see non-quasicompact schemes.
4. Let $X = \operatorname{Spec} k[x]$ and $\tilde{X}$ be the blow up at every point. Then the natural map $\varphi : \tilde{X} \to X$ is not quasicompact.

For future reference, the following definition is put down (see ref:HERE for its application):

Definition 3.4.3: Quasiseparated Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasiseparated* if every open affine subset $U \subseteq Y$ has quasicompact pre-image.

Example 3.4.4: Quasiseparated Morphism
We shall see examples of such morphisms later.

The next important type of morphism that will always have affine open pre-images:

Definition 3.4.5: Affine Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *affine* if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ is an affine scheme (interpreted as an open subscheme of X)

Affine morphisms can be thought of as being the closest types of morphisms between schemes that "behave" like morphisms of affine schemes, namely since the pre-image of every affine is affine we can just cover in affine sets and study those without needing to "think" about the gluing structure. Proposition 3.4.10 shall show we only need to check a morphism is affine on a single affine cover, which is not immediately obvious as it is not *a priori* clear that we cannot "gerrymander" a partition to force affine pre-images on the open subsets of the cover.

3.4.6 Advanced Intuition This is for the reader already familiar with all this material. Let $f : X \to Y$ is a morphism. Then f is an affine morphisms if and only if there exists a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}$ such that $X \cong \operatorname{Spec}_Y(\mathcal{A})$ where $\operatorname{Spec}_Y(\mathcal{A})$ is the relative spectrum with respect to S (this is a generalization of X as an S-algebra)

We shall give examples of affine morphisms after we prove some nice results, one important such example shall be that when the domain and codomain are affine, the morphism must be affine. Let us start by showing that if the domain/codomain are not affine, the morphism may not be affine:

Example 3.4.7: Non Affine Morphisms
The prototypical example of a non affine morphism is $\mathbb{P}^1_k \to \operatorname{Spec} k$ (i.e. the structure morphism). Then the pre-image is *not* an (as shown in proposition 2.4.6). Example of a non-affine scheme where the codomain is not affine is the embedding $\mathbb{A}^n_k \to \mathbb{P}^n_k$ for $\mathbb{A}^n_k$ embeds in a choice of U_i ($n = 1$ it is enough to just do $n = 1$ to see why).

Certainly affine morphisms are stable under composition, and are stable under base change. This condition would be nice to be affine-local on target. It takes some build-up:

Lemma 3.4.8: Affine Morphisms and Qcqs

Let $\pi : X \to Y$ be an affine morphism. Then it is quasicompact and quasiseparated.

***Proof*:**
The key is that *affine schemes* are quasi-separated (and quasicompact). By the definition of an affine morphism, for every affine open subset $V \subseteq Y$, the preimage $U = \pi^{-1}(V)$ is an affine scheme, say $U \cong \operatorname{Spec} A$.

1. *Quasicompactness:* Every affine scheme is quasicompact. Since the preimage of every affine open $V \subseteq Y$ is quasicompact, the morphism π is quasicompact.

2. *Quasiseparatedness:* A morphism is quasiseparated if the preimage of every affine open

in Y is a quasiseparated scheme. Since $U = \operatorname{Spec} A$ is an affine scheme, it is separated (the diagonal $\Delta : U \to U \times_{\mathbb{Z}} U$ is a closed immersion), and every separated scheme is quasiseparated. Therefore, π is quasiseparated.

Lemma 3.4.9: Qcqs Lemma

Let X be a quasicompact and quasiseparated scheme and $s \in \mathcal{O}_X(X)$. Then

$$\mathcal{O}_X(X)_s \cong \mathcal{O}_X(X_s)$$

This theorem can be thought of as the generalization of the result where $X = \operatorname{Spec}(R)$, we we already know that:

$$\mathcal{O}_{\operatorname{Spec}(R)}\left(\operatorname{Spec}(R)\right)_s \cong, \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)_s) = \mathcal{O}_{\operatorname{Spec}(R)}(D(s))$$

***Proof*:**
As X is quasicompact, cover it with finitely many affine open sets $U_i = \operatorname{Spec} R_i$. take $U_{ij} = U_i \cap U_j$. Then the following is exact

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{i,j} \mathcal{O}_X(U_{ij})$$

As X is quasiseparated, each U_{ij} is covered by finitely many affine open sets $U_{ijk} = \operatorname{Spec}(R_{ij})$ so that the following is exact:

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{(i,j),k} R_{ijk}$$

Now, localization (being an exact functor) gives:

$$0 \to \mathcal{O}_X(X)_s \to \left(\prod_i R_i\right)_s \to \left(\prod_{(i,j),k} R_{ijk}\right)_s$$

which is still exact. As localization commutes with finite products (importance on finiteness), we get for appropriates values:

$$0 \to \mathcal{O}_X(X)_s \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

which again is still exact, where the s_i and s_{ijk} is is the restriction to the rings R_i and R_{ijk}. Now, the scheme X_s can be covered by the affine open sets $\operatorname{Spec}(R_i)_{s_i}$ and $\operatorname{Spec}(R_i)_{s_i} \cap \operatorname{Spec}(R_{ijk})_{s_{ijk}}$ giving almost the same exact sequence:

$$0 \to \mathcal{O}_X(X_s) \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

Hence, the kernel of the map

$$\mathcal{O}_X(X)_s \to \mathcal{O}_X(X_s)$$

is the same, and we get an isomorphism.

Proposition 3.4.10: Affine Morphisms Are Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an affine morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that $\pi^{-1}(U_i)$ is affine.

***Proof*:**
We shall want to apply the affine communication lemma. For that, we must show the conditions are satisfied. First, suppose $\pi : X \to Y$ is affine over $\mathrm{Spec}\,(S)$ so that $\pi^{-1}(\mathrm{Spec}\,(S)) = \mathrm{Spec}\,(R)$. Then notice that for any $s \in S$:

$$\pi^{-1}(\mathrm{Spec}\,(S_s)) = \mathrm{Spec}\,(R)_{\pi^\# s}$$

showing the first condition. For the next, suppose that $\pi : X \to \mathrm{Spec}\,(S)$ and $(s_1, ..., s_n)$ with $X_{\pi^\# s_i} = \mathrm{Spec}\,(R_i)$. We'll show that X is affine as well. To that end, let $R = \mathcal{O}_X(X)$. Then $X \to \mathrm{Spec}\,(S)$ factors through $\alpha : X \to \mathrm{Spec}\,(R)$ (given by the isomorphism $R \to \mathcal{O}_X(X)$), giving us:

$$\begin{tikzcd} \bigcup_i X_{\pi^\# s_i} \arrow[rr, "\alpha"] \arrow[dr, "\pi"'] & & \mathrm{Spec}\,(R) \arrow[dl, "\beta"] \\ & \bigcup_i D(s_i) = \mathrm{Spec}\,(S) & \end{tikzcd}$$

Then, as we are working with scheme morphisms, the pre-image of a distinguished open set is a distinguished open set and so

$$\beta^{-1}(D(s_i)) = D(\beta^\#(s_i)) \cong \mathrm{Spec}\, R_{\beta^\# s_i}$$

and $\pi^{-1}(D(s_i)) = \mathrm{Spec}\,(R_i)$. Now, as X is quasicompact and quasiseparated, by the Qcqs lemma we have the isomorphism:

$$A_{\beta^\# s_i} = \mathcal{O}_X(X)_{\beta^\# s_i} = \mathcal{O}_X(X_{\beta^\# s_i}) = R_i$$

Hence, α induces an isomorphism $\mathrm{Spec}\,(R_i) \cong \mathrm{Spec}\,(R_{\beta^\# s_i})$, and so α is an isomorphism over each $\mathrm{Spec}\,(R_{\beta^\# s_i})$, which covers $\mathrm{Spec}\,(R)$, and hence α is an isomorphism, and so $X \cong \mathrm{Spec}\,(R)$, completing the proof.

From this, we can conclude that if the domain/codomain are affine, the morphisms must be affine

Corollary 3.4.11: Morphism Between Affine is Affine

Let $f : \mathrm{Spec}\, R \to \mathrm{Spec}\, S$ be a morphism of schemes. Then f is an affine morphism

***Proof*:**
By proposition 3.4.10, it suffices to prove it on an affine open cover of the codomain. Then naturally, take an open cover by distinguished open sets, $D(g_i)$ whose pre-image $D(f^\#(g_i))$ which is an affine scheme.

Corollary 3.4.12: Hypersurfaces and Affine Morphissm

Let $X \subseteq \operatorname{Spec}(R)$ be a closed subset which is locally cut out by one equation (Z is covered by open subsets which are each given by the quotient of one equation). Then the complement of Z, Y, is affine

Note that in the global case, $Y = \operatorname{Spec}(R_f)$, but Y need not be of this form for the local case.

***Proof*:**
exercise.

Example 3.4.13: Affine Morphisms

1. Take $\mathbb{A}^2_k \to \mathbb{P}^1_k$ given by $(a, b) \mapsto [a : b]$ which was shown to be a scheme morphism in example 3.1.11 (uhm, this is obviously not a morphism, where does the origin map too... Should probably take the blow-up). This is an affine morphism: take the usual open cover U_0, U_1. Then:

$$f^{-1}(U_0) \cong \operatorname{Spec} k[x, y]_x \qquad \text{and} \qquad f^{-1}(U_1) \cong \operatorname{Spec} k[x, y]_y$$

   hence by proposition 3.4.10 it is affine.

2. More generally let $\iota : X \to Y$ be either an open or closed immersion. Show that ι is an affine morphism.

The next two types of morphisms are affine morphisms where the pre-image has some extra properties. The next one can be thought of as the generalization of finite integral extensions (or more simply a finite extension or finite ring homomorphism):

Definition 3.4.14: Finite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finite* if for every affine open set $\operatorname{Spec}(S) \subseteq Y$, $\pi^{-1}(\operatorname{Spec}(S))$ is the spectrum of a S-algebra that is finitely generated as a S-module, i.e. it is a finite S-algebra.

Note how being a finitely generated S-module is a rather stronger condition than being a finitely generated S-algebra: it forces each element to be integral and "globally bounded" (since R is a finitely generated S-module, there can't be elements of arbitrary order). The reader should see that the composition of two finite morphisms is finite (see exercise ref:HERE)

Lemma 3.4.15: Algebra Build-up For Local On Target

Let $R \to S$ be a ring map, and $(f_1, ..., f_n) = R$ for appropriate $f_i \in R$. Then:

1. If $R_{f_i} \to S_{f_i}$ is integral, $R \to S$ is integral
2. If $R_{f_i} \to S_{f_i}$ is finite, $R \to S$ is finite

***Proof*:**

1. Take $s \in S$, and consider the ideal
$$(p \in R[x] \,|\, p(s) = 0) \subseteq R[x]$$
Then take the subset $J \subseteq R$ of the leading coefficients of the elements in I. This is in fact an ideal, as the reader could verify or see [9, section 20.5]. By clearing denominators, we get that:
$$f_i^{n_i} \in J$$
Hence, as shown in [9, section 9.6], $1 \in J$

2. The above shows that $R \to S$ is integral, for the finiteness note that we may choose a maximal N_i as the degree's of the generators are globally bounded.

Proposition 3.4.16: Finite Morphism Affine Local On Target

Let $f : X \to Y$ be a scheme morphism. Then it is finite if there is an affine cover $\bigcup_i \mathrm{Spec}\,(S_i) = Y$ such that $f^{-1}(\mathrm{Spec}\,(S_i))$ is the spectrum of a finite S_i-algebra.

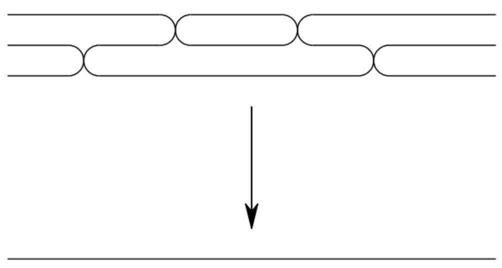
***Proof*:**

If f is finite, the result is *a fortiori* true, for the converse take $\mathrm{Spec}\, S_i \subseteq Y$, cover it in distinguished open sets, and apply corollary 3.1.15 and lemma 3.4.15.

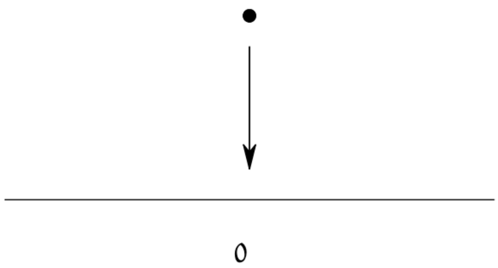
Example 3.4.17: Finite Morphism

1. If L/K is a finite extension, $\mathrm{Spec}\,(K) \to \mathrm{Spec}\,(L)$ is a finite morphism.

2. Take the map $\mathrm{Spec}\, k[t] \to \mathrm{Spec}\, k[u]$ given by $u \mapsto p(t)$ for some degree n polynomial $p(t) \in k[t]$. Then this map is finite: notice that $k[t]$ is generated as a $k[u]$-module by $1, t, ..., t^{n-1}$. This can be visualized as:



3. Let $I \subseteq R$ be an ideal and take the natural schemes and map $\mathrm{Spec}\,(R/I) \to \mathrm{Spec}\,(R)$. Then this is a finite morphism, as R/I is generated as an R-module by $1 \in R/I$. This can be visualized as:



4. More generally, all closed immersions are finite morphisms.
5. Let $f \in R[x]$ be a polynomial whose leading coefficient is *not* a unit. Show that

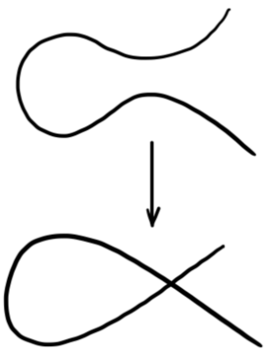
$$\operatorname{Spec} \frac{R[x]}{(f)} \to \operatorname{Spec} R$$

is *not* a finite morphism. Show that it *is* a finite morphism if the leading coefficient is a unit, making this an iff.

6. Take $\operatorname{Spec} k[t] \to \operatorname{Spec} \frac{k[x,y]}{(y^2 - x^2 - x^3)}$ corresponding to the ring map given by:

$$x \mapsto t^2 - 1 \qquad y \to t^3 - t$$

Namely, this shows this curve is rational. This is a finite morphism, since $k[t]$ is generated as a $\frac{k[x,y]}{(y^2-x^2-x^3)}$-module by 1 and t. This can be visualized as:



As the reader may have noticed, there is only one point that is 2-to-1. The reader should check that this will induce an isomorphism between $D(t^2 - 1)$ and $D(x)$, i.e. if the node is removed there is an isomorphism. This shall come up many times as an example of normalizing a curve.

7. Let $X \to \operatorname{Spec} k$ be a scheme morphism. Show that if the morphism if finite, X is the finite union of points with the discrete topology, each points having as residue field a finite extension of k. This can be visualized as:

Notice that all the fibers were finite. This is no happenstance of the chosen examples:

Proposition 3.4.18: Finite Morphism is Finite-to-one

Let $\pi : X \to Y$ be a finite morphism. Then it's fibers are finite

***Proof*:**
Since the property of the fiber is local on the target, we may assume $Y = \operatorname{Spec} B$ is affine. By the definition of a finite morphism, X is affine, say $X = \operatorname{Spec} A$, and the induced ring map $B \to A$ makes A a *finite* B-module (finitely generated as a module).

Let $y \in Y$ be a point corresponding to the prime ideal $\mathfrak{p} \subseteq B$. The fiber over y, denoted X_y, is given by the spectrum of the fiber ring:

$$A_y = A \otimes_B \kappa(y)$$

where $\kappa(y)$ is the residue field at $\mathfrak{p}$. Since A is generated by finitely many elements as a B-module, the tensor product A_y is generated by the images of those elements as a vector space over $\kappa(y)$. Thus, A_y is a finite-dimensional $\kappa(y)$-algebra.

Any finite-dimensional algebra over a field is an Artinian ring. A basic property of Artinian rings is that they have finitely many prime ideals (in fact, finitely many points, all of which are closed). Therefore, $\operatorname{Spec}(A_y)$ consists of finitely many points.

The converse is not the case

Example 3.4.19: Finite Fibers But Not Finite Morphisms

The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \hookrightarrow \mathbb{A}^2_k$ has finite fibered (they are one-to-one), but it is not affine (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ is not affine).

It also possible to have finite fibers and affine, but not finite: Take $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \hookrightarrow \mathbb{A}^1_{\mathbb{C}}$ (intuitively, localizing is "changing perspectives" to the infinitesimal).

It looks important to add 7.3.I to get a grasp on finite morphisms to affine spectrums relation to projective scheme

Definition 3.4.20: Integral Morphism

Let $\pi : X \to Y$ be a scheme morphism. The nit is an *integral morphism* if π is affine, and for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, where $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$, the induced map $S \to R$ is an integral extension.

As integral extensions must be finitely generated over the source ring, an finite morphism is a integral morphism, however an integral morphism need not be finite (think $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$). It is further easy to verify that the composition of integral maps is integral.

Proposition 3.4.21: Integral Morphisms Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is an integral morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that we get integral extension for each pre-image.

***Proof*:**
exercise.

Proposition 3.4.22: Integral Morphism Are Closed Maps

Let $\pi : X \to Y$ be a Integral morphism. Then it is a closed map: it maps closed subsets to closed subsets

As a consequence, finite morphisms are closed

***Proof*:**
idea: reduce first to ring map., and consider the induced map. Now take $I \subseteq R$ and $J = (\pi^{\#})^{-1}(I)$ and consider the integral extension $S/J \subseteq A/I$. Apply the Lying Over theorem.

Definition 3.4.23: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite type* if for every affine open subset $\operatorname{Spec}(S) \subseteq Y$, and every affine open subset $\operatorname{Spec}(R) \subseteq \pi^{-1}(\operatorname{Spec}(R))$, the induced morphism $S \to R$ makes R a finitely generated S-algebra. Equivalently, $\pi^{-1}(\operatorname{Spec}(S))$ can be covered by affine open subsets $\operatorname{Spec}(R_i)$ where each R_i is a finitely generated S-algebra.

Definition 3.4.24: Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *of finite type* if it is locally of finite type and quasicompact.

Proposition 3.4.25: Locally Finite Type Morphism Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is locally of finite type (resp. of finite type) if there is a cover of Y by affine open sets $\operatorname{Spec}(S_i)$ such that $\pi^{-1}(\operatorname{Spec}(S_i)$ is locally finite type over S_i.

***Proof*:**
exercise.

Example 3.4.26: Finite Type Morphisms

1. All Schemes that are finite type over k are example of finite type scheme morphisms $X \to \operatorname{Spec} k$
2. The map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is of finite type ($\mathbb{P}^n_R$ is covered by $n+1$ open sets)

Proposition 3.4.27: Integral and Finite Type iff Finite

1. finite morphisms are of finite type
2. A morphism is finite if it is integral and of finite type

***Proof*:**
exercise.

Definition 3.4.28: Quasifinite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *quasifinite* if it is of finite type, and for each $\mathfrak{q} \in Y$, $\pi^{-1}(\mathfrak{q})$ is a finite set.

By proposition proposition 3.4.18 and proposition 3.4.27(1), finite morphisms are quasifinite. The converse is not true (think of $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$. The finite type condition is necessary, there exists morphisms with finite fibers that are not quasifinite

Example 3.4.29: Finite Fibers, But Not Quasifinite
The morphism $\operatorname{Spec}(\mathbb{C}(t)) \to \operatorname{Spec}(\mathbb{C})$ has finite fibers, but is not quasifinite. Similarly to $\operatorname{Spec}(\overline{\mathbb{Q}}) \to \operatorname{Spec}(\mathbb{Q})$.

Quasifinite morphisms shall return when discussing Zariski Main Theorem (see ref:HERE).

Vakil promises exercise 9.3.H shows why this would naturally arise

Exercise 3.4.1

1. Let $\pi : X \to Y$ be an open embedding. Then if Y is locally Noetherian, X is locally Noetherian. If Y is Noetherian, X is Noetherian.
2. Let $\pi : X \to Y$ be an open embedding. Show that if Y is quasicompact, X need not be (there is an affine counter-example)
3. Show that the composition of two finite morphism is a finite morphism.
4. Show that being an open embedding is not local on the source.

5. Let B/A be an integral extension. Then show that any ring homomorphism $B \to C$ induces an integral extensions $C \otimes_B A/C$. Conclude that integrality is preserved by base change (see scalar extensions in [9] if you are unsure what it means to change base).
6. Show that every open embedding is locally of finite type. Hence, every quasicompact open embedding is of finite type.
7. Every open embedding into a locally Noetherian scheme is of finite type
8. The composition of two morphisms locally of finite type is locally of finite type.
9. Let $\pi : X \to Y$ be locally of finite type and Y locally Noetherian. Then X is locally Noetherian. IF $\pi : X \to Y$ is of finite type and Y is Noetherian, then X is Noetherian,
10. Show that quasifinite morphisms to $\operatorname{Spec}(k)$ are finite)
11. Let $k = \overline{\mathbb{F}_p}$. Show that the morphism $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ of k-schemes corresponding to the bijection $x_i \mapsto x_i^p$ is a bijection, and no power of F is the identity (as sets)
12. (Generalizing the above) Let X be an $\mathbb{F}_p$-scheme. Define an endomorphism $F : X \to X$ such that for each affine open subset $\operatorname{Spec}(R) \subseteq R< F$ corresponds to the map $R \to R$ given by $f \mapsto f^p$. The morphism F is called the *absolute Frobenius morphism*.

3.5 Separated and Proper Schemes

The last two major types of schemes we must cover are *separated schemes* that recover the notion of Hausdroffness for schemes, and *proper schemes* which recover a notion of compactness for schemes. As is the theme in this section, these two properties can be defined purely in terms of scheme morphisms.

From an algebraic-geometric perspective, these conditions allow for better behaviour of closed sets, whether the intersection of two closed sets is closed or whether the image of closed sets is closed is governed by separatedness and properness respectively. Both separatedness and properness can be defined in terms of morphisms, and this is in fact the more practical approach. As closed sets are locally closed set of affine scheme (i.e. vanishing sets of global sections of $\operatorname{Spec} R_i$), the preservation of closedness of these sets under various kinds of mappings is of key interest in scheme theory.

3.5.1 Separated Morphisms

Recall that if $f : X \to Y$ is a continuous map and $K \subseteq X$ is compact, then $f(K)$ is compact. In euclidean topology, this implies closed and bounded maps to closed and bounded, importantly this is a special case of when closed sets maps to closed sets. As affine schemes are quasi-compact (recall we use the term quasi-compact instead of compact due to the differing behaviour without Hausdorffness), if $K \subseteq X$ is closed it is quasi-compact, and hence its image is quasi-compact if $f : X \to Y$ is a scheme morphism (which is continuous). However, such a map *need not be closed*, namely it need not map closed sets to closed sets (recall example ref:HERE). In particular a quasicompact set need not be closed (see [12, section 3.2]).

Another example that shows the failure of separatedness is the following: Take X to be the line with double origin (recall example 2.3.17). Then the two natural inclusions $\iota_1, \iota_2 : \mathbb{A}^1_k \to X$ do not intersect on a closet set!

To recover the good properties of Hausdroffness, the notion of *separatedness* is defined. The idea is that we want to avoid cases where there can be "double points" or even more points that are intuitively arbitrarily close to each other; such points cannot be distinguished, or *separated* by (regular) functions.

In the topological setting, we saw X is Hausdorff if and only if $\Delta \subseteq X \times X$ is closed. In the case of scheme every "diagonal" (when working with fibered products) is a locally closed immersion:

Proposition 3.5.1: Diagonal Locally closed immersion

Let $f : X \to S$ be a morphism of schemes so that X is an S-scheme. Then the diagonal morphism $\delta : X \to X \times_S X$ is a locally closed immersion.

***Proof*:**

We will describe a union of open subsets of $X \times_S X$ covering the image of X, such that the image of X is a closed immersion in this union.

Say S is covered with affine open sets V_i and X is covered with affine open sets U_{ij}, with $f : U_{ij} \to V_i$. Take $W = \bigcup_{i,j} U_{ij} \times_{V_i} U_{ij}$. This is an affine open subscheme of the product $X \times_S X$ (recall this is how the product was constructed). Then the diagonal is covered by these affine open subsets $U_{ij} \times_{V_i} U_{ij}$. Any point $p \in X$ lies in some U_{ij}; then $\delta(p) \in U_{ij} \times_{V_i} U_{ij}$. Note that $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) = U_{ij}$ since certainly $U_{ij} \subset \delta^{-1}(U_{ij} \times_{V_i} U_{ij})$, and as $\pi_1 \circ \delta = \mathrm{id}_X$ (as π_1 is the first projection), $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) \subset U_{ij}$. Finally, to check that $U_{ij} \to U_{ij} \times_{V_i} U_{ij}$ is a closed immersion, say $V_i = \operatorname{Spec} B$ and $U_{ij} = \operatorname{Spec} A$. Then this corresponds to the natural ring map $A \otimes_B A \to A$ $(a_1 \otimes a_2 \mapsto a_1 a_2)$, which is certainly surjective, completing the proof.

The key is that these open subsets need not cover $X \times_S X$, hence this does not imply the embedding is closed.

In our case, a subset is closed if it is locally closed in an affine covering, this means that the diagonal can be expressed (locally) as vanishing sets. Furthermore, the product is not well-defined for schemes, as it must always be fibered over some other scheme. This leads to the following:

Definition 3.5.2: Separated Scheme

Let X be a scheme. Then X is *separated* if the map $\delta : X \to X \times_{\mathbb{Z}} X$ is a closed immersion.

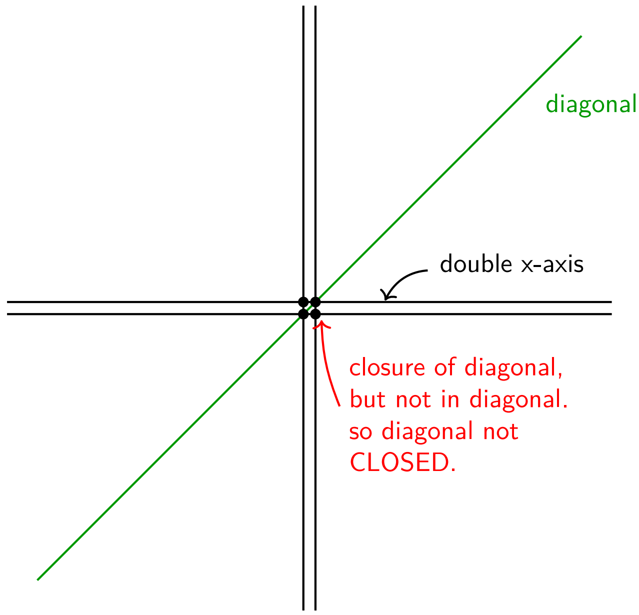
More generally, a map $X \to S$ between schemes is *separated* if the map $\delta : X \to X \times_S X$ is a closed immersion. Denote the image of δ as $\Delta(X)$ or Δ_X.

As a historical footnote, the term scheme used to mean separated scheme back before the 1960s, and the term pre-scheme was used for the modern term scheme.

Note that the first condition is equivalent to saying that the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is separated. The fact that separatedness is given by the purely topological condition of $\Delta(X)$ being closed shows its closeness to the original topological condition of Hausdorffness.

Example 3.5.3: Separated Morphisms

1. Take the line with two origins over k, label it X. Then when considering $X \times_{\operatorname{Spec} k} X$, we can think to it as in image bellow. The double origin will correspond to 4 points, but the diagonal will only have 2 points. Taking the closure of the diagonal will include all 4 points (show that any open set containing either $(0_1, 0_2)$ or $(0_2, 0_1)$ must intersect $\Delta(X)$, and hence is in the closure; see [12, section 1.5]), showing that it is *not* separated.



Note that $X \to X \times_k X$ is still a locally closed immersion. To show that it is not closed, we can take an open cover that covers *all* of $X \times_k X$ and show that for the affine open W_i containing $P_1 = (0_1, 0_2)$, $\Delta(X) \cap W_i$ is not closed. Indeed, As $P_1 \notin \Delta(X)$, $P_1 \notin \Delta(X) \cap W_i$. It is easy to see that P_1 is in the closure of this set, $P_1 \in \overline{\Delta(X) \cap W_i}$, we see that $\Delta(X) \cap W_i$ is *not* closed.

Note that if we take $X \to X$ to be the identity map, then this *is* a separated morphism, since:

$$X \cong X \times_X X$$

showing the relativeness of separatedness when picking different base schemes.

2. Let $X = \operatorname{Spec}(A) \to \operatorname{Spec}(B) = Y$ be any morphism of schemes between affine schemes. Then this is *always* a separated morphism. Unwind the definitions: when converting into rings we get the map:

$$A \otimes_B A \to A$$

which is surjective, and hence we get:

$$A \cong A \otimes_B A/\mathfrak{a}$$

that is, we are taking the quotient by an ideal, which we know will give us a closed subscheme of $X \times_Y X$ isomorphic to X.

3. More generally, if $X \to Y$ is an affine morphism, it is separated (follows from proposition 3.5.1).

4. The following scheme is a famous example of non-separatedness appearing naturally. Consider $\mathbb{C}^* \curvearrowright \mathbb{A}^2_{\mathbb{C}}$ via $t \cdot (x, y) = (tx, t^{-1}x)$; this is usually sated as "the algebraic torus acts on the affine plane with weights $(1, -1)$". Then taking $(\mathbb{A}^2_{\mathbb{C}} \setminus \{0\})/\mathbb{C}^*$ where the origin is removed as the action is non-free, the resulting space is a non-separated scheme.

   In many cases, non-separated scheme appear naturally in moduli theory.

An important example worth singling out is that of projective scheme:

Lemma 3.5.4: Projective Scheme is Separated

The scheme $\mathbb{P}^1_{\mathbb{Z}}$ is separated.

***Proof*:**
Let $X = \mathbb{P}^1_{\mathbb{Z}}$ and take a standard affine cover $U_0 = \operatorname{Spec} \mathbb{Z}[t]$ and $U_1 = \operatorname{Spec} \mathbb{Z}[u]$, glued via $u = 1/t$. To prove separatedness, we must show $\Delta(X)$ is closed in $X \times_{\mathbb{Z}} X$. It suffices to show the intersection of $\Delta(X)$ with the members of an open cover of the *product* is closed. The product is covered by the four affine patches $U_{ij} = U_i \times U_j$.

We analyze the corresponding ring maps (dual maps) $\Delta^\#$. Recall that a morphism of affine schemes is a closed immersion if and only if its ring dual map is surjective. If surjective, the image corresponds to $V(\ker(\Delta^\#))$.

- *Case 1: The standard charts* (e.g., $W_{00} = U_0 \times U_0$).
  Algebraically, $W_{00} \cong \operatorname{Spec}(\mathbb{Z}[t] \otimes_{\mathbb{Z}} \mathbb{Z}[t]) \cong \operatorname{Spec} \mathbb{Z}[t_1, t_2]$. The diagonal restricted to this chart maps U_0 into $U_0 \times U_0$. The corresponding ring dual map $\Delta^\#$ is:
  $$\varphi : \mathbb{Z}[t_1, t_2] \to \mathbb{Z}[t], \quad \text{defined by } t_1 \mapsto t,\ t_2 \mapsto t.$$
  This map is clearly surjective. By the First Isomorphism Theorem, $\mathbb{Z}[t] \cong \mathbb{Z}[t_1, t_2]/\ker(\varphi)$. The kernel is the ideal generated by $(t_1 - t_2)$. Consequently, the diagonal corresponds to the closed subscheme defined by this ideal, $V(t_1 - t_2)$, representing the line $y = x$.

- *Case 2: The mixed charts* (e.g., $W_{01} = U_0 \times U_1$).
  Algebraically, the target is $W_{01} \cong \operatorname{Spec} \mathbb{Z}[t] \times \operatorname{Spec} \mathbb{Z}[u] \cong \operatorname{Spec} \mathbb{Z}[t, u]$. The source (the part of the diagonal mapping into this chart) is the intersection $U_0 \cap U_1$. This intersection is the locus where $t \neq 0$, which is affine: $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$. The ring dual map $\Delta^\#$ maps the coordinates of the product to their values on the intersection (determined by the gluing $u = 1/t$):
  $$\psi : \mathbb{Z}[t, u] \to \mathbb{Z}[t, t^{-1}].$$
  Defined by $t \mapsto t$ and $u \mapsto t^{-1}$. Since t and t^{-1} are in the image, ψ is surjective. This implies the geometric map is a closed immersion. The algebraic geometry dictionary tells us the image is defined by the kernel of ψ:
  $$\operatorname{Spec} \mathbb{Z}[t, t^{-1}] \cong \operatorname{Spec} \left(\mathbb{Z}[t, u]/\ker(\psi)\right).$$
  Since $t(t^{-1}) - 1 = 0$ in the codomain, the element $(tu - 1)$ is in the kernel. In fact, $\ker(\psi) = (tu - 1)$. Thus, the image of the diagonal in this chart is the closed subset $V(tu - 1) \subseteq \mathbb{A}^2_{\mathbb{Z}}$, which describes the hyperbola $u = 1/t$.

Since the image is closed in every chart of an open cover of the target $X \times_{\mathbb{Z}} X$, the diagonal map is a closed immersion.

The details for the case of $\mathbb{P}^n_R$ are a bit laborious, so until I feel comfortable with them I am putting this corollary here as a refresher for my future self:

Corollary 3.5.5: Projective Space is Separated

For any ring R, the projective space $\mathbb{P}^n_R$ is separated.

***Proof*:**
Let $X = \mathbb{P}^n_R$. We use the standard affine open cover $\{U_i\}_{i=0}^n$, where $U_i = \{[x_0 : \cdots : x_n] \big| x_i \neq 0\}$. The affine ring for U_i is $R[y_0^{(i)}, \ldots, y_n^{(i)}]$ subject to $y_i^{(i)} = 1$, where the coordinates represent $y_k^{(i)} = x_k/x_i$.

To show the diagonal $\Delta : X \to X \times_R X$ is a closed immersion, we check that its image is closed in the open affine cover of the product given by $\{U_i \times U_j\}_{0 \leq i,j \leq n}$.

Consider a specific chart $W_{ij} = U_i \times U_j$.

- **Algebraic Setup:** The ring corresponding to W_{ij} is the tensor product of the rings for U_i and U_j:
$$A_{ij} = R[y_0^{(i)}, \ldots, y_n^{(i)}] \otimes_R R[z_0^{(j)}, \ldots, z_n^{(j)}]$$
where variables y come from the first factor (U_i) and z from the second (U_j), with $y_i^{(i)} = 1$ and $z_j^{(j)} = 1$.

- **The Diagonal Map:** A point lies in the image of the diagonal in this chart if it is in $U_i \cap U_j$ in both factors. The map on rings $\varphi : A_{ij} \to \mathcal{O}_X(U_i \cap U_j)$ is induced by the coordinate change formulas:
$$z_k^{(j)} = \frac{x_k}{x_j} = \frac{x_k/x_i}{x_j/x_i} = \frac{y_k^{(i)}}{y_j^{(i)}}$$
Thus, the map is defined by $y_k^{(i)} \mapsto y_k^{(i)}$ and $z_k^{(j)} \mapsto y_k^{(i)} \cdot (y_j^{(i)})^{-1}$.

- **The Closed Relations:** We can find the generators of the kernel by clearing denominators. The relations are:
$$z_k^{(j)} y_j^{(i)} - y_k^{(i)} = 0 \quad \text{for all } k = 0, \ldots, n.$$
Note specifically that when $k = i$, since $y_i^{(i)} = 1$, we get the relation $z_i^{(j)} y_j^{(i)} - 1 = 0$. This is the generalized "hyperbola" ($x_{i/j} \cdot x_{j/i} = 1$).

  The ideal generated by these polynomials $J = \langle z_k^{(j)} y_j^{(i)} - y_k^{(i)} \rangle_k$ defines a closed subset of W_{ij}. Since φ is surjective onto the localization of the polynomial ring (image is $U_i \cap U_j$), this closed set is exactly the image of the diagonal.

As the image is closed in every member of an open cover of the product, $\mathbb{P}^n_R$ is separated.

The next concept will not be too useful to us for awhile, and shall come back up when we see algebraic stacks and algebraic spaces, but is useful to define:

Definition 3.5.6: Quasi-separated

Let X be a scheme. Then X is quasi-separated if the image of $X \to X \times X$ is quasi-compact. More generally, a map $X \to S$ between schemes is quasi-separated if the image map of $X \to X \times {}_S X$ is quasi-compact.

Proposition 3.5.7: Separated Implies Quasi-separated

Let X be a separated scheme. Then X is quasi-separated.

***Proof*:**
By definition, X is separated if the diagonal morphism $\Delta : X \to X \times X$ is a closed immersion. By definition, X is quasi-separated if Δ is a quasi-compact morphism. Thus, it suffices to prove that any closed immersion is quasi-compact.

Let $f : Z \to Y$ be a closed immersion. To prove f is quasi-compact, we check the condition on an affine open cover. Let $V \subseteq Y$ be an affine open subset, say $V = \operatorname{Spec}(B)$. Because f is a closed immersion, the restriction $f|_V : f^{-1}(V) \to V$ corresponds to a surjective ring homomorphism $B \to A$ (specifically $A \cong B/I$ for the ideal I cutting out Z). Therefore, the preimage $f^{-1}(V)$ is isomorphic to $\operatorname{Spec}(A)$. Since every affine scheme is quasi-compact, $f^{-1}(V)$ is quasi-compact.

As the preimage of any affine open is quasi-compact, the map f is quasi-compact. Applying this to Δ, we conclude X is quasi-separated.

Separatedness gives us nicer behaviour on closed sets. The following results outline some important examples:

Proposition 3.5.8: Intersection of Separated affine schemes

Let $X \to S$ be a separated scheme morphism where S is affine. Then if $U, V \subseteq X$ are open affine subsets, their intersection

$$U \cap V$$

is an affine open subset.

***Proof*:**
Let $S = \operatorname{Spec} R$, $U = \operatorname{Spec} A$, and $V = \operatorname{Spec} B$.

1. **The product of affines is affine.** Recall that the product $X \times_S X$ is constructed by gluing affine charts. Specifically, the open subset $U \times_S V \subseteq X \times_S X$ is isomorphic to the spectrum of the tensor product:

   $$U \times_S V \cong \operatorname{Spec}(A \otimes_R B).$$

   Thus, $U \times_S V$ is an affine open subscheme of the product.

2. **The intersection is a preimage.** Consider the diagonal morphism $\Delta : X \to X \times_S X$.

We observe that the intersection of the open sets is exactly the pullback of their product by the diagonal:

$$\Delta^{-1}(U \times_S V) = \{x \in X \big| \Delta(x) \in U \times_S V\} = \{x \in X \big| x \in U \text{ and } x \in V\} = U \cap V.$$

3. **Separatedness implies affine.** By the definition of separatedness, Δ is a closed immersion. This means that Δ induces a closed immersion of $U \cap V$ into the open subset $U \times_S V$:

$$U \cap V \hookrightarrow U \times_S V.$$

A closed immersion into an affine scheme corresponds to a surjective ring map. Since $U \times_S V$ is affine $(\operatorname{Spec}(A \otimes_R B))$, $U \cap V$ must be isomorphic to $\operatorname{Spec}((A \otimes_R B)/I)$ for some ideal I. Since the spectrum of a quotient ring is an affine scheme, $U \cap V$ is affine.

For a counter-example, taking affine *plane* with double origin we get that taking two affine lines each containing a different origin will intersect to produce the plane without the origin, which in example 2.3.14 is not affine.

Proposition 3.5.9: Graphs Are Closed

Let X be a separated S-schemes and Y any S-scheme. Then the graph of the S-morphism $f : X \to Y$:

$$\Gamma_f = \operatorname{im}((\operatorname{id}_Y, f)Y \to Y \times_S X)$$

is closed

***Proof*:**

1. Show the graph is constructed via the pullback of the diagonal $\Delta_X \subseteq X \times_S X$ along the map $id_Y \times g : Y \times_S X \to X \times_S X$.

2. pullbacks preserve closed immersions (proposition 3.2.5(2)).

For the case where it is not separated as usual let X be the line with two origins (not separated). Let Y be the affine line $\mathbb{A}^1$. Consider the morphism $g : Y \setminus \{0\} \to X$ which is the natural inclusion into the "un-doubled" part of X. What is the graph of this morphism? It's a subset of $(Y \setminus \{0\}) \times_k X$. The key is that this map cannot be extended to the origin of Y in a unique way. Indeed, $0 \in Y$ could map to either $0_1 \in X$ or $0_2 \in X$. This ambiguity means the graph of any potential extension is *not* closed; the point $(0, 0_1)$ and $(0, 0_2)$ are "limit points" of the graph but there is no natural choice of which one is in the graph. The graph is not well-behaved at the boundary. This intuition will in fact be a central way of interpreting separatedness, as shall be explored in the next section.

Proposition 3.5.10: Morphism Determined By Dense Open Subset

Let X be a reduced S-scheme, Z any scheme, and $U \subseteq Z$ a dense open subset. Then if two morphisms $f, g : Z \to X$ agree on U, $f|_U = g|_U$, then they agree everywhere:

$$f|_U = g|_U \quad \Rightarrow \quad f = g$$

***Proof*:**
Consider the set of points in Z where f and g agree. The key is this can be described as the preimage of the diagonal of X, namely, take $(g, h) : Z \to X \times_S X$, and:

$$E = (g, h)^{-1}(\Delta_X)$$

Then, we have a closed subset E that contains U (as every point in E are points f, g agree on), and as U is open and dense, it must be that $E = Z$, completing the proof.

3.5.2 Valuation Criterion for Separatedness

We shall next characterize separated schemes and morphisms in using valuation rings. This will be analogous to the fact that in a Hausdorff space, the limit of a sequence has at most 1 point. From this result, think of a continuous map $(0, 1) \to X$. If we want to extend this to a map $[0, 1] \to X$, if X is Hausdorff, there is only one possible extension. If X wasn't, say it had two origins, then there can be more than one map that can extend (the new 0 point will have two possible places to map to).

Let us now extend this idea to a map $X \to Y$, and we have an embedding $(0, 1) \to X$ an identity map $(0, 1) \to [0, 1]$ and an embedding $(0, 1] \to Y$ where we are only adding one point. Then there should only be one lift. Note that this lift is not dependent on X and Y. Let's say Y was the double-origin scheme and X was the double y-axis scheme. Then the embedding $(0, 1)$ can be extended to two way into Y, but that will correspond to a unique lift.

With this intuition in mind, let us now look at what is the equivalent of the open interval for schemes. This would be a Discrete valuation ring (DVR) R. It has only two points. We can think of as $\operatorname{Spec}(\operatorname{Frac}(R))$ as the open interval and $\operatorname{Spec}(R)$ as adding the closed 0 (notice that the new point in $\operatorname{Spec}(R)$ is also a closed point, while the other point is "open" and is the generic point that can be thought of was closed to everywhere else). Then we have the following situation:

$$\begin{tikzcd}
\operatorname{Spec}(\operatorname{Frac}(R)) \arrow[r] \arrow[d] & Y \arrow[d] \\
\operatorname{Spec}(R) \arrow[r] \arrow[ur, dashed] & X
\end{tikzcd}$$

With the hope that having at most one lift corresponding to the map being separable. This is *sometimes* the case. To prove this, we require two technical lemmas regarding valuation rings and specialization.

Lemma 3.5.11: Lifts from Valuation Rings

Let R be a valuation ring of a field K. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$.

1. A morphism $\operatorname{Spec} K \to X$ is equivalent to giving a point $x_1 \in X$ and an inclusion of fields $\kappa(x_1) \subseteq K$.
2. A morphism $\operatorname{Spec} R \to X$ is equivalent to
   - giving two points $x_0, x_1 \in X$, with x_0 a specialization of x_1 (i.e., $x_0 \in \overline{\{x_1\}}$), and
   - an inclusion $\kappa(x_1) \subseteq K$, such that R dominates the local ring $\mathcal{O}_{x_0,Z}$ where $Z = \overline{\{x_1\}}$ with its reduced induced structure.

***Proof*:**

1. U is a single point scheme equipped with the structure sheaf K. To give a morphism $U \to X$ is to give a continuous map (picking a point $x_1 \in X$) and a morphism of sheaves $\mathcal{O}_{X,x_1} \to K$. This is a local homomorphism from a local ring to a field, which is equivalent to an inclusion of the residue field $\kappa(x_1) \subseteq K$.

2. Let $t_0 = \mathfrak{m}_R$ be the closed point of T and $t_1 = (0)$ be the generic point. Given a morphism $T \to X$, let x_0, x_1 be the images of t_0, t_1. Since the closure of t_1 contains t_0, by continuity the closure of x_1 must contain x_0, so x_0 is a specialization of x_1. Since T is reduced, the morphism factors through the reduced induced scheme structure on the closure $Z = \overline{\{x_1\}}$. Algebraically, $\kappa(x_1)$ is the function field of Z. The morphism gives a local homomorphism $\mathcal{O}_{x_0,Z} \to R$. Since the map on function fields is compatible with the inclusion $\kappa(x_1) \subseteq K$, this exactly means that the valuation ring R dominates the local ring $\mathcal{O}_{x_0,Z}$.

   Conversely, given the data of x_0, x_1 and the domination $\mathcal{O}_{x_0,Z} \to R$, we construct the map. The map of topological spaces is determined by the points. The map of sheaves is determined by the ring map $\mathcal{O}_{x_0,Z} \to R$ (since T is local, mapping to the global sections of T is sufficient), which composes with the natural map $\operatorname{Spec} \mathcal{O} \to X$ to give the desired morphism.

To make the conditions of lemma 3.5.11 concrete, let us look at a geometric example where T is a curve germ and X is the affine plane.

Example 3.5.12: Visualizing Valuation Lifts

1. Let $X = \mathbb{A}^2_k = \operatorname{Spec} k[x,y]$. Let $R = k[t]_{(t)}$ be the discrete valuation ring at the origin of the affine line. Then $K = k(t)$. Let $T = \operatorname{Spec} R = \{t_0, t_1\}$ where $t_0 = (t)$ is the closed point and $t_1 = (0)$ is the generic point.

   Consider the morphism $T \to X$ defined by the ring homomorphism:

   $$\varphi^\# : k[x,y] \to k[t]_{(t)}, \quad x \mapsto t, \quad y \mapsto t^2$$

   Geometrically, this represents a curve parameterized by t approaching the origin along the parabola $y = x^2$.

   (a) *Where the generic point maps ($U \to X$):* The generic point t_1 corresponds to the prime ideal $(0) \subset R$. Its image $x_1 \in X$ is the prime ideal $\mathfrak{p} = (\varphi^\#)^{-1}((0))$. Since $y - x^2 \mapsto t^2 - t^2 = 0$, the ideal $(y - x^2)$ is in the kernel. In fact, the kernel is exactly $(y - x^2)$. Thus, x_1 is the *generic point of the parabola* $V(y - x^2)$. The inclusion of fields is the map on function fields:

   $$\kappa(x_1) = \operatorname{Frac}\left(\frac{k[x,y]}{(y - x^2)}\right) \xrightarrow{\cong} k(t) = K$$

   (b) *Where the closed point maps:* The closed point t_0 corresponds to the maximal ideal $(t) \subset R$. Its image $x_0 \in X$ is $(\varphi^\#)^{-1}((t)) = (x,y)$, which is the origin.

   (c) *Specialization:* We see clearly that x_0 (the origin) is in the closure of x_1 (the whole parabola). This satisfies the condition $x_0 \in \overline{\{x_1\}}$.

(d) *Domination:* Let $Z = \overline{\{x_1\}}$ be the parabola. The local ring $\mathcal{O}_{x_0,Z}$ is the ring of functions on the parabola regular at the origin, which is isomorphic to $k[t]_{(t)}$. The map induced by $\varphi^{\#}$ is the identity $k[t]_{(t)} \to k[t]_{(t)}$, which is a local homomorphism (maps maximal ideal to maximal ideal), satisfying the domination condition.

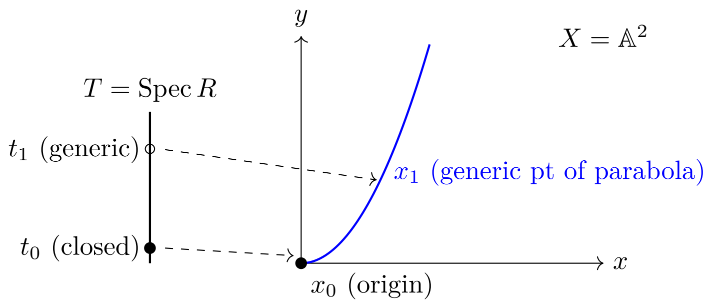


Figure 3.1: The generic point of the DVR maps to the generic point of the curve, while the closed point of the DVR maps to the limit point on the curve.

2. Let us look at a case where the generic point of T maps to the *generic point* of $X = \mathbb{A}^2_k$, rather than a curve inside it. This requires constructing a valuation on the function field $k(x, y)$.

   Let $K = k(x, y)$. We define a valuation $v : K^\times \to \mathbb{Z}$ by the "order of vanishing along the y-axis". Any rational function $f \in k(x, y)$ can be written as $f = x^n \frac{g}{h}$ where x does not divide g or h. We define $v(f) = n$.

   Let R be the valuation ring for v:

$$R = \{f \in k(x, y) \big| v(f) \geq 0\} = k(x, y)_{(x)}$$

   This is the local ring of the affine plane localized at the prime ideal (x) (the generic point of the y-axis).

   Let us analyze the map $T = \operatorname{Spec} R \to X = \operatorname{Spec} k[x, y]$ induced by the natural inclusion $k[x, y] \subset R$.

   (a) *Generic Point:* The generic point t_1 of T (ideal (0) in R) maps to the ideal (0) in $k[x, y]$. Thus, $t_1 \mapsto \eta$, the generic point of the plane.

   (b) *Closed Point:* The closed point t_0 of T (maximal ideal xR) maps to the ideal $(x) \subset k[x, y]$. This is the *generic point of the line* $x = 0$.

   (c) *Visualization:*

$$\begin{tikzcd} U = \operatorname{Spec} k(x,y) \arrow[r] \arrow[d] & \eta \text{ (generic pt of } \mathbb{A}^2) \arrow[d, dashed, "specializes"] \\ T = \operatorname{Spec} k(x,y)_{(x)} \arrow[r] & \xi \text{ (generic pt of } y\text{-axis)} \end{tikzcd}$$

   Even though t_0 is a "closed point" in the topological space T, it maps to a non-closed point in X.

(d) *Domination:* The stalk of X at the image point ξ is $\mathcal{O}_{X,\xi} = k[x,y]_{(x)}$. In this specific case, the valuation ring R is *isomorphic* to the stalk. This happens because we approached a codimension 1 point. If we approached the origin (codimension 2), R would be a further localization/quotient of the stalk.

Lemma 3.5.13: Closure and Specialization

Let $f : X \to Y$ be a quasicompact morphism of schemes (e.g., if X, Y are Noetherian). Then the subset $f(X) \subseteq Y$ is closed if and only if it is stable under specialization.

***Proof*:**
One direction is topological: closed sets are always stable under specialization (if a point is in the set, its closure is in the set). For the converse, assume $f(X)$ is stable under specialization. We want to show $\overline{f(X)} \subseteq f(X)$. We may assume X and Y are reduced, and replace Y with the reduced induced structure on $\overline{f(X)}$ so we want to show f is surjective onto Y. Let $y \in Y$. We can replace Y with an affine neighborhood of y, so assume $Y = \operatorname{Spec} B$. Since f is quasicompact, X is a finite union of affine opens $X_i = \operatorname{Spec} A_i$. Since y is in the closure of the image, $y \in \overline{f(X_i)}$ for some i. Let $Y_i = \overline{f(X_i)}$ with the reduced structure. Then $X_i \to Y_i$ corresponds to an injective ring map $B \to A_i$ (injective because the map is dominant). The point y corresponds to a prime $\mathfrak{p} \subseteq B$. Let $\mathfrak{p}' \subseteq \mathfrak{p}$ be a *minimal* prime ideal of B contained in $\mathfrak{p}$, such a prime exists by Zorn's lemma. This prime $\mathfrak{p}'$ corresponds to a point y' which specializes to y. I claim $y' \in f(X_i)$. Consider the localization $B_{\mathfrak{p}'}$. This is a field since $\mathfrak{p}'$ is minimal in a reduced ring. The map $B_{\mathfrak{p}'} \to A_i \otimes_B B_{\mathfrak{p}'}$ is an inclusion of a field into a non-zero ring as $B \to A_i$ is injective. Thus, the target ring is not the zero ring, so it has a prime ideal $\mathfrak{q}'$. This prime contracts to $\mathfrak{p}'$. Thus, y' is in the image. Since $f(X)$ is stable under specialization and $y' \rightsquigarrow y$, $y \in f(X)$ as we sought to show.

Theorem 3.5.14: Valuation Criterion for Separatedness

Let $f : X \to Y$ be a morphism of schemes, and assume that X is Noetherian. Then f is separated if and only if for any field K, and for any valuation ring R with quotient field K, $T = \operatorname{Spec} R, U = \operatorname{Spec} K$, and the morphism $i : U \to T$ is the morphism induced by the inclusion $R \subseteq K$, given a morphism of T to Y, and given a morphism of U to X, the following diagram commutes:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

and there is *at most one* morphism of T to X making the whole diagram commutative.

***Proof*:**
First, suppose f is separated. Let $h, h' : T \to X$ be two morphisms making the diagram commute. We obtain a morphism $L = (h, h') : T \to X \times_Y X$. Since the restrictions of h and h' to U are the same (given by the diagram), the generic point t_1 of T maps to the diagonal $\Delta(X) \subseteq X \times_Y X$. Since f is separated, the diagonal $\Delta(X)$ is a closed immersion, so $\Delta(X)$ is a closed set. The image of T is the closure of the image of the generic point (as $T = \operatorname{Spec} R$ has a generic point

whose closure includes the closed point). Since the generic point lands in the closed set $\Delta(X)$, the entire image $L(T)$ must lie in $\Delta(X)$. Therefore, h and h' coincide everywhere on T, so $h = h'$. The equality of the sheaf maps follows from lemma 3.5.11 as the inclusions into K are identical.

Conversely, suppose the valuation condition holds. To show f is separated, we must show that the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion. Since Δ is a locally closed immersion, it suffices to show the image $\Delta(X)$ is a closed subset of $X \times_Y X$. Since X is Noetherian, the product $X \times_Y X$ is Noetherian (or at least the morphism Δ is quasicompact). By lemma 3.5.13, it suffices to show that $\Delta(X)$ is stable under specialization.

Let $\xi_1 \in \Delta(X)$ and let $\xi_1 \rightsquigarrow \xi_0$ be a specialization in $X \times_Y X$. We must show $\xi_0 \in \Delta(X)$. Let $K = \kappa(\xi_1)$ and let $\mathcal{O}$ be the local ring of ξ_0 on the subscheme $\overline{\{\xi_1\}}$ (with reduced induced structure). By domination properties of valuation rings, there exists a valuation ring R of K which dominates $\mathcal{O}$. Let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. By lemma 3.5.11, this data induces a morphism $T \to X \times_Y X$ mapping the generic point to ξ_1 and the closed point to ξ_0. Composing this with the two projections $p_1, p_2 : X \times_Y X \to X$, we get two morphisms $h, h' : T \to X$. Since $\xi_1 \in \Delta(X)$, we have $p_1(\xi_1) = p_2(\xi_1)$, so h and h' agree on the generic point U. Furthermore, they both map to Y identically (as they factor through $X \times_Y X$). By the hypothesis of the theorem, there is at most one such lift, so $h = h'$. Consequently, the map $T \to X \times_Y X$ must factor through the diagonal, implying $\xi_0 = L(t_0) \in \Delta(X)$. Thus $\Delta(X)$ is stable under specialization and is closed.

Example 3.5.15: Valuation Criterion on $\mathbb{A}^1$

Let us demonstrate the mechanics of the valuation criterion by proving that the affine line $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ is separated over $Y = \operatorname{Spec} k$.

Suppose we are given a valuation ring R with fraction field K, and the standard diagram:

$$\begin{tikzcd} U = \operatorname{Spec} K \arrow[r, "\alpha"] \arrow[d, "\iota"'] & \operatorname{Spec} k[t] \arrow[d] \\ T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists ? \tilde{\alpha}"] & \operatorname{Spec} k \end{tikzcd}$$

We translate this geometric data into algebraic data (reversing arrows):

1. The morphism $\alpha : \operatorname{Spec} K \to \operatorname{Spec} k[t]$ corresponds to a k-algebra homomorphism $\alpha^\# : k[t] \to K$. This homomorphism is completely determined by the image of the generator t. Let $\alpha^\#(t) = \lambda \in K$.

2. The inclusion $\iota : \operatorname{Spec} K \to \operatorname{Spec} R$ corresponds to the inclusion of the valuation ring into its fraction field $R \hookrightarrow K$.

3. A lift $\tilde{\alpha} : \operatorname{Spec} R \to \operatorname{Spec} k[t]$ corresponds to a homomorphism $\tilde{\alpha}^\# : k[t] \to R$. Let $\tilde{\alpha}^\#(t) = r \in R$.

For the diagram to commute, we require that the composition $k[t] \xrightarrow{\tilde{\alpha}^\#} R \hookrightarrow K$ equals the original map $\alpha^\#$. Evaluating at t, this means we must have $r = \lambda$ inside K.

Now we analyze the possibilities based on the element $\lambda \in K$:

- *Case 1:* $\lambda \in R$. Then we must define $\tilde{\alpha}^{\#}(t) = \lambda$. Since the map is determined on generators, this is the *unique* homomorphism making the diagram commute.
- *Case 2:* $\lambda \notin R$. Then there is no element $r \in R$ such that $r = \lambda$. Thus, no lift exists.

In either case, the number of lifts is *at most one* (it is either 1 or 0). Therefore, $\mathbb{A}^1_k$ is separated.

Contrast this with the "Line with two origins" (X). If we map $\operatorname{Spec} K$ to the generic point of X, and R is a DVR, we essentially have a map sending t to 0 in the residue field. In the algebraic formulation, the gluing data allows for two distinct maps $T \to X$ (one to each origin) that become identical when restricted to the generic point U. This violates the uniqueness condition.

Corollary 3.5.16: Properties of Separated Morphisms

Assume that all schemes are Noetherian in the following statements.

1. Open and closed immersions are separated.
2. A composition of two separated morphisms is separated.
3. Separated morphisms are stable under base extension.
4. If $f : X \to Y$ and $f' : X' \to Y'$ are separated morphisms of schemes over a base scheme S, then the product morphism $f \times f' : X \times_S X' \to Y \times_S Y'$ is also separated.
5. If $f : X \to Y$ and $g : Y \to Z$ are two morphisms and if $g \circ f$ is separated, then f is separated.
6. A morphism $f : X \to Y$ is separated if and only if Y can be covered by open subsets V_i such that $f^{-1}(V_i) \to V_i$ is separated for each i.

***Proof*:**
These statements all follow immediately from the condition of the theorem (Valuation Criterion, theorem 3.5.14). We follow Hartshorne [14] and give the proof of (3) to illustrate the method.

Let $f : X \to Y$ be a separated morphism, let $Y' \to Y$ be any morphism, and let $X' = X \times_Y Y'$ be obtained by base extension. We must show that the induced map $f' : X' \to Y'$ is separated. So suppose we are given a valuation ring R with field of fractions K, let $T = \operatorname{Spec} R$ and $U = \operatorname{Spec} K$. Suppose we are given morphisms of T to Y' and U to X' as in the theorem, and two morphisms of T to X' making the diagram

$$\begin{tikzcd}
U \arrow[r] \arrow[d, "i"'] & X' \arrow[r] \arrow[d, "f'"] & X \arrow[d, "f"] \\
T \arrow[r] \arrow[ur, shift left] \arrow[ur, shift right] & Y' \arrow[r] & Y
\end{tikzcd}$$

commutative (where the double arrows represent the two potential lifts $T \to X'$). Composing with the map $X' \to X$, we obtain two morphisms of T to X. Since the outer diagram commutes with respect to f (which is separated), and the maps agree on the dense open set U, these two morphisms $T \to X$ must be the same by theorem 3.5.14.

We also have two morphisms $T \to X' \to Y'$. Since the diagram commutes, both of these must equal the given map $T \to Y'$. Thus, we have two maps $T \to X'$ such that their compositions to X are equal and their compositions to Y' are equal. But X' is the fibered product of X and Y' over Y, so by the universal property of the fibered product, a map to X' is uniquely determined by its projections to X and Y'. Therefore, the two maps of T to X' are the same. Hence f' is separated.

3.5.17 Note on Noetherian Condition You have probably noticed that in order to apply the theorem, it is not necessary to assume that all the schemes mentioned in the corollary are Noetherian. In fact, even in the theorem itself, you can get by with assuming something less than X Noetherian (see EGA I, new ed., 5.5.4).
What Hartshorne says is that if a Noetherian hypothesis will make statements and proofs substantially simpler, then he takes the hypothesis, even though it may not be necessary. This is reasonable as most cases where the valuation citation is useful is when working with stacks which will satisfy these conditions.

Rational maps to Separated Schemes

here

Exercise 3.5.1

1. add exercise 4.5 of Hartshorne: important for Weil divisors to show characterization of irred. closed subscheme of (separated noethean integral) reg. in codim 1 scheme
2. Show that $X \cong \Delta(X)$. Show that in general, $\Delta(X)$ is a locally closed immersion.
3. Show that a scheme X over S is separated if and only if it is separated over $\mathbb{Z}$.
4. If $\varphi : A \to B$ is a ring homomorphism, then the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is dominant if and only if $\ker \varphi$ is contained in the nilradical of A.

3.5.3 Universally Closed Morphisms

We now build up to the notion of properness. Proper schemes and morphisms capture the idea of compactness and “compact-preserving” maps. In topology, a map is often called *proper* if the preimage of any compact set is compact. In algebraic geometry, because the Zariski topology is so coarse (and affine schemes are almost always quasicompact), this definition is not strong enough.

Instead, we look for a property that ensures the *image* of a closed set is closed, and remains closed after any base change.

Definition 3.5.18: Universally Closed

Let $f : X \to Y$ be a morphism of schemes. Then f is *universally closed* if it is a closed map (images of closed sets are closed), and for any morphism $Y' \to Y$, the base change $f' : X \times_Y Y' \to Y'$ is also a closed map.

The intuition is that we want to avoid "missing points" at the boundary. Recall the example of $\mathbb{A}^1_k \to \operatorname{Spec} k$. This is closed, but base changing to $\mathbb{A}^1_k$ (projection of the plane to the line) maps the hyperbola $xy = 1$ to the open set $\mathbb{A}^1 \setminus \{0\}$. The "missing point" at infinity prevents the map from being universally closed.

Just as Separatedness was characterized by the *uniqueness* of lifts in the valuation criterion, Universally Closed morphisms are characterized by the *existence* of lifts.

Theorem 3.5.19: Valuation Criterion for Universally Closed Morphisms

Let $f : X \to Y$ be a quasicompact morphism of schemes. Then f is universally closed if and only if for any valuation ring R with fraction field K, and any diagram:

$$\begin{tikzcd} U = \operatorname{Spec} K \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T = \operatorname{Spec} R \arrow[r] \arrow[ur, dashed, "\exists"] & Y \end{tikzcd}$$

there exists **at least one** morphism $T \to X$ making the diagram commute.

From the perspective of functors of points, the valuative criterion is naturally a statement about extending $\operatorname{Spec}(K)$-valued points to $\operatorname{Spec}(R)$-valued points.

***Proof*:**
($\Rightarrow$) Existence implies Universally Closed: Suppose the lifting property holds. We want to show f is universally closed. Let $Y' \to Y$ be any base change. We must show $f' : X' \to Y'$ is closed. Since f is quasicompact, f' is quasicompact. By lemma 3.5.13, it suffices to show that the image $f'(X')$ is stable under specialization. Let $z_1 \in X'$ and let $y_1 = f'(z_1)$. Suppose $y_1 \rightsquigarrow y_0$ is a specialization in Y'. We need to find a $z_0 \in X'$ such that $f'(z_0) = y_0$.

Let $K = \kappa(z_1)$. We have a map $\operatorname{Spec} K \to X'$. Let $\mathcal{O}$ be the local ring of y_0 in the closure $\overline{\{y_1\}}$ (with reduced structure). By domination, there exists a valuation ring R of K dominating $\mathcal{O}$. This gives us maps $U \to X'$ and $T \to Y'$. Composing with the projection $X' \to X$, we get maps $U \to X$ and $T \to Y$ fitting the hypothesis diagram. By assumption, there exists a lift $T \to X$. By the universal property of the fibered product X', the pair of maps $T \to X$ and $T \to Y'$ induce a map $T \to X'$. Let z_0 be the image of the closed point of T in X'. Since the map commutes, $f'(z_0)$ must be the image of the closed point of T in Y', which is y_0. Thus y_0 is in the image.

($\Leftarrow$) Universally Closed implies Existence: Suppose f is universally closed. Consider the diagram with $U \to X$ and $T \to Y$. We base change by $T \to Y$ to get $X_T = X \times_Y T$. The map $U \to X$ and $U \to T$ induce a map $U \to X_T$. Let ξ_1 be the image of the unique point of U in X_T. Let $Z = \overline{\{\xi_1\}}$ with reduced structure. Since f is universally closed, the projection $X_T \to T$ is closed. Thus the image of Z is closed in T. The generic point of T is in the image (it is the image of ξ_1). Since $T = \operatorname{Spec} R$, the closure of the generic point is all of T. Thus, the image of Z is all of T. Therefore, there exists a point $\xi_0 \in Z$ that maps to the closed point $t_0 \in T$.

This gives us a local homomorphism of local rings $R \to \mathcal{O}_{\xi_0,Z}$. Since R is a valuation ring of $K = \kappa(\xi_1)$, and R dominates the local ring (by construction of the map $T \to Y$), R must actually be the local ring (or dominate it via isomorphism). By lemma 3.5.11, this domination data provides a morphism $T \to X_T$ sending $t_0 \to \xi_0$ and $t_1 \to \xi_1$. Composing $T \to X_T \to X$

gives the desired lift.

3.5.4 Proper Morphisms

We now combine the concepts. We want a scheme (or morphism) that behaves like a compact Hausdorff space.

- Hausdorff iff Separated iff Uniqueness of limits.
- Compact iff Universally Closed + Finite Type iff Existence of limits.

Definition 3.5.20: Proper Morphism

A morphism $f : X \to Y$ is called *proper* if it satisfies three conditions:

1. Separated (Hausdorff-like).
2. Universally Closed (Image of closed is closed).
3. Finite Type (Finite covering / geometrically bounded).

A scheme X is said to be *proper* if the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is proper, and an R-scheme is proper if the map $X \to \operatorname{Spec}(R)$ is proper.

Proposition 3.5.21: Properties of Proper Morphisms

Assume all schemes are Noetherian.

1. A closed immersion is proper.
2. A composition of proper morphisms is proper.
3. Proper morphisms are stable under base change.
4. Products of proper morphisms are proper.
5. If $f : X \to Y$ and $g : Y \to Z$ are morphisms, g is separated, and $g \circ f$ is proper, then f is proper.
6. Properness is local on the base (target).

***Proof*:**

1. A closed immersion is separated (diagonal is iso), finite type (surjection of rings), and universally closed (closed sets in a subspace are closed in the ambient space).
2. Let $f : X \to Y$ and $g : Y \to Z$ be proper. Separatedness and finite type are preserved under composition. For universally closed: Let $Z' \to Z$. Base change $X \to Z$ by Z' is the composition of base changing g then f. Since pullback preserves closed maps, the composition is closed.
3. Finite type and separatedness are stable. Universal closedness is stable by definition.

4. Follows formally from Base Change and Composition.

5. This is best proved using the Valuation Criterion (theorem 3.5.25). Let R be a valuation ring with field K. Suppose we have a diagram:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

We want a unique lift. Compose with $g : Y \to Z$ to get a map $\operatorname{Spec} K \to X$ and $\operatorname{Spec} R \to Z$. Since $g \circ f$ is proper, there exists a unique lift $\operatorname{Spec} R \to X$. This lift makes the big diagram commute. Does it make the triangle with Y commute? We have two maps $\operatorname{Spec} R \to Y$: one given, and one via the lift through X. Both map the generic point to the same place, and both compose with g to give the same map to Z. Since g is *separated*, by the valuation criterion for separatedness, these two maps must be equal. Thus the lift works for f.

The following two propositions give a large class of useful proper morphisms, and consequence proper schemes over Y:

Proposition 3.5.22: Finite Morphisms are Proper

Let $f : X \to Y$ be a finite morphism. Then f is proper.

***Proof*:**
A finite morphism is affine, hence separated. It is finite type (actually finite presentation if Noetherian). We need only show it is universally closed. Since properness is local on the target, assume $Y = \operatorname{Spec} R$. Then $X = \operatorname{Spec} S$ where S is a finite R-module. Let $Y' \to Y$ be a base change. The base change of a finite morphism is finite ($S \otimes_R R'$ is a finite R'-module). Thus, it suffices to show that any finite morphism is a closed map. Let $\varphi : A \to B$ be a finite ring map. Let $V(I) \subseteq \operatorname{Spec} B$ be a closed set. The image is $V(\varphi^{-1}(I))$ inside $\operatorname{Spec} A$. This follows from the *Going Up Theorem* [9] (or the Cohen-Seidenberg Theorem). Since B is integral over A, the map $\operatorname{Spec} B \to \operatorname{Spec} A$ is closed.

If the focus is put on morphisms between affine schemes (or slightly more generally affine morphisms), then properness reduces to finiteness:

Proposition 3.5.23: Affine Proper Morphisms are Finite

Let $f : X \to Y$ be an affine morphism. Then f is proper if and only if f is finite.

***Proof*:**
($\Leftarrow$) This follows immediately from proposition 3.5.22. A finite morphism is always affine and proper.

($\Rightarrow$) Suppose f is affine and proper. We must show it is finite. Since finiteness is local on the base, we may assume $Y = \operatorname{Spec} A$ is affine. Because f is affine, $X = \operatorname{Spec} B$ can also be assumed to be affine, and so f corresponds to a ring homomorphism $\varphi : A \to B$.

Since f is proper, it is of *finite type*. This means B is a finitely generated A-algebra.

To show finiteness, we must show that φ is *integral*. We use the Valuative Criterion for Universally Closed Morphisms (theorem 3.5.19). Let K be the fraction field of B (or any field containing B) and let $R \subset K$ be any valuation ring containing A (more precisely, containing $\varphi(A)$). Consider the diagram:

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\ \operatorname{Spec} R \arrow[r] & \operatorname{Spec} A \end{tikzcd}$$

Since f is universally closed, a lift $\operatorname{Spec} R \to \operatorname{Spec} B$ exists. Algebraically, this means the map $B \to K$ factors through R, i.e., $B \subseteq R$.

Thus, B is contained in every valuation ring of K that contains A. A standard theorem in commutative algebra states that the integral closure of A in K is the intersection of all valuation rings of K containing A, see lemma 5.3.7 for a slightly stronger proof. Therefore, B is integral over A.

We now have that B is a finitely generated A-algebra that is integral over A. A standard result in commutative algebra states that such a ring is a finitely generated A-module (see [9, chapter 20.5]). Since B is a finite A-module, f is a finite morphism, completing the proof.

Example 3.5.24: Non-Proper Morphisms

1. $\mathbb{A}^1_k \to \operatorname{Spec} k$ is not proper. It is not universally closed (base change to $\mathbb{A}^1$ maps the hyperbola to an open set).
2. $\mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k$ (open immersion). This is separated and finite type, but not closed (image is open).
3. The "Line with two origins" is not proper because it is not separated.

Theorem 3.5.25: Valuation Criterion for Properness

Let $f : X \to Y$ be a morphism of finite type, with X Noetherian. Then f is proper if and only if for every valuation ring R with fraction field K, and every diagram:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

there exists a *unique* lifting $T \to X$ making the diagram commute.

***Proof*:**
See derivation in previous section.

Projective Schemes and Consequences

Theorem 3.5.26: Projective Morphisms are Proper

A projective morphism between Noetherian schemes is proper.

***Proof*:**
(See previous section for proof via valuation criterion).

One of the most powerful consequences of properness is the behavior of global functions. For compact complex manifolds, Liouville's theorem states that holomorphic functions on a compact connected manifold are constant. We have the algebraic analogue:

Theorem 3.5.27: Global Sections of Proper Schemes

Let k be a field. Let X be a proper k-scheme that is reduced and connected. Then:

$$\Gamma(X, \mathcal{O}_X) = k$$

(i.e., the only global regular functions are constants).

***Proof*:**
Let $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. By proposition 3.3.7, we get a scheme morphism (that we shall also denote as f)

$$f : X \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$$

Consider the open immersion $\iota : \mathbb{A}^1_k \hookrightarrow \mathbb{P}^1_k$. Compose them to get $\varphi : X \to \mathbb{P}^1_k$. Since X is proper over k and $\mathbb{P}^1_k$ is separated over k, the image $\varphi(X)$ must be a *closed* subset of $\mathbb{P}^1_k$. However, f maps into the affine line, so the image $\varphi(X)$ does not contain the point at infinity, ∞. Thus, $\varphi(X)$ is a closed subset of $\mathbb{P}^1_k$ contained entirely in $\mathbb{A}^1_k$. The closed subsets of $\mathbb{P}^1_k$ are finite sets of points or the whole space. Since it misses ∞, it must be a finite set of points. Since X is connected, the image must be a single point. Since X is reduced, the map to a single point corresponds to a map to $\operatorname{Spec} k$. Thus, f maps every point to the same value $\lambda \in k$.

While Projective implies Proper, the converse is not true (though examples are pathological, like Hironaka's example). However, Chow's Lemma states that proper schemes are "birationally" projective.

Theorem 3.5.28: Chow's Lemma

Let $f : X \to S$ be a proper morphism of finite type over a Noetherian base S. Then there exists a scheme X' and a morphism $\pi : X' \to X$ such that:

1. π is a surjective projective morphism.
2. The composite $X' \to S$ is a projective morphism.
3. There exists a dense open $U \subseteq X$ such that $\pi^{-1}(U) \to U$ is an isomorphism.

***Proof*:**
(Sketch). The proof involves taking a finite cover of X by quasi-projective schemes U_i. We form the disjoint union $U = \coprod U_i$ and a map $U \to X$. We then take the closure of the graph of this map inside $X \times_S \mathbb{P}^N$ (for sufficiently large N). The resulting scheme X' dominates X and sits inside a projective space over S, making it projective. The dense open isomorphism comes from the fact that we started with a cover. See [14, Hartshorne II.4.10] for the complete derivation.

Exercise 3.5.2

1. Show that two copies of $\mathrm{Spec}\,(k[x_1, x_2, ...])$ minus the origin, glued at the origin, is not quasi-separated.
2. Prove that the intersection of two proper subschemes is proper.

3.6 Rational Maps

When working with [quasiprojective] varieties, we had the weaker notion of a rational function that was defined almost everywhere. A rational function was more flexible, allowing for some important classification results, the definition of Kodaira dimension, study singularities, more invariances by looking at the birational automorphisms, and some "local to global" principles. The following brings these concepts over to schemes

Definition 3.6.1: Rational Map

Let X, Y be schemes. Then a *rational map* $\pi : X \dashrightarrow Y$, is an equivalence relation:

$$(\alpha U \to Y) \sim (\beta : V \to Y) \qquad \text{if and only if} \qquad Z \subseteq U \cap V \text{ s.t. } \alpha|_Z = \beta|_Z$$

where U, V, Z are all dense open subsets[a].

If X, Y are S-schemes, then rational maps of S-schemes are defined similarly.

If the image of one of the representatives is dense, then the rational map is said to be *dominant* or *dominating*.

[a]When Y is separated, we will let $Z = U \cap V$

Note that a rational map is is an equivalence relation, not necessarily defined on a single domain.

Definition 3.6.2: Birational Map

A rational map $\pi : X \dashrightarrow Y$ is *birational* if it is dominant, and it has a dominant rational inverse. Two schemes X, Y are *birationally equivalent* if there is a birational map $\pi : X \dashrightarrow Y$.

Example 3.6.3: Rational Map

1. A scheme morphisms always gives a rational map, similarily to how all regular functions are special rational functions
2. The projection $\mathbb{P}^n_R \dashrightarrow \mathbb{P}^{n-1}_R$ projecting down a dimension is a rational map (recall the same reasoning for rational functions in [9, chapter 22.3]

Lemma 3.6.4: Dominating Rational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a rational map between irreducible schemes. Then π is dominating iff it sends the generic point of X to the generic point of Y.

***Proof*:**
recall the properties of generic points in irreducible schemes.

We may naturally want to have a result that is similar to the correspondence of rational functions and field extensions as given in [9]. The full correspondence requires irreducible varieties (integral finite type k-schemes), however we may at least always induce a map of function fields:

Proposition 3.6.5: Rational Maps of Integral Schemes induces Function Field Map

The collection of integrals schemes with dominant maps induces morphisms of function fields in the opposite direction.

***Proof*:**
First show that the form a category (importantly, composition works). Then think of how to create this correspondence.

Corollary 3.6.6: Birational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a birational map between irreducible schemes. Then the induced map of function fields is an isomorphism.

For the converse failing

Example 3.6.7: Function Field Not Inducing Rational Map
Take Spec $(k[x])$ and Spec $(k(X)$ which have the same function field $k(x)$. Then certainly there is no corresponding rational map

$$\operatorname{Spec} k[x] \dashrightarrow \operatorname{Spec} k(x)$$

of k-schemes, as such a morphism would correspond to a morphism from $U \subseteq \operatorname{Spec} k[x]$ (ex. $U = \operatorname{Spec} k[x, 1/f(x)]$) to $\operatorname{Spec} k(x)$, but not map of rings $k(X) \to k[x, 1/f(x)]$ satisfies the right properties for any $f(x)$.

Proposition 3.6.8: Reduced Schemes and Birational Maps

Let X, Y be reduced schemes. Then X and nY are birational if and only if there exists a dense open subscheme $U \subseteq X$ and $V \subseteq Y$ such that

$$U \cong V$$

***Proof*:**
Generalize the proof from [9] (If stuck, look at Vakil p. 190)

Vakil looks here at rational maps of irreducible varieties that look like translating the results over from varieties, so I will omit for now

I like his translation of the Pythagorean problem into a scheme problem

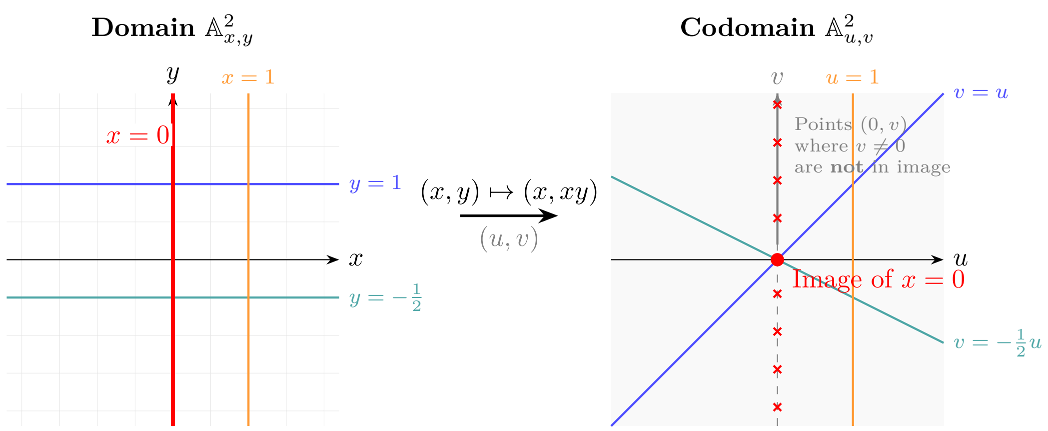
non-rational complex curve, cool presentation

3.7 Image Scheme: Chevalley's Theorem

The image of a scheme morphism need not be locally closed:

Example 3.7.1: Image That is Not Locally Closed
Let $f : \mathbb{A}^2_k \to \mathbb{A}^2_k$ given by $(x, y) \mapsto (x, xy)$. It's image is:

$$(\mathbb{A}^2_k \setminus \{(x, y) \mid x = 0\}) \cup \{(0, 0)\}$$



In general, the image of a scheme morphism shall be the union of locally closed set. As this pertains just to variety-like schemes, we limit to Noetherian spaces:

Definition 3.7.2: Constructible Set

Let X be a Noetherian topological space. Then $S \subseteq X$ is *constructible* if it belongs to the family such that:

1. every open set is in the family
2. the compliment of elements in the family are in the family
3. finite intersections of elements in this family are in the family

The first two conditions make all open and closed sets in the family. The last conditions add unions of locally closed sets.

This to me gives the vibes of a finite version of a measurable set. For completeness, this is the definition given in EGA cite:HERE

Definition 3.7.3: Constructible Set (General Case)

Let X be a topological space (not necessarily Noetherian). A *retrocompact* open set U is one where the inclusion $U \hookrightarrow X$ is quasi-compact. Then $S \subseteq X$ is *constructible* if it belongs to the family such that:

1. every *retrocompact* open set is in the family
2. the complement of elements in the family are in the family
3. finite intersections of elements in this family are in the family

This restriction ensures that constructible sets are still described by "finite amounts of data" (finitely generated ideals). For our purposes, we shall not worry about such complexities as it does not add further insight.

3.7.4 Assumption For This Section We shall assume all topological space are Noetherian for the rest of this section.

Intuitively, constructible sets can all be represents using locally closed sets:

Lemma 3.7.5: Preimages of As Locally Closed Set

Let X be a Noetherian topological space. Then any constructible set $S \subseteq X$ is a finite union of disjoint locally closed sets. As a consequence, if $f : X \to Y$ is continuous then the pre-image of a constructible set is constructible

***Proof*:**
exercise

lots more words here

Theorem 3.7.6: Chevalley's Theorem

Let $f : X \to Y$ be a finite type morphism of Noetherian schemes. Then the image of a constructible set is constructible. Thus, $f(X)$ is constructible.

***Proof*:**
here

3.8 Abstract Varieties

Let us use the tools we've developed to our usual notion of variety. We shall see that defining varieties abstractly (other common adverbs being inherently, integrally, intrinsically) shall give us many of the same benefits as defining schemes as spectrum. Most results about quasi-projectile varieties from [9] readily translate to the abstract setting with a few exceptions that we shall go over.

There is a natural way in which we can embed (quasi-projective) varieties into the category $\mathbf{Sch}_{\bar{k}}$:

Theorem 3.8.1: Varieties and Schemes: Fully Faithful Functor

Let $\bar{k}$ be an algebraically closed field. Then there exists a fully faithful functor $\mathbf{Var}_{\bar{k}} \to \mathbf{Sch}_{\bar{k}}$ where $\mathbf{Var}_{\overline{k}}$ is the usual category of varieties (over $\overline{k}$) as defined in [9].

Recall that a fully faithful means the hom-sets are in bijection, and it implies injectivity on objects (but not necessarily surjective, for example Ab $\to$ Grp is a fully faithful functor). We shall also stick with Hartshorne's notation for the functor, which is the letter t, it shall come up a few times later in the book.

***Proof*:**
For any (quasi-projective) variety, we shall first find the "points" that shall be the points of the scheme by working with the irreducible closed subsets. We shall then find the structure sheaf to put on it by finding all the regular functions, which shall be enough to define the functor.

First off, for any topological space X, and let $t(X)$ be the set of (nonempty) irreducible closed subsets of X. If Y is a closed subset of X, then $t(Y) \subseteq t(X), t\left(Y_1 \cup Y_2\right) = t\left(Y_1\right) \cup t\left(Y_2\right)$, and $t\left(\bigcap Y_i\right) = \bigcap t\left(Y_i\right)$. Hence, we can define a topology $t(X)$ by taking as closed sets the subsets of the form $t(Y)$, where Y is a closed subset of X. If $f : X_1 \to X_2$ is a continuous map, then we obtain a map $t(f) : t\left(X_1\right) \to t\left(X_2\right)$ by sending an irreducible closed subset to the closure of its image. Thus t is a functor on topological spaces. Furthermore, define a continuous map $\alpha : X \to t(X)$ by $\alpha(P) = \{P\}^-$. Note that α induces a bijection between the set of open subsets of X and the set of open subsets of $t(X)$.

Next, let k be an algebraically closed field, V be a variety over k, and let $\mathcal{O}_V$ be its sheaf of regular functions. I claim that $\left(t(V), \alpha_*\left(\mathcal{O}_V\right)\right)$ is a scheme over k. Since any variety can be covered by open affine sub-varieties, it will be sufficient to show that if V is affine, then $\left(t(V), \alpha_*\left(\mathcal{O}_V\right)\right)$ is a scheme. Hence, let V be an affine variety with affine coordinate ring A. Define a morphism of locally ringed spaces

$$\beta : (V, \mathcal{O}_V) \to X = \operatorname{Spec} A$$

as follows: for each point $P \in V$, let $\beta(P) = \mathfrak{m}_P$, the ideal of A consisting of all regular functions which vanish at P. Then β is a bijection of V onto the set of closed points of X. It is easy to see that β is a homeomorphism onto its image. Now for any open set $U \subseteq X$, we will define a homomorphism of rings $\mathcal{O}_X(U) \to \beta_*(\mathcal{O}_V)(U) = \mathcal{O}_V(\beta^{-1}U)$. Given a section $s \in \mathcal{O}_X(U)$, and given a point $P \in \beta^{-1}(U)$, we define $s(P)$ by taking the image of s in the stalk $\mathcal{O}_{X,\beta(P)}$, which is isomorphic to the local ring $A_{\mathfrak{m}_P}$, and then passing to the quotient ring $A_{\mathfrak{m}_P}/\mathfrak{m}_P$ which is isomorphic to the field k. Thus s gives a function from $\beta^{-1}(U)$ to k. It is easy to see that this is a regular function, and that this map gives an isomorphism $\mathcal{O}_X(U) \cong \mathcal{O}_V(\beta^{-1}U)$. Finally, since the prime ideals of A are in 1-1 correspondence with the irreducible closed subsets of V, these remarks show that $(X, \mathcal{O}_X)$ is isomorphic to $(t(V), \alpha_*(\mathcal{O}_V))$, so the latter is indeed an affine scheme, completing the proof.

We have defined upon enough properties to characterize the image of this embedding:

Definition 3.8.2: k-Variety

Let X be a k-scheme. Then if X is reduced, irreducible, separated, and of finite type over k, then X is said to be a *variety over* k or a k*-variety*. If X is affine/(quasi)projective, then it is an affine/(quasi)projective variety over k respectively.

It is important to note that in [9], a variety was an affine algebraic set that was irreducible, which motivates adding irreducible in the definition. However some authors drop the condition and "variety" would be "algebraic set". Note too that we have defined k-varieties and not varieties in general. Furthermore, some authors require the field $k = \overline{k}$ be algebraically closed as part of the definition to more closely reflect the geometric results of the Nullstellensatz, and also to avoid the following problem:

Example 3.8.3: Variety Not Closed Under Base Change

Take the $\mathbb{R}$-variety $\mathbb{C} \cong \mathbb{R}[x]/(x^2+1) = V$ which is a single point, hence irreducible. Then $C \times_{\mathbb{R}} V = \operatorname{Spec} \mathbb{C}[x](x^2+1)$ which now has two points, hence it is not irreducible and so *not* a variety as we have defined it.

While the abstract definition of a scheme allows for elegant categorical constructions, it introduces points that are not closed (the generic points corresponding to non-maximal prime ideals). Notice in the proof of theorem 3.8.1 that the map β is a bijection of the classical variety V onto *only* the closed points of $X = \operatorname{Spec} A$. One might naturally worry: do we lose algebraic information by only evaluating functions at the closed points?

This is where the concept of a *Jacobson scheme* is useful, serving as the exact geometric justification for why our classical intuition of evaluating polynomials at points perfectly captures the algebraic structure. Indeed, it is fair to say that Jacobson schemes are the "most general" setting in which the Nullstellensatz holds.

Definition 3.8.4: Jacobson Scheme

A topological space X is *Jacobson* if its closed points are dense in every closed subset. A scheme X is a *Jacobson scheme* if its Zariski topology is Jacobson.

Recall ([9, commutative-algebra]) that a *Jacobson ring* is a ring where the intersection of every maximal ideal of R containing $I \subseteq R$ is equal to the radical $\sqrt{I}$.

Proposition 3.8.5: Characterizing Jacobson Affine Schemes

An affine scheme $X = \operatorname{Spec} R$ is Jacobson if and only if R is *Jacobson*.

***Proof*:**
Let $X = \operatorname{Spec} R$. Any closed subset of X is of the form $V(I)$ for some ideal $I \subseteq R$. The closed points of X are exactly the maximal ideals $\operatorname{mSpec}(R)$. Therefore, the closed points residing inside $V(I)$ are exactly the maximal ideals containing I.

The closure of these points inside X is given by taking the vanishing locus of their intersection:

$$\overline{V(I) \cap \operatorname{mSpec}(R)} = V\left(\bigcap_{\substack{\mathfrak{m} \in \operatorname{mSpec}(R) \\ I \subseteq \mathfrak{m}}} \mathfrak{m}\right)$$

If R is a Jacobson ring, this intersection of maximal ideals is exactly $\sqrt{I}$. Thus, the closure becomes $V(\sqrt{I})$, which is equal to $V(I)$. This shows the closed points in $V(I)$ are dense in $V(I)$, making X a Jacobson space.

Conversely, if X is a Jacobson space, the closed points are dense in $V(I)$, so $V\left(\bigcap_{\mathfrak{m} \supset I} \mathfrak{m}\right) = V(I)$. Taking the radical of the defining ideals gives $\bigcap_{\mathfrak{m} \supset I} \mathfrak{m} = \sqrt{I}$, proving R is a Jacobson ring.

Let us formally prove that our geometric intuition of "density" translates exactly to this algebraic property of the nilradical, which ensures we don't lose information.

Corollary 3.8.6: Density of Closed Points and the Nilradical

Let R be a commutative ring and $X = \operatorname{Spec} R$. If the closed points of X are dense in X, then the intersection of all maximal ideals in R is exactly the nilradical $\operatorname{Nil}(R)$.

***Proof*:**
This is a direct application of the logic from proposition 3.8.5 using the zero ideal $I = (0)$.

The entire space X is the closed subset $V(0)$. The condition that the closed points are dense in X means that the closure of the maximal ideals is the whole space:

$$V\left(\bigcap_{\mathfrak{m} \in \operatorname{mSpec}(R)} \mathfrak{m}\right) = V(0)$$

Taking the radical of the defining ideals on both sides yields:

$$\bigcap_{\mathfrak{m} \in \operatorname{mSpec}(R)} \mathfrak{m} = \sqrt{(0)} = \operatorname{Nil}(R)$$

(Note that the left side is its own radical because an intersection of prime ideals is already a radical ideal).

Because the nilradical is zero, the geometry of the maximal ideals is dense enough to completely determine the algebra. This classically justifies viewing polynomials as functions to the base field k. Furthermore, as we saw early in the chapter in section 3.3 (Functor of Points), this classical intuition is perfectly mirrored in scheme theory, where a global section $f \in \Gamma(X, \mathcal{O}_X)$ is canonically equivalent to a scheme morphism $X \to \mathbb{A}^1_k$.

Example 3.8.7: Non-Jacobson Schemes

A scheme that is not Jacobson must have a prime ideal that is not the intersection of maximal ideals. This represents a higher dimensional point that is "not covered" by closed points. Schemes likes $\operatorname{Spec} k[[x]]$ and $\operatorname{Spec} \mathbb{Z}_{(p)}$. The scheme $\operatorname{Spec} k[x]/(x^n)$ *is* Jacobson, having only (x) as its prime and maximal ideal.

Note that:

$$k[[x]] \cong \varprojlim_n \frac{k[x]}{(x^n)}$$

giving an example of how different the geometry and the algebra can behave under limits. We shall address this in section 5.5.

Having reconciled the abstract algebraic structure with our geometric intuition of points, let us look at the standard ways to construct these varieties.

Proposition 3.8.8: Affine and Projective Varieties

1. $\mathbb{A}^n_k/I$ is an affine k-variety if and only if $I \subseteq k[x_1, ..., x_n]$ is a radical ideal
2. Let $I \subseteq k[x_1, ..., x_n]$ be a radical graded ideal. Then $\operatorname{Proj} k[x_0, ..., x_n]/I$ is a projective k-variety [a]

[a] I need not be radical, ex. $I = (x_0^2, x_0x_1, ..., x_0x_n)$

***Proof*:**
exercise.

An important class of varieties are *proper varieties*. For historical reasons, namely the 19th century Italian school, the following is a common name for proper varieties;

Definition 3.8.9: Complete Variety

Let X be a k-variety. Then it is said to be *complete* if it is proper.

The adjective "complete" for a proper variety comes from the fact that we are in some way thinking of a complete variety as an extension into projective space. It can be shown (see theorem 3.5.28[6]) that if C is complete, then there exists a morphism $C \to \mathbb{P}^n$ to a projective scheme that is isomorphic to an open dense subset. There do exist complete non-projective varieties, but they are hard to come by, partly due to how bad the singularity must be[7].

[6] see Borcherds scheme video abstract and projective varieties

[7] again see the same Borcherds video for an overview. The example he gave also allows for producing algebraic spaces that are not schemes

Before concluding this section, it is crucial to see how finite type schemes (and thus varieties) behave under categorical products, as standard geometric products naturally map to fibered products in the category of schemes.

Proposition 3.8.10: Fibered Products of Finite Type Schemes

Let S be a k-scheme of finite type. If X and Y are k-schemes of finite type over S, then their fibered product $X \times_S Y$ is a finite type k-scheme. Consequently, the product $X \times_k Y$ of two k-varieties over an algebraically closed field is a k-variety.

***Proof*:**
Because $X \to S$ and $Y \to S$ are morphisms of finite type, their base changes are also of finite type. Specifically, the projection $X \times_S Y \to Y$ is a base change of $X \to S$, so it is of finite type. Composing this with the finite type structure morphism $Y \to \operatorname{Spec} k$, we find that $X \times_S Y \to \operatorname{Spec} k$ is of finite type.

In the specific case where $S = \operatorname{Spec} k$ and X, Y are k-varieties over an algebraically closed field k, we know $X \times_k Y$ is of finite type. Furthermore, varieties are separated by definition, and the product of separated schemes is separated. Since k is algebraically closed, the tensor product of two integral k-algebras over k is again an integral domain. Because this property holds locally on affine charts, $X \times_k Y$ is reduced and irreducible. Having satisfied all conditions, $X \times_k Y$ is a k-variety.

4

Schemes II.5: Group Schemes

This chapter can comfortably skipped if the reader is not interested in studying algebraic groups or abelian varieties in the near future. The chapter is placed here since it is a natural follow up to all the constructions we have presented in the previous chapter, and (I think) does not require much in the way of quasi-coherent or coherent sheaves, and similar for sheaf cohomology.

In classical algebraic geometry, an *algebraic group* is a variety that is simultaneously a group, with the multiplication and inversion maps being morphisms of varieties. The theory of algebraic groups over algebraically closed fields is rich and well-developed, encompassing the classification of reductive groups, the structure theory of Borel and torus subgroups, and the representation theory that underpins much of modern mathematics.

Over a general base scheme, however, the correct notion is that of a *group scheme*: a scheme G equipped with morphisms encoding multiplication, identity, and inversion, subject to the usual group axioms expressed diagrammatically. This generalization is essential for arithmetic applications (where the base may be $\mathrm{Spec}\,(\mathbb{Z})$ or $\mathrm{Spec}\,(\mathbb{Z}_p)$), for the study of families of groups (where the group structure may degenerate or change character from fiber to fiber), and for capturing phenomena invisible over algebraically closed fields—such as infinitesimal group schemes in characteristic p.

This section develops the foundations of group schemes. The reader should be comfortable with fiber products (section 3.2) throughout.

4.1 Definition and First Examples

Recall that in any category $\mathcal{C}$ with finite products and a terminal object e, a *group object* is an object G equipped with morphisms

$$\mu : G \times G \to G, \qquad \epsilon : e \to G, \qquad \iota : G \to G$$

satisfying the usual group axioms—associativity, identity, and inverse—expressed as commutative diagrams. In the category **Set**, a group object is an ordinary group. In the category of smooth manifolds, a group object is a Lie group. In the category of schemes, a group object is a group scheme.

Definition 4.1.1: Group Scheme

Let S be a scheme. A *group scheme over S* is an S-scheme G together with S-morphisms

$$\mu : G \times_S G \longrightarrow G, \qquad \epsilon : S \longrightarrow G, \qquad \iota : G \longrightarrow G$$

called *multiplication*, *identity*, and *inversion*, such that the following diagrams commute:
Associativity:

$$\begin{tikzcd}
G \times_S G \times_S G \arrow[r, "\mu\times\mathrm{id}"] \arrow[d, "\mathrm{id}\times\mu"'] & G \times_S G \arrow[d, "\mu"] \\
G \times_S G \arrow[r, "\mu"'] & G
\end{tikzcd}$$

Identity:

$$\begin{tikzcd}
S \times_S G \arrow[r, "\epsilon\times\mathrm{id}"] \arrow[dr, "\sim"'] & G \times_S G \arrow[d, "\mu"] & G \times_S S \arrow[l, "\mathrm{id}\times\epsilon"'] \arrow[dl, "\sim"] \\
& G &
\end{tikzcd}$$

Inverse:

$$\begin{tikzcd}
G \arrow[r, "(\iota,\mathrm{id})"] \arrow[d, "(\mathrm{id},\iota)"'] & G \arrow[d, "\mu"] & \\
G \times_S G \arrow[r, "\mu"'] & G & S \arrow[l, "\epsilon"'] \arrow[ul, "\epsilon\circ\pi"'] \arrow[ull, bend right, "\epsilon\circ\pi"']
\end{tikzcd}$$

where $\pi : G \to S$ is the structure morphism.
A group scheme is *commutative* (or *abelian*) if additionally $\mu \circ \sigma = \mu$, where $\sigma : G \times_S G \to G \times_S G$ is the swap morphism.

The identity morphism $\epsilon : S \to G$ is a section of the structure map $\pi : G \to S$. Its image is a closed subscheme of G when G is separated over S, but in general it need not be a closed point or even a closed subscheme—it is an S-valued point of G in the sense of section 3.3.2. Over a field k, however, $\epsilon : \mathrm{Spec}\,(k) \to G$ picks out a k-rational point, which is closed if G is of finite type over k.

Proposition 4.1.2: Group Schemes over a Field are Separated

Let G be a group scheme of finite type over a field k. Then G is separated over k.

We shall assume a proposition from proven later, the intuition that translated directly from topology is that separatedness is the equivalent of Hausdorffness, and a space is Hausdorff is the diagonal is closed.

***Proof*:**
We must show the diagonal $\Delta : G \to G \times_k G$ is a closed immersion (section 3.5.1). Consider the morphism $\varphi : G \times_k G \to G$ defined by $\varphi = \mu \circ (\mathrm{id} \times \iota)$, i.e., $(g, h) \mapsto g \cdot h^{-1}$. The diagonal $\Delta(G)$ is the fiber $\varphi^{-1}(\epsilon(\mathrm{Spec}\,(k)))$. Since ϵ is a section of a morphism of finite type (and k is a field), the image of ϵ is a closed point of G. The fiber of a morphism over a closed point is a closed subscheme of the source. Hence $\Delta(G)$ is closed.

Let us now examine the fundamental examples.

Example 4.1.3: The Additive Group Scheme
The *additive group scheme* $\mathbb{G}_{a,S} = \mathbb{A}^1_S$ has underlying scheme $\mathrm{Spec}\,_S(\mathcal{O}_S[t])$, with multiplication given by the ring map

$$\mathcal{O}_S[t] \longrightarrow \mathcal{O}_S[t] \otimes_{\mathcal{O}_S} \mathcal{O}_S[t], \qquad t \longmapsto t \otimes 1 + 1 \otimes t,$$

identity $t \mapsto 0$, and inversion $t \mapsto -t$. These encode, respectively, addition, zero, and negation: for any S-scheme T,

$$\mathbb{G}_a(T) = \Gamma(T, \mathcal{O}_T)$$

with its additive group structure. Over a field k, this is simply the additive group of the coordinate ring.

Example 4.1.4: The Multiplicative Group Scheme
The *multiplicative group scheme* $\mathbb{G}_{m,S} = \mathrm{Spec}\,_S(\mathcal{O}_S[t, t^{-1}])$ has multiplication

$$t \longmapsto t \otimes t,$$

identity $t \mapsto 1$, and inversion $t \mapsto t^{-1}$. For any S-scheme T:

$$\mathbb{G}_m(T) = \Gamma(T, \mathcal{O}_T)^\times,$$

the group of invertible global sections under multiplication.

Example 4.1.5: The General Linear Group Scheme
For a positive integer n, the *general linear group scheme* is

$$\mathrm{GL}_{n,S} = \mathrm{Spec}\,_S\big(\mathcal{O}_S[x_{11}, x_{12}, \ldots, x_{nn},\, d]/(d \cdot \det(x_{ij}) - 1)\big)\,.$$

The variable d is the formal inverse of $\det(x_{ij})$, so the underlying scheme parametrizes $n \times n$ matrices with invertible determinant. The multiplication map is given by matrix multiplication: the ring map sends x_{ij} to $\sum_k x_{ik} \otimes x_{kj}$, and the identity sends $x_{ij} \mapsto \delta_{ij}$. For any S-scheme T:

$$\mathrm{GL}_n(T) \cong \mathrm{GL}_n(\Gamma(T, \mathcal{O}_T)),$$

the group of invertible $n \times n$ matrices with entries in $\Gamma(T, \mathcal{O}_T)$. When $n = 1$, we recover $\mathrm{GL}_1 \cong \mathbb{G}_m$.

Example 4.1.6: Roots of Unity

For a positive integer n, the *group scheme of n-th roots of unity* is

$$\mu_n = \operatorname{Spec}(\mathbb{Z}[t]/(t^n - 1)).$$

The group structure is inherited from $\mathbb{G}_m$: the comultiplication is $t \mapsto t \otimes t$ (the same as for $\mathbb{G}_m$, but now $t^n = 1$). For any ring A:

$$\mu_n(A) = \{a \in A \mid a^n = 1\} \subseteq A^\times.$$

The behavior of μ_n depends dramatically on the characteristic of the base. Over a field k with $\operatorname{char}(k) \nmid n$, the polynomial $t^n - 1$ has n distinct roots in $\overline{k}$, and $\mu_{n,k}$ is a reduced scheme of degree n over k—an étale group scheme. Over $\overline{k}$, it is isomorphic (as a scheme) to the disjoint union of n copies of $\operatorname{Spec}(\overline{k})$, and $\mu_n(\overline{k}) \cong \mathbb{Z}/n\mathbb{Z}$.

In characteristic $p > 0$ with $p \big| n$, the situation changes. Taking $n = p$, we have $t^p - 1 = (t-1)^p$ in $\mathbb{F}_p[t]$, so

$$\mu_{p,\mathbb{F}_p} = \operatorname{Spec}(\mathbb{F}_p[t]/((t-1)^p)) = \operatorname{Spec}(\mathbb{F}_p[s]/(s^p))$$

where $s = t - 1$. This is a *non-reduced* scheme: a single fat point at $t = 1$ with a p-dimensional tangent space. The group $\mu_p(\mathbb{F}_p) = \{1\}$ is trivial, yet the scheme μ_p is non-trivial—it carries nilpotent structure that remembers the "ghost" of the p-th roots of unity that have collapsed to 1.

Example 4.1.7: Constant Group Schemes

Let Γ be a finite (abstract) group. The *constant group scheme* associated to Γ over a base S is the scheme

$$\Gamma_S = \coprod_{\gamma \in \Gamma} S,$$

a disjoint union of copies of S indexed by elements of Γ. The group structure is defined by declaring that the component indexed by γ times the component indexed by γ' lands in the component indexed by $\gamma\gamma'$. For any connected S-scheme T:

$$\Gamma_S(T) = \Gamma.$$

The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ over a field k is always étale (ref:HERE) and reduced, regardless of the characteristic. It should be carefully distinguished from μ_n: over $\mathbb{F}_p$, the scheme $(\mathbb{Z}/p\mathbb{Z})_{\mathbb{F}_p}$ consists of p disjoint points, while $\mu_{p,\mathbb{F}_p}$ is a single fat point.

Example 4.1.8: Elliptic Curves as Group Schemes

An elliptic curve E over a field k (see section 7.1.3) is a smooth projective curve of genus 1 equipped with a k-rational point $O \in E(k)$. The chord-and-tangent construction endows E with a group scheme structure: the multiplication is the addition law on points, $\epsilon = O$, and ι is the map $P \mapsto -P$. Unlike the previous examples, E is *not* affine—it is a projective group scheme. Commutative projective group schemes (of finite type over a field) are called *abelian varieties*; elliptic curves are abelian varieties of dimension 1.

4.1.1 The Hopf Algebra Perspective

The examples above were all defined by specifying ring maps encoding multiplication, identity, and inversion. For affine group schemes, this algebraic data has a clean formalization: the coordinate ring carries the structure of a *Hopf algebra*. This perspective is particularly useful for explicit computations and for connecting group schemes to representation theory.

Let $G = \mathrm{Spec}\,(A)$ be an affine group scheme over a field k. The group operations on G dualize to k-algebra maps on A:

$$\begin{array}{rcll} \mu : G \times_k G \to G & \rightsquigarrow & \Delta : A \to A \otimes_k A & \text{(comultiplication)} \\ \epsilon : \mathrm{Spec}\,(k) \to G & \rightsquigarrow & \varepsilon : A \to k & \text{(counit)} \\ \iota : G \to G & \rightsquigarrow & S : A \to A & \text{(antipode)} \end{array}$$

The group axioms translate directly:

Definition 4.1.9: Hopf Algebra

A *commutative Hopf algebra* over k is a commutative k-algebra A equipped with k-algebra homomorphisms $\Delta : A \to A \otimes_k A$ (comultiplication) and $\varepsilon : A \to k$ (counit), and a k-algebra map $S : A \to A$ (antipode), satisfying:

1. *Coassociativity:* $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.
2. *Counit:* $(\varepsilon \otimes \mathrm{id}) \circ \Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon) \circ \Delta$.
3. *Antipode:* $\mu_A \circ (S \otimes \mathrm{id}) \circ \Delta = \varepsilon \cdot 1_A = \mu_A \circ (\mathrm{id} \otimes S) \circ \Delta$,
   where $\mu_A : A \otimes_k A \to A$ is the multiplication of A.

The coassociativity and counit axioms say that (A, Δ, ε) is a *coalgebra*; the antipode axiom says that S "undoes" the comultiplication in the same way that inversion undoes multiplication in a group. The key constraint is that Δ and ε must be *algebra* homomorphisms—this encodes the fact that multiplication $\mu : G \times G \to G$ is a morphism of schemes, not merely a set-theoretic map.

Proposition 4.1.10: Affine Group Schemes and Hopf Algebras

The functor $G \mapsto \mathcal{O}(G)$ is an anti-equivalence of categories:

$$\{\text{Affine group schemes over } k\}^{\mathrm{op}} \xrightarrow{\sim} \{\text{Commutative Hopf algebras over } k\}.$$

***Proof*:**
This follows from the anti-equivalence between affine k-schemes and commutative k-algebras. An affine group scheme is a group object in the category of affine k-schemes; dualizing, its coordinate ring becomes a cogroup object in k-**Alg**. The axioms of a cogroup object in a category of commutative algebras are exactly the axioms of a commutative Hopf algebra.

Let us revisit the earlier examples through this lens.

Example 4.1.11: Hopf Algebras

1. *The additive group.* The Hopf algebra of $\mathbb{G}_a = \operatorname{Spec}(k[t])$ is $k[t]$ with $\Delta(t) = t \otimes 1 + 1 \otimes t$, $\varepsilon(t) = 0$, and $S(t) = -t$. In the language of coalgebras, t is a *primitive* element: $\Delta(t) = t \otimes 1 + 1 \otimes t$.

2. *The multiplicative group.* The Hopf algebra of $\mathbb{G}_m = \operatorname{Spec}(k[t, t^{-1}])$ is $k[t, t^{-1}]$ with $\Delta(t) = t \otimes t$, $\varepsilon(t) = 1$, and $S(t) = t^{-1}$. Here t is a *group-like* element: $\Delta(t) = t \otimes t$.

3. *The general linear group.* The Hopf algebra of GL_n is $k[x_{ij}, \det^{-1}]$ with $\Delta(x_{ij}) = \sum_k x_{ik} \otimes x_{kj}$ (matrix multiplication), $\varepsilon(x_{ij}) = \delta_{ij}$ (the identity matrix), and $S(x_{ij})$ given by the (i,j)-cofactor of (x_{ij}) divided by det—this is the Cramer's rule formula for the entries of the inverse matrix.

4. *Roots of unity.* The Hopf algebra of μ_n is $k[t]/(t^n - 1)$ with the same comultiplication as $\mathbb{G}_m$. The quotient by the ideal $(t^n - 1)$ is compatible with the Hopf structure because $\Delta(t^n - 1) = t^n \otimes t^n - 1 \otimes 1 = (t^n - 1) \otimes t^n + 1 \otimes (t^n - 1)$, which lies in the ideal generated by $t^n - 1$ in $A \otimes A$. In Hopf algebra language, $(t^n - 1)$ is a *Hopf ideal*.

The Hopf algebra viewpoint makes certain constructions transparent. For instance, a *homomorphism* of group schemes $\varphi : G \to H$ corresponds to a homomorphism of Hopf algebras $\varphi^* : \mathcal{O}(H) \to \mathcal{O}(G)$ (respecting Δ, ε, and S). The kernel of φ is the group scheme whose Hopf algebra is $\mathcal{O}(G)/(\ker \varepsilon_H \cdot \mathcal{O}(G))$; we will formalize this in section 4.2.

4.1.12 Remark: non-commutative Hopf algebras If we drop the requirement that A be commutative, we obtain the notion of a *non-commutative Hopf algebra*, which corresponds to a "quantum group" rather than a group scheme. The theory of quantum groups, developed by Drinfeld and Jimbo, is a rich subject in its own right but lies outside our scope. In this text, all Hopf algebras are commutative (corresponding to the fact that the coordinate ring of an affine scheme is always commutative), though the group schemes themselves need not be.

4.1.2 Finite and Infinitesimal Group Schemes

Over an algebraically closed field of characteristic zero, the theory of affine algebraic groups is largely controlled by Lie theory: connected algebraic groups are smooth, their structure is determined by their Lie algebras, and finite group schemes are just finite groups viewed as zero-dimensional varieties. In characteristic $p > 0$, the situation is fundamentally richer: there exist group schemes that are "purely infinitesimal"—non-reduced, supported at a single point, yet carrying nontrivial group structure encoded in the nilpotent elements of their coordinate ring.

Definition 4.1.13: Finite Group Scheme

A group scheme G over a field k is *finite* if it is of the form $G = \operatorname{Spec}(A)$ where A is a finite-dimensional k-algebra. The *order* of G is $\dim_k A$.

A finite group scheme of order n has at most n geometric points, but may have fewer if A has nilpotents. The extreme cases are:

Étale group schemes. A finite group scheme is *étale* if its coordinate ring A is a product of separable field extensions of k. Over an algebraically closed field, an étale group scheme of order n is simply a disjoint union of n copies of $\mathrm{Spec}\,(k)$, and the group structure is that of an ordinary finite group. The constant group scheme $(\mathbb{Z}/n\mathbb{Z})_k$ is the prototypical example.

Infinitesimal (or connected) group schemes. A finite group scheme $G = \mathrm{Spec}\,(A)$ is *infinitesimal* if A is a local ring, i.e., G is supported at a single point. The group structure is then entirely encoded in the nilpotent elements of A. These exist only in positive characteristic (or over non-reduced bases).

We have already seen the simplest examples of both types.

Example 4.1.14: μ_p in Characteristic p

As computed in Example 4.1.6, over $\mathbb{F}_p$ the group scheme $\mu_p = \mathrm{Spec}\,(\mathbb{F}_p[t]/(t^p - 1)) = \mathrm{Spec}\,(\mathbb{F}_p[s]/(s^p))$ (where $s = t - 1$) is infinitesimal of order p. Its Hopf algebra structure (expressed in terms of s) is:

$$\Delta(s) = s \otimes 1 + 1 \otimes s + s \otimes s, \qquad \varepsilon(s) = 0, \qquad S(s) = \sum_{i=1}^{p-1} (-1)^i s^i.$$

The comultiplication is neither purely primitive ($s \otimes 1 + 1 \otimes s$) nor purely group-like ($s \otimes s$)—it is a mixture, reflecting the fact that μ_p is a "twisted" infinitesimal group.

Example 4.1.15: The Frobenius Kernel α_p

The group scheme $\alpha_p = \mathrm{Spec}\,(k[t]/(t^p))$, defined over a field k of characteristic $p > 0$, is the kernel of the Frobenius endomorphism on $\mathbb{G}_a$. Its Hopf algebra is $k[t]/(t^p)$ with

$$\Delta(t) = t \otimes 1 + 1 \otimes t, \qquad \varepsilon(t) = 0, \qquad S(t) = -t.$$

Note that t is primitive, just as in $\mathbb{G}_a$, but the relation $t^p = 0$ (rather than no relation) makes α_p a finite group scheme of order p. For any k-algebra A:

$$\alpha_p(A) = \{a \in A \mid a^p = 0\},$$

which is a subgroup of $(A, +)$. Over a perfect field, α_p has only the trivial k-point, yet it is a non-trivial group scheme: for instance, $\alpha_p(k[\epsilon]/(\epsilon^p)) = k \cdot \epsilon$ is a p-element group.

The schemes α_p and μ_p are both infinitesimal group schemes of order p, but they are *not* isomorphic: α_p is connected and has no non-trivial characters (homomorphisms to $\mathbb{G}_m$), while μ_p is the Cartier dual of $\mathbb{Z}/p\mathbb{Z}$ and detects p-torsion in the multiplicative group.

The contrast between α_p and μ_p illustrates a general structural result. Over a perfect field k of characteristic $p > 0$, every finite group scheme G admits a canonical filtration

$$0 \subseteq G^0 \subseteq G$$

where G^0 is the *connected component of the identity* (the largest connected subgroup scheme) and G/G^0 is étale. This is the *connected-étale sequence*:

$$1 \longrightarrow G^0 \longrightarrow G \longrightarrow G^{\text{ét}} \longrightarrow 1,$$

and over a perfect field this sequence splits (non-canonically), so $G \cong G^0 \times G^{\text{ét}}$. The connected part G^0 is an infinitesimal group scheme (its coordinate ring is local), and the étale part $G^{\text{ét}}$ is determined by the action of $\mathrm{Gal}(\bar{k}/k)$ on the geometric points $G(\bar{k})$.

This decomposition has no analogue in characteristic zero, where every finite group scheme over a field is étale. It is a manifestation of the richer arithmetic of positive characteristic, and plays a fundamental role in the theory of abelian varieties (where the p-torsion $A[p]$ of an abelian variety splits into connected and étale parts whose relative sizes determine the p-rank).

The Frobenius and Verschiebung

The characteristic-p phenomena above are organized by two fundamental endomorphisms. Let k be a perfect field of characteristic p, and let G be a group scheme over k.

The *absolute Frobenius* $F_{\text{abs}} : G \to G$ is the identity on the underlying topological space and acts as $a \mapsto a^p$ on the structure sheaf. This is a morphism of schemes but *not* a morphism of k-schemes (it does not commute with the structure map unless $k = \mathbb{F}_p$). To obtain a k-morphism, one passes to the *relative Frobenius* $F : G \to G^{(p)}$, where $G^{(p)}$ is the base change of G along the Frobenius $\operatorname{Spec}(k) \to \operatorname{Spec}(k)$. The relative Frobenius is a homomorphism of group schemes.

For a commutative group scheme, there is a dual morphism $V : G^{(p)} \to G$ called the *Verschiebung* (German for "shift"). It satisfies $V \circ F = [p]_G$ and $F \circ V = [p]_{G^{(p)}}$, where $[p]$ denotes multiplication by p in the group law. The kernels $\ker(F)$ and $\ker(V)$ are finite group schemes that capture, respectively, the "infinitesimal" and "étale" parts of the p-torsion.

For example, $\ker(F : \mathbb{G}_a \to \mathbb{G}_a^{(p)}) = \alpha_p$ (since F acts as $t \mapsto t^p$ on $k[t]$, and $\ker = \operatorname{Spec}(k[t]/(t^p))$), while $\ker(F : \mathbb{G}_m \to \mathbb{G}_m^{(p)}) = \mu_p$.

4.2 Homomorphisms, Kernels, and Exact Sequences

Definition 4.2.1: Homomorphism of Group Schemes

A *homomorphism* of S-group schemes $\varphi : G \to H$ is a morphism of S-schemes that is compatible with the group structures:

$$\varphi \circ \mu_G = \mu_H \circ (\varphi \times \varphi)$$

, $\varphi \circ \epsilon_G = \epsilon_H$, and $\varphi \circ \iota_G = \iota_H \circ \varphi$.

In the functor-of-points language, φ is a homomorphism if and only if for every S-scheme T, the induced map $\varphi(T) : G(T) \to H(T)$ is a group homomorphism.

Definition 4.2.2: Kernel of a Homomorphism

Let $\varphi : G \to H$ be a homomorphism of S-group schemes. The *kernel* of φ is the fiber product

$$\ker(\varphi) = G \times_H S$$

where the two maps to H are $\varphi : G \to H$ and $\epsilon_H : S \to H$.

By the universal property of fiber products, for any S-scheme T:

$$\ker(\varphi)(T) = \{g \in G(T) \mid \varphi(T)(g) = e_H \in H(T)\},$$

which is exactly the kernel of the group homomorphism $\varphi(T) : G(T) \to H(T)$. The kernel $\ker(\varphi)$ is a closed subgroup scheme of G (being a fiber of a morphism over a section).

Example 4.2.3: Classical Exact Sequences

1. *The n-th power map on $\mathbb{G}_m$.* The homomorphism $[n] : \mathbb{G}_m \to \mathbb{G}_m$ defined by $t \mapsto t^n$ has kernel μ_n. This gives a short exact sequence of group schemes:

$$1 \longrightarrow \mu_n \longrightarrow \mathbb{G}_m \xrightarrow{(\ )^n} \mathbb{G}_m \longrightarrow 1.$$

   This is the *Kummer sequence*. Over a field k with $\mathrm{char}(k) \nmid n$, passing to $\overline{k}$-points recovers the familiar exact sequence $1 \to \mu_n(\overline{k}) \to \overline{k}^\times \xrightarrow{n} \overline{k}^\times \to 1$.

2. *The determinant.* The determinant $\det : \mathrm{GL}_n \to \mathbb{G}_m$ is a homomorphism of group schemes with kernel SL_n:

$$1 \longrightarrow \mathrm{SL}_n \longrightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1.$$

3. *The Frobenius on $\mathbb{G}_a$ in characteristic p.* The Frobenius $F : \mathbb{G}_a \to \mathbb{G}_a$ defined by $t \mapsto t^p$ has kernel α_p:

$$0 \longrightarrow \alpha_p \longrightarrow \mathbb{G}_a \xrightarrow{F} \mathbb{G}_a \longrightarrow 0.$$

   This is the *Artin–Schreier sequence* (in its scheme-theoretic form).

A word of caution on exactness is in order. A sequence $1 \to N \xrightarrow{i} G \xrightarrow{\varphi} H \to 1$ of group schemes is said to be *exact* if: i identifies N with $\ker(\varphi)$ as a closed subgroup scheme of G, and φ is faithfully flat. The surjectivity of φ is *not* the naive statement that $\varphi(T) : G(T) \to H(T)$ is surjective for all T; this is often false even when the sequence is exact. For instance, in the Kummer sequence over $\mathbb{Q}$, the map $\mathbb{Q}^\times \xrightarrow{n} \mathbb{Q}^\times$ is not surjective (not every rational number is an n-th power), but the map of schemes $\mathbb{G}_m \xrightarrow{n} \mathbb{G}_m$ is faithfully flat. The correct notion of surjectivity for group schemes is faithfully flat descent, not surjectivity on points.

4.3 Actions of Group Schemes on Schemes

Definition 4.3.1: Action of a Group Scheme

Let G be an S-group scheme and X an S-scheme. An *action* of G on X is an S-morphism

$$\sigma : G \times_S X \longrightarrow X$$

satisfying the usual axioms: the diagram

$$\begin{tikzcd} G \times_S G \times_S X \arrow[r, "\mathrm{id}\times\sigma"] \arrow[d, "\mu\times\mathrm{id}"'] & G \times_S X \arrow[d, "\sigma"] \\ G \times_S X \arrow[r, "\sigma"'] & X \end{tikzcd}$$

commutes (associativity), and $\sigma \circ (\epsilon \times \mathrm{id}_X) = \mathrm{id}_X$ (the identity acts trivially).

In the functor-of-points language, this means: for every S-scheme T, the group $G(T)$ acts on the set $X(T)$, and these actions are compatible with base change.

Example 4.3.2: Standard Actions

1. *Translation.* Any group scheme G acts on itself by left multiplication: $\sigma(g, h) = g \cdot h$. This is the *left regular action.* Each element $g \in G(T)$ defines an automorphism $\lambda_g : G_T \to G_T$ called *left translation by* g.

2. *Conjugation.* G acts on itself by conjugation: $\sigma(g, h) = ghg^{-1}$. This is the *adjoint action* of G on itself. Its derivative at the identity gives the adjoint representation of $\mathfrak{g}$ (the Lie algebra of G) on itself; see section 4.5.

3. *Linear representations.* An action of G on $\mathbb{A}^n_S$ that is $\mathcal{O}_S$-linear corresponds to a homomorphism $G \to \mathrm{GL}_{n,S}$. This is the scheme-theoretic notion of a *linear representation.*

4. $\mathbb{G}_m$*-actions and gradings.* An action of $\mathbb{G}_m$ on an affine scheme $\mathrm{Spec}\,(A)$ corresponds to a $\mathbb{Z}$-grading on A: $A = \bigoplus_{d\in\mathbb{Z}} A_d$. The $\mathbb{G}_m$-equivariant morphisms are the graded ring homomorphisms. This is the algebraic origin of the weight decompositions ubiquitous in representation theory.

Definition 4.3.3: Orbits and Stabilizers

Let G act on X and let $x \in X(T)$ be a T-valued point. The *orbit map* is

$$\sigma_x : G_T \longrightarrow X_T, \qquad g \longmapsto g \cdot x.$$

The *stabilizer* (or *isotropy group*) of x is the fiber product

$$G_x = G_T \times_{X_T} T$$

where the two maps to X_T are $\sigma_x : G_T \to X_T$ and $x : T \to X_T$. For any T-scheme T':

$$G_x(T') = \{g \in G(T') \mid g \cdot x = x \in X(T')\}.$$

The stabilizer is a closed subgroup scheme of G_T.

4.4 Quotients by Group Actions

Having defined actions, we now ask: given an action $\sigma : G \times_S X \to X$, does the "quotient" X/G exist as a scheme? This question is considerably more subtle than its set-theoretic counterpart, and its resolution leads to geometric invariant theory (GIT).

The categorical quotient

The most basic notion is that of a *categorical quotient*. We want a scheme Y and a G-invariant morphism $\pi : X \to Y$ (meaning $\pi \circ \sigma = \pi \circ \mathrm{pr}_2$, where $\mathrm{pr}_2 : G \times X \to X$ is the second projection) that is universal: any G-invariant morphism $X \to Z$ factors uniquely through π.

Definition 4.4.1: Categorical and Geometric Quotients

Let G act on X over S. A *categorical quotient* is an S-scheme Y with a G-invariant morphism $\pi : X \to Y$ such that:

1. (Universal property) For every G-invariant morphism $f : X \to Z$, there exists a unique morphism $\overline{f} : Y \to Z$ with $f = \overline{f} \circ \pi$.
2. (Coinvariant ring, in the affine case) If $X = \mathrm{Spec}\,(A)$ and G acts on A by ring automorphisms, then $Y = \mathrm{Spec}\,(A^G)$ where A^G denotes the subring of G-invariants.

A categorical quotient $\pi : X \to Y$ is a *geometric quotient* if additionally:

(3) π is surjective,

(4) the fibers of π are exactly the G-orbits, and

(5) π is submersive (a subset $U \subseteq Y$ is open if and only if $\pi^{-1}(U)$ is open in X).

A geometric quotient, when it exists, is the best possible outcome: the scheme Y faithfully represents the orbit space. However, geometric quotients frequently fail to exist—the simplest example being the action of $\mathbb{G}_m$ on $\mathbb{A}^2$ by scalar multiplication, where the origin is a "bad" orbit (it is the only closed orbit of dimension less than 1).

The GIT approach

Geometric invariant theory, developed by Mumford, provides a systematic method for constructing quotients by reductive group actions on projective or quasi-projective varieties. The central idea is to work not with the full quotient, but with a *good quotient* obtained by restricting to "semistable" points.

Let G be a reductive group acting on a projective variety $X \subseteq \mathbb{P}^n$ via a linearization (an action that lifts to the homogeneous coordinate ring). The *semistable locus* X^{ss} consists of those points $x \in X$ for which there exists a G-invariant section $f \in \Gamma(X, \mathcal{O}_X(d))^G$ (for some $d > 0$) with $f(x) \neq 0$. The *stable locus* $X^s \subseteq X^{ss}$ consists of those semistable points with finite stabilizer and closed orbit in X^{ss}.

The fundamental theorem of GIT asserts that the categorical quotient $X^{ss} \to X^{ss}/\!/G$ exists as a projective variety, and its restriction to the stable locus $X^s \to X^s/G$ is a geometric quotient. The notation "$/\!/$" (double slash) is reserved for the GIT quotient to distinguish it from the naive set-theoretic quotient.

A thorough treatment of GIT lies outside the scope of this text, but we record the most important special case for future reference.

Proposition 4.4.2: Quotient by a Finite Group

Let G be a finite group acting on an affine scheme $X = \operatorname{Spec}(A)$ by ring automorphisms. Then the categorical quotient $X/G = \operatorname{Spec}(A^G)$ exists, and the morphism $\pi : X \to X/G$ is finite and surjective. If $|G|$ is invertible in A, then π is a geometric quotient and $X \to X/G$ is étale over the locus of points with trivial stabilizer.

***Proof*:**
The ring of invariants A^G is a subring of A, and A is integral over A^G by the theorem on integral extensions (each $a \in A$ satisfies the monic polynomial $\prod_{g \in G}(T - g \cdot a) \in A^G[T]$). Hence $\pi : \operatorname{Spec}(A) \to \operatorname{Spec}(A^G)$ is finite and surjective. The universal property of $\operatorname{Spec}(A^G)$ as a categorical quotient follows from the fact that G-invariant maps $A \to B$ factor uniquely through A^G. When $|G|$ is invertible, the Reynolds operator $R = \frac{1}{|G|}\sum_{g \in G} g$ is a projection $A \to A^G$, and the étaleness statement follows from the Jacobian criterion applied to the map π away from the ramification locus.

4.5 The Lie Algebra of a Group Scheme

In the theory of Lie groups, the tangent space at the identity carries the structure of a Lie algebra, and this infinitesimal invariant determines the local structure of the group. The same construction works for group schemes, using the Zariski tangent space from section 2.8. (In the following section, we will see that this tangent space admits a clean reformulation via dual-number valued points; see Example 3.3.5.)

Let G be a group scheme of finite type over a field k, with identity element $e \in G(k)$. The Zariski tangent space at e is:

$$T_eG = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$.

Definition 4.5.1: Lie Algebra of a Group Scheme

The *Lie algebra* of G is the tangent space $\mathfrak{g} = T_eG$, equipped with a k-linear bracket $[\ ,\ ] : \mathfrak{g} \otimes_k \mathfrak{g} \to \mathfrak{g}$ defined via the adjoint action (specified below). We write $\mathfrak{g} = \mathrm{Lie}(G)$.

To define the bracket, observe that the group structure on G induces a group structure on the set of dual-number points:

$$G(k[\epsilon]/(\epsilon^2))$$

is a group via the multiplication μ. The kernel of the natural map $G(k[\epsilon]/(\epsilon^2)) \to G(k)$ (induced by $\epsilon \mapsto 0$) is exactly $\mathfrak{g}$, and it sits in a split short exact sequence of groups:

$$0 \longrightarrow \mathfrak{g} \longrightarrow G(k[\epsilon]/(\epsilon^2)) \longrightarrow G(k) \longrightarrow 1.$$

In this sequence, $\mathfrak{g}$ is an *abelian* subgroup (the product of two tangent vectors $v, w \in \mathfrak{g}$ under the group law of $G(k[\epsilon]/(\epsilon^2))$ is $v + w$, since the quadratic terms in ϵ vanish). However, the *commutator* of elements in $G(k[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2, \epsilon_1\epsilon_2))$ is well-defined and lands in $\mathfrak{g}$; this commutator defines the Lie bracket $[v, w]$.

More concretely, the adjoint action $\mathrm{ad} : G \to \mathrm{Aut}(\mathfrak{g})$ is defined by conjugation: for $g \in G(T)$ and $v \in \mathfrak{g} \otimes_k \mathcal{O}(T)$, set $\mathrm{ad}(g)(v) = gvg^{-1}$ (computed in $G(T[\epsilon]/(\epsilon^2))$). Differentiating ad at the identity gives a map $\mathfrak{g} \to \mathrm{End}_k(\mathfrak{g})$, and the Lie bracket is $[v, w] = (\mathrm{d}_e\mathrm{ad})(v)(w)$.

Proposition 4.5.2: Lie Algebra as Primitive Elements

Let $G = \mathrm{Spec}\,(A)$ be an affine group scheme over k with Hopf algebra (A, Δ, ε) and augmentation ideal $I = \ker(\varepsilon)$. Then:

$$\mathfrak{g} = (I/I^2)^\vee$$

as a k-vector space, and the Lie bracket corresponds to the commutator in the convolution algebra $\mathrm{Hom}_k(A, k)$. Equivalently, $\mathfrak{g}$ is the space of *primitive elements* of the dual coalgebra: those $\delta \in A^\vee$ satisfying $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$ for all $a, b \in A$ (i.e., δ is a derivation $A \to k$ at ε).

***Proof*:**
By the Hopf algebra structure, the augmentation ideal $I = \ker(\varepsilon)$ is the maximal ideal at the identity, so $(I/I^2)^\vee$ is the Zariski tangent space at e as expected. A k-linear functional $\delta : A \to k$ vanishing on $k \cdot 1$ corresponds to an element of $I^\vee$; it descends to $(I/I^2)^\vee$ if and only if $\delta(I^2) = 0$, which is equivalent to the derivation (Leibniz) rule $\delta(ab) = \varepsilon(a)\delta(b) + \delta(a)\varepsilon(b)$.

Example 4.5.3: Lie Algebras of Classical Group Schemes

1. $\mathbb{G}_a$*:* The augmentation ideal of $k[t]$ (with $\varepsilon(t) = 0$) is (t), and $(t)/(t^2) \cong k$. Thus $\mathrm{Lie}(\mathbb{G}_a) = k$ with the zero bracket—the Lie algebra is one-dimensional and abelian.

2. $\mathbb{G}_m$*:* The augmentation ideal of $k[t, t^{-1}]$ (with $\varepsilon(t) = 1$) is $(t-1)$. Setting $s = t-1$, we have $(s)/(s^2) \cong k$, so $\mathrm{Lie}(\mathbb{G}_m) = k$ with the zero bracket—again one-dimensional and abelian.

3. GL_n*:* The augmentation ideal of $k[x_{ij}, \det^{-1}]$ (with $\varepsilon(x_{ij}) = \delta_{ij}$) is generated by the elements $e_{ij} := x_{ij} - \delta_{ij}$. The quotient I/I^2 is freely generated by the images of the e_{ij}, so $\mathrm{Lie}(\mathrm{GL}_n) \cong M_n(k)$, the space of all $n \times n$ matrices. The Lie bracket is the matrix commutator $[A, B] = AB - BA$, which recovers the classical Lie algebra $\mathfrak{gl}_n$.

4. SL_n*:* The additional relation $\det(x_{ij}) = 1$ imposes the linear condition $\mathrm{tr}(e_{ij}) = 0$ at the level of I/I^2. Hence $\mathrm{Lie}(\mathrm{SL}_n) = \{A \in M_n(k) : \mathrm{tr}(A) = 0\} = \mathfrak{sl}_n$.

5. μ_n *over a field with* $\mathrm{char}(k) \nmid n$*:* The augmentation ideal of $k[t]/(t^n - 1)$ (with $\varepsilon(t) = 1$) is $I = (t-1)$. Writing $t^n - 1 = (t-1)(t^{n-1} + \cdots + 1)$, and noting that $t^{n-1} + \cdots + 1$ evaluates to $n \neq 0$ at $t = 1$ (hence is coprime to $t-1$), we find that $(t-1)^2$ and $t^n - 1$ together generate $(t-1)$, so $I/I^2 = 0$. Thus $\mathrm{Lie}(\mu_n) = 0$—the Lie algebra is trivial. This reflects the fact that μ_n is étale: it has no infinitesimal structure.

6. μ_p *in characteristic* p*:* Now $t^p - 1 = (t-1)^p$, so the coordinate ring is $k[s]/(s^p)$ with $s = t-1$, and $I/I^2 = (s)/(s^2) \cong k$. Hence $\mathrm{Lie}(\mu_p) = k$ is one-dimensional. The infinitesimal group scheme μ_p carries tangent information at its unique point, unlike its étale counterpart.

The contrast between the last two examples is worth emphasizing: μ_n for $\mathrm{char}(k) \nmid n$ is étale with trivial Lie algebra, while μ_p in characteristic p is infinitesimal with a one-dimensional Lie algebra. The Lie algebra "sees" the infinitesimal structure of the group scheme, and in characteristic p this structure is genuinely richer.

This observation foreshadows a general theme: in characteristic zero, the Lie algebra functor $G \mapsto \mathrm{Lie}(G)$ is an equivalence between connected algebraic groups and their Lie algebras (this is the scheme-theoretic incarnation of Lie's third theorem). In characteristic p, the Lie algebra loses information—it cannot distinguish between α_p and $\mathbb{G}_a$, for instance, since both have Lie $= k$. The correct replacement is the theory of *p-Lie algebras* (or *restricted Lie algebras*), which equips $\mathfrak{g}$ with an additional "p-th power" operation $x \mapsto x^{[p]}$ that captures the Frobenius structure. This theory, while important for the classification of algebraic groups in positive characteristic, lies beyond our present scope.

5

Schemes III: Sheaves of Modules – Projectives, Divisors, Differentials

So far, we have only worked with the structure sheaf $\mathcal{O}_X$ over a scheme X; we can put other sheaves that shall give use many important insights, similarity to how different [vector, principal G-] bundles on manifolds provide a lot of useful information. All of these sheaves shall have an $\mathcal{O}_X$-action on them, similarly to sections of bundles have an $C^\infty(M)$-action (or $G \subseteq \mathrm{GL}_r(K)$-action) on them, making them the generalization of modules to ringed spaces. The most important of these sheaves shall be ones that are locally isomorphic to sheaves corresponding to R_i-modules M_i. Such sheaves shall be called quasi-coherent sheaves, and if they "nicely" are finitely generated they shall be called coherent shaves[1]. The term "coherent" comes from Serre who saw these sheaves as "cohering" (i.e. sticking together). The additional "quasi-" for non-finitely generated modules comes from loosing some results when omitting finite generation (for example, they are not in general preserved under pushforward).

Quasi-coherent sheaves can be interpreted as generalization of bundles. In particular, *locally free sheaves* (over X) with sheaf morphisms shall be categorically equivalent to the vector bundles (over X) with bundle morphisms. Locally free sheaves are not closed under kernels and cokernels, and hence the same holds for bundles (see [8, chapter 1] on the detail of how to form sub/quotient bundles). The category of quasi-coherent sheaves can be constructed by adding the kernels and cokernels of locally free sheaves (similarly to how we can form projective/injective resolutions of non-free modules), and so one way of thinking about quasi-coherent sheaves is that they are the natural smallest extension of the category of vector bundles to form an abelian category; all of this shall be demonstrated in section 5.1.2.

By working with quasi-coherent sheaves, we shall be able to define ideal sheaves, image schemes, twisted sheaves on projective space (which leads to recovering $S_\bullet$ from $\operatorname{Proj} S_\bullet$), sheaves remembering

[1] there are important technicalities on what "nicely" means that will be developed in this section

information at just some points point with multiplicity (the natural generalization skyscraper sheaves from example 1.1.12), and differentials (from which we recover forms and tangent bundles), among many more new interesting sheaves on schemes that will give lots more information on schemes.

5.1 Definition and Basic Properties

Let us start by defining the module-equivalent for ringed spaces:

Definition 5.1.1: $\mathscr{O}_X$-Module

Let $\mathscr{O}_X \in \mathbf{Sh}_{\mathbf{Ring}}(X)$. Then an $\mathscr{O}_X$*-module* on X is a sheaf $\mathscr{F} \in \mathbf{Sh}_{\mathbf{Ab}}(X)$ that has the following data:

1. For each open $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-module
2. if $U \subseteq V$, then the following diagram commutes:

$$\begin{tikzcd} \mathscr{O}_X(V) \times \mathscr{F}(V) \arrow[r, "\curvearrowright"] \arrow[d, "\mathrm{res}_{V,U} \times \mathrm{res}_{V,U}"'] & \mathscr{F}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\ \mathscr{O}_X(U) \times \mathscr{F}(U) \arrow[r, "\curvearrowright"] & \mathscr{F}(U) \end{tikzcd}$$

If $\mathscr{F}, \mathscr{G}$ are $\mathscr{O}_X$-modules, then a *morphism of* $\mathscr{O}_X$*-modules* $\varphi : \mathscr{F} \to \mathscr{G}$ is a sheaf-morphism such that for each open $U \subseteq X$, $\mathscr{F}(U) \to \mathscr{F}(X)$ is a $\mathscr{O}_X(U)$-module homomorphism that commutes with restrictions. Their collection is often denoted $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ or sometimes just $\mathrm{Hom}_X(\mathscr{F}, \mathscr{G})$ if it is unambiguous.

These should really be thought of as fibre bundles or even vector bundles on X. We define the pushforward and pullback of $\mathscr{O}_X$-modules on ringed spaces for future cohomological purposes:

Definition 5.1.2: $\mathscr{O}_X$-module Pushforward and Pullback

let $\left(f, f^{\#}\right) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a ringed space morphism, and $\mathscr{F}$ is a $\mathscr{O}_X$-module on X. Then $f_*\mathscr{F}$ is a $f_*\mathscr{O}_X$-module on Y. Via the map $f^{\#} : \mathscr{O}_Y \to f_*\mathscr{O}_X$, we see that this has a natural $\mathscr{O}_Y$-module structure, and it is called the direct image of $\mathscr{F}$ via f. We can do a similar chain of reasoning for $f^{-1}\mathscr{O}_Y$ creating the inverse image of $\mathscr{F}$ which is adjoint to the direct image.

Example 5.1.3: General $\mathscr{O}_X$-Modules Constructions

Fix a locally ringed space $(X, \mathscr{O}_X)$

1. Take $\mathscr{F} = \mathscr{O}_X$ with action on each section being defined by multiplication. Then then this $\mathscr{O}_X$-module as any ringed space $\mathscr{O}_X$ is a $\mathscr{O}_X$-module over itself. As each restriction map is a ring-homomorphism, it is naturally $\mathscr{O}_X(U)$-module homomorphism, and hence the action is preserved.
2. Just like how every abelian group is a $\mathbb{Z}$-module, every sheaf of abelian groups $\mathcal{G}$ is a $\underline{\mathbb{Z}}$-module

(where $\underline{\mathbb{Z}}$ is the constant sheaf associated to $\mathbb{Z}$), showing how this is a natural generalization.

3. If $\varphi : \mathscr{F} \to \mathscr{G}$ is a $\mathcal{O}_X$-module homomorphism, then the kernel, image, and cokernel are all $\mathcal{O}_X$-modules (generalizing the result form the case of sheaf morphisms). It is also closed under products and coproducts, making $\mathcal{O}_X$-modules an abelian category.

4. For cohomological purposes, we can always define the sheaf-hom $\mathcal{O}_X$ module given by

$$U \mapsto \operatorname{Hom}_{\mathcal{O}_x|_U}\left(\mathscr{F}|_U, \mathscr{G}|_U\right)$$

where $\mathcal{O}_X|_U$ is given the natural restriction to a $\mathcal{O}_X|_U$-module on U.

5. The tensor product $\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G}$ is the sheafification of the presheaf given by $U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. The sheafification is necessary and shall in fact cause very non-trivial behaviour, namely it shall not be the case that:

$$(\mathscr{F} \otimes \mathscr{G})(U) \cong \mathscr{F}(U) \otimes \mathscr{G}(U)$$

and in fact it can go spectacularly wrong, see example 5.2.13

5.1.1 Quasi-Coherent and Coherent Sheaves

A natural question then is whether the category R-Mod and the category of $\mathcal{O}_X$-Mod are equivalent when $X = \operatorname{Spec} R$. Just liked **RingedSp** and **CRing**, there are in fact too many $\mathcal{O}_X$-modules:

Example 5.1.4: Too Many $\mathcal{O}_X$-modules Over Affine Schemes

Let's say there was this correspondence so that for an R-module M, $\widetilde{M}$ is the correspondence sheaf. For this correspondence to work, we would need that $\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}}$. Let us show we can find a sheaf that does not have this property.

Let $X = \operatorname{Spec} \mathbb{Z}_{(2)}$ which consists of two (non-trivial) open sets: $\{(2), \eta\}$ and $\{\eta\}$. We have the corresponding $\mathcal{O}_X(X) = \mathbb{Z}_{(2)}$ and $\mathcal{O}_X(\{\eta\}) = \mathbb{Q}$. To define the $\mathcal{O}_X$-module on X, we shall need to define a pair of modules, an $\mathbb{Z}_{(2)}$-module M and a $\mathbb{Q}$-module N, along with a restriction module homomorphism:

$$M \to N$$

Here is a combo that works:

$$M = \mathbb{Z}_{(2)} \quad N = 0 \qquad M \twoheadrightarrow N$$

However, this cannot come from a module. Notice that:

$$M_{(0)} = M\left[\frac{1}{2}\right]$$

But the above implies that $\widetilde{M}_{(0)} = 0$.

To fix this, we require that our $\mathcal{O}_X$-module must locally look like sheaf modules. Let us start with defining the "affine components", namely the sheaf modules over $X = \operatorname{Spec} R$. Then given any

R-module M, we shall want an $\mathcal{O}_X$-modules $\widetilde{M}$ that locally looks like modules with a structure sheaf. We saw that every ring R naturally a sheaf $\mathcal{O}_{\operatorname{Spec} R}$ on $\operatorname{Spec} R$. In the same vein, every R-module is naturally a sheaf $\mathcal{O}_{\operatorname{Spec} R\text{-module}}$ on $\operatorname{Spec} R$:

Definition 5.1.5: Sheaf Associated to M

Let R be a ring and M and R-module. Then the *sheaf associated to* M on $\operatorname{Spec} R$ is a sheaf given by the basis:

$$\widetilde{M}(D(f)) = M_f \qquad (\text{an } R_f\text{-module})$$

that is, the localization of module M at the ring element f, and define the natural restriction maps. Denote this $\mathcal{O}_{\operatorname{Spec} R}$-module $\widetilde{M}$.

The judicious reader may check that this is indeed a sheaf on Spec R. As we have now introduced another sheaf structure on $\operatorname{Spec} R$, we will start being conscientious of the representation of the local and global sections of the sheaf being used and write

$$\Gamma(\operatorname{Spec} R, \widetilde{M})$$

More generally for a scheme X, denote $\Gamma(X, \mathscr{F})$ for the global sections of the sheaf $\mathscr{F}$ on X. Local section can be given by $\Gamma(U, \mathscr{F})$ for $U \subseteq X$. The usual $\mathcal{O}_X(X)$ notation for a structure sheaf can be written as $\Gamma(X, \mathcal{O}_X)$.

Definition 5.1.6: Quasi-coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. Then a $\mathcal{O}_X$-module $\mathscr{F}$ is called *quasi-coherent* if X can be covered by open affine subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists an R_i-module M_i such that:

$$\mathscr{F}|_{U_i} \cong \widetilde{M_i}$$

The category of quasi-coherent sheaves on a scheme X with locally-ringed space morphisms is denoted $\mathbf{QCoh}(X)$.

We shall also have the finitely generated equivalent of the above, however we must be a bit careful as the kernel of between two finitely generated R-modules need not be finitely generated, for example take the ring homomorphism between these modules over themselves

$$k[x_1, x_2, \dots] \to k[x_1, x_2, \dots] \quad x_i \to 0$$

for all $i \in \mathbb{N}$. Then the kernel is $(x_1, x_2, \dots)$ which is not finitely generated. The reader could check that the image and cokernel are always finitely generated, hence this is specifically a kernel problem. We simply force the required condition

Definition 5.1.7: Coherent Module

Let M be an R-module. Then if R is finitely generated and all kernel of maps $\operatorname{Hom}_R(R^n, M)$ are finitely generated for all $n \in \mathbb{N}$, then M is called *coherent*

A ring is called *coherent* if it is a coherent R-module over itself.

Note how this is stronger than being finitely presented that simply requires a choice of n, m such that $R^n \to R^m \to M$ is exact. If R is Noetherian, then the property of being coherent, finitely presented, and finitely generated are all equivalent.

Definition 5.1.8: Coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. Then a $\mathcal{O}_X$-module $\mathscr{F}$ is called *coherent* if X can be covered by open affine subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists a coherent R_i-module M_i such that:

$$\mathscr{F}|_{U_i} \cong \widetilde{M_i}$$

For the reader interested in complex geometry, coherent and quasi-coherent sheaves can be defined on ringed spaces, and this is done in particular by complex geometers (and in fact the terminology originates in complex geometry).

5.1.9 Distinction from Hartshorne

In Hartshorne, coherent sheaves are defined as sheaves over finitely generated modules, but then Hartshorne says they are a bit pathological and so he always imposes the Noetherian condition, which for us is a sufficient condition for a module to be coherent.

We will immediately show that quasi-coherence on one open cover implies quasi-coherence on any open cover:

Proposition 5.1.10: Quasi-coherence is Affine-local

Let $(X, \mathcal{O}_X)$ be a scheme. Then an $\mathcal{O}_X$-module $\mathscr{F}$ is quasi-coherent if and only if for every open affine subset $U = \operatorname{Spec} R \subseteq X$, there is an R-module M such that

$$\mathscr{F}|_U \cong_R \widetilde{M}$$

If X is coherent, then we can replace quasi-coherent with coherent and M is a coherent R-module

***Proof*:**
As $X = \bigcup_i U_i$ can be covered in affine open sets and each of these are covered by distinguished open sets, we may reduce to proving this in the case of $X = \operatorname{Spec} R$. To that end, let $M = \Gamma(\operatorname{Spec} R, \mathscr{F})$. We need to show there is an isomorphism $\varphi : \widetilde{M} \to \mathscr{F}$. As $\mathscr{F}$ is quasi-coherent, for each $g_i \in X$ we have:

$$\mathscr{F}|_{D(g_i)} \cong \widetilde{M_i}$$

for some R_{g_i}-module M_i. But then

$$\mathscr{F}(D(g_i)) \cong M_{g_i}$$

and so $M_i = M_{g_i}$, and so by corollary 1.3.4 we get that φ is an isomorphism.

By simply adding the conditions of X being Noetherian and M finitely generated we get the corresponding strengthening.

Hence, we have the equivalence of the correspondence between rings and affine schemes for [quasi] coherent sheaves:

Corollary 5.1.11: Equivalence of Categories: Module Version

If $X = \operatorname{Spec} R$, then the functor $M \mapsto \widetilde{M}$ is an equivalence of categories between R-modules and quasi-coherent $\mathcal{O}_X$-modules with inverse function $\mathscr{F} \mapsto \Gamma(\operatorname{Spec} R, \mathscr{F})$:

$$\mathbf{QCoh}(\operatorname{Spec}(R)) \underset{\widetilde{(-)}}{\overset{\Gamma(-)}{\rightleftarrows}} R\text{-Mod}^{\text{op}}$$

If R is Noetherian, then the equivalence is between finitely generated R-modules and coherent $\mathcal{O}_X$-modules.

Furthermore, there is an Adjunction:

$$\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \mathscr{F}) = \operatorname{Hom}_R(M, \Gamma(X, \mathscr{F}))$$

***Proof*:**
Follows immediately

Example 5.1.12: Quasi-Coherent Sheaves

1. Certainly, the usual structure sheaf $\mathcal{O}_X$ on X is quasi-coherent (even coherent). At each point $[\mathfrak{p}]$, we have the $R_\mathfrak{p}$-module $R_\mathfrak{p} \cong \mathcal{O}_{X,\mathfrak{p}}$, that is every point has the germ as it's stalk. We can think of all of these stalks as being one-dimension vector-spaces (as they are all $R_\mathfrak{p}$-modules over themselves).

2. Let $X = \operatorname{Spec} k[x, y]$ and Take the $k[x, y]$-module $k[x, y]/(f)$ for some irreducible f. Now take a point $[\mathfrak{p}] \in X$. If $f \notin \mathfrak{p}$, then

$$\left(\frac{k[x, y]}{(f)}\right)_\mathfrak{p} \cong 0$$

On the other hand, if $f \in \mathfrak{p}$,

$$\left(\frac{k[x, y]}{(f)}\right)_\mathfrak{p} = \frac{k[x, y]_\mathfrak{p}}{(f)_\mathfrak{p}}$$

where $(f)_\mathfrak{p}$ represents the ideal in $k[x, y]_\mathfrak{p}$; this is the local ring of the curve $V(f)$ at $\mathfrak{p}$. As the fiber of this sheaf at $\mathfrak{p}$ is a one dimensional vector-space, we essentially get an 1-dimensional module at each point. Visually, there is a line-bundle on Spec $k[x, y]/(f) \subseteq \operatorname{Spec} k[x, y]$ which is zero everywhere else.

For a concrete example, if $f = (x)$, and $x \in \mathfrak{p}$ then $(k[y])_\mathfrak{p}$ is just the germ at that point. More generally, let $Y \subseteq X$ be a closed subscheme defined by an ideal $\mathfrak{a} \subseteq R$. Then given the inclusion map $\iota : Y \to X, \iota_* \mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-module (even coherent); show it is isomorphic to $\widetilde{R/\mathfrak{a}}$.

3. Take $X = \operatorname{Spec}\mathbb{Z}$, let $M = \mathbb{Z}/2\mathbb{Z}$ be the usual $\mathbb{Z}$-module, and take $\widetilde{M}$. Then this sheaf has a single non-trivial stalk at 2 for:

$$M_{(2)} \cong \mathbb{Z}/2\mathbb{Z}$$

For any other prime p (including 0):

$$M_{(p)} \cong 0$$

Hence, we can think of this module just being a fibre bundle with only one non-trivial fiber. For another example, take the $\mathbb{Z}$-module $M = \mathbb{Z}/12\mathbb{Z}$. Then we have:

$$M_{(2)} \cong \mathbb{Z}/4\mathbb{Z} \quad M_{(3)} \cong \mathbb{Z}/3\mathbb{Z}$$

and zero everywhere else. We can picture this as a stalk at 2 with "twice" the size of the vector space (note that $\mathbb{Z}/2\mathbb{Z}$ is a vector-space over itself), and the stalk at 3 has just a vector-space.

4. Let M be a torsion R-module. Show that $\widetilde{M}_{(0)} = 0$, that is M is not "spread" across R.

5. Using schemes, we can remember the geometric multiplicities of eigenvalues of linear transformations. Let $\alpha \in \operatorname{End}_{\mathbb{C}}(V)$ be an endomorphism, and let V be the $\mathbb{C}[x]$-module of dimension n where the polynomial action is given by α: $p(x) \cdot v = p(\alpha) \cdot v$. Then as $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[x])$, this is a quasi-coherent sheaf over $\mathbb{A}^1_{\mathbb{C}}$. The stalk at η is zero as the sheaf is torsion. For closed points $\lambda \in \mathbb{A}^1_k$, we have the stalk:

$$\tilde{V}_\lambda = V_{(x-\lambda)}$$

If λ is *not* an eigenvalue, then $T - \lambda I$ is an invertible linear transformation (for $\det(T - \lambda I) \neq 0$), and hence the action $(x - \lambda) \cdot v$ on each v has an inverse $(x - \lambda)^{-1}$. From the perspective of the module, given a $w \in V$, can divide w by $(x - \lambda)$, that is we can pick a vector w' such that $(x - \lambda) \cdot w' = w$, namely $w' = (T - \lambda)^{-1} w$.

Now, let $P(x)$ be the characteristic polynomial of T so that $P(T) = 0$. Then in the $\mathbb{C}[x]_{(x-\lambda)}$-module $V_{(x-\lambda)}$ we can invert any polynomial $q(x)$ where $q(\lambda) \neq 0$. As α is not a root of $P(x)$, $P(x)$ is invertible. But then for any $v \in V$

$$v = 1 \cdot v = (P(x)^{-1} \cdot P(x)) \cdot v = P(x)^{-1} \cdot (P(T) \cdot v)) = P(x)^{-1} \cdot 0 = 0$$

showing the stalk is empty. By linear algebra, we know that $P(x)$ is up to multiplicity the minimal polynomial that kills the entire module, hence if λ is an eigenvalue we cannot repeat this trick with a different polynomial. Hence, the stalks are only at the eigenvalues λ.

Now, we can decompose V into:

$$V \cong V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_m}$$

where

$$V_{\lambda_i} = \{v \in V \big| (x - \lambda_i)^k \cdot v = 0\}$$

When localizing at λ_i, the other pieces are killed, and we are left with:

$$\tilde{M}_{\lambda_i} \cong V_{\lambda_i} \cong \frac{\mathbb{C}[x]}{\langle (x-\lambda_i)^{k_1} \rangle} \oplus \frac{\mathbb{C}[x]}{\langle (x-\lambda_i)^{k_2} \rangle} \oplus \cdots \oplus \frac{\mathbb{C}[x]}{\langle (x-\lambda_i)^{m_r} \rangle}$$

where m_i is the geometric multiplicities of each Jordan block. We can in fact make this sheaf a bit cleaner if we descent to $\operatorname{Spec}(\mathbb{C}[x]/m_T(x))$ where $m_T(x)$ is the minimal polynomial of T; in this case we forget all the zero stalks and focus on the intrinsic data of T.

6. Let $U \subseteq X$ be an open subscheme with inclusion map $\iota : U \to X$. Inspired by exercise 1.4.1, define the sheaf $\iota_!(\mathcal{O}_U)$ by extending $\mathcal{O}_U$ by zero outside of 0 . Then this is an $\mathcal{O}_X$-module, but not in general quasi-coherent! To see this, let X be any integral scheme so that each $\mathcal{O}_X(W)$ is an integral domain. Take $V = \operatorname{Spec} R$ to be an open affine subset of X not contained in U but $U \cap V \neq \emptyset$. Then $\iota_!(\mathcal{O}_U)|_V$ contains no global sections of V, but is not the zero sheaf. But then it cannot be of the form $\widetilde{M}$ for any R-module M.

7. It is important the we took the pushforward in the above example. Let Y be a closed subscheme of X (not necessarily from an ideal). Then $\mathcal{O}_X|_Y$ is not in general quasi-coherent on Y, and usually not even a $\mathcal{O}_Y$-module. Let $X = \operatorname{Spec} k[x,y]$ and Y be the closed subscheme given by $\left(y^2\right)$, so that $Y = \operatorname{Spec} k[x,y]/\left(y^2\right)$. Then in $\mathcal{O}_Y(U)$ where $U \subseteq Y$, we have that the function y is nilpotent: $y^2 = 0$. However, on $\mathcal{O}_X|_Y(U)$ there is no nilpotence! In particular, when restricting, we are restricting to Y as a set! Hence, we see that $\mathcal{O}_X|_Y$ is not even a $\mathcal{O}_Y$-module

8. Some of the simplest quasi-coherent sheaves are free sheaves where $\mathscr{F} \cong \mathcal{O}_X^{\oplus I}$, and locally free sheaves, where there is an open cover such that $\mathscr{F}|_{U_i} \cong \mathcal{O}_X^{\oplus I}$. In fact, we can think of quasi-coherent sheaves as the smallest sub-category of $\mathcal{O}_X$-Mod that is an abelian category and includes locally-free sheaves (show that the category of locally-free sheaves along with $\mathcal{O}_X$-homomorphisms is not abelian, take the trivial line bundle on $\mathbb{A}^1_{\mathbb{C}}$ with coordinate t and multiply it by t). The way we take the closure is simply by adding the kernels and cokernels of of locally-free sheaves, showing that they can be thought of as the "building blocks" of quasi-coherent sheaves.

9. Let us give the scheme equivalent of the Hedgehog (or Hairy Sphere[a]) Theorem. Take $X = \operatorname{Spec} \mathbb{R}[x,y,z]/\left(x^2+y^2+z^2-1\right)$. All the closed points of this spectrum correspond to the sphere, and there are many non-closed points. Labeling this ring as R, take the R-module M to be the submodule of $R \oplus R \oplus R$ of elements (X,Y,Z) where $xX + yY + zZ = 0$ (note that x,y,z here represent the indeterminates of the ring). This module represent all vectors that are orthogonal to the points of the sphere. Then this sheaf will on each open set look like $\mathcal{O}_X \oplus \mathcal{O}_X$ (similar to the sphere example), but it cannot look like it globally (or else its change of basis to $\mathbb{C}$ and then analytification would contradict the hairy ball theorem).

10. As mentioned, quasi-coherent sheaves are the (proper) closure of locally free sheaves, which means there must be non-locally free sheaves. We shall see that for the curve $y^2 - x^3 - x = 0$ in $\mathbb{A}^2_{\mathbb{C}}$ (more generally, for elliptic curve), every nonempty set has a nontrivial invertible sheaves, see section 6.1.1.

11. (If you know some number theory) Let K be a number field and $\mathcal{O}_K$ the ring of integers. Then the fractional ideals form an invertible sheaf. Show that two invertible sheaves that differ by a principal ideal yield the same invertible sheaf.
Let us do a concrete example: Take $R = \mathbb{Z}[\sqrt{-5}]$ which the reader should recall is not a UFD as
$$2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$
It was shown in number theory that $J = (2, 1+\sqrt{-5})$ is not principal. We shall use this to find a non-trivial line bundle. First, if $I \subseteq R$ is principal, it is easy to see that $I \cong_R R$, and this is not the case if I is not principal[b]. Next, $D(2), D(3)$ over R. Over both of these, J becomes principal:
$$D(2) : (2, 1+\sqrt{-5}) = R[1/2] \quad \text{as 2 is now a unit}$$
$$D(3) : (2, 1+\sqrt{-5}) = (1+\sqrt{-5}) \quad \text{as} \quad 2 = \frac{(1+\sqrt{-5})(1-\sqrt{-5})}{3}$$
12. Let $\mathfrak{m}_a \subseteq k[x_1, ..., x_m]$ be the maximal ideal of $a \in \mathbb{A}^n_k$. Interpret the map
$$\mathfrak{m}_a \to \frac{\mathfrak{m}_a}{\mathfrak{m}_a^2}$$
13. Take $\mathbb{A}^1_k$ and let $\mathscr{F}$ be the sky-scrapper sheaf supported at the generic point with (abelian) group $k(t)$. Show this is a quasi-coherent sheaf. Show that this sheaf is not locally free. More generally, let X be an integral Noetherian scheme. Put the constant sheaf $\kappa(X)$ on it. Then this is a quasi-coherent $\mathcal{O}_X$-module, but not in general finitely generated on open subsets (and hence not coherent, unless X is restricted to a point).
14. Let $X = \mathbb{P}^n_k$. Then there is a natural line bundle on X where each point $p \in X$ is associated to the 1-dimensional sub-space in $\mathbb{A}^{n+1}_k$ of the points that make up the equivalence class. We shall later see that all line bundles on $\mathbb{P}^n_k$ will be isomorphic to this line bundle, an n-times tensor product of this line bundle or a n-times tensor product of the dual of the bundle (which can be thought of as the inverse).
15. On affine space $\mathbb{A}^n_k$ locally free $\mathcal{O}_X$-modules must be trivial. It is highly nontrivial to prove (the Quillen-Suslin Theorem, formally known as Serre's Theorem[c]), but it is enlighetning: Every projective module over a polynomial ring is free. This translates to every vector bundle over affine space is globally trivial. We shall define non-trivial quasi-coherent sheaves on $\mathbb{A}^n_k$ by using the kernel of maps.
16. Let $X = \operatorname{Spec} R$ be an and let M be a projective R-module. Recall projective modules over local rings are free. Then the germ $\widetilde{M}_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module (see [9]). From this perspective, projective modules are one of the closest modules to how we usually view non-trivial vector-bundles
17. let $X = \operatorname{Spec} R$ again and let M be a flat R-module. Then taking $\widetilde{M}$, the localization at each point remains flat (see [9]). This can geometrically be interpreted as the fibers of M over X varying smoothly!

[a]Also known more humorously as the Hairy Ball Theorem

[b]Notice that this means we can take $\widetilde{I}$, something we can't do with schemes as ideals are not rings. We shall return to this in ref:HERE

[c]The title got moved as there are enough important things named Serre's Theorem

Let us establish some basic properties of quasi-coherent sheaves. The following proposition is a direct translation of the results from structure sheaves and affine schemes and is boxed here for reference:

Proposition 5.1.13: Properties of Sheaf Module

Let M be an R-module, $X = \operatorname{Spec} R$, $\widetilde{M}$ the associated sheaf of M, and $f : R \to S$ a ring homomorphism with $f^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ Then:

1. The stalks of $\widetilde{M}$ at $\mathfrak{p}$ is isomorphic to $M_\mathfrak{p}$
2. $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$
3. $M \to \widetilde{M}$ is an exact fully faithful functor from R-**Mod** to $\mathcal{O}_{\operatorname{Spec} R}$-**Mod**.
4. If N is another R-module
$$(\widetilde{M \otimes_R N}) \cong \widetilde{M} \otimes_{\mathcal{O}_{\operatorname{Spec} R}} \widetilde{N}$$
5. If $\{M_i\}_{i\in I}$ is a collection of R modules:
$$\widetilde{\left(\bigoplus_{i\in I} M_i\right)} \cong \bigoplus_{i\in I} \widetilde{M_i}$$
6. If N is an S-module,
$$f_*(\widetilde{N}) = \widetilde{{}_RN}$$
where ${}_RN$ is where we consider N with the natural R-module structure given by f. Furthermore, $f_*(-)$ is an exact functor (note: X must be affine)
7. If $\widetilde{N}$ is the associated sheaf of N and $\widetilde{M}$ is some sheaf module, then
$$f^*(\widetilde{M}) = \widetilde{M \otimes_R S}$$

It is important for exactness that $f_*(-)$ is pushing a quasi-coherent sheaf over an *affine scheme*. This fails for non-affine schemes, as we shall see with some of the simplest cases with projective schemes. In chapter 6, we shall find right-derived functors for this map.

***Proof*:**

1. Direct generalization of proposition 2.3 .14
2. Direct generalization of corollary 3.1.5
3. The fully faithful part can be deduced by generalizing corollary 3.1.5. For exactness, recall from [9] that localization is exact.

4. Recall from [9] that $(M \otimes_R N)_f \cong M_f \otimes_{R_f} N_f$.
5. exercise
6. exercise
7. exercise

Let us next show that being 0 on an open subset is almost globally zero, and that it is also relatively easy to find global sections form local section on (distinguished) open subsets:

Lemma 5.1.14: Quasi-coherence and Extending From $D(f)$

Let $X = \operatorname{Spec} R$ be an affine scheme, $f \in R$, $D(f) \subseteq X$, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then:

1. If $s \in \Gamma(\operatorname{Spec} R, \mathscr{F})$ is a global section of $\mathscr{F}$ whose restriction to $D(f)$ is zero, then there exists some $n > 0$ such that $f^n s = 0$
2. If $t \in \mathscr{F}(D(f))$, then or some $n > 0$, $f^n t \in \mathscr{F}(\operatorname{Spec} R)$.

The idea behind (2) is that we can eliminate the denominator to have a globally defined function; think $1/x \in \mathcal{O}_X(\mathbb{A}^1_k \setminus \{0\})$. For part (1), think of this as reflecting the same property as analytic continuation of complex holomorphic functions, but having to take into consideration nilpotent elements. This idea works for "vector bundles" on affine schemes.

***Proof*:**

1. Suppose $s \in \Gamma(\operatorname{Spec} R, \mathscr{F})$ such that $s|_{D(f)} = 0$. Cover $\operatorname{Spec} R$ in finitely many $D(g_i)$ (as $\operatorname{Spec} R$ is quasicompact), and consider $s|_{g_i} = s_i \in M_i = \mathscr{F}(D(g_i))$. Then as $D(f) \cap D(g_i) = D(fg_i)$,
$$\mathscr{F}|_{D(fg_i)} = \widetilde{M_{i\,f}}$$
Then the homomorphism restriction map must imply $s_i = 0 \in (M_i)_f$, which by definition of localization implies $f^{n_i} s = 0$. Doing this for each M_i and taking the $n = \max_i n_i$, by gluing we get that $f^n s = 0$.

2. Let $t \in \mathscr{F}(D(g))$, and like above consider $t_i \in \mathscr{F}(D(fg_i)) = (M_i)_f$. Then by definition, for some $n > 0$ there is some element $t_i \in M_i = \mathscr{F}(D(g_i))$ that restricts to $f^{n_i} t$ on $D(fg_i)$. Again, take $n = \max_i n_i$ to make it independent of choice.

   With n chosen to be large enough, on $D(g_i) \cap D(g_j) = D(fg_ig_j)$, the two sections t_i, t_j restrict to $f^n t$. Hence, by part a there exists an $m > 0$ such that
$$f^{m_{ij}}(t_i - t_j) = 0 \qquad \text{on} \qquad D(g_ig_j)$$
   Naturally take $m = \max_{i,j} m_{ij}$. Now the local sections $f^m t_i \in \mathscr{F}(D(g_i))$ glue to give a global section $s \in \mathscr{F}(\operatorname{Spec} R)$ where $s|_{D(f)} = f^{n+m} t$.

Next, recall that the global section functor need not be exact. In the case where $X = \operatorname{Spec} R$, and we are working with quasi-coherent sheaves, it is exact:

Proposition 5.1.15: Exactness of Global Section with sheaves of Modules

Let $X = \operatorname{Spec} R$ be an and $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ an exact sequence of $\mathcal{O}_X$-modules, where at least $\mathscr{F}'$ is quasi-coherent. Then:

$$0 \to \Gamma(\operatorname{Spec} R, \mathscr{F}') \to \Gamma(\operatorname{Spec} R, \mathscr{F}) \to \Gamma(\operatorname{Spec} R, \mathscr{F}'') \to 0$$

is exact

***Proof*:**
Let $X = \operatorname{Spec} R$ be an affine scheme. By proposition 1.2.16, Γ is left-exact, hence it suffices to show it is right-exact when working with $\mathcal{O}_X$-modules and at least $\mathscr{F}'$ is quasi-coherent, namely we must show the map is surjective.

Let $s \in \Gamma(X, \mathscr{F}'')$. As $\mathscr{F} \to \mathscr{F}''$ is surjective, by exercise 1.2.1[1] there is an open neighborhood $D(f)$ of x such that $s|_{D(f)} \in \mathscr{F}''(D(f))$ lifts to a section $t_x \in \mathscr{F}(D(f))$. I claim now that there exists an appropriate $n > 0$ such that $f^n s \in \Gamma(\mathscr{F}'', X)$ lifts to a global section $t \in \mathscr{F}(X)$.

As X is an affine scheme, it is quasicompact, and so cover it by a finite principal open sets $D(g_i)$, and for g_i $s|_{D(g_i)}$ lifts to a section $t_i \in \mathscr{F}(D(g_i))$. Then, on $D(f) \cap D(g_i) = D(fg_i)$, there is $t_x, t_i \in \mathscr{F}(D(fg_i))$ which both lift $s_{D(fg_i)}$.

$$t_x - t_i \in \mathscr{F}'(D(fg_i))$$

As $\mathscr{F}'$ is quasi-coherent, by lemma 5.1.14[1], there exists some $n_i > 0$ such that $f^{n_i}(t_x - t_i)$ extends to a section:

$$u_i \in \mathscr{F}'(D(g_i))$$

Pick $n = \max_i n_i$. Now, taking the image of u_i (as the function is injective), consider $t_i' = f^n t_i + u_i \in \mathscr{F}(D(g_i))$. Then t_i' is a lifting of $f^n s|_{D(g_i)}$, and

$$t_i'|_{D(g_i)} = f^n t_x|_{D(g_i)}$$

Then on $D(g_i g_i)$, t_i', t_j' both lift $f^n s$, and so $t_i' - t_j' \in \mathscr{F}'(D(g_i g_j))$. As $t_i'|_{D(fg_ig_j)} = t_j'|_{D(fg_ig_j)}$, by lemma 5.1.14[2] there exists an m_{ij} such that $f^{m_{ij}}(t_i' - t_j') = 0$ on $D(g_i g_j)$. As usual, take $m = \max_{i,j} m_{ij}$. Then the section $f^m t_i' \in \mathscr{F}(D(g_i))$ glue to a global section $t'' \in \mathscr{F}(X)$, which is a lift of $f^{n+m}s$.

Now, to show each s has a nonempty pre-image. As X is quasicompact, cover X by $D(f_1), ..., D(f_r)$. Then we know that each $s|_{D(f_i)} \in \mathscr{F}''(D(f_i))$ lifts to a section in $\mathscr{F}(D(f_i))$. Then by what was just shown there exists a global section $t_i \in \mathscr{F}(X)$ such that t_i is a lift of $f_i^n s$ where n is the maximum value of all lifts.

Now, As $D(f_i)$ cover X, $(f_1^n, ..., f_r^n) = R$, so let

$$1 = \sum_i a_i f_i^n$$

Take $t = \sum_i a_i t_i \in \mathscr{F}(X)$. Then the image t is

$$\sum_i a_i f_i^n s = s$$

showing each s has a nonempty preimage, as we sought to show.

The reader could think of the above result as something special about affine schemes. In chapter 6 we shall see that it is exactly the non-affine schemes that will have a non-exact global-section function (over quasi-coherent sheaves), see theorem 5.1.17

Proposition 5.1.16: All Objects Are Quasi-Coherent

Let X be a scheme. Then

1. the kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent.
2. Any extension of quasi-coherent by quasi-coherent is quasi-coherent.

If X is Noetherian, the same is true for coherent. If $f : X \to Y$ is a morphism of schemes then:

1. If $\mathscr{G}$ is a quasi-coherent $\mathcal{O}_Y$-module, then $f^*\mathscr{G}$ is a quasi-coherent $\mathcal{O}_X$-module
2. If X, Y and Noetherian, $\mathscr{G}$ is coherent,then $f^*\mathscr{G}$ is coherent
3. If either X is Noetherian or f is quasicompact and quasiseparated, then if $\mathscr{F}$ is quasi-coherent, $f_*\mathscr{F}$ is quasi-coherent.

sheaves.

Proof:
As shown in chapter 1, these properties can be shown locally, so it suffices to assume X is affine. Then as $M \mapsto \widetilde{M}$ is exact and fully faithful, kernels, cokernels, and images of quasi-coherent sheaves are preserved. We must still show that the extension of two quasi-coherent sheaves is quasi-coherent, so consider an exact sequence of $\mathcal{O}_X$-modules where $\mathscr{F}', \mathscr{F}''$ is quasi-coherent:

$$0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$$

Then by proposition 5.1.15, the global sections are exact as $\Gamma(X, -)$ is an exact functor:

$$0 \to \Gamma\left(X, \mathscr{F}'\right) \to \Gamma(X, \mathscr{F}) \to \Gamma\left(X, \mathscr{F}'\right) \to 0$$

relabel for simplicity to:

$$0 \to M' \to M \to M'' \to 0$$

Now apply the $\widetilde{(-)}$ exact functor. Then there is a natural commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{F}' \arrow[r] & \mathscr{F} \arrow[r] & \mathscr{F}'' \arrow[r] & 0
\end{tikzcd}$$

As $\mathscr{F}', \mathscr{F}''$ are quasi-coherent, Γ and $\widetilde{(-)}$ returns itself by corollary 5.1.11, we have that $\mathscr{F}' \to M'$

and $\mathscr{F}'' \to M''$ are isomorphism. By the 5-lemma, $M \to \mathscr{F}$ is an isomorphism. Naturally, if X is Noetherian, all sheaves are coherent.

Now let $f : X \to Y$ be a scheme morphism. As these properties are local we may assume X, Y are affine. As Γ and $\widetilde{(-)}$ are equivalence of categories and inverses of each other, and $f^*(\widetilde{M}) \cong \left(\widetilde{M \otimes_R S}\right)$, a quasi-coherent $\mathcal{O}_Y$-module $\mathscr{G}$ get's pushed out to a quasi-coherent $\mathcal{O}_X$-module $f^*\mathscr{G}$. If Y is Noetherian, everything becomes coherent.

the push-forward requires more conditions (see example 4.5). If X is Noetherian of f is quasicompact, X can be covered by finitely many affine open subsets, and as X is Noetherian of f is separated, the intersection of two affine open subsets is either affine or the union of affine open subsets. Label the affine open U_i and the affine local subsets whose union is the intersections $U_i \cap U_j$ as U_{ijk}. Now for any $s \in \mathscr{F}\left(f^{-1}(V)\right), V \subseteq Y$ open, there is a unique collection of $s_i \in \mathscr{F}\left(f^{-1}(V) \cap U_i\right)$ whose restriction to $f^{-1}(V) \cap U_{ijk}$ must all be equal. Thus, we get the following exact sequence:

$$0 \to f_*\mathscr{F} \to \bigoplus_i f_*\left(\mathscr{F}|_{U_i}\right) \to \bigoplus_i f_*\left(\mathscr{F}|_{U_{ijk}}\right)$$

where f_* is a bit overloaded for the second and third position, As it should be the function induced by $f : U_i \to Y$ and $f : U_{ijk} \to Y$. Now, as U_i are affine, by proposition 4.1.11[4], $f_*\left(\mathscr{F}|_{U_i}\right)$ and $f_*\left(\mathscr{F}|_{U_{ijk}}\right)$ are quasi-coherent. As the kernel of a $\mathcal{O}_X$-module morphism of a quasi-coherent sheaf is quasi-coherent, $f_*(\mathscr{F})$ is quasi-coherent, completing the proof.

If X were not Noetherian or f not quasi-compact and separated, the pushforward need not be quasi-coherent:

Example 5.1.17: Pushforward Not Quasi-Coherent

Let $X = \coprod_{i \in \mathbb{N}} \operatorname{Spec}\mathbb{Z}, Y = \operatorname{Spec}\mathbb{Z}$, and $f : X \to Y$ be the identity for each disjoint element of $\operatorname{Spec}\mathbb{Z}$. Take $\mathscr{F} = \mathcal{O}_X$.

Now, for any quasi-coherent sheaf $\mathscr{G}$ on Y, as we have a change of basis for each open set we must have:

$$\mathscr{G}(Y) \otimes_{\mathcal{O}_Y(Y)} \mathcal{O}_X(U) \xrightarrow{\sim} \mathscr{G}(U)$$

We'll show this is not the case when $\mathscr{G} = f_*\mathcal{O}_X$ and $U = D(2) \subseteq \operatorname{Spec}\mathbb{Z}$. In this case we get:

$$\left(\prod_{i \in \mathbb{N}} \mathbb{Z}\right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{1}{2}\right] \to \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right]$$

Next, for $t \in \left(\prod_{i \in \mathbb{N}} \mathbb{Z}\right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{1}{2}\right]$, we have hat:

$$t = \left(\frac{a_i}{2^n}\right)_{i \in \mathbb{N}}$$

for some fixed n. The image ought to be the identity. However, the codomain contains elements that are not of this form, namely there are:

$$s = \frac{b_i}{2^{n_i}} \in \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right]$$

where n_i grows arbitrarily large and cannot be an image the image for any t !

If X were not Noetherian or f not quasi-compact and separated, the pushforward need not be quasi-coherent:

If X, Y are Noetherian, it is not necessarily the case that f_* of a coherent is a coherent sheaf!

Example 5.1.18: Pushforward of Noetherian Not Coherent

Take $f : \operatorname{Spec} k[x] \to \operatorname{Spec} k$ given by $k \to k[x]$. Then $f_*\left(\mathcal{O}_{\operatorname{Spec} k[x]}\right) = k[x]$ is not a finitely generated k-module.
There are however many sufficient properties that can be added to f to make the image coherent, for example finite, projective, or proper. (see ref:HERE)

(This was shoved at the end of the last proof, I want it separate)

Proposition 5.1.19: Pushforward Q.C. if Qcqs

the pushforward of a quasi-coherent sheaf is quasi-coherent if X is qcqs

***Proof*:**
Prove that:

$$\Gamma(f^{-1}(U), \mathcal{F}) \otimes_A A_g \xrightarrow{\cong} \Gamma(f^{-1}(D(g)), \mathcal{F})$$

5.1.2 Bundles and Q.C. Sheaves

An important class of examples (and historically the first studied class of examples) were inspired by vector bundles. In this section, we shall show how $\mathbf{QCoh}(S)$ can be interpreted as the closure with respect to kernels and images of the category of vector bundles on S. The way we shall construct this closure

Recall that a vector-bundle can be thought of as a local-trivialization, that is for every point $x \in M$ there is a open neighborhood $x \in U \subseteq M$ such that there is a map $\pi^{-1}(U) \cong_{\mathbf{Top}} U \times \mathbb{R}^k$ respecting projecting and isomorphic on fibers. In other words, it is *locally free*. If there is a global trivialization, than it is called *free*. As a reminder, here is the definition of a vector bundle, for simplicity defined over a quasi-projective variety

Definition 5.1.20: Vector Bundle

Let X be a quasi-projective variety over k. Then a *vector bundle* E of rank r is a surjective morphism $\pi : E \to X$ along with an open covering $\{U_i\}$ and isomorphism:

$$\varphi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{A}^r_k$$

such that for every intersection $U_i \cap U_j$:

$$\varphi_j \circ \varphi_i^{-1}|_{U_i \cap U_j} = (\mathrm{id}, \varphi_{ij}) \qquad \text{for} \qquad \varphi_{ij} \in \mathrm{GL}_r(k)$$

A vector bundle has a natural topology impose by the pre-images and gluing. A *section* is a continuous map $s : X \to E$ such that $\pi \circ s = \mathrm{id}$. On $U \subseteq X$, we can define a local section $s : U \to E$ such that $\pi|_U \circ s = \mathrm{id}_U$.

Next, we define the algebraic counter-part:

Definition 5.1.21: Free and Locally-Free $\mathcal{O}_X$-module

Let X be a ringed space and $\mathscr{F}$ a $\mathcal{O}_X$-module. Then $\mathscr{F}$ is free if

$$\mathscr{F} \cong \mathcal{O}_X^n$$

$\mathscr{F}$ is locally free if X can be covered in open sets U such that $\mathscr{F}|_U$ is a free $\mathcal{O}_X|_U$-module. The rank of each $U \subseteq \mathscr{F}$ is the number of copies of $\mathcal{O}_X$ (the "dimension", whether finite or infinite).

If X is connected then the rank must be globally the same.

Proposition 5.1.22: Locally Free if and only if Stalk is Free

Let $\mathscr{F}$ be an quasi-coherent sheaf over X. Then $\mathscr{F}$ is locally free iff each stalk $\mathscr{F}_x$ is free.

***Proof*:**
Exercise: key is to remember localization preserves exactness, and that if all local modules are 0 then the global module is 0. Conclude the right result after by restricting at distinguished open sets, i.e. localizing at f.

Let us link locally free sheaves and vector bundles. Starting with a vector-bundles, consider an open cover $U_i \subseteq X$ that are locally trivializable. Then each $U_i \cap U_j \hookrightarrow U_i$ and $U_i \cap U_j \hookrightarrow U_j$ are two different local trivializations of rank r. Therefore, there is an invertible $r \times r$ matrices A_{ij} of functions given an isomorphism:

$$(\mathcal{O}(U_i \cap U_j))^r \to (\mathcal{O}(U_i \cap U_j))^r$$

Naturally, $A_{ii} = \mathrm{id}$ and

$$A_{jk}A_{ij} = A_{ik}$$

Notice these satisfy the cocycle condition. Hence, we can define a sheaf using these. We first build-up important the required machinery:

Lemma 5.1.23: Coherent Sheaf Locally Free Criterion

Let $\mathcal{E}$ be a coherent sheaf on X. Then

1. A subsheaf of a locally free sheaf is locally free
2. If the rank is 1, a nonzero map between locally free sheaves is injective

***Proof*:**
exercise.

Naturally, the quotient of a locally free sheaf need not be locally free (which is why it is not an abelian category); the problem is the torsion of the module. However, there is a natural way of fixing this:

Lemma 5.1.24: Natural Quotient of Locally Free Sheaf

Let $\mathcal{E}$ be a locally free sheaf of rank r and $\mathcal{E}' \subseteq \mathcal{E}$ a locally free sheaf of rank $r' \leq r$. Then there exists a subsheaf $\overline{\mathcal{E}}' \subseteq \mathcal{E}$ of rank r' containing $\mathcal{E}'$ such that $\mathcal{E}/\overline{\mathcal{E}}'$ is a locally free sheaf of rank $r - r'$. if $\mathcal{E}'$ is a maximal subsheaf of rank r', then the quotient is free.

***Proof*:**
Let T be the torsion submodule of $M = \mathcal{E}/\mathcal{E}'$. The fiber of M at a point of C is a finitely generated module over a principal ideal domains. If $T = 0$, then the fibers of M are torsion-free and therefore free. Then, using lemma 5.1.23(1), M is free.

Now define $\bar{\mathcal{E}}'$ to be the kernel of the natural composition

$$\mathcal{E} \to M \to M/T$$

The new quotient of $\mathcal{E}$ by $\bar{\mathcal{E}}'$ is M/T which has no torsion, hence it is locally free, hence By definition of $\bar{\mathcal{E}}'$, it is clear that $\mathcal{E} \subset \bar{\mathcal{E}}'$. Moreover, from the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{E}' & \longrightarrow & \mathcal{E} & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bar{\mathcal{E}}' & \longrightarrow & \mathcal{E} & \longrightarrow & M/T & \longrightarrow & 0 \end{array}$$

thus by the snake's lemma, $\bar{\mathcal{E}}'/\mathcal{E}' \cong T$. Hence, $\bar{\mathcal{E}}'$ and $\mathcal{E}'$ have the same rank, completing the proof.

Theorem 5.1.25: Vector-bundle/Locally Free Sheaf Correspondence

The category of Vector bundles with bundle morphism is equivalent to the category of locally free sheaves with sheaf morphisms

$$\left\{ \begin{array}{c} \text{Vector Bundles } V \\ \text{on } X \end{array} \right\} \underset{\text{Spec Sym}(\mathscr{F}^\vee) \leftarrow\!\shortmid \mathscr{F}}{\overset{V \mapsto \mathscr{F}}{\rightleftarrows}} \left\{ \begin{array}{c} \text{locally free} \\ \text{sheaves } \mathscr{F} \text{ on } X \end{array} \right\}$$

The base space can be more general than what we do in the proof, however generalities add complications not intrinsic to the problem and so is left to the reader to sort out the extra details if they so wish.

***Proof*:**
Throughout, fix X to be a complete k-variety where $k = \overline{k}$ is algebraically and let $\mathcal{O}_X$ be the structure sheaf. Let us start with a vector bundle and construct a locally free sheaf. Let E be a vector bundle. Define a sheaf $\mathcal{E}$ by letting

$$\mathcal{E}(U) = \{\text{local sections on E}\}$$

This certainly respects the sheaf axioms of restriction and gluing. Define an $\mathcal{O}_X$-action on $\mathcal{E}$ by taking:

$$(f \cdot s)_p = f(p) \cdot s_p$$

and addition is clear, turning it into a $\mathcal{O}_X$-module. We need to show that $\mathcal{E}$ is locally free. Take an locally-trivializable open covering of $E \to X$ so that

$$\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^r_k$$

Define the canonical local "coordinate" sections:

$$\begin{gathered} x_i : U_i \to U_i \times \mathbb{A}^r_k \\ p \mapsto (p, (0, ..., 1, ..., 0) \end{gathered}$$

Then every section $s \in \mathcal{E}(U_i)$ can be written as:

$$s = f_1 x_1 + \cdots + f_r x_r$$

for some functions $f_i \in \mathcal{O}_X$. As each U_i is locally trivial it follows that there exists a unique s for each r-tuple $(f_1, ..., f_r)$, hence the map:

$$\mathcal{E}(U_i) \to (\mathcal{O}(U_i))^r s \mapsto (f_1, ..., f_r)$$

is bijective, and is certainly a $\mathcal{O}_X(U_i)$-module morphism, hence an isomorphism.

Conversely, the direction requires a bit more work due to non-closed points. Let us first show what we are aiming for by demonstrating the result for a curve. Suppose that $\mathcal{E}$ is a locally free sheaf over a curve C. Define a vector bundle by taking

$$E = \{(p, t) \mid p \in X,\, p \text{ closed point}\}\, t \in \mathcal{E}_p / \mathfrak{m}_p \mathcal{E}_p \tag{5.1}$$

Then the map $\pi : E \to X$ projecting onto the first component shall have the required properties. This same construction does not generalize as we miss the non-closed points. We shall thus create a more general construction, and show how this generalizes the above[a]. First, a $\mathcal{O}_X$-algebra $\mathcal{A}$ is defined in the same way a $\mathcal{O}_X$-module were each $\mathcal{A}(U)$ is a $\mathcal{O}_X(U)$-algebra. Then we have $\operatorname{Spec}(\mathcal{A}(U))$. We shall glue these together to form a new scheme $\operatorname{Spec}_X(\mathcal{A})$ call the *relative scheme* $\mathcal{A}$ with respect to X. Namely, for any $V \subseteq U$, we have a ring homomorphism $\mathcal{A}(U) \to \mathcal{A}(V)$ giving spectrum maps $\operatorname{Spec}(\mathcal{A}(V)) \to \operatorname{Spec}(\mathcal{A}(U))$ that give natural gluing conditions. This forms a scheme, denoted $\operatorname{Spec}_X(\mathcal{A})$ with the natural projection:

$$\pi : \operatorname{Spec}_X(\mathcal{A}) \to X$$

where if $U \subseteq X$, $\pi^{-1}(U) \cong \operatorname{Spec}(\mathcal{A}(U))$. In particular, we can characterize the set $\operatorname{Spec}_X(\mathcal{A})$ as:

$$\operatorname{Spec}_X(\mathcal{A}) = \{(p, \mathfrak{q}) \mid p \in X,\ \mathfrak{q} \in \operatorname{Spec}(\mathcal{A}_p)\}$$

In this way, we defined a natural “bundle” on X. This construction is even universal (see exercise ref:HERE). For us, it is important that the fibers are:

$$\operatorname{Spec}_X(\mathcal{A})_x \cong \operatorname{Spec}(\mathcal{A}_x \otimes \mathcal{O}_X, \kappa(x))$$

where $\mathcal{A}_x$ is the stalk at x and $\kappa(x)$ is the residue field at x. What shall be important to us is that:

$$\operatorname{Spec}_X(\operatorname{Sym}(\mathcal{E}^\vee))$$

shall be our desired vector bundle. To see this, the first thing is that Sym is the usual symmetric algebra; if $\mathscr{F}$ is a $\mathcal{O}_X$-algebra then:

$$\operatorname{Sym}\mathscr{F} = \mathcal{O}_X \oplus (\mathscr{F}) \oplus (\mathscr{F} \otimes \mathscr{F} / \sim) \oplus \cdots$$

where $\sim$ is the usual symmetric tensor relations. Then by some elementary ring theory we see that if $\mathscr{F}|_U$ has generators $\{s_1, ..., s_n\}$ then:

$$\operatorname{Sym}(\mathscr{F})|_U \cong \mathcal{O}_X(U)[t_1, ..., t_n]$$

for some indeterminates t_i. Next, we take the dual $\mathcal{E}^\vee = \mathcal{H}om(\mathcal{E}, \mathcal{O}_X)$ simply to reverse the directions of the arrows in the right way (you the intuition from differential geometry [5]). But as wee see in the above, the localization give us n-dimension affine space over the appropriate ring. Looking at this spectrum as set, we get:

$$\{(p, v) \mid p \in X,\ \mathcal{E}_p \otimes \kappa(p)\}$$

which we can compare with equation 5.1 to see this is indeed the right generalization!

We may now return to our discussion. By letting $E = \operatorname{Spec}_X(\operatorname{Sym}(\mathcal{E}^\vee))$, we have the surjection $E \to X$ onto the first component. Let us show that we get a vector-bundle structure. Choose an open cover U_i which is locally trivializable so that $\mathcal{E}(U_i) \cong (\mathcal{O}_X(U_i))^r$. As $\overline{k}$ is algebraically closed, we have that each fiber is isomorphic to $\mathbb{A}^r_{\overline{k}}$.

Now, on $U_i \cap U_j$, we get a transition function on $\mathcal{E}(U_i \cap U_j)$ given by the $r \times r$ matrix A_{ij}. Then it is clear that these respect the cocycle condition and $A_{ii} = I$. Now, fiber-wise, we see that reducing mod $\mathfrak{m}_p$ where $\mathfrak{m}_p$ is the maximal ideal $\mathcal{E}_p$ gives isomorphisms, and tensoring preserves isomorphisms. Certainly the cocycle condition is preserved, and hence we have that E is a vector-space.

Finally, for the maps, let us start with a map of locally free sheaves. Then the vector-bundle map can be defined by fiber-wise by reducing and tensoring. Conversely, if $\varphi : E \to E'$ is a vector bundle map and a section $s : X \to E$, composing gives a section $\varphi(s) : X \to E'$. Certainly these maps respect restrictions, giving the desired result.

[a] this is not a one-off definition, we focus on this notion further in ref:HERE

For section 5.2, rank 1 locally free sheaves shall be boxed for their importance in generalizing the notion of functions on affine schemes to projective schemes

Definition 5.1.26: Line Bundle

A locally free $\mathcal{O}_X$-module of rank 1 (over a locally ringed space) is called a *line bundle*, or *invertible sheaf.*

Theorem 5.1.25 justifies this name, and in most literature the term "line bundle" is used instead of "locally free sheaf of rank 1". But why "invertible sheaf"? This is because under the tensor product, these sheaves shall have an inverse, namely if SL is an invertible sheaf, there exists an invertible sheaf $\mathcal{L}$ such that $\mathrm{SL} \otimes \mathcal{L} = \mathcal{O}_X$; this is the identity in the group. We shall look at this in deph in section 5.3.

5.1.27 K-Theory If the reader knows some K theory (see [8]) to quickly draw the connection recall that we can take the monoid of $\mathcal{O}_X$-module along with the tensor product. The prototypical example in algebra is the monoid of k-modules. Then if $\mathscr{F} \otimes \mathscr{F}^\vee \cong \mathcal{O}_X$ (or for R-modules $M \otimes_R M^\vee \cong R$), then these would be the invertible elements in this monoid. Then every locally free sheaf of rank 1 is invertible. If X is a locally ringed space (not just a ringed space) then this becomes and iff. Hence, it is common to call them a line bundle as well.

5.1.3 Ideal Sheaves and Closed Subscheme

Let us return to closed subschemes, which were defined to be closed subsets $X \subseteq Y$ where $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is a surjective map. As the map is surjective, the kernel at each open set must give an ideal. Given a closed immersion $\pi : X \to Y$, there is an ideal sheaf that we can put on Y that is given by the kernel of the map:

$$\mathcal{O}_Y \to \pi_* \mathcal{O}_X$$

Definition 5.1.28: Ideal Sheaf

Let $\mathcal{O}_X$ be a structure sheaf for a scheme X. Then an ideal sheaf is a $\mathcal{O}_X$-submodule of $\mathcal{O}_X$.

Lemma 5.1.29: Kernel and Ideal Sheaves

Let Y be a closed subscheme of X and $\iota : Y \to X$ the inclusion map. Then the kernel of $\iota^\# : \mathcal{O}_X \to \iota_* \mathcal{O}_Y$ is an ideal sheaf, and is usually denoted $\mathcal{I}_Y$.

***Proof*:**
exercise

The reader should find no difficulty in seeing this is a coherent sheaf over $\mathcal{O}_Y$. We get the following short exact sequence of $\mathcal{O}_Y$-modules:

$$0 \to \mathcal{K} \to \mathcal{O}_Y \to \pi_* \mathcal{O}(X) \to 0$$

For each affine open subset with a closed immersion $\operatorname{Spec}(R) \to Y$, let $I(R)$ be the corresponding ideal (they define a base for the sheaf). We may ask if we can study the closed immersion condition locally using $\mathcal{K}$ and the ideals $I(R)$. The first thing we may want is an equivalent of the fact that every ideal sheaf should correspond to a closed subscheme (i.e. make lemma 4.1.16 and iff). This is not the case, though mainly since we have not limited ourselves to quasi-coherent sheaves. The following necessary property for ideal sheaves given by kernels shall be shown to not apply to all ideal sheaves:

Proposition 5.1.30: Sheaf Ideal For closed immersion

Let $X \to Y$ be a scheme morphism. Then $X \to Y$ is a closed immersion if and only if for each affine open subset $\operatorname{Spec} S \subseteq Y$ and corresponding inclusion $\operatorname{Spec} S \to Y$ giving the ideals $I(S) \subseteq S$, the restriction morphism $S \to S_f$ induces an isomorphism:

$$I(S)_f \cong I(S_f)$$

for all $f \in S$. In other words the closed immersions $\operatorname{Spec} S/I \to \operatorname{Spec} S$ glue to get a closed immersion $X \to Y$.

***Proof*:**
The ($\Rightarrow$) is the easier direction. For the ($\Leftarrow$) direction, take an ideal $I(S) \subseteq S$ corresponding to an affine open subset $\operatorname{Spec}(S) \subseteq X$. Suppose that for each closed immersion $\operatorname{Spec}(S) \to Y$ and $f \in S$, the restriction $S \to S_f$ induces an isomorphism $I(S)_f \cong (S_f)$. Using this, show you can glue together closed immersions $\operatorname{Spec}(S/I) \to \operatorname{Spec}(S)$ in a well-defined manner to get a closed immersion $X \to Y$.

Example 5.1.31: Not All Ideal Sheaves Correspond To closed immersions
Take $X = \operatorname{Spec} k[x]_{(x)}$. This scheme has two points: the closed point and the generic point, say η. Then take:

$$\mathcal{K}(X) = 0 \subseteq \mathcal{O}_X(X) = k[x]_{(x)}$$
$$\mathcal{K}(\eta) = k(x) = \mathcal{O}_X(\eta)$$

Then using the above proposition, we see that this sheaf cannot define a closed immersion.

We recover a correspondence if we limit to quasi-coherent sheaves:

Proposition 5.1.32: Quasi-coherent Ideal Sheaves

Let X be a scheme. Then for any closed subscheme $Y \subseteq X$, the corresponding ideal sheaf $\mathcal{I}_Y$ is a quasi-coherent sheaf of ideals on X. If X is Noetherian, $\mathcal{I}_Y$ is coherent.
Conversely, any quasi-coherent sheaf of ideals on X corresponds uniquely to a closed subscheme of X.

***Proof*:**
This follows by the Qcqs lemma (lemma 3.3.5) to get the required condition for proposition 4.1.17

Corollary 5.1.33: Ideal Sheaves On Affine Schemes

Let $\operatorname{Spec} R$ be an affine scheme. Then there is a one-to-one correspondence between the ideals $\mathfrak{a} \subseteq R$ and the closed subscheme $Y \subseteq X$ given by

$$\mathfrak{a} \mapsto \operatorname{Spec} R/\mathfrak{a}$$

With further control of closed subschemes, we can define the following interesting analogy to vanishing sets:

Definition 5.1.34: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$. Then a vanishing scheme corresponding to $s, V(s)$, is the scheme corresponding to gluing $\operatorname{Spec}(R/(s_R))$ where $s_R = s|_{\operatorname{Spec}(R)}$. More generally, we can take the ideal $(S) \subseteq \mathcal{O}_X(X)$ generated by the elements S.

The reader should use proposition 4.1.17 to insure this is well-defined. The advantage of working with vanishing schemes over vanishing sets is that it remembers extra information, such as the multiplicity of intersection.

Proposition 5.1.35: Properties of Vanishing Schemes

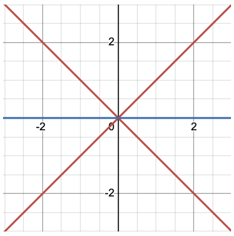
Let X be a scheme. Then:

1. $\cup_i^n V(S_i) = V(\cap_i^n S_i)$
2. $\cap_{i \in \mathcal{I}} V(S_i) = V\left(\left\langle S_i \middle| i \in \mathcal{I} \right\rangle\right)$

***Proof*:**
Note that the underlying sets of the schemes will just be finite unions and intersections.
The harder point to verify is that the union becomes intersection. In the case of vanishing sets, the proof takes advantage of the fact that $I \cap J \subseteq IJ \subseteq (I \cap J)^2$. Note that it must be the intersection, as we would need that that $V(S) \cap V(S) = V(S)$ and not $V(S^2)$.

Example 5.1.36: Vanishing Scheme Intersection

1. Take $V(y - x^2), V(y) \subseteq \mathbb{A}_k^2$. Then with vanishing sets, this would just be the point $(0, 0)$. However, this point now has multiplicity 2.
2. Take $V(y^2 - x^2), V(y) \subseteq \mathbb{A}_k^2$. The visual of these two are:



Use this to show that if X, Y, Z are closed subscheme of a scheme W, then

$$(X \cap Z) \cup (Y \cap Z) \neq (X \cup Y) \cap Z$$

Hence, not all properties of unions and intersection can carry over!

Closed subscheme are in analogy to closed sets, which can be made very precise with the following generalization of the vanishing set

Definition 5.1.37: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$. Then a *vanishing scheme* corresponding to s, $V(s)$, is the scheme corresponding to gluing $\operatorname{Spec}(R/(s_R))$ where $s_R = s|_{\operatorname{Spec}(R)}$. More generally, we can take the ideal $(S) \subseteq \mathcal{O}_X(X)$ generated by the elements S.

The advantage of working with vanishing schemes over vanishing sets is that it remembers extra information, such as the multiplicity of intersection.

Proposition 5.1.38: Properties of Vanishing Schemes

Let X be a scheme. Then:

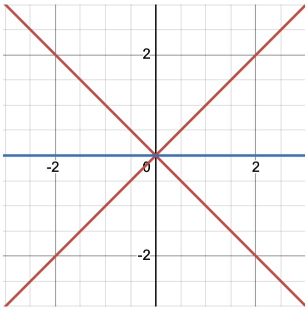
1. $\cup_i^n V(S_i) = V(\cap_i^n S_i)$
2. $\cap_{i \in \mathcal{I}} V(S_i) = V(\langle S_i \mid i \in \mathcal{I} \rangle)$

***Proof*:**
Note that the underlying sets of the schemes will just be finite unions and intersections.

The harder point to verify is that the union becomes intersection. In the case of vanishing sets, the proof takes advantage of the fact that $I \cap J \subseteq IJ \subseteq (I \cap J)^2$. Note that it must be the intersection, as we would need that that $V(S) \cap V(S) = V(S)$ and not $V(S^2)$.

Example 5.1.39: Vanishing Scheme Intersection

1. Take $V(y - x^2), V(y) \subseteq \mathbb{A}_k^2$. Then with vanishing sets, this would just be the point $(0, 0)$. However, this point now has multiplicity 2.
2. Take $V(y^2 - x^2), V(y) \subseteq \mathbb{A}_k^2$. The visual of these two are:



Use this to show that if X, Y, Z are closed subscheme of a scheme W, then

$$(X \cap Z) \cup (Y \cap Z) \neq (X \cup Y) \cap Z$$

Hence, not all properties of unions and intersection can carry over!

Locally closed immersion

A lot on when it is nicer, on locally closed immersions, etc. Vakil p.238-240

Recall that the line without the origin is not a closed subscheme of $\mathbb{A}^2_k$. This is This is however still a pretty reasonable subset, and almost a closed subscheme: it is a closed subscheme of an open subscheme of $\mathbb{A}^2_k$ (namely the subscheme given by $k[x, 0]$). This motivates the following:

Definition 5.1.40: Locally closed immersion

Let $\pi : X \to Y$ be a scheme morphism. Then it is a *locally closed immersion* if π can be factored into:

$$X \xrightarrow{\alpha} Z \xrightarrow{\beta} Y$$

where α is a closed immersion, and b is an open embedding.

The above is such an example: more formally the morphism $\operatorname{Spec} k[t, t^{-1}] \to \operatorname{Spec} k[x, y]$ is a locally closed immersion. Note that we could have also taken an open subset of a closed subset (in exercise ref:HERE you make sure these are equivalent)

Proposition 5.1.41: Locally closed immersion is Locally Finite

Let $\pi : X \to Y$ be a locally closed immersion. Then it is locally finite.

***Proof*:**
exercise.

Give a characterization of intersection of locally closed immersions as you've moved the fibred product to earlier

5.1.4 Image Scheme

In general, the image of a scheme is badly behaved, for example check that $(x, y) \mapsto (x, xy)$ does not give a scheme as an image.

Definition 5.1.42: Image Scheme

Let $\iota : Z \to Y$ be a closed subscheme, which gives the short exact sequence:

$$0 \to \mathcal{K}_{Z/Y} \to \mathcal{O}_Y \to \iota_* \mathcal{O}_Z \to 0$$

Then the *image* of $\pi : X \to Y$ *lies in* Z if the composition

$$\mathcal{K}_{Z/Y} \to \mathcal{O}_Y \to \pi_* \mathcal{O}_X$$

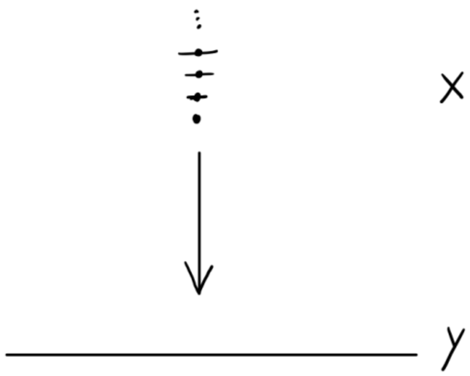
is zero. Then the *scheme-theoretic image* of π (a closed subscheme of Y) is the smallest closed subscheme containing the image

Intuitively, locally functions vanishing on Z pullback to zero functions on X. If the image of π line sin some subschemes Z_j, it lies in their intersection, and hence it is the scheme-theoretic image. Note that the scheme theoretic image may not necessarily be an open subscheme:

Example 5.1.43: Scheme-theoretic Image

1. Take
$$\pi : \text{Spec}\,(k[\epsilon]/(\epsilon^2)) \to \text{Spec}\,(k)$$
given by $x \mapsto \epsilon$. Then the scheme-theoretic image of π ks
$$\text{Spec}\,(k[x]/(x^2))$$
as the polynomials pulling back to 0 are precisely the multiples of x^2.
2. Now take
$$\pi : \text{Spec}\,(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k$$
given by $x \mapsto 0$. Then the scheme-theoretic image is
$$\text{Spec}\,(k[x]/(x))$$
which is reduced.
3. Take
$$\text{Spec}\,k[t, t^{-1}] = \mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k = \text{Spec}\,(k[u])$$
given by $u \mapsto t$. Then any function $g(u)$ which pulls back to 0 as a function of t must be the zero function. Hence,the scheme-theoretic image is
$$\text{Spec}\,(k[u])$$
4. Consider the following more pathological example:
$$\pi : \coprod \text{Spec}\left(\frac{k[\epsilon_n]}{(\epsilon_n)^n}\right) \to \text{Spec}\,k[x]$$

which well denote $\pi : X \to Y$, and it is given by $x \mapsto \epsilon_n$ on the nth component. Now, if $g(x)$ on Y pulls back to 0 on X, then its Taylor expansion must be 0. But then the scheme-theoretic image is $V(0)$ on Y, that is it is Y, even though the scheme-theoretic image is the origin (the closed point 0). Visually:



5.1.5 Relative Spectrum

moved from somewhere else The following should also give a strong intuition on locally free $\mathcal{O}_X$-modules;

figure this out here

Definition 5.1.44: $\mathcal{O}_S$-algebra

A $\mathcal{O}_X$-module over X is a $\mathcal{O}_X$-algebra if each $\mathcal{O}_X(U)$ is a $\mathcal{O}_X(U)$-algebra.

Note that $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ and $\varphi_{ii} = \text{id}$. This is equivalent to the usual local trivialization definition of vector bundle as the reader can verify. This construction is given by gluing the disjoint union of $U_i \times \mathbb{A}^r_k$. From this, we can define a morphism of vector bundles $E \to E'$ to be a morphism of varieties such that the natural commutative diagram projecting to X holds and the morphism restricts to a linear morphism on each fiber. We can define the direct sum and tensor product of vector bundles fiber-wise as well.

Really want to put this down HERE:

Theorem 5.1.45: Quasi-coherence and Relative Spectrum

$$\text{Mod}(\mathcal{A}) \cong \mathbf{QCoh}(\underline{\text{Spec}}_S \mathcal{A})$$

***Proof*:**
here

5.2 Projective Schemes with Twisted Sheaves

Let us now explore quasi-coherent shaves on projective schemes. The main result of this section is to recover the graded ring of a projective scheme as the global sections of a special kind of sheaf that is in Adjunction with $\operatorname{Proj}(-)$. Let us start by defining the natural way we can have a sheaf over a projective scheme:

Definition 5.2.1: Sheaf Associated to M [on projective scheme]

Let S be a graded ring and M a graded S-module. Then the sheaf associated to M on $\operatorname{Proj}(S)$, denoted by $\widetilde{M}$, is given by:

$$\widetilde{M}\Big|_{D_+(f)} = \left(\widetilde{M\left[f^{-1}\right]_0}\right)$$

for homogeneous $f \neq 0$.

Like in projective schemes, if S is the usual polynomial ring with grading by degree, then open sets $D(x_i) \subseteq \operatorname{Proj} S$ are isomorphic to $\operatorname{Spec}\,(S_{x_i})_0$, and so the module $\widetilde{M}(D(x_i))$ becomes a module over an affine scheme.

Proposition 5.2.2: Sheaf on Projective Scheme is Quasi-Coherent

Let S be a graded ring, M a graded S-module, and $X = \operatorname{Proj} S$. Then:

1. $\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}}$
2. $\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module, and if S is Noetherian and M finitely generated, it is coherent.

***Proof*:**
a repeat of similar result earlier

Let us start by showing how we can define some natural sheaves on $\operatorname{Proj} S$ whose direct sum of global sections will recover S :

Definition 5.2.3: Twisting Sheaf

Let $S_\bullet = \bigoplus_n S_n$ be a graded ring and $X = \operatorname{Proj} S$. Then for every $n \in \mathbb{Z}$, define the sheaf:

$$\mathcal{O}_X(n) := \widetilde{S(n)}$$

where $S(n)$ is the graded ring $S(n)_i = S_{n+i}$. The sheaf $\mathcal{O}_X(1)$ will be called the twisting sheaf.

Note that $S(0) = S$, so $\mathcal{O}_X(0) = \mathcal{O}_X$ which implies $\Gamma(\operatorname{Proj} S, S(0)) = \Gamma\left(\operatorname{Proj} S, \mathcal{O}_{\operatorname{Proj} S}\right)$ represents the constant global sections. Let us next consider $\widetilde{S(n)}$. Let us work on our usual $S = R\left[x_0, \ldots, x_n\right]$ (proposition 4.2.5 shall generalize this discussion). On $D(x_i)$ we will then have our usual localization, except we will not have the degree 0 elements but degree m elements:

$$\widetilde{S(m)}\left(D_{+}\left(x_{i}\right)\right)= \text{ degree } m \text{ of } R\left[x_{0}, \ldots, x_{n}, x_{i}^{-1}\right]$$

Note that this is no longer a ring, but a $R\left[x_{0}/x_{i}, \ldots, x_{n}/x_{i}\right]$-module (it is closed under the action as the elements are degree 0). As an abelian group, it is generated by:

$$x_{0}^{m}, x_{0} x_{1}^{m-1}, x_{0} x_{2}^{m-1}, x_{3}^{m-6} x_{2}^{3} x_{8}^{3}, \ldots, x_{n}^{m}$$

Is this module any different from $\widetilde{S}\left(D_{+}\left(x_{i}\right)\right)$ as modules? In fact they are not! Note that on $D_{+}\left(x_{i}\right)$, multiplying by x_{i} is an isomorphism from the degree m component of $R\left[x_{0}, \ldots, x_{n} x_{i}^{-1}\right]$ to its degree $m+1$ component! Thus, $\widetilde{S(n)}$ is locally isomorphic to $\widetilde{S}$, and hence it is locally free of rank 1 , that is a *line bundle* (or invertible sheaf).
These sheaves are not be isomorphic, because the gluing of these sheaves is subtly different and shall motivate why it is called the twisted sheaves. Let us first distinguish the $\widetilde{S(m)}$ by showing they have different global sections. Going about it a bit informally, let's consider $\widetilde{S(m)}$ and cover it with $D\left(x_{i}\right)$ so that $\Gamma\left(\widetilde{S(m)}, D\left(x_{i}\right)\right)=\left(R\left[x_{0}, \ldots, x_{n} x_{i}^{-1}\right]\right)_{m}$. Then whenever gluing two $D\left(x_{i}\right), D\left(x_{i}\right)$, their intersection shall have polynomials in degree n that have no x_{i}, x_{j} term. The gluing is very similar to what we've done before: we can think of the transition from $D\left(x_{i}\right)$ to $D\left(x_{j}\right)$ on $D\left(x_{i}\right) \cap D\left(x_{j}\right)$ would map $x_{k} \mapsto x_{k}$ and

$$\frac{x_{k}}{x_{i}} \mapsto \frac{x_{k}}{x_{j}}$$

For a concrete example, if $X=\operatorname{Proj} R[x, y]$, then homogeneous polynomial of degree 1 are of the form $F(x, y)=a x+b y$, which on the two distinguished open sets $D_{+}(x)$, $D_{+}(y)$ are represented as can be (where $t=x/y$ and $s=y/x$):

$$f(t)=a t+b \qquad g(s)=a+b s$$

and have transition function

$$f(t)=t \cdot t g(1/t)$$

Contrast this with the regular gluing which doesn't have the t factor. More generally, on $S(n)$, we would have:

$$f(t)=t^{n} g(1/t)$$

Then that means that the global section is:

$$\Gamma(\widetilde{S(n)}, \operatorname{Proj} S)=S_{n}$$

and so we can use dimension counting to show they are distinct for each classical projective space!

$\dim \Gamma(\widetilde{S(n)}, \operatorname{Proj} S)$	$\mathbb{P}_{R}^{0}$	$\mathbb{P}_{R}^{1}$	$\mathbb{P}_{R}^{2}$	$\mathbb{P}_{R}^{3}$	$\mathbb{P}_{R}^{4}$
$S(0)$	1	1	1	1	1
$S(1)$	1	2	3	4	5
$S(2)$	1	3	6	10	15

For negative twisted sheaves, we would need that:

$$f(t)=t^{-n} g(1/t)$$

for which there cannot be any regular functions for any $\mathbb{P}_{R}^{m}$, and hence:

$\dim\Gamma(\widetilde{S(n)}, \operatorname{Proj} S)$	$\mathbb{P}^0_R$	$\mathbb{P}^1_R$	$\mathbb{P}^2_R$	$\mathbb{P}^3_R$	$\mathbb{P}^4_R$
$S(-2)$	1	0	0	0	0
$S(-1)$	1	0	0	0	0
$S(0)$	1	1	1	1	1
$S(1)$	1	2	3	4	5
$S(2)$	1	3	6	10	15

After introducing some properties of twisted sheaves, we shall show the $\widetilde{S(-n)}$ (given by the multiplicity of poles, or similarly by the dual space of $S(n)$) are not isomorphic as well
We should take a moment to justify the term "twisting". First off, only $\widetilde{S(0)}$ is globally free, hence there is something going on for each $\widetilde{S(n)}$. To see that it is a twisting, let us look more closely at the transition for $\widetilde{S(1)}$ on $\mathbb{P}^1_R$. On $D(x_0)\cap D(x_1)$, we have only degree 1 terms.Then we see from the map that

$$\frac{x_0}{x_1} \mapsto \frac{x_0x_0}{x_1x_0} = \frac{x_1}{x_0}$$

as we see, we would have to “flip twice”, to get back to the original. In higher twists, the power would be greater, showing the number of twists increases. Let us now introduce the grading to any sheaf defined on projective space:

Definition 5.2.4: n-twisted sheaf of $\mathscr{F}$

For any $\mathcal{O}_X$-module $\mathscr{F}$, let:

$$\mathscr{F}(n) := \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$$

we call this sheaf the n-twisted sheaf of $\mathscr{F}$.

Proposition 5.2.5: Properties of Twisted Sheaves

Let S be a graded ring and $X = \operatorname{Proj} S$. Let S be generated by S_1 as an S_0-algebra. Then

1. The sheaf $\mathcal{O}_X(n)$ is an invertible sheaf and $\mathcal{O}_X(n) \ncong \mathcal{O}_{\mathcal{O}_X}\mathcal{O}_X(m)$ for $n \neq m$
2. For any graded S-module M, $\widetilde{M}(n) = \widetilde{M(n)}$
3. $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$
4. Let T be a graded ring generated by T_1 as a T_0-algebra. Let $\varphi : S \to T$ be a homogeneous homomorphism, and take $U \subseteq \operatorname{Proj} T$. Given $f : U \to X$:

$$f^*\left(\mathcal{O}_X(n)\right) \cong \mathcal{O}_Y(n)|_U \quad \text{and} \quad f_*\left(\mathcal{O}_Y(n)|_U\right) \cong \left(f_*\mathcal{O}_U\right)(n)$$

***Proof*:**

1. We must show that $\mathcal{O}_X(n)$ is an invertible sheaf, i.e. locally free of rank 1 . We shall use the fact that S is generated by S_1 as an S_0-algebra, meaning it is finitely generated and hence will be "locally S_n ". Take $\mathcal{O}_X(n)|_{D(f)} \cong \widetilde{(S_n)_f}$ on $\operatorname{Spec}(S_f)_0$. Now, notice that $(S_n)_f$ is isomorphic to $(S_0)_f$ via the map $s \mapsto f^n s$. As S is generated by S_1 as an S_0-algebra, we see that each open subsets are isomorphic to $\widetilde{S_n}$, and hence it is an

invertible sheaf (i.e. a line bundle).
For the second claim, formalize the global section calculations from earlier.

2. immediate

3. Show that for any two graded modules $\widetilde{M \otimes_S N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$ (not too difficult generalization of an earlier result), and that $\left((M \otimes_S N)_f\right)_0 = (M_f)_0 \otimes_{(S_f)_0} (N_f)_0$.

4. Generalize form the affine case.

Let us now show there is a natural graded ring associated to each projective scheme:

Definition 5.2.6: Graded s-module Associated to $\mathscr{F}$

Let S be a graded ring, $X = \operatorname{Proj} S$, and $\mathscr{F}$ a sheaf of $\mathcal{O}_X$-modules. Then the graded S-module associated to $\mathscr{F}$ as:

$$\Gamma_*(\mathscr{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathscr{F}(n))$$

where the product is given by the properties of proposition 4.2.5.

We can now naturally recover S in the case of it being a polynomial ring:

Proposition 5.2.7: Recovering Graded Ring

Let R be a ring, $S = R[x_0, \ldots, x_n]$, and $X = \operatorname{Proj} S$. Then:

$$\Gamma_*(\mathcal{O}_X) \cong S$$

***Proof*:**
We cover X with the open sets $D_+(x_i)$. Then to give a section $t \in \Gamma(X, \mathcal{O}_X(n))$ is the same as giving sections $t_i \in \mathcal{O}_X(n)(D_+(x_i))$ for each i, which agree on the intersections $D_+(x_i x_j)$. Now t_i is just a homogeneous element of degree n in the localization S_{x_i}, and its restriction to $D_+(x_i x_j)$ is just the image of that element in S_{x_i, x_j}. Summing over all n, we see that $\Gamma_*(\mathcal{O}_X)$ can be identified with the set of $(r+1)$-tuples $(t_0, \ldots, t_r)$ where for each $i, t_i \in S_{x_i}$, and for each i, j, the images of t_i and t_j in S_{x_i, x_j} are the same.
Now the x_i are not zero divisors in S, so the localization maps $S \to S_{x_i}$ and $S_{x_i} \to S_{x_i, x_j}$ are all injective, and these rings are all subrings of $S' = S_{x_0 \cdots x_r}$. Hence $\Gamma_*(\mathcal{O}_X)$ is the intersection $\bigcap S_{x_i}$ taken inside S'. Now any homogeneous element of S' can be written uniquely as a product $x_0^{i_0} \cdots x_r^{i_r} f(x_0, \ldots, x_r)$, where the $i_j \in \mathbb{Z}$, and f is a homogeneous polynomial not divisible by any x_i. This element will be in S_{x_i} if and only if $i_j \geq 0$ for $j \neq i$. It follows that the intersection of all the S_{x_i} (in fact the intersection of any two of them) is exactly S.

Note that if S is not a polynomial ring, this result need not hold, though it is be related (see Hartshorne exercise II.5.14). A case where it fails is if we have that $M_0 = k$ (or some finite dimensional vector space) and $M_n = 0$ for all other integers; then $\tilde{M} = 0$. Hence, this shows that can't start with an M, do $\widetilde{M}$ and recover M by Γ_*. Unfortunately, the map $M \mapsto \Gamma_*(\widetilde{M})$ is neither injective nor surjective.

However, if you start with a coherent sheaf $\mathscr{F}$, apply Γ_* to get $\Gamma_*(\mathscr{F})$, and then do $\widetilde{\Gamma_*(\mathscr{F})}$, then:

$$\widetilde{\Gamma_*(\mathscr{F})} \cong \mathscr{F}$$

To prove this, we first require a technical lemma:

Lemma 5.2.8: Extending Sections for Line Bundles

Let X be a scheme, $\mathcal{L}$ an invertible sheaf on $X, f \in \Gamma(X, \mathcal{L}), X_f$ be the open set of points $x \in X$ where $f_x \notin \mathfrak{m}_x \mathcal{L}_x$, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Now:

1. Suppose that X is quasi-compact, and $s \in \Gamma(X, \mathscr{F})$ is a global section of $\mathscr{F}$ whose restriction to X_f is 0 . Then for some $n > 0$, we have $f^n s = 0$, where $f^n s$ is considered as a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

2. Suppose furthermore that X has a finite covering by open affine subsets U_i, such that $\mathcal{L}|_{U_i}$ is free for each i, and such that $U_i \cap U_j$ is quasi-compact for each i, j (it is quasi-separated). Given a section $t \in \Gamma\left(X_f, \mathscr{F}\right)$, then for some $n > 0$, the section $f^n t \in \Gamma\left(X_f, \mathscr{F} \otimes \mathcal{L}^{\otimes n}\right)$ extends to a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

***Proof*:**
This lemma is a direct generalization of lemma 4.1.12, with an extra twist due to the presence of the invertible sheaf $\mathcal{L}$.

1. First, cover X with a finite number (possible since X is quasi-compact) of open affines $U = \operatorname{Spec} A$ such that $\mathcal{L}|_U$ is free. Let $\psi : \mathcal{L}|_U \cong \mathcal{O}_U$ be an isomorphism expressing the freeness of $\mathcal{L}|_U$. Since $\mathscr{F}$ is quasi-coherent, by proposition 4.1 .13 there is an A-module M with $\mathscr{F}|_U \cong \widetilde{M}$. Our section $s \in \Gamma(X, \mathscr{F})$ restricts to give an element $s \in M$. On the other hand, our section $f \in \Gamma(X, \mathcal{L})$ restricts to give a section of $\mathcal{L}|_U$, which in turn gives rise to
an element $g = \psi(f) \in A$. Clearly $X_f \cap U = D(g)$. Now $s|_{X_f}$ is zero, so $g^n s = 0$ in M for some $n > 0$, just as in the proof of lemma4.1.12. Using the isomorphism

$$\mathrm{id} \times \psi^{\otimes n} : \mathscr{F} \otimes \mathcal{L}^n|_U \cong \mathscr{F}|_U ,$$

we conclude that $f^n s \in \Gamma\left(U, \mathscr{F} \otimes \mathcal{L}^n\right)$ is zero. This statement is intrinsic (i.e., independent of ψ). So now we do this for each open set of the covering, pick one n large enough to work for all the sets of the covering, and we find $f^n s = 0$ on X.
2. Proceed as in the proof of lemma 4.1.12, keeping track of the twist due to $\mathcal{L}$ as above. The hypothesis $U_i \cap U_j$ quasi-compact is used to be able to apply part (1) there.

Proposition 5.2.9: Partial Equivalence For Graded Rings

Let S be a graded ring, which is finitely generated by S_1 as an S_0-algebra. Let $X = \operatorname{Proj} S$ and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then there is a natural isomorphism

$$\beta : \widetilde{\Gamma_*(\mathscr{F})} \to \mathscr{F}$$

In particular, all quasi-coherent sheaves on projective schemes come from a graded module. Note the important of S being generated by S_1 and not any other S_n; if it were then the grading of the two rings would be different and we won't recover the result.

***Proof*:**

First, define the morphism β for any $\mathcal{O}_X$-module $\mathscr{F}$. Let $f \in S_1$. Since $\widetilde{\Gamma_*(\mathscr{F})}$ is quasi-coherent in any case, to define β, it is enough to give the image of a section of $\widetilde{\Gamma_*(\mathscr{F})}$ over $D_+(f)$. Such a section is represented by a fraction m/f^d, where $m \in \Gamma(X, \mathscr{F}(d))$, for some $d \geq 0$. We can think of f^{-d} as a section of $\mathcal{O}_X(-d)$, defined over $D_+(f)$. Taking their tensor product, we obtain $m \otimes f^{-d}$ as a section of $\mathscr{F}$ over $D_+(f)$. This defines β.
Now let $\mathscr{F}$ be quasi-coherent. To show that β is an isomorphism we have to identify the module $\Gamma_*(\mathscr{F})_{(f)}$ with the sections of $\mathscr{F}$ over $D_+(f)$. We apply lemma 4.2 .8 . considering f as a global section of the invertible sheaf $\mathcal{L} = \mathcal{O}(1)$. Since we have assumed that S is finitely generated by S_1 as an S_0-algebra, we can find finitely many elements $f_0, \ldots, f_r \in S_1$ such that X is covered by the open affine subsets $D_+(f_i)$. The intersections $D_+(f_i) \cap D_+(f_j)$ are also affine, and $\mathcal{L}|_{D_+(f_i)}$ is free for each i, so the hypotheses of lemma 4.2.8 are satisfied. The conclusion of lemma 4.2.8 tells us that $\mathscr{F}(D_+(f)) \cong \Gamma_*(\mathscr{F})_{(f)}$, which is just what we wanted.

We shall later show this that all resolution of quasi-coherent sheaves by locally free sheaves must have finite resolutions, see ref:HERE. For now, this shows we can translate our characterization of closed subschemes with ideal sheaves to projective schemes, and that

Corollary 5.2.10: Characterizing Projective sub-Schemes

Let R be a ring. Then:

1. If Y is a closed subscheme of $\mathbb{P}^r_R$, then there is a homogeneous ideal $I \subseteq S = R[x_0, \ldots, x_r]$ such that Y is the closed subscheme determined by I.

2. A scheme Y over $\operatorname{Spec} R$ is projective if and only if it is isomorphic to $\operatorname{Proj} S$ for some graded ring S, where $S_0 = R$, and S is finitely generated by S_1 as an S_0-algebra.

***Proof*:**

1. Let $\mathcal{I}_Y$ be the ideal sheaf of Y on $X = \mathbb{P}^r_R$. Now $\mathcal{I}_Y$ is a subsheaf of $\mathcal{O}_X$; the twisting functor is exact; the global section functor Γ is left exact; hence $\Gamma_*(\mathcal{I}_Y)$ is a submodule of $\Gamma_*(\mathcal{O}_X)$. But by proposition 4.2.9, $\Gamma_*(\mathcal{O}_X) = S$. Hence $\Gamma_*(\mathcal{I}_Y)$ is a homogeneous ideal of S, which we will call I. Now I determines a closed subscheme of X, whose sheaf of ideals will be $\widetilde{I}$. Since $\mathcal{I}_Y$ is quasi-coherent, we have $\mathcal{I}_Y \cong \widetilde{I}$ by proposition 4.2.9, and hence Y is the subscheme determined by I. In fact, $\Gamma_*(\mathcal{I}_Y)$ is the largest ideal in S defining Y.

2. Recall that by definition Y is projective over $\operatorname{Spec} R$ if it is isomorphic to a closed subscheme of $\mathbb{P}^r_R$ for some r. By part (1), any such Y is isomorphic to $\operatorname{Proj} S/I$, and we can take I to be contained in $S_+ = \bigoplus_{d>0} S_d$, so that $(S/I)_0 = R$. Conversely, any such graded ring S is a quotient of a polynomial ring, so $\operatorname{Proj} S$ is projective.

Corollary 5.2.11: Characterizing Hypersurfaces

Let R be a UFD. Then any hypersurface $X \subseteq \mathbb{P}^n_R$ corresponds to a principal ideal

***Proof*:**
The key result is that in a UFD, every height-1 prime ideal (i.e. minimal ideal) is principal. In fact, R is a UFD if and only if every height 1 prime ideal is principal. If R is a UFD, let $\mathfrak{p}$ be a height 1 prime ideal. Choose $x \in \mathfrak{p}$. As R is a UFD, $x = up_1 \cdots p_k$ for some choice of unit u. As $\mathfrak{p}$ is prime, at least one $p_i \in \mathfrak{p}$. Then $(p_i) \subseteq \mathfrak{p}$, and as $\mathfrak{p}$ has height 1 $(p_i) = \mathfrak{p}$. Conversely, assume $(p_i) = \mathfrak{p}$ for all height-1 prime ideals. Let r be an irreducible element. We want to show (r) is prime. As Let $\mathcal{P}_r$ be the collection of all prime ideals containing r. As R is Noetherian, there is a minimal element; say $\mathfrak{p}$. As $\mathfrak{p}$ is a minimal prime ideal, it has height-1 (exercise 5.2.1(1)). Then by assumption, $\mathfrak{p}$ is principal so $\mathfrak{p} = (p)$ for some $p \in R$. Then $(r) \subseteq \mathfrak{p} = (p)$, hence $r = sp$ for some $s \in R$. But as r is irreducible, either s or p must be a unit, but p cannot be (as (p) is prime) so s is a unit. As R is an integral domain, $(r) = (p)$ are associates, and (r) is thus prime.

It is now a matter of putting everything together. As R is a UFD, $R[x_1, ..., x_n]$ is a UFD, and the graded structure respects unique decomposition (see [9]). Then by corollary 5.2.10, an irreducible codimension 1 subscheme $X \subseteq \mathbb{P}^n_R$ corresponds to a height-1 prime ideal $\mathfrak{p}$, which by the above must be principal; let $p \in R$ so that $(p) = \mathfrak{p}$. Then $V((p)) = X$, completing the proof.

Definition 5.2.12: Degree of Projective Hypersurface

Let R be a UFD and $X \subseteq \mathbb{P}^n_R$ be a hypersurface. By corollary 5.2.11, let $f \in R[x_0, ..., x_n]$ be the function corresponding to X. Then

$$\deg X := \deg f$$

We conclude this section by finally showing why tensoring quasi-coherent sheaves is not closed:

Example 5.2.13: Lack of Closure For Quasi-coherent Sheaves
Take $\mathscr{F} = \mathcal{O}(-1)$ and $\mathscr{G} = \mathcal{O}(1)$. Then let us compare:

$$(\mathscr{F} \otimes \mathscr{G})(U) \quad \text{and} \quad (\mathscr{F}(U) \otimes \mathscr{G}(U)$$

The left hand side is $\tilde{R}$ and hence is 1 dimensional, while the right hand side is 0 dimensional (tensoring a 0 dimensional space with a 2 dimensional space).

Another example I came across is very interesting:

Example 5.2.14: Free Sheaves Need Not Be Projective Objects
Really cool! here

Exercise 5.2.1

1. (Krull Principal Ideal Theorem) Let R be a ring and $\emptyset \neq (x) \subsetneq R$ a nontrivial proper ideal. Show that if $\mathfrak{p}$ is a minimal prime ideal containing (x), $\mathfrak{p}$ has height-1. (hint: you will localize, and at some point use Nakayama's lemma)

5.2.1 Maps to Projective Space

It has long been foreshadowed that many of the common schemes that are worked with embed into projective space. In this section, we shall give the right conditions for finding such a mapping. We shall see that a morphism of a scheme X to a projective space is determined by finding an invertible sheaf $\mathcal{L}$ on X and a set of global sections.

Note that maps from projective space are better determined using cohomological properties (see chapter 6); theorem 3.5.28 determined that all proper schemes can be covered by projective schemes, showing how they are natural in some sense of the word.

Build-up: Global Sections and Finite Generation

Definition 5.2.15: Twisting Sheaf On $\mathbb{P}^n_X$

Let X be a scheme. Then the twisting sheaf $\mathcal{O}(1)$ on $\mathbb{P}^n_X$ is $g^*(\mathcal{O}(1))$ where:

$$g : \mathbb{P}^n_X \to \mathbb{P}^n_{\mathbb{Z}}$$

where $\mathbb{P}^n_X = X \times_{\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$ by definition 3.2.3.

Definition 5.2.16: Very Ample Sheaf

Let X be a Y-scheme, and $\mathcal{L}$ a line bundle on X. Then $\mathcal{L}$ is said to be very ample relative to Y if there is a locally closed immersion $\iota : X \to \mathbb{P}^n_Y$ for some n such that

$$\iota^*(\mathcal{O}(1)) \cong \mathcal{L}$$

Using very ample sheaves, we can in fact characterize projective schemes.

Lemma 5.2.17: Characterizing Projective using Ample Sheaf

if Y is Noetherian, then X is projective if and only if X is both proper and has a very ample sheaf relative to Y. Indeed, if X is projective over Y.

***Proof*:**
If X is projective, then certainly X is proper and there exists a very ample sheaf on X relative to Y, and there is a closed immersion $\iota : X \to \mathbb{P}^n_Y$ for some n, and so $\iota^*(\mathcal{O}(1))$ is very ample invertible sheaf on X.

Conversely, if X is proper over Y and $\mathcal{L}$ is very ample invertible sheaf, then $\mathcal{L} \cong \iota^*(\mathcal{O}(1))$ for some immersion $\iota : X \to \mathbb{P}^n_Y$. Then by exercise:HERE (4.4 in Hartshorne) the image of X is closed, so ι is a closed immersion, and hence X is projective over Y.

Recall proposition 4.1 .28 that showed that if if X a R-scheme and $f_0, \dots, f_n$ are functions on X (global sections) with no common zeros. Then there exists a scheme morphism $[f_0 : \cdots f_n] : X \to \mathbb{P}^n_R$. It was commented that not all maps to projective space arise in this form, for example $\mathrm{id} : \mathbb{P}^n_k \to \mathbb{P}^n_k$. In this section, we remedy that. To start, we shall look for sheaves that are generated by their global section. The most important example will be when $S = k[x_0, \dots, x_n]$ and $X = \operatorname{Proj} S$; then the global section of a coherent sheaf must be finitely generated.

Definition 5.2.18: Generated By Global Sections

Let X be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is generated by global sections if there is a family of global sections $\{s_i\}_{i\in I} \subseteq \Gamma(X, \mathscr{F})$ such that for each $x \in X$, the images of s_i in the stalk $\mathscr{F}_x$ generate $\mathscr{F}_x$ as an $\mathcal{O}_{X,x}$-module.

Naturally, $\mathscr{F}$ is generated by global sections iff $\mathscr{F}$ can be written as the quotient of a free-sheaf.

Example 5.2.19: Generated By Global Sections

1. Let $X = \operatorname{Spec} R$. Then any $\mathscr{F} = \widetilde{M}$ is generated by global sections, simply pick generators for M as an R-module
2. Let $X = \operatorname{Proj} S$ for a graded ring S generated by S_1 as an S_0-algebra. Then the elements of S_1 can give global sections of $\mathcal{O}_X(1)$ that generate it.
3. The scheme $\mathcal{O}_X(-1)$ is not generated by global sections, as it has an empty global section.

Theorem 5.2.20: Global Section Generation of twisted Sheaf over Proj Noetherian

Let X be a projective scheme over a Noetherian ring R, let $\mathcal{O}(1)$ be a very ample invertible sheaf on X, and let $\mathscr{F}$ be a coherent $\mathcal{O}_X$-module. Then there is an integer n_0 such that for all $n \geq n_0$, the sheaf $\mathscr{F}(n)$ can be generated by a finite number of global sections.

***Proof*:**
Let $i : X \to \mathbb{P}^r_R$ be a closed immersion of X into a projective space over R, such that $i^*(\mathcal{O}(1)) = \mathcal{O}_X(1)$. Then $i_*\mathscr{F}$ is coherent on $\mathbb{P}^r_R$ as closed immersion are finite morphisms and projective schemes are Noetherian, and $i_*(\mathscr{F}(n)) = (i_*\mathscr{F})(n)$ by proposition 4.2.5 and $\mathscr{F}(n)$ is generated by global sections if and only if $i_*(\mathscr{F}(n))$ is (in fact, their global sections are the same), so this can be reduced to the case $X = \mathbb{P}^r_R = \operatorname{Proj} R[x_0, \dots, x_r]$.
Now cover X with the open sets $D_+(x_i), i = 0, \dots, r$. Since $\mathscr{F}$ is coherent, for each i there is a finitely generated module M_i over $B_i = R[x_0/x_i, \dots, x_n/x_i]$ such that $\mathscr{F}|_{D_+(x_i)} \cong \tilde{M}_i$. For each i, take a finite number of elements $s_{ij} \in M_i$ which generate this module. By lemma 4.1.12 there is an integer n such that $x_i^n s_{ij}$ extends to a global section t_{ij} of $\mathscr{F}(n)$. As usual, we take one n to work for all i, j. Now $\mathscr{F}(n)$ corresponds to a B_i-module M_i' on $D_+(x_i)$, and the map $x_i^n : \mathscr{F} \to \mathscr{F}(n)$ induces an isomorphism of M_i to M_i'. So the sections $x_i^n s_{ij}$ generate M_i', and hence the global sections $t_{ij} \in \Gamma(X, \mathscr{F}(n))$ generate the sheaf $\mathscr{F}(n)$ everywhere, as we sought to show.

Corollary 5.2.21: Coherent Sheaf is quotient of "free twisted sheaves"

Let X be projective R-scheme. Then any coherent sheaf $\mathscr{F}$ on X can be written as a quotient of a sheaf $\mathcal{E}$, where $\mathcal{E}$ is a finite direct sum of twisted structure sheaves $\mathcal{O}(n_i)$ for various integers n_i.

***Proof*:**
Let $\mathscr{F}(n)$ be generated by a finite number of global sections. Then we have a surjection $\bigoplus_{i=1}^r \mathcal{O}_X \to \mathscr{F}(n) \to 0$. Tensoring with $\mathcal{O}_X(-n)$ we obtain a surjection $\bigoplus_{i=1}^r \mathcal{O}_X(-n) \to \mathscr{F} \to 0$ as required.

From this, we can now classify a large number of coherent sheaves of on projective space:

Theorem 5.2.22: Finite Generation of Coherent Sheaf Over Projective Scheme

Let k be a field, R a finitely generated k-algebra, $X = \mathbb{P}^n_R$, and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathscr{F})$ is a finitely generated R-module. If $R = k$, then $\Gamma(X, \mathscr{F})$ is a finite dimensional k-vector space.

This proof can be thought of the generalization of the proof that $\mathcal{O}_{\mathbb{P}^n_k}(\mathbb{P}^n_k) = k$. Note too that this is not true if the space is not projective, for example over $\mathbb{A}^n_k$ for any n the resulting space is not a finite dimensional k-vector space, as $\Gamma\left(\mathbb{A}^n_k, \mathcal{O}_{\mathbb{A}^n_k}\right) = k[x]$ which is infinite dimensional!

***Proof*:**
Let $X = \operatorname{Proj} S$, where S is a graded ring with $S_0 = R$ which is finitely generated by S_1 as an S_0-algebra, corollary 4.2 .10 . Let M be the graded S-module $\Gamma_*(\mathscr{F})$. Then by proposition 4.2.9, $\tilde{M} \cong \mathscr{F}$. On the other hand, by theorem 4.4.4 for n sufficiently large, $\mathscr{F}(n)$ is generated by a finite number of global sections in $\Gamma(X, \mathscr{F}(n))$. Let M' be the submodule of M generated by these sections. Then M' is a finitely generated S-module. Furthermore, the inclusion $M' \hookrightarrow M$ induces an inclusion of sheaves $\tilde{M}' \hookrightarrow \tilde{M} = \mathscr{F}$. Twisting by n we have an inclusion $\tilde{M}'(n) \hookrightarrow \mathscr{F}(n)$ which is actually an isomorphism, because $\mathscr{F}(n)$ is generated by global sections in M'. Twisting by $-n$ we find that $\tilde{M}' \cong \mathscr{F}$. Thus $\mathscr{F}$ is the sheaf associated to a finitely generated S-module, and so we have reduced to showing that if M is a finitely generated S-module, then $\Gamma(X, \tilde{M})$ is a finitely generated R-module.
Now, recall from [9] that there is a finite filtration

$$0 = M^0 \subseteq M^1 \subseteq \ldots \subseteq M^r = M$$

of M by graded submodules, where for each $i, M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogeneous prime ideal $\mathfrak{p}_i \subseteq S$, and some integer n_i. This filtration gives a filtration of $\tilde{M}$, and the short exact sequences

$$0 \to \widetilde{M^{i-1}} \to \widetilde{M^i} \to \widetilde{M^i/M^{i-1}} \to 0$$

give rise to left-exact sequences

$$0 \to \Gamma\left(X, \widetilde{M^{i-1}}\right) \to \Gamma\left(X, \widetilde{M^i}\right) \to \Gamma\left(X, M^i \widetilde{/M^i} - 1\right)$$

Thus to show that $\Gamma(X, \tilde{M})$ is finitely generated over R, it will be sufficient to show that $\Gamma\left(X, (S/\mathfrak{p})^{\sim}(n)\right)$ is finitely generated, for each $\mathfrak{p}$ and n. Thus we have reduced to the following special case: Let S be a graded integral domain, finitely generated by S_1 as an S_0-algebra, where $S_0 = R$ is a finitely generated integral domain over k. Then $\Gamma\left(X, \mathcal{O}_X(n)\right)$ is a finitely generated R-module, for any $n \in \mathbf{Z}$.
Let $x_0, \ldots, x_r \in S_1$ be a set of generators of S_1 as an R-module. Since S is an integral domain, multiplication by x_0 gives an injection $S(n) \to S(n+1)$ for any n. Hence there is an injection $\Gamma\left(X, \mathcal{O}_X(n)\right) \to \Gamma\left(X, \mathcal{O}_X(n+1)\right)$ for any n. Thus it is sufficient to prove $\Gamma\left(X, \mathcal{O}_X(n)\right)$ finitely generated for all sufficiently large n, say $n \geq 0$.
Let $S' = \bigoplus_{n \geq 0} \Gamma\left(X, \mathcal{O}_X(n)\right)$. Then S' is a ring, containing S, and contained in the intersection $\bigcap S_{x_i}$ of the localizations of S at the elements $x_0, \ldots, x_r$.
Next, let's show that S' is integral over S. Let $s' \in S'$ be homogeneous of degree $d \geq 0$. Since $s' \in S_{x_i}$ for each i, we can find an integer n such that $x_i^n s' \in S$. Choose one n that works for all i. Since the x_i generate S_1, the monomials in the x_i of degree m generate S_m for any m. So by taking a larger n, we may assume that $ys' \in S$ for all $y \in S_n$. In fact, since s' has positive degree, we can say that for any $y \in S_{\geq n} = \bigoplus_{e > n} S_e, ys' \in S_{\geq n}$. Now it follows inductively, for any $q \geq 1$ that $y \cdot (s')^q \in S_{\geq n}$ for any $y \in S_{\geq n}$. Take for example $y = x_0^n$. Then for every $q \geq 1$ we have $(s')^q \in (1/x_0^n)\, S$. This is a finitely generated sub- S-module of the quotient field of S'. It follows by a well-known criterion for integral dependence (see [9]), that s' is integral over S. Thus S' is contained in the integral closure of S in its quotient field.
To complete the proof, since S is a finitely generated k-algebra, by commutative algebra we know that S' will be a finitely generated S-module. It follows that for every n, S_n is a finitely generated S_0-module, as we sought to show.
In fact, this proof shows that $S_n = S_n'$ for all sufficiently large n, see exercises ref:HERE

Note that it was not technically needed that R is a finitely generated k-algebra, but just a "Nagata ring". See ref:HERE for a generalization. As almost all (Noetherian) rings are Nagata, this isn't much of a generalization.

Corollary 5.2.23: Pushforward of Coherent Sheaf by Projective Morphism

Let $f : X \to Y$ be a projective morphism of schemes of finite type over a field k. Let $\mathscr{F}$ be a coherent sheaf on X. Then $f_*\mathscr{F}$ is coherent on Y.

Recall this is not true if the map is not projective, see example 4.6

***Proof*:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$, where R is a finitely generated k-algebra. Then in any case, $f_*\mathscr{F}$ is quasi-coherent by proposition 4.1.14, so $f_*\mathscr{F} = \Gamma\left(Y, f_*\mathscr{F}\right)^{\sim} = \Gamma(X, \mathscr{F})^{\sim}$. But $\Gamma(X, \mathscr{F})$ is a finitely generated R-module by the theorem, so $f_*\mathscr{F}$ is coherent.

This result shall have a cohomological proof which will allow us to generalize to proper morphisms, see ref:HERE.

Characterizing Morphisms to Projective Space

Let R be a fixed ring and $\mathbb{P}^n_R = \operatorname{Proj} R[x_1, ..., x_n]$ be our projective space over R. Then $\mathbb{P}^n_R$ has an invertible sheaf, denoted $\mathcal{O}(1)$. Let $x_1, ..., x_n$ be the homogeneous coordinates that generate the global section $\Gamma(\mathbb{P}^n_R, \mathcal{O}(1))$. Note that the images of $x_1, ..., x_n$ generate the stalks $\mathcal{O}(1)_p$ of the sheaf $\mathcal{O}(1)$ as a module over the local ring $\mathcal{O}_p$.

Now, let X be any R-scheme, and assume the existence of an R-morphism $\varphi : X \to \mathbb{P}^n_R$. Then it must be that $\varphi^*(\mathcal{O}(1)) = \mathcal{L}$ is an invertible sheaf on X, and the global sections $s_1, ..., s_n$ where $s_i = \varphi^*(x_i)$ generate $\mathcal{L}$.

The converse, that an invertible sheaf $\mathcal{L}$ with chosen global sections s_i determine φ, is also true:

Theorem 5.2.24: Determining Map To Projective Space

Let R be a ring, X an R-scheme. Then:

1. If $\varphi : X \to \mathbb{P}^n_R$ is an R-morphism, then $\varphi^*(\mathcal{O}(1))$ is an invertible sheaf on X which is generated by global sections ${}_i = \varphi^*(x_i)$ for $i = 0, 1, ..., n$.
2. If $\mathcal{L}$ is an invertible sheaf on X and $s_0, ..., s_n \in \Gamma(X, \mathcal{L})$ are global sections which generate $\mathcal{L}$, then there exist a unique R-morphism $\varphi : X \to \mathbb{P}^n_R$ such that
$$\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$$
and $s_i = \varphi^*(x_i)$ under this isomorphism.

***Proof*:**
part (1) was proven above the theorem, so we focus on (2).

let $\mathcal{L}$ be a line bundle with global sections $s_0, ..., s_n$ that generate it. For each i, let:

$$X_i“ = ”D(s_i) = \{P \in X \mid (s_i)_P \notin \mathfrak{m}_P \mathcal{L}_P\}$$

be the natural open sets that cover X. Define the morphism from X_i to the standard open subset $U_i \cong \operatorname{Spec} R[y_0, ..., y_n] \subseteq \mathbb{P}^n_R$ (where $y_j = x_j/x_i$) by sending:

$$R[y_0, ..., y_n] \to \Gamma(X_i, \mathcal{O}_X) \qquad y_j \to \frac{s_j}{s_i}$$

and extending R-linearly. This is well-defined since for each $P \in X_i$, $(s_i)_P \notin \mathfrak{m}_P \mathcal{L}_P$ and $\mathbb{C}$ is locally free of rank 1 making s_j/s_i well-defined in $\Gamma(X_i, \mathcal{O}_{X_i})$. But now, by proposition 1.3.3 these morphisms determine a morphism $\varphi : X \to \mathbb{P}^n_R$, and clearly $\mathcal{L} \cong \varphi^*(\mathcal{O}(1))$. Certainly any morphism with these property uniquely determines φ, which completes the proof.

Example 5.2.25: Maps Into Projective Space

1. (characterizing automorphism)
2. (open subsets)

5.3 Divisors: Geometric Representation of Line Bundles

Having proven the connection between line bundles and maps to/from projective space, we focus on their "geometry". In a similar that functions create zero sets, we shall see that *divisors* arise as zero-sets of *sections* of line bundles. Note that functions are sections of a trivial line bundle and hence divisors are a generalization of zero-sets of functions.

In fact, divisors are a way of measuring how far away the geometry of your space is constructed by "functions" (i.e. elements of the global section of the structure sheaf). There are two ways of approaching this problem. For the first, recall that if V is a variety and $f \in \mathcal{O}_V$, then $V(f)$ is a hypersurface. Recall further that a hypersurface is defined to be a codimension 1 subvariety (more generally a codimension 1 subscheme). Then we may ask which hypersurfaces are given as the zero's of global sections. The first approach shall yield *Weil Divisors* while the second approach shall yield *Cartier Divisors*. In most nice cases (think Integral Noetherian Schemes), these two shall give the same notion of divisor. If you have read [9, chapter 22.7], then in this section we shall generalize the notion of divisors to general schemes and show hey they relate to a *Picard Scheme*. We shall also show how divisors naturally correspond to line bundles which shall let us use divisors to gain further important results about schemes[2].

The intuition for divisors works well when starting with examples from complex analysis (in particular, see [4, chapter 3 and 6]). In this case, the two notions of divisors shall coincide and so we shall be finding the divisors for Riemann surface X. Starting with $X = \mathbb{C}$, consider the free-abelian group generated by the points of X, so that each element of this group is of the form:

$$\sum_i n_i p_i \qquad n_i \in \mathbb{Z},\ p_i \in X$$

We may ask if there is a meromorphic function associated to this function. As we are interested in algebraic constructions, we limit ourselves to function with a finite number of poles and zero, that is all *rational functions*. Then We get that for each $\sum_i n_i p_i$, we can associated to it the meromorphic function;

$$(z - p_i)^{n_i}$$

Let us call the set of meromorphic functions the *principle divisors*, and the finite formal $\mathbb{Z}$-linear combinations the *divisors*. Then the quotient of these two groups is zero:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} = 0$$

Let us now consider X to be the Riemann surface. We shall once again take the divisor group to be the free group. This time, the principal divisor group (the group of meromorphic functions) has a fundamental limitations placed on it: recall that the sum of the zero's and pole of a meromorphic function is zero[3]. In this case, the sum of each divisor is some integer $n \in \mathbb{Z}$. Using this, you can show that:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} \cong \mathbb{Z}$$

Finally, consider an elliptic curve given by $\mathbb{C}/L$ where L is a lattice. Richard Borcherds, finish off!

[2]For example, this section is foundational in the establishment of the *Riemann-Roch Theorem* which is one of the fundamental results in roughly classifying curves

[3]Recall that by adding the point at infinite, we shall add the "compliment" of each zero/pole; convince yourself that z^2 has two zeros and two poles

5.3.1 Weil Divisors

Weil divisors generalize the above discussion on (compact) Riemann surfaces. We shall requires that there is as notion of "degree of zero" at a point. There are many ways of doing this, and for clarity of presentation of the material we shall limit to scheme that have a valuation rings for each stalk to have a well-defined degree. For this, we shall require the schemes to be nice enough for this (we shall present schemes where this is not so obvious in example ref:HERE). Naturally, if we are working with curves, it suffices to work with nonsingular schemes as each stalk is a DVR. To emulate this in higher dimensions, we want the principal components to be associated to functions which vanish, but in higher dimensions such components would be codimension 1. Thus we shall want our schemes to be at least regular in the codimension components, that is we are going to be working over some relatively nice schemes:

Definition 5.3.1: Regular Codimension 1 Scheme

Let X be a scheme. Then X is said to be regular of codimension 1 if every local ring $\mathcal{O}_{X,\mathfrak{p}}$ of X of dimension 1 is regular

Example 5.3.2: Regular Codimension 1 Scheme

1. Naturally, all regular curves (and regular schemes) satisfy this criterion, as the codimension 1 local rings are the local rings of points, which are all regular by definition
2. if X is a Noetherian regular scheme, any local ring of dimension 1 is an integrally closed domain (a normal domain), and hence by [9] it is a DVR.

As all local rings at each prime of codimension 1 is regular, there exists a natural notion of valuation, which is the key tool we want to use. If they were not regular, we can instead take the length of $R/(f)$; we still have the main important property which is that the order of fg is the order of f plus the order of g by doing the following short exact sequence:

$$0 \to R/g \xrightarrow{\times f} R/fg \to R/f \to 0$$

which is key as it gives a homomorphism between K^* and the Weil divisor groupie shall defined shortly. It is a bit easier to work with valuations and it gives us the core information we need, hence we still with them. Note further that if X was not integral, we would also insure to take f to be a nonzero divisor.

For the sake of narrative clarity, we shall be adding some more conditions on our schemes: Noetherian is to keep things finite, integral is to have a generic point for the entire scheme to work with, and separated is to have injective restriction maps to not have any issue picking functions:

Definition 5.3.3: Weil Divisor

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then a prime divisor is a closed integral subscheme. A Weil divisor is an element of the free abelian group generated by the prime divisors. This group is denoted:

$$\operatorname{WDiv} X$$

If $D = \sum_i n_i Y_i$ and all $n_i \geq 0$, we say that D is effective.

Let $Y \subseteq X$ be as in he above definition. Then Y has a generic point, say η, and hence it has a stalk $\mathcal{O}_{Y,\eta}$ which has dimension 1 by choice of X and Y and hence is a DVR. Label the valuation ν_Y. As X is separated, Y is uniquely determined by its valuation (see exercise 3.3.2.)

Now, let $k(X)$ denote the function field of X (remember we are assuming X is integral). Pick $f \in (k(X))^*$, a nonzero rational function. Then $\nu_Y(f)$ is an integer. If it is positive, it represents the order of vanishing of f on Y, and if it is negative, it represents the order of the pole along Y (we shall take the order of the pole to be the absolute value of the valuation). As we are dealing with the codimension 1 case, by differential geometry we would expect that almost all function do not vanish along Y, and indeed:

Lemma 5.3.4: Almost All Function Non Vanishing

Let X be a Noetherian integral separated scheme, regular in codimension 1. Let $f \in (k(X))^*$. Then $\nu_Y(f) = 0$ for all except finitely many prime divisors Y

***Proof*:**
As rational function must be regular on some open subset, let's say f is regular on $U = \operatorname{Spec} R \subseteq X$. Then $C = X \backslash U$ is a proper closed subset of X. As X is Noetherian,

$$C = \bigcup_i^n Z_i$$

for irreducible closed subsets Z_i which each must have codimension ≥ 1. Then any irreducible component contained in C, i.e. any prime divisors, would have to be one of the Z_i 's, and so C must contain at most finitely many prime divisors (the rest meet U). Thus, if we show here are only finitely many primes divides Y of U for which $\nu_Y(f) \neq 0$, we are done.

As f is regular on $U, \nu_Y(f) \geq 0$ and $\nu_Y(f) > 0$ if and only if Y is contained in a closed subset of U defined by the principal ideal (f). As $f \neq 0$, this closed subset is proper, and so it can only contains finitely many irreducible subsets of codimension 1 on U, but then we have that there are only finitely many prime divisors Y of U for which $\nu_Y(f) \neq 0$, as we sought to show.

Thus, our sum is well-defined:

Definition 5.3.5: Principal Divisor

Let $f \in (\kappa(X))^*$. Then the principal divisor associated to f is:

$$\operatorname{Div}(f) = \sum_i \nu_Y(f)Y \in \operatorname{Div} X$$

This group is sometimes denoted PDiv X.

Note that $\operatorname{Div}(f/g) = \operatorname{Div} f - \operatorname{Div} g$ by the properties of discrete valuations, and hence this is even a group homomorphism from the multiplicative group $(\kappa(X))^*$ to the additive group $\operatorname{Div} X$!

Definition 5.3.6: Divisor Class Group

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then the divisor class group of X is:

$$\mathrm{Cl}X = \frac{\operatorname{WDiv} X}{\operatorname{PDiv} X}$$

We shall give examples of the divisor class group for a few schemes, however let us first generalize a very important result from commutative algebra that will allows us to find many schemes with trivial divisor class groups. First, a technical lemma:

Lemma 5.3.7: Normal Domain: Intersection Height One Localizations

Let R be a normal domain (an integrally closed domain) Then

$$R = \bigcap_{\mathrm{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$$

***Proof*:**
By [9], we already have $R = \bigcap_{\mathfrak{p}} R_{\mathfrak{p}}$. If R is normal, then $R_{\mathfrak{p}}$ is normal, and since $\mathfrak{p}$ is of height 1 , then $R_{\mathfrak{p}}$ is a local normal domain of dimension 1, making it a DVR by [9]. Then it is not too hard to see that the height 1 prime ideals correspond to DVR's containing R, so we have:

$$R = \bigcap_{R \subseteq R_{\mathfrak{P}}} R_{\mathfrak{P}}$$

Then this reduces to a standard argument by contradiction: if $x \in \operatorname{Frac}(R)\backslash R$, as R is normal we must have x not integral over R. As x is not integral over R, $1/x$ must be contained in some maximal ideal (consider the map $R[s,t] \to R[x, 1/x]$ mapping $s \mapsto x$ and $t \mapsto 1/x$; find the maximal idea via the image of the map); call this ideal $\mathfrak{m}$. Then we have $R_{\mathfrak{m}}$, and it certainly contains R, but then $x \in R$, hence our original assumption is void, and we get the equality.

Proposition 5.3.8: Characterizing UFD Geometrically

Let R be a Noetherian domain. Then R is a UFD if and only if $\operatorname{Spec} R$ is normal and Cl Spec $R = 0$.

***Proof*:**
As R is a UFD, it is normal (common algebraic exercise, or see [9]), hence $\operatorname{Spec}(R)$ is normal. It was shown in the proof of corollary corollary 5.2.11 that R is a UFD iff every height-1 prime ideal is principal. From this, we'll show that when every height-1 prime is principal, $\mathrm{ClSpec} R = 0$. If every prime of height 1 is principal, then every prime divisor Y is mapped to by a p, and so $\operatorname{Cl}\operatorname{Spec} R = 0$.

Conversely, given R Is normal and $\operatorname{Cl}\operatorname{Spec} R = 0$, it suffices to show that every height-1 prime $\mathfrak{p}$ is principal. First, for choose a prime divisor Y. As ClSpec $R = 0$, we know that there is some $p \in (\kappa(X))^*$ that maps to Y; it corresponds to some prime $\mathfrak{p}$. We must show that $p \in R^*$, and that $(p) = \mathfrak{p}$. As $\mathfrak{p}$ is the prime associated to the prime divisor Y and p map to it,

$$\nu_Y(p) = 1$$

Hence $p \in R_{\mathfrak{p}}$ (as it is ≥ 0) and p generates $\mathfrak{p}R_{\mathfrak{p}}$ (as $\nu_Y(p) = 1$). Next, if $\mathfrak{p}'$ is any other height 1 prime ideal, it correspond to a prime divisor Y', and as $p \in \kappa(X)^*$ we have

$$\nu_{Y'}(p) = 0$$

and so $p \in R_{\mathfrak{p}'}$. Then by lemma 4.3 .6 we have $p \in R$. Finally, let us show that p generates $\mathfrak{p}$. Let $g \in \mathfrak{p}$. Then:

$$\nu_Y(g) \geq 1 \quad \text{and} \quad \nu_{Y'}(g) \geq 0$$

Then for all prime divisors Y' (including Y):

$$\nu_{Y'}(g/p) \geq 0$$

Thus, $g/p \in R_{\mathfrak{p}'}$ for all height 1 prime ideals, implying by lemma 4.3.6 that $g/p \in R$, that is $g \in pR = (p)$. Hence $\mathfrak{p} \subseteq (p)$, and so $\mathfrak{p} = (p)$, as we sought to show.

Example 5.3.9: Divisor Class Groups

1. Affine space has trivial class group, $\mathrm{Cl}\mathbb{A}_k^n = 0$, as it is affine and $k[x_1, \ldots, x_n]$ is a UFD.
2. Let R be a Dedekind domain and consider Spec R. Then it is not a UFD, hence it must have a non-trivial divisor class group. The height 1 primes are precisely the primes of R, which as we know generate an abelian group known as the ideal class group. But this is just the divisor class group under a different name, and similarly with the principal divisors corresponding to elements of $\mathrm{Frac}(R)$. Hence, the ideal class group corresponds with the divisor class group!

It is natural to look for the divisor class group for projective space. The reader familiar with some differential geometry would hope that we recover the group of integers, and this is indeed the case. Recall that the degree of a hypersurface in $\mathbb{P}^n_k$ is the degree of the homogeneous polynomials defining it

Proposition 5.3.10: Characterizing Divisors of Projective Space

Let $X = \mathbb{P}^n_k$. For any divisor $D = \sum_i n_i Y_i$, define $\deg D = \sum_i n_i \deg Y_i$, and let H be the hyperplane given by x_0. Then:

1. for any $f \in \kappa(X)^*, \operatorname{Div} f = 0$
2. if D is a divisor of degree d, then $[D] = [dH]$ in $\operatorname{Cl}\mathbb{P}^n_k$
3. The degree functions give an isomorphism $\deg : \operatorname{Cl}\mathbb{P}^n_k \to \mathbb{Z}$

***Proof*:**

1. Let $S = k[x_0, \ldots, x_n]$ be the homogeneous coordinate ring of X and $f \in S$ a homogeneous element of order d. Then f factors into irreducibles
$$f = f_1^{n_1} \cdots f_r^{n_r}$$
Each f_i defines a hypersurface Y_i of degree d_i (i.e. an irreducible codimension 1 subscheme), and so
$$\operatorname{div} f = \sum n_y Y_i$$
hence $\deg f = d$. Note that f is not a rational function, it is a global section of the line bundle $\mathcal{O}(D)$, but $\operatorname{div} f$ is still well-defined by corollary 5.2.11. Now if $F = f/g$ is a rational function on $\mathbb{P}^n_k$ so that the degree of f, g are equal, then by the homomorphism property we have $\operatorname{div} F = \operatorname{div} f - \operatorname{div} g$, and so $\deg F = 0$

2. let $D = \sum_i n_i Y_i$ be any divisor of degree d. Then D can be written as the difference of effective divisors with degrees d_1, d_2, $D = D_1 - D_2$ where $d = d_1 - d_2$. As the irreducible hypersurfaces in $\mathbb{P}^n_k$ correspond to homogeneous prime ideals of height 1 in S and these are principal (corollary 5.2.11), let $f_{1,j}$ correspond to $Y_{1,j}$ and $f_{2,j}$ correspond to $Y_{2,j}$, then let $f_i = \prod_{j=1}^{n_i} f_{i,j}$ for $i = 1, 2$. Then $D_1 = \operatorname{Div} f_1$ and $D_2 = \operatorname{Div} f_2$ so that $D = \operatorname{div} f_1/f_2$. Then:
$$D - dH = \operatorname{div} \frac{f_1}{f_2 x_0^d}$$
Hence the two are off by a principal divisor, meaning $[D] = [dH]$.
3. To complete the proof, if $[dH] \neq [d'H]$ then $d \neq d'$. Since $\deg H = 1$, we get an isomorphism:
$$\deg : \operatorname{Cl} X \to \mathbb{Z}$$
completing the proof.

Proposition 5.3.11: Divisor Class Group and Closed Subsets

Let X be a Noetherian integral separated scheme, regular in codimension 1. Let $Z \subseteq X$ be a proper closed subset and $U = X \setminus Z$. Then:

1. There is a surjective homomorphism $\operatorname{Cl} X \to \operatorname{Cl} U$ given by:

$$\sum n_i Y_i \mapsto \sum n_i (Y_i \cap U)$$

2. If $\operatorname{codim}(Z, X) \geq 2$, then $\operatorname{Cl} X \to \operatorname{Cl} U$ is an isomorphism

3. If Z is an irreducible subset of codimension 1, then there is an exact sequence:

$$\mathbb{Z} \to \operatorname{Cl} X \to \operatorname{Cl} U \to 0$$

   where the first map is given by $1 \mapsto 1 \cdot Z$

***Proof*:**

1. If Y is a prime divisor, then $Y \cap U$ is either empty or a prime-divisor. Then if $f \in \kappa(X)^*$ and $\operatorname{div} f = \sum n_i Y_i$, then we can consider f as as rational function on U by:

$$\operatorname{div}_U f = \sum n_i (Y_i \cap U)$$

   Then this is certainly a homomorphism, and is surjective as the prime divisors of U are the restriction of its closure in X.

2. As $\operatorname{div} X$ is generated on the codimension 1 subsets, and removing a codimension 2 subset does not effect the Dimensionality of the other subsets, we have an isomorphism.

3. The kernel of $\operatorname{Cl} X \to \operatorname{Cl} U$ are all divisors whose support is contained in Z. Then as Z is irreducible, the kernel is the subgroup of $\operatorname{Cl} X$ generated by $1 \cdot Z$, completing the proof.

Example 5.3.12: More Divisor Class Groups

let $Y \subseteq \mathbb{P}^2_k$ be a degree d irreducible curve. Then combining the above observations we get:

$$\operatorname{Cl}(\mathbb{P}^2_k \setminus Y) = \mathbb{Z}/d\mathbb{Z}$$

Intuitively, By Bézout's theorem we have that any line L intersects a curve of degree d a total of d, counting multiplicity. Using this, can you piece together what the torsion in this group represents?

For a concrete example, let $\operatorname{Spec} R = \operatorname{Spec} k[x, y, z]/(xy - z^2)$ be the affine cone of $xy = 1$. Let's show that $\operatorname{Cl} \operatorname{Spec} R = \mathbb{Z}/2\mathbb{Z}$, which will be generated by the "ruling of the cone" $y = z = 0$. Let Y represent this subscheme. Visually:

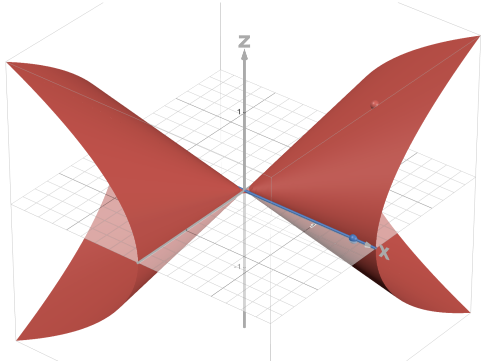


Figure 5.1: A ruling on the quadratic cone

As Y is irreducible and 1-codimension in $\operatorname{Spec} R$, Y is a prime divisor. Hence we can consider:

$$\mathbb{Z} \to \operatorname{Cl} X \to \operatorname{Cl}(\operatorname{Spec} R \setminus Y) \to 0$$

Now, on $\operatorname{Spec} R$, notice that Y can be cut out by the function y: $Y = V(y)$. It may then seem that $\operatorname{div} y = 1 \cdot Y$, however we in fact have

$$\operatorname{div} y = 2 \cdot Y$$

To see this, notice that $y = 0$ implies $z^2 = 0$. Then as z generates the maximal ideal at the generic point of Y, we see that $\nu_Y(y) = 2$. Now, certainly:

$$\operatorname{Spec} R \setminus Y \cong \operatorname{Spec} R_y \qquad R_y = \frac{k[x, y, z, y^{-1}]}{(xy - z^2)}$$

Then as $x = y^{-1}z^2$, we can eliminate x and reduce to:

$$R_y = k[y, z, y^{-1}]$$

This is now UFD, an thus:

$$\operatorname{Cl}(\operatorname{Spec} R \setminus Y) = 0$$

And so $\operatorname{Cl} X$ must be generated by Y, where $2 \cdot Y = 0$. Then:

$$\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z} \qquad \text{or} \qquad \operatorname{Cl} X = 0$$

If we show Y is not principal, then we have the former. As R is normal (classical exercise to show $k[x_1, ..., x_n, z]/(z^2 - f)$ is integrally closed where $\operatorname{char} k \neq 2$ and f is a square-free nonconstant polynomial), it is enough to show that the prime ideal of Y is not principal, that is $\mathfrak{p} = (y, z)$ is not principal. We reduce to the 3-dimensional vector space $\mathfrak{m}/\mathfrak{m}^2$ where $\mathfrak{m} = (x, y, z)$. Then as $\mathfrak{p} \subseteq \mathfrak{m}$, it is easy to see that $\mathfrak{p}$ in $\mathfrak{m}/\mathfrak{m}^2$ must be generated by the basis elements $\overline{y}$ and $\overline{z}$. But then $\mathfrak{p}$ cannot be principal, and hence we can conclude that

$$\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z}$$

Proposition 5.3.13: Class Group Stably Isomorphic

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then $X \times \mathbb{A}^1_{\mathbb{Z}}$ also satisfies all these conditions, and:

$$\operatorname{Cl} X \cong \operatorname{Cl}(X \times \mathbb{A}^1_{\mathbb{Z}})$$

***Proof*:**
Certainly $X \times \mathbb{A}^1$ is Noetherian, integral, and separated (verify this if this is not clear). To see that it is regular in codimension one, first notice that there are two kinds of points of codimension one on $X \times \mathbb{A}^1$.

1. A point x whose image in X is a point y of codimension one. In this case x is the generic point of $\pi^{-1}(y)$, where $\pi : X \times \mathbb{A}^1 \to X$ is the projection. Its local ring is $\mathcal{O}_x \cong \mathcal{O}_y[t]_{\mathfrak{m}}$, which is clearly a discrete valuation ring, since $\mathcal{O}_y$ is. The corresponding prime divisor $\overline{\{x\}}$ is just $\overline{\pi^{-1}(\{y\})}$.
2. A point $x \in X \times \mathbb{A}^1$ of codimension one, whose image in X is the generic point of X. In this case $\mathcal{O}_x$ is a localization of $K[t]$ at some maximal ideal, where K is the function field of X. It is a discrete valuation ring because $K[t]$ is a principal ideal domain.

Thus $X \times \mathbb{A}^1$ also regular in codimension 1. For the rest of this proof, let's call these points *type 1* and *type 2*.

Now, define a map

$$\operatorname{Cl} X \to \operatorname{Cl}(X \times \mathbb{A}^1) \qquad \sum n_i Y_i \mapsto \pi^* D = \sum n_i \pi^{-1}(Y_i)$$

If $f \in \kappa(X)^*$, then $\pi^*((f))$ is the divisor of f considered as an element of $\kappa(X)(t)$, the function field of $X \times \mathbb{A}^1$ (from now on, let $K = \kappa(X)$). Thus we have a homomorphism $\pi^* : \operatorname{Cl} X \to \operatorname{Cl}(X \times \mathbb{A}^1)$.

To show π^* is injective, suppose $D \in \operatorname{div} X$, and $\pi^* D = (f)$ for some $f \in K(t)$. Since $\pi^* D$ involves only prime divisors of type 1, f must be in K, otherwise we could write $f = g/h$, with $g, h \in K[t]$, relatively prime. If g, h are both not in K, then $\operatorname{div} f$ will involve some prime divisor of type 2 on $X \times \mathbb{A}^1$. Now if $f \in K$, it is clear that $D = \operatorname{div} f$, so π^* is injective.

To show that π^* is surjective, it will be sufficient to show that any prime divisor of type 2 on $X \times \mathbb{A}^1$ is linearly equivalent to a linear combination of prime divisors of type 1. So let $Z \subseteq X \times \mathbb{A}^1$ be a prime divisor of type 2. Localizing at the generic point of X, we get a prime divisor in $\operatorname{Spec} K[t]$, which corresponds to a prime ideal $\mathfrak{p} \subseteq K[t]$. This is principal, so let f be a generator. Then $f \in K(t)$, and the divisor of f consists of Z plus perhaps something purely of type 1. It cannot involve any other prime divisors of type 2. Thus Z is linearly equivalent to a divisor purely of type 1, completing the proof.

4 more examples in Hartshorne

5.3.2 Cartier Divisors

We shall now generalize this to scheme. The problem with working over a general scheme is that we don't have a function field, and we can contain zero-divisors. We must thus be a bit more careful with our definition. The key intuition we want to preserve is how far away a scheme is from being "principal". To that end, we shall define a divisor as elements that are "locally principal". In most cases, we focus only on divisors that have no poles. For full generality, we shall give the general definition that includes poles and build up to the most common use case.

Definition 5.3.14: Maximal Quotient Sheaf

Let X be a scheme. Then for each affine open $\operatorname{Spec} R = U \subseteq X$, take all non zero-divisors $S \subseteq \Gamma(U, \mathcal{O}_X)$ and take $S^{-1}\Gamma(U, \mathcal{O}_X) = K(U)$, the *maximal quotient field* of $\Gamma(U, \mathcal{O}_X)$. Then this forms a presheaf, whose sheafification is called the *total quotient ring* of $\mathcal{O}_X$. This sheaf will be denoted $\mathscr{K}_X$,

Let $\mathscr{K}_X^*$ denote the abelian group sheaf of invertible elements for each open subset, and let $\mathcal{O}_X^*$ denote the abelian group sheaf of all invertible elements.

If X is integral, then this recovers the usual rational functions we'd define on each open subsets.

Definition 5.3.15: Cartier Divisor

Let $(X, \mathcal{O}_X)$ be a scheme with maximal quotient sheaf $\mathscr{K}$. Then a *Cartier divisor* is an element of the global section of $\mathscr{K}^*/\mathcal{O}_X^*$:

$$\Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$$

The *principal Cartier divisors* are given by the image of The natural map

$$\Gamma(X, \mathscr{K}^*) \to \Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$$

An equivalent way of describing Cartier divisors that is very useful is by taking an open cover $\{U_i\}$ of X and then choosing elements $f_i \in \Gamma(U_i, \mathscr{K}^*)$ such that for each overlap $U_i \cap U_j$

$$\frac{f_i}{f_j} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^*)$$

or equivalently, $f_i = f_j$ up to a multiple of a section in $\Gamma(U_i \cap U_j, \mathcal{O}_Z^*)$. The idea is that we can loosen the definition of a function to only need to be equal on overlap by a unit, as units do no contribute to the "zero set".

Proposition 5.3.16: Cartier Divisors To Weil Divisors

Then there exists a homomorphism:

$$\operatorname{CDiv}(X) \to \operatorname{WDiv}(X)$$

***Proof*:**
Let $f \in \mathrm{CDiv}\,(X) = \Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$. By the above discussion, f can be given by a collection not $f_i \in K_i = \Gamma(U_i, \mathscr{K}^*)$. Then we can map each f_i to their corresponding Weil Divisor $\mathrm{div} f$ in U_i. Since K_i and K_j have the same zero's on the intersection, we get that we can glue these to get a Weil divisor D on X. Then it is routine to check this is a homomorphism, completing the proof.

Proposition 5.3.17: Weil and Cartier Equivalence Under Right Conditions

Let X be an integral separated Noetherian schemes, where all the local rings are UFD's. Then

$$\mathrm{WDiv}\,(X) \cong \mathrm{CDiv}\,(X)$$

and, under this same isomorphism, the principal Weil divisors correspond to the principal Cartier divisors

***Proof*:**
As X is a UFD at each stalk, it must be normal, and hence as shown in example 5.3.2 it is regular in codimension 1. As X is integral $\mathscr{K}$ is just the constant sheaf that corresponds to the function field $\kappa(X)$ of X, which we shall simply denote as K in this proof.

Let us start by showing a Cartier divisor gives a Weil divisor. By our remark, we can choose an open cover $\{U_i\}$ of X an element $f_i \in \Gamma(U_i, \mathscr{K}^*)$ that agree on intersections up to a multiple of $\mathcal{O}_X^*$. Now, for each prime divisor Y, take it's coefficient to be $\nu_Y(f_i)$ if $Y \cap U_i$. If $Y \cap U_j$ for some other j, then notice that f_i/f_j is invertible on $U_i \cap U_j$, and so must have trivial valuation, $\nu_Y(f_i/f_j)$, hence $\nu_Y(f_i) = \nu_Y(f_j)$. Thus, we can get:

$$D = \sum_i \nu_Y(f_i) Y$$

where the sum is finite as X is Noetherian.

Conversely, let D be a Weil divisor. Take any point $x \in X$. Then D induces a Weil divisor D_x on $\mathrm{Spec}\,\mathcal{O}_{X,x}$ ($\sum_i n_i Y$ becomes $\sum_i n_y Y \cap \mathrm{Spec}\,\mathcal{O}_{X,x}$). As $\mathcal{O}_{X,x}$ is a UFD, we have that this divisor be principal, and so $D_x = \mathrm{div} f_x$ for some $f_x \in K$ (as we are working with the constant sheaf). Now, $\mathrm{div} f_x$ and D on X both restrict to $\mathrm{div} f_x$ on $\mathrm{Spec}\,\mathcal{O}_{X,x}$, thus they can only differ by prime divisors that do not pass through x. By the Noetherian property and that they are codimension 1, hence can only be finitely many nonzero such divisors, so there is an open neighborhood U_x of x such that D and $\mathrm{div} f_x$ have the same restriction to U_x. Finally, we may cover X with such U_x. I claim that the functions f_x now give the Cartier divisor. Indeed, if f, f' gave the same Weil divisor on an open set U, then $f/f' \in \Gamma(U, \mathcal{O}_X^*)$ as X is normal (recall from the proof of proposition 5.3.8), and so these are indeed well defined Cartier divisors.

Now, it should be clear that these two construction are inverses of each other and certainly map principal to principal, giving us our bijective correspondence as we sought to show.

If the conditions are not met, then we may get a different groups, as we shall soon show. Note that regular local ring are UFD's (see [9]) and so this equivalence applies to regular integral separated Noetherian schemes. If X is a normal scheme, then it is not necessarily locally factorial. Instead, the correspondence goes thusly:

Definition 5.3.18: Locally Principal Weil Divisor

let X be a normal scheme. Then D is *locally principal* if X can be covered by open sets U such that $D|_U$ is principal for each U.

In this case, Cartier divisors correspond to locally principal Weil Divisors.

Definition 5.3.19: Cartier Divisor Class Group

Let X be a scheme. Then the *Cartier divisor class group* is the Cartier divisors divided by the principal Cartier divisors, and is denoted

$$\operatorname{CaCl} X$$

Not to be confused with calcium-chloride which almost has the same symbol: CaCl_2.

Example 5.3.20: Cartier Divisors

1. Go over Hartshorne p.142
2. This should ring bells of the definition of the ideal group; recall that the ideal group of $\mathbb{Z}$ was isomorphic to $\mathbb{Q}^*/\mathbb{Z}^* \cong \mathbb{Q}^*_{\geq 0}$. If we take $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$, Then it's total field is $\mathbb{Q}(\sqrt{-5})$ and it's group of units is ± 1, hence elements of $\Gamma(\operatorname{Spec}\mathbb{Z}[\sqrt{-5}], \mathbb{Q}(\sqrt{-5})_{>0})$ correspond to Cartier divisors. To see this in action, consider $(2, 1+\sqrt{-5}) \subseteq \mathbb{Z}[\sqrt{-5}]$ which we know is not principal. Taking the affine cover $D(2)$ and $D(1+\sqrt{-5})$ (as these correspond to coprime ideals), we can choose:

   $$\frac{2}{1+\sqrt{-5}} \in \Gamma(D(1+\sqrt{-5}), \mathscr{K}^*) \qquad \text{and} \qquad \frac{1+\sqrt{-5}}{2} \in \Gamma(D(2), \mathscr{K}^*)$$

   and we see that on the intersection:

   $$\frac{2}{1+\sqrt{-5}} = \frac{1+\sqrt{-5}}{2}$$

   up to a multiplication of -1 (which is in $\mathcal{O}^*_{\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]}$), and hence this gives a Cartier divisor.

   If we impose the same conditions as we did for Weil divisors, we shall see that these two groups are equivalent. In fact, we can weaken the codimension 1 regularity condition a bit:

5.3.3 Picard Group

Throughout the discussion of the divisors on projective schemes, it was clear that there is a strong connection to the line bundles $\mathcal{O}(d)$. In this section, we shall make this precise. This shall also be familiar to number theorists who recall that fractional ideals are finitely generated R-module J such $rJ \subseteq R$.

Proposition 5.3.21: Line Bundles K-theory Properties

let $\mathcal{L}, \mathcal{M}$ be invertible sheaves (line bundles) on a ringed space X. Then:

1. $\mathcal{L} \otimes \mathcal{M}$ is invertible
2. There exists an invertible sheaf $\mathcal{L}^{-1}$ such that $\mathcal{L} \otimes \mathcal{L}^{-1} \cong \mathcal{O}_X$

The way to think about the tensor product between line bundles is that they happen stalk-wise:

$$(\mathcal{L} \otimes \mathcal{M})_p \cong \mathcal{L}_P \otimes_p \mathcal{M}_p$$

which can be thought of as "multiplying" the values:

$$(s \otimes t)(p) = s(p) \otimes t(p)$$

As line bundles are one-dimensional, this shall not add dimensions (think of multiplying scalar vs wedging differential forms), hence why the tensor product stays the same dimension.

***Proof*:**

1. As both are locally free, on the appropriate restriction we get:
$$\mathcal{O}_X \otimes \mathcal{O}_X \cong \mathcal{O}_X$$
2. Take $\mathcal{L}^{-1} = \mathit{Hom}(\mathcal{L}, \mathcal{O}_X)$. Then it is an elementary algebra exercise to verify that:
$$\mathcal{L}^{-1} \otimes \mathcal{L} \cong \mathit{Hom}(\mathcal{L}, \mathcal{L}) \cong \mathcal{O}_X$$

Definition 5.3.22: Picard Group

Let X be a ringed space. Then the *picard group* is the group of invertible sheaves under $\otimes_{\mathcal{O}_X}$. The picard group is denoted

$$\operatorname{Pic} X$$

We shall see in ref:HERE that $\operatorname{Pic} X = H^1(X, \mathcal{O}_X^*)$.

Definition 5.3.23: Line Bundles Associated To Cartier Divisor

Let D be a Cartier divisor on a scheme X, represent by $\{U_i, f_i\}$. Then the *sheaf associated to* D as a sub-sheaf $\mathcal{L}(D) \subseteq \mathcal{K}$ for each U_i, $\mathcal{L}(D)(U_i)$ is a sub $\mathcal{O}_X$-module of $\mathcal{K}(U)$[a] generated by f_i^{-1} on U_i.

[a]Not that X need not be integral, hence we need not have a global rational function field K

Note that this is well-defined since on intersections $U_i \cap U_j$, f_i/f_j is invertible, and hence f_i^{-1} and f_j^{-1} will generate the same $\mathcal{O}_X$-module. To return to the usual $\operatorname{Spec} \mathbb{Z}[\sqrt{-5}]$, we see that $(2, 1+\sqrt{-5})$ is a $\mathcal{O}_X$-modules that is generated by $(\frac{2}{1+\sqrt{-5}})$ and $(\frac{1+\sqrt{-5}}{2})$ on their respective open sets, and hence we have a sheaf associated to the divisor mentioned earlier! It is often simpler to picture $\mathcal{L}(D)$ as follows:

$$\mathcal{L}(D)(U) = \{f \in \kappa(X)^* \mid \operatorname{div}(f) + D \geq 0 \text{ in } U\} \cup \{0\}$$

Intuitively, we see that $\mathcal{L}(D)$ at an open set U contains all functions that have have poles that are bounded by D, and zeros' that are prescribed by D (it has at most as many poles as D, and at least as many zeros as D); in fact this was historically the use of divisors on Riemann surfaces.

Proposition 5.3.24: Correspondence Between Line Bundles and Cartier Divisors

Let X be a scheme. Then $\mathcal{L}(D)$ is an invertible sheaf, and the map the map $D \mapsto \mathcal{L}(D)$ is an injective homomorphism that maps that respects principal ideals.

1. $\mathcal{L}(D_1 - D_2) = \mathcal{L}(D_1) \otimes \mathcal{L}(D_2)^{-1}$
2. $D_1 \sim D_2$ if and only if $\mathcal{L}(D_1) \cong \mathcal{L}(D_2)$ as invertible sheaves

As a consequence, we have an injective homomorphism:

$$\operatorname{CaCl} X \to \operatorname{Pic} X$$

Note how this pairs well with the intuition that the tensor product multiplies the values of the functions: it shall "gain" the zero's poles prescribed by each divisor.

***Proof*:**
Let's show the map $D \mapsto \mathcal{L}(D)$ is injective and maps to an invertible sheaf. First showing a Cartier divisor, let $\{U_i\}$ be an open covering and take $f_i \in \Gamma(U_i, \mathcal{K}^*)$ that respects the conditions of Cartier divisor. Then we can define:

$$\mathcal{O}_{U_i} \to \mathcal{L}(D)|_{U_i} \qquad 1 \mapsto f_i^{-1}$$

As $\mathcal{L}(D)|_{U_i}$ is generated by f_i^{-1} this is certainly an isomorphism. Then $\mathcal{L}(D)$ is indeed an invertible sheaf. Then for any invertible sheaf $\mathcal{L}(D)$, we can recover the Cartier divisor D (and its embedding in $\mathcal{K}$) by taking f_i on U_i to be the inverse of the local generators of $\mathcal{L}(D)$ (which is unique up to element of $\mathcal{O}_X^*$). Then this will by construction give us Cartier divisors, showing injectivity

Let us next show that the group structure is preserved. Let D_1 and D_2 be locally defined by collections f_i and g_i respectively. Then $\mathcal{L}(D - 1 - D_2)$ is locally generated by $f_i^{-1} g_i$, and so by some module theory we see that:

$$\mathcal{L}(D_1 - D_2) = \mathcal{L}(D_1) \cdot \mathcal{L}(D_2)^{-1}$$

Then with some elementary algebra we see that $\mathcal{L}(D_1) \cdot \mathcal{L}(D_2)^{-1} \cong \mathcal{L}(D_1) \otimes \mathcal{L}(D_2)^{-1}$

Finally, to show that taking the quotient by principals induces isomorphism on the sheaves of divisors (and vice-versa), by our above work it suffices to show that $D = D_1 - D_2$ is principal if and only if $\mathcal{L}(D) \cong \mathcal{O}_X$. If D is principal, then it is defined by $f \in \Gamma(X, \mathcal{K}^*)$, hence $\mathcal{L}(D)$ is globally generated by f^{-1}, so that $1 \mapsto f^{-1}$ gives the isomorphism:

$$\mathcal{O}_X \cong \mathcal{L}(D)$$

Conversely, given such an isomorphism, certainly the image of 1 gives an element of $\Gamma(X, \mathcal{K}^*)$, and so we have that D is principal, completing the proof.

Note the map need not be surjective, as there may be invertible sheaves on X which are not isomorphic to any invertible sheaf of $\mathcal{K}$ (Hartshorne gives a reference to check). If X is projective over a field or if X is integral, then this is an isomorphism.

Proposition 5.3.25: Isomorphism of Cartier Class Group and Picard Group

Let X be an integral scheme. Then the homomorphism $\operatorname{CaCl} X \to \operatorname{Pic} X$ given in proposition 5.3.24 is an isomorphism.

***Proof*:**
What's left to show is surjectivity. Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{O}_X$. We must show it is isomorphic to a sub-sheaf of $\mathcal{K}$. As X is integral, we have a function field K for X. Consider:

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{K}$$

On any open U where $\mathcal{L} \cong \mathcal{O}_X$, w have

$$\mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

so this is a constant sheaf on U. As X is irreducible, any sheaf whose restriction to each open set of a covering of$X$ is constant *must* be a constant sheaf (by the injectivity of the restriction maps). Hence $\mathcal{L} \otimes \mathcal{K}_*$ is isomorphic to the constant sheaf $\mathcal{K}$, and so:

$$\mathcal{L} \to \mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

gives $\mathcal{L}$ as a sub-sheaf of $\mathcal{K}$, as we sought to show.

Thus, under the right conditions on X:

Corollary 5.3.26: Divisor Class Group and Picard Group

Let X be a Noetherian, integral, separated, locally factorial scheme. Then

$$\operatorname{Cl} X \cong \operatorname{Pic} X$$

***Proof*:**
Follow from proposition 5.3.25 and proposition 5.3.17.

Corollary 5.3.27: Classifying Line Bundles On Projective Space

Let $X = \mathbb{P}^n_k$ for some field k. Then every invertible sheaf on X is isomorphic to $\mathcal{O}(d)$ for some $d \in \mathbb{Z}$

***Proof*:**
By proposition 5.3.10, $\operatorname{Cl} X \cong \mathbb{Z}$, and by corollary 5.3.26 $\operatorname{Pic} X \cong \mathbb{Z}$. As the generators of $\operatorname{Cl} X$ is a hyperplane, which corresponds to the invertible sheaf $\mathcal{O}(1)$, $\operatorname{Pic} X$ is a free group generated by $\mathcal{O}(1)$, and every invertible sheaf $\mathcal{L}$ is isomorphic to $\mathcal{O}(d)$ for some $d \in \mathbb{Z}$, as we sought to show.

This level of generality is what Hartshorne covers, however the Stacks project (as of writing this book) focus on *effect Cartier divisors*. Hence we shall take a moment to look at those:

Definition 5.3.28: Effective Cartier Divisor

Let X be a scheme and D a Cartier divisor. Then D is called *effective* if it can be represented by $\{(U_i, f_i)\}$ where all $f_i \in \Gamma(U_i, \mathcal{O}_{U_i})$.

The *associated subscheme of codimension* 1 is the closed subscheme defined by the sheaf of ideals $\mathscr{I}$ which is locally generated by f_i.

By the definition, we see almost immediately that there is an injective correspondence between effective Cartier divisors on X and *locally principal* closed subschemes (subschemes whose sheaf of ideals is locally generated by a single element). If X is an integral separated Noetherian locally factorial Scheme, then the Cartier divisors correspond to the Wald divisors, and it is not too hard to show that the effective Cartier divisors correspond to the effective Weil divisors.

Proposition 5.3.29: Effective Cartier Divisors and Line Bundles

Let D be an effective Cartier divisor on a scheme X and Y the associated locally principal closed subscheme. Then:

$$\mathscr{I}_Y \cong \mathcal{L}(-D)$$

***Proof*:**
As $\mathcal{L}(-D)$ is the sub sheaf of $\mathcal{K}$ generated by f_i, this is actually a sub-sheaf of $\mathcal{O}_X$, and it is by construction the ideal sheaf of $\mathscr{I}_Y$, completing the proof.

5.4 Differentials

We next build-up to defining differential forms on a scheme. This shall give us a lot of information about the structure of a scheme and is one of the first computational tools you can use when studying a new algebraic object (similarly to how differential forms give a lot of information on manifolds)

Definition 5.4.1: A-derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. Then an A-derivation of B into M is a map $d : B \to M$ such that

1. $d(b + b') = d(b) + d(b')$
2. $d(bb') = d(b)b' + bd(b')$
3. $d(a) = 0$ for all $a \in A$

Definition 5.4.2: Module of Relative Differential Forms

Let A be a commutative ring, B an A-algebra. Then a module of relative differential form is a B-module $\Omega_{B/A}$ with an A-derivation $d : B \to \Omega_{B/A}$ which satisfies the universal property of relative differential forms: if M is another B-module and $d' : B \to M$ an A-derivation, then there is a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that the following diagram commutes:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, "f"] \\ & M \end{tikzcd}$$

Certainly $\Omega_{B/A}$ exists via the usual construction: take the free B-module F generated by the primitives $d(b)$ for $b \in B$ and quotient by the submodule generated by the relationships:

$$d(b+b') - d(b) - d(b') \qquad \text{and} \qquad d(bb') - d(b)b' - bd(b') \qquad \text{and} \qquad d(a)$$

Then the map $d : B \to \Omega_{B/A}$ is given by $b \mapsto d(b)$. Then it is an exercise in algebra to see that this satisfies the universal property and $\left(\Omega_{B/A}, d\right)$ is unique up to unique isomorphism. The following gives an alternative insightful construction:

Proposition 5.4.3: Natural Differential Structure

Let B be an A-algebra, $\delta : B \otimes_A B \to B$ the diagonal map given by $f(b \otimes b') = bb'$, and let $I = \ker \delta$. Then I/I^2 has the structure of a left B-module, and we can define a map

$$d : B \to I/I^2 \qquad d(b) = 1 \otimes b - b \otimes 1 \bmod I^2$$

In which case $(I/I^2, d)$ is a module of relative differential for B/A.

***Proof*:**
Hartshorne referenced Matsumura.

Example 5.4.4: Relative Differential

1. Take $A = \mathbb{R}$ and $B = \mathbb{R}[x]$. Then $I = \ker \delta$. Then as $\mathbb{R}[x]$ is an integral domain, it can be shown that any element is a linear combination of elements of the form:

$$f(x) \otimes g(x) - f(x)g(x) \otimes 1 \in I$$

Then it should be verified any element in I can be written as:

$$f(x)(1 \otimes g(x) - g(x) \otimes 1)$$

The reader should then verify that I/I^2 is generated by $f'(x)(x \otimes 1 - 1 \otimes x)$[a]. Let dx

represent this element. Notice that this is the image of $d(x)$. Now, let's consider:

$$d(f(x) = d\left(\sum_i a_i x^i\right) = \sum_i a_i d(x^i)$$

Hence, let's compute $d(x^i)$. For $i = 2$, we get:

$$d(x^2) = d(x \cdot x) = xd(x) - xd(x) = 2xdx$$

Repeating this pattern, we can see that:

$$d(x^n) = nx^{n-1}dx$$

2. For example, taking $B = \mathbb{R}[x_1, ..., x_n]$, Then $\Omega_{B/A}$ is a free B-module generated by $dx_1, ..., dx_n$.
3. In proposition 5.4.9, we shall see that if E/F is a finitely generated separable extension (of characteristic 0), then
$$\Omega_{B/A} = \bigoplus_{b \in T} B db$$
where T is a transcendental A-basis for B.
4. What if we take $B = A[\epsilon]/(\epsilon^2)$?
5. Let $A = \mathbb{R}$ and $B = C^\infty(M)$ for some smooth n-manifold M. Then $C^\infty(M)$. Then $\Omega_{C^\infty(M)/\mathbb{R}}$ is a locally free $C^\infty(M)$-module (based off of the patches of M), consisting of the space of sections of the cotangent bundle T^*M.

[a] Hint: what is $(1 \otimes x^2 - x^2 \otimes 1)$. Repeat for higher powers

Proposition 5.4.5: Base Change For Differentials

Let A', B be A-algebras and $B' = B \otimes_A A'$. Then:

$$\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$$

Furthermore, if S is a multiplicative subset of of S, then:

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$$

***Proof*:**
Hartshorne referenced Matsumura.

Proposition 5.4.6: Differentials and Exact Sequence I

Let $A \to B \to C$ be a (not necessarily exact nor a complex) sequence of rings. Then there is an exact sequence of C-modules:

$$\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

***Proof*:**
Hartshorne referenced Matsumura.

Proposition 5.4.7: Differentials and Exact Sequence II

Let B be an A-algebra, I be an ideal of B, and $C = B/I$. Then there is a natural exact sequence of C-modules

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0,$$

where for any $b \in I$, if $\bar{b}$ is its image in I/I^2, then $\delta\bar{b} = db \otimes 1$. In particular, I/I^2 has a natural structure of a C-module, and δ is a C-linear map, even though it is defined via the derivation d.

***Proof*:**
Hartshorne referenced Matsumura.

Corollary 5.4.8: Finite Generation of Differential Forms

If B is a finitely generated A-module or a localization of a finitely generated A-module, then $\Omega_{B/A}$ is a finitely generated B-module

***Proof*:**
As B is a finitely generated A-module, it is the quotient of polynomial rings. Hence, it follows from our direct computations from example 5.4.4 and, proposition 5.4.7, and the localization result comes from proposition 5.4.5.

Proposition 5.4.9: Differentials and Field Extensions

Let K/k be a finitely generated field extension. Then:

$$\dim_k \Omega_{K/k} \geq \mathrm{trdeg} K/k$$

where equality holds iff K is separably generated over k. Then if K/k is a finite algebraic extension, $\Omega_{K/k} = 0$ if and only if K/k is separable.

***Proof*:**
Hartshorne referenced Matsumura.

Proposition 5.4.10: Local Ring and Relative Differential

Let B be a local ring containing a field k isomorphic to $B/\mathfrak{m}$. Then given the map δ from proposition 5.4.7:

$$\delta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\cong} \Omega_{B/k} \otimes_B k$$

***Proof*:**
By proposition 5.4.7 $\operatorname{coker}(\delta) = 0$, so δ is surjective. To show that δ is injective, we shall show the following dual vector space map is surjective:

$$\delta' : \operatorname{Hom}_k(\Omega_{B/k} \otimes k, k) \to \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$$

The term on the left is isomorphic to $\operatorname{Hom}_B(\Omega_{B/k}, k)$, which by definition of the differentials, can be identified with the set $\operatorname{Der}_k(B, k)$ of k-derivations of B to k. If $d : B \to k$ is a derivation, then $\delta'(d)$ is obtained by restricting to $\mathfrak{m}$, and noting that $d(\mathfrak{m}^2) = 0$. Now, to show that δ' is surjective, let $h \in \operatorname{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$. For any $b \in B$, we can write $b = \lambda + c$, $\lambda \in k$, $c \in \mathfrak{m}$, in a unique way. Define $db = h(\bar{c})$, where $\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$ is the image of c. Then one verifies immediately that d is a k-derivation of B to k, and that $\delta'(d) = h$. Thus δ' is surjective, as required.

We finish off this brief summary of the essential results about relative differentials by showing the relation between regular local rings and relative differentials. We require the following lemma:

Lemma 5.4.11: Characterizing Free Modules using Residue and Quotient Field

Let A be a Noetherian local integral domain, with residue field k and quotient field K. Then if M is a finitely generated A-module and $\dim_k M \otimes_A k = \dim_k M \otimes_A K = r$, then M is free of rank r.

***Proof*:**
Since $\dim_k M \otimes k = r$, by Nakayama's lemma M can be generated by r elements. Hence, there is a surjective map $\varphi : A^r \to M$. Let R be its kernel. Then we obtain an exact sequence

$$0 \to R \otimes K \to K^r \to M \otimes K \to 0,$$

As $\dim_K M \otimes K = r$, we have $R \otimes K = 0$. But R is torsion-free, so $R = 0$, and M is isomorphic to A^r, completing the proof.

Theorem 5.4.12: Characterizing Regular Local Rings using Differential

Let B be the localization of a finitely generated k-algebra where $B/\mathfrak{m}$ is prefect (or it contains a field k isomorphic to $B/\mathfrak{m}$ where k is perfect). Then B is a regular local ring if and only if $\Omega_{B/k}$ is a free B-module of rank equal to $\dim B$.

***Proof*:**
If $\Omega_{B/k}$ is free of rank $\dim B$, then by proposition 5.4.10

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim B$$

which is an equivalent characterization of a regular local ring.

Conversely, suppose that B is a regular local ring of dimension r. Then $\dim_k \mathfrak{m}/\mathfrak{m}^2 = r$, so by proposition 5.4.10 we have $\dim_k \Omega_{B/k} \otimes k = r$. On the other hand, let K be the quotient field of B. Then by proposition 5.4.5, $\Omega_{B/k} \otimes_B K = \Omega_{K/k}$. As k is perfect, K is a separably generated extension field of k, and so $\dim_K \Omega_{K/k} = \operatorname{trdeg} K/k$ by 5.4.9. But we also have $\dim B = \operatorname{trdeg} K/k$. Finally, note that by corollary 5.4.8, $\Omega_{B/k}$ is a finitely generated B-module. But then by lemma 5.4.11 $\Omega_{B/k}$ is a free B module of rank r, as we sought to show.

5.4.1 Sheaves of Differentials

Let us now define sheaves of differentials on schemes. Let $f : X \to Y$ be a scheme morphism and $\Delta : X \to X \times_Y X$ the diagonal morphism. Then $X \cong \Delta(X)$ (see exercise ref:HERE, section 3.5.1). Namely, it is isomorphic to a closed subscheme of an open subset $W \subseteq X \times_Y X$.

Definition 5.4.13: Sheaf of Differentials

Let $\mathscr{I}$ be the sheaf of ideals of $\Delta(X)$ in W. Then the *sheaf of differentials* of X over Y is

$$\Omega_{X/Y} = \Delta^*(\mathscr{I}/\mathscr{I}^2)$$

Intuitively, this sheaf of modules ought to be a quasi-coherent $\mathcal{O}_X$-module. Indeed, it is first of all naturally a $\mathcal{O}_{\Delta(X)}$-module, and by the isomorphism $X \cong \Delta(X)$ we get a natural $\mathcal{O}_X$-module structure on it. By proposition 5.1.32 it is quasi-coherent, and by proposition 5.1.16 if Y is Noetherian and f is of finite-type it is coherent.

Reducing to the affine case, if $\operatorname{Spec} S = V \subseteq Y$ and $\operatorname{Spec} R = U \subseteq X$ such that $f(U) \subseteq V$, then $V \times_U V \subseteq X \times_Y X$ is an affine open subset isomorphic to $\operatorname{Spec} S \otimes_R S$, and $\Delta(X) \cap (V \times_U V)$ is a closed subscheme given by the kernel of the diagonal map $S \times_R S \to S$. Thus, $\mathscr{I}/\mathscr{I}^2$ is the usual I/I^2, and we get:

$$\Omega_{V/U} \cong \widetilde{\Omega_{S/R}}$$

Hence, $\Omega_{X/Y}$ could also be defined by giving an affine covering of X and Y which is glued together corresponding to the sheaves of modules $\widetilde{\Omega_{S/R}}$. Similarly, the maps $d : S \to \Omega_{S/R}$ glue to create a map $d : \mathcal{O}_X \to \Omega_{X/Y}$ of sheaves of abelian groups. This map is a derivation of local rings at each point. From this, we shall carry most of our results over to schemes.

(The other remark)

Proposition 5.4.14: Sheaf of Differentials: Base Change

Let $f : X \to Y$ be a morphism, let $g : Y' \to Y$ be another morphism, and let $f' : X' = X \times_Y Y' \to Y'$ be obtained by base extension. Then $\Omega_{X'/Y'} \cong g'^*(\Omega_{X/Y})$ where $g' : X' \to X$ is the first projection.

***Proof*:**
Follows from proposition 5.4.5

Proposition 5.4.15: Sheaf of Differentials: Short Exact Sequence I

Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes. Then there is an exact sequence of sheaves on X,

$$f^*\Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0.$$

***Proof*:**
Follows from proposition 5.4.6

Proposition 5.4.16: Sheaf of Differentials: Short Exact Sequence II

Let $f : X \to Y$ be a morphism, and let Z be a closed subscheme of X, with ideal sheaf $\mathscr{I}$. Then there is an exact sequence of sheaves on Z,

$$\mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X/Y} \otimes \mathscr{O}_Z \to \Omega_{Z/Y} \to 0.$$

***Proof*:**
Follows from proposition 5.4.7

The next exact sequence will be specific to sheaves of differentials on projective space, and will be important for computations in the rest of the section:

Theorem 5.4.17: Sheaves of Differentials On Projective Space

Let R be a ring, $Y = \operatorname{Spec} R$, and $X = \mathbb{P}^n_R$. Then there exists an exact sequence of sheaves on X:

$$0 \to \Omega_{X/Y} \to \mathcal{O}_X(-1)^{\oplus n+1} \to \mathcal{O}_X \to 0$$

Recall that $\mathcal{O}(-1) \cong \operatorname{Hom}(\mathcal{O}(1), \mathcal{O})$, and so they are naturally interpreted as linear functionals. the final (non-trivial) map is naturally an evaluation map, giving us a great way of understanding the differentials $\mathcal{O}_{X/Y}$!

***Proof*:**
Let $S = R[x_0, \ldots, x_n]$ be the homogeneous coordinate ring of X and E be the graded S-module $S(-1)^{n+1}$, with basis $e_0, \ldots, e_n$ in degree 1. Define a degree 0 homomorphism of graded S-modules $E \to S$ by sending $e_i \mapsto x_i$, and let M be the kernel. Then the exact sequence

$$0 \to M \to E \to S$$

of graded S-modules gives rise to an exact sequence of sheaves on X,

$$0 \to \tilde{M} \to \mathcal{O}_X(-1)^{n+1} \to \mathcal{O}_X \to 0.$$

Note that $E \to S$ is not surjective, but it is surjective in all degrees ≥ 1, and hence is surjective in all stalks, and so the corresponding map of sheaves is surjective!

Thus, if we show $\tilde{M} \cong \Omega_{X/Y}$, we're done. We shall show that we we get an isomorphism on patches that glue together. First note that if we localize at x_i, then $E_{x_i} \to S_{x_i}$ is a surjective homomorphism of free S_{x_i}-modules, so M_{x_i} is free of rank n, generated by $\{e_j - (x_j/x_i)e_i \mid j \neq i\}$. Then if U_i is the standard open set of X defined by x_i, $\tilde{M}|_{U_i}$ is a free $\mathcal{O}_{U_i}$-module generated by the sections $(1/x_i)e_j - (x_j/x_i^2)e_i$ for $j \neq i$. (Here we need the additional factor $1/x_i$ to get elements of degree 0 in the module M_{x_i}.)

To define the map $\varphi_i : \Omega_{X/Y}|_{U_i} \to \tilde{M}|_{U_i}$, recall that $U_i \cong \operatorname{Spec} R[x_0/x_i, \ldots, x_n/x_i]$, so $\Omega_{X/Y}|_{U_i}$ is a free $\mathcal{O}_{U_i}$-module generated by $d(x_0/x_i), \ldots, d(x_n/x_i)$. Thus, define φ_i by

$$\varphi_i(d(x_j/x_i)) = (1/x_i^2)(x_i e_j - x_j e_i).$$

Then φ_i is an isomorphism. Let us now show the isomorphisms φ_i glue together to give an isomorphism $\varphi : \Omega_{X/Y} \to \tilde{M}$ on all of X. But this is a simple calculation: on $U_i \cap U_j$, we have, for any k, $(x_k/x_i) = (x_k/x_j) \cdot (x_j/x_i)$. Hence in $\Omega|_{U_i \cap U_j}$ we have

$$d\left(\frac{x_k}{x_i}\right) - \frac{x_k}{x_j} d\left(\frac{x_j}{x_i}\right) = \frac{x_j}{x_i} d\left(\frac{x_k}{x_j}\right).$$

Now applying φ_i to the left-hand side and φ_j to the right-hand side, we get the same thing both ways, namely $(1/x_i x_j)(x_j e_k - x_k e_j)$. Thus the isomorphisms φ_i glue, which completes our proof.

Let us now focus down our attention to nonsingular varieties, as they are the most concrete example and most common place to apply these techniques. Let us first properly define the notion:

Definition 5.4.18: Abstract Nonsingular Variety

Let X be an abstract variety over an algebraically closed field k. Then X is *nonsingular* if all of its local rings are *regular*

5.4.19 Non Closed Points? Note that it we require the local rings at all points, not just closed points. It may thus seem that this condition is stronger, however as each local ring at a non-maximal prime ideal is the localization of a local ring at a maximal ideal, if localization preserves regularity the condition is equivalent, and this is indeed the case (see Matsumura p.199).

We can finally connect being nonsingular to the differential sheaves we've been working with:

Proposition 5.4.20: Nonsingular Varieties and Differential Sheaf

Let X be an irreducible (not necessarily reduced) separated scheme of finite type over an algebraically closed field k. Then $\mathcal{O}_{X/k}$ is locally free of rank $n = \dim X$ iff X is a nonsingular variety over k.

***Proof*:**
If $x \in X$ is closed, then as $\mathcal{O}_{X/k}$ is locally free by proposition 5.1.22 $\mathcal{O}X, x$ is free of dimension n, has a residue field k, and its localization is a k-algebra of finite type. Let $B = \mathcal{O}_{X,x}$. Then the module $\mathcal{O}_{B/k}$ is equal to the stalks $(\mathcal{O}_{X/k})_x$ for the sheaf $\mathcal{O}_{X/k}$. Thus, by theorem 5.4.12

$(\mathcal{O}_{X/k})_x$ if free of rank n iff B is a regular local ring. Then remark 5.4.19 gives us the desired result.

From this result, we get a lovely new view of seeing why the set of nonsingular points are open and dense:

Corollary 5.4.21: Nonsingular Points Open and Dense using Differentials

Let X be a variety over k. Then there is an open dense subset of X which is nonsingular

***Proof*:**
Let $n = \dim X$. Then the function field $k(X)$ has transcendence degree n over k and $k(X)/k$ is separable. Then by proposition 5.4.9, $\Omega_{K/k}$ is a $K(X)$-vector space of dimension n.

Now, notice that $\mathcal{O}_{K/k}$ is just the stalk of $\mathcal{O}_{X/k}$ at the generic point of X! Then by proposition 5.1.22 $\mathcal{O}_{X/k}$ is locally free in some neighbourhood of the generic point, that is some nonempty open dense set U. By proposition 5.4.20, the points of U are nonsingular, completing the proof.

An important result is to be able to cake closed subschemes whose points are nonsingular:

Theorem 5.4.22: Differentials: Nonsingular Closed Subschemes

Let X be a nonsingular variety over k and let $Y \subseteq X$ be an irreducible closed subscheme defined by a sheaf of ideals $\mathscr{I}$. Then Y is nonsingular if and only if

1. $\Omega_{Y,k}$ is locally free
2. the sequence of proposition 5.4.16 is exact on the left as well:

$$0 \to \mathscr{I}/\mathscr{I}^2 \to \Omega_{X,k} \otimes \mathcal{O}_Y \to \Omega_{Y,k} \to 0.$$

Given Y is nonsingular, then $\mathscr{I}$ is locally generated by $r = \operatorname{codim}_X Y$ elements, and $\mathscr{I}/\mathscr{I}^2$ is a locally free sheaf of rank r on Y.

***Proof*:**
First suppose the two conditions hold so that $\Omega_{Y,k}$ is locally free, and the given sequence is also left exact. Then by proposition 5.4.20 it suffices to show that rank $\Omega_{Y,k} = \dim Y$. Let rank $\Omega_{Y,k} = q$. We know that $\Omega_{X,k}$ is locally free of rank n, so it follows from (2) that $\mathscr{I}/\mathscr{I}^2$ is locally free on Y of rank $n - q$. Hence by Nakayama's lemma, $\mathscr{I}$ can be locally generated by $n - q$ elements, and it follows that $\dim Y \geq n - (n - q) = q$ (see [9, chapter 21.5]). On the other hand, considering any closed point $y \in Y$, by proposition 5.4.10 we have $q = \dim(\mathfrak{m}/\mathfrak{m}^2)$, and so $q \geq \dim Y$ by (see [9, chapter 21.5.3]), but then by what we saw earlier $q = \dim Y$. From this, we conclude that that Y is nonsingular and that $n - q = \operatorname{codim}_X Y$ giving the desired result for the generators of $\mathscr{I}$.

Conversely, assume that Y is nonsingular. Then (1) is immediate as $\Omega_{Y,k}$ is locally free of rank $q = \dim Y$. By proposition 5.4.16 we have the exact sequence

$$\mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y \xrightarrow{\varphi} \Omega_{Y,k} \to 0.$$

We need to show exactness on the left. Consider a closed point $y \in Y$. Then $\ker \varphi$ is locally free of rank $r = n - q$ at y, so it is possible to choose sections $x_1, \ldots, x_r \in \mathscr{I}$ in an appropriate neighborhood of y such that $dx_1, \ldots, dx_r$ generate $\ker \varphi$. Let $\mathscr{I}'$ be the ideal sheaf generated by $x_1, \ldots, x_r$, and let Y' be the corresponding closed subscheme. Then by construction, the $dx_1, \ldots, dx_r$ generate a free sub-sheaf of rank r of $\Omega_{X,k} \otimes \mathcal{O}_Y$ in a neighborhood of y. It follows that in the exact sequence of proposition 5.4.16 for Y',

$$\mathscr{I}'/\mathscr{I}'^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y \to \Omega_{Y',k} \to 0,$$

Now, since its image is free of rank r and $\Omega_{Y',k}$ is locally free of rank $n-r$, δ injective. Then we demonstrates that Y' is irreducible and nonsingular of dimension $n-r$ (in a neighborhood of y). But $Y \subseteq Y'$, both are integral schemes of the same dimension, so we must have $Y = Y'$, $\mathscr{I} = \mathscr{I}'$, and this shows that

$$0 \to \mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y$$

is injective, as we sought to show.

(Hartshorne here covers the very interesting *Bernini's Theorem* which is some Morse Theory on nonsingular projective varieties, but I will omit if for now)

We are now ready to bring in the notion of the tangent space of a nonsingular variety:

Definition 5.4.23: Tangent Sheaf and Canonical Sheaf

Let X be a nonsingular variety over k. Then the *tangent sheaf* of X is defined as:

$$\mathscr{T}_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$$

The *canonical sheaf* of X is the invertible sheaf:

$$\omega_X = \bigwedge^n \Omega_{X/k}$$

where $n = \dim X$.

Note that $\mathscr{T}_X$ is a locally free sheaf of rank $n = \dim X$. If is projective X, then the dimension of the global section of ω_X over X will give an important invariant:

Definition 5.4.24: Geometric Genus

Let X be projective and nonsingular. Then the *geometric genus* of X is:

$$p_g = \dim_k \Gamma(X, \omega_X)$$

5.4.25 Foreshadowing In [9, chapter 21.5.2], we introduced the Hilbert function and the *arithmetic genus*. This shall be the dual of the geometric genus in the case of curves, as we shall show in section 6.3. Note however these these two need not be equal in higher dimensions, see example ref:HERE.

When working over varieties, the Geometric genus is a rather stable result, it is in fact more stable than isomorphisms:

Theorem 5.4.26: Birational Invariance of Geometric Genus

Let X, X' be two nonsingular projective varieties. THen if X is birationally equivalent to X', then:

$$p_g(X) = p_g(X')$$

Note that the converse is not true: the geometric genus is in general not enough to determine a variety (this is only the case in dimension 1, see ref:HERE).

***Proof*:**
As X and X' are birationally equivalent, there exists a rational map $F : X \dashrightarrow X'$ that has a rational inverse. Let $V \subseteq X$ be the largest open set for which $f : V \to X'$ represents the rational map F. Then by proposition 5.4.15 we get $f^*\Omega_{X'/k} \to \Omega_{V/k}$. As X, X' are birational, these are locally free sheaves of the same rank, say $n = \dim X$, so there is an induced map on the exterior powers $f^*\omega_{X'} \to \omega_V$. This map in turn induces a map on the space of global sections

$$f^* : \Gamma(X', \omega_{X'}) \to \Gamma(V, \omega_V)$$

Now, since f is birational there is an open set $U \subseteq V$ such that $f(U)$ is open in X' and f induces an isomorphism from U to $f(U)$. Thus f induces:

$$\omega_V|_U \cong \omega_{X'}|_{f(U)}$$

Since a nonzero global section of an invertible sheaf cannot vanish on a dense open set, the map of vector spaces

$$f^* : \Gamma(X', \omega_{X'}) \to \Gamma(V, \omega_V)$$

is injective.

Next we will compare $\Gamma(V, \omega_V)$ with $\Gamma(X, \omega_X)$. First by the valuation criterion for properness (theorem 3.5.25), $X - V$ has codimension ≥ 2 in X. For, if $P \in X$ is a point of codimension 1, then $\mathcal{O}_{P,X}$ is a discrete valuation ring (as X is nonsingular). We already have a map of the generic point of X to X', and X' is projective, and hence proper over k, so there exists a unique morphism $\operatorname{Spec} \mathcal{O}_{P,X} \to X'$ compatible with the given birational map. This extends to a morphism of some neighborhood of P to X', so we must have $P \in V$ by definition of V.

Now, let us show there that the natural restriction map $\Gamma(X, \omega_X) \to \Gamma(V, \omega_V)$ is surjective, giving us bijectivity. This shall be an immediate consequence of lemma 5.3.7. It is enough to show that for any open affine subset $U \subseteq X$ such that $\omega_X|_U \cong \mathcal{O}_U$, that $\Gamma(U, \mathcal{O}_U) \to \Gamma(U \cap V, \mathcal{O}_{U\cap V})$ is bijective. Since X is nonsingular, it is normal, and since $U - U \cap V$ has codimension ≥ 2 in U, but this is now an immediate consequence of lemma 5.3.7.

Combining our results, we see that $p_g(X') \leq p_g(X)$. The other inequality is given by by symmetry, and thus conclude that

$$p_g(X) = p_g(X')$$

as we sought to show

We finish off this section by looking at the relation between tangent sheaves and their dual (normal sheaves):

Definition 5.4.27: Normal and Conormal Sheaf

Let $Y \subseteq X$ be a nonsingular subvariety of a variety X over k. Then $\mathscr{I}/\mathscr{I}^2$ is called the *conormal sheaf* of Y in X, and $\mathcal{N}_{Y/X} = \mathcal{H}om_{\mathcal{O}_Y}(\mathscr{I}/\mathscr{I}^2, \mathcal{O}_Y)$ is called the *normal sheaf* of Y in X.

Note that normal sheaf is locally free of rank $r = \operatorname{codim}_X(Y)$. The normal sheaf does indeed mimic the notion of a normal bundle on a manifold as we get the dual exact sequence:

$$0 \to \mathscr{T}_Y \to \mathscr{T}_X \otimes \mathcal{O}_Y \to \mathcal{N}_{Y/X} \to 0$$

Proposition 5.4.28: Subvarieties and Normal Sheaf

Let $Y \subseteq X$ be a nonsingular subvariety of codimension r from a nonsingular variety X over k. Then

$$\omega_Y \cong \omega_X \otimes \bigwedge^r \mathcal{N}_{Y/X}.$$

In case $r = 1$, considering Y as a divisor $\mathcal{L}$ be the associated invertible sheaf on X, Then

$$\omega_Y \cong \omega_X \otimes \mathcal{L} \otimes \mathcal{O}_Y$$

***Proof*:**
First, take the highest exterior powers of the locally free sheaves in the exact sequence (by exercise ref:HERE)

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_X \otimes \mathcal{O}_Y \to \Omega_Y \to 0$$

Thus we find that $\omega_X \otimes \mathcal{O}_Y \cong \omega_Y \otimes \bigwedge^r(\mathcal{I}/\mathcal{I}^2)$. Since formation of the highest exterior power commutes with taking the dual sheaf, we get

$$\omega_Y \cong \omega_X \otimes \bigwedge^r \mathcal{N}_{Y/X}$$

In the special case $r = 1$, $\mathcal{I}_Y \cong \mathcal{L}^{-1}$ by proposition 5.3.29. Thus, $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{L}^{-1} \otimes \mathcal{O}_Y$, and $\mathcal{N}_{Y/X} \cong \mathcal{L} \otimes \mathcal{O}_Y$. So applying the previous result with $r = 1$, we obtain

$$\omega_Y \cong \omega_X \otimes \mathcal{L} \otimes \mathcal{O}_Y$$

completing the proof.

(Hartshorne gives many good examples here! One important one was showing that there exists non-rational varieties in all dimensions!

5.5 *Formal Schemes and Local Geometry

As we have seen, one feature which clearly distinguishes the theory of schemes from the older classical theory of varieties is the possibility of having nilpotent elements in the structure sheaf. In particular, if $Y \subseteq X$ is a closed subvariety defined by a sheaf of ideals $\mathcal{I}$, then for any $n \geq 1$ we can consider the closed subscheme Y_n defined by the n-th power $\mathcal{I}^n$. For $n \geq 2$, this is a scheme with nilpotent elements. It carries information about Y together with the infinitesimal properties of the embedding of Y in X.

We wish to construct an object which carries information about all the infinitesimal neighborhoods Y_n of Y simultaneously. Such an object would be “thicker” than any individual Y_n, but contained inside any actual open neighborhood of Y in X. We might naturally call this the formal neighborhood of Y in X.

To see why a new geometric construction is necessary to capture this, recall that the algebraic construction of the formal power series ring $k[[x]]$ is given by the limit of the truncated polynomial rings $k[x]/(x^n)$. We shall now explore the geometric consequences of this algebraic operation.

Topologically, the scheme $\operatorname{Spec}(k[x]/(x^n))$ consists of exactly one point, as the ring is Artinian and possesses a unique prime ideal (x). Geometrically, one views this as a “fat point” – a point equipped with infinitesimal nilpotents that remember $n-1$ orders of derivatives. Since $k[[x]]$ is the algebraic limit of these rings as $n \to \infty$, one might intuitively expect the geometric limit of a sequence of one-point spaces to remain a one-point space. However, passing to the infinite limit crosses a structural threshold: the limit of infinite nilpotents coalesces into a genuine coordinate variable, making $k[[x]]$ a domain of Krull dimension 1. Consequently, the affine scheme $\operatorname{Spec}(k[[x]])$ possesses *two* points. One should visualize $\operatorname{Spec}(k[[x]])$ as the “germ” of a curve, an unimaginably small neighborhood that has lost all global properties but is just large enough to have a center and an infinitesimal edge.

Formal Completions

The fact that the Spec functor transforms a limit of one-point spaces into a two-point space demonstrates that Spec does not commute nicely with infinite limits. To rectify this topological mismatch, Grothendieck introduced formal schemes. We shall follow Hartshorne and define this globally as a completion along a closed subscheme.

Definition 5.5.1: Formal Completion

Let X be a Noetherian scheme, and let Y be a closed subscheme defined by a sheaf of ideals $\mathcal{I}$. Then we define the *formal completion* of X along Y, denoted $(\hat{X}, \mathcal{O}_{\hat{X}})$, to be the following ringed space: we take the topological space Y, and on it the sheaf of rings $\mathcal{O}_{\hat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^n$. Here we consider each $\mathcal{O}_X/\mathcal{I}^n$ as a sheaf of rings on Y, and make them into an inverse system in the natural way.

The reader should notice that the structure sheaf $\mathcal{O}_{\hat{X}}$ actually depends only on the closed subset Y, and not on the particular scheme structure on Y. For if $\mathcal{J}$ is another sheaf of ideals defining a closed subscheme structure on Y, then since X is a Noetherian scheme, there are integers m, n such that $\mathcal{I} \supseteq \mathcal{J}^m$ and $\mathcal{J} \supseteq \mathcal{I}^n$. Thus the inverse systems $(\mathcal{O}_X/\mathcal{I}^n)$ and $(\mathcal{O}_X/\mathcal{J}^m)$ are cofinal with each other, and hence have the same inverse limit.

Furthermore, one sees easily that the stalks of the sheaf $\mathcal{O}_{\hat{X}}$ are local rings, so in fact $(\hat{X}, \mathcal{O}_{\hat{X}})$ is a locally ringed space. If $U = \operatorname{Spec} A$ is an open affine subset of X, and if $I \subseteq A$ is the ideal $\Gamma(U, \mathcal{I})$, then evaluating the sections reveals that $\Gamma(\hat{X} \cap U, \mathcal{O}_{\hat{X}}) = \hat{A}$, the I-adic completion of A. Thus the process of completing X along Y is strictly analogous to the I-adic completion of a ring. However, one should note that the local rings of $\hat{X}$ are in general *not* complete, and their dimension (which is equal to $\dim X$) is *not* equal to the dimension of the underlying topological space Y.

We shall call a formal scheme obtained by completing a single Noetherian scheme X along a closed subscheme Y *algebraizable*.[4]

Example 5.5.2: Formal Completions

1. Take X to be a Noetherian scheme, and let $Y = X$. Then clearly $\hat{X} = X$. Thus the category of Noetherian formal schemes naturally includes all Noetherian schemes.
2. Take X to be a Noetherian scheme, and let $Y = \{P\}$ be a closed point. Then $\hat{X}$ is a one point space $\{P\}$ with the completion $\hat{\mathcal{O}}_P$ of the local ring at P as its structure sheaf. An $\hat{\mathcal{O}}_P$-module M, considered as a sheaf on $\hat{X}$, is coherent if and only if M is a finitely generated module. This directly resolves our earlier algebraic paradox: $\hat{X}$ is topologically a single point, capturing the geometric intuition of a true limit of fat points without accidentally spawning a generic point.

5.5.3 Comparing Local Geometries To summarize the distinctions between these deeply related objects for a point P on a curve:

1. $\operatorname{Spec}(k[x]/(x^n))$ is a single point representing a "fat point" of finite thickness.
2. $\hat{X}$ (the formal completion at P) is a single point representing the true geometric limit of fat points (infinite formal thickness).
3. $\operatorname{Spec}(k[[x]])$ is a two-point space representing a "formal disk" or germ (a center point plus a surrounding generic neighborhood).

Universal Smoothness and Singularities

The utility of formal schemes shines when studying the local properties of varieties. In classical algebraic geometry, curves can exhibit wildly different global shapes, such as a line $y = x$ or a parabola $y = x^2$. However, if one selects a smooth point on any such curve over a field k and takes the formal completion of the local ring at that point, the resulting ring is always isomorphic to $k[[t]]$, where t is a local parameter.

Because their local algebra is identical, their formal geometry is identical. Thus, $\operatorname{Spec}(k[[x]])$ does not merely represent *a* curve; it represents the fundamental atomic structure of *any* smooth curve at a smooth point. Under an infinite-power formal microscope, every smooth curve is indistinguishable from a straight line.

This universal behavior fails precisely when the geometry ceases to be smooth.

Example 5.5.4: Singularities and Formal Completions

If a curve possesses a singularity, its formal completion at that point acts as a perfect algebraic memory of the singularity, remaining distinct from $k[[x]]$.

1. *The Cusp:* Consider the curve cut out by $y^2 = x^3$, which has a sharp cusp at the origin. The

[4] It is apparently quite difficult to find non-algebraizable formal schemes, maybe add an exercise based on Hartshorne p.194 later?

formal completion of the local ring at the origin is

$$\frac{k[[x,y]]}{(y^2 - x^3)}$$

This ring is not isomorphic to $k[[t]]$. Even infinitesimally, the formal scheme remembers the sharp turn.

2. *The Node:* Consider a curve with a self-intersection at the origin, locally modeled by $xy = 0$. Its formal completion at the crossing is

$$\frac{k[[x,y]]}{(xy)}$$

This represents two distinct microscopic branches crossing each other. Topologically, the formal completion is still a single point, but the underlying sheaf of topological rings completely captures the crossing branches.

6

Cohomology of Schemes

Let X be a scheme with a sheaf $\mathscr{F}$. A natural question to ask is if we can quantify the obstruction to the gluing local sections to create global sections? This is precisely what *sheaf cohomology* shall study by taking the derived functors of the global-section functor $X \to \Gamma(X, \mathscr{F})$ for a sheaf $\mathscr{F}$. Important sheaves this will apply include (but are not limited to) the structure sheaf $\mathcal{O}_X$, the twisted sheaves $\mathcal{O}_X(n)$, the differential sheaves $\Omega_{B/A}$ and there wedge products, and the pushforward of closed sheaves (in particular to study a scheme X given information about closed subschemes $S \subseteq X$). Though this is theoretically nice, it is computationally difficult as it will require an injective resolution that uses acyclic sheaves that are generally difficult to compute. For computations, *Čech cohomology* is introduced as a more "geometric" approach in computing cohomology. From this we shall compute the cohomology of projective space $\mathbb{P}^n_R$ over twisted sheaves $\mathcal{O}_X(n)$.

We then bring over the generalization of deRham cohomology, and generalize Poincaré's duality to *Serre duality.* Through this, we can take $\mathscr{F}$ is a constant sheaf (ex. $\mathbb{R}$ or $\mathbb{C}$), and equate sheaf cohomology to the usual *singular cohomology* as studied in a first course in algebraic topology (see [3]).

The reader may want to review [9, Part VII] to get a refresher on the derived functor approach to (co)homology theory.

6.1 Definition and Basic Properties

Recall that a category $\mathbf{C}$ is pre-additive (or enriched over the category of abelian groups) if for every $A, B \in \mathrm{Obj}(\mathbf{C})$, $\mathrm{Hom}_{\mathbf{C}}(A, B)$ is an abelian group and composition of morphisms is bilinear:

$$f \circ (g + h) = (f \circ g) + (f \circ h) \qquad \text{and} \qquad (f + g) \circ h = (f \circ h) + (g \circ h)$$

An *additive category* is a pre-additive category that is closed under finite product and coproduct, and furthermore products are the same as coproducts (this is sometimes called *biproducts*).

An abelian category is additive and has the "right" conditions to do cohomology[1]. In particular, a category $\mathbf{A}$ is *abelian* if:

1. for every $A, B \in \mathbf{A}$, $\mathrm{Hom}_{\mathbf{A}}(A, B)$ is an abelian group under point-wise multiplication
2. finite products and coproducts exist, and they equal each other
3. every morphism $f \in \mathrm{Hom}_{\mathbf{A}}(A, B)$ has a kernel and cokernel.
4. every monomorphism is the kernel of its cokernel, every epimorphism is the cokernel of its kernel.
5. Image is defined as $\mathrm{im}(f) = \ker(\mathrm{coker}(f))$. By the first isomorphism theorem

$$\mathrm{im}(f) = \ker(\mathrm{coker}(f)) = \mathrm{coker}(\ker(f))$$

   As a consequence, every morphism factors into an epimorphism follows by a monomorphism.

The category R-**Mod** is the typical abelian category, and by the Freyd-Mitchell theorem any (small) abelian category $\mathbf{A}$ fully-faithfully embeds into R-**Mod** via an exact functor for appropriate (not necessarily commutative ring) R. This is useful theoretically, however it is more practical to think of the abelian category $\mathbf{Sh}_{\mathbf{A}}(X)$ as a new type of abelian category. The "sheafification" of R-**Mod** gives $\mathcal{O}_X$-**Mod**. The important full subcategories $\mathbf{QCoh}(X)$ and $\mathbf{Coh}(X)$ are both abelian, and $\mathbf{QCoh}(X)$ has enough injectives (as we'll soon see). Though not particularly important at the moment, it should be pointed out that $\mathbf{QCoh}(X)$ is not closed under internal Hom's, while $\mathbf{Coh}(X)$ is closed under internal Hom's.

A functor between abelian categories $F : \mathbf{A} \to \mathbf{B}$ is *additive* if it preserves Hom's, that is for any $X, Y \in \mathrm{Obj}(\mathbf{A})$

$$\mathrm{Hom}_{\mathbf{A}}(X, Y) \to \mathrm{Hom}_{\mathbf{B}}(FX, FY)$$

is a (group) homomorphism. A functor F is *left/right* exact if it is additive and preserves kernel/cokernels respectively. In other words, it left exact if a short exact sequence preserves exactness on the left:

$$0 \to FX \to FY \to FZ$$

and right-exact if it preserves exactness on the right:

$$FX \to FY \to FZ \to 0$$

If F is both left and right exact, it is called *exact*. The most common (and arguably important) left-exact functor are the covariant functor $\mathrm{Hom}(X, -)$ and contravariant $\mathrm{Hom}(-, Y)$. The most important right-exact functor is $- \otimes Y$ (note $X \otimes -$ is also covariant and adjoint to $\mathrm{Hom}(X, -)$).

Given a left- or right-exact functor $F : \mathbf{A} \to \mathbf{B}$, we can induced a *derived functor* $D(F) : D(\mathbf{A}) \to D(\mathbf{B})$ between the derived categories that gives the right information to fix exactness. The derived categories is abstractly defined by taking the category of complexes $\mathrm{Kom}(\mathbf{A})$, passing to the homotopy category $K(\mathbf{A}^\bullet)$, and then localizing on quasi-isomorphisms. Such a category does not lends itself well to computations. If $\mathbf{A}$ has enough projective or enough injectives, then projective/injective resolution are *universal* with respect to resolutions up to homotopy, and furthermore $D(\mathbf{A}) = K(\mathbf{A}^\bullet)$. Hence,

[1]Technically, you can do cohomology on exact categories that are not abelian, such as vector bundles, but it is more natural to see this as a special generalization of abelian categories for the cases like vector bundles

it is much more desirable to work with abelian categories with either enough projective or enough injectives. The details of all of these constructions can be found in [9, chapter 27]. A "classical" approach using δ-functors is briefly introduced at the end of the chapter 27 of the aforementioned book.

Recall that if R is a ring, then every R-module is isomorphic to a submodule of an injective module (see [9, chatper 16.3]). From this, we shall be able to conclude that ringed space have enough injectives:

Proposition 6.1.1: Ringed Space Has Enough Injectives

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category of sheaves of $\mathcal{O}_X$-modules, ${}_{\mathcal{O}_X}\mathbf{Mod}$ has enough injectives.

***Proof*:**
Let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then for each $x \in X$, the stalk $\mathscr{F}_x$ is an $\mathcal{O}_{x,X}$-module. Then there is an injection

$$\mathscr{F}_x \to I_x$$

where I_x is an injective $\mathcal{O}_{x,X}$-module. We can do this for each point. Now, let $j : \{x\} \to X$ denote the inclusion of the one-point space for each point, let $\{x\}$ have I_x as a sheaf, and define the sheaf:

$$\mathscr{I} = \prod_{x \in X} j_*(I_x)$$

where j_* is the usual pushforward. This shall be our injective object.

For any $\mathcal{O}_X$-module $\mathscr{G}$, consider

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, \mathscr{I}) \cong \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, j_*(I_x))$$

Then for each point $x \in X$, notice that:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, j_*(I_x)) \cong \mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathscr{G}_x, I_x)$$

From this, if $\mathscr{G} = \mathscr{F}$ we a natural injective $\mathcal{O}_X$-module map $\mathscr{F} \to \mathscr{I}$ given by $\mathscr{F}_x \to I_x$, and furthermore $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact as it is the product $\mathrm{Hom}_{\mathcal{O}_{x,X}}(-, I_x)$ which is exact as I_x is injective, and the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$ which is exact by proposition 1.2.22. Hence, $\mathrm{Hom}(-, \mathscr{I})$ is exact, making $\mathscr{I}$ an injective $\mathcal{O}_X$-module, completing the proof.

Corollary 6.1.2: Sheaves of Abelian's Has Enough Injectives

Let X be a topological space. Then the category of sheaves of abelian groups on X has enough injectives

***Proof*:**
Take $\mathcal{O}_X = \mathbb{Z}$ and apply proposition 6.1.1

Definition 6.1.3: Global Section Functor and Sheaf Cohomology

Let X be a topological space. Recall the *global section functor*

$$\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab}$$

is left- but not right-exact (by proposition 1.2.16 and example 1.2.7 respectively). Then the cohomology functors $H^i(X, -)$, i.e. the right derived functors of $\Gamma(X, -)$, are the *cohomology groups* of the global section functor (with respect to X).

To be able to create resolutions, recall that it is enough to find acyclic objects ([9, chapter 27.4]). We shall show that the following types of objects shall be acyclic:

Definition 6.1.4: Flasque Sheaf

Let X be a topological space. Then a sheaf $\mathscr{F}$ on X is called *flasque* (or *flabby*) if for every inclusion $U \subseteq V$, $\text{res} : \mathscr{F}(V) \to \mathscr{F}(U)$ is surjective.

Proposition 6.1.5: Injective Sheaf Modules Are Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space. Then any injective $\mathcal{O}_X$-module is flasque

***Proof*:**
For any open subset $V \subseteq X$, let $j : V \to X$ be the open immersion and $\mathcal{O}_V = j_!(\mathcal{O}_X|_V)$, which is the restriction of $\mathcal{O}_X$ to V, extended by zero outside V[a]. Now let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module, and let $V \subseteq U$ be open sets. Then we have an inclusion $0 \to \mathcal{O}_U \to \mathcal{O}_V$ of sheaves of $\mathcal{O}_X$-modules. Since $\mathscr{I}$ is injective, we get a surjection $\text{Hom}(\mathcal{O}_V, \mathscr{I}) \to \text{Hom}(\mathcal{O}_U, \mathscr{I}) \to 0$. But $\text{Hom}(\mathcal{O}_U, \mathscr{I}) = \mathscr{I}(U)$ and $\text{Hom}(\mathcal{O}_V, \mathscr{I}) = \mathscr{I}(V)$, so $\mathscr{I}$ is flasque.

[a]Recall how this is different from j_*, namely $(j_!\mathcal{F})(U) = \{s \in \mathcal{F}(U \cap V) \big| \text{supp}(s) \text{ is closed in } U\}$

Corollary 6.1.6: Trivial Homology For Flask Sheaf

Let $\mathscr{F}$ be a flasque sheaf on X. Then

$$H^i(X, \mathscr{F}) = 0 \qquad i > 0$$

***Proof*:**
As there are enough injective, embed $\mathscr{F}$ into $\mathscr{I}$ and take it's quotient $\mathscr{G}$. Most importantly, check that the quotient sheaf of two flasque sheaves is flasque, and so we have a short exact sequence of flasque sheaves:

$$0 \to \mathscr{F} \to \mathscr{I} \to \mathscr{G} \to 0$$

As $\mathscr{F}$ is flasque, we have the $\Gamma(X, -)$ will preserve this exact sequence:

$$0 \to \Gamma(X, \mathscr{F}) \to \Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{G}) \to 0$$

Showing $H^1(X, \mathscr{F}) = 0$. As $\mathscr{I}$ is injective, $H^i(X, \mathscr{I}) = 0$ for $i > 0$, and so by taking the long exact sequence of cohomology:

$$H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{G}),\ i \geq 2$$

As $\mathscr{G}$ is also injective, we get by induction that $H^i(X, \mathscr{F}) = 0$ for $i > 0$, completing the proof.

Hence, flasque sheaves are acyclic for $\Gamma(X, -)$, and so we may compute cohomology by taking Flasque resolutions!

Example 6.1.7: Flasque Sheaves

1. It was just shown that all injective sheaves are flasque
2. Show that the constant sheaf is flasque on an irreducible space (this need not be the case on reducible spaces).
3. Let $\mathscr{F}$ be any sheaf. Then we can always construct a Flasque resolution known as the *Godement resolution*. Define:
$$C^0(\mathscr{F}) = \prod_{x \in X} \iota_{x,*}(\mathscr{F}_x)$$
So:
$$\Gamma(X, C^0(\mathscr{F})) = \prod_{x \in X} \mathscr{F}_x$$
Note this sheaf *need not be injective*. For example if $\mathscr{F} = \underline{\mathbb{Z}}$, then this sheaf is $\mathbb{Z}$ at each stalk, but it is easy to verify that in the category of sheaves over abelian groups all stalks should be divisible groups. Nevertheless this sheaf is certainly flasque as there is no gluing restriction that is applied to the product! We may then take:
$$0 \to \mathscr{F} \xrightarrow{\epsilon} C^0(\mathscr{F})$$
Let $Q = C^0(\mathscr{F})/\mathrm{im}(\epsilon)$. We now have another sheaf that we can apply this construction; label this sheaf $C^1(\mathscr{F})$. Repeating this process, we get:
$$0 \to \mathscr{F} \to C^0(\mathscr{F}) \to C^1(\mathscr{F}) \to C^2(\mathscr{F}) \to \cdots$$
Note that these sheaves are *not* quasi-coherent!
4. a very similar sheaf is the sheaf of Weil divisors, namely let X be a normal scheme and:
$$Div \cong \bigoplus_{x \in X^{(1)}} i_{x*}(\mathbb{Z})$$
where $X^{(1)}$ is the set of codimension-1 points. Then:
$$1 \to \mathcal{O}^* \to \mathcal{K}^* \to Div \to 0$$
Note that $\mathcal{O}^*$ are invertible regular functions, $\mathcal{K}^*$ is in fact the constant sheaf (hence *flasque*) and Div is flasque. Hence
$$\Gamma(X, \mathcal{K}^*) \to \Gamma(X, Div) \to H^1(X, \mathcal{O}^*) \to 0$$
i.e.
$$H^1(X, \mathcal{O}^*) \cong \frac{\mathrm{WDiv}\,(X)}{\mathrm{PDiv}(X)}$$
recovering the result from section 5.3 as cohomological information.

Note that acyclic doesn't imply flasque. Though not important for our discussion, it is worth pointing out the notion of *fine sheaf* which is acyclic:

Example 6.1.8: Acyclic, But Not Injective Or Flasque

(sheaves with partitions of unity. Not flasque because consider $1/x$ on $(0,1)$ that cannot extend to $(-1,1)$. Not injective over the wrong coefficients. But will be shown to be acyclic using Čech cohomology.

Finally, let us link this to ringed spaces. If $(X, \mathcal{O}_X)$ is a ringed space, by taking $R = \Gamma(X, \mathcal{O}_X)$, we always have a natural $\overline{\mathbb{Z}}$-module structure on any $\mathcal{O}_X$-modules $\mathscr{F}$. Thus, the cohomology groups of $\mathscr{F}$ have a natural $\mathbb{Z}$-module structure on them, and the associated exact sequence are naturally $\mathbb{Z}$-modules. Then for R-schemes, the cohomology groups of any $\mathcal{O}_X$-module will have natural R-modules structure. We immediately get:

Corollary 6.1.9: Section and Hom Functor

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the derived functors of $\Gamma(X, -)$ coincide with $H^i(X, -)$

***Proof*:**
immediate from above.

6.1.1 Computation: Čech Cohomology

(I want to move this here because using Čech cohomology, we can generalize the vanishing theorem for Noetherian schemes, namely we can drop the Noetherian condition, which feels appropriate for a book with this level of generality).

The Čech Cohomology computes to what degree a sections on an open cover $\mathcal{U}$ of topological space X can be extended to the larger space X, or more generally from an intersection of open sets to a larger open set. In the case where X is a Noetherian separated scheme with a quasi-coherent sheaf and the covering is an affine open covering, then the Čech cohomology groups shall coincide with the cohomology groups derived by the global section functor. Čech cohomology shall provide a practical way to compute sheaf cohomology. The following construction will be familiar to those who have seen the computation of group cohomology or have covered algebraic topology (see [3, chapter 4])

Let X be a topological space and let $\mathcal{U} = (U_i)_{i \in I}$ be an open covering. Give I a well-ordering[2]. Then for any finite set of indices $i_0, ..., i_p \in I$, let

$$U_{i_0,\dots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$$

Let $\mathscr{F}$ be a sheaf of abelian groups on X. Define $C^*(\mathcal{U}, \mathscr{F})$ to be:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\dots,i_p})$$

[2]Alternatively, make sure that if indices repeat then the element is 0 , and rearranging the indices gives multiplies the function by $(-1)^\sigma$

where each element $\alpha \in C^p(\mathcal{U}, \mathscr{F})$ is given choosing an element $\alpha_{i_0,\dots,i_p} \in \mathscr{F}(U_{i_0,\dots,i_p})$. For each $(p+1)$-tuple $i_0 < \cdots < i_p$. Define the coboundary map $d : C^p \to C^{p+1}$ to be:

$$(d\alpha)_{i_0,\dots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \alpha_{i_0,\dots,\widehat{i_k},\dots,i_{p+1}}|_{U_{i_0,\dots,i_{p+1}}}$$

As usual, $d^2 = 0$, meaning we have a complex and hence a cohomology:

Definition 6.1.10: Čech Cohomology

Let X be a topological space and let $\mathcal{U}$ be an open covering of X. For any sheaf of abelian groups $\mathscr{F}$ on X, define the *Čech cohomology group* of $\mathscr{F}$ with respect to $\mathcal{U}$ to be:

$$\check{\mathrm{H}}^p(\mathcal{U}, \mathscr{F}) = h^p(C^*(\mathcal{U}, \mathscr{F})) = \frac{\ker d^{+1}}{\operatorname{im} d}$$

Note that $\check{\mathrm{H}}^p(\mathfrak{U}, -)$ is not in general a δ-functor, that is if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of sheaves of abelian groups on X, $\check{H}^p(\mathfrak{U}, -)$ is not a δ-functor:

Example 6.1.11: Čech cohomology not δ-functor

Say $\mathfrak{U}$ consists of only X. Then $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$ for $p > 0$ because all intersection are trivial. Furthermore, $\check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) = \check{\mathrm{H}}^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$. Hence, if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, then applying $\check{\mathrm{H}}^*(\mathfrak{U}, -)$ we should get the "long exact sequence":

$$0 \to \Gamma(X, \mathscr{F}') \to \Gamma(X, \mathscr{F}) \to \Gamma(X, \mathscr{F}'') \to 0$$

But $\Gamma(X, -)$ need not be right-exact, and hence this is not a long exact sequence!

This is one of the few cohomologies theories which does not work with derived functors for any choice of covering, even though they are central in most of homological algebra. This is fixed by choosing an appropriate open covering given the cohomology of interest. In our case, we shall be interested in affine open coverings.

Example 6.1.12: Čech Cohomology

1. Let $X = \mathbb{P}^1_k$, $\mathscr{F}$ be sheaf of differentials Ω, and $\mathfrak{U}$ the usual affine open covering of two $\mathbb{A}^1_k$ with coordinates x and $1/x$; label these sets U, V. Then we get two terms:

$$\begin{aligned} C^0 &= \Gamma(U, \Omega) \times \Gamma(V, \Omega) \\ C^1 &= \Gamma(U \cap V, \Omega) \end{aligned}$$

Then:

$$\begin{aligned} \Gamma(U, \Omega) &= k[x]dx \\ \Gamma(V, \Omega) &= k[y]dy \\ \Gamma(U \cap V, \Omega) &= k\left[x, \frac{1}{x}\right] dx \end{aligned}$$

The differential map $d : C^0 \to C^1$ is given by:

$$x \mapsto x$$
$$y \mapsto \frac{1}{x}$$
$$dy \mapsto \frac{1}{x^2}dx$$

Hence, ker d is all elements $(f(x)dx, g(y)dy) \in C^0$ such that

$$f(x) = \frac{-1}{x^2} g\left(\frac{1}{x}\right)$$

But this is true if and only if $f = g = 0$. Hence:

$$\check{\mathrm{H}}^0(\mathfrak{U}, \Omega) = 0$$

Next, for $\check{\mathrm{H}}^1$, the image of d is all expression of the form:

$$\left(f(x) = \frac{1}{x^2} g\left(\frac{1}{x}\right)\right) dx$$

This is a sub-space of $k[x, 1/x]dx$ generated by all $x^n dx$ with $-1 \neq n \in \mathbb{Z}$. Therefor $C^1/\mathrm{im}(d) = k$, that is:

$$\check{\mathrm{H}}^1(\mathfrak{U}, \Omega) = k$$

which is generated by the image of $x^{-1}dx$ (the space is 1-dimensional over k)

2. Let us compute the Čech cohomology in a more classical setting: take S^1 with the usual euclidean topology and put the constant $\mathbb{Z}$ sheaf on it, $\underline{\mathbb{Z}}$. Let U, V be two open semi-circle where $U \cap V$ is the disjoint small open intervals. Then:

$$C^0 = \Gamma(U, \underline{\mathbb{Z}}) \times \Gamma(V, \underline{\mathbb{Z}}) = \mathbb{Z} \times \mathbb{Z}$$
$$C^1 = \Gamma(U \cap V, \underline{\mathbb{Z}}) = \mathbb{Z} \times \mathbb{Z}$$

Then $d : C^0 \to C^1$ is given by

$$d(f_U, f_V) = f_V|U \cap V - f_U|U \cap V(a, b) \qquad \mapsto (b - a, b - a)$$

Thus:

$$\check{\mathrm{H}}^0(\{U, V\}, \underline{\mathbb{Z}}) = \mathbb{Z} \qquad \text{and} \qquad \check{\mathrm{H}}^1(\{U, V\}, \underline{\mathbb{Z}}) = \mathbb{Z}$$

which aligns with our intuition from singular cohomology.

Let us now build-up towards showing the cohomology given by the derived functor on Γ matches the Čech cohomology. The main idea is that each open subset must be cohomologically trivial in order for the main obstruction comes from their "combinations".

Lemma 6.1.13: 0th Cohomology Matching

Let X be a topological space, $\mathfrak{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Then:

$$\check{\mathrm{H}}^0(\mathfrak{U},\mathscr{F})=\Gamma(X,\mathscr{F})$$

***Proof*:**

By definition, $\check{\mathrm{H}}^0(\mathfrak{U},\mathscr{F})=\ker\left(d:C^0(\mathfrak{U},\mathscr{F})\to C^1(\mathfrak{U},\mathscr{F})\right)$. If $\alpha\in C^0$ is given by $\{\alpha_i\in\mathcal{F}(U_i)\}$, then for each $i<j,(d\alpha)_{ij}=\alpha_j-\alpha_i$. So $d\alpha=0$ says the sections α_i and α_j agree on $U_i\cap U_j$. But then by the sheaf axioms $\ker d=\Gamma(X,\mathcal{F})$.

We next work towards fixing the lack of a δ-functor. Let $X,\mathfrak{U},\mathscr{F}$ be as above, $V\subseteq X$ an open subset, and $\iota:V\to X$ the natural inclusion map. Define the complex $\mathscr{C}(\mathfrak{U},\mathscr{F})$ of sheaves of on X for each $p\geq 0$ as:

$$\mathscr{C}(\mathfrak{U},\mathscr{F})=\prod_{i_0<\dots<i_p}f_*\left(\mathscr{F}|_{U_{i_0,\dots,i_p}}\right)$$

and define the differential map $d:\mathscr{C}^p\to\mathscr{C}^{p+1}$ in the same way. Importantly, this was defined so that:

$$\Gamma\left(X,\mathscr{C}^p(\mathfrak{U},\mathscr{F})\right)=C^p(\mathfrak{U},\mathscr{F})$$

Lemma 6.1.14: Čhech Cohomology Resolution

Let X be a topological space, $\mathfrak{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups. Then the complex $\mathscr{C}^*(\mathfrak{U},\mathscr{F})$ is a resolution of $\mathscr{F}$, namely we have a long exact sequence:

$$0\to\mathscr{F}\xrightarrow{\epsilon}\mathscr{C}^0(\mathfrak{U},\mathscr{F})\to\mathscr{C}^1(\mathfrak{U},\mathscr{F})\to\cdots$$

for appropriate natural map $\epsilon:\mathscr{F}\to\mathscr{C}^0$

***Proof*:**

Define $\epsilon:\mathscr{F}\to\mathfrak{U}^0$ by taking the product of the natural maps $\mathscr{F}\to f_*\left(\mathscr{F}|_{U_i}\right)$ for $i\in I$. Then the exactness at the first step follows from the sheaf axioms for $\mathscr{F}$.

For the exactness of the complex $\mathscr{C}$ for $p\geq 1$, it is enough to check exactness on the stalks. Let $x\in X$, and suppose $x\in U_j$. For each $p\geq 1$, define the map $k:\mathscr{C}^p(\mathfrak{U},\mathscr{F})_x\to\mathcal{C}^{p-1}(\mathfrak{U},\mathscr{F})_x$ as follows: for $\alpha_x\in\mathscr{C}^p(\mathfrak{U},\mathscr{F})_x$, it is represented by a section $\alpha\in\Gamma\left(V,\mathscr{C}^p(\mathfrak{U},\mathscr{F})\right)$ over a neighborhood V of x, which we may choose so small that $V\subseteq U_j$. Now for any p-tuple $i_0<\ldots<i_{p-1}$, set

$$(k\alpha)_{i_0,\dots,i_{p-1}}=\alpha_{j,i_0,\dots,i_{p-1}}$$

This makes sense because $V\cap U_{i_0,\dots,i_{p-1}}=V\cap U_{j,i_0,\dots,i_{p-1}}$. Then take the stalk of $k\alpha$ at x to get the required map k. Now one checks that for any $p\geq 1,\alpha\in\mathscr{C}^p_x$,

$$(dk + kd)(\alpha) = \alpha$$

Thus k is a homotopy operator for the complex $\mathscr{C}_x$, showing that the identity map is homotopic to the zero map! It follows that the cohomology groups $h^p(\mathscr{C}_x)$ of this complex are 0 for $p \geq 1$.

Lemma 6.1.15: Čhech and Flasque Sheaves

Let X be a topological space, $\mathfrak{U}$ be an open covering, and let $\mathscr{F}$ be a flasque sheaf of abelian groups on X. Then for all $p > 0$ we have $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$.

***Proof*:**
Consider the resolution $0 \to \mathscr{F} \to \mathcal{C}(\mathfrak{U}, \mathscr{F})$ given by lemma 6.1.14. As $\mathscr{F}$ is flasque, the sheaves $\mathcal{C}^p(\mathfrak{U}, \mathscr{F})$ are flasque for each $p \geq 0$. Indeed, for any $i_0, \dots, i_p$, $\mathscr{F}|_{U_{i_0,\dots,i_p}}$ is a flasque sheaf on $U_{i_0,\dots,i_p}$, f_* preserves flasque sheaves, and a product of flasque sheaves is flasque. Thus, we can use this resolution to compute the usual cohomology groups of $\mathscr{F}$. But $\mathscr{F}$ is flasque, so $H^p(X, \mathscr{F}) = 0$ for $p > 0$. On the other hand, the answer given by this resolution is

$$h^p(\Gamma(X, \mathcal{C} \cdot (\mathfrak{U}, \mathscr{F}))) = \check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F})$$

But then $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$ for $p > 0$, as we sought to show.

Lemma 6.1.16: Chech Cohomology Functorial Map

Let X be a topological space and $\mathfrak{U}$ an open covering. Then for each $p \geq 0$ there is a natural map, functorial in $\mathscr{F}$,

$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$$

***Proof*:**
This follows from comparing two acyclic resolution (see [9]), namely:

$$0 \to \mathscr{F} \to \mathscr{I}^*$$
$$0 \to \mathscr{F} \to \mathscr{C}^*(\mathfrak{U}, \mathscr{F})$$

which gives a morphism of complexes $\mathscr{C}^*(\mathfrak{U}, \mathscr{F}) \to \mathscr{I}^*$ which induces the identity map on $\mathscr{F}$ and is unique up to homotopy. By applying the functors $\Gamma(X, -)$ and h^p, we extract the desired maps.

Theorem 6.1.17: Čech Cohomology Equivalence

Let X be a Noetherian separated scheme, $\mathfrak{U}$ an open affine cover of X, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then for all $p \geq 0$, the natural maps

$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$$

are isomorphisms.

***Proof*:**
For $p = 0$ we have an isomorphism by 6.1.13. For $p > 0$, by corollary 5.1.16 embed $\mathscr{F}$ in a flasque, quasi-coherent sheaf $\mathcal{G}$, and let $\mathcal{Q}$ be the quotient:

$$0 \to \mathscr{F} \to \mathcal{G} \to \mathcal{Q} \to 0$$

For each $i_0 < \ldots < i_p$, the open set $U_{i_0,\ldots,i_p}$ is affine, since it is an intersection of affine open subsets of a separated scheme. Since $\mathscr{F}$ is quasi-coherent, we therefore have an exact sequence

$$0 \to \mathscr{F}\left(U_{i_0,\ldots,i_p}\right) \to \mathcal{G}\left(U_{i_0,\ldots,i_p}\right) \to \mathcal{Q}\left(U_{i_0,\ldots,i_p}\right) \to 0$$

of abelian groups by proposition 4.1.13. Taking products, we find that the corresponding sequence of Čech complexes

$$0 \to C^{\cdot}(\mathfrak{U}, \mathscr{F}) \to C^{\cdot}(\mathfrak{U}, \mathcal{G}) \to C^{\cdot}(\mathfrak{U}, \mathcal{Q}) \to 0$$

is exact. Therefore we get a long exact sequence of Čech cohomology groups. Since $\mathcal{G}$ is flasque, its Čech cohomology vanishes for $p > 0$, so we have an exact sequence

$$0 \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{G}) \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{Q}) \to \check{\mathrm{H}}^1(\mathfrak{U}, \mathscr{F}) \to 0$$

and isomorphisms

$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathcal{Q}) \to \check{\mathrm{H}}^{p+1}(\mathfrak{U}, \mathscr{F})$$

for each $p \geq 1$. Now comparing with the long exact sequence of usual cohomology for the above short exact sequence, using the case $p = 0$, the fact it is flasque, we conclude that the natural map

$$\check{\mathrm{H}}^1(\mathfrak{U}, \mathscr{F}) \to H^1(X, \mathscr{F})$$

is an isomorphism. But $\mathcal{Q}$ is also quasi-coherent (proposition 4.1.14), so we obtain the result for all p by induction, completing the proof.

6.1.2 Application: Sheaf Cohomology of Projective Spaces

Let us now compute the cohomology for the twisted sheaves on projective space. Let R be a Noetherian ring, $S_\bullet = R\left[x_0, \ldots, x_n\right]$, and $X = \operatorname{Proj} S = \mathbb{P}^n_R$. Let $\mathcal{O}_X(1)$ be the twisted sheaf of Serre. If $\mathscr{F}$ is a sheaf of $\mathcal{O}_X$-module, let $\Gamma_*(\mathscr{F})$ be the graded S-module

Theorem 6.1.18: Twisted Sheaf Cohomology

Let R be a Noetherian ring, and let $X = \mathbb{P}^n_R$, with $n \geq 1$. Then:

1. $H^0(X, \mathcal{O}_X(-d)) \cong 0$ for $d \in \mathbb{N}$
2. $H^n(X, \mathcal{O}_X(-n-1)) \cong R$.
3. The natural map

$$H^0(X, \mathcal{O}_X(n)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \to H^n(X, \mathcal{O}_X(-n-1)) \cong R$$

is a perfect pairing (isomorphic bilinear map) of finitely generated free R-modules, for each $d \in \mathbb{Z}$.
4. $H^i(X, \mathcal{O}_X(d)) = 0$ for $0 < i < n$ and all $d \in \mathbb{Z}$

Overall:

$$H^i\left(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}(d)\right) = \begin{cases} (R[x_0, \dots, x_n])_d & i = 0, d \geq 0 \\ \left(\frac{1}{x_0 x_1 \cdots x_n} R[1/x_0, \dots, 1/x_n]\right)_d & i = n, d < 0 \\ 0 & \text{otherwise} \end{cases}$$

Notice how this is different from the singular cohomology of projective space, namely the higher order cohomology groups when $R = \mathbb{C}$ and $d = 0$ are trivial! See [3, section 4.1] for computations of the singular cohomology of $\mathbb{P}^n_{\mathbb{C}}$.

***Proof*:**

Let us first setup the Čech cohomology. Let $\mathscr{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_X(d)$. As cohomology commutes with direct sums on Noetherian space, the cohomology on $\mathscr{F}$ will be the direct sum of the cohomology on $\mathcal{O}_X(d)$ for $d \in \mathbb{Z}$. Hence, we can compute $\mathscr{F}$ to find the cohomologies.

Let $D(x_i)$ cover $\operatorname{Proj} S, 0 \leq i \leq n$: label this open cover $\mathcal{U}$. Then for any direct subset of indices, we have:

$$\mathscr{F}\left(U_{i_0, \dots, i_p}\right) = S_{x_{i_0}, \dots, x_{i_p}}\Big)$$

Thus, the grading of $\mathscr{F}$ is given by the grading of S, and we get:

$$C^*(\mathcal{U}, \mathscr{F}) : \prod S_{x_{i_0}} \to \prod S_{x_{i_0}, x_{i_1}} \to \cdots \to S_{x_{i_0}, \dots, x_{i_n}}$$

1. $H^0(X, \mathscr{F})$ is the kernel of d^0 (the first map), which is just S by proposition 4.2.7.
2. $H^0(X, \mathcal{O}_X(-d))$ must be 0 as we have the isomorphism above.
3. $H^n(X, \mathscr{F})$ is the cokernel of of d^{n-1} (the last map). To see what this is, notice that $S_{x_{i_0}, \dots, x_{i_n}}$ is a free R-module generated by monomial $\left\{x_0^{l_0} x_1^{l_1} \cdots x_n^{l_n} \middle| l_i \in \mathbb{Z}\right\}$. The image of d^{n-1} is generated by the sub-basis where at least one $l_i \geq 0$. Hence, $H^n(\mathscr{F}, X)$ is a free R-module generated by the negative monomials:

$$\left\{x_0^{l_0} x_1^{l_1} \cdots x_n^{l_n} \middle| l_i \in -\mathbb{N}\right\}$$

Note that this is graded by $\sum_i l_i$. Now there is only one monomial of degree $-n-1$, namely:

$$x_0^{-1} \cdots x_n^{-1}$$

hence:

$$H^n\left(X, \mathcal{O}_X(-n-1)\right) \cong R$$

4. Note that $H^n\left(X, \mathcal{O}_X(-d-n-1)\right) \cong 0$ since there are no negative monomials of degree bigger than 0 for $m < -n-1$. For $d \geq 0, H^0\left(X, \mathcal{O}_X(d)\right)$ has a basis consisting of the usual monomials of degree d, i.e., $\left\{x_0^{m_0} \cdots x_n^{m_n} \middle| m_i \geq 0 \text{ and } \sum m_i = d\right\}$. The natural pairing with $H^n\left(X, \mathcal{O}_X(-n-d-1)\right)$ into $H^n\left(X, \mathcal{O}_X(-n-1)\right)$ is determined by

$$\left(x_0^{m_0} \cdots x_n^{m_n}\right) \cdot \left(x_0^{l_0} \cdots x_n^{l_n}\right) = x_0^{m_0+l_0} \cdots x_n^{m_n+l_n}$$

where $\sum l_i = -d-n-1$, and the object on the right becomes 0 if any $m_i + l_i \geq 0$. So it is clear that there is a perfect pairing under which $x_0^{-m_0-1} \cdots x_n^{-m_n-1}$ is the dual basis element of $x_0^{m_0} \cdots x_n^{m_n}$.

5. Let us do induction on n. If $n=1$ there is nothing to prove, so let $n>1$. If we localize the complex $C^*(\mathfrak{U}, \mathscr{F})$ with respect to x_n, as graded S-modules, we get the Čech complex for the sheaf $\mathscr{F}|_{U_n}$ on the space U_n, with respect to the open affine covering $\left\{U_i \cap U_n \middle| i = 0, \ldots, n\right\}$. By (4.5), this complex gives the cohomology of $\mathscr{F}|_{U_n}$ on U_n, which is 0 for $i > 0$ by (3.5). Since localization is an exact functor, we conclude that $H^i(X, \mathscr{F})_{x_n} = 0$ for $i > 0$. In other words, every element of $H^i(X, \mathscr{F})$, for $i > 0$, is annihilated by some power of x_n.
Next let's show that for $0 < i < n$, multiplication by x_n induces a bijective map of $H^i(X, \mathscr{F})$ into itself. Then it will follow that this module is 0 . Consider the exact sequence of graded S-modules

$$0 \to S(-1) \xrightarrow{x_n} S \to S/\left(x_n\right) \to 0$$

This gives the exact sequence of sheaves

$$0 \to \mathcal{O}_X(-1) \to \mathcal{O}_X \to \mathcal{O}_H \to 0$$

on X, where H is the hyperplane $x_n = 0$. Twisting by all $d \in \mathbb{Z}$ and taking the direct sum, we have

$$0 \to \mathscr{F}(-1) \to \mathscr{F} \to \mathscr{F}_H \to 0$$

where $\mathscr{F}_H = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_H(d)$. Taking cohomology, we get a long exact sequence

$$\cdots \to H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}_H) \to \cdots$$

Considered as graded S-modules, $H^i(X, \mathscr{F}(-1))$ is just $H^i(X, \mathscr{F})$ shifted one place, and the map $H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F})$ of the exact sequence is multiplication by x_n.
Now H is isomorphic to $\mathbb{P}_R^{n-1}$, and $H^i(X, \mathscr{F}_H) = H^i(H, \bigoplus \mathcal{O}_H(d))$ by lemma 5.1.10. So we can apply our induction hypothesis to $\mathscr{F}_H$, and find that $H^i(X, \mathscr{F}_H) = 0$ for $0 < i < n-1$. Furthermore, for $i = 0$ we have an exact sequence

$$0 \to H^0(X, \mathscr{F}(-1)) \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H) \to 0$$

by (1), since $H^0(X, \mathscr{F}_H)$ is just $S/(x_n)$. At the other end of the exact sequence we have

$$0 \to H^{n-1}(X, \mathscr{F}_H) \xrightarrow{\partial} H^n(X, \mathscr{F}(-1)) \xrightarrow{x_n} H^n(X, \mathscr{F}) \to 0$$

Indeed, we have described $H^n(X, \mathscr{F})$ above as the free R-module with basis formed by the negative monomials in $x_0, \ldots, x_n$. So it is clear that x_n is surjective. On the other hand, the
kernel of x_n is the free submodule generated by those negative monomials $x_0^{l_0} \cdots x_n^{l_n}$ with $l_n = -1$. Since $H^{n-1}(X, \mathscr{F}_H)$ is the free R-module with basis consisting of the negative monomials in $x_0, \ldots, x_{n-1}$, and ∂ is division by x_n, the sequence is exact. In particular, ∂ is injective.
Putting these results all together, the long exact sequence of cohomology shows that the map multiplication by $x_n : H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F})$ is bijective for $0 < i < n$, as we sought to show.

6.2 Vanishing Theorems

(moved this section here because we can prove the dimension theorem for schemes using Čech cohomology *wihtout* the Noetherian condition without extra difficulty)

Two fundamental results we want to find:

1. An important result of a valid cohomology theory is that the top-degree cohomology shouldn't be higher than the dimension of the base space, however we appropriately define dimension (see [9, chapter 27] for a cursory overview of cohomology theory). We shall prove exactly this.
2. On affine space with a Q.C. sheaf, we expect and will prove we get trivial cohomology

6.2.1 Dimension Theorem for Schemes

Let us now prove our first standard sheaf cohomology result, in particular that the homological dimension and "geometric" dimension match in suitably nice contexts. Naturally if a space X is not Noetherian, the dimension can be infinite, and even if X is finite dimension in the appropriate sense it may behave pathologically on open components. Hence let us restrict to X being Noetherian. To prove this, let us first build-up the appropriate results for the cohomology of Noetherian space

Lemma 6.2.1: Noetherian Space and Flasque Sheaves

Let X be a Noetherian topological space. Then the colimit of flasque sheaves is flasque.

***Proof*:**
Let $(\mathscr{F}_\alpha)$ be a directed (filtered) system of flasque sheaves. Then for any inclusion $V \subseteq U$, for each α we have $\mathscr{F}_\alpha(U) \to \mathscr{F}_\alpha(V)$ is surjective. As $\varinjlim_\alpha$ is an right-exact functor (or as $(\mathscr{F}_\alpha)$ is filtered, an exact functor), we get that the map:

$$\varinjlim_\alpha \mathscr{F}_\alpha(U) \to \varinjlim_\alpha \mathscr{F}_\alpha(V)$$

is surjective. Then on a Noetherian topological space:

$$\varinjlim \left(\mathscr{F}_\alpha(U)\right) = \left(\varinjlim \mathscr{F}_\alpha\right)(U)$$

This is *not* necessarily true if X is not Noetherian: it is always true for a presheaf but for a general sheaf you may need to sheafify, however the Noetherian property we can find a single index where every section comes from the "maximal" section. This simplification implies:

$$\left(\varinjlim_\alpha \mathscr{F}_\alpha\right)(U) \to (\varinjlim_\alpha \mathscr{F}_\alpha)(V)$$

is surjective, hence $\lim_\rightarrow \mathscr{F}_\alpha$ is flasque, completing the proof.

Lemma 6.2.2: Colimit and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)$ a direct system of abelian sheaves. Then there is a natural isomorphism for each $i \geq 0$:

$$\varinjlim H^i\left(X, \mathscr{F}_\alpha\right) \cong H^i\left(X, \varinjlim \mathscr{F}_\alpha\right)$$

***Proof*:**
We proceed using the definition of cohomology via resolutions. Since the direct system is filtered, we may choose a functorial flasque resolution for the system. For each α, let:

$$0 \to \mathscr{F}_\alpha \to \mathcal{I}_\alpha^\bullet$$

be a flasque resolution of $\mathscr{F}_\alpha$. By definition, the cohomology groups $H^i(X, \mathscr{F}_\alpha)$ are computed as the cohomology of the of global sections $\Gamma(X, \mathcal{I}_\alpha^\bullet)$. Since filtered colimits are exact in the category of abelian sheaves, taking the colimit of these resolutions yields an exact sequence:

$$0 \to \varinjlim \mathscr{F}_\alpha \to \varinjlim \mathcal{I}_\alpha^\bullet$$

Thus, the complex $\varinjlim \mathcal{I}_\alpha^\bullet$ is a resolution of $\varinjlim \mathscr{F}_\alpha$. By lemma 6.2.1, since X is Noetherian, the filtered colimit of flasque sheaves is flasque. Therefore, each term in the complex $\varinjlim \mathcal{I}_\alpha^\bullet$ is $\Gamma(X, -)$-acyclic, and this complex can be used to compute the cohomology of the limit sheaf.

We now compare the computations. For the left-hand side, since filtered colimits of abelian groups are exact, they commute with the cohomology functor H^i of a complex:

$$\varinjlim H^i(X, \mathscr{F}_\alpha) = \varinjlim H^i\left(\Gamma(X, \mathcal{I}_\alpha^\bullet)\right) \cong H^i\left(\varinjlim \Gamma(X, \mathcal{I}_\alpha^\bullet)\right)$$

For the right-hand side, using the resolution constructed above:

$$H^i(X, \varinjlim \mathscr{F}_\alpha) \cong H^i\left(\Gamma(X, \varinjlim \mathcal{I}_\alpha^\bullet)\right)$$

It remains only to show that $\varinjlim \Gamma(X, \mathcal{I}_\alpha^\bullet) \cong \Gamma(X, \varinjlim \mathcal{I}_\alpha^\bullet)$, but since X is Noetherian, global sections commute with filtered colimits of sheaves, and so:

$$\varinjlim H^i\left(X, \mathscr{F}_\alpha\right) \cong H^i\left(X, \varinjlim \mathscr{F}_\alpha\right)$$

as we sought to show.

Corollary 6.2.3: Direct sum and cohomology commuting

Let X be a Noetherian topological space. Then there is a natural isomorphism for each $i \geq 0$:

$$\bigoplus_\alpha H^i\left(X, \mathscr{F}_\alpha\right) \cong H^i\left(X, \bigoplus_\alpha \mathscr{F}_\alpha\right)$$

***Proof*:**
The direct is a special case of a colimit.

This proof technique works whenever applying a functor keeps our shaves flasque. An important example of this is with sheaves induced on closed subsets:

Proposition 6.2.4: Cohomology of Closed Subsets

Let $Y \subseteq X$ be a closed subset of $X, \mathscr{F}$ a sheaf of abelian groups, and $j : Y \to X$ the natural inclusion. Then:

$$H^i(Y, \mathscr{F}) = H^i\left(X, j_* \mathscr{F}\right)$$

where $j_* \mathscr{F}$ is the pushforward (which as Y is closed is an extension of $\mathscr{F}$ by zero)

***Proof*:**
If $\mathscr{I}*$ is a flasque resolution of $\mathscr{F}, j_* \mathscr{F}^*$ is a flasque resolution of $j_* \mathscr{F}$, and so for each i :

$$\Gamma\left(Y, \mathscr{I}^i\right) = \Gamma\left(X, j_* \mathscr{I}^*\right)$$

But then

$$H^i(Y, \mathscr{F}) = H^i\left(X, j_* \mathscr{F}\right)$$

as we sought to show.

Due to this lemma, we shall often write $\mathscr{F}$ instead of $j_*\mathscr{F}$ without worrying about ambiguity. With these lemmas, we can prove our first main result:

Theorem 6.2.5: Noetherian Space and Dimension

Let X be a Noetherian topological space of dimension n. Then for all $i > n$, all sheaves of abelian groups $\mathscr{F}$ on X vanish:

$$H^i(X, \mathscr{F}) = 0 \qquad i > n$$

***Proof*:**
Let us first set some notation: let $Y \subseteq X$ be a closed subset, $\mathscr{F}$ be any sheaf on X, and $\mathscr{F}_Y = j_*\left(\mathscr{F}|_Y\right)$ where $j : Y \to X$ is the inclusion. Similarly for open subset $U \subseteq X$ where $\mathscr{F}_U = i_*\left(\mathscr{F}|_U\right)$ with the inclusion $i : U \to X$. If $U = X \backslash Y$, we get the short exact sequence:

$$0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_Y \to 0$$

With this setup, we shall prove the result by induction on the dimension of X.

Step 1. Reduction to the case X irreducible. If X is reducible, let Y be one of its irreducible components, and let $U = X - Y$. Then for any $\mathscr{F}$ we have an exact sequence

$$0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_Y \to 0.$$

From the long exact sequence of cohomology, it will be sufficient to prove that $H^i\left(X, \mathscr{F}_Y\right) = 0$ and $H^i\left(X, \mathscr{F}_U\right) = 0$ for $i > n$. But Y is closed and irreducible, and $\mathscr{F}_U$ can be regarded as a sheaf on the closed subset $\bar{U}$, which has one fewer irreducible components than X. Thus using proposition 6.2.4 and induction on the number of irreducible components, we reduce to the case X irreducible.

Step 2. Suppose X is irreducible of dimension 0 . Then the only open subsets of X are X and the empty set. For otherwise, X would have a proper irreducible closed subset, and $\dim X$ would be ≥ 1. Thus $\Gamma(X, -)$ induces an equivalence of categories $\mathbf{Ab}(X) \to \mathbf{Ab}$. In particular, $\Gamma(X, -)$ is an exact functor, so $H^i(X, \mathscr{F}) = 0$ for $i > 0$, and for all $\mathscr{F}$.

Step 3. Now let X be irreducible of dimension n, and let $\mathscr{F} \in \mathbf{Ab}(X)$. Let $B = \bigcup_{U \subset X} \mathscr{F}(U)$, and let A be the set of all finite subsets of B. For each $\alpha \in A$, let $\mathscr{F}_\alpha$ be the subsheaf of $\mathscr{F}$ generated by the sections in α (over various open sets). Then A is a directed set, and $\mathscr{F} = \varinjlim \mathscr{F}_\alpha$. So by lemma 6.2.2, it will be sufficient to prove vanishing of cohomology for each $\mathscr{F}_\alpha$. If α' is a subset of α, then we have an exact sequence

$$0 \to \mathscr{F}_{\alpha'} \to \mathscr{F}_\alpha \to \mathscr{G} \to 0$$

where $\mathscr{G}$ is a sheaf generated by $\#\left(\alpha - \alpha'\right)$ sections over suitable open sets. Thus, using the long exact sequence of cohomology, and induction on $\#(\alpha)$, we reduce to the case that $\mathscr{F}$ is generated by a single section over some open set U. In that case $\mathscr{F}$ is a quotient of the sheaf $\mathbb{Z}_U$ (where $\mathbb{Z}$ denotes the constant sheaf $\mathbb{Z}$ on X). Letting $\mathcal{K}$ be the kernel, we have an exact sequence

$$0 \to \mathcal{K} \to \mathbb{Z}_U \to \mathscr{F} \to 0$$

Again using the long exact sequence of cohomology, it will be sufficient to prove vanishing for $\mathcal{K}$ and for $\underline{\mathbb{Z}}_U$.

Step 4. Let U be an open subset of X and let $\mathcal{K}$ be a subsheaf of $\underline{\mathbb{Z}}_U$. For each $x \in U$, the stalk $\mathcal{K}_x$ is a subgroup of $\underline{\mathbb{Z}}$. If $\mathcal{K} = 0$, skip to Step 5 . If not, let d be the least positive integer which occurs in any of the groups $\mathcal{K}_x$. Then there is a nonempty open subset $V \subseteq U$ such that $\mathcal{K}|_V \cong d \cdot \underline{\mathbb{Z}}|_V$ as a subsheaf of $\underline{\mathbb{Z}}|_V$. Thus $\mathcal{K}_V \cong \underline{\mathbb{Z}}_V$ and we have an exact sequence

$$0 \to \underline{\mathbb{Z}}_V \to \mathcal{K} \to \mathcal{K}/\underline{\mathbb{Z}}_V \to 0$$

Now the sheaf $\mathcal{K}/\underline{\mathbb{Z}}_V$ is supported on the closed subset $(U - V)^-$of X, which has dimension $< n$, since X is irreducible. So using lemma proposition 6.2.4and the induction hypothesis, we know $H^i\left(X, \mathcal{K}/\underline{Z}_V\right) = 0$ for $i \geq n$. So by the long exact sequence of cohomology, we need only show vanishing for $\underline{\mathbb{Z}}_V$. Step 5. To complete the proof, we need only show that for any open subset $U \subseteq X$, we have $H^i\left(X, \underline{Z}_U\right) = 0$ for $i > n$. Let $Y = X - U$. Then we have an exact sequence

$$0 \to \underline{\mathbb{Z}}_U \to \underline{\mathbb{Z}} \to \underline{\mathbb{Z}}_Y \to 0$$

Now $\dim Y < \dim X$ since X is irreducible, so using proposition 6.2.4 and the induction hypothesis, we have $H^i\left(X, \underline{Z}_Y\right) = 0$ for $i \geq n$. On the other hand, $\underline{\mathbb{Z}}$ is flasque, since it is a constant sheaf on an irreducible space (II, Ex. 1.16a). Hence $H^i(X, \underline{\mathbb{Z}}) = 0$ for $i > 0$ as $\underline{\mathbb{Z}}$ is flasque. So from the long exact sequence of cohomology we have $H^i\left(X, \underline{\mathbb{Z}}_U\right) = 0$ for $i > n$.

6.2.2 Vanishing Theorem for Affine Schemes

In this section, we shall make rigorous the intuition that the cohomology of schemes measure some form of gluing by showing that if $X = \operatorname{Spec} R$, is a Noetherian, then for any quasi-coherent sheaf $\mathscr{F}$ over X,

$$H^i(\operatorname{Spec}(R), \mathscr{F}) = 0 \quad \forall i > 0$$

The key insight is that if I is an injective R-module, $\tilde{I}$ on $\operatorname{Spec} R$ is flasque. This will motivate the idea that cohomology of schemes should be thought of how the gluing of schemes can be measured, which shall motivate introducing Čech cohomology in the next section and showing it is equivalent to the cohomology given by flasque resolutions.

First, let M be an R-module and $\mathfrak{a} \subseteq R$ be an ideal. Define the $\mathfrak{a}$-torsion submodule

$$\operatorname{Tor}_{\mathfrak{a}}(M) = \left\{m \in M \middle| \mathfrak{a}^n m = 0, \text{ for some } n > 0\right\}$$

Lemma 6.2.6: $\mathfrak{a}$-Torsion Module Injective

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, and I an injective R-module. Then the submodule

$$J = \operatorname{Tor}_{\mathfrak{a}}(I)$$

is injective

6.2.7 Note This result is in fact true without the Noetherian condition, however it is harder to prove. The extra effort for building it up is not necessary for the main theory, for the curious see ref:HERE.

***Proof*:**
To show that J is injective, it will be sufficient to prove Baer's Criterion, namely to show that for any ideal $\mathfrak{b} \subseteq A$, and for any homomorphism $\varphi : \mathfrak{b} \to J$, there exists a homomorphism $\psi : A \to J$ extending φ. Since A is noetherian, $\mathfrak{b}$ is finitely generated. On the other hand, every element of J is annihilated by some power of $\mathfrak{a}$, so there exists an $n > 0$ such that $\mathfrak{a}^n\varphi(\mathfrak{b}) = 0$, or equivalently, $\varphi(\mathfrak{a}^n\mathfrak{b}) = 0$. Now applying Artin-Rees lemma (see [9]) to the inclusion $\mathfrak{b} \subseteq A$, we find that there is an $n' \geq n$ such that $\mathfrak{a}^n\mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^{n'}$. Hence $\varphi\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right) = 0$, and so the map $\varphi : \mathfrak{b} \to J$ factors through $\mathfrak{b}/\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right)$. Now we consider the following diagram:

$$\begin{tikzcd}
A \arrow[r] & A/\mathfrak{a}^{n'} \arrow[dr, dashed, "\psi'"] & & \\
\mathfrak{b} \arrow[u, hook] \arrow[r] \arrow[rrr, bend right, "\varphi"'] & \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \arrow[u, hook] \arrow[r] & J \arrow[r] & I
\end{tikzcd}$$

Since I is injective, the composed map of $\mathfrak{b}/\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right)$ to I extends to a map $\psi' : A/\mathfrak{a}^{n'} \to I$. But the image of ψ' is annihilated by $\mathfrak{a}^{n'}$, so it is contained in J. Composing with the natural map $A \to A/\mathfrak{a}^{n'}$, we obtain the required map $\psi : A \to J$ extending φ, as we sought to show.

Lemma 6.2.8: Flasque on I

Let I be an injective R-module over a Noetherian ring R. Then for any $f \in R$, the natural map $I \to I_f$ is surjective

***Proof*:**
Let $f \in R$ and $\varphi : I \to I_f$ be the natural map. I claim this map is surjective. Pick $x \in I_f$. By definition of localization and of φ :

$$\frac{\varphi(y)}{f^n} = x$$

We shall find an element that maps to the left hand side. To find this map, for each $i > 0$, let $\mathfrak{b}_i$ be the annihilator of f^i. Then $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$, and as R is Noetherian there exists an m such that $\mathfrak{b}_m = \mathfrak{b}_{m+1} = \cdots$. Now, define from the ideal to the injective R-module $\psi : (f^{n+m}) \to I$ where $f^{n+m} \mapsto f^m y$. As I is injective, by Baer's criterion this map extends to $\Psi : A \to I$. Label $\psi(1)$ as z. Then we have $f^{n+m}z = f^m z$. But then we see that Ψ and φ match, in particular we have:

$$\varphi(z) = \frac{\varphi(y)}{f^n} = x$$

showing surjectivity, and completing the proof.

Lemma 6.2.9: Flasque on II

Let I be an injective R-module over a Noetherian ring R. Then $\tilde{I}$ on $X = \operatorname{Spec} R$ is flasque.

Note that $\widetilde{I}$ is flasque even if R is not Noetherian, however this proof would be adds technicality without adding insight.

***Proof*:**
We will use Noetherian induction on $Y = (\operatorname{Supp} \tilde{I})$ (see section 0.3 for an overview of Noetherian induction). If Y consists of a single closed point of X, then I is a skyscraper sheaf which is certainly flasque.
In the general case, it suffices to show that for any open set $U \subseteq X, \Gamma(X, \tilde{I}) \to \Gamma(U, \tilde{I})$ is surjective. If $Y \cap U = \emptyset$, we are done, so say $Y \cap U \neq \emptyset$. Then there is an $f \in A$ such that the open set $X_f = D(f)$ is contained in U and $X_f \cap Y \neq \emptyset$. Let $Z = X - X_f$, and consider the following diagram:

$$\begin{tikzcd}
\Gamma(X, \widetilde{I}) \arrow[r] & \Gamma_Z(U, \widetilde{I}) \arrow[r] & \Gamma(X_f, \widetilde{I}) \\
\Gamma_Z(X, \widetilde{I}) \arrow[r] \arrow[u, hook] & \Gamma_Z(U, \widetilde{I}) \arrow[u, hook] &
\end{tikzcd}$$

where Γ_Z denotes sections with support in Z. Now given a section $s \in \Gamma(U, \tilde{I})$, consider its image s' in $\Gamma\left(X_f, \tilde{I}\right)$. By proposition 4.1.11 $\Gamma\left(X_f, \tilde{I}\right) = I_f$, so by lemma 5.1.13 there is a $t \in I = \Gamma(X, \tilde{I})$ restricting to s'. Let t' be the restriction of t to $\Gamma(U, \tilde{I})$. Then $s - t'$ goes to 0 in $\Gamma\left(X_f, \tilde{I}\right)$, so it has support in Z. Thus to complete the proof, it will be sufficient to show that $\Gamma_Z(X, \tilde{I}) \to \Gamma_Z(U, \tilde{I})$
is surjective.
Now, let $J = \Gamma_Z(X, \tilde{I})$. If $\mathfrak{a}$ is the ideal generated by f, then $J = \Gamma_{\mathfrak{a}}(I)$ so by lemma 5.1.12, J is also an injective A-module. Furthermore, the support of $\tilde{J}$ is contained in $Y \cap Z$, which is strictly smaller than Y. Hence by our induction hypothesis, $\tilde{J}$ is flasque. Since $\Gamma(U, \tilde{J}) = \Gamma_Z(U, \tilde{I})$, we conclude that

$$\Gamma_Z(X, \tilde{I}) \to \Gamma_Z(U, \tilde{I})$$

is surjective, as we sought to show.

Theorem 6.2.10: Affine Noetherian Scheme: Trivial Cohomology

Let $X = \operatorname{Spec} R$ be the spectrum of a Noetherian ring R. Then for all quasi-coherent sheaves $\mathscr{F}$ on X,

$$H^i(X, \mathscr{F}) = 0 \quad \forall i > 0$$

***Proof*:**
Let $\mathscr{F}$ be a quasi-coherent sheaf on the $X = \operatorname{Spec} R$ so that we have a module $M = \Gamma(X, \mathscr{F})$ and by corollary 4.1.10 $\mathscr{F} = \widetilde{M}$. Then from [9], we know we can take the injective resolution $0 \to M \to I^*$. Then as $\widetilde{(-)}$ is exact, we have that $0 \to \widetilde{M} \to \widetilde{I}^*$ is exact. By lemma 5.1.14, each $\widetilde{I}^i$ is flasque, hence by the principles of homology we can use this sequence to compute the cohomology groups. But then, applying the (in this case exact) Γ we have the sequence of R-modules $0 \to M \to I^*$ and we immediately get:

$$H^0(X, \mathscr{F}) = M \quad H^i(X, \mathscr{F}) = 0 \quad i > 0$$

as we sought to show.

Corollary 6.2.11: Injective Embedding of Sheaves On Neotherian Schemes

Let X be a Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. Then $\mathscr{F}$ can embed in a quasi-coherent flasque sheaf.

This corollary shall in fact use the Noetherian condition in a meaningful way, as shall be pointed out in the proof

***Proof*:**
As X is Noetherian, let $X = \bigcup_i^n \operatorname{Spec} R_i = \bigcup_i^n U_i$ be a finite open affine cover, and let $\mathscr{F}|_U = \widetilde{M_i}$ for each i. Embed each M_i into an injective R_i-module I_i, and define:

$$\mathscr{G} = \bigoplus_i^n \iota_* \left(\widetilde{I_i}\right)$$

where ι is the inclusion of $U_i \to X$ for each U_i (where we are overloading notation letting ι represent each inclusion). As for each i we have the injective map $\mathscr{F}|_{U_i} \to \widetilde{I_i}$, we get a map the injective maps $\mathscr{F} \to f_* \left(\widetilde{I_i}\right)$. Then taking the direct sum, we get the injective map $\mathscr{F} \to \mathscr{G}$. Now, each $\widetilde{I_i}$ is flasque by lemma 5.1 .14 and by definition quasi-coherent on U_i. Then $f_* \left(\widetilde{I_i}\right)$ is also
flasque (exercise) and quasi-coherent (proposition 4.1.14 this uses the Noetherian assumption!). Then certainly the direct sum is flasque, completing the proof.

The final theorem shall show that having trivial cohomology > 0 characterizes affine schemes (with the necessary Noetherian condition to be consistent with what we've proven):

Theorem 6.2.12: Characterizing affine schemes using Cohomology

Let X be a Noetherian scheme. Then the following are equivalent:

1. X is affine
2. $H^i(X, \mathscr{F}) = 0$ for $i > 0$ and all for all quasi-coherent sheaves $\mathscr{F}$
3. $H^1(X, \mathscr{I}) = 0$ for $i > 0$ and all for all coherent sheaves of ideals $\mathscr{I}$

***Proof*:**
(1) $\Rightarrow$ (2) Theorem 6.2.10 (2) $\Rightarrow$ (3) A-fortiori
(3) $\Rightarrow$ (1) We shall prove X is affine using corollary 3.1.15. First we show that X can be covered by open affine subsets of the form X_f, with $f \in R = \Gamma(X, \mathcal{O}_X)$. Let $\mathfrak{p}$ be a closed point of X, let U be an open affine neighborhood of $\mathfrak{p}$, and let $Y = X - U$. Then we have an exact sequence

$$0 \to \mathscr{I}_{Y \cup \{\mathfrak{p}\}} \to \mathscr{I}_Y \to k(\mathfrak{p}) \to 0$$

where $\mathscr{I}_Y$ and $\mathscr{I}_{Y \cup \{\mathfrak{p}\}}$ are the ideal sheaves of the closed sets Y and $Y \cup \{\mathfrak{p}\}$, respectively. The quotient is the skyscraper sheaf $k(\mathfrak{p}) = \mathcal{O}_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ at $\mathfrak{p}$. Now from the exact sequence of cohomology, by (3) we get an exact sequence

$$\Gamma(X, \mathscr{I}_Y) \to \Gamma(X, k(\mathfrak{p})) \to H^1\left(X, \mathscr{I}_{Y \cup \{\mathfrak{p}\}}\right) = 0.$$

So there is an element $f \in \Gamma(X, \mathscr{I}_Y)$ which goes to 1 in $k(\mathfrak{p})$, i.e., $f_{\mathfrak{p}} \equiv 1 \pmod{\mathfrak{m}_{\mathfrak{p}}}$. Since $\mathscr{I}_Y \subseteq \mathcal{O}_X$, we can consider f as an element of R. Then by construction, $\mathfrak{p} \in X_f \subseteq U$. Furthermore, $X_f = U_{\bar{f}}$, where $\bar{f}$ is the image of f in $\Gamma(U, \mathcal{O}_U)$, so X_f is affine.
Thus every closed point of X has an open affine neighborhood of the form X_f. By quasi-compactness, we can cover X with a finite number of these, corresponding to $f_1, \dots, f_r \in R$. Let us verify that $(f_1, \dots, f_r) = R$. We use $f_1, \dots, f_r$ to define a map $\alpha : \mathcal{O}_X^r \to \mathcal{O}_X$ by sending $\langle a_1, \dots, a_r \rangle$ to $\sum f_i a_i$. Since the X_{f_i} cover X, this is a surjective map of sheaves. Let $\mathscr{F}$ be the kernel:

$$0 \to \mathscr{F} \to \mathcal{O}_X^r \xrightarrow{\alpha} \mathcal{O}_X \to 0$$

Filter $\mathscr{F}$ as follows:

$$\mathscr{F} = \mathscr{F} \cap \mathcal{O}_X^r \supseteq \mathscr{F} \cap \mathcal{O}_X^{r-1} \supseteq \cdots \supseteq \mathscr{F} \cap \mathcal{O}_X$$

for a suitable ordering of the factors of $\mathcal{O}_X^r$. Each of the quotients of this filtration is a coherent sheaf of ideals in $\mathcal{O}_X$. Thus by (3) and the long exact sequence of cohomology, we climb up the filtration and deduce that $H^1(X, \mathscr{F}) = 0$. But then $\Gamma(X, \mathcal{O}_X^r) \to \Gamma(X, \mathcal{O}_X)$ is surjective, which tells us that $f_1, \dots, f_r$ generates R, as we sought to show.

6.3 Serre Duality Theorem

Recall from [3, Chapter 4.3] that a compact n-dimensional smooth manifold satisfies *Poincaré Duality*:

$$H^i(M;\mathbb{Z}) \cong H_{n-i}(M;\mathbb{Z})$$

In other words, cycles on a manifold and cocycles are in fact dual concepts to each other (some concrete consequence of this is that looking at the "boundary" and the derivative are in fact two sides of the same coin). By using sheaves of differentials, we recover a similar duality for schemes. In this section, we shall start by building up some more cohomological theory which will

6.3.1 Ext Group on Sheaves

Recall from [9, chatper 27.3] that $\mathrm{Hom}(M,-)$ is a left-exact functor that gives rise to right-derived functors $\mathrm{Ext}^i(M,-)$. Let us translate these results to sheaves. First, recall that if $\mathscr{F},\mathscr{G}$ are sheaves on X, $\mathrm{Hom}(\mathscr{F},\mathscr{G})$ is the group of $\mathcal{O}_X$-module homomorphism, and $\mathcal{H}om(\mathscr{F},\mathscr{G})$ is the *hom-sheaf*. From these, we get the following:

Definition 6.3.1: Ext Group For Sheaves

Let $(X,\mathcal{O}_X)$ be a ringed space and $\mathscr{F}$ a $\mathcal{O}_X$-module. Then the *right-derived functors* of $\mathrm{Hom}(\mathscr{F},-)$ are denoted as $\mathrm{Ext}^i(\mathscr{F},-)$, and the right-derived functors of $\mathcal{H}om(\mathscr{F},-)$ is $\mathcal{E}xt^i(\mathscr{F},-)$.

For both Ext and $\mathcal{E}xt$, we have that $\mathrm{Ext}^0 = \mathrm{Hom}$, and if $\mathscr{G}$ is injective than Ext^i and $\mathcal{E}xt^i$ is trivial for $i>0$. One natural result we need to establish is that an in injective shear $\mathscr{I}$ on X restricts to an injective sheaf $\mathscr{I}|_U$ on U for any open subset:

Lemma 6.3.2: Injective Sheaves and Restriction

Let $\mathscr{I}$ be an injective sheaf on X. Then $\mathscr{I}|_U$ is an injective sheaf on U.

***Proof*:**
The key is to remember that injective naturally extend to global maps (see [14, p. III 6.1])

Proposition 6.3.3: Open Subset and Ext Group

Let $(X,\mathcal{O}_X)$ be a ringed space and $U\subseteq X$ an open subset. Then:

$$\mathcal{E}xt^i_X(\mathscr{F},\mathscr{G})|_U \cong \mathcal{E}xt^i_U(\mathscr{F}|_U,\mathscr{G}|_U)$$

***Proof*:**
To outline: first, by [9, Chapter 27] from $\mathscr{G}$ we get a δ-functor from X-**Mod** to U-**Mod**, which by the comments above agree at $i=0$. With $\mathscr{G}$ injective, both sides vanish on $i>0$, and the rest follows from lemma 6.3.2.

For computational purposes, we state the usual functor-representation result:

Proposition 6.3.4: Functor Representation and Ext

Let $\mathscr{G}$ be any $\mathcal{O}_X$-module on X. Then:

1. $\mathcal{E}xt^0(\mathcal{O}_X, \mathscr{G}) \cong \mathscr{G}$
2. $\mathcal{E}xt^i(\mathcal{O}_X, \mathscr{G}) = 0$ for $i > 0$
3. $\mathrm{Ext}^i(\mathcal{O}_X, \mathscr{G}) = H^i(X, \mathscr{G})$ for $i > 0$

***Proof*:**
For (1) and (2), $\mathcal{H}om(\mathcal{O}_X, -)$ is the identity functor. Point (3) follows from $\Gamma(X, -)$ and $\mathrm{Hom}(\mathcal{O}_X, -)$ being the same (see corollary 6.1.9).

6.3.2 The Theorem

I will write down these results in more detail later. For now, I will put down the important theorems for reference:

Theorem 6.3.5: Duality for Projective Space

Let $X = \mathbb{P}^n_k$ over a field k. Then:

1. $H^n(X, \omega_X) \cong k$. Fix one such isomorphism;
2. for any coherent sheaf $\mathcal{F}$ on X, the natural pairing
$$\mathrm{Hom}(\mathcal{F}, \omega) \times H^n(X, \mathcal{F}) \to H^n(X, \omega) \cong k$$
is a perfect pairing of finite-dimensional vector spaces over k;
3. for every $i \geq 0$ there is a natural functorial isomorphism
$$\mathrm{Ext}^i(\mathcal{F}, \omega) \to H^{n-i}(X, \mathcal{F})',$$
where the tick, $'$, denotes the dual vector space, which for $i = 0$ is the one induced by the pairing of (2).

***Proof*:**
Hartshorne III 7.1 (p.239)

Theorem 6.3.6: Duality for a Projective Scheme

Let X be a projective scheme of dimension n over an algebraically closed field k, ω_X be a dualizing sheaf on X, and let $\mathcal{O}(1)$ be a very ample sheaf on X. Then:

1. for all $i \geq 0$ and $\mathcal{F}$ coherent on X, there are natural functorial maps

$$\theta^i : \operatorname{Ext}^i(\mathcal{F}, \omega_X) \to H^{n-i}(X, \mathcal{F})',$$

   such that θ^0 is the map given in the definition of dualizing sheaf above;

2. the following conditions are equivalent:
   (a) X is Cohen-Macaulay and equidimensional (that is, all irreducible components have the same dimension);
   (b) for any $\mathcal{F}$ locally free on X, we have $H^i(X, \mathcal{F}(-q)) = 0$ for $i < n$ and $q \gg 0$;
   (c) the maps θ^i of (1) are isomorphisms for all $i \geq 0$ and all $\mathcal{F}$ coherent on X.

***Proof*:**
Hartshorne III 7.6 (p.241)

Corollary 6.3.7: Serre Duality: Nonsingular Projective Variety

Let X be a nonsingular projective variety over an algebraically closed field k. Then the dualizing sheaf ω_X° is isomorphic to the canonical sheaf ω_X.

***Proof*:**
As X is projective, embed it into $\mathbb{P}^n_k$. Then by theorem 5.4.22, X is a local complete intersection in $\mathbb{P}^n_k$, and so by proposition 5.4.28

$$\omega_X \cong \Omega_P \otimes \bigwedge^r \widetilde{\mathscr{I}/\mathscr{I}^2}$$

as we sought to show.

6.4 Families of Schemes

here

Recall that the fibers of a scheme morphism is a scheme. We may consider what type of families these fibers for.

Definition 6.4.1: Family of Schemes

Let X, B be schemes. Then a *Family of schemes* is given by a morphism $\pi : X \to B$. The scheme B is sometimes called the *parameter space*.

The notion of a family of schemes as it stands is much too general: consider any family $\pi : X \to B$, choose $b \in B$, and create a new family by taking the disjoint union of $X \setminus \pi^{-1}(b)$ and Y for some arbitrary scheme Y where Y is sent to b. From this, it can be deduced that there is no real properties that can be deduced, and further constrictions must be given.

Higher Direct Image Sheaves

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6.4.1 Flat Morphisms

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6.4.2 Smooth Morphisms

here

6.4.3 On Proper Morphisms: prelude to Zariski Main Theorem here

here

6.5 Dimension and Semi-continuity

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6.6 *GAGA

I want to put here the most important GAGA results so that I may freely use them.

7

Application: Curves and Surfaces

In this chapter, we shall use the techniques we have taken the time to build to understand better low-dimensional algebraic-geometric objects, namely *curves* and *surfaces*.

7.1 Algebraic Curves

Intuitively, an algebraic curve ought to be a one-dimensional object given by a set of equations. In geometric settings, we shall want to work over algebraically closed fields, while in the number-theoretic setting we shall want to consider other fields and rings. For a moment let's consider curves over $\mathbb{C}$. As we saw in ref:HERE, all such curves can be projectivised and birationally into projective space. Furthermore, by the process of normalization, there is a birational map from the non-singular projective curves the projective curve. Hence, up birational equivalence we can consider algebraic curves as complete smooth objects; this can be thought of as the *geometric* way of looking at curves. If we go the analytic route, we shall see that curves correspond to compact Riemann surfaces due to GAGA. It is not too bad to show that smooth complete curves over $\mathbb{C}$ create compact Riemann surfaces, however the that all compact Riemann surface are given and characterized by smooth complete curves is a little harder to prove (it is non-trivial to show that a compact Riemann surface has a non-trivial meromorphic function defined on it; see [13]). On the other hand, looking at curves *algebraically*, we saw in [9] they correspond to fields $K/\mathbb{C}$ of transcendence degree 1. These three perspective shall constantly inspire each other to push forward the properties of we can define for curves.

Let us first state the definition of a curve in terms of scheme-theoretic language:

Definition 7.1.1: Curve

Let X be a k-scheme. Then if X is Noetherian, integral, separated dimension 1 scheme of finite type over k, then X is called a *curve*. If X is proper over k, it is called *complete*, and if all local rings are regular, it is called *regular* or *nonsingular*.

In this section, a curve will mean complete nonsingular curve over an algebraically closed field, or in terms of schemes it is an regular integral scheme of dimension 1 that is proper over k . When we say point, we shall mean closed point (the only other point is the generic point as the scheme is integral and dimension 1).

The following result will be important for being able to show that *nonsingular* curves that are rational are in fact isomorphic to the projective line. Note that this is *not* true for singular rational curves, and in general two rationally equivalent curves are *not* isomorphic[1].

Proposition 7.1.2: Proper Curve Equivalence

Let X be a nonsingular curve over k with function field K. Then the following are equivalent:

($\Rightarrow$) X is projective

($\Rightarrow$) X is proper

($\Rightarrow$) $X \cong t(C_K)$ where C_K is an abstract nonsingular curve and t is the functor from varieties to schemes (see theorem 3.8.1)

***Proof*:**
(1) follows from (2) and (3) follows from (1) from classical algebraic geometry, see section 0.2. For (2) $\Rightarrow$ (3), as X is complete every DVR of K/k has a unique centre. Then as the closed points of X are all DVR are in one-to-one correspondence with the DVR's of K/k, we have a natural bijection $X \cong t(C_K)$, which is certainly an isomorphism.

We next generalize one more classical result about the image of projective varieties:

Proposition 7.1.3: Image of Complete Nonsingular Curve is Closed

Let X be a complete nonsingular curve over k, Y any curve over k, and $f : X \to Y$ a morphism. Then either the image of f is a point, or f is surjective, and so Y is complete. In the latter case, $\kappa(X)$ is a finite extension of $\kappa(Y)$, and so f is a finite morphism.

***Proof*:**
As X is complete, $f(X)$ must be closed and proper over k, and $f(X)$ is irreducible. It follows that it either maps to a point, or to all of Y (recall we defined curves to be integral). Hence Y is complete as well.

To show f is finite, As f is dominant we have $\kappa(Y) \subseteq \kappa(X)$. As both have transcendence degree 1, $\kappa(X)$ is a finite extension of $\kappa(Y)$. Now, taking $\operatorname{Spec} S = V \subseteq Y$, and $U = f^{-1}(V)$, then it should be clear that $U = \operatorname{Spec} R$ where R is the integral closure of S, and as integral closure

[1] See section 7.1.3 for more examples

form finite S-modules, we have the desired result.

7.1.1 Divisors and Riemann-Roch

As all nontrivial maps between nonsingular complete curves must be finite, we have the following natural definition:

Definition 7.1.4: Degree of Morphism of Curves

Let $f : X \to Y$ be a finite morphism of curves. Then the *degree* of f is the degree of the finite extension of $[\kappa(X) : \kappa(Y)]$.

From this, can pull-back the divisor class group. Let $f : X \to Y$ be a finite morphism of curves. Define $f^* : \operatorname{WDiv} Y \to \operatorname{WDiv} X$ as follows: for each point $y \in Y$ choose $t \in \mathcal{O}_{Y,y}$ such that $\nu_y(t) = 1$ (where we choose the natural valuation on $\mathcal{O}_{Y,y}$). Then map it to:

$$f^*(y) = \sum_{f(x)=y} \nu_x(t) \cdot x$$

As f is a finite morphism, this is a finite sum. The reader should verify that this is indeed a well-defined function (namely, for another choice $t' = ut$ where u is unit, we get the same result). Then we see that f will map principal divisors to principal divisor, and hence we have:

$$f^* : \operatorname{Cl} Y \to \operatorname{Cl} X$$

Let us compute some examples of divisors on curves. To that end, we prove two useful results:

Proposition 7.1.5: Finite Map and Degrees

Let $f : X \to Y$ be a finite morphism of nonsingular curves. Then for any divisor D of Y,

$$\deg f^*(D) = \deg f \cdot \deg D$$

***Proof*:**
As we are working with nonsingular curves, it suffice to show that for every (closed) point $y \in Y$, $\deg f^*(y) = \deg f$. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open subset containing y, and let R be the integral closure of S in $K(X)$. Then as we saw in proposition 7.1.3, $U = \operatorname{Spec} R$ is equal to $f^{-1}(V) \subseteq X$. Let $\mathfrak{m}_y$ be the maximal ideal of y in S. Then we can localize S and R with respect to $S = B \setminus \mathfrak{m}_y$, and we get a ring extension:

$$\mathcal{O}_{Y,y} \hookrightarrow R'$$

where R' is a finitely generated $\mathcal{O}_{Y,y}$-module. Now, R' is torsion free and has rank $r = [K(X) : K(Y)]$, and so $'$ i a free $\mathcal{O}_{Y,y}$-module of rank $r = \deg f$. If t is a local parameter of y, then R'/tR' is a K-vector pace of dimension r

On the other hand, the points $x_i \in X$ such that $f(x_i) = y$ are in one-to-one correspondence with the maximal ideals $\mathfrak{m}_i \subseteq R'$, and for each i $R'_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}$. Now, by construction we have:

$$tR' = \bigcap_i (tR'_{\mathfrak{m}_i} \cap R')$$

Then as these are co-prime, we can apply the Chinese remainder theorem and see that the dimension of the quotients are:

$$\dim_k R'/tR' = \sum_i \dim_k \frac{R'}{tR'_{\mathfrak{m}_i} \cap R'}$$

and as $R'/(tR'_{\mathfrak{m}_i} \cap R') \cong R'_{\mathfrak{m}_i}/tR'_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}/t\mathcal{O}_{X,x_i}$, we have the dimensions in the sum are equal to $\nu_{x_i}(t)$. Then as $f^*(y) = \sum \nu_{x_i}(t)x_i$, we have that

$$\deg f^*(q) = \deg f$$

as we sought to show.

Corollary 7.1.6: Degree Map and Integers

Lt X be a complete nonsingular curve. Then the principal divisors have degree 0, and the degree function induces a surjective morphism:

$$\deg : \operatorname{Cl} X \twoheadrightarrow \mathbb{Z}$$

***Proof*:**
Take $f \in \kappa(X)^*$. If $f \in k$, then $\operatorname{div} f = 0$, so assume $f \notin k$. Then we have the field extension $\kappa(X)/k(f)$ which induces a morphism $\varphi : X \to \mathbb{P}^1_k$ that is finite by proposition 7.1.3.Then $\operatorname{div} f = \varphi^*(\{0\} - \{\infty\})$, and as $\{0\} - \{\infty\}$ is a divisor of degree 0 in $\mathbb{P}^1_k$, we get that $\operatorname{div} f = 0$.

Thus, the degree map is independent of principal divisors, and so by proposition 5.3.10 we get the surjection, completing the proof.

Example 7.1.7: Divisors On Curves and Groups on Elliptic Curves

1. Recall that a (complete) curve is *rational* if it is birational to $\mathbb{P}^1_k$, and furthermore as X is a nonsingular curve, they are in fact *isomorphic* (by proposition 7.1.2). Then as all points correspond to principal divisors. I claim that X is a complete nonsingular curve is rational if and only if there exist two distinct points $x, y \in X$ such that $[x] = [y]$. Then there is a rational function $f \in \kappa(X)$ where $\operatorname{div} f = 1 \cdot x - 1 \cdot y$. Now, take the morphism $\varphi : X \to \mathbb{P}^1_k$ defined in corollary 7.1.6. Then $\varphi^*(\{0\}) = 1 \cdot x$, and so φ is of degree 1. But then it is invertible, that is it is birational, and so X is rational.

2. Let $X \subseteq \mathbb{P}^2_k$ be the nonsingular curve curve $y^2z = x^3 - xz^2$ where $\operatorname{char}(k) \neq 2$. Then as we saw in [9, chapter 21.3], this curve is not rational[a]. Take K be the kernel of $\operatorname{Cl} X \to \mathbb{Z}$, which by corollary 7.1.6 is nonempty. We shall show that there is a one-to-one correspondence between the closed points of X and the elements of K, which shows that this curve has a natural group structure on it! In fact *all* elliptic curves will have a natural group structure on them, making them natural group varieties! The following visual illustrates how we shall

define the multiplication structure:

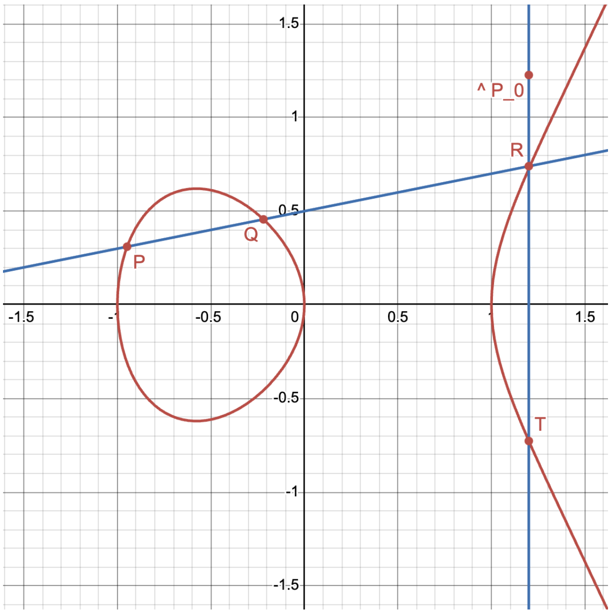

Figure 7.1: Group law on X in $z \neq 0$

Let $P_0 = [0:1:0]$ on X. This is an inflection point, hence the tangent line $z = 0$ at the point meets the curve in the divisor $3P_0$ (check this). If $L \subseteq \mathbb{P}^2_k$ is any other line meeting X in three points, say P, Q, R (which my coincide), then L is linearly equivalent to $z = 0$ in $\mathbb{P}^2_k$, that is we have

$$[P + Q + R] = [3P_0]$$

Now, to any closed point $P \in X$, associate to it the divisor $P - P_0 \in K$. his map is injective, since if $[P - P_0] = [Q - P_0]$ implies $[P] = [Q]$, and so X would be rational by example (1), a contradiction.

Then let's show that the map from the closed points of X to K is surjective. Take $D = \sum n_i P_i \in K$. Let us first take it where $\sum_i n_i = 0$. Then we can write

$$D = \sum_i n_i(P_i - P_0)$$

Now, for any point R, the line P_0R meet X also at the point T (where we always count intersection with multiplicity so that if $R = P_0$, then the line P_0R is the tangent line at P_0 and then $T = P_0$). Then $[P_0 + R + T] = [3P_0$, so $[R - P_0] = -[T - P_0]$. Now, if i is an index where $n_i < 0$, then take $P_i = R$. Then replacing P_i by T, we get divisors in the same cosets with the ith coefficient $-n_i > 0$. By repeating this process, we can assume that $D = \sum_i n_i(P_i - P_0)$ where all $n_i > 0$. If $\sum_i n_i = 1$, we are done, so suppose $\sum_i n_i \geq 2$. Let P, Q be two (not necessarily distinct) points P_i which occur in D. Let the line PQ meet X in R let the line P_0R meet X in T. Then:

$$[P + Q + R] = [3P_0] \qquad \text{and} \qquad [P_0 + R_0 + T] = [3P_0]$$

and so:

$$[P - P_0] + [Q - P_0] = [T - P_0]$$

By replacing P and Q by T, we get another equivalent divisor of the same form whose $\sum_i n_i$ is one less, and so by induction:

$$[D] = [P - P_0]$$

giving us the bijective correspondence. Finally, it is not too hard o show that this addition and inverse morphisms $+ : X \times X \to X$ and $(-)^{-1} : X \to X$, and hence we have a natural group variety!

More generally, if X is a complete nonsingular curve, then K is isomorphic to the group of closed points of an abelian variety known as the *Jacobian variety* of X. Then an *abelian variety*, is a complete group variety over a field k. The dimension of a Jacobean variety shall be shown in section ref:HERE to be related to the *genus* of the curve (see section 7.1). In general, the divisor class group of such a curve is an extension of $\mathbb{Z}$ by the group of closed points of the Jacobian variety of X Notice it has a continuous and a discrete component.

This process to higher dimensional nonsingular varieties: if X is a nonsingular projective variety of dimension ≥ 2 then we can define the subgroup$K \subseteq \operatorname{Cl} X$ such that $\operatorname{Cl} X/K$ is a finitely generated abelian group known as the *Néron-Severe group* of X, and K is isomorphic to the group of closed points of an abelian variety known as the *Picard Variety* (generalizing the Picard group from Number Theory!)

[a] We shall return a lot to this curve in section 7.1.3

If we are working over a characterize 0 curve then as we are smooth and proper, then the analytification of curve define a Riemann surface, and in fact a compact oriented surface as it is proper and it is complex (giving a natural orientation). Recall compact oriented surfaces are classified up to diffeomorphism (and since we are in low enough dimension, this is equivalent up to homeomorphism[2]) by their genus. These split up curves into a natural countable family of curves:

Definition 7.1.8: Genus of Curve

Let X be a curve. Then

1. The *arithmetic genus* of X is $1 - P_X(0)$ where P_X is the Hilbert polynomial of X. It is denoted $p_a(X)$
2. The *geometric genus* of X is $\dim_k \Gamma(X, \omega_X)$ where ω_X is the canonical sheaf. It is denoted $p_g(X)$.

These are in fact all equivalence, and naturally is also measured by the 1st cohomology group (see [3]). We first need a bit of build-up:

[2] Recall in real dimension $1, 2, 3$ that homeomorphic and diffeomorphic are essentially equivalent

Definition 7.1.9: Euler Characteristic

Let X be a projective scheme over k, $\mathscr{F}$ a coherent sheaf on X. Then the *Euler characteristic* of $\mathscr{F}$ is:

$$\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathscr{F})$$

The Euler Characteristic is additive, meaning that if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, then

$$\chi(\mathscr{F}) = \chi(\mathscr{F}') + \chi(\mathscr{F}'')$$

as the reader should verify.

Lemma 7.1.10: Hilbert Polynomial and Euler Characteristic

let X be a projective scheme over k, $\mathcal{O}_X(1)$ a very ample invertible sheaf on X over k, and $\mathscr{F}$ a coherent sheaf on X. Then there exists a polynomial $P(z) \in \mathbb{Q}[z]$ such that:

$$\chi(\mathscr{F}(n)) = P(n)$$

for all $n \in \mathbb{Z}$. This polynomial is called the *Hilbert polynomial* of $\mathscr{F}$.

To see the connection, if you let $X = \mathbb{P}^n_k$ and $M = \Gamma_*(\mathscr{F})$, then the Hilbert polynomial of $\mathscr{F}$ is just the Hilbert polynomial of M.

***Proof*:**
exercise

Lemma 7.1.11: Hilbert Polynomial and Cohomology

Let X be a projective scheme of dimension r over k. If X is integral and k algebraically closed, then:

$$p_a(X) = \sum_{i=1}^{r-1} (-1)^i \dim_k H^{r-1}(X, \mathcal{O}_X)$$

In particular, if X is a curve, then:

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$$

***Proof*:**
exercise

Proposition 7.1.12: Equivalence of Genus

Let X be a curve. Then:

$$p_a(X) = p_g(X) = \dim_k H^1(X, \mathcal{O}_X)$$

Hence, this number is simply called the *genus* of X and will be denoted g.

***Proof*:**
We have $p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$ by lemma 7.1.11. Next by Serre duality, we have that if X is a projective nonsingular curve, $H^1(X, \mathcal{O}_X)$ and $H^0(X, \omega_X)$ are dual vector spaces, and hence $p_a = \dim_k H^1(X, \mathcal{O}_X) = \dim_k \Gamma(X, \omega_X) = p_g$, completing the proof.

From this, we can deduce the genus must be non-negative (as we hope it would by our topological understanding of the genus). Furthermore, for each genus $g \geq 0$, we can associate to it a curve of genus g: Take a (primitive will) divisor of a quadratic nonsingular surface of type $g+1$ or $g+2$[3]. Then as we saw in section 6.1.2, we get a corresponding genus g curve.

One of the most important results for classifying curves comes from the Riemann-Roch theorem which relates the genus, the divisors, and the degree of a curve. As we are working over curves, the Weil and Cartier divisors coincide, and so for each divisor D we have a natural map $D \mapsto \mathcal{L}(D)$ to an invertible sheaf defined in a natural way using the principal parts of the covering of the curve. Recall that a divisor $D = \sum_i n_i P_i$ is called *effective* if $n_i \geq 0$. These don't form a group, however we still have the notion of "cosets": two divisors are said to be *linearly equivalent* and if they differ by a principal divisor, and all the divisors in an equivalent class is called a *complete linear system*; let $|D|$ denote this set.

As shown in (Harthsorne II 7.7, didn't put it down yet), if we have $\mathrm{Cl}\,(C)$, then if $|D| = |D'|$ we have $D = D' + \mathrm{div}(f)$ for some rational function f. Any other such principal divisor differs by a constant multiple (if we had $D = D' + \mathrm{div}(f) = D' + \mathrm{div}(f')$, then f/f' would be a polynomial and hence constant on projective space). We thus get that there is a one-to-one correspondence between the effective Cartier divisors and:

$$\frac{H^0(X, \mathcal{L}(D)) - \{0\}}{k^*}$$

Let $\ell(D) = \dim_k H^0(X, \mathcal{L}(D))$ (we may thus say that the dimension of $|D|$ is $\ell(D) - 1$). This number is finite by theorem 5.2.22.

Lemma 7.1.13: Divisors and Degree

Let D be a divisor on a curve X. Then if $\ell(D) \neq 0$, we must have

$$\deg D \geq 0.$$

Furthermore, if $\ell(D) \neq 0$ and $\deg D = 0$, then $D \sim 0$, that is,

$$\mathcal{L}(D) \cong \mathcal{O}_X$$

***Proof*:**
If $\ell(D) \neq 0$, then the complete linear system $|D|$ is nonempty. Hence D is linearly equivalent to some effective divisor. Since the degree depends only on the linear equivalence class, and the degree of an effective divisor is nonnegative, we find

$$\deg D \geq 0.$$

If $\deg D = 0$, then D is linearly equivalent to an effective divisor of degree 0. But there is only one such, namely the zero divisor.

[3] see ref:HERE as to why

Finally, we establish one more convention: let $\Omega_{X/k}$ or Ω_X represent the sheaf of relative differentials of X over k Since X has dimension 1, it is an invertible sheaf on X, and so is equal to the canonical sheaf ω_X on X. We call any divisor in the corresponding linear equivalence class a *canonical divisor*, and denote it by K. Then we finally get:

Theorem 7.1.14: Riemann-Roch Theorem

Let D be a divisor on a curve X of genus g. Then:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

***Proof*:**
The divisor $K - D$ corresponds to the invertible sheaf $\omega_X \otimes \mathscr{L}(D)^\vee$. Since X is projective (by proposition 7.1.2), we can apply Serre duality (namely, proposition 5.4.28) to conclude that the vector space $H^0(X, \omega_X \otimes \mathscr{L}(D)^\sim)$ is dual to $H^1(X, \mathscr{L}(D))$. Thus we have to show that for any D,

$$\chi(\mathscr{L}(D)) = \deg D + 1 - g,$$

where for any coherent sheaf $\mathscr{F}$ on X, $\chi(\mathscr{F})$ is the Euler characteristic

$$\chi(\mathscr{F}) = \dim H^0(X, \mathscr{F}) - \dim H^1(X, \mathscr{F}).$$

First, consider the case where $D = 0$. Then the formula gives

$$\dim H^0(X, \mathscr{O}_X) - \dim H^1(X, \mathscr{O}_X) = 0 + 1 - g.$$

since $H^0(X, \mathscr{O}_X) = k$ for any projective variety, and $\dim H^1(X, \mathscr{O}_X) = g$ by proposition 7.1.12.

Next, let D be any divisor, and let P be any point. Let's start by showing that the formula is true for D if and only if it is true for $D + P$. Since any divisor can be reached from 0 in a finite number of steps by adding or subtracting a point each time, this will show the result holds for all D.

Consider P as a closed subscheme of X. Its structure sheaf is a skyscraper sheaf k sitting at the point P, which we denote by $k(P)$, and its ideal sheaf is $\mathscr{L}(-P)$ by proposition 5.3.29. Therefore we get a short exact sequence

$$0 \to \mathscr{L}(-P) \to \mathscr{O}_X \to k(P) \to 0.$$

Tensoring with $\mathscr{L}(D + P)$ we get

$$0 \to \mathscr{L}(D) \to \mathscr{L}(D + P) \to k(P) \to 0.$$

Since $\mathscr{L}(D + P)$ is locally free of rank 1, tensoring by it does not affect the sheaf $k(P)$. Now, the Euler characteristic is additive on short exact sequences, and $\chi(k(P)) = 1$, so we have

$$\chi(\mathscr{L}(D + P)) = \chi(\mathscr{L}(D)) + 1.$$

On the other hand, $\deg(D+P) = \deg D + 1$, so our formula is true for D if and only if it is true for $D+P$. But, we get

$$\chi(\mathscr{L}(D)) = \deg D + 1 - g,$$

from which we can conclude that:

$$\ell(D) - \ell(K-D) = \deg D + 1 - g$$

as we sought to show.

Hartshorne goes through many examples here!

7.1.2 Hurwitz's Theorem

Recall that a finite morphism $f : X \to Y$ between schemes preserves dimension, has finite fibers, that normalization maps are finite morphisms, and that such maps are proper. In the case of curves, we can think of finite morphisms are *covering maps* (think of the Riemann surfaces defined to make square-roots maps well-defined). We endeavor to link the genus of Y to X given a finite morphism via *Hurwitz's Theorem.*

Recall that the *degree* of a finite morphism $f : X \to Y$ is $[K(X) : K(Y)]$. As we are working with smooth curves, each local ring is a DVR. Choose a point $p \in X$ and let $q = f(p)$. Take a uniformizer $t \in \mathcal{O}_Q$, and identify t with $f_q^\#(t)$ where $f_q^\# : \mathcal{O}_q \to \mathcal{O}_p$ is the natural map induces by f. This may no longer be a uniformizer (consider $f(x) = x^2$ on the affine line). This value is independent of choice of uniformizer, and hence we define the following:

Definition 7.1.15: Ramification Index

With the setup above, the *ramification index* of f at the point $p \in X$ is given by:

$$\nu_p(t) = e_p$$

where ν_p is the discrete valuation at p. If $e_p > 1$, we say that f is *ramified* p if $e_p > 1$, and unramified if $e_p = 1$.

If f is unramified everywhere, it is easy to see that it is étale. If $\operatorname{char} k = 0$ or $\operatorname{char} k = p$ and $p \nmid e_p$, we say that the ramification is *tame*, and it is *wild* otherwise.

7.1.3 Elliptic Curves

here

7.2 Surfaces

Here

Part II

Intersection Theory

7.2.1 Inspiration For now, this part will very heavily be inspired by Fulton [19] and Vakil [18]

Throughout most this chapter, schemes shall always be $\overline{k}$-schemes of finite type over an algebraically closed field. A point shall be a closed point unless stated otherwise. If needed, such a scheme shall be called an *algebraic scheme*.

8

Rational Equivalence

Recall from [9] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f, g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$
2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$
3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:
$$i(P, F' \cdot G) = i(P, F \cdot G)$$
4. If F, G have have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.
5. If $P \notin F \cap G$:
$$i(P, F \cdot G) = 0$$
6. if $f(x, y) = x - a$ and $g(x, y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at p) then:
$$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H)$$

8.1 Basic Definitions and Examples

The notion of order, regular co-dimension 1 subscheme, divisor, and so forth were all defined in section 5.3. As a quick refresher:

1. put the refresher here

Definition 8.1.1: K-cycle

Let X be an algebraic scheme. Then a *k-cycle* is a finite formal sum:

$$\sum_i n_i[V_i]$$

where the V_i are k-dimensional subvarieties of X and $n_i \in \mathbb{Z}$.

The group of k-cycle is denoted $Z_k(X)$.

By definition, the $(n-1)$-cycles of an n-dimensional algebraic scheme is a Weil divisor. Let $W \subseteq X$ be a $(k+1)$-dimensional variety and $f \in k(W)^*$ be a nonzero rational function. Then:

$$[\operatorname{div}(f)] = \sum_{\substack{V \subseteq W \\ \operatorname{codim}(V)=1}} \operatorname{ord}_V(f)[V]$$

It is easy to check this is a finite sum (see lemma 5.3.4).

Definition 8.1.2: Rational Equivalence

A k-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to 0* if they exists a finite number of $(k+1)$-dimensional subvareities W_i and $f_i \in k(W_i)^*$ such that

$$\alpha = \sum_{W_i} [\operatorname{div}(f_i)]$$

This equivalence is denoted $\alpha \sim 0$. Two k-cycles $\alpha, \beta \in Z_k(X)$ are *rationally equivalent* if there exists their difference is rationally equivalent to 0:

$$\alpha \sim \beta \qquad \text{iff} \qquad \alpha - \beta \sim 0$$

As $[\operatorname{div}(f^{-1})] = -[\operatorname{div}(f)]$ for a rational function f, the quotient is a well-defined group:

Definition 8.1.3: Chow Group

Let $Z_k(X)$ denote the k-cycles of X and $\mathrm{Rat}_k(X)$ the k-cycles rationally equivalent to 0. Then the *Chow group* is:

$$A_k(X) = \frac{Z_k(X)}{\mathrm{Rat}_k(X)}$$

An element of $A_k(X)$ is called a *k-cycle class*. In cohomological context, this group is often denoted $CH_i(X)$.

By taking the direct sum $Z_*(X) = \sum_i^\infty A_k(X)$ and $\mathrm{Rat}_*(X) \sum_i^\infty$, we call the elements of $Z_*(X)$ a *cycle* and define the group $A_*(X)$ to be the quotient of these two groups.

A cycle is *positive* if it is not zero and given $\alpha = \{\alpha_k\}_k$ each coefficient of α_k is positive. A *cycle class* is positive it there there is a cycle representative that is positive.

Example 8.1.4: Chow Group

Chow groups are in general difficult to compute. The following are elementary examples:

1. $A_k(X) = A_k(X_{\mathrm{red}})$
2. disjoint unions become direct sums
3. If $X_1, X_2 \subseteq X$ are closed subschemes then we get an exact sequence (note the lack of left-exactness)

$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k\left(X_1 \cup X_2\right) \to 0$$

4. If X is n-dimensional, it has no $(n+1)$-dimensional irreducible components, so:

$$A_n(X) = Z_n(X)$$

   Thus, it has two be that two rationally equivalent cycles in A_n will have the same coefficients (including possibly 0). Thus, the *coefficient of V in α* for an irreducible component $V \subseteq X$ and cycle class α is the coefficient of $[V]$ in α.

Let us now look at how these classes change under the usual cohomological morphisms.

Recall that if $f : X \to Y$ is a scheme morphism, the image of a closed subvariety need not be a closed subvariety ($f : V(xy-1) \to \mathbb{A}^1_k$ where $(x, y) \mapsto x$ being the canonical counter-example, see example 3.5.24). If f is proper, this is the case as the image of an irreducible set is always irreducible, and by definition f is universally closed and hence the image of a closed set is closed. It thus makes sense to pushforward the Chow group. If $V \subseteq X$ is a subvariety and $W = f(X)$, there is an induced embedding $k(W) \hookrightarrow k(V)$. If $\dim_k V = \dim_k W$, this is a finite field extension.

Definition 8.1.5: Pushforward of K-cycle

Let:

$$\deg(V/W) = \begin{cases} [k(V) : k(W)] & \dim_k(V) = \dim_k(W) \\ 0 & \text{otherwise} \end{cases}$$

and $f : X \to Y$ be a morphism. Then:

$$f_*[V] = \deg(V/W)[W]$$

Giving a map:

$$f_* : Z_k(X) \to Z_k(Y)$$

where if f, g are proper morphisms that compose then $(fg)_* = f_*g_*$. If $\dim_{\mathbb{C}}(V) = \dim_{\mathbb{C}}(W)$, then V is generically a covering of W with $\deg(V/W)$ sheets, and the pushforward agrees with the pushforward in topology. To prove this, we require the following build-up result:

Proposition 8.1.6: Pushforward of Divisor

Let $f : X \to Y$ be a proper surjective morphisms between varieties, and let $r \in k(X)^*$. Then:

1. if $\dim(Y) < \dim(X)$, then
$$f_*[\operatorname{div}(r)] = 0$$
2. if $\dim(Y) = \dim(X)$, then
$$f_*[\operatorname{div}(r)] = [\operatorname{div}(N(r))]$$
where $N(r)$ is the *norm* of r (computed as the determinant of the $R(Y)$-linear endomorphism of $R(X)$ given by multiplying by r).

***Proof*:**
As the order function is a homomorphism, without loss of generality we may assume $r \in k(X)$ is an irreducible polynomial (split over the rational part and then the irreducible parts). Let us break it into cases:

1. (case 1) Let $Y = \operatorname{Spec} k$ and $X = \mathbb{P}^1_k$. Then we know that $k(X) = k(t)$ where by choosing coordinates we can write $t = x_1/x_0$. Then r will be some degree d-polynomial in $k[t]$. Now, (r) is a prime ideal in $k[t]$, hence corresponds to a point, and $\operatorname{ord}_p r = 1$. By [9] we see that the only other point on which it has a nonzero order is the point at infinite ∞ where $\operatorname{ord}_\infty r = d$ (the degree); this is because whentaking $s = 1/t$ ew get:
$$\frac{1}{s^d}(a_d + \cdots + a_0 s^d)$$
Then:
$$[\operatorname{div}(r)] = [P] - d[\infty]$$
Now (letting $s = 1/t$):
$$\mathcal{O}_P = k[t]/(r) \qquad \text{and} \qquad \mathcal{O}_\infty = k[s]/(s) = k$$
Hence:
$$f_*[\operatorname{div}(r)] = d[Y] - d[Y] = 0$$

2. (case 2) Let $f : X \to Y$ be finite so that $K = k(Y)$ and $L = k(X)$ are a finite extension. We want to know which codimension 1 subvarieties of X map to a codimension 1 subvariety of Y, so let $W \subseteq Y$ be codimension 1. Then let $A = \mathcal{O}_{Y,W}$ be the associated local ring. Then it is an exercise in commutative algebra that the maximal ideals of the finite extension B/A where $L = B \otimes_A K$ correspond to subvarieties $V_i \subseteq X$ mapping onto W, and $B_{\mathfrak{m}_i} \cong \mathcal{O}_{X,V_i}$[a]. Then:
$$f_*[\operatorname{div}(r)] = \sum_i \operatorname{ord}_{V_i}(r)[k(V_i) : k(W)]$$
We need to show that:
$$\sum_i \operatorname{ord}_{V_i}(r)[k(V_i) : k(W)] = \operatorname{ord}_W(N(r))$$
As N and ord are homomorphisms, it suffices to show $r \in B$. Then he quotes a bunch of results I'll come back to.

Now, for the general case, we can prove (2) as there is always a B as in case 2 (Fulton also quickly demonstrates a more elementary proof using normalization).

For the general case of (a), if $\dim(Y) < \dim(X) - 1$, then there are no divisors and automatically $f_*[\operatorname{div}(r)] = 0$, so assume $\dim(Y) = \dim(X) - 1$. Now, algebraically we have:
$$f_*[\operatorname{div}(r)] = \sum_V \operatorname{ord}_V(r)[k(V) : k(Y)]$$
BY letting $K = k(Y)$, notice that the equation is invariant if we take $Y = \operatorname{Spec} K$ and replace X with $X_k = X \otimes_Y \operatorname{Spec}(K)$. Namely, the above equation only deals with the "top-dimension" information and hence we can look simply look at X as a curve over $Y = \operatorname{Spec}(K)$.

Now, we want to use the finite-morphism case, so take the normalization $h : \tilde{X} \to X$ and the finite morphism $g : \tilde{X} \to \mathbb{P}^1_K$. Letting $\pi : \mathbb{P}^1_K \to Y$ be the projection we get:
$$f \circ h = \pi \circ g$$
We can now take the image of r by the isomorphism $k(X) \cong k(\tilde{X})$ so that we take $\tilde{r}k(\tilde{X})$. Then by functoriality:
$$f_*[\operatorname{div}(r)] = f_* h_*[\operatorname{div}(\tilde{r}] = \pi_* g_*[\operatorname{div}(\tilde{r})]$$
The latter is zero by an application of Case 2 to g and Case 1 to π, completing the proof.

[a] if X, Y are affine, they have coordinate rings Γ_X, Γ_Y, so then A is the localization of Γ_Y at the prime ideal corresponding to W, and $B = \Gamma_X \otimes_{\Gamma_Y} A$

Theorem 8.1.7: Proper Morphism Preserves Rational Equivalence

Let $f : X \to Y$ be a proper morphism and $\alpha \in Z_k(X)$. Then if $\alpha \sim 0$, $f_*(\alpha) \sim 0 \sim f_*(0)$.

Thus, there is an induced homomorphism:

$$f_* A_k(X) \to A_k(Y)$$

making f_* a covariant functor

***Proof*:**
It suffices to show that zero maps to zero, so it suffices to understand $\alpha = [\mathrm{div}(r)]$ for a rational function r on a subvariety $V \subseteq X$. Then we may replace X by V and Y by $f(X)$, and so f is surjective. Then proposition 8.1.6 gives the desired result.

Definition 8.1.8: Degree of zero cycle

Let X be a proper scheme over $S = \operatorname{Spec} k$ for a ground field k (i.e. a complete scheme) and $\alpha = \sum_P n_P[P]$ a zero-cycle on X, then the *degree* of α is :

$$\deg(\alpha) = \int_X \alpha = \sum_P n_P[k(P) : k]$$

that is:

$$\deg(\alpha) = \pi_*(\alpha)$$

where $\pi : X \to \operatorname{Spec} k$ is the structure morphism and we identify $A_0(S) = \mathbb{Z}[S]$ with $\mathbb{Z}$ TBU. If it is unambiguous, we often write $\int \alpha$ instead of $\int_X \alpha$.

By theorem 8.1.7, rationally equivalent cycles have the same degree. This morphism can be extend to A_*X by defining $\int_X \alpha = 0$ for $\alpha \in A_kX$, $k > 0$. For any morphism of complete schemes $f : X \to Y$ and any $\alpha \in \int_X \alpha$, we get:

$$\int_X \alpha = \int_Y f_*(\alpha)$$

which is just a special case of functoriality.

8.1.9 Assuming Pushforward with subschemes Let $Y_1, \ldots, Y_r$ be closed subschemes of a scheme X. Let Y be a closed subscheme of X which contains all the Y_i. Given $\alpha_i \in A_*Y_i$, $i = 1, \ldots, r$, and $\beta \in A_*Y$, we will usually write "$\beta = \sum_{i=1}^r \alpha_i$ in A_*Y" in place of the precise equation $\beta = \sum_{i=1}^r \varphi_{i*}(\alpha_i)$ where φ_i is the inclusion of Y_i in Y.

Example 8.1.10: Pushforward Via Proper Maps

1. (Bézout's Theorem)
2. (lack of properness)

3. (curve of genus g)
4. (abelian variety)

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